

Territory Analyses with Year as Factor

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For this report, we generated 50 different simulated data sets varying probability of mate fidelity, a pair's territory fidelity, a single male's territory fidelity, and a single female's territory fidelity. Latin hyper cube sampling (LHS) was used to generate these probabilities for the 50 different runs. In addition to the 50 runs generated from the LHS, we also generated two simulated data sets, where probabilities were chosen to either show the territory effect, or to make the territory effect more difficult to detect.

I am using GLMMs instead of GAMMs. The reason being for this is that we did try modelling year with both cyclic and cubic regression splines. However, edfs on the year smoother were large and the k-indices were often too small, meaning our basis dimension was inadequate. Since we only have 20 years in the data set, we cannot get the adequate dimension for our basis.

In addition to computing AIC and BIC for models, we also compute Cohen's Kappa for the binomial models and marginal and conditional R^2 for the Poisson models.

This report does not include models fit with a z-score response. All models seemed to violate assumptions.

The probability of fledgling were randomly drawn from a beta(14, (14/0.65 - 14)) distribution for good territories, a beta(9, (9/0.55 - 9)) distribution for mediocre territories, and a beta(14, (5/0.45 - 5)) distribution for bad territories. Note that maybe playing with priors could be interesting. I initially ran the simulation where priors were very different and the models seemed to have an easier time picking up on the territory effect.

Simulation 1

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	condtional_R2	marginal_R2	Family	RE
340	1	- 464.5323	971.0645	1051.472	353.1209	319	0.9232408	0.9110579	Poisson	Terr
340	1	- 464.5195	973.0390	1057.276	350.3942	318	0.9216720	0.9089455	Poisson	BirdID+Terr
340	1	- 466.5900	975.1800	1055.588	372.4453	319	0.9211131	0.9124503	Poisson	BirdID

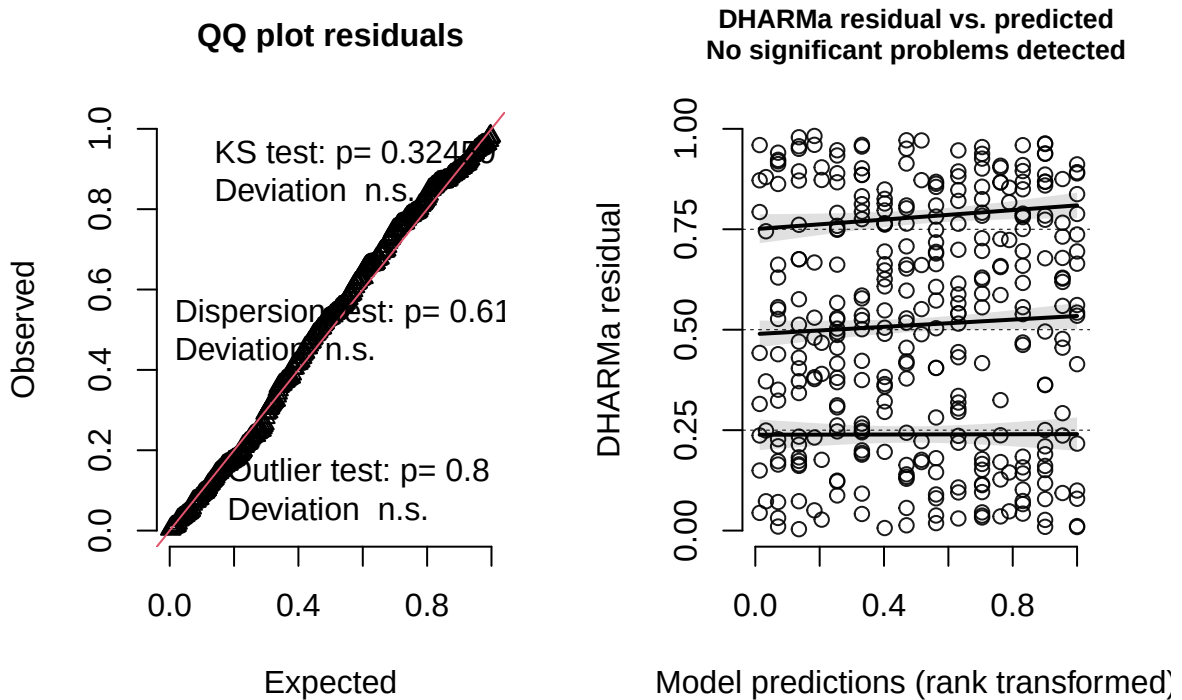
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.2581616	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.3087890	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.0858927	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.3012479	Poisson	BirdID+Terr

effect	group	term	estimate	Family	RE
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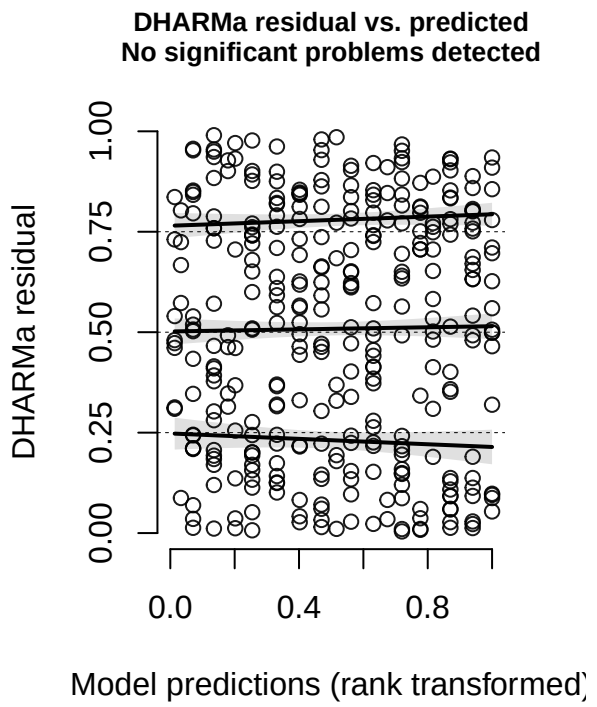
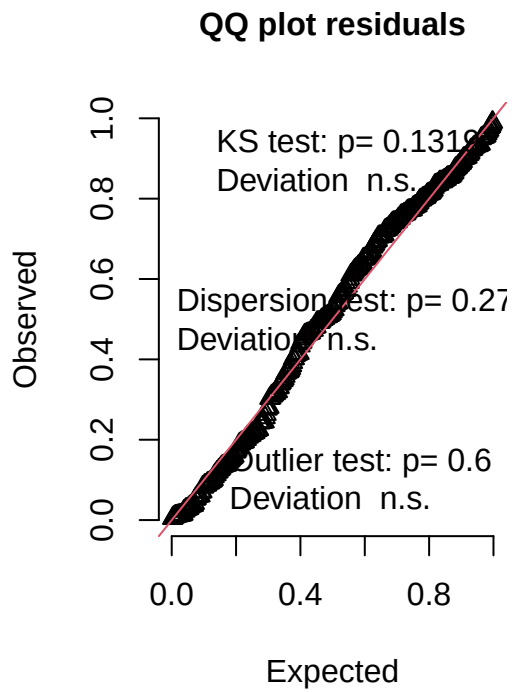
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.5362760	0.2211406	2.4250453	0.0153065	Poisson	BirdID
fixed	factor(fac.yr)2	-1.1520757	0.3750773	-3.0715691	0.0021294	Poisson	BirdID
fixed	factor(fac.yr)3	-0.8420693	0.3153415	-2.6703407	0.0075774	Poisson	BirdID
fixed	factor(fac.yr)4	-0.2365260	0.2806434	-0.8427992	0.3993408	Poisson	BirdID
fixed	factor(fac.yr)5	-0.0520277	0.2754872	-0.1888570	0.8502049	Poisson	BirdID
fixed	factor(fac.yr)6	-0.0837035	0.2780875	-0.3009968	0.7634169	Poisson	BirdID
fixed	factor(fac.yr)7	-0.4986123	0.3048589	-1.6355513	0.1019335	Poisson	BirdID
fixed	factor(fac.yr)8	-1.0024195	0.3444921	-2.9098477	0.0036160	Poisson	BirdID
fixed	factor(fac.yr)9	-0.1266382	0.2772187	-0.4568171	0.6478025	Poisson	BirdID
fixed	factor(fac.yr)10	-0.2942070	0.2932976	-1.0031003	0.3158125	Poisson	BirdID
fixed	factor(fac.yr)11	-0.5319607	0.3005152	-1.7701623	0.0767001	Poisson	BirdID
fixed	factor(fac.yr)12	-0.4370052	0.3009886	-1.4518994	0.1465296	Poisson	BirdID
fixed	factor(fac.yr)13	0.1268883	0.2852884	0.4447721	0.6564845	Poisson	BirdID
fixed	factor(fac.yr)14	-0.1159866	0.3191713	-0.3633992	0.7163067	Poisson	BirdID
fixed	factor(fac.yr)15	-0.9646833	0.4744255	-2.0333715	0.0420150	Poisson	BirdID
fixed	factor(fac.yr)16	-17.6746645	1884.0208565	-0.0093814	0.9925149	Poisson	BirdID
fixed	factor(fac.yr)17	-0.9062554	0.5597596	-1.6190082	0.1054455	Poisson	BirdID
fixed	factor(fac.yr)18	-0.3192143	0.3978443	-0.8023599	0.4223448	Poisson	BirdID
fixed	factor(fac.yr)19	-2.1720261	1.0387996	-2.0909000	0.0365370	Poisson	BirdID
fixed	factor(fac.yr)20	-0.0908690	0.4332096	-0.2097575	0.8338570	Poisson	BirdID
fixed	(Intercept)	0.4767244	0.2238764	2.1294085	0.0332205	Poisson	Terr
fixed	factor(fac.yr)2	-1.1039264	0.3803959	-2.9020464	0.0037073	Poisson	Terr
fixed	factor(fac.yr)3	-0.7689952	0.3216809	-2.3905533	0.0168230	Poisson	Terr
fixed	factor(fac.yr)4	-0.1489640	0.2840661	-0.5243992	0.6000009	Poisson	Terr
fixed	factor(fac.yr)5	0.0086050	0.2798342	0.0307504	0.9754686	Poisson	Terr
fixed	factor(fac.yr)6	0.0195915	0.2812087	0.0696690	0.9444572	Poisson	Terr
fixed	factor(fac.yr)7	-0.4544224	0.3072137	-1.4791739	0.1390939	Poisson	Terr
fixed	factor(fac.yr)8	-0.9890679	0.3448782	-2.8678761	0.0041324	Poisson	Terr
fixed	factor(fac.yr)9	-0.0683848	0.2774144	-0.2465078	0.8052891	Poisson	Terr
fixed	factor(fac.yr)10	-0.2330265	0.2939114	-0.7928461	0.4278675	Poisson	Terr
fixed	factor(fac.yr)11	-0.4712504	0.2997757	-1.5720103	0.1159482	Poisson	Terr
fixed	factor(fac.yr)12	-0.3885213	0.3035456	-1.2799435	0.2005650	Poisson	Terr
fixed	factor(fac.yr)13	0.2148276	0.2861072	0.7508640	0.4527345	Poisson	Terr
fixed	factor(fac.yr)14	-0.0174852	0.3230938	-0.0541179	0.9568412	Poisson	Terr
fixed	factor(fac.yr)15	-0.8792337	0.4780132	-1.8393501	0.0658637	Poisson	Terr
fixed	factor(fac.yr)16	-17.7469564	2040.6728318	-0.0086966	0.9930612	Poisson	Terr
fixed	factor(fac.yr)17	-0.9601733	0.5595740	-1.7159005	0.0861803	Poisson	Terr
fixed	factor(fac.yr)18	-0.2879616	0.3972686	-0.7248536	0.4685418	Poisson	Terr
fixed	factor(fac.yr)19	-2.0659972	1.0396674	-1.9871713	0.0469034	Poisson	Terr
fixed	factor(fac.yr)20	-0.0923975	0.4406516	-0.2096838	0.8339145	Poisson	Terr
fixed	(Intercept)	0.4774747	0.2247288	2.1246712	0.0336141	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-1.1047762	0.3807616	-2.9014908	0.0037139	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.7730790	0.3221709	-2.3995932	0.0164133	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.1551458	0.2847374	-0.5448733	0.5858406	Poisson	BirdID+Terr
fixed	factor(fac.yr)5	0.0022379	0.2807788	0.0079703	0.9936407	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	0.0136484	0.2820362	0.0483924	0.9614035	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.4578710	0.3081207	-1.4860118	0.1372760	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.9889354	0.3458607	-2.8593463	0.0042452	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)9	-0.0713911	0.2785688	-0.2562781	0.7977361	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.2369374	0.2950490	-0.8030440	0.4219493	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.4745220	0.3009407	-1.5767957	0.1148425	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.3906491	0.3045616	-1.2826604	0.1996111	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	0.2085428	0.2873639	0.7257096	0.4680168	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.0253015	0.3242670	-0.0780267	0.9378068	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.8825350	0.4790996	-1.8420703	0.0654649	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-17.5985225	1894.4394763	-0.0092896	0.9925881	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.9525323	0.5611590	-1.6974373	0.0896140	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.2868703	0.3987792	-0.7193714	0.4719121	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-2.0719750	1.0412572	-1.9898782	0.0466044	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.0906613	0.4421545	-0.2050445	0.8375374	Poisson	BirdID+Terr

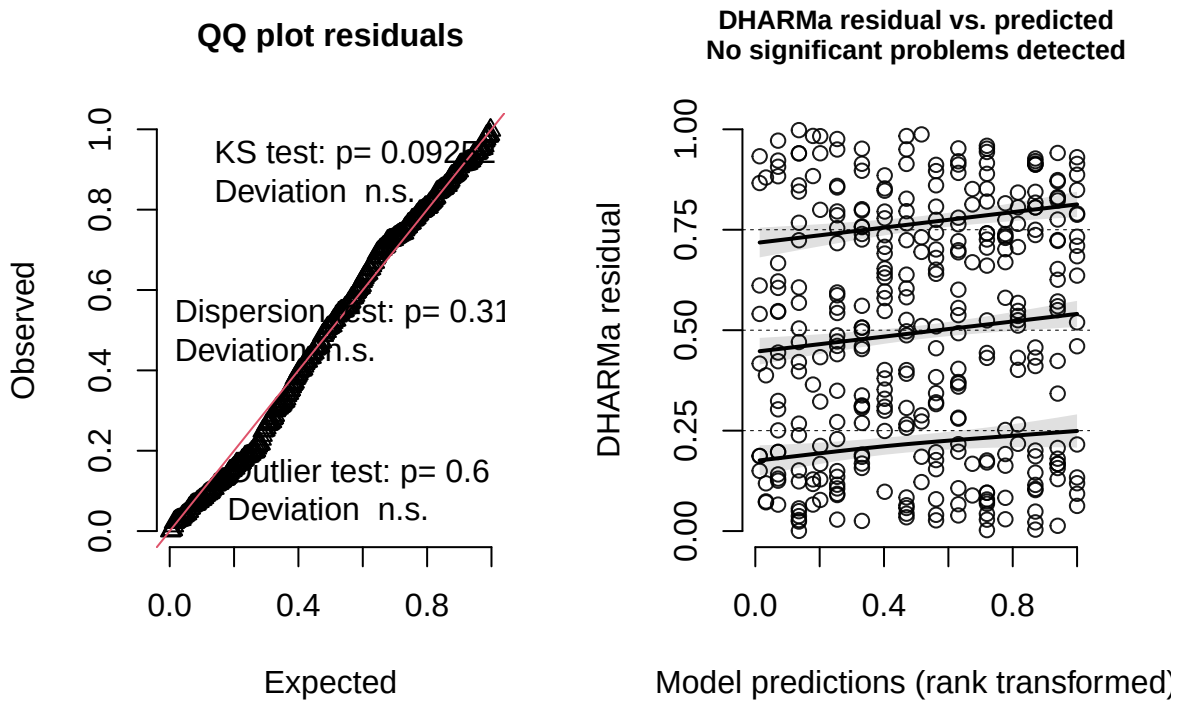
DHARMA residual



DHARMa residual



DHARMa residual



The territory only model performed best on all model selection criteria.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
340	1	-195.1793	434.3585	518.5953	258.1039	318	0.689	Binomial	BirdID+Terr
340	1	-196.6810	435.3620	515.7699	303.3099	319	0.536	Binomial	BirdID
340	1	-198.1773	438.3546	518.7624	318.5018	319	0.503	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.1108776	Binomial	BirdID
ran_pars	Terr	sd__(Intercept)	0.9552550	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.0875024	Binomial	BirdID+Terr
ran_pars	Terr	sd__(Intercept)	0.8627287	Binomial	BirdID+Terr

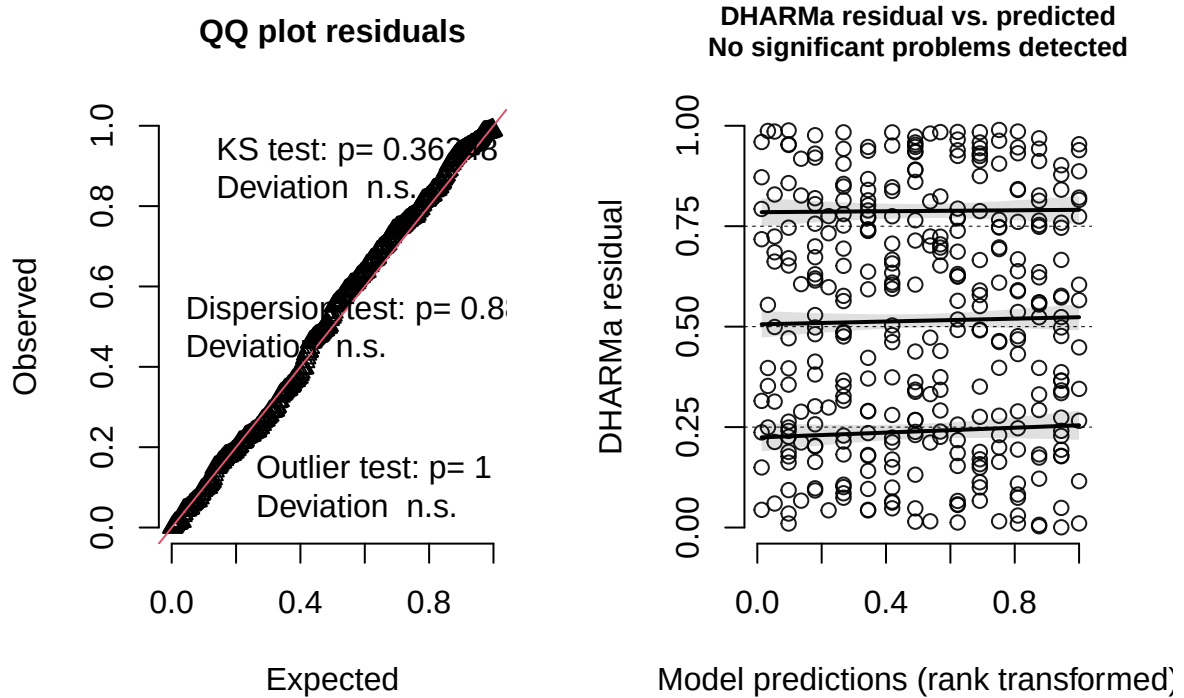
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	2.8137616	1.158421	2.4289632	0.0151421	Binomial	BirdID
fixed	factor(fac.yr)2	-3.4687102	1.255564	-2.7626716	0.0057330	Binomial	BirdID
fixed	factor(fac.yr)3	-2.4237677	1.226179	-1.9766835	0.0480774	Binomial	BirdID
fixed	factor(fac.yr)4	-1.1568201	1.274750	-0.9074876	0.3641490	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-2.2138787	1.248045	-1.7738780	0.0760833	Binomial	BirdID
fixed	factor(fac.yr)6	-1.6435119	1.260103	-1.3042682	0.1921422	Binomial	BirdID
fixed	factor(fac.yr)7	-1.2691406	1.296399	-0.9789735	0.3275931	Binomial	BirdID
fixed	factor(fac.yr)8	-3.3803187	1.256628	-2.6899921	0.0071454	Binomial	BirdID
fixed	factor(fac.yr)9	-2.2165408	1.249004	-1.7746471	0.0759562	Binomial	BirdID
fixed	factor(fac.yr)10	-1.5204240	1.280952	-1.1869485	0.2352479	Binomial	BirdID
fixed	factor(fac.yr)11	-2.3246458	1.249979	-1.8597473	0.0629213	Binomial	BirdID
fixed	factor(fac.yr)12	-1.9771177	1.263443	-1.5648649	0.1176146	Binomial	BirdID
fixed	factor(fac.yr)13	-1.5512594	1.332457	-1.1642098	0.2443390	Binomial	BirdID
fixed	factor(fac.yr)14	-2.0293348	1.348923	-1.5044105	0.1324757	Binomial	BirdID
fixed	factor(fac.yr)15	-3.7400702	1.415277	-2.6426424	0.0082262	Binomial	BirdID
fixed	factor(fac.yr)16	-16.1055882	278.129777	-0.0579067	0.9538229	Binomial	BirdID
fixed	factor(fac.yr)17	-3.4335062	1.524413	-2.2523462	0.0243004	Binomial	BirdID
fixed	factor(fac.yr)18	-2.1536232	1.434840	-1.5009497	0.1333686	Binomial	BirdID
fixed	factor(fac.yr)19	-4.8430777	1.728832	-2.8013583	0.0050888	Binomial	BirdID
fixed	factor(fac.yr)20	-2.7274336	1.558609	-1.7499151	0.0801330	Binomial	BirdID
fixed	(Intercept)	2.5803370	1.110398	2.3237948	0.0201365	Binomial	Terr
fixed	factor(fac.yr)2	-3.3291863	1.219436	-2.7301041	0.0063314	Binomial	Terr
fixed	factor(fac.yr)3	-2.1246261	1.189861	-1.7856079	0.0741628	Binomial	Terr
fixed	factor(fac.yr)4	-0.8065632	1.233465	-0.6539003	0.5131760	Binomial	Terr
fixed	factor(fac.yr)5	-1.9264962	1.204136	-1.5998992	0.1096209	Binomial	Terr
fixed	factor(fac.yr)6	-1.2776894	1.212726	-1.0535681	0.2920807	Binomial	Terr
fixed	factor(fac.yr)7	-1.1822551	1.244815	-0.9497438	0.3422424	Binomial	Terr
fixed	factor(fac.yr)8	-3.1919862	1.195993	-2.6688999	0.0076100	Binomial	Terr
fixed	factor(fac.yr)9	-1.9585314	1.192960	-1.6417412	0.1006436	Binomial	Terr
fixed	factor(fac.yr)10	-1.1982914	1.229273	-0.9747964	0.3296612	Binomial	Terr
fixed	factor(fac.yr)11	-2.0122538	1.193652	-1.6857959	0.0918351	Binomial	Terr
fixed	factor(fac.yr)12	-1.7138512	1.206180	-1.4208912	0.1553484	Binomial	Terr
fixed	factor(fac.yr)13	-1.1295639	1.270027	-0.8894015	0.3737873	Binomial	Terr
fixed	factor(fac.yr)14	-1.4134021	1.300537	-1.0867837	0.2771324	Binomial	Terr
fixed	factor(fac.yr)15	-3.1850945	1.349046	-2.3609973	0.0182259	Binomial	Terr
fixed	factor(fac.yr)16	-20.2553097	2273.083731	-0.0089109	0.9928902	Binomial	Terr
fixed	factor(fac.yr)17	-3.3852222	1.454675	-2.3271325	0.0199582	Binomial	Terr
fixed	factor(fac.yr)18	-2.0661046	1.359862	-1.5193486	0.1286748	Binomial	Terr
fixed	factor(fac.yr)19	-4.2254655	1.666385	-2.5357076	0.0112220	Binomial	Terr
fixed	factor(fac.yr)20	-2.2507096	1.493748	-1.5067529	0.1318740	Binomial	Terr
fixed	(Intercept)	2.9409429	1.241356	2.3691371	0.0178296	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-3.6961541	1.345825	-2.7463854	0.0060256	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-2.4403953	1.314150	-1.8570142	0.0633091	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-1.0225945	1.358920	-0.7525055	0.4517471	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-2.2884731	1.335535	-1.7135249	0.0866160	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-1.5017567	1.343729	-1.1176037	0.2637363	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-1.2566690	1.389747	-0.9042433	0.3658664	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-3.6134772	1.339282	-2.6980696	0.0069743	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-2.1608440	1.336699	-1.6165526	0.1059749	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-1.3770751	1.369434	-1.0055800	0.3146176	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-2.3183394	1.338222	-1.7324030	0.0832018	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-1.9995464	1.346368	-1.4851414	0.1375064	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-1.5160299	1.418625	-1.0686618	0.2852221	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-1.9282958	1.447776	-1.3319016	0.1828925	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-3.7848346	1.509757	-2.5069157	0.0121790	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-17.6543943	528.351430	-0.0334141	0.9733444	Binomial	BirdID+Terr

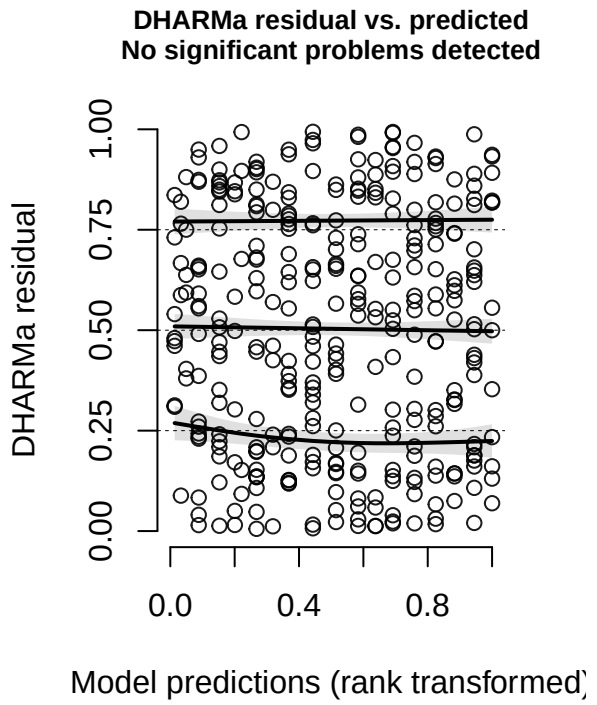
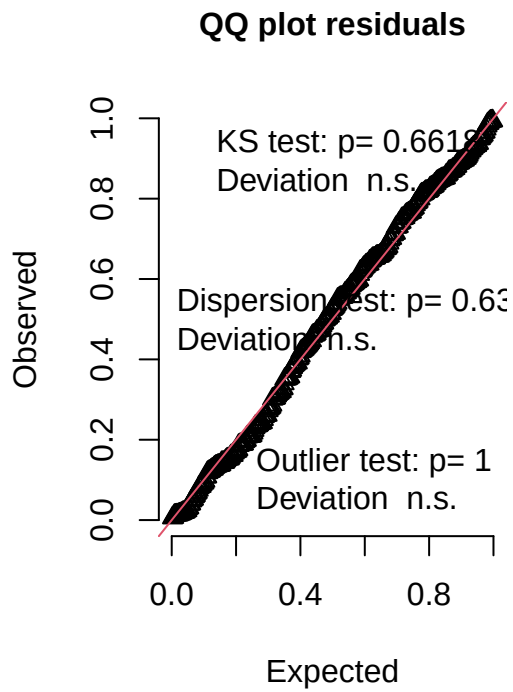
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-3.5250000	1.624924	-2.1693323	0.0300575	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-2.2923549	1.518460	-1.5096572	0.1311309	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-5.1945510	1.882545	-2.7593229	0.0057921	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-2.8654549	1.667147	-1.7187778	0.0856548	Binomial	BirdID+Terr

The territory only model performed worst on model selection criteria

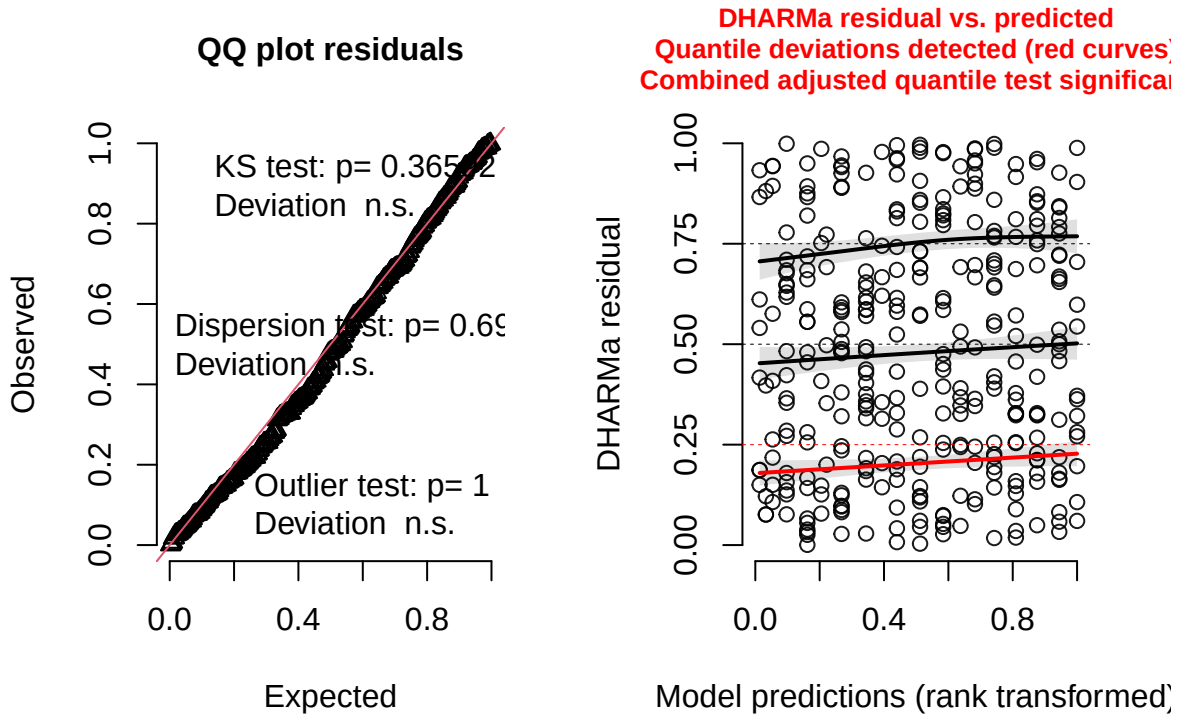
DHARMA residual



DHARMA residual



DHARMA residual



A slight violation of model assumptions.

Probabilities used to generate simulation 1

Simulation 1
 Probability of mate fidelity: 0.958
 Probability of pair dispersal: 0.77
 Probability of single male dispersal: 0.182
 Probability of single female dispersal: 0.154

Simulation 2

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Family	RE
363	1	-477.8737	997.7475	1079.530	292.2337	342	Poisson	Terr
363	1	-477.8735	999.7470	1085.424	292.3158	341	Poisson	BirdID+Terr
363	1	-486.5275	1015.0551	1096.838	395.8302	342	Poisson	BirdID

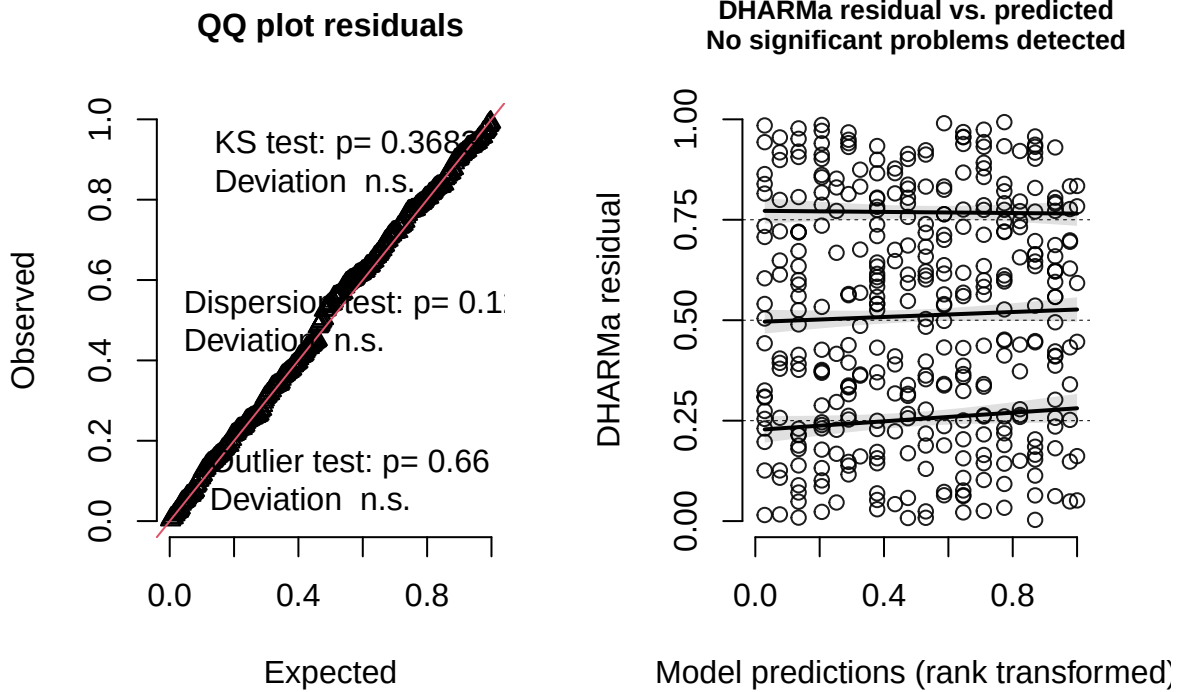
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.0000002	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.3962638	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.0000527	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.3958857	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.4054651	0.2357023	1.7202428	0.0853883	Poisson	BirdID
fixed	factor(fac.yr)2	-0.3541718	0.3248931	-1.0901178	0.2756612	Poisson	BirdID
fixed	factor(fac.yr)3	-0.5877867	0.3248931	-1.8091692	0.0704247	Poisson	BirdID
fixed	factor(fac.yr)4	-0.2087548	0.3021090	-0.6909917	0.4895707	Poisson	BirdID
fixed	factor(fac.yr)5	0.1144104	0.2873719	0.3981265	0.6905369	Poisson	BirdID
fixed	factor(fac.yr)6	-0.2451225	0.3042903	-0.8055546	0.4204997	Poisson	BirdID
fixed	factor(fac.yr)7	-0.6931472	0.3212080	-2.1579385	0.0309326	Poisson	BirdID
fixed	factor(fac.yr)8	-0.4054651	0.2981424	-1.3599713	0.1738390	Poisson	BirdID
fixed	factor(fac.yr)9	0.2657032	0.2788867	0.9527281	0.3407279	Poisson	BirdID
fixed	factor(fac.yr)10	-0.0752234	0.2946278	-0.2553168	0.7984785	Poisson	BirdID
fixed	factor(fac.yr)11	-0.8362480	0.3639761	-2.2975353	0.0215883	Poisson	BirdID
fixed	factor(fac.yr)12	-0.3101549	0.3178209	-0.9758797	0.3291241	Poisson	BirdID
fixed	factor(fac.yr)13	-0.4626235	0.3381998	-1.3679002	0.1713433	Poisson	BirdID
fixed	factor(fac.yr)14	-0.3144933	0.3146964	-0.9993548	0.3176229	Poisson	BirdID
fixed	factor(fac.yr)15	-0.7419373	0.3944053	-1.8811545	0.0599509	Poisson	BirdID
fixed	factor(fac.yr)16	-0.4054651	0.3944053	-1.0280417	0.3039302	Poisson	BirdID
fixed	factor(fac.yr)17	0.2876821	0.3248931	0.8854667	0.3759050	Poisson	BirdID
fixed	factor(fac.yr)18	-0.5232481	0.4249183	-1.2314088	0.2181700	Poisson	BirdID
fixed	factor(fac.yr)19	-0.4054651	0.4249183	-0.9542190	0.3399728	Poisson	BirdID
fixed	factor(fac.yr)20	0.3677248	0.3639761	1.0102991	0.3123520	Poisson	BirdID
fixed	(Intercept)	0.4163941	0.2602456	1.6000045	0.1095976	Poisson	Terr
fixed	factor(fac.yr)2	-0.4692124	0.3504258	-1.3389781	0.1805778	Poisson	Terr
fixed	factor(fac.yr)3	-0.6687993	0.3509143	-1.9058766	0.0566662	Poisson	Terr
fixed	factor(fac.yr)4	-0.2639223	0.3291452	-0.8018415	0.4226446	Poisson	Terr
fixed	factor(fac.yr)5	-0.0222089	0.3145915	-0.0705960	0.9437193	Poisson	Terr
fixed	factor(fac.yr)6	-0.2944356	0.3288220	-0.8954255	0.3705597	Poisson	Terr
fixed	factor(fac.yr)7	-0.7740173	0.3449201	-2.2440482	0.0248293	Poisson	Terr
fixed	factor(fac.yr)8	-0.5269755	0.3208033	-1.6426746	0.1004503	Poisson	Terr
fixed	factor(fac.yr)9	0.1983460	0.3035936	0.6533272	0.5135453	Poisson	Terr
fixed	factor(fac.yr)10	-0.1908165	0.3215612	-0.5934065	0.5529091	Poisson	Terr
fixed	factor(fac.yr)11	-0.8813050	0.3877072	-2.2731197	0.0230190	Poisson	Terr
fixed	factor(fac.yr)12	-0.3691895	0.3456827	-1.0680012	0.2855200	Poisson	Terr
fixed	factor(fac.yr)13	-0.4777798	0.3624140	-1.3183260	0.1873945	Poisson	Terr
fixed	factor(fac.yr)14	-0.2698226	0.3383794	-0.7973964	0.4252208	Poisson	Terr
fixed	factor(fac.yr)15	-0.7192910	0.4194755	-1.7147389	0.0863931	Poisson	Terr
fixed	factor(fac.yr)16	-0.4013536	0.4246363	-0.9451702	0.3445720	Poisson	Terr
fixed	factor(fac.yr)17	0.1922216	0.3595379	0.5346351	0.5929022	Poisson	Terr
fixed	factor(fac.yr)18	-0.5761332	0.4570605	-1.2605184	0.2074824	Poisson	Terr
fixed	factor(fac.yr)19	-0.4538672	0.4531669	-1.0015453	0.3165632	Poisson	Terr
fixed	factor(fac.yr)20	0.1735199	0.4065240	0.4268379	0.6694974	Poisson	Terr
fixed	(Intercept)	0.4161586	0.2602374	1.5991496	0.1097874	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.4683697	0.3503650	-1.3368049	0.1812863	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.6690638	0.3509350	-1.9065179	0.0565830	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.2644750	0.3291687	-0.8034633	0.4217070	Poisson	BirdID+Terr

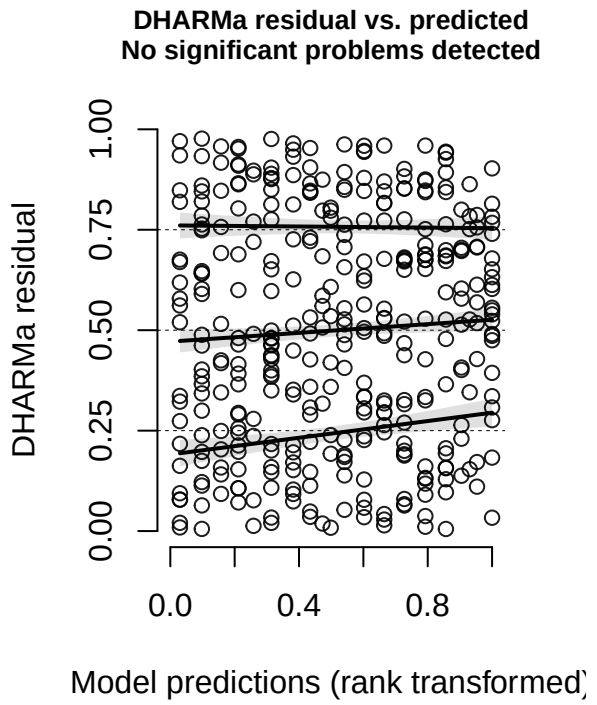
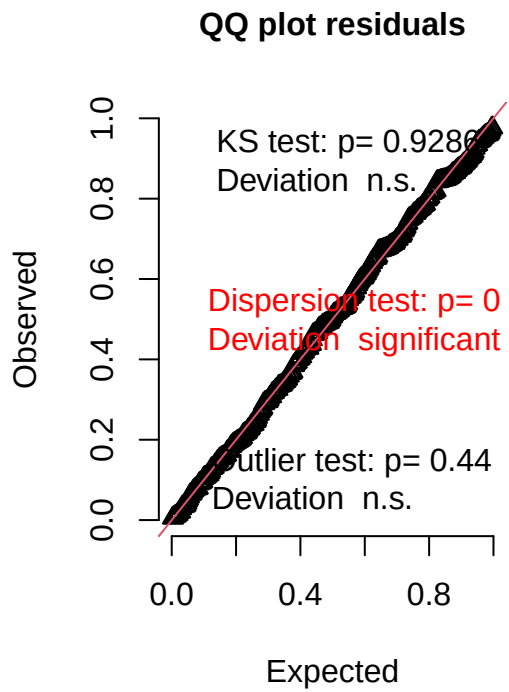
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.0255438	0.3147157	-0.0811648	0.9353109	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.2948691	0.3288464	-0.8966771	0.3698913	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.7740627	0.3449265	-2.2441381	0.0248235	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.5250862	0.3207167	-1.6372276	0.1015829	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	0.1990639	0.3035737	0.6557352	0.5119945	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.1891969	0.3214963	-0.5884884	0.5562045	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.8772772	0.3873418	-2.2648657	0.0235209	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.3701754	0.3457436	-1.0706643	0.2843204	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.4756993	0.3622610	-1.3131398	0.1891358	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.2692767	0.3383482	-0.7958570	0.4261152	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.7182591	0.4193439	-1.7128163	0.0867463	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.4036843	0.4249017	-0.9500651	0.3420792	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	0.1931309	0.3594862	0.5372415	0.5911008	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.5814670	0.4577490	-1.2702749	0.2039867	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-0.4556462	0.4534029	-1.0049478	0.3149220	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	0.1744422	0.4064446	0.4291906	0.6677845	Poisson	BirdID+Terr

Territory only model is performing best on AIC and BIC.

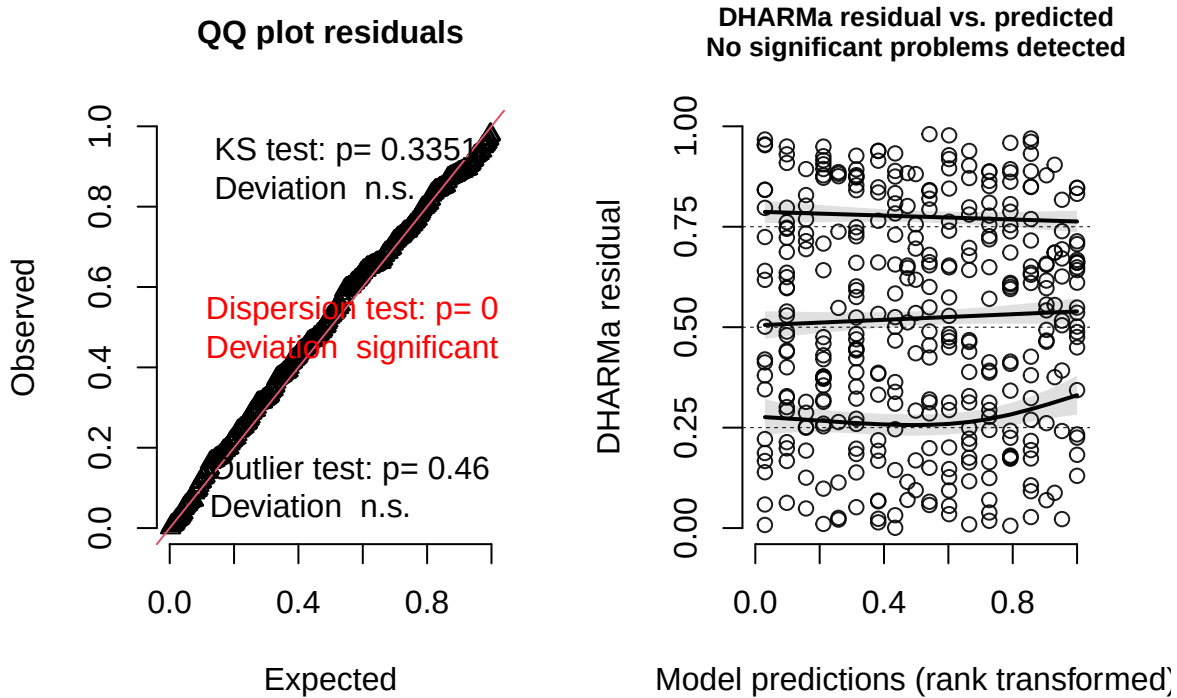
DHARMA residual



DHARMa residual



DHARMA residual



Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
363	1	-195.2656	432.5312	514.3137	252.1550	342	0.670	Binomial	Terr
363	1	-194.8000	433.5999	519.2768	226.8081	341	0.713	Binomial	BirdID+Terr
363	1	-211.2862	464.5723	546.3548	349.0808	342	0.442	Binomial	BirdID

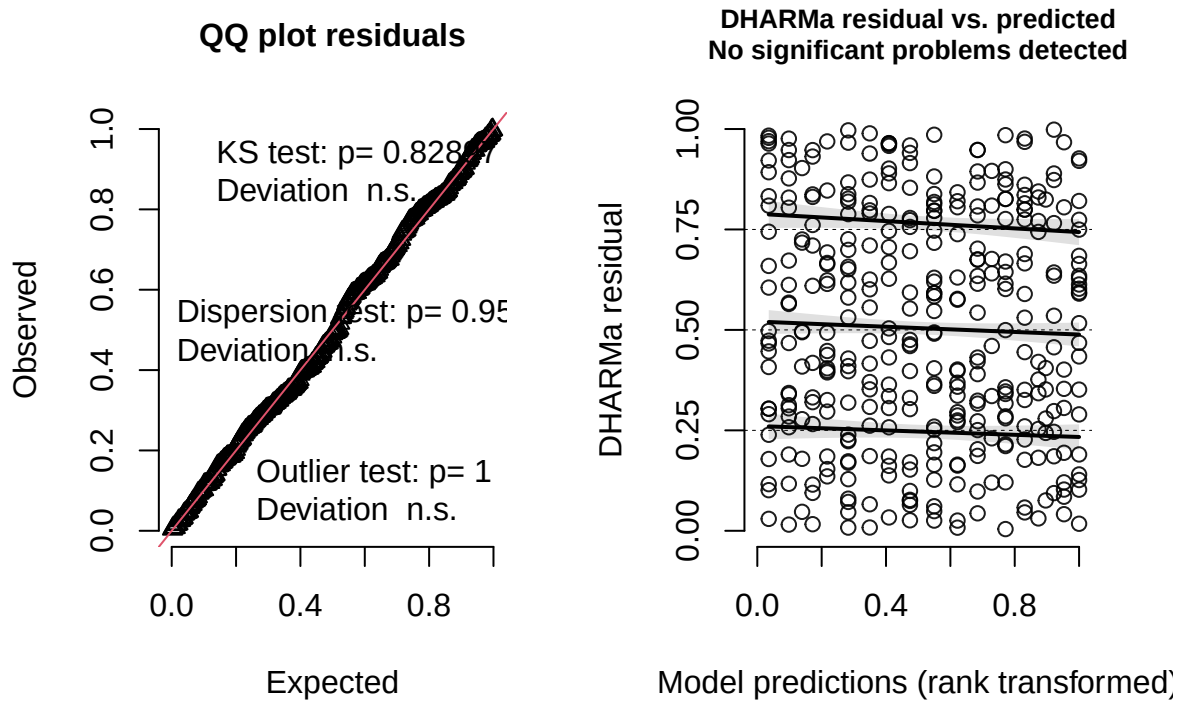
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.8981144	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.7392691	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	0.5972489	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.8488186	Binomial	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	2.3293530	1.131853	2.0579994	0.0395902	Binomial	BirdID
fixed	factor(fac.yr)2	-1.7290653	1.237697	-1.3970025	0.1624128	Binomial	BirdID
fixed	factor(fac.yr)3	-2.5334484	1.212419	-2.0895822	0.0366553	Binomial	BirdID
fixed	factor(fac.yr)4	-1.6006873	1.227899	-1.3035983	0.1923706	Binomial	BirdID
fixed	factor(fac.yr)5	-1.2624979	1.238081	-1.0197213	0.3078606	Binomial	BirdID
fixed	factor(fac.yr)6	-1.5050028	1.231260	-1.2223273	0.2215839	Binomial	BirdID

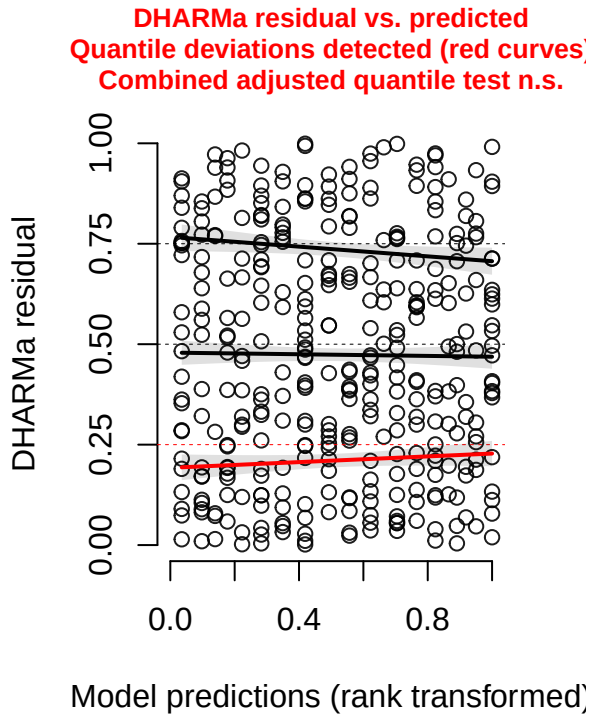
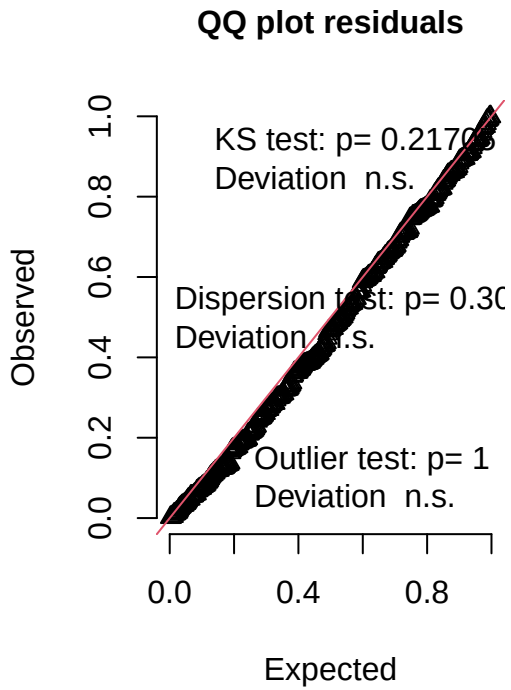
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)7	-1.9396156	1.207388	-1.6064562	0.1081737	Binomial	BirdID
fixed	factor(fac.yr)8	-1.4482195	1.212615	-1.1942944	0.2323628	Binomial	BirdID
fixed	factor(fac.yr)9	0.8654975	1.600583	0.5407389	0.5886876	Binomial	BirdID
fixed	factor(fac.yr)10	-1.3005499	1.239820	-1.0489827	0.2941861	Binomial	BirdID
fixed	factor(fac.yr)11	-2.5116273	1.238775	-2.0275085	0.0426104	Binomial	BirdID
fixed	factor(fac.yr)12	-1.2464400	1.256901	-0.9916768	0.3213552	Binomial	BirdID
fixed	factor(fac.yr)13	-2.0807837	1.247237	-1.6683141	0.0952534	Binomial	BirdID
fixed	factor(fac.yr)14	-1.3600579	1.244541	-1.0928185	0.2744735	Binomial	BirdID
fixed	factor(fac.yr)15	-2.1611395	1.277741	-1.6913750	0.0907652	Binomial	BirdID
fixed	factor(fac.yr)16	-1.1921035	1.368757	-0.8709389	0.3837875	Binomial	BirdID
fixed	factor(fac.yr)17	0.2888621	1.633328	0.1768549	0.8596223	Binomial	BirdID
fixed	factor(fac.yr)18	-2.2432502	1.346678	-1.6657663	0.0957600	Binomial	BirdID
fixed	factor(fac.yr)19	-1.2940127	1.436473	-0.9008262	0.3676807	Binomial	BirdID
fixed	factor(fac.yr)20	-0.7525338	1.648976	-0.4563643	0.6481280	Binomial	BirdID
fixed	(Intercept)	3.4250489	1.203732	2.8453593	0.0044361	Binomial	Terr
fixed	factor(fac.yr)2	-2.5894214	1.338480	-1.9345989	0.0530395	Binomial	Terr
fixed	factor(fac.yr)3	-3.7597905	1.313159	-2.8631644	0.0041943	Binomial	Terr
fixed	factor(fac.yr)4	-2.0432244	1.317481	-1.5508565	0.1209361	Binomial	Terr
fixed	factor(fac.yr)5	-2.0876527	1.321064	-1.5802815	0.1140424	Binomial	Terr
fixed	factor(fac.yr)6	-2.0751221	1.301380	-1.5945545	0.1108119	Binomial	Terr
fixed	factor(fac.yr)7	-2.6417465	1.291363	-2.0457039	0.0407855	Binomial	Terr
fixed	factor(fac.yr)8	-2.2048259	1.296176	-1.7010237	0.0889385	Binomial	Terr
fixed	factor(fac.yr)9	0.9406549	1.767787	0.5321086	0.5946508	Binomial	Terr
fixed	factor(fac.yr)10	-2.0587007	1.316718	-1.5635089	0.1179329	Binomial	Terr
fixed	factor(fac.yr)11	-3.5251730	1.324205	-2.6621061	0.0077653	Binomial	Terr
fixed	factor(fac.yr)12	-1.9815388	1.358832	-1.4582666	0.1447671	Binomial	Terr
fixed	factor(fac.yr)13	-3.0249716	1.335625	-2.2648361	0.0235227	Binomial	Terr
fixed	factor(fac.yr)14	-1.6276255	1.316412	-1.2364101	0.2163062	Binomial	Terr
fixed	factor(fac.yr)15	-2.6420178	1.387742	-1.9038254	0.0569329	Binomial	Terr
fixed	factor(fac.yr)16	-1.1411025	1.489066	-0.7663211	0.4434853	Binomial	Terr
fixed	factor(fac.yr)17	-0.2761776	1.936746	-0.1425987	0.8866071	Binomial	Terr
fixed	factor(fac.yr)18	-3.0334872	1.476069	-2.0551114	0.0398682	Binomial	Terr
fixed	factor(fac.yr)19	-1.5494242	1.548675	-1.0004835	0.3170766	Binomial	Terr
fixed	factor(fac.yr)20	-2.0542188	1.804925	-1.1381187	0.2550709	Binomial	Terr
fixed	(Intercept)	3.6202755	1.325686	2.7308702	0.0063167	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-2.7567433	1.458783	-1.8897554	0.0587907	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-4.0113612	1.436570	-2.7923187	0.0052332	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-2.1614526	1.434751	-1.5065004	0.1319388	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-2.1555841	1.438785	-1.4981978	0.1340819	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-2.1937941	1.427755	-1.5365343	0.1244074	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-2.7897989	1.413798	-1.9732654	0.0484653	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-2.3886348	1.415986	-1.6869052	0.0916216	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	0.8906146	1.906126	0.4672381	0.6403295	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-2.1734787	1.440607	-1.5087246	0.1313692	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-3.7609187	1.447710	-2.5978393	0.0093812	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-2.0778233	1.483442	-1.4006772	0.1613106	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-3.0661952	1.459459	-2.1009117	0.0356487	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-1.6320273	1.442877	-1.1310928	0.2580160	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-2.7196745	1.510741	-1.8002260	0.0718250	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-1.1950130	1.615746	-0.7396043	0.4595402	Binomial	BirdID+Terr
fixed	factor(fac.yr)17	-0.2066317	2.070113	-0.0998166	0.9204899	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-3.2173506	1.603105	-2.0069499	0.0447550	Binomial	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)19	-1.7861463	1.658753	-1.0768004	0.2815695	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-2.2775854	1.920451	-1.1859637	0.2356366	Binomial	BirdID+Terr

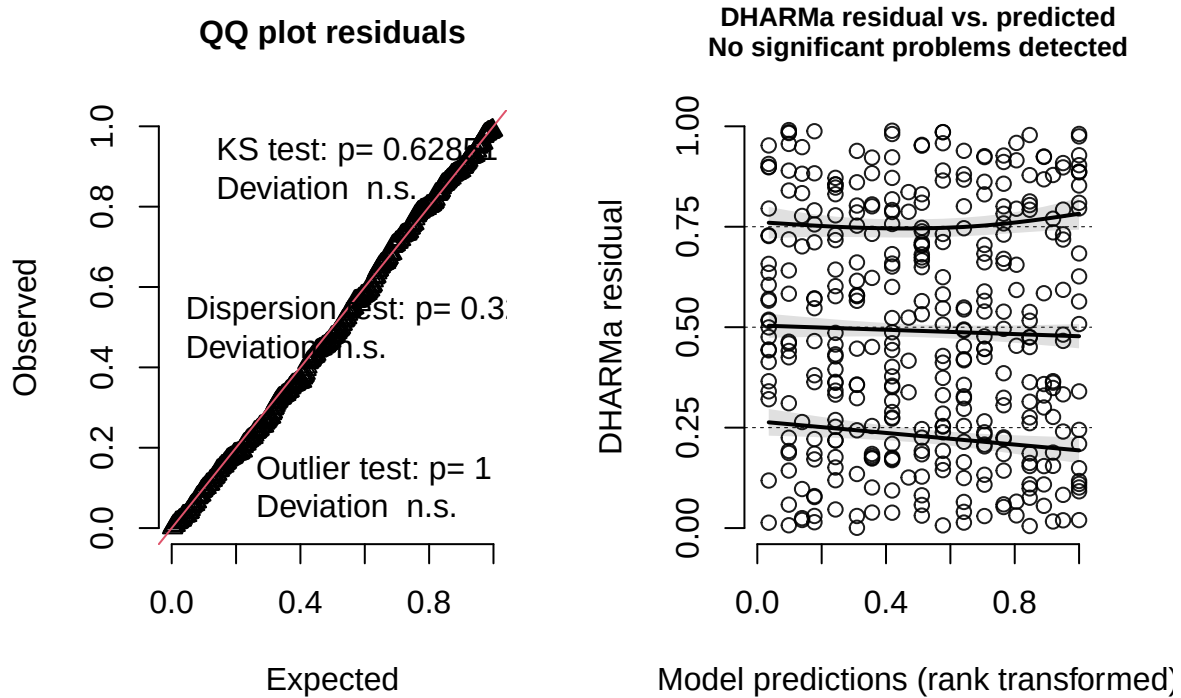
DHARMA residual



DHARMa residual



DHARMa residual



A slight violation of model assumptions for the territory only model.

Probabilites used to generate simulation 2

Simulation 2
 Probability of mate fidelity: 0.012
 Probability of pair dispersal: 0.036
 Probability of single male dispersal: 0.005
 Probability of single female dispersal: 0.728

Simulation 3

Number of fledgling as the response (Poisson)

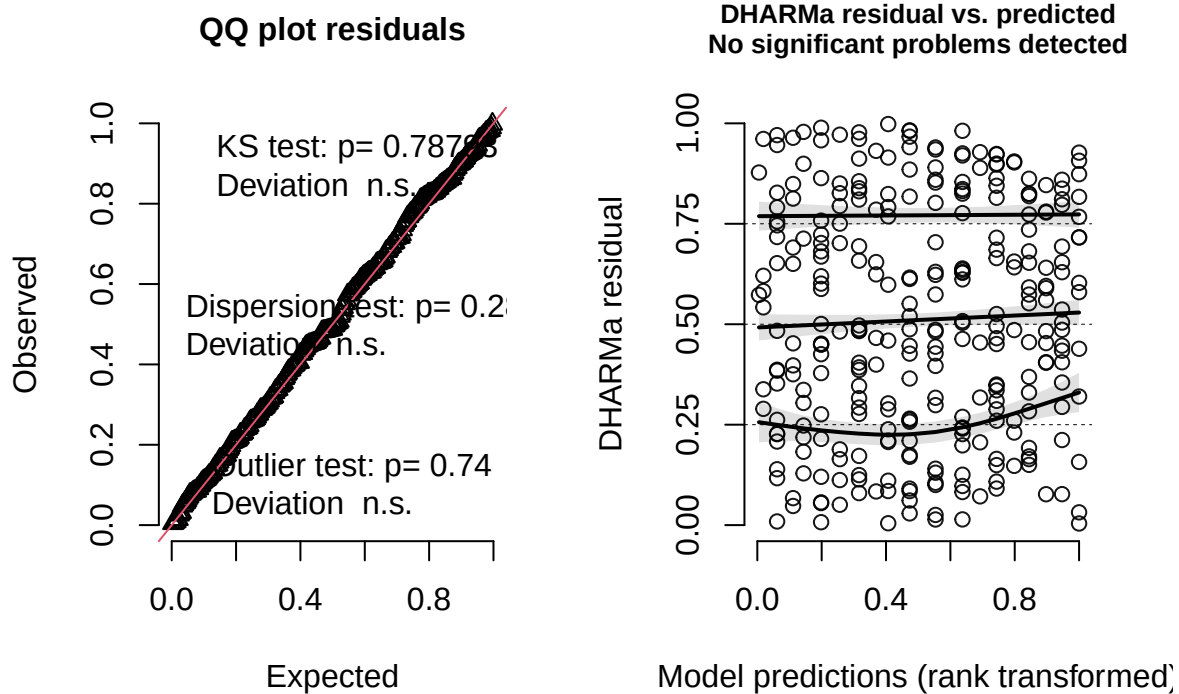
nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
299	1	-404.4838	850.9675	928.6768	294.3871	278	0.7724340	0.7400006	Poisson	Terr
299	1	-404.3547	852.7094	934.1191	285.8474	277	0.7649668	0.7268502	Poisson	BirdID+Terr
299	1	-405.4300	852.8600	930.5693	307.7891	278	0.7609744	0.7365056	Poisson	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.2507472	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.2933198	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.1555390	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.2726930	Poisson	BirdID+Terr

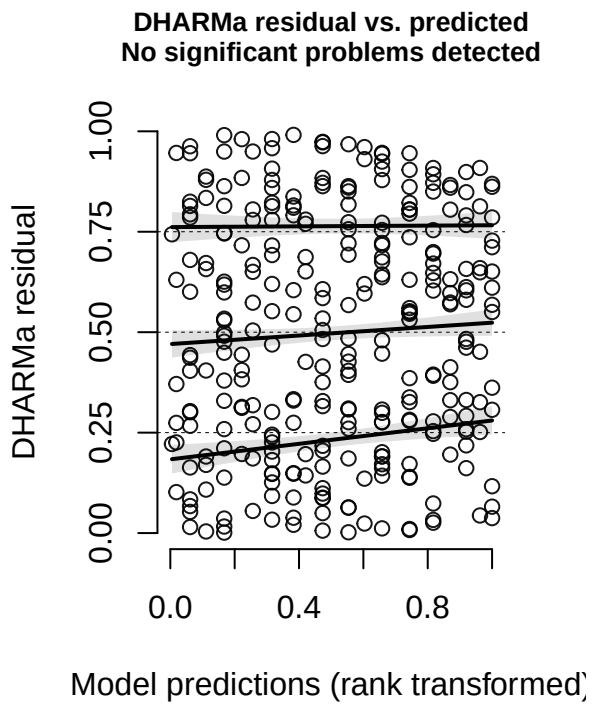
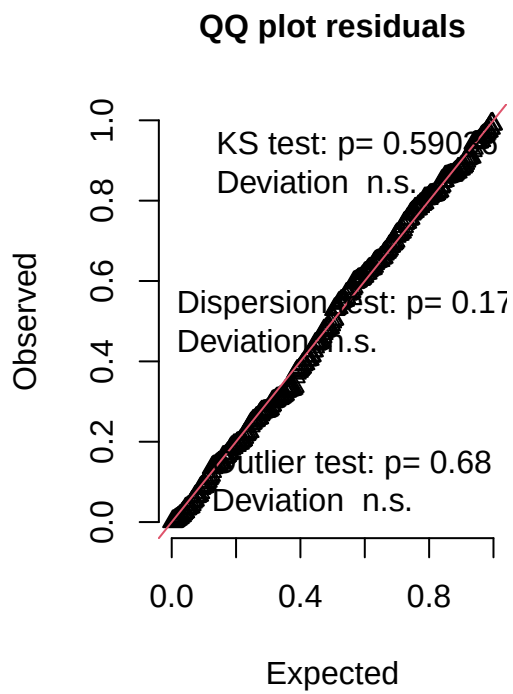
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.5345172	0.2309919	2.3140086	0.0206673	Poisson	BirdID
fixed	factor(fac.yr)2	-0.0885128	0.2938197	-0.3012486	0.7632249	Poisson	BirdID
fixed	factor(fac.yr)3	-0.9633014	0.3497301	-2.7544135	0.0058797	Poisson	BirdID
fixed	factor(fac.yr)4	-0.1518424	0.2872643	-0.5285808	0.5970963	Poisson	BirdID
fixed	factor(fac.yr)5	-0.2089564	0.2865696	-0.7291645	0.4659011	Poisson	BirdID
fixed	factor(fac.yr)6	-0.4172342	0.3024447	-1.3795389	0.1677287	Poisson	BirdID
fixed	factor(fac.yr)7	-0.6843755	0.3184577	-2.1490308	0.0316320	Poisson	BirdID
fixed	factor(fac.yr)8	-0.2724777	0.2970630	-0.9172388	0.3590175	Poisson	BirdID
fixed	factor(fac.yr)9	0.0101555	0.2999581	0.0338565	0.9729916	Poisson	BirdID
fixed	factor(fac.yr)10	-0.6342120	0.3601485	-1.7609734	0.0782429	Poisson	BirdID
fixed	factor(fac.yr)11	-0.9507498	0.4305657	-2.2081409	0.0272345	Poisson	BirdID
fixed	factor(fac.yr)12	-1.2316963	0.3981129	-3.0938365	0.0019759	Poisson	BirdID
fixed	factor(fac.yr)13	0.1665458	0.3004892	0.5542488	0.5794086	Poisson	BirdID
fixed	factor(fac.yr)14	-0.6487718	0.4539677	-1.4291143	0.1529714	Poisson	BirdID
fixed	factor(fac.yr)15	-1.0264411	0.4788167	-2.1437035	0.0320567	Poisson	BirdID
fixed	factor(fac.yr)16	-1.1228021	0.4783006	-2.3474820	0.0189008	Poisson	BirdID
fixed	factor(fac.yr)17	-0.1060488	0.3726676	-0.2845667	0.7759761	Poisson	BirdID
fixed	factor(fac.yr)18	-0.1854033	0.4194216	-0.4420451	0.6584565	Poisson	BirdID
fixed	factor(fac.yr)19	-1.6116723	0.7577757	-2.1268461	0.0334329	Poisson	BirdID
fixed	factor(fac.yr)20	-16.4496215	1949.5982961	-0.0084374	0.9932680	Poisson	BirdID
fixed	(Intercept)	0.4814778	0.2335134	2.0618852	0.0392187	Poisson	Terr
fixed	factor(fac.yr)2	-0.0698511	0.2991767	-0.2334778	0.8153904	Poisson	Terr
fixed	factor(fac.yr)3	-0.9136747	0.3536010	-2.5839145	0.0097686	Poisson	Terr
fixed	factor(fac.yr)4	-0.1149478	0.2890614	-0.3976589	0.6908817	Poisson	Terr
fixed	factor(fac.yr)5	-0.1824729	0.2870498	-0.6356837	0.5249826	Poisson	Terr
fixed	factor(fac.yr)6	-0.3501635	0.3045263	-1.1498630	0.2502003	Poisson	Terr
fixed	factor(fac.yr)7	-0.6211626	0.3199689	-1.9413219	0.0522192	Poisson	Terr
fixed	factor(fac.yr)8	-0.2475246	0.2980543	-0.8304682	0.4062741	Poisson	Terr
fixed	factor(fac.yr)9	0.0405917	0.3031429	0.1339030	0.8934793	Poisson	Terr
fixed	factor(fac.yr)10	-0.5463288	0.3625811	-1.5067768	0.1318679	Poisson	Terr
fixed	factor(fac.yr)11	-0.8701211	0.4346966	-2.0016746	0.0453197	Poisson	Terr
fixed	factor(fac.yr)12	-1.1567908	0.4014013	-2.8818807	0.0039531	Poisson	Terr
fixed	factor(fac.yr)13	0.3105750	0.3044223	1.0202112	0.3076283	Poisson	Terr
fixed	factor(fac.yr)14	-0.3906234	0.4587534	-0.8514888	0.3944979	Poisson	Terr
fixed	factor(fac.yr)15	-0.8726400	0.4804713	-1.8162168	0.0693371	Poisson	Terr
fixed	factor(fac.yr)16	-0.9666813	0.4826588	-2.0028255	0.0451960	Poisson	Terr
fixed	factor(fac.yr)17	0.0467779	0.3751222	0.1247005	0.9007607	Poisson	Terr
fixed	factor(fac.yr)18	-0.1898335	0.4214405	-0.4504395	0.6523935	Poisson	Terr
fixed	factor(fac.yr)19	-1.6176814	0.7576289	-2.1351895	0.0327455	Poisson	Terr
fixed	factor(fac.yr)20	-16.7533220	2450.1554687	-0.0068377	0.9945444	Poisson	Terr
fixed	(Intercept)	0.4801661	0.2365659	2.0297354	0.0423834	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.0675927	0.3009729	-0.2245807	0.8223055	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.9145498	0.3560851	-2.5683465	0.0102185	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.1175178	0.2924321	-0.4018636	0.6877844	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.1864330	0.2907254	-0.6412685	0.5213482	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.3599654	0.3080065	-1.1686942	0.2425269	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.6377982	0.3232826	-1.9728815	0.0485091	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.2525830	0.3019455	-0.8365183	0.4028634	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	0.0342407	0.3069765	0.1115418	0.9111867	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.5614463	0.3663708	-1.5324536	0.1254105	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.8999333	0.4382598	-2.0534243	0.0400314	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-1.1726015	0.4046785	-2.8976122	0.0037602	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	0.2718074	0.3086758	0.8805597	0.3785562	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.4524989	0.4628705	-0.9775929	0.3282757	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.9142648	0.4846524	-1.8864343	0.0592365	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-1.0060833	0.4863652	-2.0685757	0.0385859	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	0.0176874	0.3801877	0.0465227	0.9628936	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.1916859	0.4268798	-0.4490395	0.6534032	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.6073485	0.7623426	-2.1084332	0.0349935	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-16.3708352	1983.8362524	-0.0082521	0.9934158	Poisson	BirdID+Terr

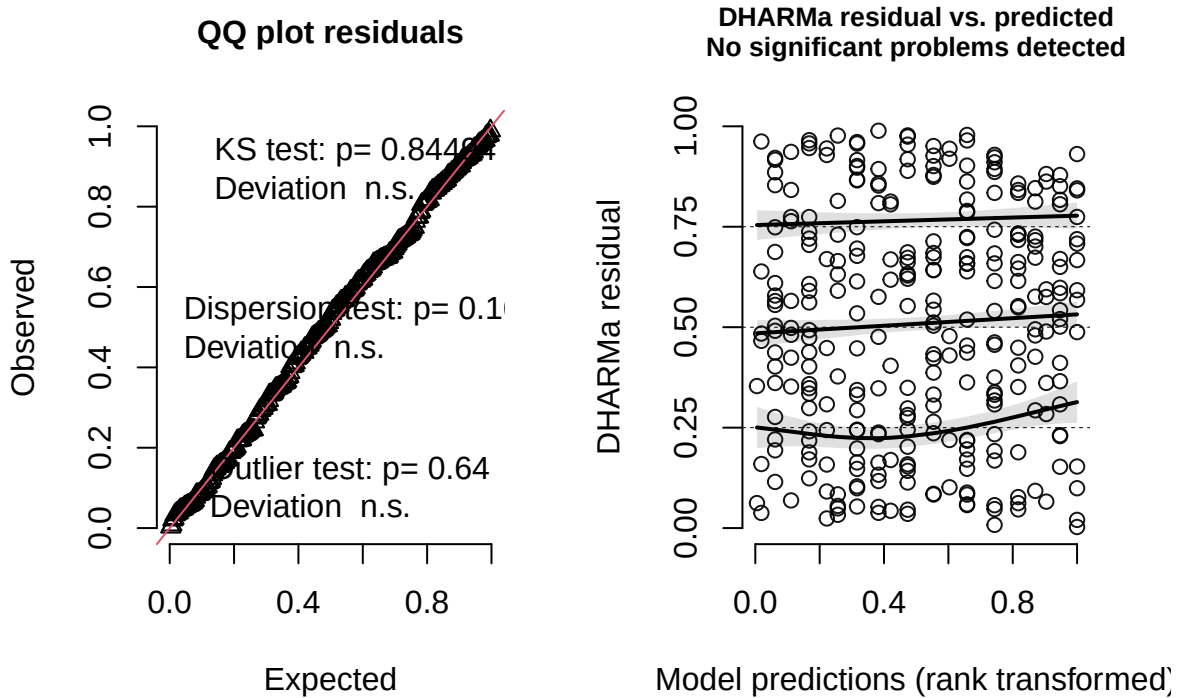
DHARMA residual



DHARMa residual



DHARMA residual



Assumptions look good and the territory only model is performing the best on all of the model selection criteria.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
299	1	-165.5590	375.1181	456.5278	205.6924	277	0.742	Binomial	BirdID+Terr
299	1	-166.6307	375.2615	452.9708	245.6715	278	0.650	Binomial	BirdID
299	1	-168.8620	379.7239	457.4332	268.6901	278	0.481	Binomial	Terr

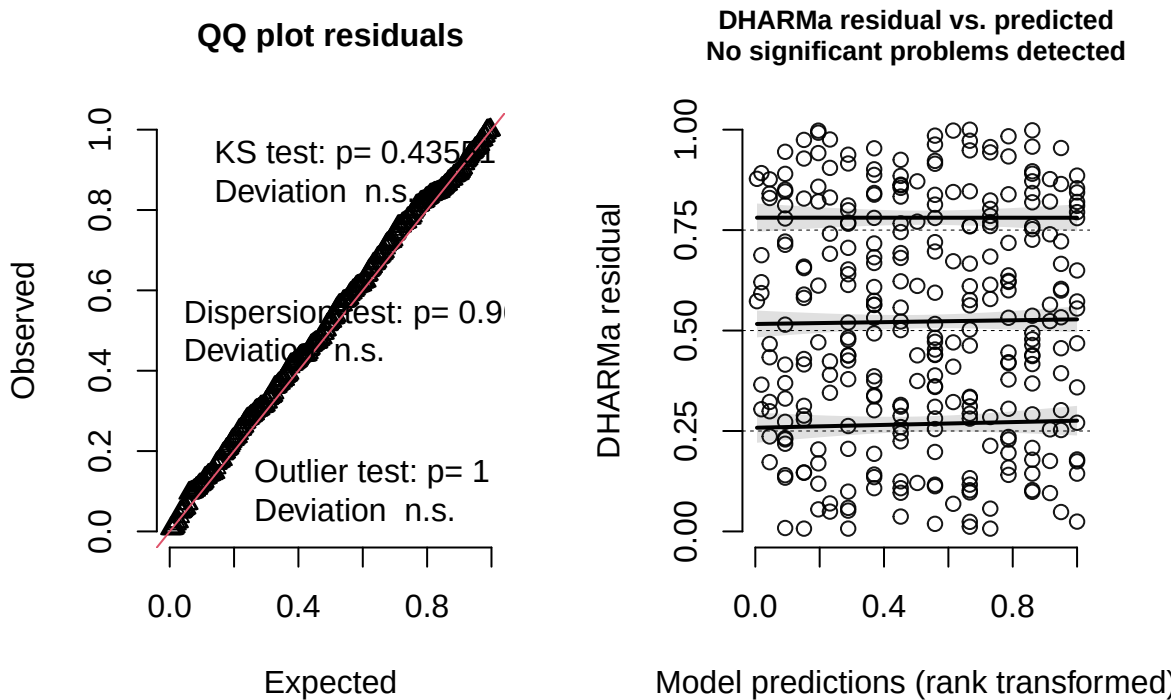
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.1684020	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	0.9546470	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.1748278	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.8964986	Binomial	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	2.7532174	1.173196	2.3467678	0.0189371	Binomial	BirdID
fixed	factor(fac.yr)2	-1.1548111	1.320696	-0.8743958	0.3819028	Binomial	BirdID
fixed	factor(fac.yr)3	-2.7451536	1.261499	-2.1761045	0.0295474	Binomial	BirdID

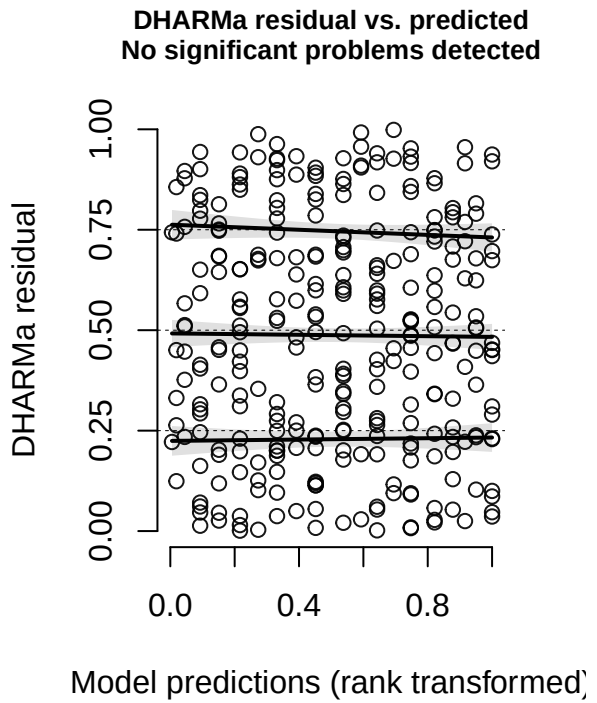
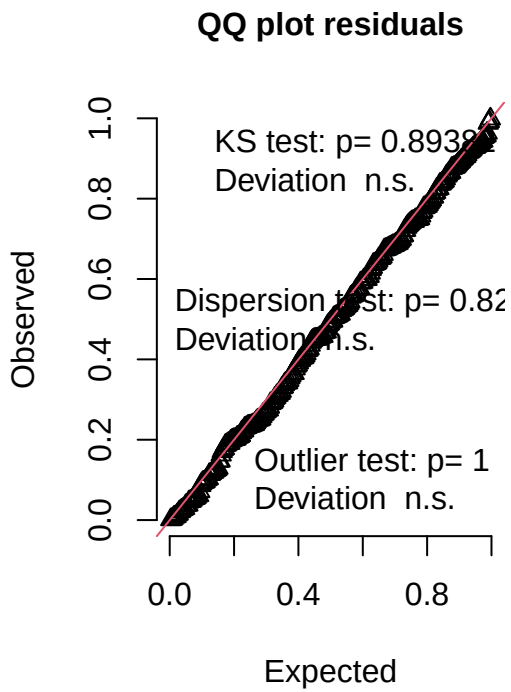
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)4	-1.0811433	1.297636	-0.8331639	0.4047523	Binomial	BirdID
fixed	factor(fac.yr)5	-1.8733297	1.257479	-1.4897504	0.1362899	Binomial	BirdID
fixed	factor(fac.yr)6	-2.3515431	1.261227	-1.8644880	0.0622532	Binomial	BirdID
fixed	factor(fac.yr)7	-2.7069530	1.260279	-2.1478989	0.0317218	Binomial	BirdID
fixed	factor(fac.yr)8	-1.5101386	1.296364	-1.1649035	0.2440580	Binomial	BirdID
fixed	factor(fac.yr)9	0.5340909	1.695119	0.3150757	0.7527042	Binomial	BirdID
fixed	factor(fac.yr)10	-2.9999240	1.314684	-2.2818597	0.0224976	Binomial	BirdID
fixed	factor(fac.yr)11	-2.9447347	1.366909	-2.1543012	0.0312166	Binomial	BirdID
fixed	factor(fac.yr)12	-3.2749469	1.301153	-2.5169583	0.0118373	Binomial	BirdID
fixed	factor(fac.yr)13	-1.3578506	1.419806	-0.9563637	0.3388885	Binomial	BirdID
fixed	factor(fac.yr)14	-1.8688995	1.585381	-1.1788326	0.2384648	Binomial	BirdID
fixed	factor(fac.yr)15	-4.1196646	1.452386	-2.8364815	0.0045614	Binomial	BirdID
fixed	factor(fac.yr)16	-2.9386655	1.409476	-2.0849349	0.0370752	Binomial	BirdID
fixed	factor(fac.yr)17	-0.4124463	1.741172	-0.2368786	0.8127510	Binomial	BirdID
fixed	factor(fac.yr)18	-2.2201519	1.547206	-1.4349428	0.1513034	Binomial	BirdID
fixed	factor(fac.yr)19	-4.3154936	1.745682	-2.4720956	0.0134324	Binomial	BirdID
fixed	factor(fac.yr)20	-16.5084027	395.724303	-0.0417169	0.9667244	Binomial	BirdID
fixed	(Intercept)	2.5210609	1.164154	2.1655744	0.0303437	Binomial	Terr
fixed	factor(fac.yr)2	-0.8515075	1.315930	-0.6470767	0.5175823	Binomial	Terr
fixed	factor(fac.yr)3	-2.4655685	1.248883	-1.9742185	0.0483569	Binomial	Terr
fixed	factor(fac.yr)4	-0.8967747	1.284031	-0.6984059	0.4849234	Binomial	Terr
fixed	factor(fac.yr)5	-1.8073087	1.243044	-1.4539375	0.1459636	Binomial	Terr
fixed	factor(fac.yr)6	-1.9680797	1.248615	-1.5762101	0.1149774	Binomial	Terr
fixed	factor(fac.yr)7	-2.4220080	1.244275	-1.9465219	0.0515921	Binomial	Terr
fixed	factor(fac.yr)8	-1.3468070	1.273535	-1.0575341	0.2902679	Binomial	Terr
fixed	factor(fac.yr)9	0.5681039	1.655205	0.3432227	0.7314309	Binomial	Terr
fixed	factor(fac.yr)10	-2.5368947	1.301141	-1.9497464	0.0512064	Binomial	Terr
fixed	factor(fac.yr)11	-2.3612526	1.341978	-1.7595318	0.0784872	Binomial	Terr
fixed	factor(fac.yr)12	-2.7467797	1.275136	-2.1541069	0.0312318	Binomial	Terr
fixed	factor(fac.yr)13	-0.7149956	1.374764	-0.5200860	0.6030036	Binomial	Terr
fixed	factor(fac.yr)14	-0.9557382	1.494181	-0.6396401	0.5224066	Binomial	Terr
fixed	factor(fac.yr)15	-3.0956886	1.401814	-2.2083444	0.0272203	Binomial	Terr
fixed	factor(fac.yr)16	-2.4244108	1.363460	-1.7781308	0.0753824	Binomial	Terr
fixed	factor(fac.yr)17	-0.0195733	1.682470	-0.0116337	0.9907179	Binomial	Terr
fixed	factor(fac.yr)18	-1.7224185	1.490029	-1.1559632	0.2476962	Binomial	Terr
fixed	factor(fac.yr)19	-4.3886750	1.681205	-2.6104335	0.0090428	Binomial	Terr
fixed	factor(fac.yr)20	-13.3367110	173.463053	-0.0768850	0.9387150	Binomial	Terr
fixed	(Intercept)	2.7597785	1.253235	2.2021234	0.0276566	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-0.8668013	1.417397	-0.6115445	0.5408392	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-2.6510840	1.350015	-1.9637443	0.0495598	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-0.8793619	1.385249	-0.6348043	0.5255561	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-1.8408971	1.337619	-1.3762491	0.1687445	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-2.1746470	1.347444	-1.6139057	0.1065479	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-2.8043661	1.349056	-2.0787625	0.0376392	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-1.4459453	1.382127	-1.0461738	0.2954808	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	0.7131117	1.824180	0.3909217	0.6958551	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-2.9486928	1.411090	-2.0896568	0.0366486	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-2.9841534	1.459260	-2.0449771	0.0408571	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-3.2107236	1.395362	-2.3009969	0.0213918	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-0.9926665	1.524032	-0.6513424	0.5148255	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-1.3918616	1.698119	-0.8196488	0.4124163	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-4.1283437	1.561966	-2.6430430	0.0082165	Binomial	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)16	-3.0331444	1.502475	-2.0187658	0.0435116	Binomial	BirdID+Terr
fixed	factor(fac.yr)17	0.0467054	1.915432	0.0243837	0.9805465	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-2.0806923	1.650492	-1.2606495	0.2074352	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-4.5545414	1.864476	-2.4427995	0.0145738	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-19.8600091	2182.568285	-0.0090994	0.9927398	Binomial	BirdID+Terr

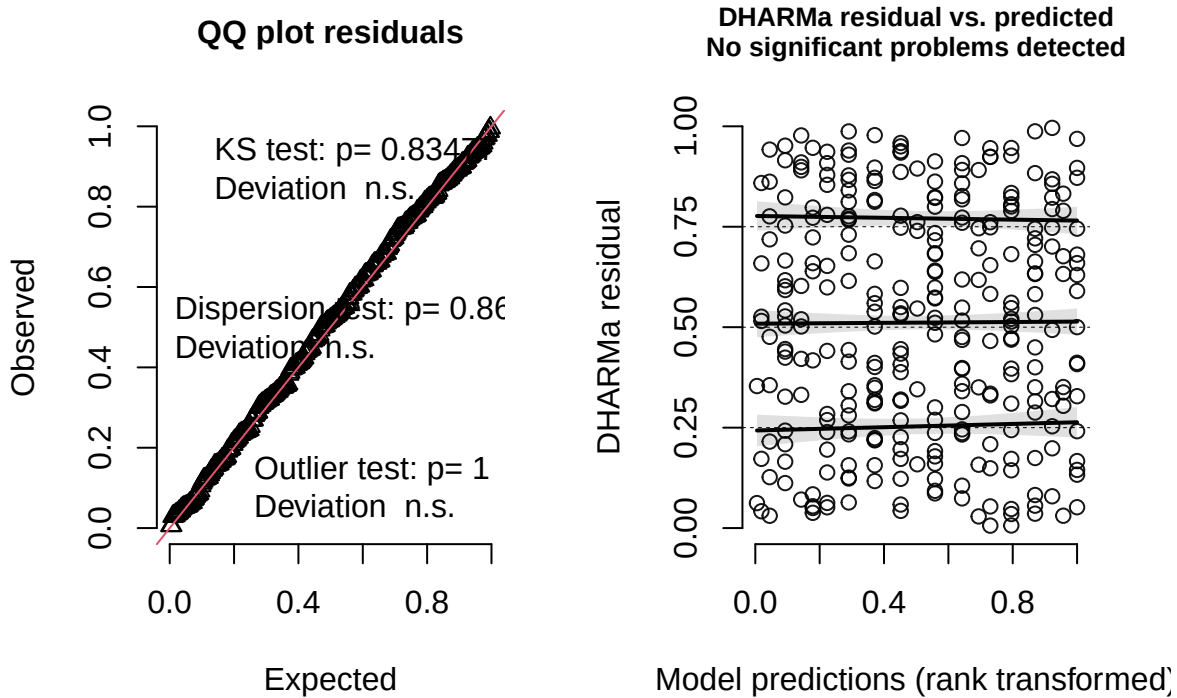
DHARMA residual



DHARMa residual



DHARMa residual



Model diagnostics look reasonable and the territory only model performed the worst on the model selection criteria.

Probabilities used to generate simulation 3

Simulation 3
 Probability of mate fidelity: 0.073
 Probability of pair dispersal: 0.33
 Probability of single male dispersal: 0.022
 Probability of single female dispersal: 0.242

Simulation 4

Number of fledgling as the response (Poisson)

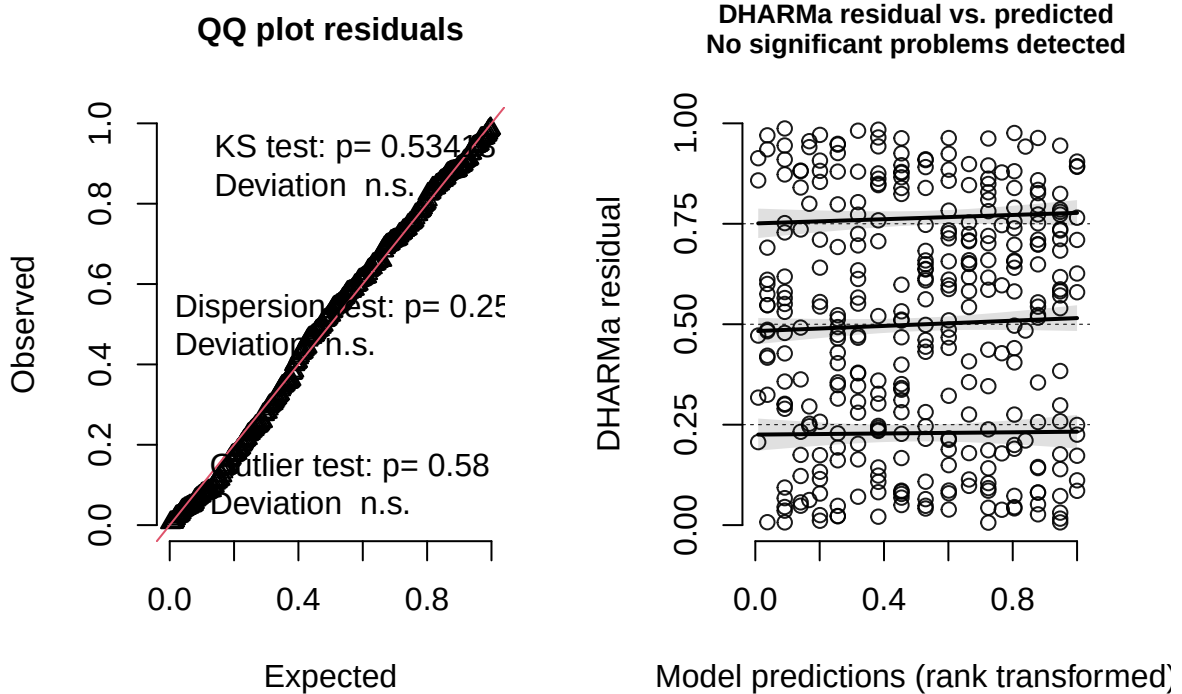
nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	REfamily	RE
339	1	-471.7699	985.5397	1065.886	351.4062	318	0.2678586	0.1474535	Poisson	BirdID
339	1	-471.7698	987.5395	1071.712	351.3071	317	0.2682123	0.1476102	Poisson	BirdID+Terr
339	1	-474.0697	990.1393	1070.485	381.8447	318	0.2184748	0.1568940	Poisson	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.3130924	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.2166995	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.3129613	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.0141697	Poisson	BirdID+Terr

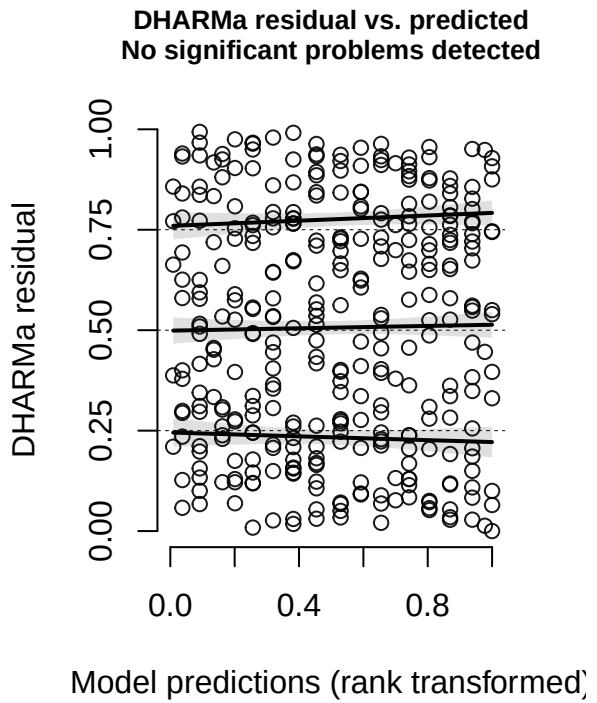
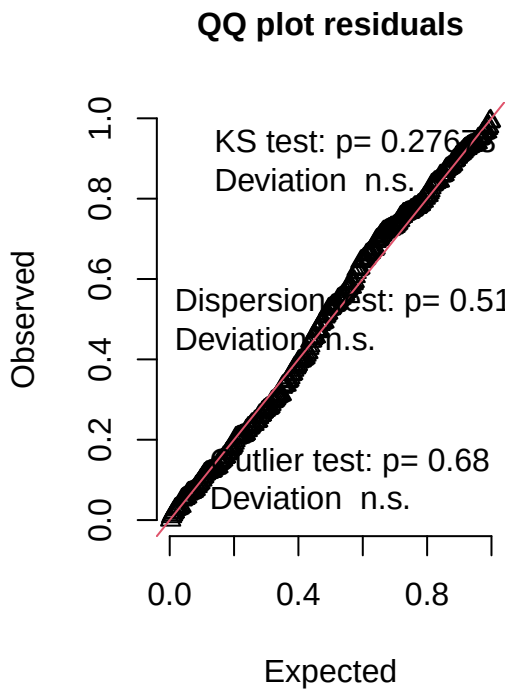
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.6531942	0.2229137	2.9302562	0.0033868	Poisson	BirdID
fixed	factor(fac.yr)2	-0.5895212	0.3009334	-1.9589754	0.0501157	Poisson	BirdID
fixed	factor(fac.yr)3	-0.7849619	0.3167059	-2.4785202	0.0131929	Poisson	BirdID
fixed	factor(fac.yr)4	-0.4819374	0.2901361	-1.6610738	0.0966986	Poisson	BirdID
fixed	factor(fac.yr)5	-0.1200299	0.2783042	-0.4312904	0.6662572	Poisson	BirdID
fixed	factor(fac.yr)6	-0.1535222	0.2784157	-0.5514133	0.5813504	Poisson	BirdID
fixed	factor(fac.yr)7	-1.0590858	0.3465473	-3.0561073	0.0022423	Poisson	BirdID
fixed	factor(fac.yr)8	-0.5618958	0.2930066	-1.9176900	0.0551503	Poisson	BirdID
fixed	factor(fac.yr)9	-0.4755132	0.2927014	-1.6245677	0.1042547	Poisson	BirdID
fixed	factor(fac.yr)10	-0.2904410	0.2935502	-0.9894083	0.3224634	Poisson	BirdID
fixed	factor(fac.yr)11	-0.6203184	0.3159152	-1.9635599	0.0495812	Poisson	BirdID
fixed	factor(fac.yr)12	-0.4571648	0.3101916	-1.4738142	0.1405316	Poisson	BirdID
fixed	factor(fac.yr)13	-0.3328941	0.2922814	-1.1389507	0.2547237	Poisson	BirdID
fixed	factor(fac.yr)14	-0.8099734	0.3660584	-2.2126892	0.0269191	Poisson	BirdID
fixed	factor(fac.yr)15	-1.3432745	0.4496990	-2.9870521	0.0028168	Poisson	BirdID
fixed	factor(fac.yr)16	-0.9801037	0.4301035	-2.2787627	0.0226812	Poisson	BirdID
fixed	factor(fac.yr)17	-0.3287485	0.4400985	-0.7469885	0.4550705	Poisson	BirdID
fixed	factor(fac.yr)18	-0.9739000	0.5189951	-1.8765111	0.0605851	Poisson	BirdID
fixed	factor(fac.yr)19	-1.6017978	0.7640976	-2.0963262	0.0360533	Poisson	BirdID
fixed	factor(fac.yr)20	-0.2108785	0.5343764	-0.3946253	0.6931194	Poisson	BirdID
fixed	(Intercept)	0.6508416	0.2139436	3.0421180	0.0023492	Poisson	Terr
fixed	factor(fac.yr)2	-0.5531915	0.2985868	-1.8526989	0.0639255	Poisson	Terr
fixed	factor(fac.yr)3	-0.7453111	0.3127350	-2.3832033	0.0171627	Poisson	Terr
fixed	factor(fac.yr)4	-0.4209846	0.2847330	-1.4785238	0.1392676	Poisson	Terr
fixed	factor(fac.yr)5	-0.1068520	0.2693833	-0.3966542	0.6916225	Poisson	Terr
fixed	factor(fac.yr)6	-0.1515475	0.2698677	-0.5615623	0.5744143	Poisson	Terr
fixed	factor(fac.yr)7	-1.0351165	0.3386702	-3.0564146	0.0022400	Poisson	Terr
fixed	factor(fac.yr)8	-0.5338592	0.2829134	-1.8870059	0.0591595	Poisson	Terr
fixed	factor(fac.yr)9	-0.4098588	0.2811857	-1.4576087	0.1449484	Poisson	Terr
fixed	factor(fac.yr)10	-0.2718179	0.2822936	-0.9628909	0.3356023	Poisson	Terr
fixed	factor(fac.yr)11	-0.5683605	0.3044441	-1.8668794	0.0619184	Poisson	Terr
fixed	factor(fac.yr)12	-0.4209135	0.2990855	-1.4073350	0.1593281	Poisson	Terr
fixed	factor(fac.yr)13	-0.3147213	0.2797873	-1.1248594	0.2606486	Poisson	Terr
fixed	factor(fac.yr)14	-0.8097867	0.3549047	-2.2817019	0.0225069	Poisson	Terr
fixed	factor(fac.yr)15	-1.3480180	0.4403001	-3.0615889	0.0022017	Poisson	Terr
fixed	factor(fac.yr)16	-0.9711974	0.4198360	-2.3132780	0.0207074	Poisson	Terr
fixed	factor(fac.yr)17	-0.3898624	0.4214385	-0.9250754	0.3549267	Poisson	Terr
fixed	factor(fac.yr)18	-0.9893192	0.5038064	-1.9636892	0.0495661	Poisson	Terr
fixed	factor(fac.yr)19	-1.5181691	0.7505447	-2.0227563	0.0430983	Poisson	Terr
fixed	factor(fac.yr)20	-0.0724817	0.5125805	-0.1414056	0.8875496	Poisson	Terr
fixed	(Intercept)	0.6548481	0.2227917	2.9392839	0.0032897	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.5909619	0.3008487	-1.9643160	0.0494935	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.7874054	0.3167007	-2.4862758	0.0129088	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.4834975	0.2900566	-1.6669071	0.0955329	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.1217467	0.2782214	-0.4375894	0.6616840	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.1555583	0.2783424	-0.5588739	0.5762478	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-1.0602283	0.3464296	-3.0604437	0.0022101	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.5638800	0.2929323	-1.9249497	0.0542357	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.4773445	0.2926208	-1.6312736	0.1028326	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.2920352	0.2934520	-0.9951720	0.3196526	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.6218701	0.3158277	-1.9690169	0.0489512	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.4587095	0.3100985	-1.4792380	0.1390767	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.3348662	0.2922046	-1.1459992	0.2517955	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.8111818	0.3659497	-2.2166486	0.0266471	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-1.3448192	0.4496407	-2.9908748	0.0027818	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.9822121	0.4301262	-2.2835441	0.0223983	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.3297632	0.4399595	-0.7495309	0.4535373	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.9749267	0.5188395	-1.8790523	0.0602373	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.6052883	0.7647979	-2.0989708	0.0358195	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.2112932	0.5342259	-0.3955128	0.6924645	Poisson	BirdID+Terr

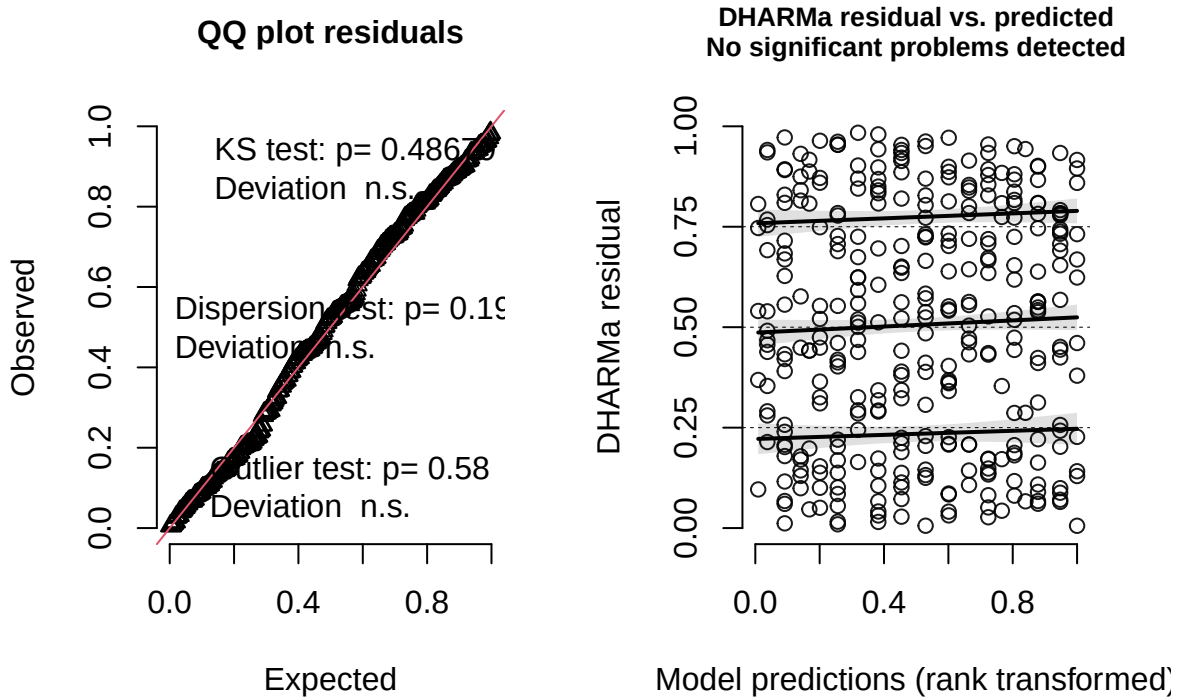
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics all look okay and the territory only model performed the worst on the model selection criteria.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
339	1	-203.3321	448.6641	529.0101	313.9716	318	0.585	Binomial	BirdID
339	1	-202.9741	449.9481	534.1201	291.6194	317	0.607	Binomial	BirdID+Terr
339	1	-207.8207	457.6414	537.9874	351.9582	318	0.444	Binomial	Terr

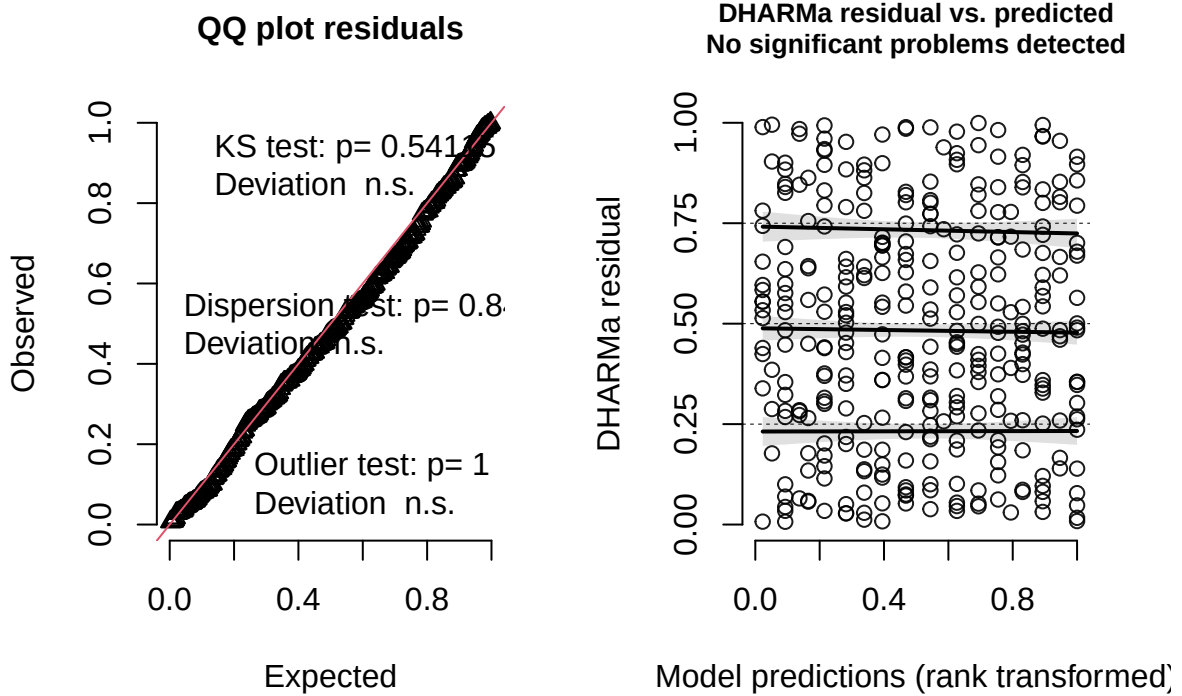
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.1054517	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	0.7994774	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.0938777	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.5548885	Binomial	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	1.4497957	0.7637755	1.8981962	0.0576702	Binomial	BirdID
fixed	factor(fac.yr)2	-1.3530761	0.8694856	-1.5561800	0.1196653	Binomial	BirdID
fixed	factor(fac.yr)3	-1.2058707	0.8793993	-1.3712437	0.1702990	Binomial	BirdID
fixed	factor(fac.yr)4	-0.2717172	0.9058088	-0.2999719	0.7641986	Binomial	BirdID

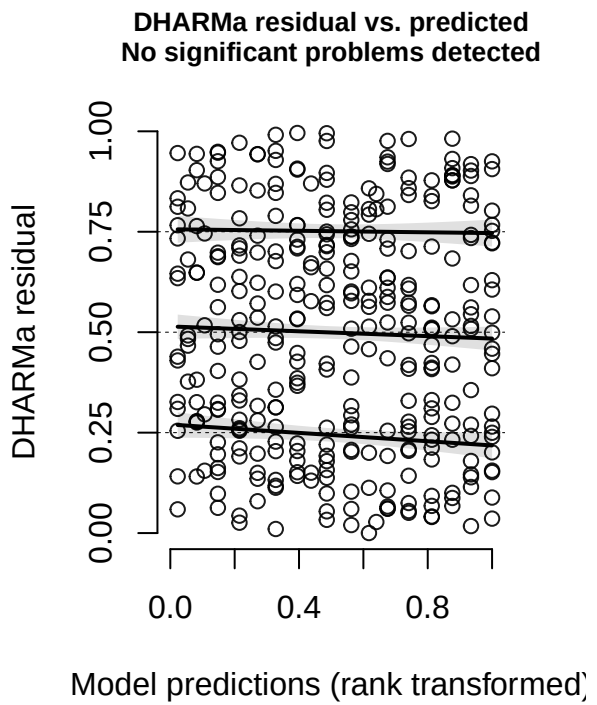
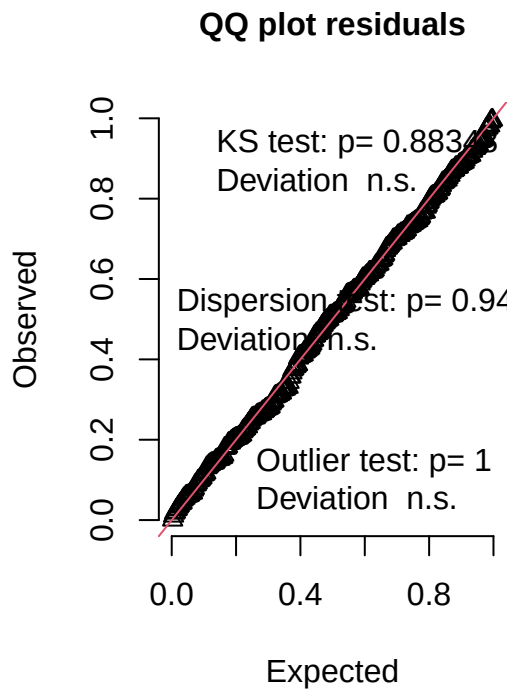
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.2228036	0.9526760	0.2338714	0.8150848	Binomial	BirdID
fixed	factor(fac.yr)6	-0.0653506	0.9332306	-0.0700262	0.9441728	Binomial	BirdID
fixed	factor(fac.yr)7	-1.6154756	0.9021900	-1.7906156	0.0733550	Binomial	BirdID
fixed	factor(fac.yr)8	-0.6418044	0.8939072	-0.7179765	0.4727718	Binomial	BirdID
fixed	factor(fac.yr)9	-0.4183400	0.9214466	-0.4540035	0.6498263	Binomial	BirdID
fixed	factor(fac.yr)10	-0.2309352	0.9613559	-0.2402182	0.8101611	Binomial	BirdID
fixed	factor(fac.yr)11	-0.1629902	0.9612783	-0.1695557	0.8653595	Binomial	BirdID
fixed	factor(fac.yr)12	-0.1042785	0.9893496	-0.1054011	0.9160576	Binomial	BirdID
fixed	factor(fac.yr)13	-0.7334161	0.9180601	-0.7988759	0.4243624	Binomial	BirdID
fixed	factor(fac.yr)14	-1.1054092	0.9740001	-1.1349169	0.2564101	Binomial	BirdID
fixed	factor(fac.yr)15	-2.0075840	1.0059848	-1.9956405	0.0459731	Binomial	BirdID
fixed	factor(fac.yr)16	-1.5130452	1.0473245	-1.4446766	0.1485488	Binomial	BirdID
fixed	factor(fac.yr)17	-0.1454627	1.2871169	-0.1130144	0.9100192	Binomial	BirdID
fixed	factor(fac.yr)18	-1.5656692	1.2040827	-1.3003004	0.1934980	Binomial	BirdID
fixed	factor(fac.yr)19	-1.9255456	1.3563160	-1.4196880	0.1556985	Binomial	BirdID
fixed	factor(fac.yr)20	-0.3494197	1.7385194	-0.2009870	0.8407088	Binomial	BirdID
fixed	(Intercept)	1.1044740	0.7201494	1.5336734	0.1251100	Binomial	Terr
fixed	factor(fac.yr)2	-0.9721618	0.8437454	-1.1521981	0.2492397	Binomial	Terr
fixed	factor(fac.yr)3	-0.7512452	0.8507397	-0.8830494	0.3772096	Binomial	Terr
fixed	factor(fac.yr)4	0.0572022	0.8653688	0.0661015	0.9472970	Binomial	Terr
fixed	factor(fac.yr)5	0.4284715	0.9068621	0.4724771	0.6365863	Binomial	Terr
fixed	factor(fac.yr)6	0.0164861	0.8833299	0.0186636	0.9851094	Binomial	Terr
fixed	factor(fac.yr)7	-1.1467013	0.8523192	-1.3453895	0.1784995	Binomial	Terr
fixed	factor(fac.yr)8	-0.3084879	0.8401514	-0.3671813	0.7134838	Binomial	Terr
fixed	factor(fac.yr)9	-0.0519716	0.8595033	-0.0604670	0.9517837	Binomial	Terr
fixed	factor(fac.yr)10	0.0144974	0.9102413	0.0159270	0.9872926	Binomial	Terr
fixed	factor(fac.yr)11	0.1672896	0.9001167	0.1858532	0.8525599	Binomial	Terr
fixed	factor(fac.yr)12	0.0687715	0.9421604	0.0729934	0.9418113	Binomial	Terr
fixed	factor(fac.yr)13	-0.5015526	0.8589457	-0.5839165	0.5592765	Binomial	Terr
fixed	factor(fac.yr)14	-0.9252345	0.9051500	-1.0221891	0.3066914	Binomial	Terr
fixed	factor(fac.yr)15	-1.7497745	0.9388536	-1.8637352	0.0623589	Binomial	Terr
fixed	factor(fac.yr)16	-1.2629802	0.9721719	-1.2991326	0.1938984	Binomial	Terr
fixed	factor(fac.yr)17	-0.3362278	1.1721862	-0.2868382	0.7742362	Binomial	Terr
fixed	factor(fac.yr)18	-1.4398201	1.0880291	-1.3233287	0.1857261	Binomial	Terr
fixed	factor(fac.yr)19	-1.1923661	1.2164940	-0.9801661	0.3270041	Binomial	Terr
fixed	factor(fac.yr)20	0.0024615	1.4820193	0.0016609	0.9986748	Binomial	Terr
fixed	(Intercept)	1.4358292	0.7850702	1.8289183	0.0674119	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-1.3125685	0.8921680	-1.4712122	0.1412337	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-1.1152159	0.9038968	-1.2337867	0.2172824	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-0.1465385	0.9320403	-0.1572233	0.8750688	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	0.3639261	0.9832221	0.3701363	0.7112810	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-0.0364473	0.9580140	-0.0380446	0.9696521	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-1.5633084	0.9275447	-1.6854265	0.0919063	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-0.5782901	0.9175079	-0.6302835	0.5285091	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-0.3312584	0.9446988	-0.3506497	0.7258512	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-0.2043022	0.9892679	-0.2065186	0.8363859	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-0.0706651	0.9856672	-0.0716927	0.9428465	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-0.1113472	1.0170898	-0.1094763	0.9128248	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-0.6909734	0.9424408	-0.7331744	0.4634521	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-1.0663708	0.9985440	-1.0679257	0.2855540	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-2.0054079	1.0370658	-1.9337326	0.0531460	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-1.4975885	1.0823386	-1.3836599	0.1664626	Binomial	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-0.0617364	1.3228361	-0.0466698	0.9627764	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-1.5822730	1.2389910	-1.2770658	0.2015790	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-1.7634117	1.3829542	-1.2751050	0.2022721	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-0.2115532	1.7614236	-0.1201035	0.9044012	Binomial	BirdID+Terr

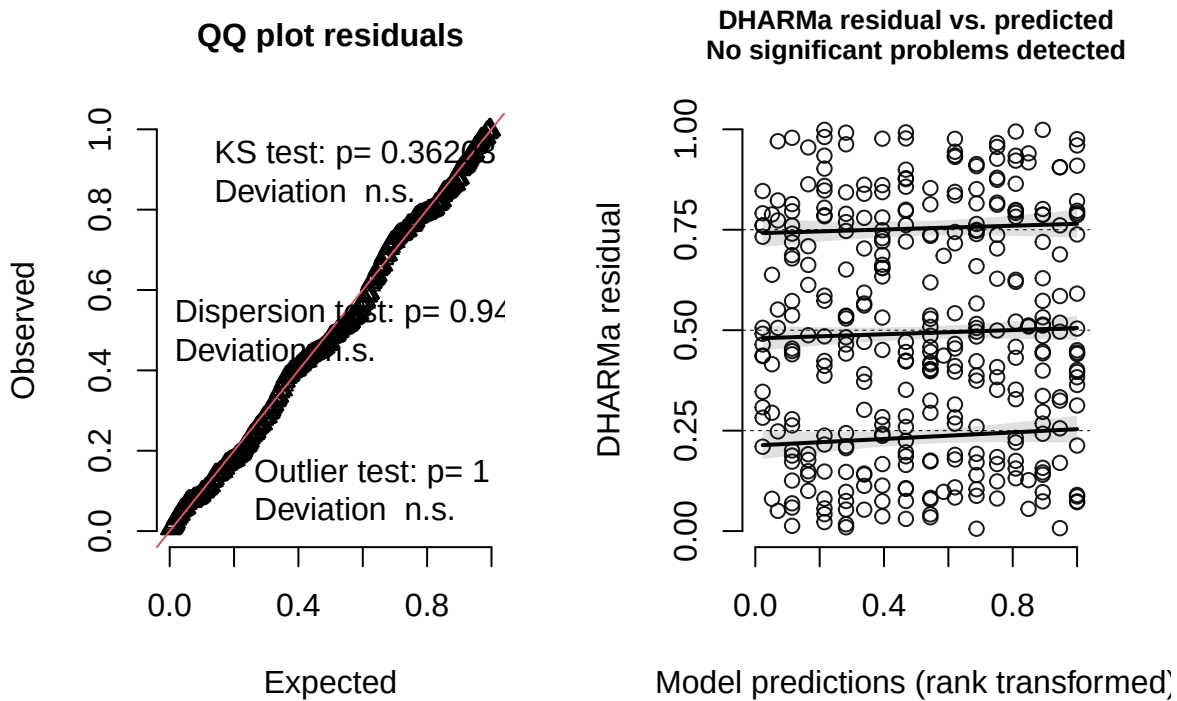
DHARMA residual



DHARMa residual



DHARMA residual



Diagnostics look reasonable and the territory only model is performing the worst on the selection criteria.

Probabilities used to generate simulation 4

Simulation 4
 Probability of mate fidelity: 0.449
 Probability of pair dispersal: 0.102
 Probability of single male dispersal: 0.615
 Probability of single female dispersal: 0.371

Simulation 5

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
389	1	-506.0481	1056.096	1143.295	358.2129	367	0.3758426	0.1433754	Poisson	BirdID+Terr
389	1	-508.4673	1058.935	1142.170	406.5580	368	0.3307784	0.1564340	Poisson	Terr
389	1	-511.6407	1065.281	1148.516	421.4911	368	0.3164467	0.1652093	Poisson	BirdID

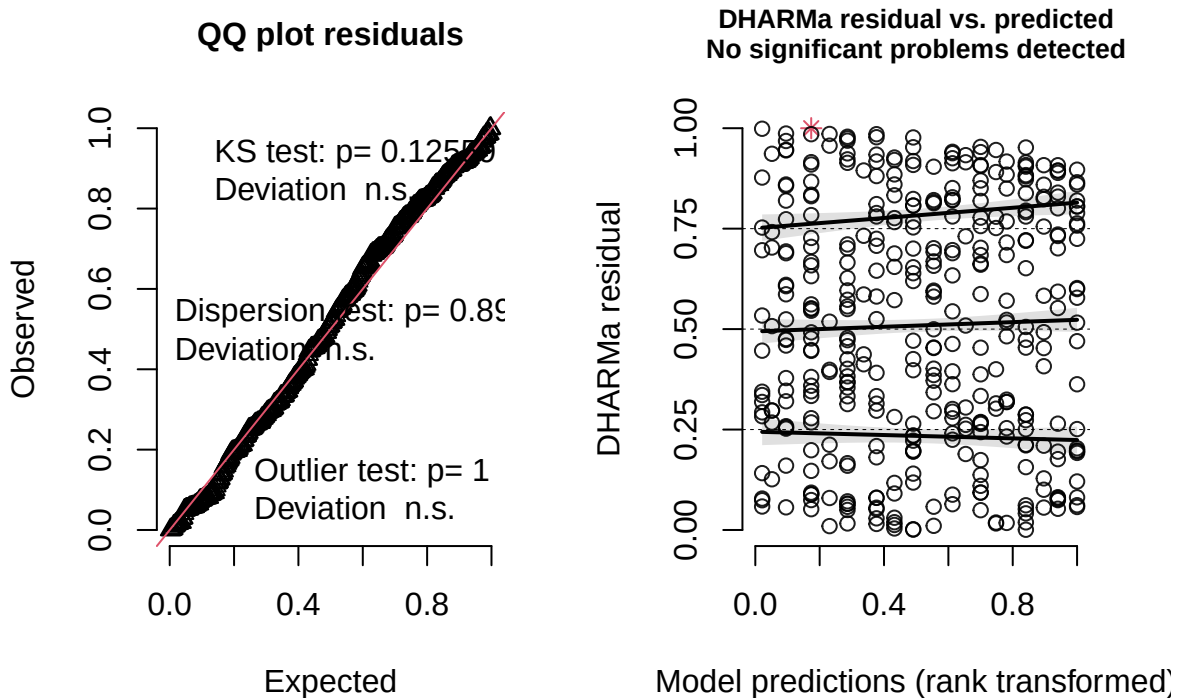
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.3942436	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.4189202	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.3157652	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.3984110	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.1966044	0.2758889	0.7126217	0.4760799	Poisson	BirdID
fixed	factor(fac.yr)2	-0.1090133	0.3599223	-0.3028801	0.7619812	Poisson	BirdID
fixed	factor(fac.yr)3	-0.2745953	0.3568985	-0.7693932	0.4416599	Poisson	BirdID
fixed	factor(fac.yr)4	-0.1392998	0.3336843	-0.4174597	0.6763422	Poisson	BirdID
fixed	factor(fac.yr)5	0.3173090	0.3179906	0.9978564	0.3183490	Poisson	BirdID
fixed	factor(fac.yr)6	-0.3021724	0.3456375	-0.8742465	0.3819840	Poisson	BirdID
fixed	factor(fac.yr)7	-0.7607871	0.3580726	-2.1246726	0.0336139	Poisson	BirdID
fixed	factor(fac.yr)8	-0.6057225	0.3495136	-1.7330440	0.0830879	Poisson	BirdID
fixed	factor(fac.yr)9	-0.0275103	0.3264063	-0.0842825	0.9328318	Poisson	BirdID
fixed	factor(fac.yr)10	-0.5003991	0.3565939	-1.4032747	0.1605349	Poisson	BirdID
fixed	factor(fac.yr)11	-0.9243643	0.3860754	-2.3942584	0.0166540	Poisson	BirdID
fixed	factor(fac.yr)12	-0.2185844	0.3431944	-0.6369115	0.5241825	Poisson	BirdID
fixed	factor(fac.yr)13	0.1427788	0.3364043	0.4244261	0.6712551	Poisson	BirdID
fixed	factor(fac.yr)14	-0.3415596	0.3740841	-0.9130556	0.3612133	Poisson	BirdID
fixed	factor(fac.yr)15	-1.4284399	0.5377190	-2.6564804	0.0078961	Poisson	BirdID
fixed	factor(fac.yr)16	-0.1333363	0.4279353	-0.3115805	0.7553594	Poisson	BirdID
fixed	factor(fac.yr)17	-0.2129006	0.4528128	-0.4701736	0.6382310	Poisson	BirdID
fixed	factor(fac.yr)18	-0.7096230	0.5123818	-1.3849497	0.1660678	Poisson	BirdID
fixed	factor(fac.yr)19	-1.2571791	0.6590771	-1.9074840	0.0564579	Poisson	BirdID
fixed	factor(fac.yr)20	-0.6049339	0.5955941	-1.0156815	0.3097811	Poisson	BirdID
fixed	(Intercept)	0.2120948	0.2748813	0.7715868	0.4403592	Poisson	Terr
fixed	factor(fac.yr)2	-0.1456354	0.3655654	-0.3983840	0.6903471	Poisson	Terr
fixed	factor(fac.yr)3	-0.2888200	0.3566192	-0.8098836	0.4180071	Poisson	Terr
fixed	factor(fac.yr)4	-0.1169472	0.3345563	-0.3495593	0.7266695	Poisson	Terr
fixed	factor(fac.yr)5	0.2815916	0.3187432	0.8834435	0.3769967	Poisson	Terr
fixed	factor(fac.yr)6	-0.3094709	0.3416463	-0.9058223	0.3650299	Poisson	Terr
fixed	factor(fac.yr)7	-0.8041606	0.3546530	-2.2674577	0.0233623	Poisson	Terr
fixed	factor(fac.yr)8	-0.6206897	0.3454370	-1.7968247	0.0723635	Poisson	Terr
fixed	factor(fac.yr)9	-0.0034406	0.3213124	-0.0107079	0.9914565	Poisson	Terr
fixed	factor(fac.yr)10	-0.3896331	0.3535176	-1.1021603	0.2703920	Poisson	Terr
fixed	factor(fac.yr)11	-0.7099859	0.3840123	-1.8488624	0.0644777	Poisson	Terr
fixed	factor(fac.yr)12	-0.0789745	0.3393443	-0.2327268	0.8159736	Poisson	Terr
fixed	factor(fac.yr)13	0.2455434	0.3348546	0.7332835	0.4633855	Poisson	Terr
fixed	factor(fac.yr)14	-0.2519981	0.3731407	-0.6753433	0.4994577	Poisson	Terr
fixed	factor(fac.yr)15	-1.3767326	0.5399687	-2.5496527	0.0107830	Poisson	Terr
fixed	factor(fac.yr)16	-0.2231446	0.4240540	-0.5262175	0.5987371	Poisson	Terr
fixed	factor(fac.yr)17	-0.3126146	0.4482225	-0.6974540	0.4855187	Poisson	Terr
fixed	factor(fac.yr)18	-0.7959800	0.5080240	-1.5668156	0.1171578	Poisson	Terr
fixed	factor(fac.yr)19	-1.0927084	0.6590128	-1.6580988	0.0972975	Poisson	Terr
fixed	factor(fac.yr)20	-0.3208494	0.5972644	-0.5371983	0.5911307	Poisson	Terr
fixed	(Intercept)	0.2032196	0.2879182	0.7058240	0.4802976	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.1880573	0.3764465	-0.4995591	0.6173855	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.3404837	0.3696820	-0.9210177	0.3570412	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.1716208	0.3479881	-0.4931802	0.6218853	Poisson	BirdID+Terr

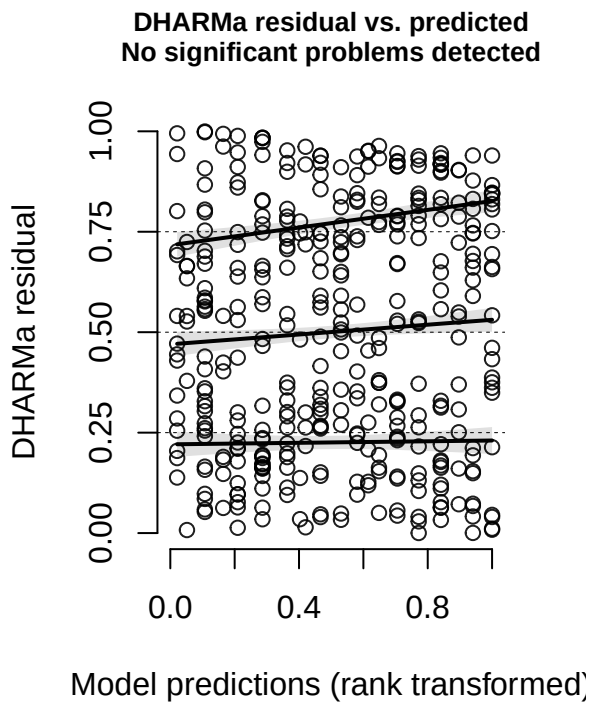
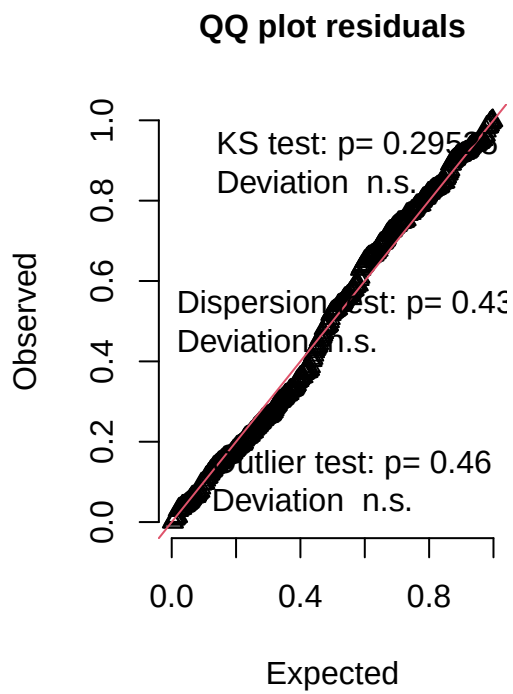
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.2229352	0.3328857	0.6697051	0.5030458	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.3482411	0.3566569	-0.9764036	0.3288645	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.8384193	0.3686824	-2.2740966	0.0229602	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.6701920	0.3603592	-1.8597888	0.0629154	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.0779544	0.3373639	-0.2310692	0.8172611	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.4962567	0.3685018	-1.3466874	0.1780809	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.8560397	0.3988193	-2.1464352	0.0318383	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.2222195	0.3565131	-0.6233135	0.5330785	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	0.1295202	0.3525622	0.3673684	0.7133442	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.3848834	0.3896577	-0.9877475	0.3232763	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-1.4604917	0.5518655	-2.6464633	0.0081338	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.2918354	0.4460743	-0.6542304	0.5129634	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.3837783	0.4708178	-0.8151313	0.4149971	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.8911164	0.5284174	-1.6863871	0.0917213	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.1899978	0.6767945	-1.7582853	0.0786990	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.4774060	0.6167731	-0.7740384	0.4389081	Poisson	BirdID+Terr

The territory only model is performing the best on BIC and marginal R^2 , while the model with both random effects is performing better on AIC and conditional R^2 .

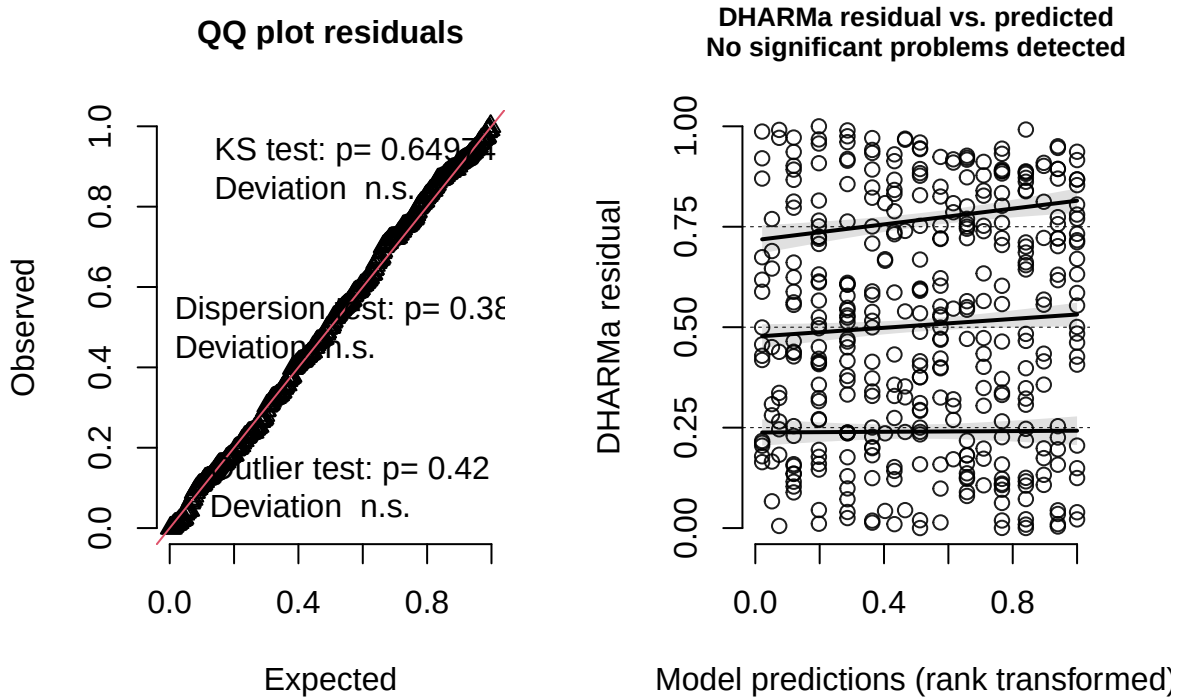
DHARMA residual



DHARMa residual



DHARMa residual



model diagnostics look reasonable.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
389	1	-245.0317	532.0635	615.2987	410.3034	368	0.508	Binomial	Terr
389	1	-244.2084	532.4168	619.6156	374.2976	367	0.593	Binomial	BirdID+Terr
389	1	-247.8206	537.6411	620.8763	435.7544	368	0.413	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.6913398	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	0.8511136	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	0.5903204	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.8579476	Binomial	BirdID+Terr

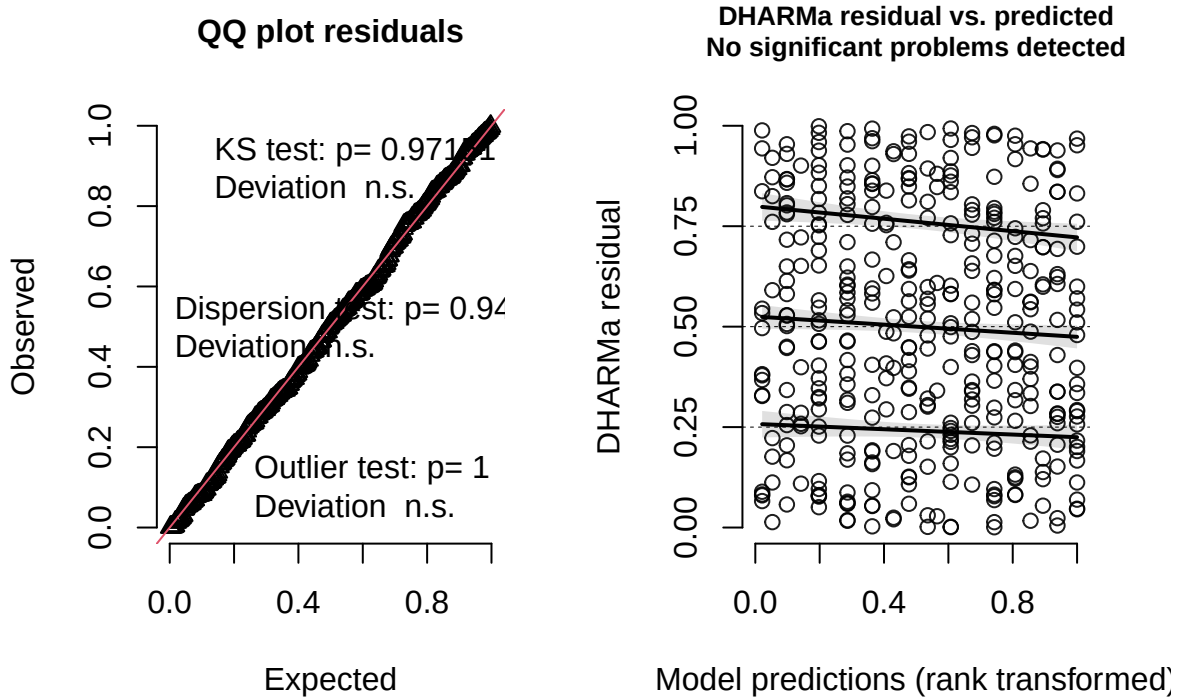
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	1.0102805	0.6402767	1.5778811	0.1145929	Binomial	BirdID
fixed	factor(fac.yr)2	-0.1608376	0.8415158	-0.1911284	0.8484250	Binomial	BirdID
fixed	factor(fac.yr)3	0.2560891	0.8315747	0.3079569	0.7581152	Binomial	BirdID
fixed	factor(fac.yr)4	-0.8207752	0.7540001	-1.0885611	0.2763475	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.7377401	0.8612885	0.8565541	0.3916914	Binomial	BirdID
fixed	factor(fac.yr)6	-0.7531524	0.7591649	-0.9920802	0.3211584	Binomial	BirdID
fixed	factor(fac.yr)7	-1.4206022	0.7385795	-1.9234247	0.0544267	Binomial	BirdID
fixed	factor(fac.yr)8	-1.2368707	0.7393243	-1.6729745	0.0943323	Binomial	BirdID
fixed	factor(fac.yr)9	-0.6855174	0.7549996	-0.9079706	0.3638938	Binomial	BirdID
fixed	factor(fac.yr)10	-1.0485034	0.7679892	-1.3652580	0.1721720	Binomial	BirdID
fixed	factor(fac.yr)11	-1.6386642	0.7729650	-2.1199721	0.0340084	Binomial	BirdID
fixed	factor(fac.yr)12	-0.7076518	0.7855674	-0.9008162	0.3676860	Binomial	BirdID
fixed	factor(fac.yr)13	-0.6668470	0.8043194	-0.8290823	0.4070579	Binomial	BirdID
fixed	factor(fac.yr)14	-0.8064713	0.8272624	-0.9748675	0.3296260	Binomial	BirdID
fixed	factor(fac.yr)15	-2.6277526	0.9411946	-2.7919332	0.0052394	Binomial	BirdID
fixed	factor(fac.yr)16	-0.8249596	0.9648672	-0.8549981	0.3925522	Binomial	BirdID
fixed	factor(fac.yr)17	-1.0301516	0.9957083	-1.0345918	0.3008596	Binomial	BirdID
fixed	factor(fac.yr)18	-2.4120018	1.0778089	-2.2378752	0.0252292	Binomial	BirdID
fixed	factor(fac.yr)19	-1.6199817	1.0143552	-1.5970557	0.1102533	Binomial	BirdID
fixed	factor(fac.yr)20	-0.7712547	1.1708191	-0.6587309	0.5100686	Binomial	BirdID
fixed	(Intercept)	1.0539106	0.6492508	1.6232719	0.1045313	Binomial	Terr
fixed	factor(fac.yr)2	-0.0947892	0.8681229	-0.1091887	0.9130528	Binomial	Terr
fixed	factor(fac.yr)3	0.2640021	0.8480834	0.3112926	0.7555782	Binomial	Terr
fixed	factor(fac.yr)4	-0.7656078	0.7672994	-0.9977954	0.3183786	Binomial	Terr
fixed	factor(fac.yr)5	0.7280110	0.8794447	0.8278076	0.4077795	Binomial	Terr
fixed	factor(fac.yr)6	-0.6975002	0.7644576	-0.9124118	0.3615520	Binomial	Terr
fixed	factor(fac.yr)7	-1.4132427	0.7447976	-1.8974855	0.0577639	Binomial	Terr
fixed	factor(fac.yr)8	-1.2583861	0.7446422	-1.6899206	0.0910431	Binomial	Terr
fixed	factor(fac.yr)9	-0.6641598	0.7606024	-0.8732024	0.3825528	Binomial	Terr
fixed	factor(fac.yr)10	-0.8742405	0.7718993	-1.1325836	0.2573892	Binomial	Terr
fixed	factor(fac.yr)11	-1.3310201	0.7808102	-1.7046653	0.0882569	Binomial	Terr
fixed	factor(fac.yr)12	-0.3724252	0.7936170	-0.4692757	0.6388726	Binomial	Terr
fixed	factor(fac.yr)13	-0.4822631	0.8198257	-0.5882508	0.5563640	Binomial	Terr
fixed	factor(fac.yr)14	-0.6534382	0.8427944	-0.7753234	0.4381486	Binomial	Terr
fixed	factor(fac.yr)15	-2.5249953	0.9615453	-2.6259764	0.0086401	Binomial	Terr
fixed	factor(fac.yr)16	-0.9214030	0.9805870	-0.9396444	0.3474000	Binomial	Terr
fixed	factor(fac.yr)17	-1.0717125	1.0111398	-1.0599054	0.2891877	Binomial	Terr
fixed	factor(fac.yr)18	-2.4853422	1.1097878	-2.2394751	0.0251250	Binomial	Terr
fixed	factor(fac.yr)19	-1.2952304	1.0291428	-1.2585527	0.2081920	Binomial	Terr
fixed	factor(fac.yr)20	-0.1485647	1.1681763	-0.1271767	0.8988006	Binomial	Terr
fixed	(Intercept)	1.1409327	0.6715026	1.6990743	0.0893052	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-0.1653258	0.8881233	-0.1861519	0.8523257	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	0.2474325	0.8760927	0.2824273	0.7776159	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-0.8792708	0.7919325	-1.1102849	0.2668763	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	0.6802109	0.9096613	0.7477628	0.4546032	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-0.7686446	0.7921078	-0.9703788	0.3318577	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-1.5527594	0.7714042	-2.0128999	0.0441252	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-1.3889157	0.7722306	-1.7985763	0.0720857	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-0.7945274	0.7901061	-1.0055959	0.3146100	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-1.0794697	0.8004988	-1.3484963	0.1774988	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-1.6065072	0.8100720	-1.9831660	0.0473489	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-0.5789129	0.8236517	-0.7028613	0.4821422	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-0.6507741	0.8515655	-0.7642092	0.4447426	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-0.8872034	0.8730193	-1.0162471	0.3095117	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-2.8046253	0.9996835	-2.8055132	0.0050237	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-1.0501736	1.0246997	-1.0248598	0.3054293	Binomial	BirdID+Terr

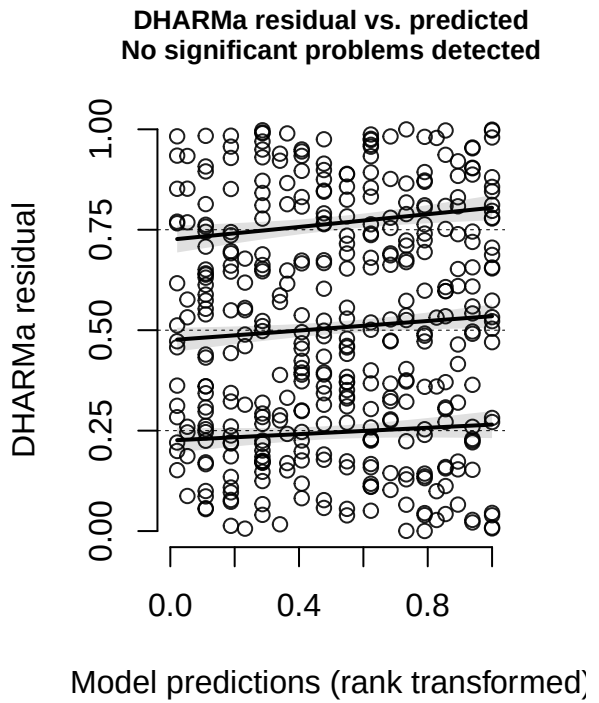
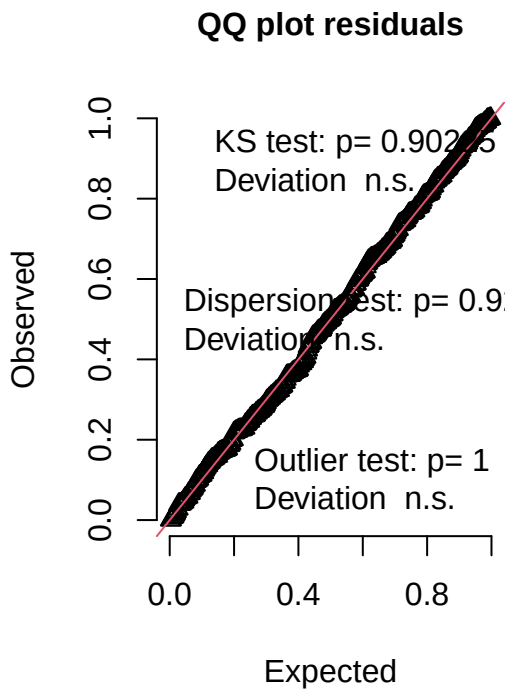
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-1.1940337	1.0539477	-1.1329155	0.2572497	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-2.7442379	1.1773795	-2.3308015	0.0197638	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-1.5028839	1.0850347	-1.3851022	0.1660212	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-0.3465865	1.2315831	-0.2814155	0.7783918	Binomial	BirdID+Terr

The territory only model is performing well on AIC and BIC.

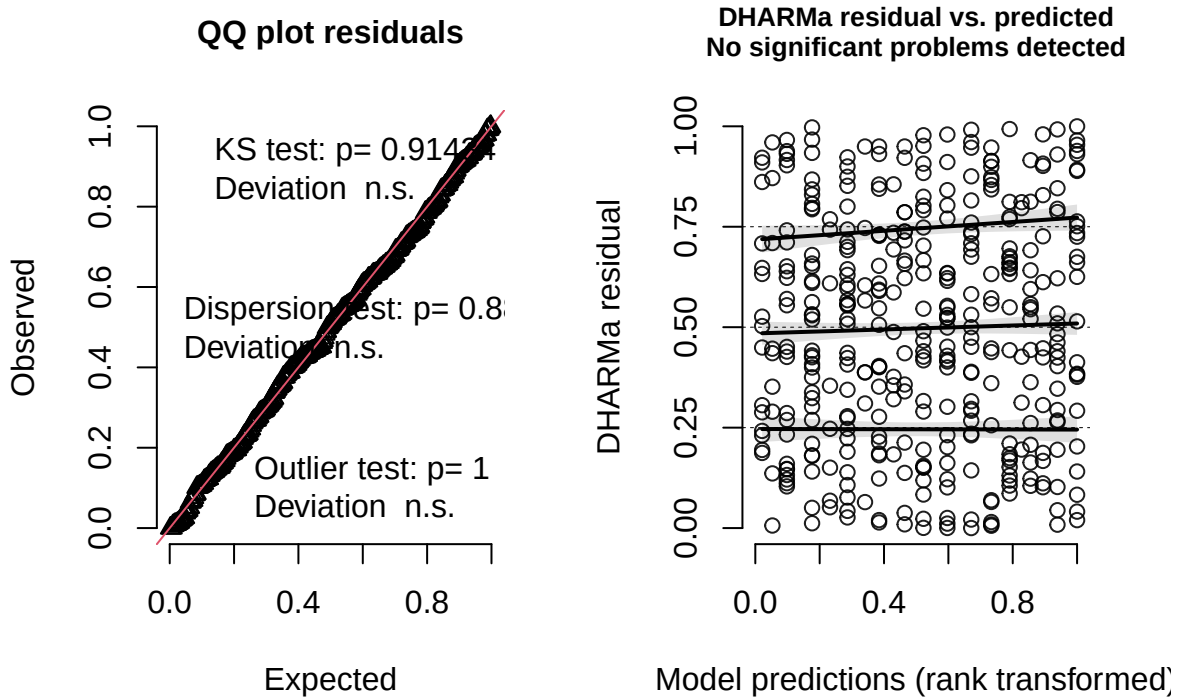
DHARMA residual



DHARMa residual



DHARMA residual



Probabilites used to generate simulation 5

Simulation 5

Probability of mate fidelity: 0.974

Probability of pair dispersal: 0.085

Probability of single male dispersal: 0.461

Probability of single female dispersal: 0.177

Simulation 6

Number of fledgling as the response (Poisson)

boundary (singular) fit: see `help('isSingular')`

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	condtional_R2	marginal_R2	family	RE
310	1	-425.4797	892.9595	971.4275	309.7348	289	0.2652298	0.0961449	Poisson	Terr
310	1	-425.4799	894.9597	977.1643	309.6761	288	0.2655248	0.0962933	Poisson	BirdID+Terr
310	1	-429.5458	901.0916	979.5596	352.4694	289	0.1951081	0.1163631	Poisson	BirdID

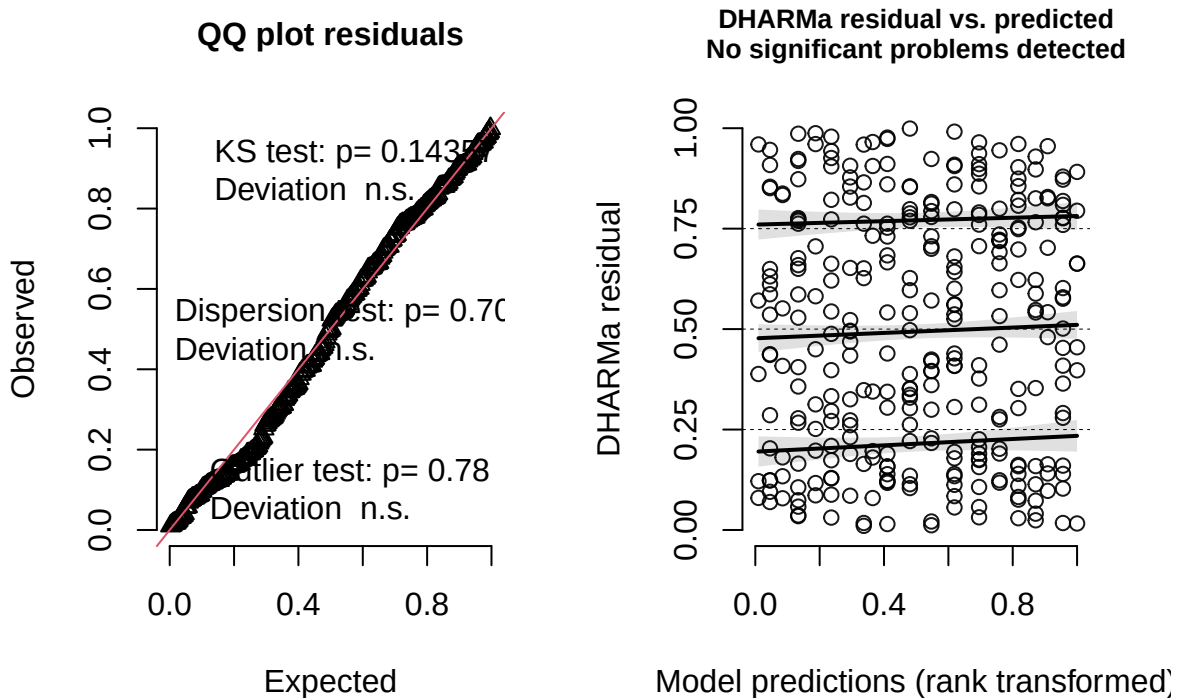
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.2471817	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.3744046	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.0025602	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.3746336	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.3985558	0.2300600	1.7323994	0.0832025	Poisson	BirdID
fixed	factor(fac.yr)2	-0.1034126	0.3107720	-0.3327604	0.7393151	Poisson	BirdID
fixed	factor(fac.yr)3	-0.7776546	0.3740905	-2.0787875	0.0376369	Poisson	BirdID
fixed	factor(fac.yr)4	-0.3396531	0.3115392	-1.0902421	0.2756065	Poisson	BirdID
fixed	factor(fac.yr)5	-0.1700393	0.2935880	-0.5791766	0.5624700	Poisson	BirdID
fixed	factor(fac.yr)6	-0.1119044	0.2982081	-0.3752559	0.7074702	Poisson	BirdID
fixed	factor(fac.yr)7	-0.6970500	0.3339566	-2.0872471	0.0368658	Poisson	BirdID
fixed	factor(fac.yr)8	-0.6032623	0.3393519	-1.7776895	0.0754549	Poisson	BirdID
fixed	factor(fac.yr)9	0.0500579	0.2933879	0.1706201	0.8645225	Poisson	BirdID
fixed	factor(fac.yr)10	-0.3458524	0.3159158	-1.0947616	0.2736211	Poisson	BirdID
fixed	factor(fac.yr)11	-0.5759741	0.3590039	-1.6043670	0.1086332	Poisson	BirdID
fixed	factor(fac.yr)12	-0.2000912	0.3131754	-0.6389109	0.5228809	Poisson	BirdID
fixed	factor(fac.yr)13	-0.0603037	0.3058151	-0.1971901	0.8436788	Poisson	BirdID
fixed	factor(fac.yr)14	-0.5544568	0.4133657	-1.3413229	0.1798157	Poisson	BirdID
fixed	factor(fac.yr)15	-0.6599936	0.4310587	-1.5310990	0.1257449	Poisson	BirdID
fixed	factor(fac.yr)16	0.1666539	0.3726134	0.4472568	0.6546897	Poisson	BirdID
fixed	factor(fac.yr)17	0.0444471	0.3630726	0.1224193	0.9025669	Poisson	BirdID
fixed	factor(fac.yr)18	-0.7220849	0.5134515	-1.4063353	0.1596246	Poisson	BirdID
fixed	factor(fac.yr)19	-0.3882826	0.4541887	-0.8548926	0.3926106	Poisson	BirdID
fixed	factor(fac.yr)20	-1.2858517	0.7578144	-1.6967898	0.0897364	Poisson	BirdID
fixed	(Intercept)	0.3663639	0.2392737	1.5311496	0.1257324	Poisson	Terr
fixed	factor(fac.yr)2	-0.1137100	0.3236036	-0.3513868	0.7252982	Poisson	Terr
fixed	factor(fac.yr)3	-0.7038169	0.3842683	-1.8315770	0.0670145	Poisson	Terr
fixed	factor(fac.yr)4	-0.2955511	0.3225513	-0.9162917	0.3595139	Poisson	Terr
fixed	factor(fac.yr)5	-0.1967349	0.3002452	-0.6552474	0.5123085	Poisson	Terr
fixed	factor(fac.yr)6	-0.1340980	0.3086231	-0.4345040	0.6639225	Poisson	Terr
fixed	factor(fac.yr)7	-0.6632618	0.3433631	-1.9316632	0.0534011	Poisson	Terr
fixed	factor(fac.yr)8	-0.5327463	0.3474097	-1.5334816	0.1251572	Poisson	Terr
fixed	factor(fac.yr)9	0.1034061	0.3041782	0.3399523	0.7338925	Poisson	Terr
fixed	factor(fac.yr)10	-0.3478274	0.3217856	-1.0809291	0.2797287	Poisson	Terr
fixed	factor(fac.yr)11	-0.5890548	0.3686739	-1.5977664	0.1100950	Poisson	Terr
fixed	factor(fac.yr)12	-0.2054592	0.3234348	-0.6352415	0.5252709	Poisson	Terr
fixed	factor(fac.yr)13	-0.0581529	0.3147424	-0.1847634	0.8534146	Poisson	Terr
fixed	factor(fac.yr)14	-0.4493816	0.4278744	-1.0502653	0.2935961	Poisson	Terr
fixed	factor(fac.yr)15	-0.5677816	0.4386423	-1.2944070	0.1955249	Poisson	Terr
fixed	factor(fac.yr)16	0.1786043	0.3819266	0.4676403	0.6400418	Poisson	Terr
fixed	factor(fac.yr)17	0.0759742	0.3731939	0.2035784	0.8386829	Poisson	Terr
fixed	factor(fac.yr)18	-0.7169845	0.5262470	-1.3624486	0.1730564	Poisson	Terr
fixed	factor(fac.yr)19	-0.3246605	0.4695179	-0.6914763	0.4892663	Poisson	Terr
fixed	factor(fac.yr)20	-1.2030182	0.7697037	-1.5629627	0.1180614	Poisson	Terr
fixed	(Intercept)	0.3672853	0.2392187	1.5353531	0.1246971	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.1145126	0.3235579	-0.3539168	0.7234012	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.7031458	0.3840876	-1.8306915	0.0671466	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.2943778	0.3223933	-0.9131014	0.3611892	Poisson	BirdID+Terr

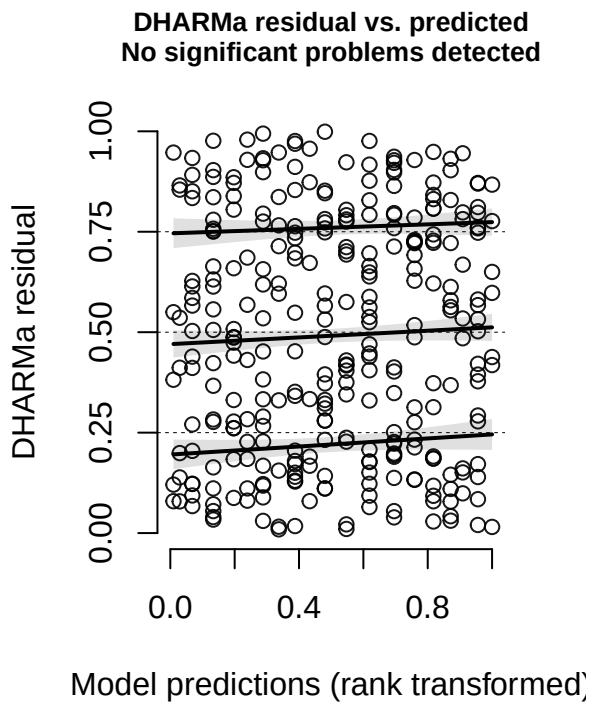
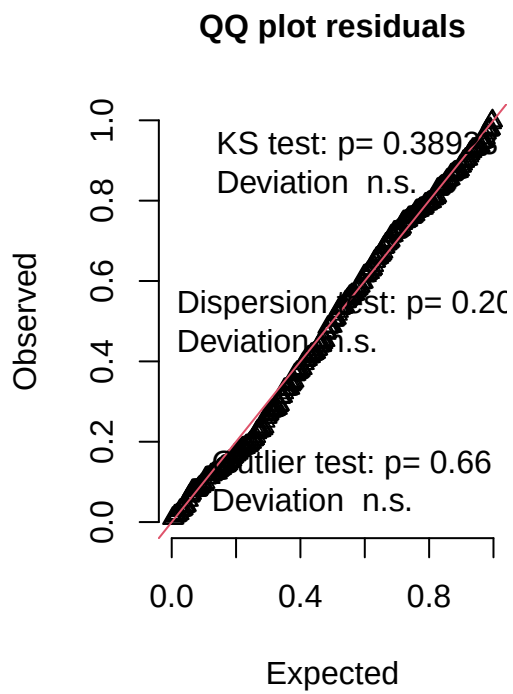
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.1980162	0.3002201	-0.6595700	0.5095298	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.1362810	0.3086491	-0.4415403	0.6588219	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.6651592	0.3434016	-1.9369718	0.0527488	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.5333180	0.3473486	-1.5353970	0.1246863	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	0.1026618	0.3041349	0.3375534	0.7356997	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.3479574	0.3216963	-1.0816330	0.2794156	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.5901205	0.3686519	-1.6007528	0.1094317	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.2065735	0.3234064	-0.6387427	0.5229903	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.0590423	0.3147050	-0.1876116	0.8511811	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.4497491	0.4277886	-1.0513348	0.2931048	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.5677793	0.4385018	-1.2948165	0.1953835	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	0.1800890	0.3817028	0.4718042	0.6370665	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	0.0748209	0.3731922	0.2004890	0.8410982	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.7240486	0.5273564	-1.3729775	0.1697593	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-0.3265526	0.4696378	-0.6953287	0.4868493	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-1.2052307	0.7700591	-1.5651145	0.1175561	Poisson	BirdID+Terr

The territory only model is performing the best on AIC and BIC, and the R^2 s are similar for the territory only model and the model with both random effects.

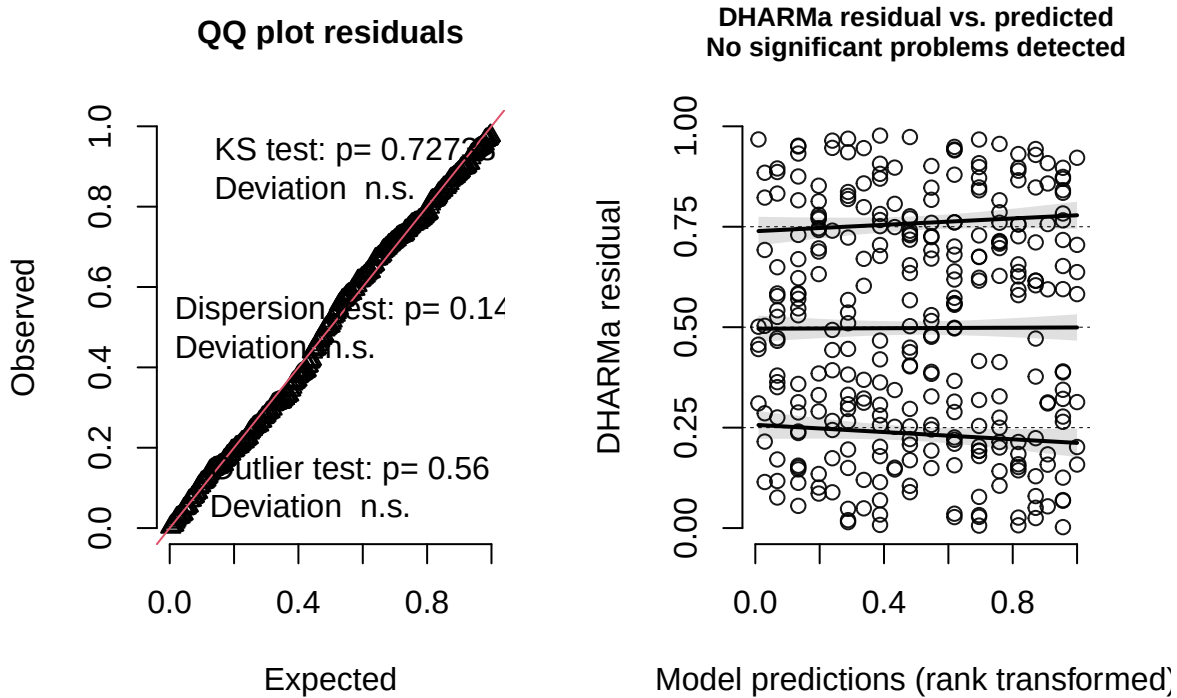
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
310	1	-192.3010	426.6020	505.0700	319.3393	289	0.444	Binomial	BirdID
310	1	-191.9789	427.9578	510.1624	303.2563	288	0.493	Binomial	BirdID+Terr
310	1	-193.6669	429.3339	507.8019	336.8596	289	0.393	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.8678188	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	0.7066644	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	0.8134659	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.5001300	Binomial	BirdID+Terr

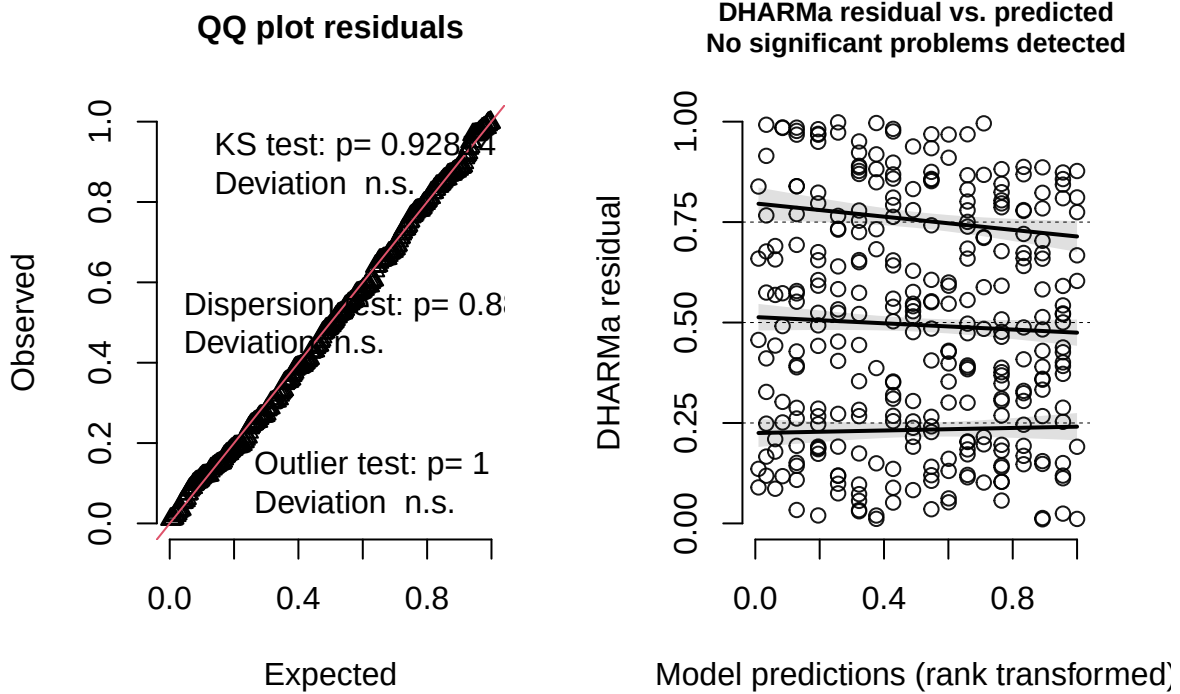
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.7763029	0.6135672	1.2652288	0.2057893	Binomial	BirdID
fixed	factor(fac.yr)2	0.2655096	0.8143936	0.3260212	0.7444083	Binomial	BirdID
fixed	factor(fac.yr)3	-0.6720373	0.8016770	-0.8382894	0.4018682	Binomial	BirdID
fixed	factor(fac.yr)4	0.0764616	0.7839507	0.0975337	0.9223025	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.2464828	0.7771441	0.3171649	0.7511185	Binomial	BirdID
fixed	factor(fac.yr)6	-0.3723180	0.7716642	-0.4824871	0.6294599	Binomial	BirdID
fixed	factor(fac.yr)7	-0.5808526	0.7648458	-0.7594376	0.4475908	Binomial	BirdID
fixed	factor(fac.yr)8	-0.7915771	0.7837334	-1.0100082	0.3124914	Binomial	BirdID
fixed	factor(fac.yr)9	0.6828150	0.8624074	0.7917547	0.4285037	Binomial	BirdID
fixed	factor(fac.yr)10	-0.7366862	0.7781642	-0.9466977	0.3437928	Binomial	BirdID
fixed	factor(fac.yr)11	-0.3556531	0.8319163	-0.4275107	0.6690074	Binomial	BirdID
fixed	factor(fac.yr)12	0.6491476	0.8722578	0.7442153	0.4567462	Binomial	BirdID
fixed	factor(fac.yr)13	-0.1142288	0.8075583	-0.1414496	0.8875148	Binomial	BirdID
fixed	factor(fac.yr)14	-0.4055721	0.9426058	-0.4302669	0.6670015	Binomial	BirdID
fixed	factor(fac.yr)15	-1.2974733	0.9393741	-1.3812105	0.1672142	Binomial	BirdID
fixed	factor(fac.yr)16	1.1998754	1.3565008	0.8845371	0.3764063	Binomial	BirdID
fixed	factor(fac.yr)17	0.1574975	1.0006153	0.1574007	0.8749291	Binomial	BirdID
fixed	factor(fac.yr)18	-1.0181117	1.0545302	-0.9654647	0.3343121	Binomial	BirdID
fixed	factor(fac.yr)19	-0.9779452	1.0532201	-0.9285288	0.3531333	Binomial	BirdID
fixed	factor(fac.yr)20	-2.2299291	1.3927374	-1.6011124	0.1093520	Binomial	BirdID
fixed	(Intercept)	0.7470863	0.5937128	1.2583294	0.2082727	Binomial	Terr
fixed	factor(fac.yr)2	0.2152537	0.8085299	0.2662285	0.7900632	Binomial	Terr
fixed	factor(fac.yr)3	-0.4600481	0.7796655	-0.5900583	0.5551515	Binomial	Terr
fixed	factor(fac.yr)4	0.2062432	0.7660329	0.2692354	0.7877485	Binomial	Terr
fixed	factor(fac.yr)5	0.3118774	0.7539014	0.4136846	0.6791051	Binomial	Terr
fixed	factor(fac.yr)6	-0.2766195	0.7502886	-0.3686842	0.7123631	Binomial	Terr
fixed	factor(fac.yr)7	-0.4190098	0.7385979	-0.5673044	0.5705074	Binomial	Terr
fixed	factor(fac.yr)8	-0.6345352	0.7535172	-0.8420978	0.3997332	Binomial	Terr
fixed	factor(fac.yr)9	0.8868320	0.8415616	1.0537933	0.2919775	Binomial	Terr
fixed	factor(fac.yr)10	-0.6317341	0.7472649	-0.8453952	0.3978902	Binomial	Terr
fixed	factor(fac.yr)11	-0.1915353	0.8062545	-0.2375618	0.8122210	Binomial	Terr
fixed	factor(fac.yr)12	0.6760044	0.8416293	0.8032092	0.4218538	Binomial	Terr
fixed	factor(fac.yr)13	-0.0796185	0.7763236	-0.1025583	0.9183135	Binomial	Terr
fixed	factor(fac.yr)14	-0.1079652	0.9101570	-0.1186226	0.9055744	Binomial	Terr
fixed	factor(fac.yr)15	-0.9801656	0.9020619	-1.0865835	0.2772209	Binomial	Terr
fixed	factor(fac.yr)16	1.3084405	1.3034313	1.0038431	0.3154542	Binomial	Terr
fixed	factor(fac.yr)17	0.1363357	0.9593472	0.1421130	0.8869908	Binomial	Terr
fixed	factor(fac.yr)18	-0.9668405	1.0154966	-0.9520865	0.3410531	Binomial	Terr
fixed	factor(fac.yr)19	-0.9628929	1.0117463	-0.9517138	0.3412421	Binomial	Terr
fixed	factor(fac.yr)20	-2.0318259	1.3602553	-1.4937093	0.1352516	Binomial	Terr
fixed	(Intercept)	0.8186306	0.6256742	1.3083975	0.1907385	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	0.2272878	0.8293394	0.2740588	0.7840394	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-0.6616386	0.8149382	-0.8118881	0.4168559	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	0.1189881	0.7999049	0.1487528	0.8817487	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	0.2362295	0.7904716	0.2988463	0.7650573	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-0.3734667	0.7853893	-0.4755179	0.6344179	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-0.5899949	0.7784637	-0.7578965	0.4485129	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-0.7691602	0.7966236	-0.9655252	0.3342818	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	0.7671211	0.8811387	0.8706020	0.3839715	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-0.7554808	0.7910973	-0.9549785	0.3395886	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-0.3546686	0.8474590	-0.4185083	0.6755755	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	0.6291108	0.8864812	0.7096720	0.4779076	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-0.1222676	0.8217191	-0.1487949	0.8817155	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-0.3251881	0.9606193	-0.3385192	0.7349720	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-1.2475990	0.9545295	-1.3070304	0.1912024	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	1.2372592	1.3909981	0.8894759	0.3737474	Binomial	BirdID+Terr

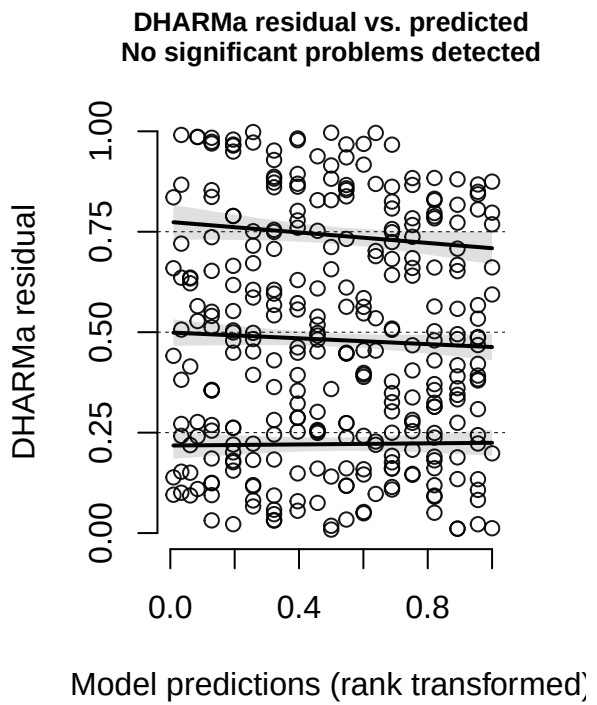
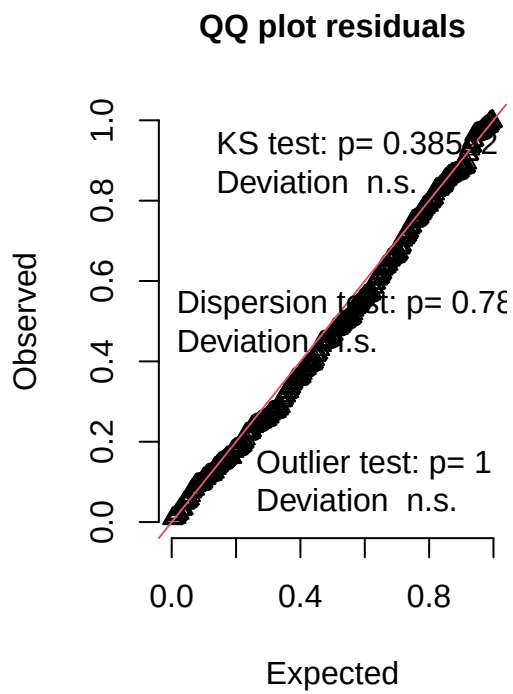
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	0.1506149	1.0190987	0.1477922	0.8825067	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-1.0833131	1.0836777	-0.9996635	0.3174734	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-1.0363137	1.0782256	-0.9611289	0.3364874	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-2.2620971	1.4327180	-1.5788852	0.1143624	Binomial	BirdID+Terr

The band ID only model performed best on AIC and BIC, while the model that had both random effects had the best kappa value.

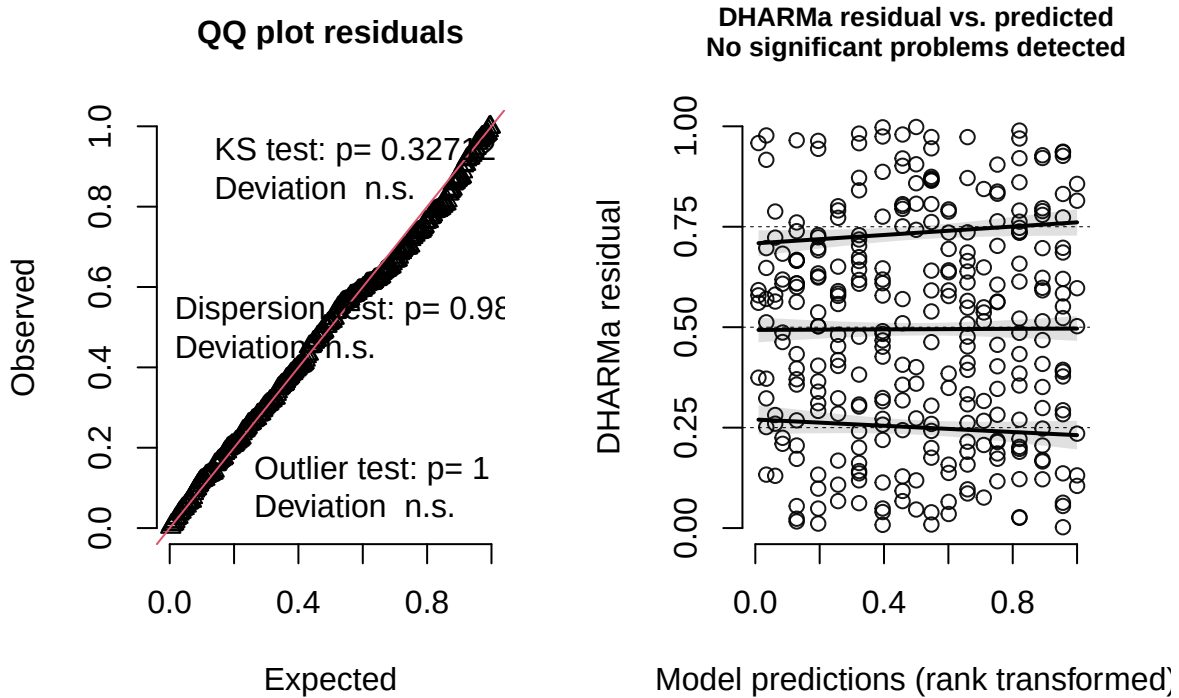
DHARMA residual



DHARMa residual



DHARMA residual



Diagnostics look good.

Probabilities used to generate simulation 6

Simulation 6
 Probability of mate fidelity: 0.826
 Probability of pair dispersal: 0.226
 Probability of single male dispersal: 0.794
 Probability of single female dispersal: 0.382

Simulation 7

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
380	1	-516.3822	1076.764	1163.448	329.3788	358	0.3865058	0.1400295	Poisson	BirdID+Terr
380	1	-519.2152	1080.430	1163.174	377.6559	359	0.3491965	0.1618638	Poisson	Terr
380	1	-520.2352	1082.470	1165.214	383.9354	359	0.3214326	0.1368396	Poisson	BirdID

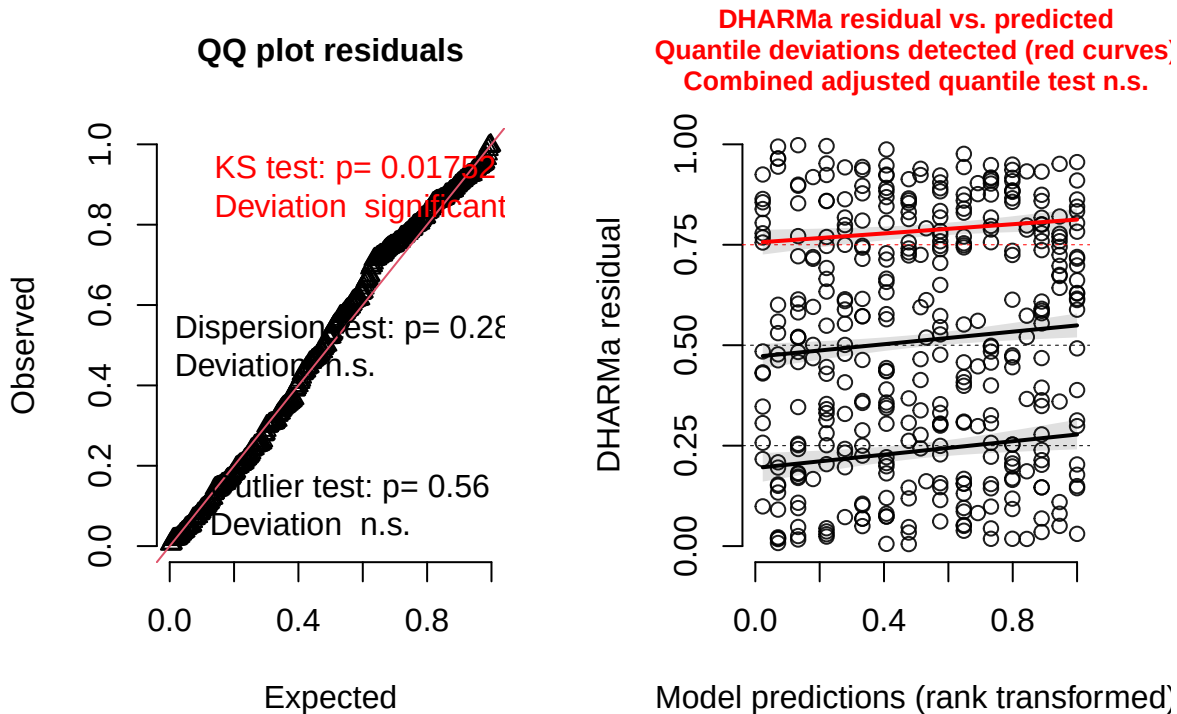
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.4141763	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.4169141	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.3447756	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.3597288	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	-0.6593376	0.3366551	-1.9584959	0.0501719	Poisson	BirdID
fixed	factor(fac.yr)2	0.5007230	0.4088766	1.2246311	0.2207142	Poisson	BirdID
fixed	factor(fac.yr)3	0.0880477	0.4252347	0.2070566	0.8359656	Poisson	BirdID
fixed	factor(fac.yr)4	0.7616694	0.3882024	1.9620422	0.0497576	Poisson	BirdID
fixed	factor(fac.yr)5	0.9641778	0.3805469	2.5336637	0.0112877	Poisson	BirdID
fixed	factor(fac.yr)6	1.0816130	0.3744301	2.8886915	0.0038685	Poisson	BirdID
fixed	factor(fac.yr)7	0.5484399	0.3850919	1.4241792	0.1543946	Poisson	BirdID
fixed	factor(fac.yr)8	0.7237473	0.3845787	1.8819225	0.0598465	Poisson	BirdID
fixed	factor(fac.yr)9	1.1888740	0.3736304	3.1819519	0.0014629	Poisson	BirdID
fixed	factor(fac.yr)10	0.8435588	0.3874750	2.1770662	0.0294756	Poisson	BirdID
fixed	factor(fac.yr)11	0.4291776	0.4078090	1.0523985	0.2926167	Poisson	BirdID
fixed	factor(fac.yr)12	0.4352827	0.4227756	1.0295832	0.3032057	Poisson	BirdID
fixed	factor(fac.yr)13	1.1573638	0.3940884	2.9368128	0.0033160	Poisson	BirdID
fixed	factor(fac.yr)14	0.6313977	0.4146585	1.5226932	0.1278355	Poisson	BirdID
fixed	factor(fac.yr)15	-0.0155024	0.4851897	-0.0319511	0.9745110	Poisson	BirdID
fixed	factor(fac.yr)16	0.6397145	0.4775715	1.3395156	0.1804029	Poisson	BirdID
fixed	factor(fac.yr)17	1.0349559	0.4541038	2.2791174	0.0226601	Poisson	BirdID
fixed	factor(fac.yr)18	0.3685414	0.5460489	0.6749239	0.4997241	Poisson	BirdID
fixed	factor(fac.yr)19	0.8258666	0.5142617	1.6059267	0.1082900	Poisson	BirdID
fixed	factor(fac.yr)20	0.6471343	0.6437465	1.0052626	0.3147704	Poisson	BirdID
fixed	(Intercept)	-0.6591917	0.3322923	-1.9837703	0.0472815	Poisson	Terr
fixed	factor(fac.yr)2	0.4562113	0.4061636	1.1232204	0.2613439	Poisson	Terr
fixed	factor(fac.yr)3	0.0129643	0.4198199	0.0308806	0.9753648	Poisson	Terr
fixed	factor(fac.yr)4	0.8016255	0.3826810	2.0947616	0.0361922	Poisson	Terr
fixed	factor(fac.yr)5	1.0301545	0.3760980	2.7390587	0.0061615	Poisson	Terr
fixed	factor(fac.yr)6	1.1798382	0.3694296	3.1936751	0.0014047	Poisson	Terr
fixed	factor(fac.yr)7	0.5800713	0.3794767	1.5286087	0.1263615	Poisson	Terr
fixed	factor(fac.yr)8	0.8213236	0.3761001	2.1837897	0.0289777	Poisson	Terr
fixed	factor(fac.yr)9	1.2197333	0.3680722	3.3138422	0.0009202	Poisson	Terr
fixed	factor(fac.yr)10	0.8411966	0.3819538	2.2023519	0.0276405	Poisson	Terr
fixed	factor(fac.yr)11	0.4677093	0.4034942	1.1591477	0.2463960	Poisson	Terr
fixed	factor(fac.yr)12	0.5612529	0.4189358	1.3397111	0.1803393	Poisson	Terr
fixed	factor(fac.yr)13	1.3128547	0.3866587	3.3953835	0.0006853	Poisson	Terr
fixed	factor(fac.yr)14	0.6768488	0.4076007	1.6605685	0.0968001	Poisson	Terr
fixed	factor(fac.yr)15	0.0388215	0.4798811	0.0808983	0.9355229	Poisson	Terr
fixed	factor(fac.yr)16	0.9580657	0.4718509	2.0304417	0.0423117	Poisson	Terr
fixed	factor(fac.yr)17	1.0289115	0.4517176	2.2777759	0.0227399	Poisson	Terr
fixed	factor(fac.yr)18	0.4054057	0.5461135	0.7423469	0.4578772	Poisson	Terr
fixed	factor(fac.yr)19	0.7724592	0.5014104	1.5405729	0.1234208	Poisson	Terr
fixed	factor(fac.yr)20	0.5049981	0.6162127	0.8195190	0.4124904	Poisson	Terr
fixed	(Intercept)	-0.6733702	0.3417267	-1.9704934	0.0487819	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	0.4571841	0.4154199	1.1005349	0.2710991	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	0.0055859	0.4303669	0.0129795	0.9896442	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	0.7654025	0.3948744	1.9383445	0.0525812	Poisson	BirdID+Terr

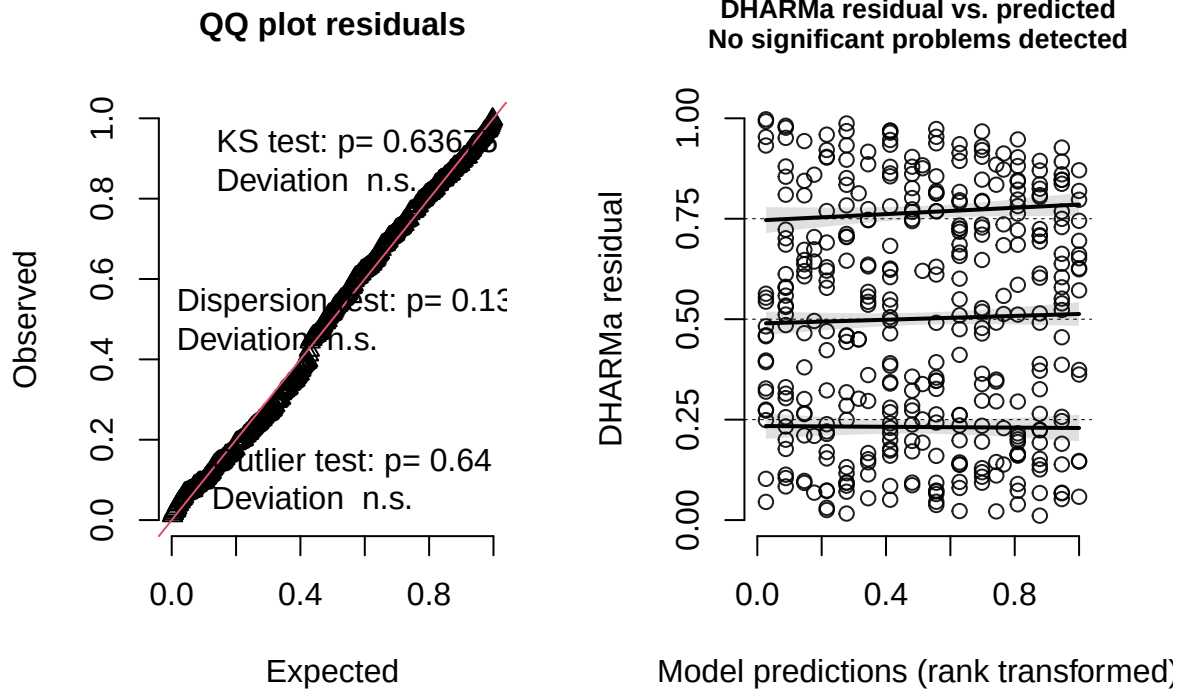
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.9862120	0.3877496	2.5434248	0.0109772	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	1.1367338	0.3808771	2.9845163	0.0028403	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	0.5479607	0.3906611	1.4026499	0.1607213	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	0.7595889	0.3888081	1.9536344	0.0507445	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	1.1935124	0.3806581	3.1353925	0.0017162	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	0.7996042	0.3948282	2.0251951	0.0428473	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	0.4071131	0.4154700	0.9798855	0.3271426	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	0.4772340	0.4310752	1.1070783	0.2682601	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	1.2033218	0.4018811	2.9942234	0.0027514	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	0.6333427	0.4214026	1.5029396	0.1328546	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.0049276	0.4930395	-0.0099943	0.9920259	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	0.8307850	0.4881643	1.7018555	0.0887825	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	1.0186623	0.4689871	2.1720477	0.0298521	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	0.3618250	0.5604619	0.6455835	0.5185491	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	0.7441971	0.5248553	1.4179091	0.1562173	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	0.5433198	0.6478376	0.8386667	0.4016564	Poisson	BirdID+Terr

The models seem to be performing similarly.

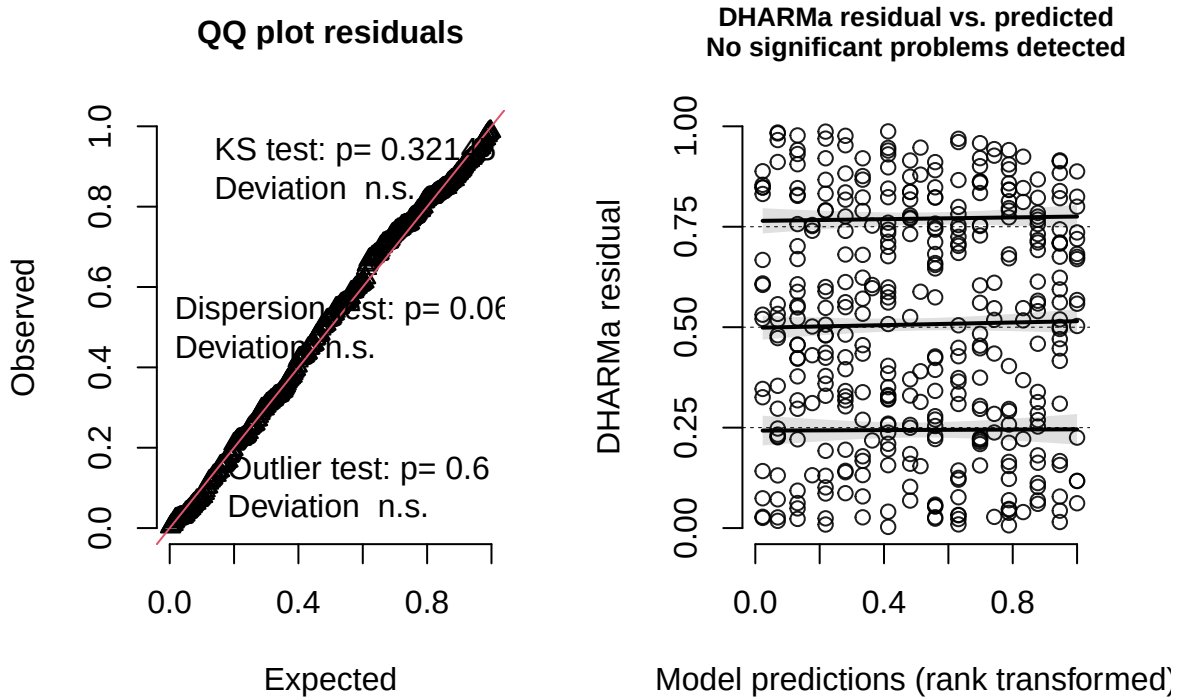
DHARMA residual



DHARMa residual



DHARMa residual



There are some model assumption violations with the band ID only model, I would not consider using it.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
380	1	-224.6300	493.2600	579.9438	259.9359	358	0.783	Binomial	BirdID+Terr
380	1	-229.0744	500.1488	582.8924	337.9935	359	0.653	Binomial	BirdID
380	1	-231.4290	504.8580	587.6016	356.9222	359	0.571	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.277175	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.129523	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.349438	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.120672	Binomial	BirdID+Terr

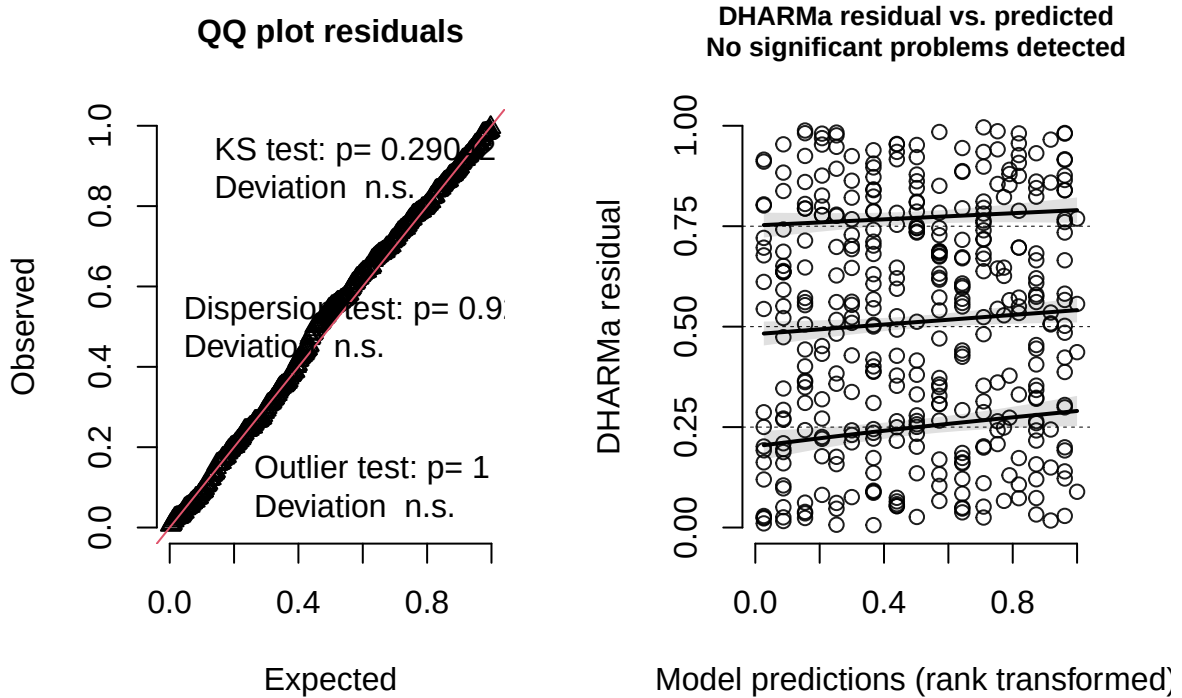
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	-1.3861907	0.6927405	-2.0010245	0.0453897	Binomial	BirdID
fixed	factor(fac.yr)2	1.9964537	0.8307921	2.4030727	0.0162580	Binomial	BirdID
fixed	factor(fac.yr)3	1.0013017	0.8283184	1.2088367	0.2267256	Binomial	BirdID
fixed	factor(fac.yr)4	2.1273559	0.8413873	2.5283907	0.0114587	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	2.0216569	0.8405917	2.4050402	0.0161707	Binomial	BirdID
fixed	factor(fac.yr)6	2.9336416	0.8766710	3.3463429	0.0008189	Binomial	BirdID
fixed	factor(fac.yr)7	1.6449338	0.8008350	2.0540235	0.0399734	Binomial	BirdID
fixed	factor(fac.yr)8	2.1167456	0.8285850	2.5546511	0.0106294	Binomial	BirdID
fixed	factor(fac.yr)9	3.0875791	0.9086967	3.3978104	0.0006793	Binomial	BirdID
fixed	factor(fac.yr)10	2.1860666	0.8591452	2.5444669	0.0109445	Binomial	BirdID
fixed	factor(fac.yr)11	1.3704084	0.8408957	1.6297009	0.1031647	Binomial	BirdID
fixed	factor(fac.yr)12	1.5800625	0.8829325	1.7895621	0.0735243	Binomial	BirdID
fixed	factor(fac.yr)13	2.6336714	1.0081047	2.6124979	0.0089883	Binomial	BirdID
fixed	factor(fac.yr)14	1.5062179	0.9002286	1.6731504	0.0942977	Binomial	BirdID
fixed	factor(fac.yr)15	1.4145479	0.9272503	1.5255297	0.1271270	Binomial	BirdID
fixed	factor(fac.yr)16	2.2728422	1.1295911	2.0120929	0.0442101	Binomial	BirdID
fixed	factor(fac.yr)17	3.0363301	1.2585672	2.4125291	0.0158423	Binomial	BirdID
fixed	factor(fac.yr)18	2.6140585	1.2751311	2.0500312	0.0403614	Binomial	BirdID
fixed	factor(fac.yr)19	2.5896120	1.3085584	1.9789808	0.0478182	Binomial	BirdID
fixed	factor(fac.yr)20	3.3763548	1.6452370	2.0521996	0.0401503	Binomial	BirdID
fixed	(Intercept)	-1.4557262	0.6557962	-2.2197843	0.0264334	Binomial	Terr
fixed	factor(fac.yr)2	1.7984175	0.7924849	2.2693398	0.0232477	Binomial	Terr
fixed	factor(fac.yr)3	0.9050117	0.7819947	1.1573117	0.2471450	Binomial	Terr
fixed	factor(fac.yr)4	2.1629538	0.7900496	2.7377445	0.0061862	Binomial	Terr
fixed	factor(fac.yr)5	2.2005183	0.7995832	2.7520817	0.0059218	Binomial	Terr
fixed	factor(fac.yr)6	3.1775180	0.8355739	3.8027971	0.0001431	Binomial	Terr
fixed	factor(fac.yr)7	1.7404956	0.7588711	2.2935327	0.0218174	Binomial	Terr
fixed	factor(fac.yr)8	2.3796228	0.7820863	3.0426600	0.0023450	Binomial	Terr
fixed	factor(fac.yr)9	3.1698575	0.8629920	3.6731018	0.0002396	Binomial	Terr
fixed	factor(fac.yr)10	2.3853790	0.8261524	2.8873352	0.0038852	Binomial	Terr
fixed	factor(fac.yr)11	1.6102847	0.7988892	2.0156545	0.0438361	Binomial	Terr
fixed	factor(fac.yr)12	2.2046974	0.8525814	2.5859085	0.0097123	Binomial	Terr
fixed	factor(fac.yr)13	3.5011389	0.9855836	3.5523510	0.0003818	Binomial	Terr
fixed	factor(fac.yr)14	1.8973172	0.8431157	2.2503639	0.0244259	Binomial	Terr
fixed	factor(fac.yr)15	1.5802289	0.8752666	1.8054257	0.0710081	Binomial	Terr
fixed	factor(fac.yr)16	3.0770947	1.0272329	2.9955182	0.0027398	Binomial	Terr
fixed	factor(fac.yr)17	2.6342516	1.1336634	2.3236629	0.0201436	Binomial	Terr
fixed	factor(fac.yr)18	2.6347049	1.1760549	2.2402908	0.0250720	Binomial	Terr
fixed	factor(fac.yr)19	2.1557515	1.1733007	1.8373393	0.0661598	Binomial	Terr
fixed	factor(fac.yr)20	2.7655286	1.4334228	1.9293182	0.0536914	Binomial	Terr
fixed	(Intercept)	-1.1153050	0.7269615	-1.5342011	0.1249802	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	1.7530585	0.8629204	2.0315415	0.0422001	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	0.4755435	0.8848219	0.5374454	0.5909600	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	2.0296410	0.8944201	2.2692256	0.0232546	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	2.0488222	0.8965637	2.2851944	0.0223014	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	3.0388542	0.9258126	3.2823642	0.0010294	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	1.4443242	0.8451003	1.7090566	0.0874405	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	2.1687281	0.8748961	2.4788406	0.0131810	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	3.0735904	0.9622997	3.1940054	0.0014031	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	2.0740341	0.9305808	2.2287524	0.0258304	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	1.2181524	0.8991965	1.3547121	0.1755093	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	1.6506725	0.9537388	1.7307386	0.0834984	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	3.1942361	1.1044527	2.8921440	0.0038262	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	1.4574485	0.9762350	1.4929279	0.1354561	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	1.2746765	1.0175733	1.2526630	0.2103284	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	2.5255717	1.1875127	2.1267746	0.0334388	Binomial	BirdID+Terr

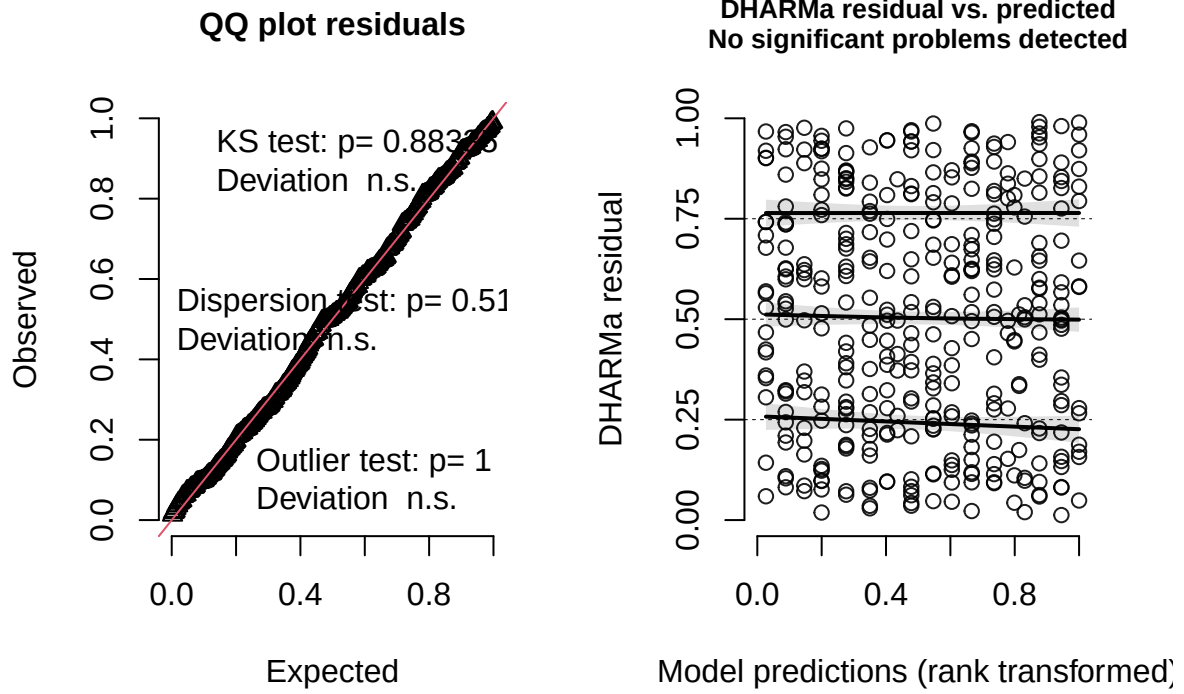
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	2.7178956	1.3423808	2.0246830	0.0428999	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	2.3413525	1.3769816	1.7003512	0.0890649	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	2.1890616	1.4612380	1.4980870	0.1341106	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	3.2100655	1.7926949	1.7906368	0.0733516	Binomial	BirdID+Terr

The model with both random effects is performing best on all selection criteria.

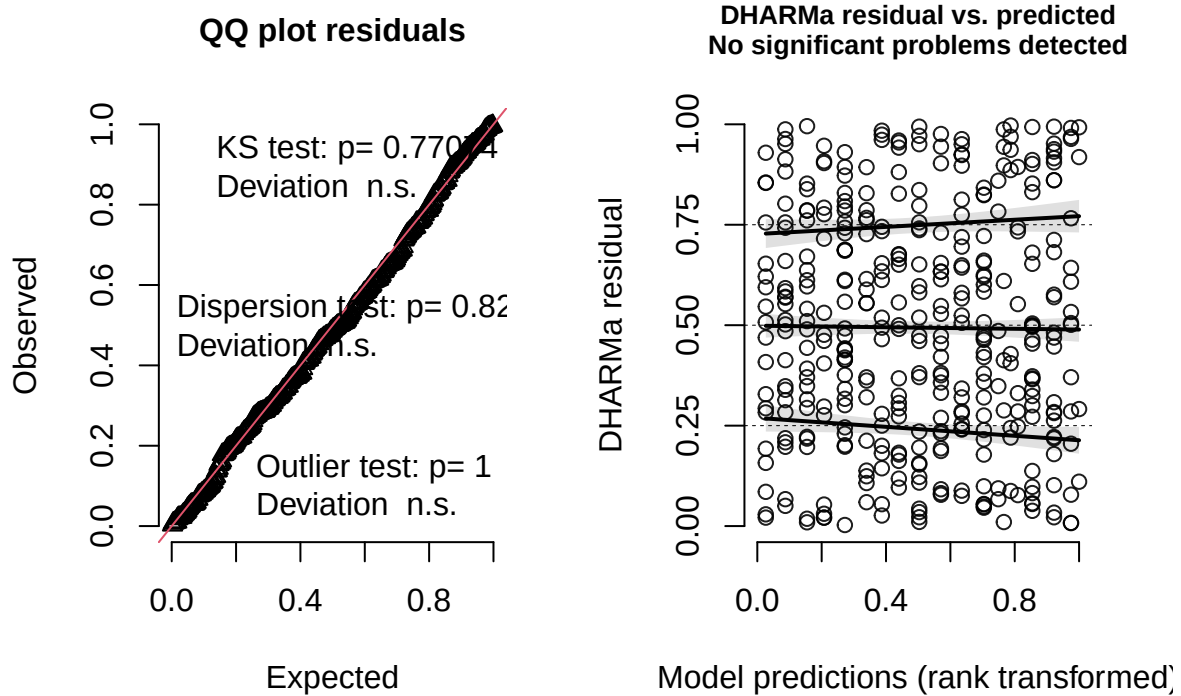
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Probabilites used to generate simulation 7

Simulation 7
 Probability of mate fidelity: 0.632
 Probability of pair dispersal: 0.076
 Probability of single male dispersal: 0.302
 Probability of single female dispersal: 0.618

Simulation 8

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	REfamily	RE
310	1	-420.8132	883.6264	962.0944	313.4095	289	0.2255325	0.1393022	Poisson	Terr
310	1	-420.8133	885.6266	967.8312	313.3672	288	0.2247592	0.1387190	Poisson	BirdID+Terr
310	1	-422.3729	886.7458	965.2138	338.0531	289	0.1807820	0.1436686	Poisson	BirdID

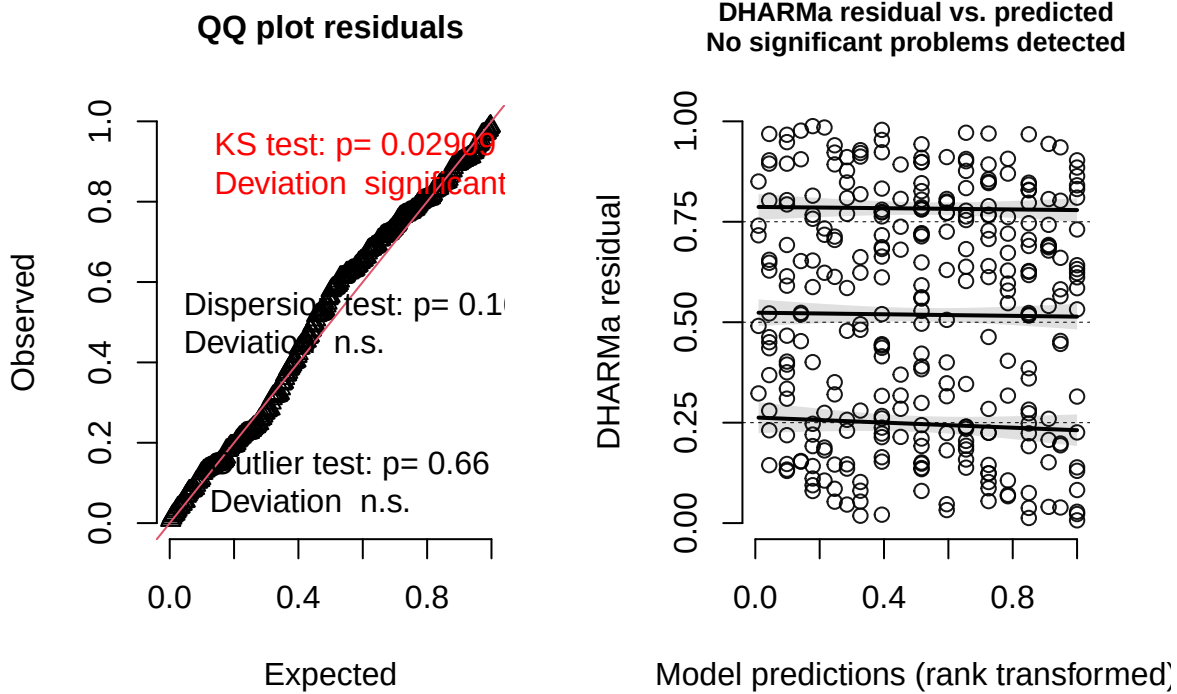
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.1672539	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.2606009	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.0026269	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.2607854	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.1183082	0.3406662	0.3472849	0.7283773	Poisson	BirdID
fixed	factor(fac.yr)2	0.2275753	0.4069723	0.5591911	0.5760313	Poisson	BirdID
fixed	factor(fac.yr)3	-0.4706241	0.4668566	-1.0080701	0.3134208	Poisson	BirdID
fixed	factor(fac.yr)4	-0.6330643	0.4787573	-1.3223075	0.1860658	Poisson	BirdID
fixed	factor(fac.yr)5	0.3507562	0.3831279	0.9155067	0.3599257	Poisson	BirdID
fixed	factor(fac.yr)6	0.2807279	0.3787833	0.7411308	0.4586141	Poisson	BirdID
fixed	factor(fac.yr)7	-0.0386620	0.3881891	-0.0995957	0.9206653	Poisson	BirdID
fixed	factor(fac.yr)8	0.1356469	0.3762285	0.3605438	0.7184405	Poisson	BirdID
fixed	factor(fac.yr)9	0.2074983	0.3870805	0.5360596	0.5919173	Poisson	BirdID
fixed	factor(fac.yr)10	0.1626068	0.3934333	0.4133021	0.6793853	Poisson	BirdID
fixed	factor(fac.yr)11	-0.6299673	0.4575438	-1.3768458	0.1685599	Poisson	BirdID
fixed	factor(fac.yr)12	0.1425249	0.4071617	0.3500449	0.7263050	Poisson	BirdID
fixed	factor(fac.yr)13	0.3216196	0.4262401	0.7545502	0.4505189	Poisson	BirdID
fixed	factor(fac.yr)14	0.3055144	0.4139294	0.7380834	0.4604638	Poisson	BirdID
fixed	factor(fac.yr)15	-0.2299484	0.4684398	-0.4908814	0.6235103	Poisson	BirdID
fixed	factor(fac.yr)16	-0.1294737	0.4425221	-0.2925812	0.7698423	Poisson	BirdID
fixed	factor(fac.yr)17	-0.2122355	0.4584536	-0.4629379	0.6434089	Poisson	BirdID
fixed	factor(fac.yr)18	-0.4111333	0.5362949	-0.7666180	0.4433087	Poisson	BirdID
fixed	factor(fac.yr)19	-0.5919606	0.5668834	-1.0442370	0.2963757	Poisson	BirdID
fixed	factor(fac.yr)20	-0.6412531	0.6766479	-0.9476909	0.3432869	Poisson	BirdID
fixed	(Intercept)	0.1338028	0.3469367	0.3856693	0.6997416	Poisson	Terr
fixed	factor(fac.yr)2	0.1693904	0.4149312	0.4082373	0.6830995	Poisson	Terr
fixed	factor(fac.yr)3	-0.4917943	0.4752694	-1.0347696	0.3007765	Poisson	Terr
fixed	factor(fac.yr)4	-0.6491935	0.4843730	-1.3402761	0.1801556	Poisson	Terr
fixed	factor(fac.yr)5	0.3495903	0.3896673	0.8971508	0.3696385	Poisson	Terr
fixed	factor(fac.yr)6	0.2630812	0.3842752	0.6846166	0.4935859	Poisson	Terr
fixed	factor(fac.yr)7	-0.0481942	0.3934228	-0.1224999	0.9025031	Poisson	Terr
fixed	factor(fac.yr)8	0.0940066	0.3820558	0.2460547	0.8056399	Poisson	Terr
fixed	factor(fac.yr)9	0.1940389	0.3930238	0.4937077	0.6215127	Poisson	Terr
fixed	factor(fac.yr)10	0.1237099	0.3992419	0.3098620	0.7566659	Poisson	Terr
fixed	factor(fac.yr)11	-0.6417498	0.4637891	-1.3837103	0.1664472	Poisson	Terr
fixed	factor(fac.yr)12	0.1213414	0.4153285	0.2921576	0.7701661	Poisson	Terr
fixed	factor(fac.yr)13	0.3035143	0.4347629	0.6981145	0.4851056	Poisson	Terr
fixed	factor(fac.yr)14	0.3164841	0.4226457	0.7488166	0.4539677	Poisson	Terr
fixed	factor(fac.yr)15	-0.2770932	0.4728568	-0.5859982	0.5578767	Poisson	Terr
fixed	factor(fac.yr)16	-0.1311189	0.4494695	-0.2917192	0.7705013	Poisson	Terr
fixed	factor(fac.yr)17	-0.2235586	0.4637939	-0.4820214	0.6297907	Poisson	Terr
fixed	factor(fac.yr)18	-0.5121567	0.5423963	-0.9442482	0.3450428	Poisson	Terr
fixed	factor(fac.yr)19	-0.6005933	0.5748713	-1.0447439	0.2961413	Poisson	Terr
fixed	factor(fac.yr)20	-0.6744253	0.6873900	-0.9811392	0.3265241	Poisson	Terr
fixed	(Intercept)	0.1306500	0.3474416	0.3760345	0.7068912	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	0.1724399	0.4153539	0.4151637	0.6780221	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.4874917	0.4755222	-1.0251713	0.3052824	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.6465328	0.4847888	-1.3336381	0.1823225	Poisson	BirdID+Terr

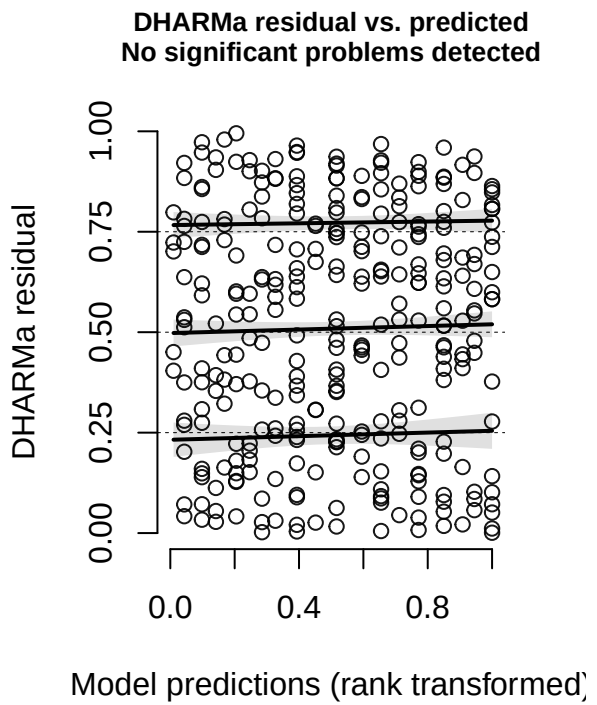
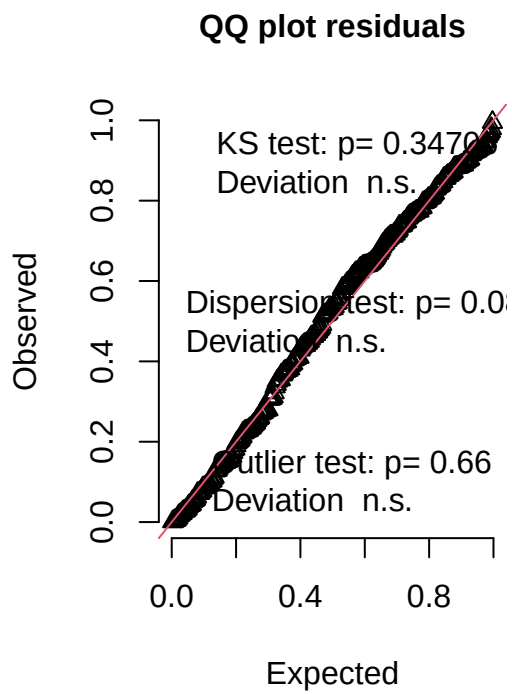
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.3531238	0.3901062	0.9051990	0.3653600	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	0.2661621	0.3847329	0.6918099	0.4890567	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.0444704	0.3938444	-0.1129136	0.9100990	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	0.0969899	0.3825187	0.2535560	0.7998386	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	0.1973099	0.3934659	0.5014664	0.6160429	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	0.1268930	0.3996780	0.3174880	0.7508734	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.6390239	0.4642088	-1.3765873	0.1686399	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	0.1242363	0.4157698	0.2988103	0.7650848	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	0.3060826	0.4352105	0.7032978	0.4818702	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	0.3191819	0.4230936	0.7544002	0.4506090	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.2744495	0.4732724	-0.5798975	0.5619837	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.1282813	0.4498897	-0.2851394	0.7755373	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.2206331	0.4641919	-0.4753059	0.6345690	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.5091645	0.5427318	-0.9381511	0.3481667	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-0.5965863	0.5750348	-1.0374787	0.2995128	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.6696106	0.6872549	-0.9743264	0.3298945	Poisson	BirdID+Terr

territory only model is best on all criteria.

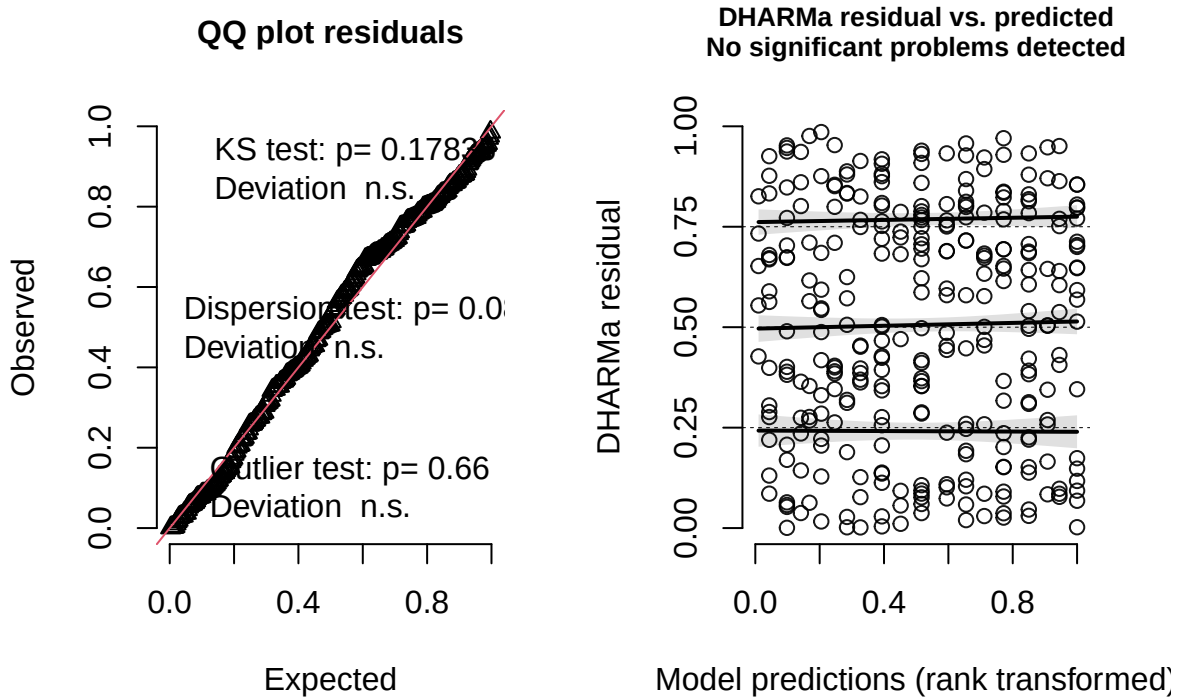
DHARMA residual



DHARMa residual



DHARMa residual



The band ID model may not be valid.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
310	1	-181.8170	407.6340	489.8386	209.3902	288	0.774	Binomial	BirdID+Terr
310	1	-184.9101	411.8202	490.2882	283.3504	289	0.518	Binomial	Terr
310	1	-186.2771	414.5541	493.0221	279.5715	289	0.570	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.212110	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.113610	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.238666	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.199395	Binomial	BirdID+Terr

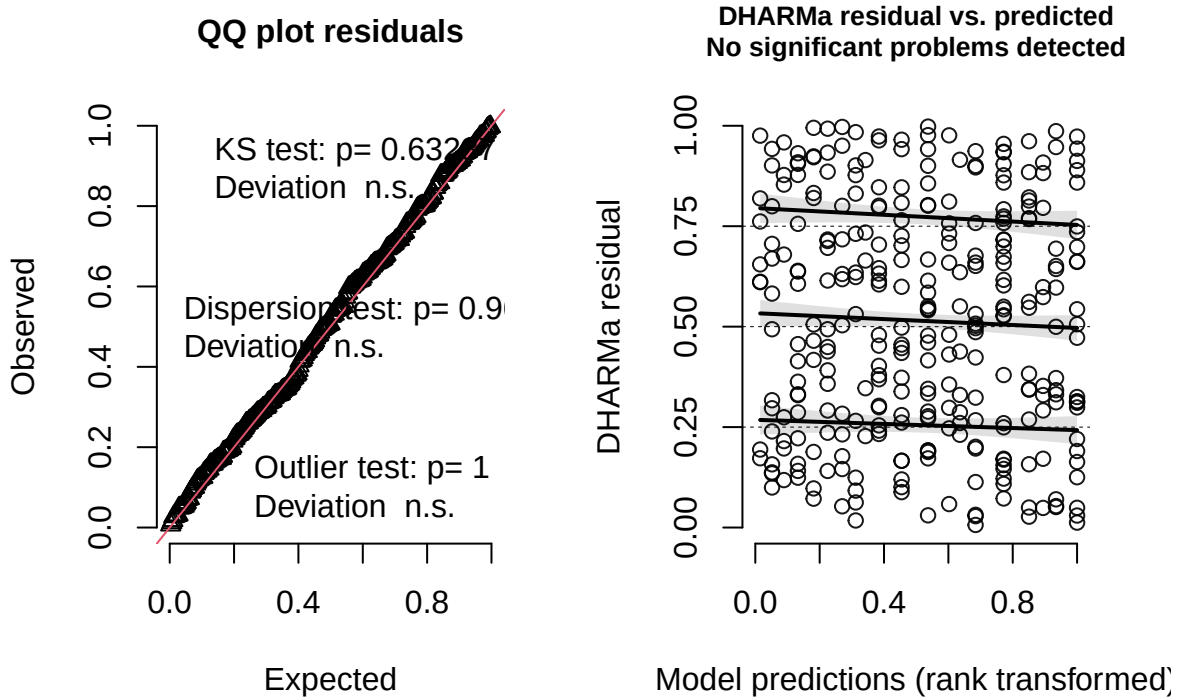
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.9569665	0.8894406	1.0759195	0.2819632	Binomial	BirdID
fixed	factor(fac.yr)2	0.8248017	1.1146157	0.7399875	0.4593076	Binomial	BirdID
fixed	factor(fac.yr)3	-1.4063133	1.0760545	-1.3069164	0.1912411	Binomial	BirdID
fixed	factor(fac.yr)4	-1.0125305	1.0758662	-0.9411305	0.3466380	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.3057289	1.0518100	0.2906693	0.7713043	Binomial	BirdID
fixed	factor(fac.yr)6	0.8273530	1.0588860	0.7813429	0.4346009	Binomial	BirdID
fixed	factor(fac.yr)7	-0.1605734	1.0078439	-0.1593237	0.8734139	Binomial	BirdID
fixed	factor(fac.yr)8	0.3499038	1.0115236	0.3459176	0.7294046	Binomial	BirdID
fixed	factor(fac.yr)9	-0.1937212	1.0323815	-0.1876450	0.8511549	Binomial	BirdID
fixed	factor(fac.yr)10	-0.4446171	1.0383336	-0.4282025	0.6685037	Binomial	BirdID
fixed	factor(fac.yr)11	-1.1196563	1.0480155	-1.0683586	0.2853588	Binomial	BirdID
fixed	factor(fac.yr)12	0.5916448	1.1178586	0.5292662	0.5966208	Binomial	BirdID
fixed	factor(fac.yr)13	0.6812026	1.2865072	0.5294977	0.5964602	Binomial	BirdID
fixed	factor(fac.yr)14	-0.0437711	1.1702698	-0.0374026	0.9701640	Binomial	BirdID
fixed	factor(fac.yr)15	-1.0258582	1.1269861	-0.9102669	0.3626818	Binomial	BirdID
fixed	factor(fac.yr)16	-0.5005528	1.1106051	-0.4507028	0.6522038	Binomial	BirdID
fixed	factor(fac.yr)17	-0.6902806	1.1203475	-0.6161308	0.5378082	Binomial	BirdID
fixed	factor(fac.yr)18	-1.1542412	1.2017194	-0.9604915	0.3368079	Binomial	BirdID
fixed	factor(fac.yr)19	-1.7264342	1.2239609	-1.4105305	0.1583831	Binomial	BirdID
fixed	factor(fac.yr)20	-0.4960008	1.3587139	-0.3650517	0.7150728	Binomial	BirdID
fixed	(Intercept)	0.9396693	0.8397880	1.1189364	0.2631673	Binomial	Terr
fixed	factor(fac.yr)2	0.7788368	1.1044914	0.7051543	0.4807142	Binomial	Terr
fixed	factor(fac.yr)3	-1.1525073	1.0484122	-1.0992883	0.2716423	Binomial	Terr
fixed	factor(fac.yr)4	-0.7119728	1.0125058	-0.7031789	0.4819442	Binomial	Terr
fixed	factor(fac.yr)5	0.5434168	0.9931330	0.5471742	0.5842590	Binomial	Terr
fixed	factor(fac.yr)6	0.9121818	1.0055614	0.9071368	0.3643345	Binomial	Terr
fixed	factor(fac.yr)7	0.1221846	0.9398747	0.1300009	0.8965657	Binomial	Terr
fixed	factor(fac.yr)8	0.3874847	0.9563838	0.4051561	0.6853628	Binomial	Terr
fixed	factor(fac.yr)9	-0.1543004	0.9736329	-0.1584790	0.8740794	Binomial	Terr
fixed	factor(fac.yr)10	-0.3636690	0.9749280	-0.3730214	0.7091325	Binomial	Terr
fixed	factor(fac.yr)11	-1.0120633	0.9906043	-1.0216625	0.3069407	Binomial	Terr
fixed	factor(fac.yr)12	0.6492160	1.0671460	0.6083666	0.5429443	Binomial	Terr
fixed	factor(fac.yr)13	0.9678041	1.2326701	0.7851283	0.4323784	Binomial	Terr
fixed	factor(fac.yr)14	0.5904226	1.1412268	0.5173578	0.6049064	Binomial	Terr
fixed	factor(fac.yr)15	-0.7883349	1.0642114	-0.7407691	0.4588334	Binomial	Terr
fixed	factor(fac.yr)16	-0.0580595	1.0500241	-0.0552935	0.9559046	Binomial	Terr
fixed	factor(fac.yr)17	-0.3269953	1.0607529	-0.3082672	0.7578791	Binomial	Terr
fixed	factor(fac.yr)18	-1.3402205	1.1549830	-1.1603811	0.2458937	Binomial	Terr
fixed	factor(fac.yr)19	-1.4667245	1.1682500	-1.2554885	0.2093015	Binomial	Terr
fixed	factor(fac.yr)20	-0.4118901	1.3180154	-0.3125078	0.7546546	Binomial	Terr
fixed	(Intercept)	1.4711710	1.0025602	1.4674141	0.1422634	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	0.7017174	1.2364217	0.5675389	0.5703481	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-1.9135430	1.2231402	-1.5644511	0.1177117	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-1.1835323	1.2008234	-0.9856006	0.3243292	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	0.3260534	1.1856668	0.2749958	0.7833195	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	0.8600918	1.1909319	0.7222006	0.4701711	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-0.3463798	1.1141219	-0.3108994	0.7558771	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	0.0653438	1.1376232	0.0574388	0.9541956	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-0.6103713	1.1570275	-0.5275339	0.5978229	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-0.8287872	1.1580231	-0.7156914	0.4741819	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-1.5014266	1.1778145	-1.2747565	0.2023955	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	0.5599582	1.2652522	0.4425665	0.6580793	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	0.8435681	1.4577761	0.5786678	0.5628133	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-0.0308518	1.3184933	-0.0233993	0.9813318	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-1.5372354	1.2464819	-1.2332593	0.2174790	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-0.5500386	1.2389994	-0.4439378	0.6570876	Binomial	BirdID+Terr

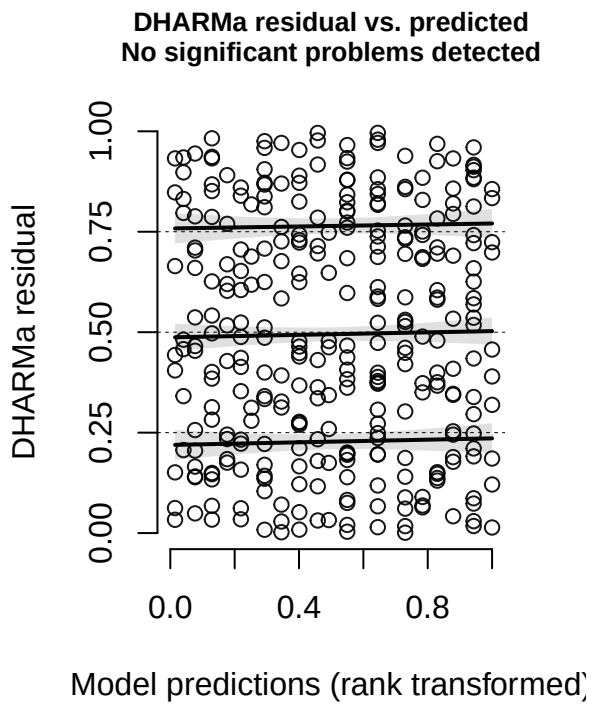
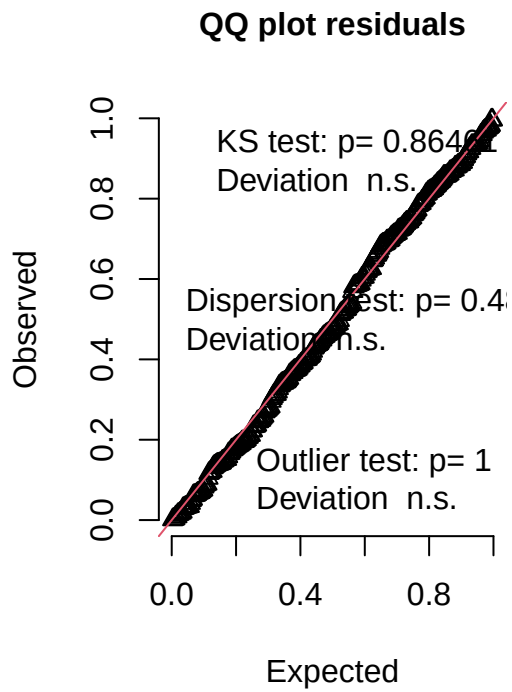
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-0.9822872	1.2448003	-0.7891123	0.4300464	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-2.0651043	1.3597929	-1.5186903	0.1288405	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-2.2067836	1.3835569	-1.5950075	0.1107105	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-0.8802886	1.5346996	-0.5735902	0.5662452	Binomial	BirdID+Terr

The model fitting both random effects seems to be best optimizing model selection criteria.

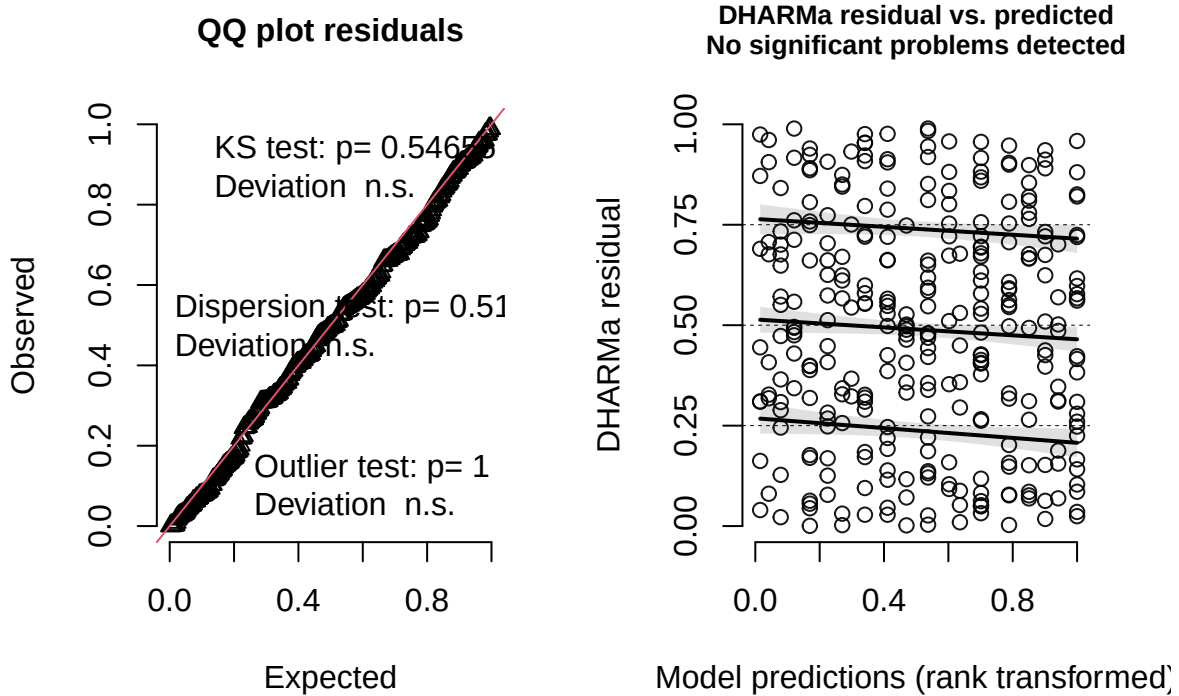
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Probabilities used to generate simulation 8

Simulation 8

Probability of mate fidelity: 0.704

Probability of pair dispersal: 0.942

Probability of single male dispersal: 0.713

Probability of single female dispersal: 0.04

Simulation 9

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
354	1	-483.7292	1009.458	1090.714	358.8003	333	0.2929704	0.1024262	Poisson	Terr
354	1	-483.5912	1011.182	1096.307	348.3313	332	0.3057354	0.1031938	Poisson	BirdID+Terr
354	1	-490.1081	1022.216	1103.471	413.3622	333	0.2143035	0.1251971	Poisson	BirdID

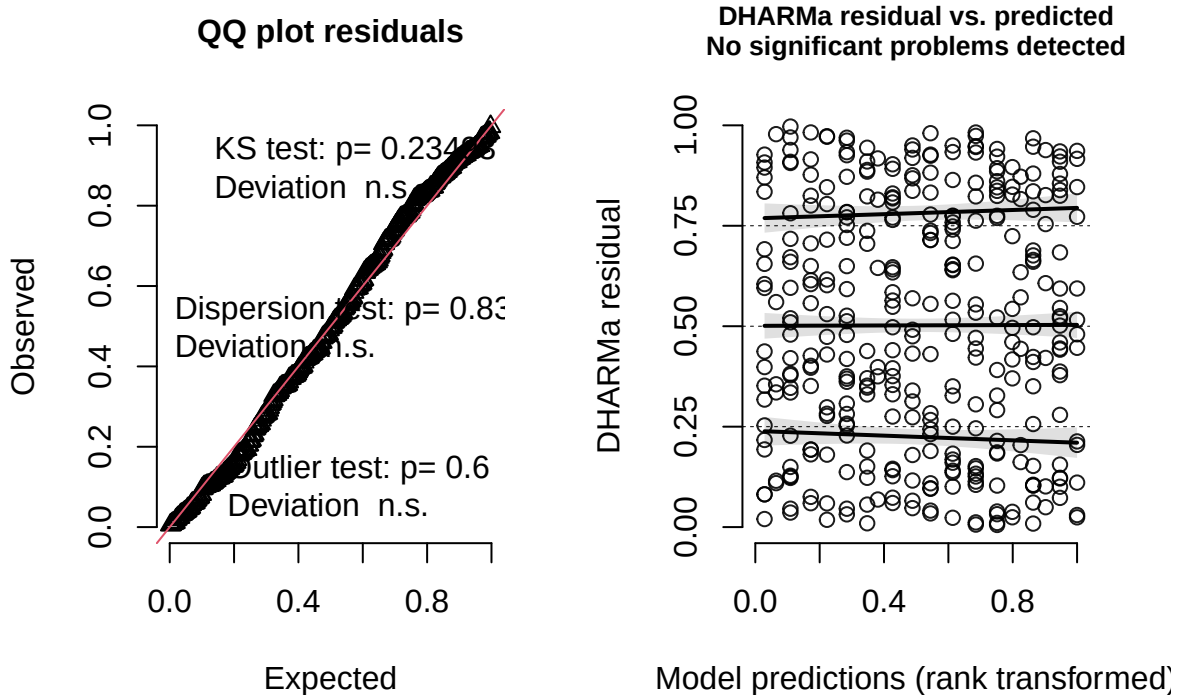
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.2678218	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.4101174	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.1482966	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.4002049	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.3141969	0.2638706	1.1907234	0.2337622	Poisson	BirdID
fixed	factor(fac.yr)2	-0.0943753	0.3191658	-0.2956937	0.7674640	Poisson	BirdID
fixed	factor(fac.yr)3	-0.5578382	0.3403487	-1.6390196	0.1012092	Poisson	BirdID
fixed	factor(fac.yr)4	-0.6398071	0.3512178	-1.8216818	0.0685033	Poisson	BirdID
fixed	factor(fac.yr)5	0.2313065	0.3062278	0.7553413	0.4500442	Poisson	BirdID
fixed	factor(fac.yr)6	-0.1150726	0.3177754	-0.3621193	0.7172629	Poisson	BirdID
fixed	factor(fac.yr)7	-0.5001218	0.3433591	-1.4565562	0.1452389	Poisson	BirdID
fixed	factor(fac.yr)8	-0.1502219	0.3340781	-0.4496609	0.6529550	Poisson	BirdID
fixed	factor(fac.yr)9	0.0807131	0.3266635	0.2470832	0.8048438	Poisson	BirdID
fixed	factor(fac.yr)10	-0.0859491	0.3279696	-0.2620642	0.7932720	Poisson	BirdID
fixed	factor(fac.yr)11	-0.7457613	0.3888506	-1.9178610	0.0551286	Poisson	BirdID
fixed	factor(fac.yr)12	-0.6021785	0.3752076	-1.6049208	0.1085112	Poisson	BirdID
fixed	factor(fac.yr)13	-0.2087524	0.3525126	-0.5921842	0.5537272	Poisson	BirdID
fixed	factor(fac.yr)14	-0.5470603	0.3822850	-1.4310273	0.1524224	Poisson	BirdID
fixed	factor(fac.yr)15	-0.5946293	0.3896779	-1.5259507	0.1270221	Poisson	BirdID
fixed	factor(fac.yr)16	0.2351479	0.3442554	0.6830624	0.4945674	Poisson	BirdID
fixed	factor(fac.yr)17	0.2108799	0.4039939	0.5219878	0.6016788	Poisson	BirdID
fixed	factor(fac.yr)18	0.0603815	0.4252044	0.1420058	0.8870754	Poisson	BirdID
fixed	factor(fac.yr)19	-0.5211360	0.5320749	-0.9794410	0.3273621	Poisson	BirdID
fixed	factor(fac.yr)20	-0.7371205	0.5784195	-1.2743701	0.2025323	Poisson	BirdID
fixed	(Intercept)	0.2921709	0.2751370	1.0619108	0.2882762	Poisson	Terr
fixed	factor(fac.yr)2	-0.0820291	0.3368522	-0.2435167	0.8076051	Poisson	Terr
fixed	factor(fac.yr)3	-0.6078954	0.3537944	-1.7182167	0.0857571	Poisson	Terr
fixed	factor(fac.yr)4	-0.5757156	0.3643241	-1.5802293	0.1140544	Poisson	Terr
fixed	factor(fac.yr)5	0.2249410	0.3179136	0.7075539	0.4792223	Poisson	Terr
fixed	factor(fac.yr)6	-0.1838935	0.3290477	-0.5588660	0.5762532	Poisson	Terr
fixed	factor(fac.yr)7	-0.5261347	0.3536520	-1.4877188	0.1368251	Poisson	Terr
fixed	factor(fac.yr)8	-0.2037939	0.3457586	-0.5894110	0.5555856	Poisson	Terr
fixed	factor(fac.yr)9	0.0072588	0.3325530	0.0218276	0.9825855	Poisson	Terr
fixed	factor(fac.yr)10	-0.1264304	0.3382646	-0.3737619	0.7085815	Poisson	Terr
fixed	factor(fac.yr)11	-0.7301125	0.3996901	-1.8266966	0.0677454	Poisson	Terr
fixed	factor(fac.yr)12	-0.6013569	0.3843794	-1.5644880	0.1177030	Poisson	Terr
fixed	factor(fac.yr)13	-0.1723224	0.3563890	-0.4835232	0.6287243	Poisson	Terr
fixed	factor(fac.yr)14	-0.4804583	0.3935230	-1.2209156	0.2221180	Poisson	Terr
fixed	factor(fac.yr)15	-0.5252640	0.4017450	-1.3074562	0.1910578	Poisson	Terr
fixed	factor(fac.yr)16	0.2176580	0.3524028	0.6176399	0.5368127	Poisson	Terr
fixed	factor(fac.yr)17	0.1337183	0.4118386	0.3246863	0.7454185	Poisson	Terr
fixed	factor(fac.yr)18	0.0691585	0.4404746	0.1570090	0.8752377	Poisson	Terr
fixed	factor(fac.yr)19	-0.4338992	0.5533795	-0.7840897	0.4329875	Poisson	Terr
fixed	factor(fac.yr)20	-0.8177389	0.5967892	-1.3702308	0.1706149	Poisson	Terr
fixed	(Intercept)	0.2887023	0.2779688	1.0386140	0.2989843	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.0865901	0.3388475	-0.2555429	0.7983038	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.6133134	0.3564156	-1.7207816	0.0852905	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.5858668	0.3668545	-1.5970003	0.1102656	Poisson	BirdID+Terr

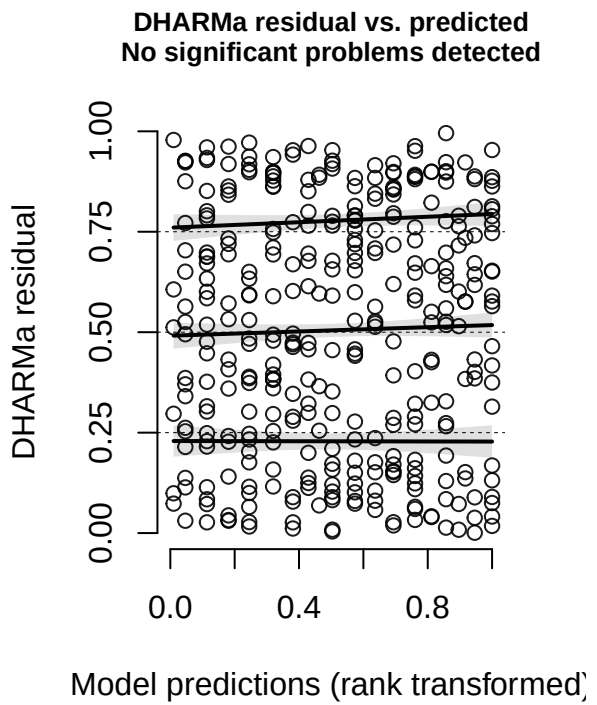
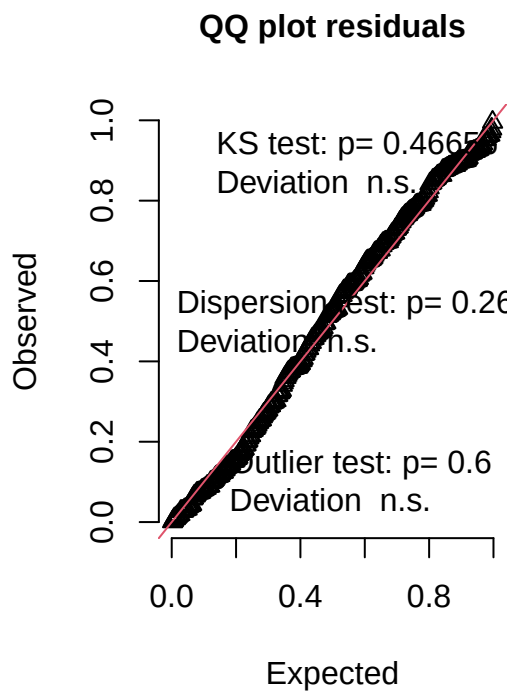
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.2203690	0.3206277	0.6873050	0.4918905	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.1859884	0.3320068	-0.5601944	0.5753469	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.5240350	0.3568728	-1.4684085	0.1419933	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.1953073	0.3492669	-0.5591919	0.5760307	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	0.0088574	0.3367637	0.0263016	0.9790167	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.1360221	0.3421822	-0.3975136	0.6909887	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.7421976	0.4034125	-1.8397981	0.0657979	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.6136473	0.3882703	-1.5804641	0.1140006	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.1826525	0.3604343	-0.5067567	0.6123255	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.4968161	0.3972154	-1.2507475	0.2110266	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.5461166	0.4055389	-1.3466441	0.1780949	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	0.2251867	0.3567511	0.6312151	0.5278999	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	0.1343467	0.4173733	0.3218862	0.7475389	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	0.0729020	0.4451022	0.1637871	0.8698987	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-0.4460667	0.5580830	-0.7992839	0.4241258	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.8290409	0.6030350	-1.3747808	0.1691994	Poisson	BirdID+Terr

Territory only model is performing well on AIC and BIC.

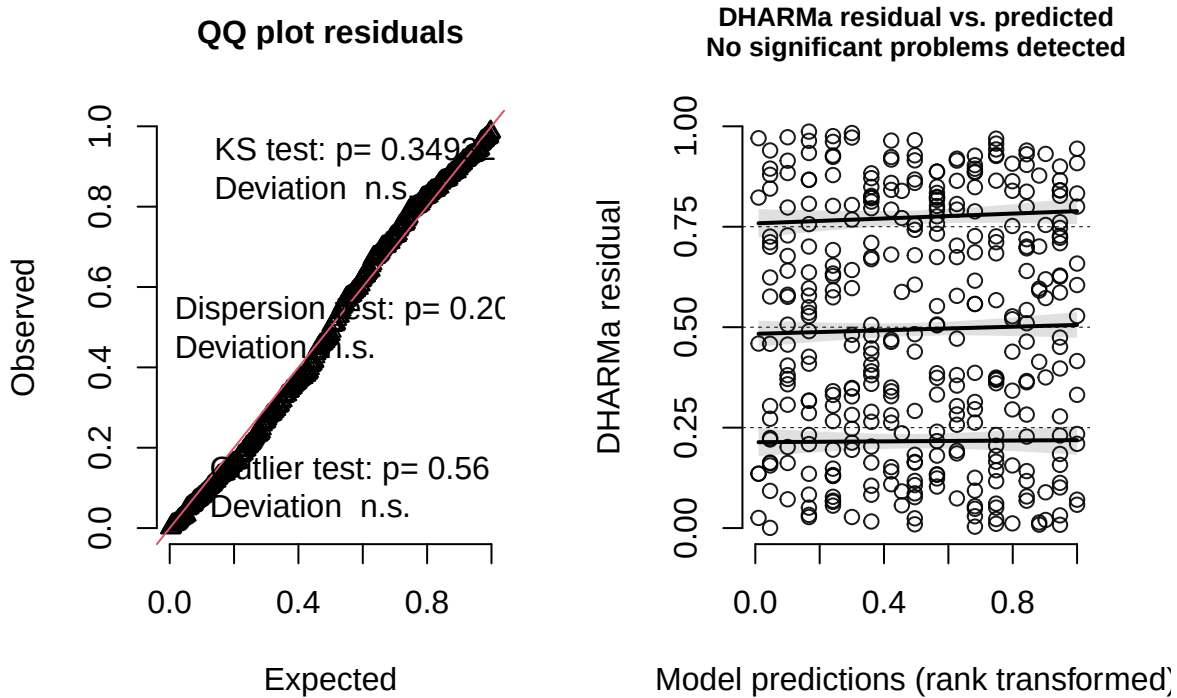
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
354	1	-216.8532	477.7064	562.8309	266.2747	332	0.731	Binomial	BirdID+Terr
354	1	-219.7844	481.5688	562.8240	337.3611	333	0.516	Binomial	Terr
354	1	-223.1027	488.2054	569.4607	354.9883	333	0.552	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.030895	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.119672	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.080187	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.160984	Binomial	BirdID+Terr

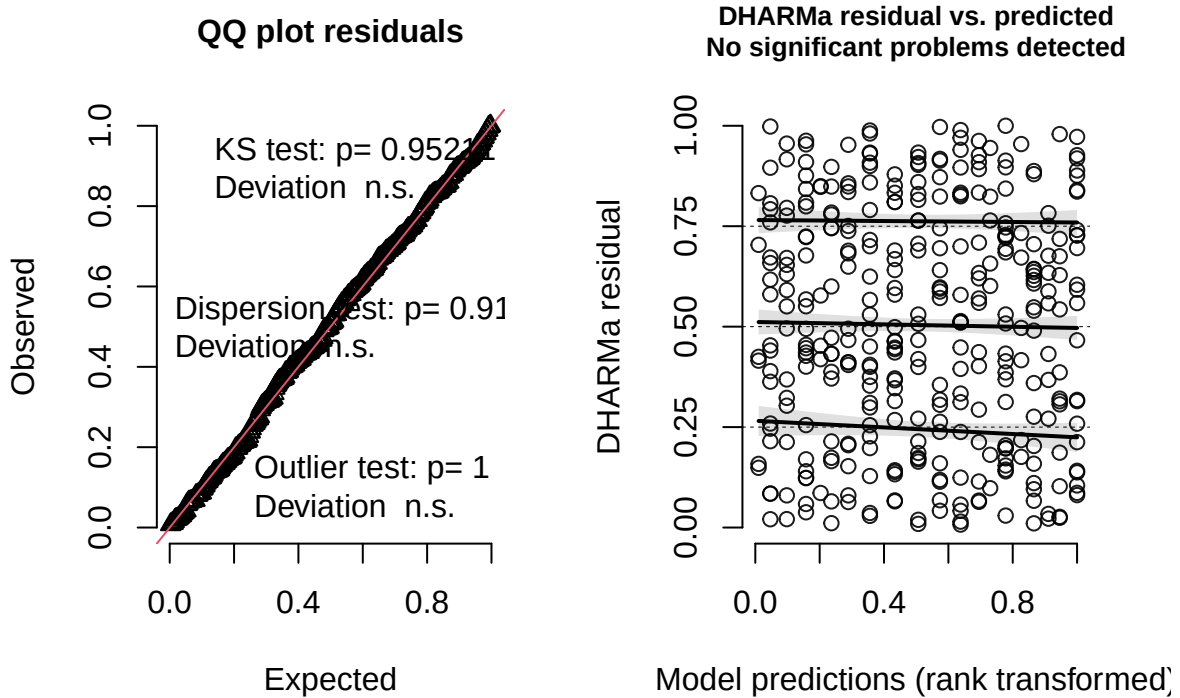
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	1.4714524	0.8600472	1.7108973	0.0871001	Binomial	BirdID
fixed	factor(fac.yr)2	-1.0948092	0.9577873	-1.1430609	0.2530133	Binomial	BirdID
fixed	factor(fac.yr)3	-1.1627680	0.9498574	-1.2241501	0.2208956	Binomial	BirdID
fixed	factor(fac.yr)4	-1.5679993	0.9562604	-1.6397199	0.1010634	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.2539648	1.0204165	0.2488834	0.8034509	Binomial	BirdID
fixed	factor(fac.yr)6	-0.4712458	0.9731822	-0.4842319	0.6282213	Binomial	BirdID
fixed	factor(fac.yr)7	-1.1351534	0.9629912	-1.1787785	0.2384864	Binomial	BirdID
fixed	factor(fac.yr)8	-1.0648748	0.9847182	-1.0814005	0.2795190	Binomial	BirdID
fixed	factor(fac.yr)9	-0.1853517	1.0182210	-0.1820348	0.8555554	Binomial	BirdID
fixed	factor(fac.yr)10	-0.9935899	0.9896997	-1.0039308	0.3154120	Binomial	BirdID
fixed	factor(fac.yr)11	-1.6606655	1.0075629	-1.6482003	0.0993116	Binomial	BirdID
fixed	factor(fac.yr)12	-1.3990072	1.0085760	-1.3871113	0.1654078	Binomial	BirdID
fixed	factor(fac.yr)13	-0.8021034	1.0383125	-0.7725068	0.4398143	Binomial	BirdID
fixed	factor(fac.yr)14	-1.5332322	1.0253075	-1.4953876	0.1348133	Binomial	BirdID
fixed	factor(fac.yr)15	-1.6285331	1.0305019	-1.5803301	0.1140313	Binomial	BirdID
fixed	factor(fac.yr)16	0.0826492	1.1359405	0.0727584	0.9419984	Binomial	BirdID
fixed	factor(fac.yr)17	-0.5100010	1.2812163	-0.3980601	0.6905859	Binomial	BirdID
fixed	factor(fac.yr)18	-0.3310749	1.2796847	-0.2587160	0.7958544	Binomial	BirdID
fixed	factor(fac.yr)19	-1.5652340	1.2678648	-1.2345433	0.2170005	Binomial	BirdID
fixed	factor(fac.yr)20	-2.2349684	1.3096230	-1.7065739	0.0879013	Binomial	BirdID
fixed	(Intercept)	1.8886826	0.8862366	2.1311269	0.0330787	Binomial	Terr
fixed	factor(fac.yr)2	-1.3996560	0.9966472	-1.4043645	0.1602103	Binomial	Terr
fixed	factor(fac.yr)3	-1.7406801	0.9886614	-1.7606432	0.0782988	Binomial	Terr
fixed	factor(fac.yr)4	-1.8633433	0.9939331	-1.8747169	0.0608317	Binomial	Terr
fixed	factor(fac.yr)5	-0.2439217	1.0506870	-0.2321545	0.8164180	Binomial	Terr
fixed	factor(fac.yr)6	-1.1262390	0.9974759	-1.1290889	0.2588603	Binomial	Terr
fixed	factor(fac.yr)7	-1.6941146	0.9848417	-1.7201898	0.0853980	Binomial	Terr
fixed	factor(fac.yr)8	-1.5477029	1.0077397	-1.5358161	0.1245835	Binomial	Terr
fixed	factor(fac.yr)9	-0.8088809	1.0414185	-0.7767107	0.4373295	Binomial	Terr
fixed	factor(fac.yr)10	-1.3529513	1.0101491	-1.3393581	0.1804541	Binomial	Terr
fixed	factor(fac.yr)11	-2.0460366	1.0321675	-1.9822719	0.0474488	Binomial	Terr
fixed	factor(fac.yr)12	-1.7943503	1.0343985	-1.7346799	0.0827975	Binomial	Terr
fixed	factor(fac.yr)13	-0.8471675	1.0433539	-0.8119657	0.4168113	Binomial	Terr
fixed	factor(fac.yr)14	-1.6619781	1.0429404	-1.5935505	0.1110368	Binomial	Terr
fixed	factor(fac.yr)15	-1.7219418	1.0501358	-1.6397325	0.1010608	Binomial	Terr
fixed	factor(fac.yr)16	-0.2320428	1.1299397	-0.2053585	0.8372920	Binomial	Terr
fixed	factor(fac.yr)17	-0.6211638	1.2751252	-0.4871394	0.6261595	Binomial	Terr
fixed	factor(fac.yr)18	-0.5241133	1.2866167	-0.4073578	0.6837452	Binomial	Terr
fixed	factor(fac.yr)19	-1.4675130	1.3245039	-1.1079718	0.2678740	Binomial	Terr
fixed	factor(fac.yr)20	-2.7062475	1.3583786	-1.9922631	0.0463422	Binomial	Terr
fixed	(Intercept)	1.8623002	0.9563597	1.9472800	0.0515012	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-1.3269998	1.0615225	-1.2500911	0.2112663	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-1.7045263	1.0606055	-1.6071257	0.1080268	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-1.9235542	1.0668582	-1.8030083	0.0713869	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	0.1256247	1.1318554	0.1109901	0.9116242	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-0.8737218	1.0801206	-0.8089113	0.4185662	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-1.4578122	1.0705958	-1.3616831	0.1732979	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-1.3139204	1.0970853	-1.1976465	0.2310547	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-0.3563987	1.1409010	-0.3123835	0.7547491	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-1.2658737	1.1046952	-1.1459032	0.2518352	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-2.1107134	1.1306028	-1.8668920	0.0619167	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-1.7789941	1.1337311	-1.5691500	0.1166130	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-0.7634450	1.1475948	-0.6652565	0.5058864	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-1.6904313	1.1483394	-1.4720660	0.1410031	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-1.7774700	1.1546988	-1.5393365	0.1237222	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	0.1581808	1.2497376	0.1265712	0.8992798	Binomial	BirdID+Terr

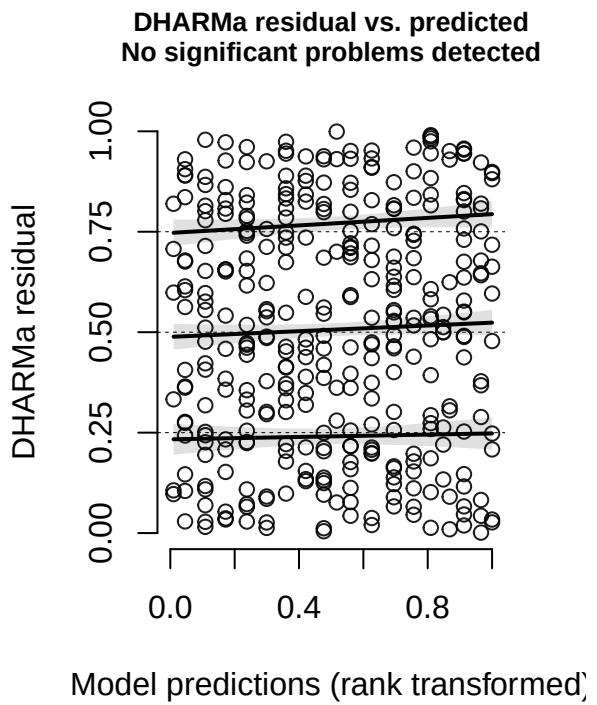
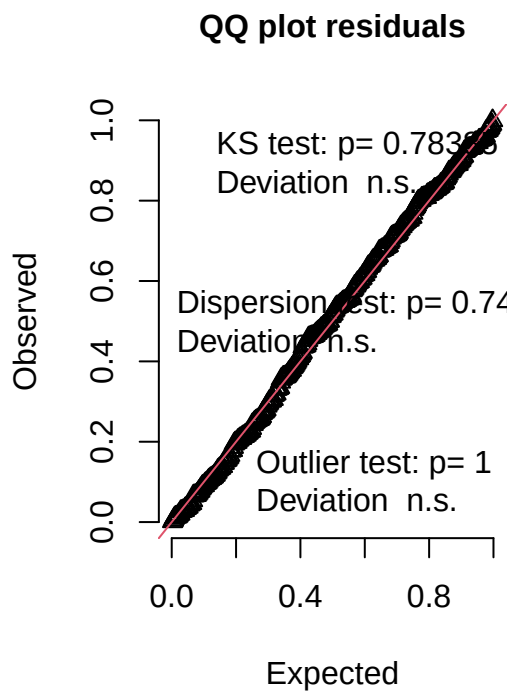
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-0.5048493	1.4390974	-0.3508097	0.7257311	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-0.4270975	1.4036957	-0.3042664	0.7609249	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-1.5987026	1.4716852	-1.0863075	0.2773430	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-2.7349947	1.5192098	-1.8002745	0.0718173	Binomial	BirdID+Terr

The territory only model is performing the best on BIC, but not by a significant margin.

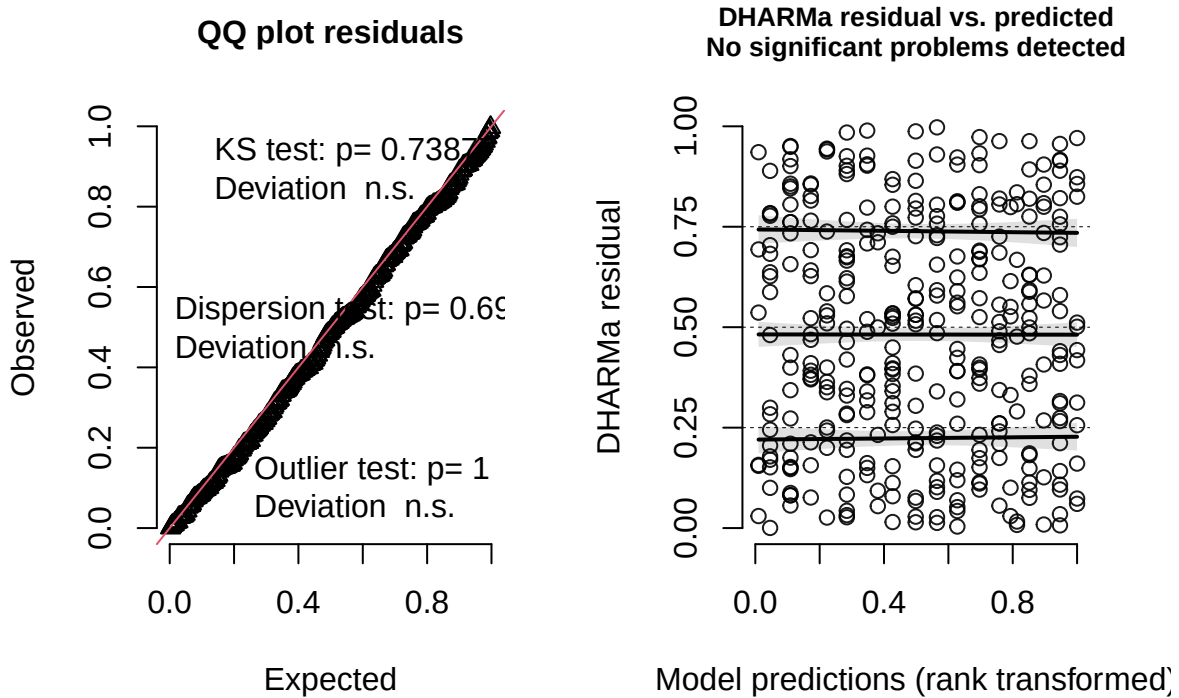
DHARMA residual



DHARMa residual



DHARMA residual



Diagnostics look good.

Probabilities used to generate simulation 9

Simulation 9

Probability of mate fidelity: 0.102

Probability of pair dispersal: 0.547

Probability of single male dispersal: 0.172

Probability of single female dispersal: 0.31

Simulation 10

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
357	1	-452.6809	949.3619	1034.672	302.4077	335	0.8968757	0.8510694	Poisson	BirdID+Terr
357	1	-454.8132	951.6264	1033.059	341.2851	336	0.8968704	0.8602692	Poisson	BirdID
357	1	-456.8379	955.6758	1037.108	354.3390	336	0.8950821	0.8642772	Poisson	Terr

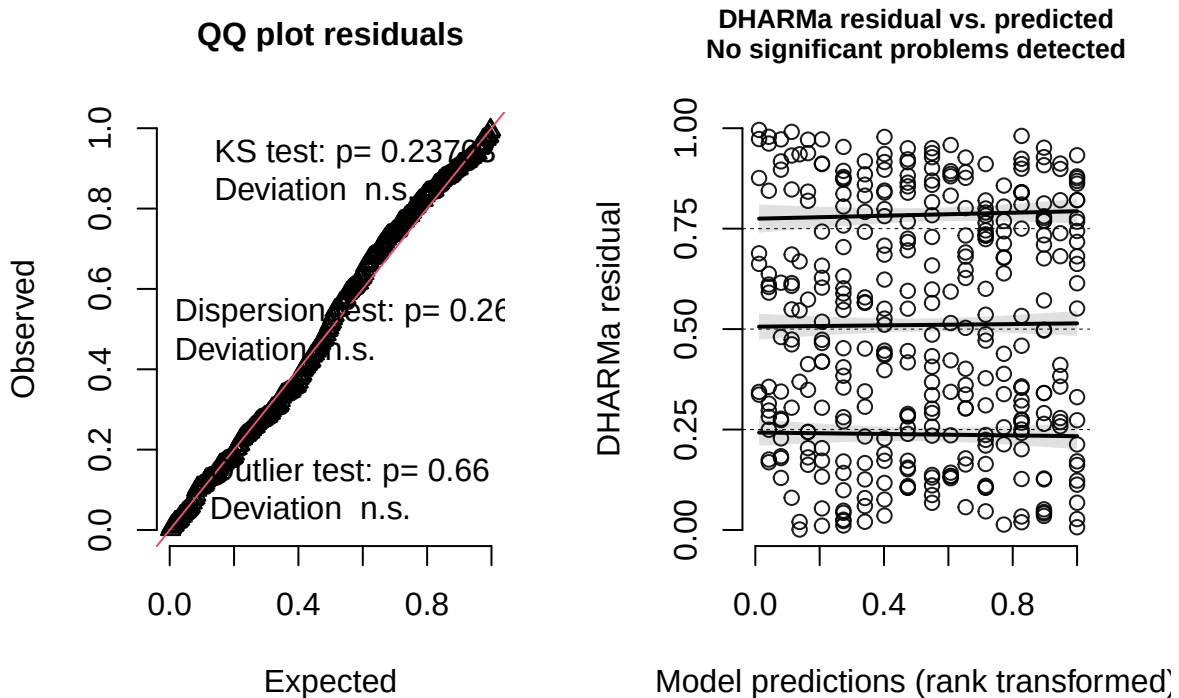
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.4916155	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.4453814	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.4207051	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.3522915	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.1577154	0.2962126	0.5324399	0.5944214	Poisson	BirdID
fixed	factor(fac.yr)2	-0.1377398	0.3875393	-0.3554215	0.7222738	Poisson	BirdID
fixed	factor(fac.yr)3	-0.3875912	0.3987144	-0.9721025	0.3309996	Poisson	BirdID
fixed	factor(fac.yr)4	-0.0406015	0.3567913	-0.1137962	0.9093994	Poisson	BirdID
fixed	factor(fac.yr)5	0.2827889	0.3368192	0.8395867	0.4011401	Poisson	BirdID
fixed	factor(fac.yr)6	-0.2718544	0.3632967	-0.7482985	0.4542802	Poisson	BirdID
fixed	factor(fac.yr)7	-0.5893856	0.3760468	-1.5673198	0.1170399	Poisson	BirdID
fixed	factor(fac.yr)8	-0.1732334	0.3603540	-0.4807312	0.6307075	Poisson	BirdID
fixed	factor(fac.yr)9	-0.0622189	0.3616521	-0.1720407	0.8634055	Poisson	BirdID
fixed	factor(fac.yr)10	-0.1505400	0.3624516	-0.4153381	0.6778944	Poisson	BirdID
fixed	factor(fac.yr)11	-0.3762167	0.3742252	-1.0053217	0.3147420	Poisson	BirdID
fixed	factor(fac.yr)12	-0.1583178	0.3825469	-0.4138520	0.6789825	Poisson	BirdID
fixed	factor(fac.yr)13	-0.1696691	0.4044431	-0.4195129	0.6748413	Poisson	BirdID
fixed	factor(fac.yr)14	-0.6419675	0.4301689	-1.4923615	0.1356044	Poisson	BirdID
fixed	factor(fac.yr)15	-0.7895584	0.4993690	-1.5811123	0.1138524	Poisson	BirdID
fixed	factor(fac.yr)16	-1.3296158	0.5554453	-2.3937834	0.0166756	Poisson	BirdID
fixed	factor(fac.yr)17	-0.6558535	0.4842015	-1.3545052	0.1755753	Poisson	BirdID
fixed	factor(fac.yr)18	-0.8069189	0.5034248	-1.6028591	0.1089658	Poisson	BirdID
fixed	factor(fac.yr)19	-17.3034121	2232.2893308	-0.0077514	0.9938153	Poisson	BirdID
fixed	factor(fac.yr)20	-0.7615996	0.6974337	-1.0920028	0.2748319	Poisson	BirdID
fixed	(Intercept)	0.1872226	0.2875105	0.6511852	0.5149270	Poisson	Terr
fixed	factor(fac.yr)2	-0.1190964	0.3839225	-0.3102094	0.7564017	Poisson	Terr
fixed	factor(fac.yr)3	-0.4377937	0.3969135	-1.1029952	0.2700293	Poisson	Terr
fixed	factor(fac.yr)4	-0.0947933	0.3526402	-0.2688104	0.7880756	Poisson	Terr
fixed	factor(fac.yr)5	0.3200377	0.3258903	0.9820411	0.3260796	Poisson	Terr
fixed	factor(fac.yr)6	-0.2629740	0.3569638	-0.7366964	0.4613070	Poisson	Terr
fixed	factor(fac.yr)7	-0.5524336	0.3672633	-1.5041896	0.1325325	Poisson	Terr
fixed	factor(fac.yr)8	-0.0838738	0.3510176	-0.2389447	0.8111485	Poisson	Terr
fixed	factor(fac.yr)9	-0.0512241	0.3504793	-0.1461545	0.8837994	Poisson	Terr
fixed	factor(fac.yr)10	-0.1579012	0.3501183	-0.4509938	0.6519940	Poisson	Terr
fixed	factor(fac.yr)11	-0.3227056	0.3557687	-0.9070658	0.3643720	Poisson	Terr
fixed	factor(fac.yr)12	-0.0115704	0.3715666	-0.0311395	0.9751583	Poisson	Terr
fixed	factor(fac.yr)13	0.0155284	0.3940644	0.0394057	0.9685669	Poisson	Terr
fixed	factor(fac.yr)14	-0.7050576	0.4169376	-1.6910387	0.0908294	Poisson	Terr
fixed	factor(fac.yr)15	-0.7627148	0.4887057	-1.5606832	0.1185985	Poisson	Terr
fixed	factor(fac.yr)16	-1.2218590	0.5449401	-2.2421894	0.0249491	Poisson	Terr
fixed	factor(fac.yr)17	-0.6424159	0.4735000	-1.3567392	0.1748641	Poisson	Terr
fixed	factor(fac.yr)18	-0.7553176	0.4892788	-1.5437367	0.1226521	Poisson	Terr
fixed	factor(fac.yr)19	-17.0900362	1865.5238769	-0.0091610	0.9926907	Poisson	Terr
fixed	factor(fac.yr)20	-1.0853709	0.6606577	-1.6428641	0.1004110	Poisson	Terr
fixed	(Intercept)	0.1192790	0.3052051	0.3908160	0.6959333	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.1019527	0.3936625	-0.2589850	0.7956468	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.3688655	0.4108415	-0.8978291	0.3692766	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.0143482	0.3687964	-0.0389056	0.9689657	Poisson	BirdID+Terr

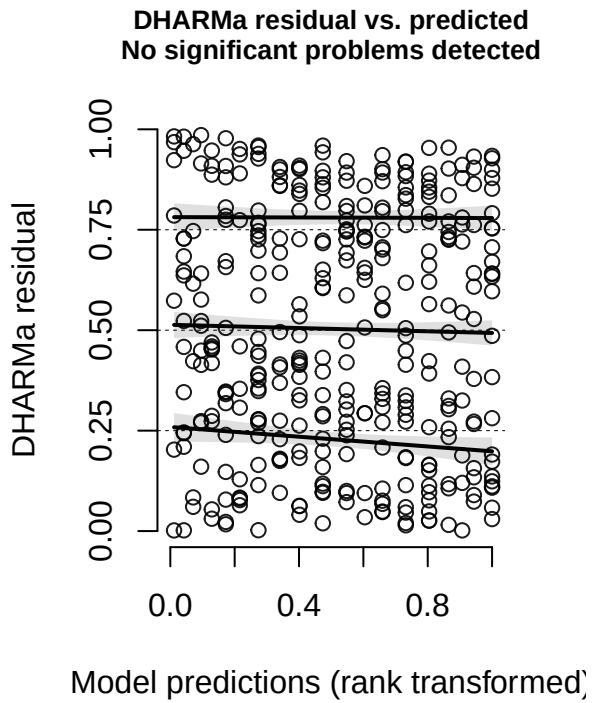
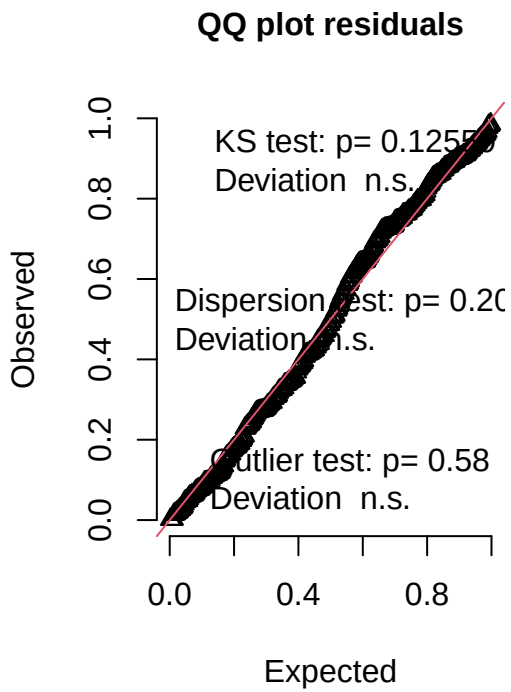
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.3177669	0.3465878	0.9168438	0.3592245	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.2267648	0.3752186	-0.6043539	0.5456084	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.5586923	0.3856147	-1.4488356	0.1473835	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.0972806	0.3715370	-0.2618330	0.7934502	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.0418243	0.3716656	-0.1125320	0.9104016	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.1484459	0.3721303	-0.3989084	0.6899607	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.3851617	0.3809504	-1.0110547	0.3119902	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.0928276	0.3950134	-0.2349986	0.8142098	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.0905515	0.4167433	-0.2172837	0.8279872	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.6758332	0.4400704	-1.5357387	0.1246025	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.8189723	0.5101143	-1.6054682	0.1083908	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-1.2851838	0.5649591	-2.2748262	0.0229164	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.6732033	0.4972823	-1.3537650	0.1758114	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.7973949	0.5125861	-1.5556313	0.1197958	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-17.1348094	2093.7038791	-0.0081840	0.9934702	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.8697931	0.6994834	-1.2434793	0.2136913	Poisson	BirdID+Terr

The only metric that the territory only model is performing well in is marginal R^2 , and it is certainly not by a significant margin.

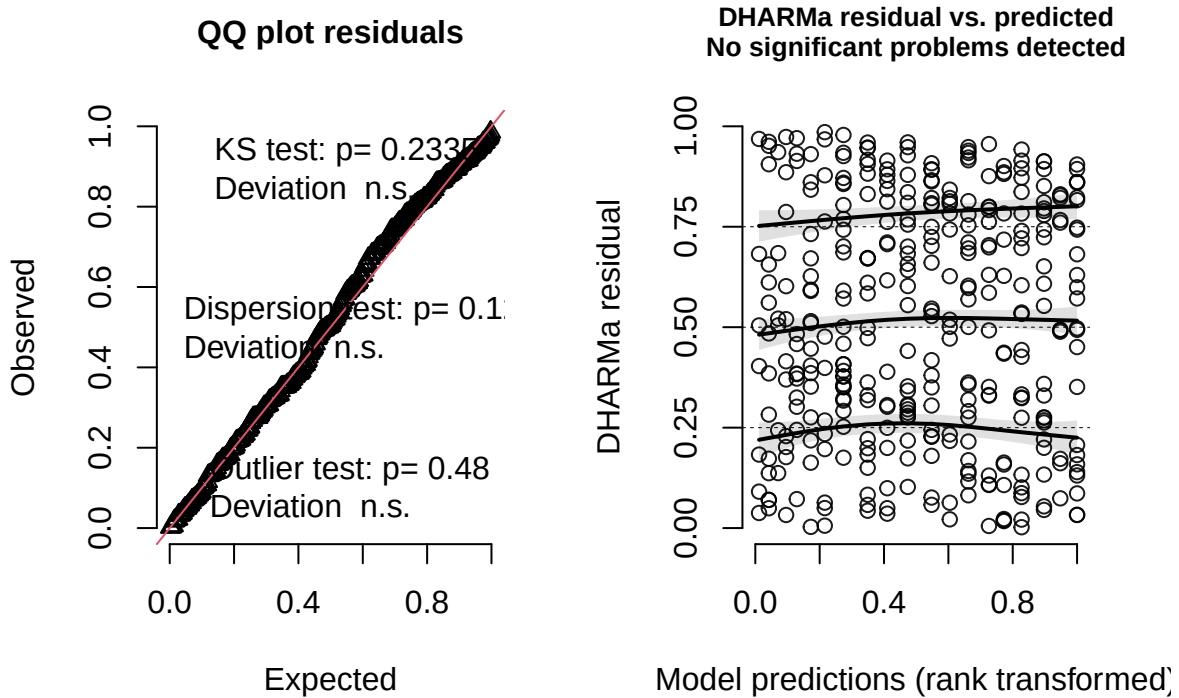
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
357	1	-206.0346	456.0692	541.3794	178.0812	335	0.853	Binomial	BirdID+Terr
357	1	-214.2249	470.4498	551.8822	298.0765	336	0.683	Binomial	Terr
357	1	-217.3777	476.7554	558.1879	324.1065	336	0.688	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.232150	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.429676	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.679733	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.859451	Binomial	BirdID+Terr

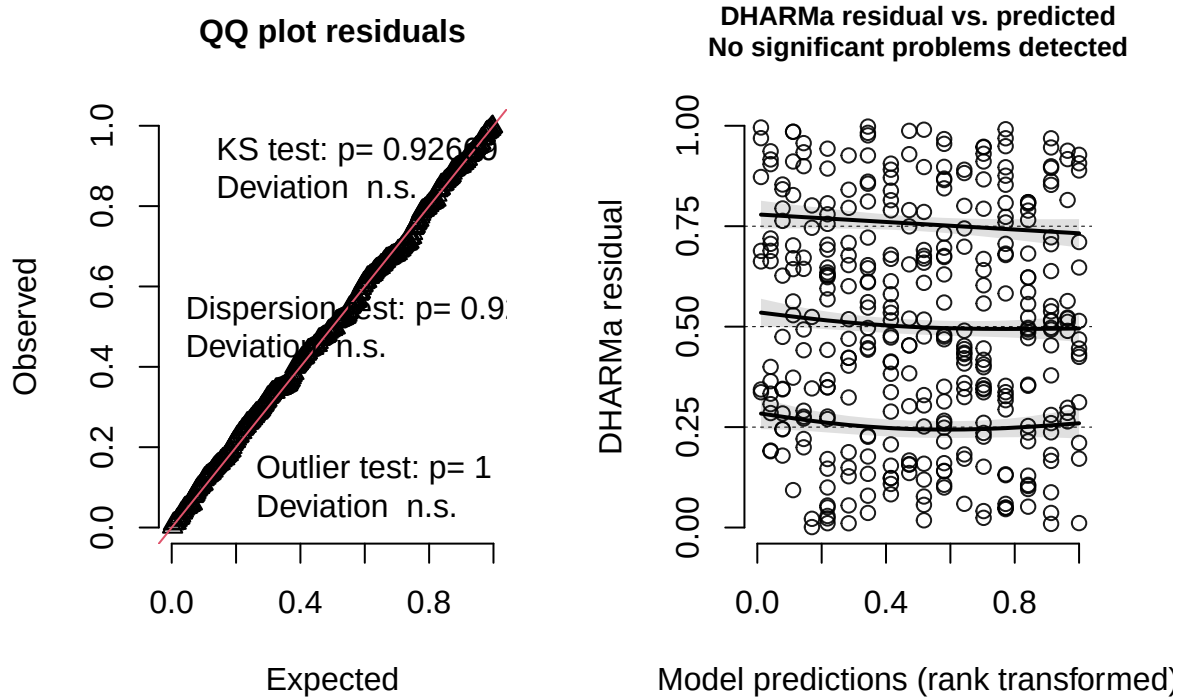
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	1.0596778	0.8018137	1.3216010	0.1863011	Binomial	BirdID
fixed	factor(fac.yr)2	0.0699552	1.0372360	0.0674439	0.9462284	Binomial	BirdID
fixed	factor(fac.yr)3	-0.7173287	0.9559486	-0.7503842	0.4530233	Binomial	BirdID
fixed	factor(fac.yr)4	-0.4558097	0.9280945	-0.4911242	0.6233386	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.1825627	0.9376303	-0.1947065	0.8456227	Binomial	BirdID
fixed	factor(fac.yr)6	-0.6849290	0.9227245	-0.7422898	0.4579118	Binomial	BirdID
fixed	factor(fac.yr)7	-1.1275781	0.9116621	-1.2368378	0.2161473	Binomial	BirdID
fixed	factor(fac.yr)8	-0.7324981	0.9290046	-0.7884762	0.4304182	Binomial	BirdID
fixed	factor(fac.yr)9	-0.5157596	0.9433986	-0.5467038	0.5845823	Binomial	BirdID
fixed	factor(fac.yr)10	-0.3229859	0.9432832	-0.3424061	0.7320453	Binomial	BirdID
fixed	factor(fac.yr)11	-0.8974638	0.9431076	-0.9516028	0.3412985	Binomial	BirdID
fixed	factor(fac.yr)12	-0.6262508	0.9916186	-0.6315440	0.5276849	Binomial	BirdID
fixed	factor(fac.yr)13	-0.7233072	1.0384384	-0.6965335	0.4860948	Binomial	BirdID
fixed	factor(fac.yr)14	-1.1191733	0.9858872	-1.1351941	0.2562940	Binomial	BirdID
fixed	factor(fac.yr)15	-1.9296350	1.1100460	-1.7383379	0.0821513	Binomial	BirdID
fixed	factor(fac.yr)16	-2.7936783	1.1257553	-2.4816034	0.0130793	Binomial	BirdID
fixed	factor(fac.yr)17	-1.8721843	1.1057463	-1.6931409	0.0904286	Binomial	BirdID
fixed	factor(fac.yr)18	-2.4915832	1.1503387	-2.1659561	0.0303145	Binomial	BirdID
fixed	factor(fac.yr)19	-9.5656044	30.5928551	-0.3126745	0.7545280	Binomial	BirdID
fixed	factor(fac.yr)20	-1.4553905	1.3008911	-1.1187643	0.2632407	Binomial	BirdID
fixed	(Intercept)	1.2694599	0.8465667	1.4995391	0.1337338	Binomial	Terr
fixed	factor(fac.yr)2	-0.1598129	1.1007361	-0.1451873	0.8845630	Binomial	Terr
fixed	factor(fac.yr)3	-0.7110487	1.0099262	-0.7040601	0.4813954	Binomial	Terr
fixed	factor(fac.yr)4	-0.7168754	0.9720322	-0.7375017	0.4608173	Binomial	Terr
fixed	factor(fac.yr)5	-0.0467810	0.9728767	-0.0480852	0.9616483	Binomial	Terr
fixed	factor(fac.yr)6	-0.7551402	0.9697449	-0.7786998	0.4361566	Binomial	Terr
fixed	factor(fac.yr)7	-1.3536324	0.9526610	-1.4208962	0.1553469	Binomial	Terr
fixed	factor(fac.yr)8	-0.4301003	0.9685302	-0.4440753	0.6569882	Binomial	Terr
fixed	factor(fac.yr)9	-0.6204085	0.9739533	-0.6370003	0.5241246	Binomial	Terr
fixed	factor(fac.yr)10	-0.2135807	0.9869088	-0.2164138	0.8286652	Binomial	Terr
fixed	factor(fac.yr)11	-0.8090734	0.9807697	-0.8249372	0.4094073	Binomial	Terr
fixed	factor(fac.yr)12	-0.1367948	1.0205035	-0.1340464	0.8933659	Binomial	Terr
fixed	factor(fac.yr)13	-0.3932528	1.0745439	-0.3659718	0.7143861	Binomial	Terr
fixed	factor(fac.yr)14	-1.6744252	1.0136949	-1.6518039	0.0985745	Binomial	Terr
fixed	factor(fac.yr)15	-1.9059042	1.1284751	-1.6889200	0.0912348	Binomial	Terr
fixed	factor(fac.yr)16	-2.6527950	1.1530834	-2.3006099	0.0214137	Binomial	Terr
fixed	factor(fac.yr)17	-1.9665944	1.1311290	-1.7386119	0.0821030	Binomial	Terr
fixed	factor(fac.yr)18	-2.2534320	1.1468519	-1.9648849	0.0494276	Binomial	Terr
fixed	factor(fac.yr)19	-10.5716350	50.1855005	-0.2106512	0.8331595	Binomial	Terr
fixed	factor(fac.yr)20	-2.6223455	1.3342663	-1.9653840	0.0493698	Binomial	Terr
fixed	(Intercept)	1.4475137	1.1461444	1.2629417	0.2066101	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	0.1523179	1.4495385	0.1050803	0.9163121	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-0.5842286	1.3113093	-0.4455307	0.6559362	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-0.3791021	1.2828763	-0.2955095	0.7676047	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	0.2342329	1.2855935	0.1821982	0.8554272	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-0.5479475	1.2867225	-0.4258475	0.6702190	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-1.6053222	1.2715757	-1.2624669	0.2067808	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-0.5090019	1.2903805	-0.3944588	0.6932423	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-0.3514651	1.3114283	-0.2680018	0.7886979	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	0.0728907	1.3312696	0.0547528	0.9563354	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-1.4253484	1.3261855	-1.0747730	0.2824764	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-0.3535753	1.3805447	-0.2561129	0.7978637	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-0.6840070	1.4506698	-0.4715111	0.6372758	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-1.9715754	1.3791756	-1.4295318	0.1528514	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-3.1540888	1.5217878	-2.0726206	0.0382076	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-3.9757396	1.5589087	-2.5503350	0.0107619	Binomial	BirdID+Terr

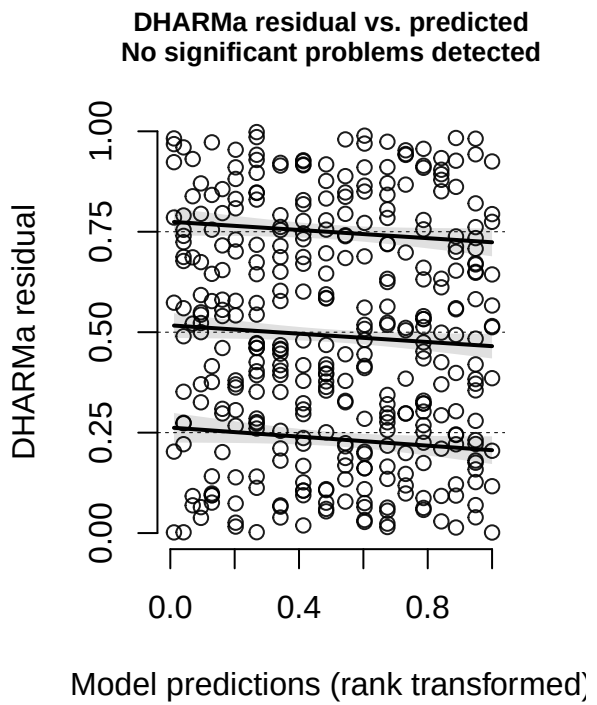
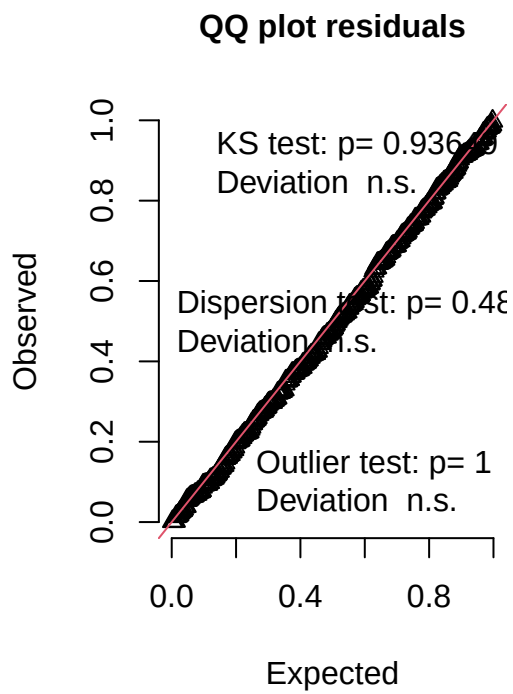
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-2.6902625	1.6149380	-1.6658612	0.0957411	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-3.2043452	1.5888506	-2.0167693	0.0437196	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-14.2306465	330.2461943	-0.0430910	0.9656290	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-2.9208000	1.7680780	-1.6519633	0.0985420	Binomial	BirdID+Terr

The model fitting both random effects is the model best optimizing selection criteria.

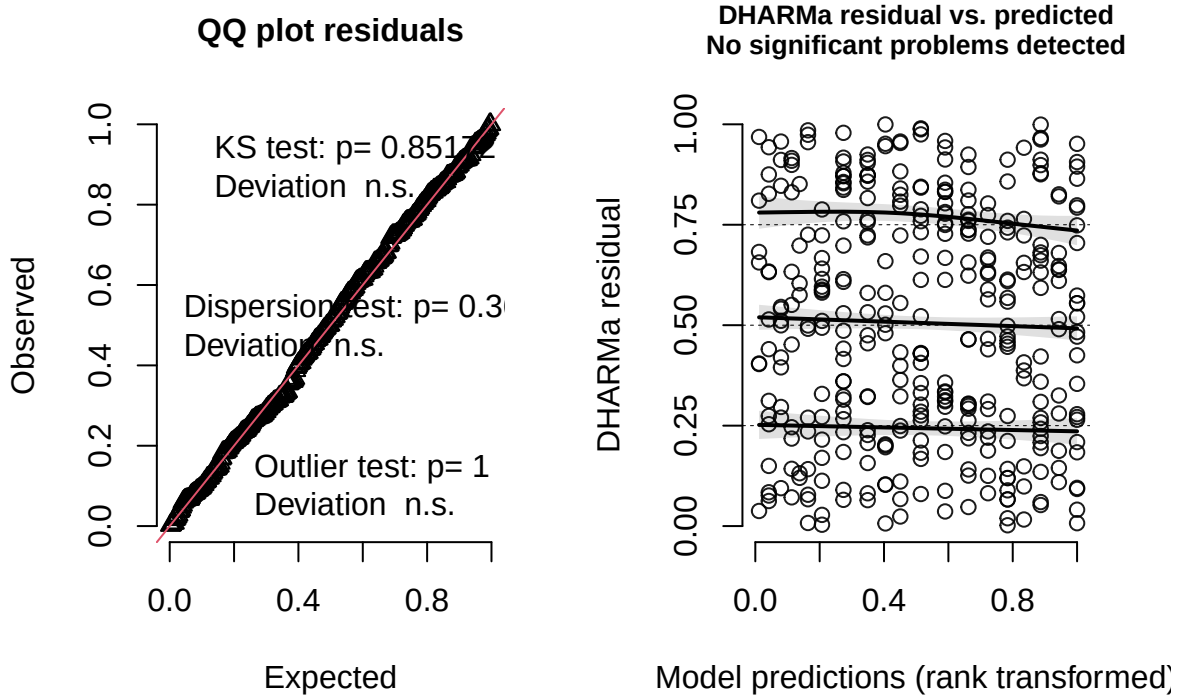
DHARMA residual



DHARMa residual



DHARMa residual



Probabilites used to generate simulation 10

Simulation 10
 Probability of mate fidelity: 0.616
 Probability of pair dispersal: 0.14
 Probability of single male dispersal: 0.489
 Probability of single female dispersal: 0.569

Simulation 11

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	condtional_R2	marginal_R2	REfamily	RE
411	1	-542.5400	1127.080	1211.470	455.9347	390	0.2573280	0.1617170	Poisson	Terr
411	1	-543.0756	1128.151	1212.542	448.9637	390	0.2691386	0.1580232	Poisson	BirdID
411	1	-542.2258	1128.452	1216.861	439.1124	389	0.2733433	0.1572217	Poisson	BirdID+Terr

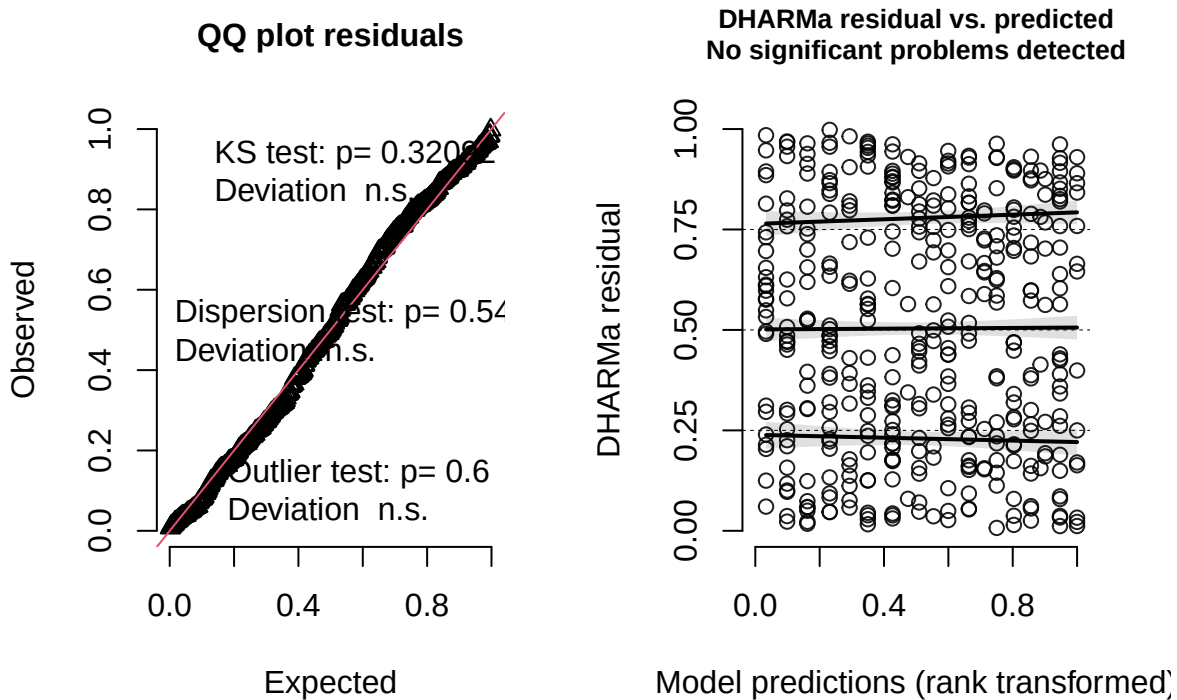
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.3179948	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.2935023	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.2153227	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.2454054	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.5078654	0.2226167	2.2813451	0.0225280	Poisson	BirdID
fixed	factor(fac.yr)2	-0.1944092	0.2901993	-0.6699163	0.5029112	Poisson	BirdID
fixed	factor(fac.yr)3	-1.3369451	0.3686956	-3.6261488	0.0002877	Poisson	BirdID
fixed	factor(fac.yr)4	-0.4701293	0.3000509	-1.5668317	0.1171540	Poisson	BirdID
fixed	factor(fac.yr)5	-0.0504343	0.2689325	-0.1875351	0.8512411	Poisson	BirdID
fixed	factor(fac.yr)6	-0.6184031	0.2892226	-2.1381562	0.0325041	Poisson	BirdID
fixed	factor(fac.yr)7	-1.2714671	0.3619074	-3.5132389	0.0004427	Poisson	BirdID
fixed	factor(fac.yr)8	-0.8790778	0.3102391	-2.8335495	0.0046034	Poisson	BirdID
fixed	factor(fac.yr)9	-0.4195170	0.2900619	-1.4463018	0.1480926	Poisson	BirdID
fixed	factor(fac.yr)10	-0.6193337	0.3002533	-2.0627042	0.0391407	Poisson	BirdID
fixed	factor(fac.yr)11	-0.9437150	0.3423147	-2.7568641	0.0058359	Poisson	BirdID
fixed	factor(fac.yr)12	-0.5028628	0.3098026	-1.6231714	0.1045528	Poisson	BirdID
fixed	factor(fac.yr)13	-0.2287814	0.2939792	-0.7782232	0.4364375	Poisson	BirdID
fixed	factor(fac.yr)14	-0.2805690	0.3079097	-0.9112056	0.3621870	Poisson	BirdID
fixed	factor(fac.yr)15	-0.7370389	0.3497506	-2.1073272	0.0350892	Poisson	BirdID
fixed	factor(fac.yr)16	-0.4821929	0.3588055	-1.3438837	0.1789860	Poisson	BirdID
fixed	factor(fac.yr)17	-0.3403213	0.3609974	-0.9427250	0.3458216	Poisson	BirdID
fixed	factor(fac.yr)18	-0.1388446	0.3807720	-0.3646396	0.7153804	Poisson	BirdID
fixed	factor(fac.yr)19	-0.1694238	0.4895253	-0.3460982	0.7292689	Poisson	BirdID
fixed	factor(fac.yr)20	-0.5494233	0.5214203	-1.0537051	0.2920180	Poisson	BirdID
fixed	(Intercept)	0.5287493	0.2182989	2.4221343	0.0154296	Poisson	Terr
fixed	factor(fac.yr)2	-0.2486174	0.2915815	-0.8526515	0.3938526	Poisson	Terr
fixed	factor(fac.yr)3	-1.3329806	0.3664317	-3.6377332	0.0002750	Poisson	Terr
fixed	factor(fac.yr)4	-0.4382284	0.2961144	-1.4799292	0.1388921	Poisson	Terr
fixed	factor(fac.yr)5	0.0003920	0.2637484	0.0014862	0.9988142	Poisson	Terr
fixed	factor(fac.yr)6	-0.6422681	0.2838943	-2.2623490	0.0236758	Poisson	Terr
fixed	factor(fac.yr)7	-1.2955582	0.3574748	-3.6241949	0.0002899	Poisson	Terr
fixed	factor(fac.yr)8	-0.8740660	0.3058288	-2.8580241	0.0042629	Poisson	Terr
fixed	factor(fac.yr)9	-0.4169199	0.2829350	-1.4735533	0.1406019	Poisson	Terr
fixed	factor(fac.yr)10	-0.6090883	0.2951208	-2.0638613	0.0390309	Poisson	Terr
fixed	factor(fac.yr)11	-0.9619381	0.3344757	-2.8759585	0.0040280	Poisson	Terr
fixed	factor(fac.yr)12	-0.5062551	0.3033429	-1.6689201	0.0951332	Poisson	Terr
fixed	factor(fac.yr)13	-0.2842746	0.2846221	-0.9987793	0.3179016	Poisson	Terr
fixed	factor(fac.yr)14	-0.3530592	0.3012305	-1.1720565	0.2411744	Poisson	Terr
fixed	factor(fac.yr)15	-0.7884741	0.3432479	-2.2970984	0.0216132	Poisson	Terr
fixed	factor(fac.yr)16	-0.4957036	0.3516974	-1.4094605	0.1586990	Poisson	Terr
fixed	factor(fac.yr)17	-0.2815309	0.3549589	-0.7931365	0.4276983	Poisson	Terr
fixed	factor(fac.yr)18	-0.1265783	0.3762660	-0.3364064	0.7365644	Poisson	Terr
fixed	factor(fac.yr)19	-0.3670201	0.4822945	-0.7609876	0.4466645	Poisson	Terr
fixed	factor(fac.yr)20	-0.6466463	0.5134395	-1.2594401	0.2078714	Poisson	Terr
fixed	(Intercept)	0.5151799	0.2233949	2.3061401	0.0211028	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.2298219	0.2943214	-0.7808533	0.4348888	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-1.3336320	0.3705169	-3.5993828	0.0003190	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.4536672	0.3014734	-1.5048334	0.1323669	Poisson	BirdID+Terr

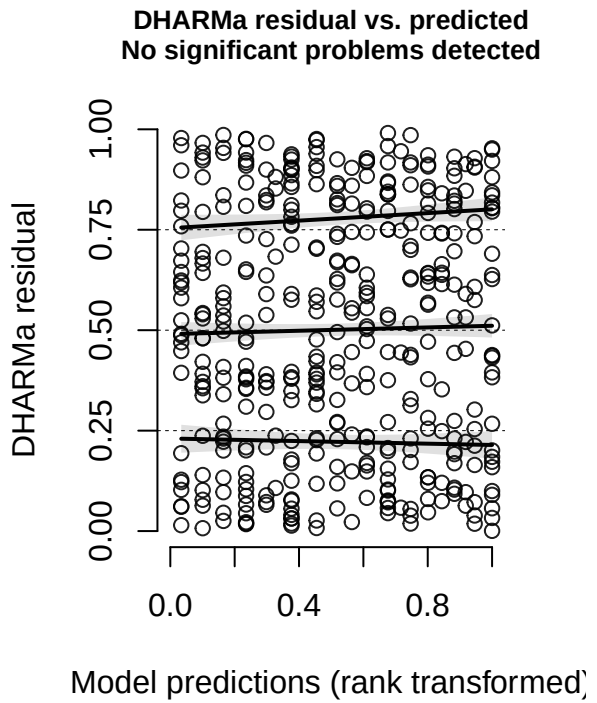
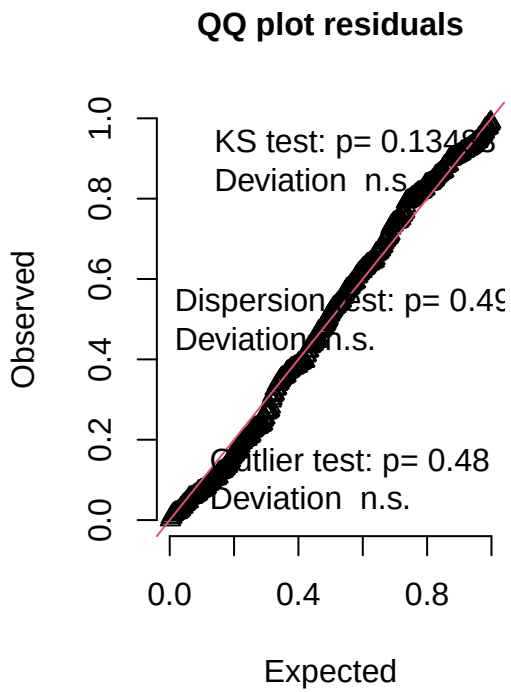
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.0199478	0.2696050	-0.0739889	0.9410193	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.6388759	0.2895081	-2.2067630	0.0273306	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-1.2862322	0.3624922	-3.5483030	0.0003877	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.8750359	0.3110520	-2.8131496	0.0049059	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.4190892	0.2892561	-1.4488520	0.1473789	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.6188920	0.3009543	-2.0564317	0.0397409	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.9596486	0.3410024	-2.8141990	0.0048899	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.5044017	0.3096717	-1.6288272	0.1033496	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.2689522	0.2919907	-0.9210983	0.3569991	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.3255387	0.3077653	-1.0577500	0.2901695	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.7797306	0.3495248	-2.2308304	0.0256924	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.4992400	0.3586968	-1.3918162	0.1639781	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.3061451	0.3620613	-0.8455616	0.3977973	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.1310640	0.3835583	-0.3417056	0.7325725	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-0.3143524	0.4932661	-0.6372877	0.5239375	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.6086717	0.5222866	-1.1653979	0.2438580	Poisson	BirdID+Terr

The only selection criteria that the territory only model didn't do best in was conditional R^2 , but it still wasn't far off from the best conditional R^2 .

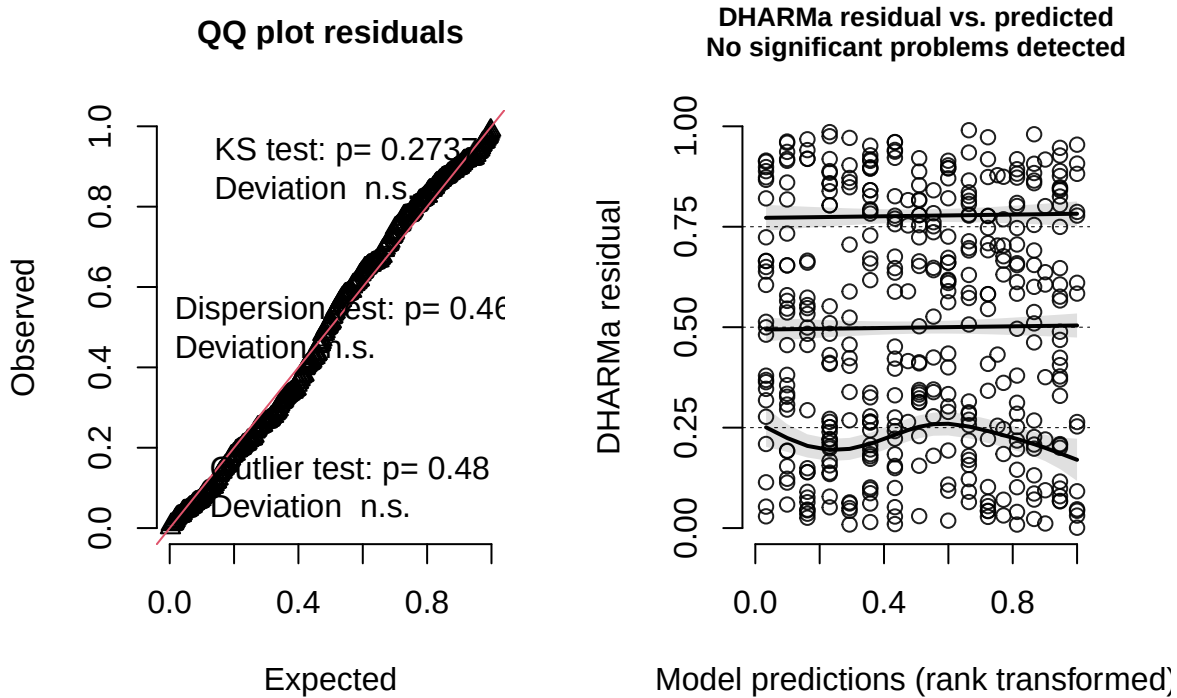
DHARMA residual



DHARMa residual



DHARMa residual



Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
411	1	-259.8949	561.7899	646.1803	423.8499	390	0.487	Binomial	BirdID
411	1	-259.8802	563.7603	652.1694	419.6743	389	0.513	Binomial	BirdID+Terr
411	1	-263.5613	569.1225	653.5130	466.0152	390	0.429	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.9446483	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	0.6750049	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	0.9379687	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.2050393	Binomial	BirdID+Terr

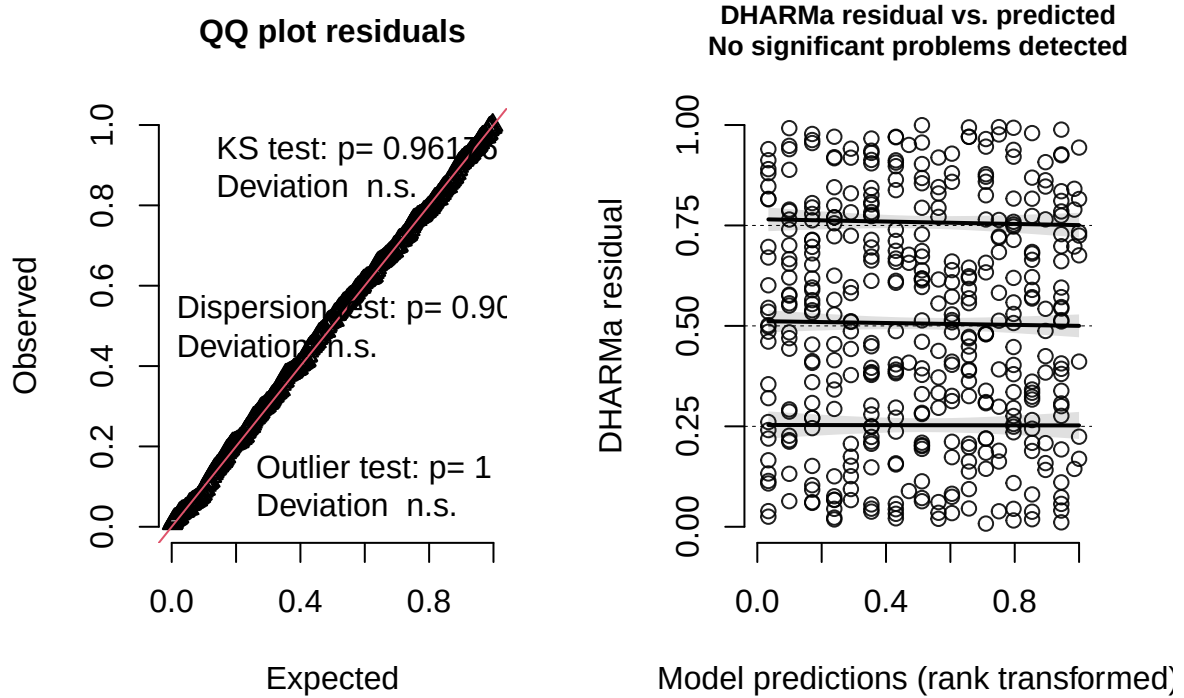
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.7042494	0.6286414	1.1202720	0.2625979	Binomial	BirdID
fixed	factor(fac.yr)2	0.0503921	0.7854681	0.0641555	0.9488464	Binomial	BirdID
fixed	factor(fac.yr)3	-1.3494651	0.7580937	-1.7800769	0.0750634	Binomial	BirdID
fixed	factor(fac.yr)4	0.2044951	0.7968317	0.2566352	0.7974604	Binomial	BirdID
fixed	factor(fac.yr)5	0.4858379	0.7859501	0.6181536	0.5364741	Binomial	BirdID
fixed	factor(fac.yr)6	-0.7372895	0.7332514	-1.0055071	0.3146527	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)7	-1.5229649	0.7736847	-1.9684567	0.0490155	Binomial	BirdID
fixed	factor(fac.yr)8	-1.1251524	0.7441541	-1.5119884	0.1305368	Binomial	BirdID
fixed	factor(fac.yr)9	-0.3486475	0.7594026	-0.4591076	0.6461569	Binomial	BirdID
fixed	factor(fac.yr)10	-0.6306734	0.7572072	-0.8328941	0.4049045	Binomial	BirdID
fixed	factor(fac.yr)11	-0.8208294	0.7775829	-1.0556166	0.2911434	Binomial	BirdID
fixed	factor(fac.yr)12	0.1089601	0.7983642	0.1364792	0.8914425	Binomial	BirdID
fixed	factor(fac.yr)13	0.3945366	0.8104194	0.4868302	0.6263787	Binomial	BirdID
fixed	factor(fac.yr)14	0.1788110	0.8217930	0.2175864	0.8277514	Binomial	BirdID
fixed	factor(fac.yr)15	-0.7622409	0.8218676	-0.9274498	0.3536930	Binomial	BirdID
fixed	factor(fac.yr)16	0.1979288	0.9187418	0.2154346	0.8294285	Binomial	BirdID
fixed	factor(fac.yr)17	0.4625958	0.9845962	0.4698330	0.6384743	Binomial	BirdID
fixed	factor(fac.yr)18	0.5225843	1.1027434	0.4738948	0.6355749	Binomial	BirdID
fixed	factor(fac.yr)19	0.5152380	1.4098509	0.3654557	0.7147713	Binomial	BirdID
fixed	factor(fac.yr)20	-0.3572865	1.1805249	-0.3026506	0.7621562	Binomial	BirdID
fixed	(Intercept)	0.7539570	0.5927792	1.2719020	0.2034079	Binomial	Terr
fixed	factor(fac.yr)2	-0.0262226	0.7630606	-0.0343650	0.9725861	Binomial	Terr
fixed	factor(fac.yr)3	-1.2130196	0.7251735	-1.6727303	0.0943804	Binomial	Terr
fixed	factor(fac.yr)4	0.2492227	0.7570879	0.3291859	0.7420152	Binomial	Terr
fixed	factor(fac.yr)5	0.5828747	0.7453823	0.7819809	0.4342258	Binomial	Terr
fixed	factor(fac.yr)6	-0.7368355	0.6903228	-1.0673783	0.2858010	Binomial	Terr
fixed	factor(fac.yr)7	-1.5182733	0.7311848	-2.0764562	0.0378518	Binomial	Terr
fixed	factor(fac.yr)8	-1.0781407	0.7017088	-1.5364502	0.1244280	Binomial	Terr
fixed	factor(fac.yr)9	-0.3316482	0.7110826	-0.4663990	0.6409299	Binomial	Terr
fixed	factor(fac.yr)10	-0.5362161	0.7120587	-0.7530504	0.4514197	Binomial	Terr
fixed	factor(fac.yr)11	-0.8751343	0.7268667	-1.2039819	0.2285966	Binomial	Terr
fixed	factor(fac.yr)12	0.0095237	0.7493050	0.0127101	0.9898591	Binomial	Terr
fixed	factor(fac.yr)13	0.1737639	0.7562363	0.2297747	0.8182669	Binomial	Terr
fixed	factor(fac.yr)14	0.0072288	0.7671654	0.0094227	0.9924819	Binomial	Terr
fixed	factor(fac.yr)15	-0.7500839	0.7675282	-0.9772722	0.3284344	Binomial	Terr
fixed	factor(fac.yr)16	0.1175079	0.8617221	0.1363640	0.8915335	Binomial	Terr
fixed	factor(fac.yr)17	0.4443873	0.9267725	0.4794998	0.6315831	Binomial	Terr
fixed	factor(fac.yr)18	0.5416702	1.0472367	0.5172376	0.6049903	Binomial	Terr
fixed	factor(fac.yr)19	0.1650687	1.3619452	0.1212007	0.9035321	Binomial	Terr
fixed	factor(fac.yr)20	-0.4786610	1.1238299	-0.4259194	0.6701666	Binomial	Terr
fixed	(Intercept)	0.7057010	0.6300069	1.1201481	0.2626507	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	0.0565314	0.7869731	0.0718339	0.9427341	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-1.3515048	0.7600123	-1.7782671	0.0753600	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	0.2165013	0.7992843	0.2708689	0.7864919	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	0.5052111	0.7882398	0.6409358	0.5215644	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-0.7383099	0.7349470	-1.0045758	0.3151012	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-1.5283581	0.7755151	-1.9707652	0.0487507	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-1.1242987	0.7458451	-1.5074158	0.1317041	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-0.3406905	0.7608481	-0.4477773	0.6543140	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-0.6244040	0.7590910	-0.8225681	0.4107536	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-0.8213556	0.7791539	-1.0541635	0.2918081	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	0.1173024	0.8001436	0.1466017	0.8834464	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	0.3966141	0.8120955	0.4883836	0.6252781	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	0.1862464	0.8234723	0.2261720	0.8210676	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-0.7655676	0.8235522	-0.9295921	0.3525823	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	0.1997682	0.9207309	0.2169669	0.8282341	Binomial	BirdID+Terr
fixed	factor(fac.yr)17	0.4899139	0.9882542	0.4957368	0.6200801	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	0.5365287	1.1059071	0.4851480	0.6275713	Binomial	BirdID+Terr

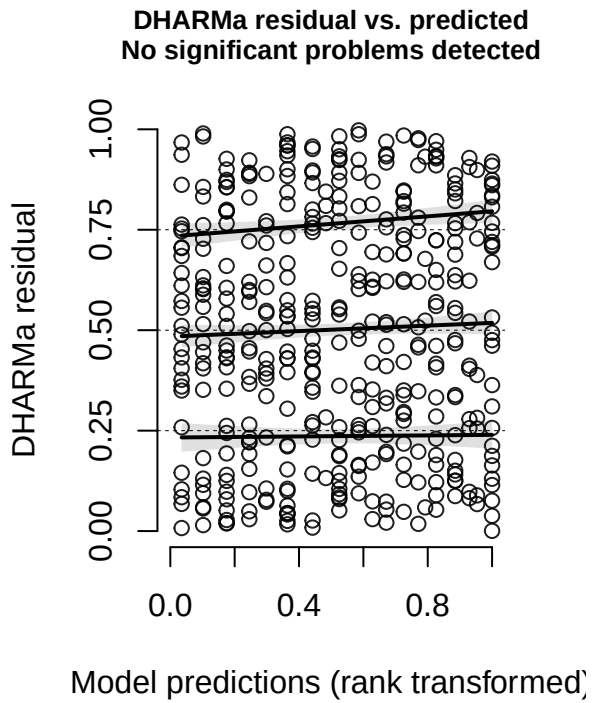
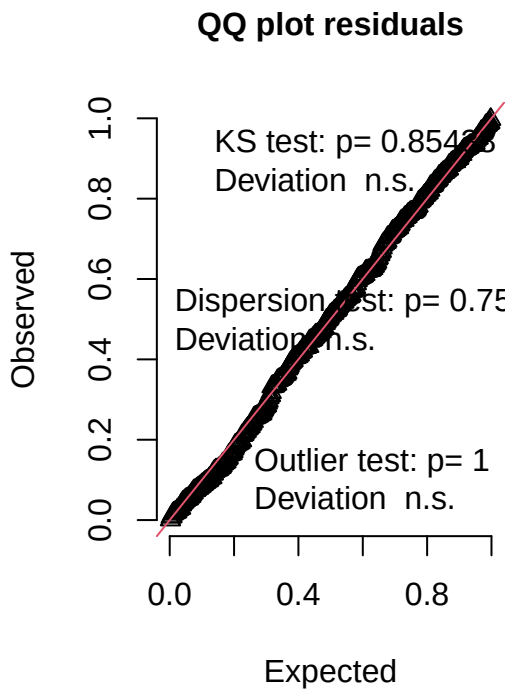
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)19	0.5005817	1.4161681	0.3534762	0.7237315	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-0.3612355	1.1838399	-0.3051388	0.7602605	Binomial	BirdID+Terr

Territory only performed poorly on selection criteria, comparatively to the other 2 models.

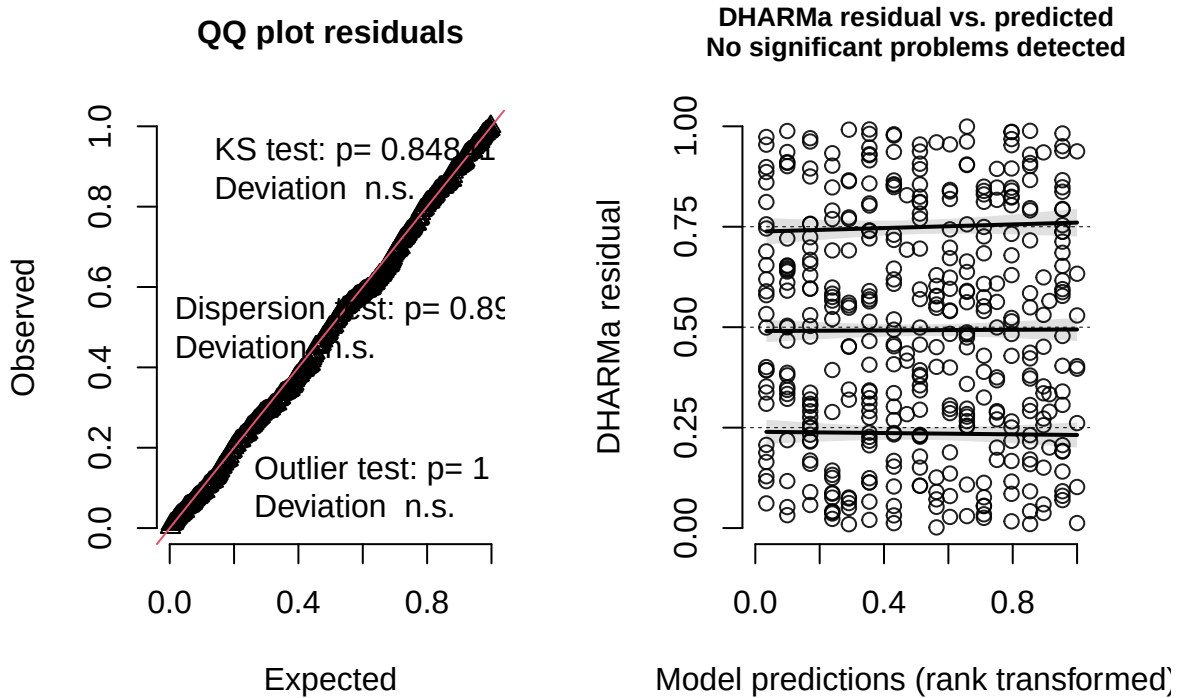
DHARMa residual



DHARMa residual



DHARMA residual



Diagnostics look good.

Probabilities used to generate simulation 11

Simulation 11
 Probability of mate fidelity: 0.676
 Probability of pair dispersal: 0.595
 Probability of single male dispersal: 0.766
 Probability of single female dispersal: 0.275

Simulation 12

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
331	1	-423.2833	888.5667	968.4112	264.8882	310	0.9154635	0.8805371	Poisson	Terr
331	1	-423.1874	890.3747	974.0213	257.2829	309	0.9197071	0.8864044	Poisson	BirdID+Terr
331	1	-429.0185	900.0369	979.8814	304.0441	310	0.9003511	0.8757391	Poisson	BirdID

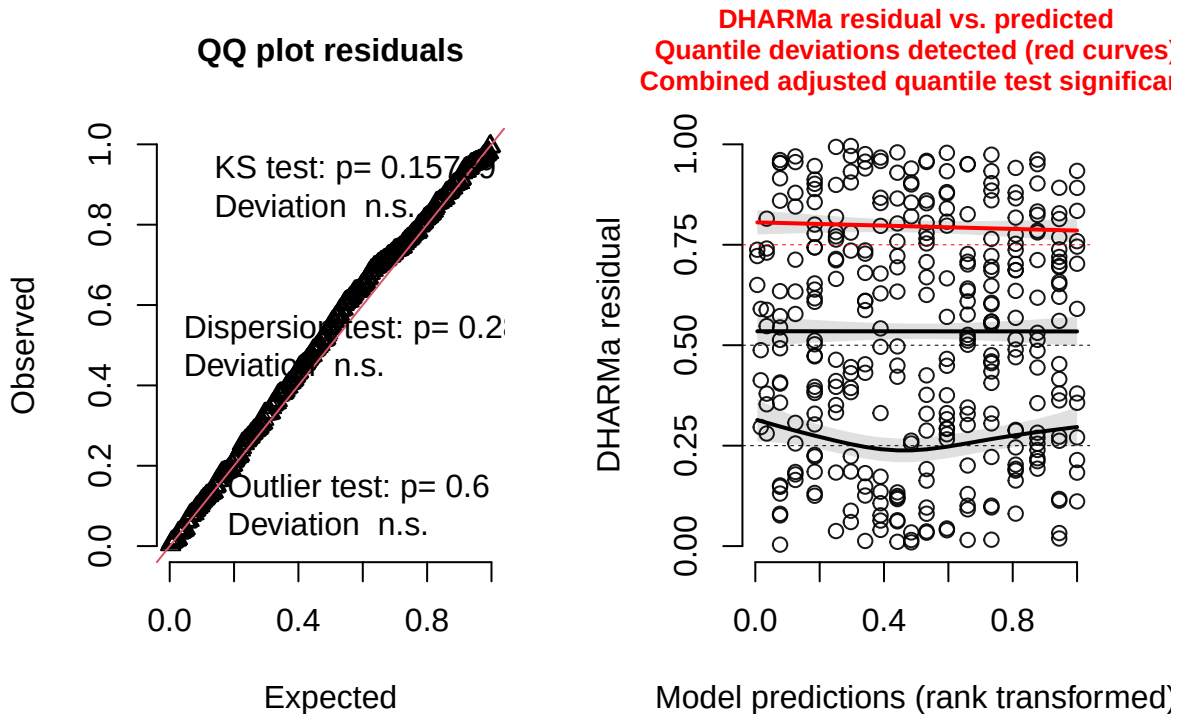
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.4068579	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.5097943	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.1588043	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.4901451	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.4812364	0.2345607	2.0516503	0.0402037	Poisson	BirdID
fixed	factor(fac.yr)2	-0.0501258	0.2791454	-0.1795686	0.8574912	Poisson	BirdID
fixed	factor(fac.yr)3	-0.4148759	0.3127358	-1.3266018	0.1846404	Poisson	BirdID
fixed	factor(fac.yr)4	-0.2484243	0.2996918	-0.8289326	0.4071425	Poisson	BirdID
fixed	factor(fac.yr)5	-0.1368869	0.3008745	-0.4549634	0.6491355	Poisson	BirdID
fixed	factor(fac.yr)6	-0.3515902	0.2984366	-1.1781067	0.2387541	Poisson	BirdID
fixed	factor(fac.yr)7	-0.8847557	0.3316137	-2.6680310	0.0076297	Poisson	BirdID
fixed	factor(fac.yr)8	-0.5282389	0.3221732	-1.6396114	0.1010860	Poisson	BirdID
fixed	factor(fac.yr)9	-0.5760950	0.3440865	-1.6742737	0.0940768	Poisson	BirdID
fixed	factor(fac.yr)10	-0.6070283	0.3566976	-1.7018010	0.0887927	Poisson	BirdID
fixed	factor(fac.yr)11	-0.8521848	0.3783410	-2.2524249	0.0242954	Poisson	BirdID
fixed	factor(fac.yr)12	-0.6717054	0.3637714	-1.8465039	0.0648191	Poisson	BirdID
fixed	factor(fac.yr)13	-0.5268603	0.3341808	-1.5765726	0.1148939	Poisson	BirdID
fixed	factor(fac.yr)14	-1.3241342	0.4217665	-3.1394957	0.0016924	Poisson	BirdID
fixed	factor(fac.yr)15	-0.7007658	0.3803497	-1.8424252	0.0654130	Poisson	BirdID
fixed	factor(fac.yr)16	-0.9928652	0.4263094	-2.3289778	0.0198602	Poisson	BirdID
fixed	factor(fac.yr)17	-0.5643925	0.4049088	-1.3938754	0.1633552	Poisson	BirdID
fixed	factor(fac.yr)18	-1.5692792	0.6475392	-2.4234506	0.0153738	Poisson	BirdID
fixed	factor(fac.yr)19	-16.6995938	1760.9164342	-0.0094835	0.9924334	Poisson	BirdID
fixed	factor(fac.yr)20	-17.8760328	3038.2806349	-0.0058836	0.9953056	Poisson	BirdID
fixed	(Intercept)	0.3133910	0.2466826	1.2704220	0.2039344	Poisson	Terr
fixed	factor(fac.yr)2	0.0443135	0.3062134	0.1447145	0.8849363	Poisson	Terr
fixed	factor(fac.yr)3	-0.3060553	0.3267273	-0.9367301	0.3488974	Poisson	Terr
fixed	factor(fac.yr)4	-0.0208845	0.3191662	-0.0654346	0.9478280	Poisson	Terr
fixed	factor(fac.yr)5	0.0437402	0.3113784	0.1404727	0.8882865	Poisson	Terr
fixed	factor(fac.yr)6	-0.1101858	0.3119420	-0.3532254	0.7239195	Poisson	Terr
fixed	factor(fac.yr)7	-0.6569326	0.3414740	-1.9238144	0.0543778	Poisson	Terr
fixed	factor(fac.yr)8	-0.2537982	0.3336951	-0.7605693	0.4469144	Poisson	Terr
fixed	factor(fac.yr)9	-0.3853566	0.3519886	-1.0947987	0.2736048	Poisson	Terr
fixed	factor(fac.yr)10	-0.4840214	0.3627719	-1.3342307	0.1821283	Poisson	Terr
fixed	factor(fac.yr)11	-0.6014882	0.3872989	-1.5530336	0.1204151	Poisson	Terr
fixed	factor(fac.yr)12	-0.2930681	0.3754658	-0.7805455	0.4350699	Poisson	Terr
fixed	factor(fac.yr)13	-0.3328038	0.3399125	-0.9790865	0.3275372	Poisson	Terr
fixed	factor(fac.yr)14	-1.1740565	0.4230520	-2.7752063	0.0055167	Poisson	Terr
fixed	factor(fac.yr)15	-0.4370409	0.3908429	-1.1182009	0.2634812	Poisson	Terr
fixed	factor(fac.yr)16	-0.7100156	0.4346418	-1.6335649	0.1023502	Poisson	Terr
fixed	factor(fac.yr)17	-0.2888448	0.4043901	-0.7142725	0.4750587	Poisson	Terr
fixed	factor(fac.yr)18	-1.3844215	0.6427710	-2.1538330	0.0312533	Poisson	Terr
fixed	factor(fac.yr)19	-18.0739326	3719.8742121	-0.0048587	0.9961233	Poisson	Terr
fixed	factor(fac.yr)20	-17.8135964	3714.5585756	-0.0047956	0.9961737	Poisson	Terr
fixed	(Intercept)	0.3168223	0.2489839	1.2724606	0.2032095	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	0.0393935	0.3066099	0.1284808	0.8977685	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.3093430	0.3290305	-0.9401651	0.3471329	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.0363798	0.3215382	-0.1131430	0.9099172	Poisson	BirdID+Terr

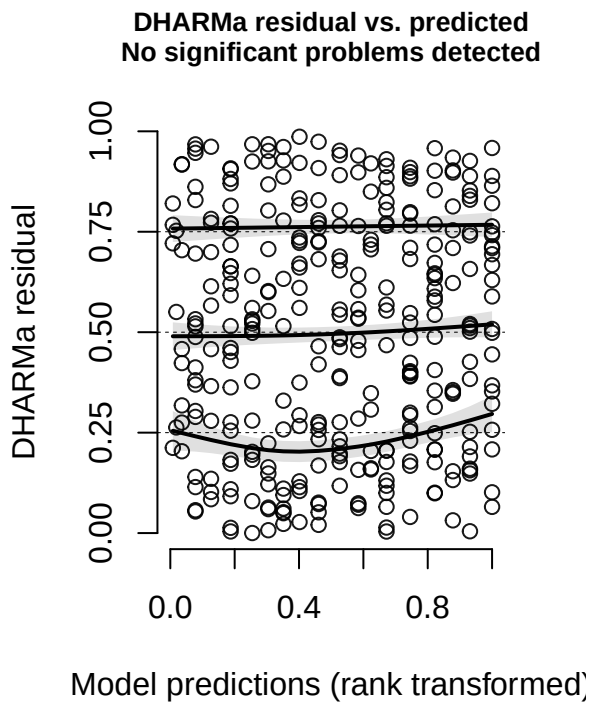
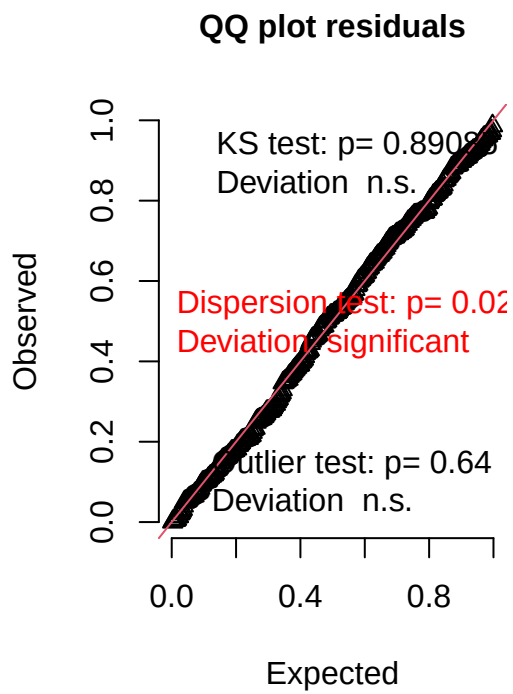
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.0286053	0.3142412	0.0910299	0.9274688	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.1258041	0.3146045	-0.3998801	0.6892449	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.6709957	0.3443313	-1.9486923	0.0513322	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.2735479	0.3365016	-0.8129173	0.4162655	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.4062865	0.3556604	-1.1423440	0.2533111	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.5013093	0.3666287	-1.3673486	0.1715161	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.6250056	0.3906370	-1.5999650	0.1096064	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.3310992	0.3786562	-0.8744059	0.3818972	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.3503797	0.3436353	-1.0196264	0.3079057	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-1.1868416	0.4267964	-2.7808145	0.0054223	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.4666866	0.3945672	-1.1827810	0.2368959	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.7447668	0.4385892	-1.6980964	0.0894896	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.3134825	0.4099274	-0.7647268	0.4444343	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-1.4005487	0.6466450	-2.1658694	0.0303212	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-18.4309567	4489.9239309	-0.0041050	0.9967247	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-18.9766592	6404.2204880	-0.0029631	0.9976358	Poisson	BirdID+Terr

The territory only model is performing the best on AIC and BIC, and performs competitively on the R^2 s.

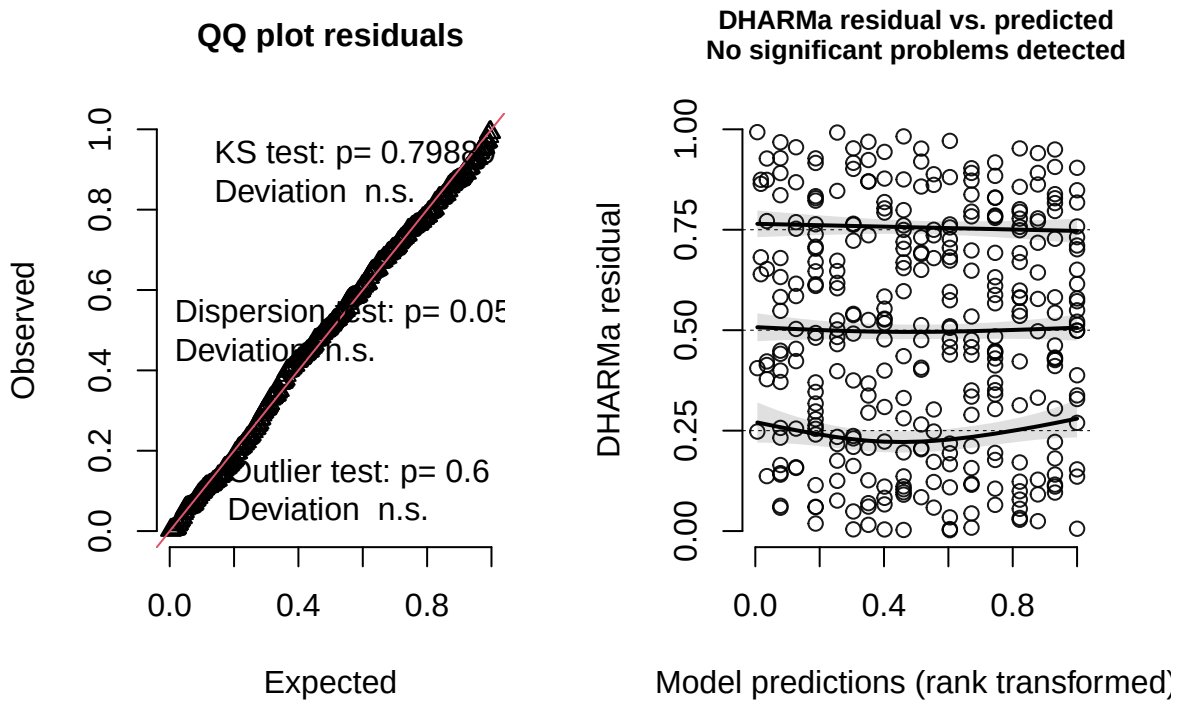
DHARMA residual



DHARMa residual



DHARMa residual



I would be a bit more cautious trusting the territory only model.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
331	1	-185.7744	415.5488	499.1954	165.4490	309	0.865	Binomial	BirdID+Terr
331	1	-192.0023	426.0046	505.8491	258.9417	310	0.678	Binomial	Terr
331	1	-192.3617	426.7234	506.5679	264.0757	310	0.583	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.463166	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.577062	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.629627	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.759176	Binomial	BirdID+Terr

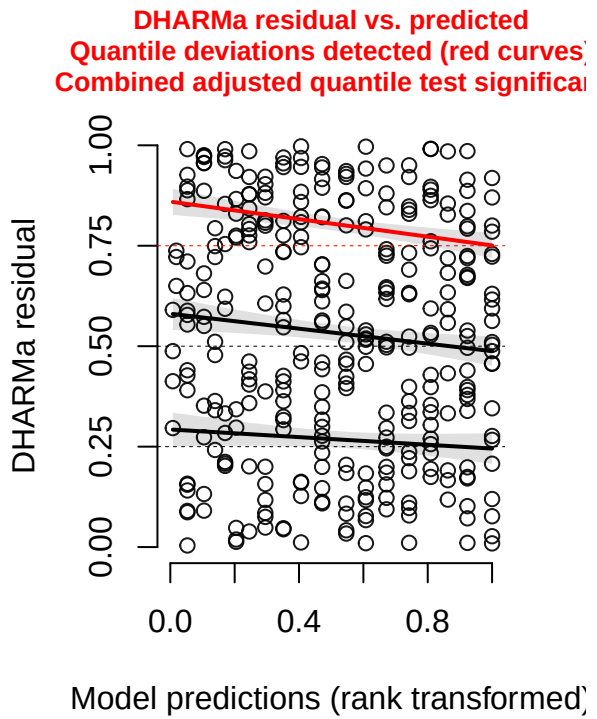
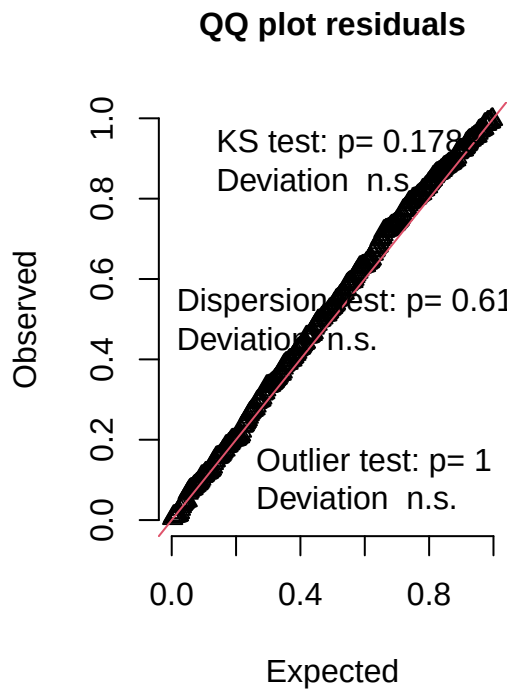
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	1.5258068	7.871784e-01	1.9383241	0.0525837	Binomial	BirdID
fixed	factor(fac.yr)2	0.6958208	9.822488e-01	0.7083957	0.4786996	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)3	-0.2216507	9.254573e-01	-0.2395040	0.8107148	Binomial	BirdID
fixed	factor(fac.yr)4	-0.2543608	9.333363e-01	-0.2725285	0.7852157	Binomial	BirdID
fixed	factor(fac.yr)5	-0.1158758	9.633348e-01	-0.1202861	0.9042565	Binomial	BirdID
fixed	factor(fac.yr)6	0.1574807	9.607594e-01	0.1639127	0.8697998	Binomial	BirdID
fixed	factor(fac.yr)7	-1.1274731	9.114999e-01	-1.2369427	0.2161084	Binomial	BirdID
fixed	factor(fac.yr)8	-1.1232229	9.450746e-01	-1.1885018	0.2346358	Binomial	BirdID
fixed	factor(fac.yr)9	-1.0173591	9.865482e-01	-1.0312311	0.3024325	Binomial	BirdID
fixed	factor(fac.yr)10	-1.3164236	1.007238e+00	-1.3069636	0.1912251	Binomial	BirdID
fixed	factor(fac.yr)11	-1.5430672	1.002034e+00	-1.5399348	0.1235763	Binomial	BirdID
fixed	factor(fac.yr)12	-1.1982440	1.042714e+00	-1.1491588	0.2504905	Binomial	BirdID
fixed	factor(fac.yr)13	-1.2953559	9.820592e-01	-1.3190202	0.1871624	Binomial	BirdID
fixed	factor(fac.yr)14	-2.7146006	1.021253e+00	-2.6581088	0.0078581	Binomial	BirdID
fixed	factor(fac.yr)15	-2.4060278	1.072356e+00	-2.2436831	0.0248528	Binomial	BirdID
fixed	factor(fac.yr)16	-2.1390930	1.104657e+00	-1.9364318	0.0528148	Binomial	BirdID
fixed	factor(fac.yr)17	-1.6452055	1.165354e+00	-1.4117643	0.1580194	Binomial	BirdID
fixed	factor(fac.yr)18	-2.1787528	1.291687e+00	-1.6867501	0.0916514	Binomial	BirdID
fixed	factor(fac.yr)19	-	3.252449e+02	-0.0446445	0.9643907	Binomial	BirdID
		14.5204056					
fixed	factor(fac.yr)20	-	3.279964e+01	-0.3335799	0.7386966	Binomial	BirdID
		10.9413012					
fixed	(Intercept)	0.9690245	8.229135e-01	1.1775533	0.2389747	Binomial	Terr
fixed	factor(fac.yr)2	0.9615769	1.053635e+00	0.9126277	0.3614384	Binomial	Terr
fixed	factor(fac.yr)3	0.3280369	9.875789e-01	0.3321627	0.7397664	Binomial	Terr
fixed	factor(fac.yr)4	0.8816495	9.977602e-01	0.8836287	0.3768967	Binomial	Terr
fixed	factor(fac.yr)5	0.5946386	1.012879e+00	0.5870778	0.5571514	Binomial	Terr
fixed	factor(fac.yr)6	1.1282757	1.005758e+00	1.1218160	0.2619407	Binomial	Terr
fixed	factor(fac.yr)7	-0.3793703	9.432603e-01	-0.4021904	0.6875439	Binomial	Terr
fixed	factor(fac.yr)8	-0.2913127	9.956725e-01	-0.2925789	0.7698440	Binomial	Terr
fixed	factor(fac.yr)9	-0.2081636	1.003478e+00	-0.2074422	0.8356645	Binomial	Terr
fixed	factor(fac.yr)10	-0.5551228	1.031189e+00	-0.5383327	0.5903474	Binomial	Terr
fixed	factor(fac.yr)11	-0.3348715	1.027724e+00	-0.3258381	0.7445468	Binomial	Terr
fixed	factor(fac.yr)12	0.3116244	1.054672e+00	0.2954705	0.7676345	Binomial	Terr
fixed	factor(fac.yr)13	-0.6142174	9.910656e-01	-0.6197545	0.5354194	Binomial	Terr
fixed	factor(fac.yr)14	-1.9786446	1.021530e+00	-1.9369429	0.0527523	Binomial	Terr
fixed	factor(fac.yr)15	-0.9219226	1.047407e+00	-0.8801951	0.3787536	Binomial	Terr
fixed	factor(fac.yr)16	-0.5385725	1.081883e+00	-0.4978104	0.6186177	Binomial	Terr
fixed	factor(fac.yr)17	-0.8338180	1.108086e+00	-0.7524847	0.4517596	Binomial	Terr
fixed	factor(fac.yr)18	-1.6464436	1.216777e+00	-1.3531188	0.1760177	Binomial	Terr

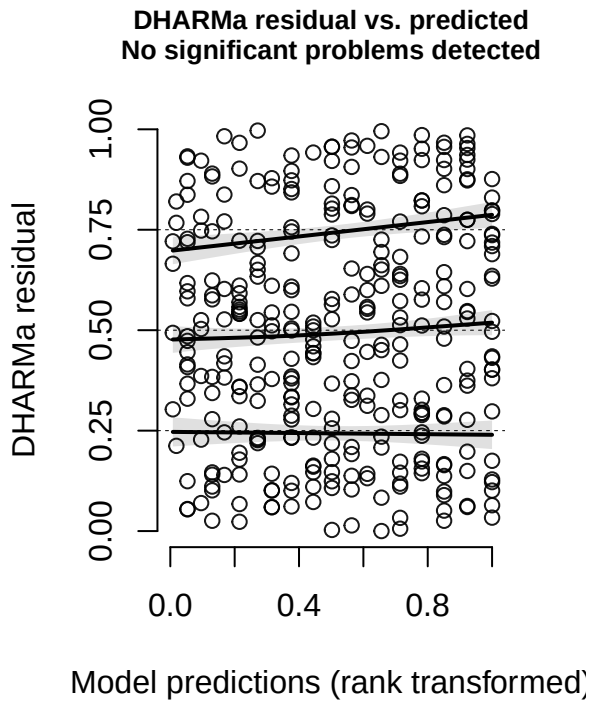
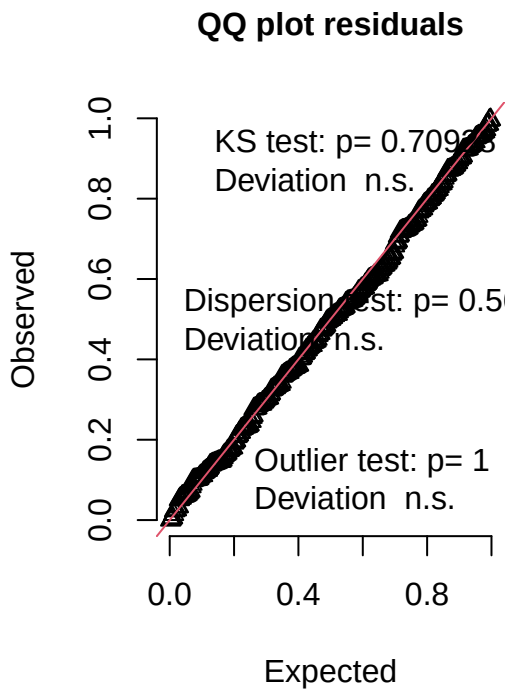
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)19	- 16.4955892	1.263714e+03	-0.0130533	0.9895853	Binomial	Terr
fixed	factor(fac.yr)20	- 15.5503982	7.556960e+02	-0.0205776	0.9835826	Binomial	Terr
fixed	(Intercept)	1.4822840	1.013147e+00	1.4630497	0.1434538	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	1.1166247	1.265441e+00	0.8823996	0.3775608	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	0.2545013	1.173122e+00	0.2169437	0.8282522	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	0.7080763	1.205613e+00	0.5873165	0.5569911	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	0.5933835	1.217062e+00	0.4875543	0.6258656	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	1.2514050	1.241197e+00	1.0082243	0.3133468	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-0.7920756	1.159585e+00	-0.6830682	0.4945638	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-0.5575521	1.219007e+00	-0.4573824	0.6473962	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-0.8984304	1.245188e+00	-0.7215217	0.4705886	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-1.1760578	1.276954e+00	-0.9209870	0.3570572	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-1.0693013	1.277775e+00	-0.8368464	0.4026789	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-0.5934933	1.304270e+00	-0.4550386	0.6490814	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-1.0509207	1.255370e+00	-0.8371405	0.4025136	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-3.0442772	1.309450e+00	-2.3248510	0.0200799	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-2.1200706	1.362359e+00	-1.5561758	0.1196663	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-1.5225186	1.379592e+00	-1.1036002	0.2697666	Binomial	BirdID+Terr
fixed	factor(fac.yr)17	-1.3764702	1.427541e+00	-0.9642246	0.3349333	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-2.2687032	1.666891e+00	-1.3610383	0.1735016	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	- 26.6412635	1.328764e+05	-0.0002005	0.9998400	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	- 19.7715182	1.805843e+03	-0.0109486	0.9912644	Binomial	BirdID+Terr

The territory only model is consistently the second best on the selection criteria.

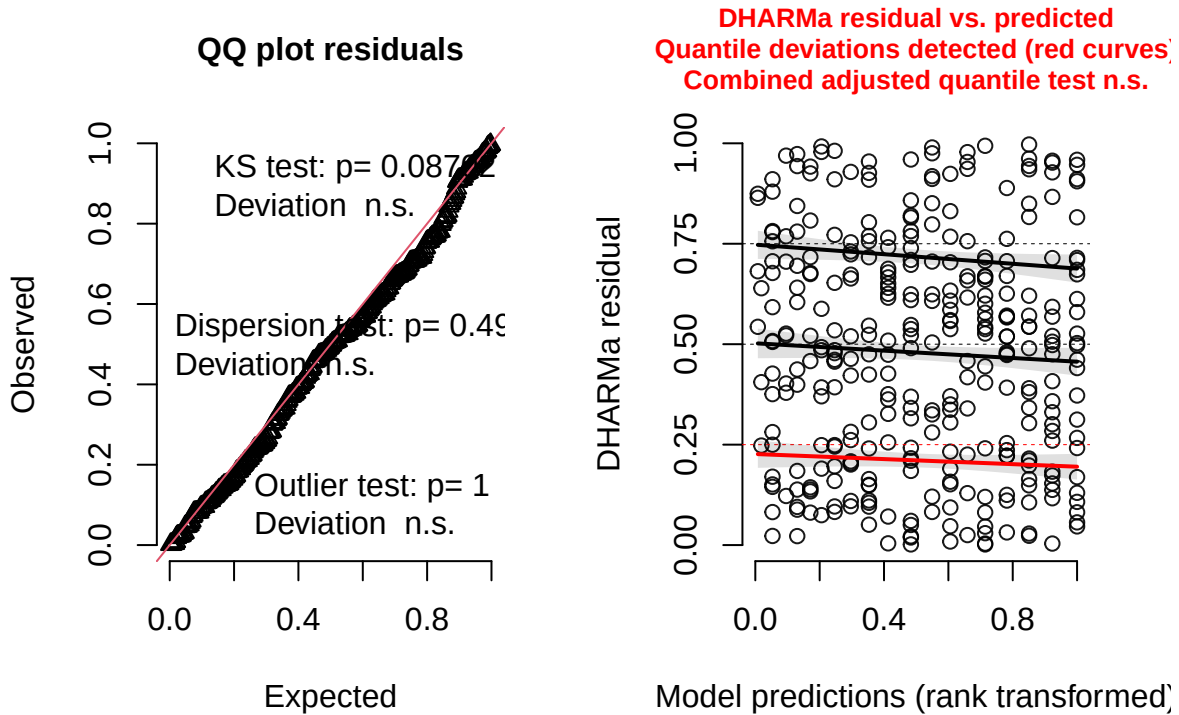
DHARMa residual



DHARMa residual



DHARMa residual



Definitely some mildly concerning diagnostics on the non territory only models.

Probabilities used to generate simulation 12

Simulation 12
 Probability of mate fidelity: 0.926
 Probability of pair dispersal: 0.694
 Probability of single male dispersal: 0.245
 Probability of single female dispersal: 0.105

Simulation 13

Number of fledgling as the response (Poisson)

boundary (singular) fit: see `help('isSingular')`

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
333	1	-451.3577	944.7153	1024.686	288.3440	312	0.3976961	0.0805993	Poisson	Terr
333	1	-451.3581	946.7161	1030.495	288.2473	311	0.3977285	0.0800499	Poisson	BirdID+Terr

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	condtional_R2	marginal_R2	Family	RE
333	1	-466.2463	974.4926	1054.464	396.3608	312	0.2147865	0.1120874	Poisson	BirdID

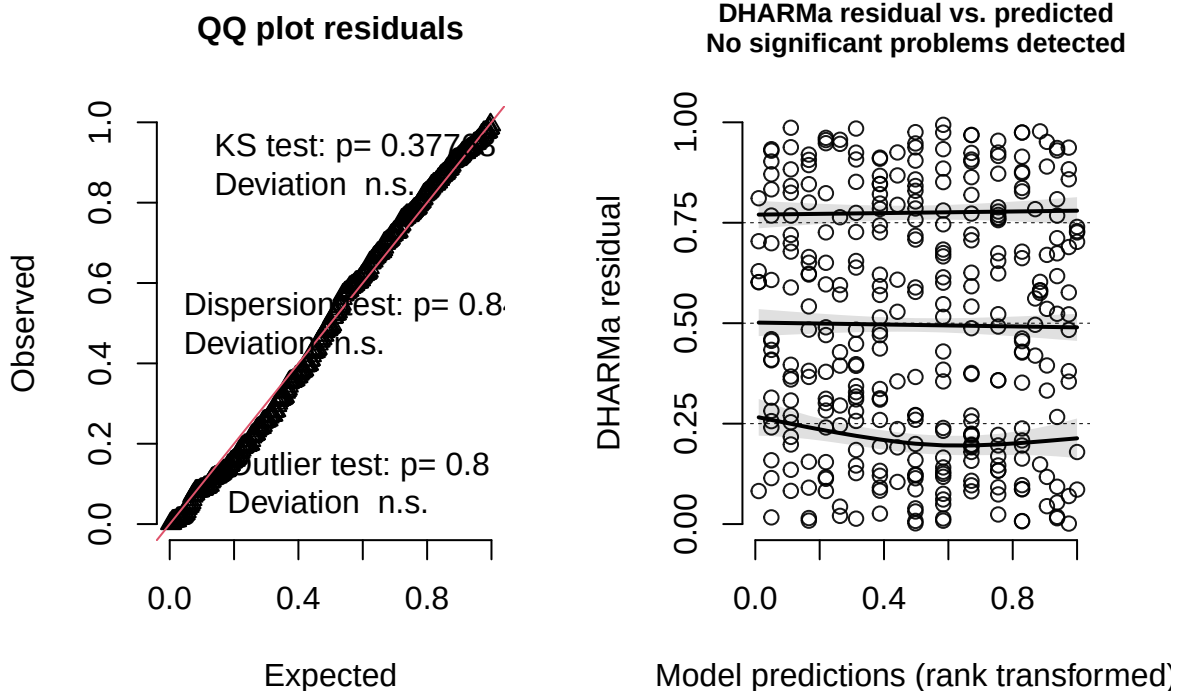
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.2859919	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.5665142	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.0005412	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.5670509	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.6192830	0.2311549	2.6790820	0.0073824	Poisson	BirdID
fixed	factor(fac.yr)2	-0.7558352	0.3408131	-2.2177409	0.0265725	Poisson	BirdID
fixed	factor(fac.yr)3	-1.2625528	0.3840379	-3.2875736	0.0010105	Poisson	BirdID
fixed	factor(fac.yr)4	-0.4629533	0.2941418	-1.5739121	0.1155078	Poisson	BirdID
fixed	factor(fac.yr)5	-0.4647765	0.2914398	-1.5947598	0.1107659	Poisson	BirdID
fixed	factor(fac.yr)6	-0.4513587	0.3040950	-1.4842687	0.1377377	Poisson	BirdID
fixed	factor(fac.yr)7	-0.5471452	0.2995690	-1.8264415	0.0677838	Poisson	BirdID
fixed	factor(fac.yr)8	-0.4737263	0.2941386	-1.6105548	0.1072768	Poisson	BirdID
fixed	factor(fac.yr)9	-0.4968544	0.2977207	-1.6688609	0.0951449	Poisson	BirdID
fixed	factor(fac.yr)10	-0.5627460	0.3177637	-1.7709575	0.0765678	Poisson	BirdID
fixed	factor(fac.yr)11	-0.9915298	0.3603183	-2.7518165	0.0059266	Poisson	BirdID
fixed	factor(fac.yr)12	-0.6898166	0.3422866	-2.0153191	0.0438712	Poisson	BirdID
fixed	factor(fac.yr)13	-0.0875620	0.3240378	-0.2702215	0.7869899	Poisson	BirdID
fixed	factor(fac.yr)14	-0.5766424	0.3803053	-1.5162615	0.1294532	Poisson	BirdID
fixed	factor(fac.yr)15	-0.5172100	0.4038155	-1.2808076	0.2002612	Poisson	BirdID
fixed	factor(fac.yr)16	-0.3194916	0.4378772	-0.7296374	0.4656119	Poisson	BirdID
fixed	factor(fac.yr)17	0.0658847	0.3872262	0.1701453	0.8648959	Poisson	BirdID
fixed	factor(fac.yr)18	-0.2240749	0.3845353	-0.5827162	0.5600844	Poisson	BirdID
fixed	factor(fac.yr)19	-1.3712832	0.6366232	-2.1539949	0.0312406	Poisson	BirdID
fixed	factor(fac.yr)20	-0.4434086	0.5229809	-0.8478486	0.3965223	Poisson	BirdID
fixed	(Intercept)	0.5275995	0.2518145	2.0951912	0.0361540	Poisson	Terr
fixed	factor(fac.yr)2	-0.6734061	0.3602548	-1.8692493	0.0615881	Poisson	Terr
fixed	factor(fac.yr)3	-1.2113760	0.3993966	-3.0330149	0.0024212	Poisson	Terr
fixed	factor(fac.yr)4	-0.5539581	0.3064996	-1.8073696	0.0707047	Poisson	Terr
fixed	factor(fac.yr)5	-0.4797192	0.3099298	-1.5478318	0.1216628	Poisson	Terr
fixed	factor(fac.yr)6	-0.4418520	0.3184184	-1.3876458	0.1652449	Poisson	Terr
fixed	factor(fac.yr)7	-0.4795148	0.3172837	-1.5113124	0.1307089	Poisson	Terr
fixed	factor(fac.yr)8	-0.4922142	0.3134685	-1.5702186	0.1163643	Poisson	Terr
fixed	factor(fac.yr)9	-0.5284625	0.3149370	-1.6779941	0.0933482	Poisson	Terr
fixed	factor(fac.yr)10	-0.5712885	0.3349327	-1.7056815	0.0880674	Poisson	Terr
fixed	factor(fac.yr)11	-0.9287004	0.3774774	-2.4602806	0.0138828	Poisson	Terr
fixed	factor(fac.yr)12	-0.5798724	0.3652898	-1.5874314	0.1124150	Poisson	Terr
fixed	factor(fac.yr)13	-0.1654435	0.3477036	-0.4758176	0.6342043	Poisson	Terr
fixed	factor(fac.yr)14	-0.6074353	0.4061456	-1.4956098	0.1347554	Poisson	Terr
fixed	factor(fac.yr)15	-0.5011721	0.4310388	-1.1627077	0.2449481	Poisson	Terr
fixed	factor(fac.yr)16	-0.2561887	0.4651398	-0.5507779	0.5817859	Poisson	Terr
fixed	factor(fac.yr)17	0.0004551	0.4060019	0.0011210	0.9991055	Poisson	Terr
fixed	factor(fac.yr)18	-0.2243182	0.4182575	-0.5363160	0.5917402	Poisson	Terr

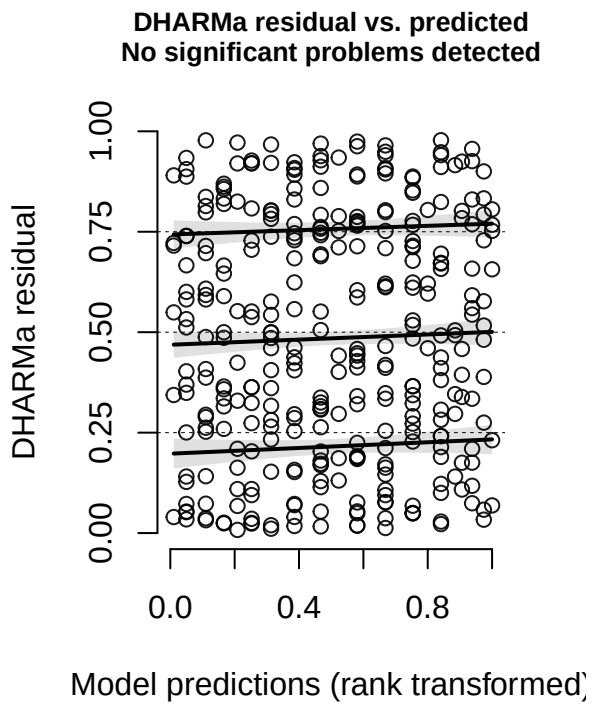
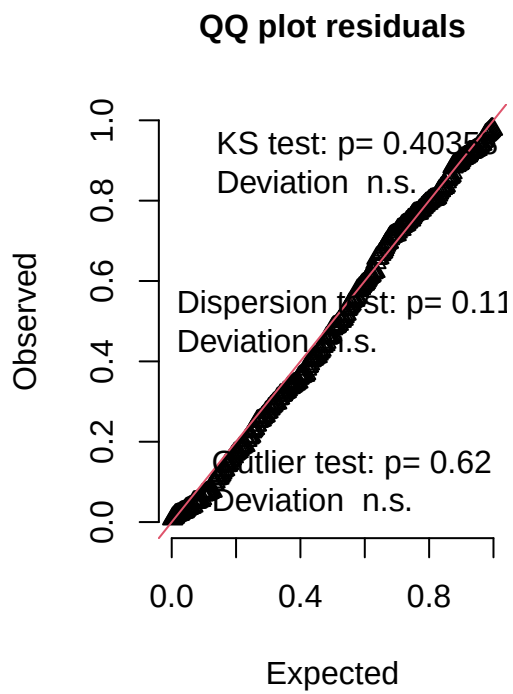
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)19	-1.5503006	0.6467961	-2.3968922	0.0165348	Poisson	Terr
fixed	factor(fac.yr)20	-0.4431589	0.5707842	-0.7764037	0.4375107	Poisson	Terr
fixed	(Intercept)	0.5251562	0.2520437	2.0835915	0.0371973	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.6694352	0.3603088	-1.8579485	0.0631763	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-1.2048145	0.3991253	-3.0186370	0.0025391	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.5512475	0.3066652	-1.7975548	0.0722476	Poisson	BirdID+Terr
fixed	factor(fac.yr)5	-0.4758026	0.3100629	-1.5345357	0.1248979	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.4383174	0.3185496	-1.3759783	0.1688283	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.4759807	0.3174219	-1.4995208	0.1337386	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.4888299	0.3136203	-1.5586676	0.1190751	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.5255437	0.3150967	-1.6678809	0.0953394	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.5679520	0.3350746	-1.6950015	0.0900751	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.9266069	0.3776764	-2.4534415	0.0141497	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.5765242	0.3654249	-1.5776820	0.1146387	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.1620467	0.3478369	-0.4658699	0.6413087	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.6080330	0.4065912	-1.4954405	0.1347995	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.5020393	0.4315377	-1.1633730	0.2446782	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.2521868	0.4651672	-0.5421422	0.5877205	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	0.0038328	0.4061159	0.0094376	0.9924700	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.2244118	0.4186480	-0.5360393	0.5919314	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.5418875	0.6453989	-2.3890459	0.0168922	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.4433281	0.5713412	-0.7759427	0.4377828	Poisson	BirdID+Terr

Similar response as on the batch of previous models.

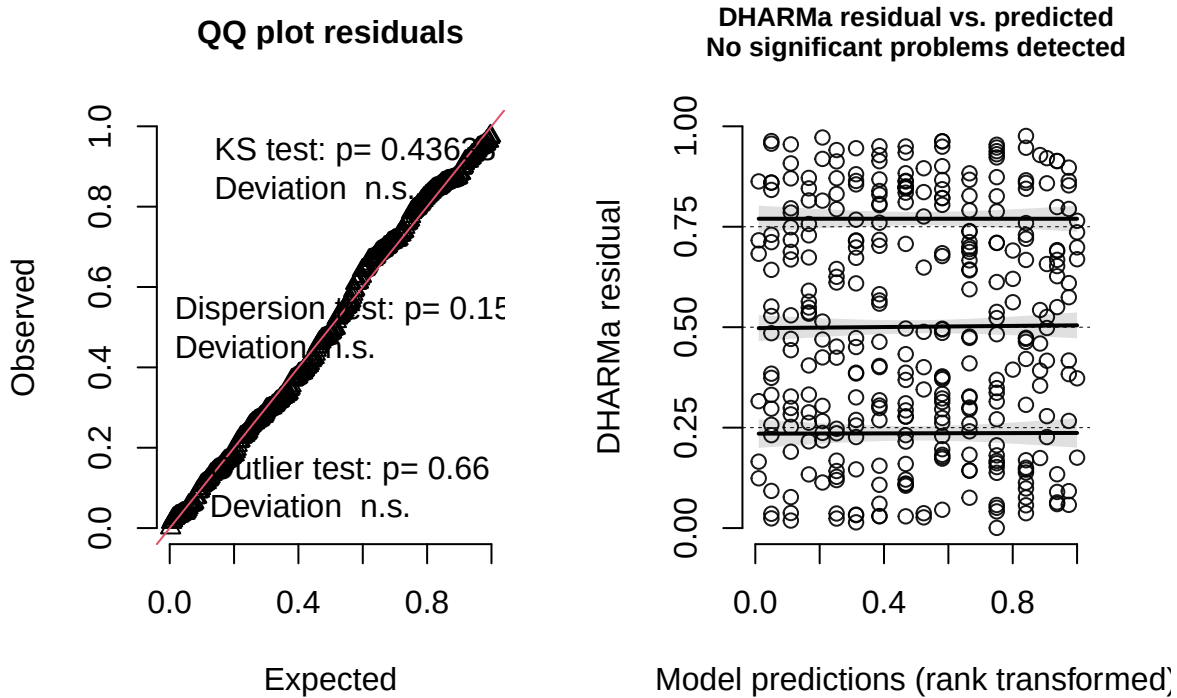
DHARMa residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
333	1	-201.5072	445.0144	524.9854	276.3058	312	0.628	Binomial	Terr
333	1	-201.5073	447.0145	530.7936	276.3559	311	0.628	Binomial	BirdID+Terr
333	1	-213.5292	469.0585	549.0295	379.6899	312	0.346	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.6521215	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.5004279	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	0.0354126	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.4985386	Binomial	BirdID+Terr

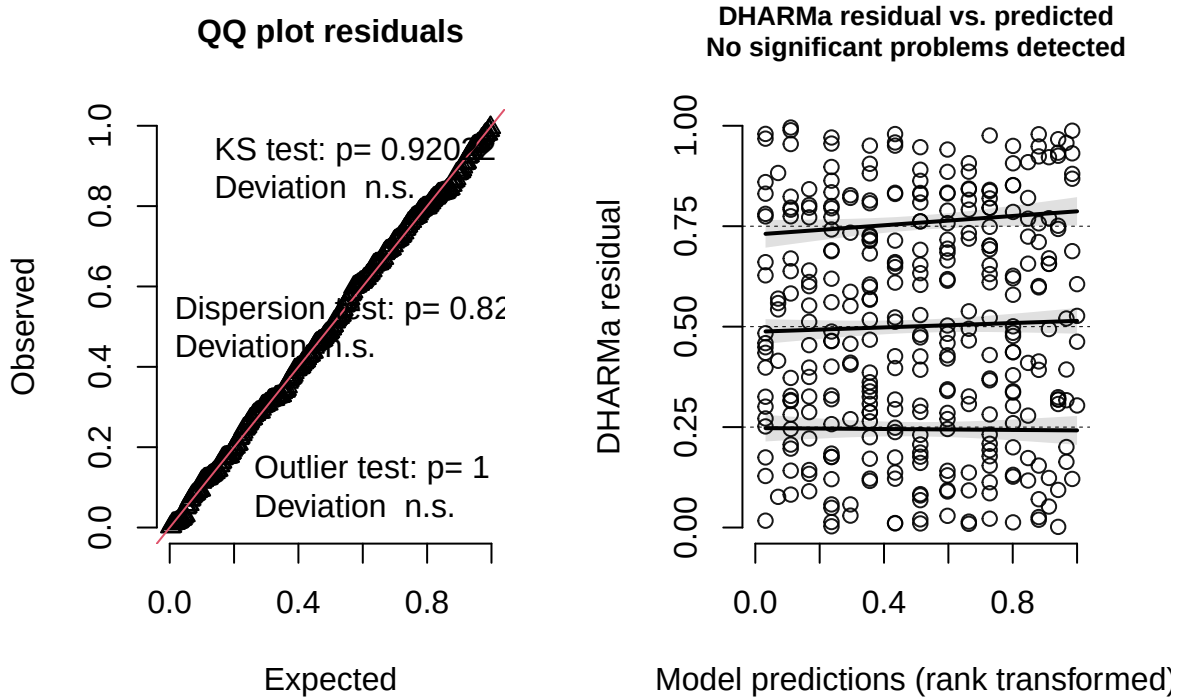
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	1.5800605	8.322642e-01	1.8985083	0.0576292	Binomial	BirdID
fixed	factor(fac.yr)2	-0.9945713	9.763326e-01	-1.0186808	0.3083545	Binomial	BirdID
fixed	factor(fac.yr)3	-2.0819833	9.549446e-01	-2.1802137	0.0292416	Binomial	BirdID
fixed	factor(fac.yr)4	-0.8918118	9.362403e-01	-0.9525458	0.3408202	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-1.2809545	9.185947e-01	-1.3944719	0.1631751	Binomial	BirdID
fixed	factor(fac.yr)6	-1.2729377	9.415749e-01	-1.3519239	0.1763997	Binomial	BirdID
fixed	factor(fac.yr)7	-0.9998810	9.313021e-01	-1.0736376	0.2829851	Binomial	BirdID
fixed	factor(fac.yr)8	-1.0801016	9.218662e-01	-1.1716468	0.2413389	Binomial	BirdID
fixed	factor(fac.yr)9	-1.4629561	9.222327e-01	-1.5863198	0.1126668	Binomial	BirdID
fixed	factor(fac.yr)10	-0.8562643	9.553555e-01	-0.8962782	0.3701042	Binomial	BirdID
fixed	factor(fac.yr)11	-1.5011630	9.581200e-01	-1.5667797	0.1171662	Binomial	BirdID
fixed	factor(fac.yr)12	-1.4828956	9.697314e-01	-1.5291818	0.1262194	Binomial	BirdID
fixed	factor(fac.yr)13	-0.5271528	1.046170e+00	-0.5038882	0.6143399	Binomial	BirdID
fixed	factor(fac.yr)14	-1.3216802	1.049384e+00	-1.2594819	0.2078563	Binomial	BirdID
fixed	factor(fac.yr)15	-0.6663205	1.117856e+00	-0.5960699	0.5511285	Binomial	BirdID
fixed	factor(fac.yr)16	0.2957191	1.423618e+00	0.2077236	0.8354448	Binomial	BirdID
fixed	factor(fac.yr)17	0.2374245	1.426452e+00	0.1664441	0.8678075	Binomial	BirdID
fixed	factor(fac.yr)18	-0.3652510	1.200439e+00	-0.3042644	0.7609265	Binomial	BirdID
fixed	factor(fac.yr)19	-1.6398988	1.197221e+00	-1.3697547	0.1707635	Binomial	BirdID
fixed	factor(fac.yr)20	19.5167350	1.822942e+04	0.0010706	0.9991458	Binomial	BirdID
fixed	(Intercept)	1.7829893	9.345117e-01	1.9079369	0.0563994	Binomial	Terr
fixed	factor(fac.yr)2	-0.9030563	1.081314e+00	-0.8351469	0.4036350	Binomial	Terr
fixed	factor(fac.yr)3	-2.0055469	1.081057e+00	-1.8551730	0.0635715	Binomial	Terr
fixed	factor(fac.yr)4	-1.0349861	1.024601e+00	-1.0101363	0.3124300	Binomial	Terr
fixed	factor(fac.yr)5	-1.2807948	1.032815e+00	-1.2401014	0.2149379	Binomial	Terr
fixed	factor(fac.yr)6	-1.2725140	1.061999e+00	-1.1982256	0.2308292	Binomial	Terr
fixed	factor(fac.yr)7	-0.5827014	1.047385e+00	-0.5563394	0.5779789	Binomial	Terr
fixed	factor(fac.yr)8	-1.0520936	1.033472e+00	-1.0180187	0.3086691	Binomial	Terr
fixed	factor(fac.yr)9	-1.5313175	1.037601e+00	-1.4758251	0.1399908	Binomial	Terr
fixed	factor(fac.yr)10	-0.6253570	1.061899e+00	-0.5889041	0.5559256	Binomial	Terr
fixed	factor(fac.yr)11	-1.4629426	1.063263e+00	-1.3758988	0.1688530	Binomial	Terr
fixed	factor(fac.yr)12	-1.3104886	1.095322e+00	-1.1964415	0.2315243	Binomial	Terr
fixed	factor(fac.yr)13	-0.6139017	1.184522e+00	-0.5182695	0.6042703	Binomial	Terr
fixed	factor(fac.yr)14	-1.4675986	1.199191e+00	-1.2238244	0.2210185	Binomial	Terr
fixed	factor(fac.yr)15	-0.3944835	1.266493e+00	-0.3114769	0.7554381	Binomial	Terr
fixed	factor(fac.yr)16	0.4521222	1.583915e+00	0.2854460	0.7753025	Binomial	Terr
fixed	factor(fac.yr)17	0.8162621	1.650885e+00	0.4944390	0.6209962	Binomial	Terr
fixed	factor(fac.yr)18	0.0634790	1.391206e+00	0.0456287	0.9636062	Binomial	Terr
fixed	factor(fac.yr)19	-2.2325634	1.322102e+00	-1.6886468	0.0912871	Binomial	Terr
fixed	factor(fac.yr)20	28.1036379	1.575008e+06	0.0000178	0.9999858	Binomial	Terr
fixed	(Intercept)	1.7842202	9.345100e-01	1.9092576	0.0562289	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-0.8914805	1.081709e+00	-0.8241408	0.4098595	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-2.0039505	1.080897e+00	-1.8539699	0.0637435	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-1.0258520	1.024783e+00	-1.0010434	0.3168058	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-1.2849753	1.032736e+00	-1.2442440	0.2134098	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-1.2703697	1.061947e+00	-1.1962643	0.2315934	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-0.5869652	1.047326e+00	-0.5604417	0.5751782	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-1.0601174	1.033392e+00	-1.0258624	0.3049565	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-1.5387163	1.037564e+00	-1.4830085	0.1380722	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-0.6306801	1.061799e+00	-0.5939732	0.5525300	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-1.4717784	1.063340e+00	-1.3841093	0.1663250	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-1.3176217	1.095284e+00	-1.2029954	0.2289781	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-0.6209132	1.184017e+00	-0.5244124	0.5999917	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-1.4593293	1.199161e+00	-1.2169587	0.2236199	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-0.4076530	1.265966e+00	-0.3220095	0.7474455	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	0.4567103	1.585284e+00	0.2880936	0.7732751	Binomial	BirdID+Terr

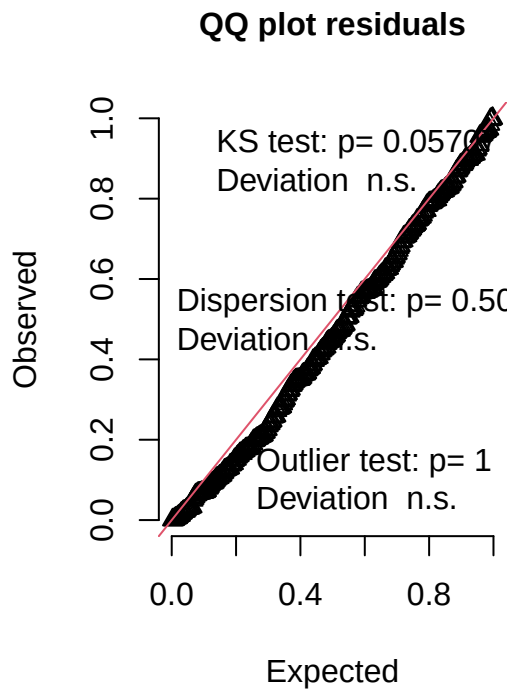
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	0.8194966	1.652704e+00	0.4958520	0.6199988	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	0.0684278	1.391922e+00	0.0491607	0.9607912	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-2.2304777	1.322079e+00	-1.6870982	0.0915845	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	18.1721770	1.098823e+04	0.0016538	0.9986805	Binomial	BirdID+Terr

Territory only model is best on all selection criteria.

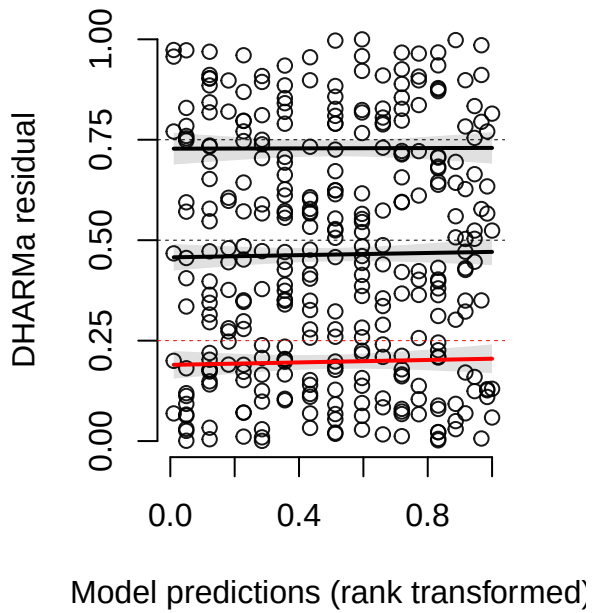
DHARMA residual



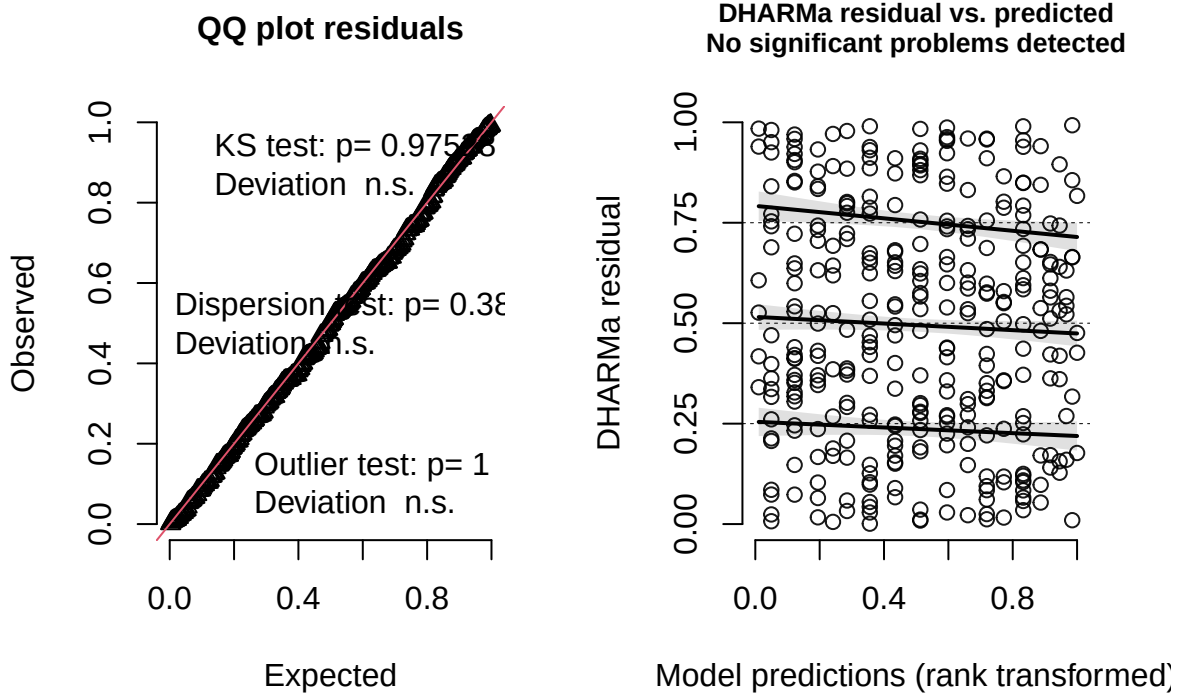
DHARMa residual



DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significance



DHARMa residual



Mild diagnostic issue on the territory only model.

Probabilities used to generate simulation 13

Simulation 13
 Probability of mate fidelity: 0.365
 Probability of pair dispersal: 0.207
 Probability of single male dispersal: 0.813
 Probability of single female dispersal: 0.764

Simulation 14

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
327	1	-411.0848	864.1696	943.7588	311.3070	306	0.3222063	0.1926871	Poisson	Terr
327	1	-410.1715	864.3429	947.7221	284.9438	305	0.3500531	0.1807419	Poisson	BirdID+Terr
327	1	-411.3057	864.6114	944.2006	297.6065	306	0.3413424	0.1791210	Poisson	BirdID

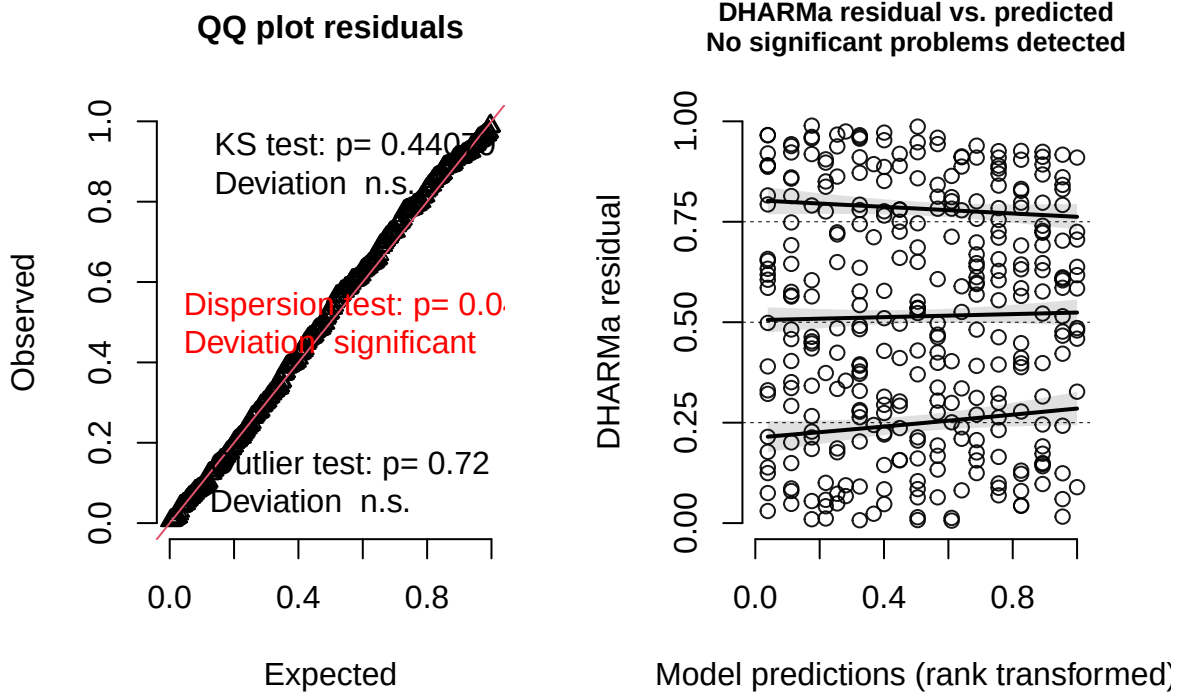
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.4069877	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.3602565	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.3073339	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.2854538	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.6211373	0.2459391	2.5255735	0.0115510	Poisson	BirdID
fixed	factor(fac.yr)2	-0.4275209	0.3182898	-1.3431811	0.1792133	Poisson	BirdID
fixed	factor(fac.yr)3	-1.5797581	0.4067556	-3.8838016	0.0001028	Poisson	BirdID
fixed	factor(fac.yr)4	-0.9443118	0.3460756	-2.7286283	0.0063598	Poisson	BirdID
fixed	factor(fac.yr)5	-0.3723921	0.3066375	-1.2144377	0.2245806	Poisson	BirdID
fixed	factor(fac.yr)6	-0.3655473	0.3200450	-1.1421745	0.2533815	Poisson	BirdID
fixed	factor(fac.yr)7	-0.7787431	0.3446477	-2.2595334	0.0238502	Poisson	BirdID
fixed	factor(fac.yr)8	-0.7466243	0.3574424	-2.0887964	0.0367261	Poisson	BirdID
fixed	factor(fac.yr)9	-0.2094131	0.3195700	-0.6552966	0.5122768	Poisson	BirdID
fixed	factor(fac.yr)10	-0.2395721	0.3102725	-0.7721347	0.4400347	Poisson	BirdID
fixed	factor(fac.yr)11	-1.3223516	0.3916098	-3.3767074	0.0007336	Poisson	BirdID
fixed	factor(fac.yr)12	-0.8569489	0.3956421	-2.1659701	0.0303135	Poisson	BirdID
fixed	factor(fac.yr)13	-0.8759789	0.3857542	-2.2708214	0.0231578	Poisson	BirdID
fixed	factor(fac.yr)14	-1.2969628	0.4286585	-3.0256316	0.0024811	Poisson	BirdID
fixed	factor(fac.yr)15	-1.0449994	0.4312717	-2.4230651	0.0153902	Poisson	BirdID
fixed	factor(fac.yr)16	-0.6717557	0.4100959	-1.6380455	0.1014122	Poisson	BirdID
fixed	factor(fac.yr)17	-1.1317573	0.4720949	-2.3973088	0.0165160	Poisson	BirdID
fixed	factor(fac.yr)18	-0.6148205	0.4122059	-1.4915374	0.1358205	Poisson	BirdID
fixed	factor(fac.yr)19	-0.9315110	0.6629677	-1.4050624	0.1600027	Poisson	BirdID
fixed	factor(fac.yr)20	-0.9681409	0.6657073	-1.4543042	0.1458620	Poisson	BirdID
fixed	(Intercept)	0.6697183	0.2377996	2.8163136	0.0048578	Poisson	Terr
fixed	factor(fac.yr)2	-0.4806082	0.3172515	-1.5149125	0.1297946	Poisson	Terr
fixed	factor(fac.yr)3	-1.5842583	0.4018370	-3.9425393	0.0000806	Poisson	Terr
fixed	factor(fac.yr)4	-0.9803848	0.3404155	-2.8799649	0.0039772	Poisson	Terr
fixed	factor(fac.yr)5	-0.4078820	0.2980247	-1.3686183	0.1711186	Poisson	Terr
fixed	factor(fac.yr)6	-0.3745925	0.3053901	-1.2266030	0.2199718	Poisson	Terr
fixed	factor(fac.yr)7	-0.7724226	0.3326795	-2.3218220	0.0202425	Poisson	Terr
fixed	factor(fac.yr)8	-0.7925874	0.3488866	-2.2717623	0.0231009	Poisson	Terr
fixed	factor(fac.yr)9	-0.2127832	0.3051176	-0.6973812	0.4855643	Poisson	Terr
fixed	factor(fac.yr)10	-0.3016966	0.2974834	-1.0141626	0.3105051	Poisson	Terr
fixed	factor(fac.yr)11	-1.3939989	0.3815803	-3.6532255	0.0002590	Poisson	Terr
fixed	factor(fac.yr)12	-0.8185951	0.3852469	-2.1248582	0.0335984	Poisson	Terr
fixed	factor(fac.yr)13	-0.9614812	0.3749666	-2.5641785	0.0103420	Poisson	Terr
fixed	factor(fac.yr)14	-1.4367392	0.4185643	-3.4325415	0.0005980	Poisson	Terr
fixed	factor(fac.yr)15	-1.1329259	0.4189069	-2.7044812	0.0068411	Poisson	Terr
fixed	factor(fac.yr)16	-0.6434127	0.3997244	-1.6096407	0.1074763	Poisson	Terr
fixed	factor(fac.yr)17	-1.1015862	0.4616476	-2.3862056	0.0170232	Poisson	Terr
fixed	factor(fac.yr)18	-0.6235721	0.4001803	-1.5582281	0.1191792	Poisson	Terr
fixed	factor(fac.yr)19	-1.0432030	0.6435994	-1.6208887	0.1050415	Poisson	Terr
fixed	factor(fac.yr)20	-0.9898597	0.6486967	-1.5259207	0.1270296	Poisson	Terr
fixed	(Intercept)	0.6323992	0.2484266	2.5456181	0.0109084	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.4497326	0.3244194	-1.3862689	0.1656648	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-1.5835266	0.4104625	-3.8579078	0.0001144	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.9536603	0.3494286	-2.7291993	0.0063488	Poisson	BirdID+Terr

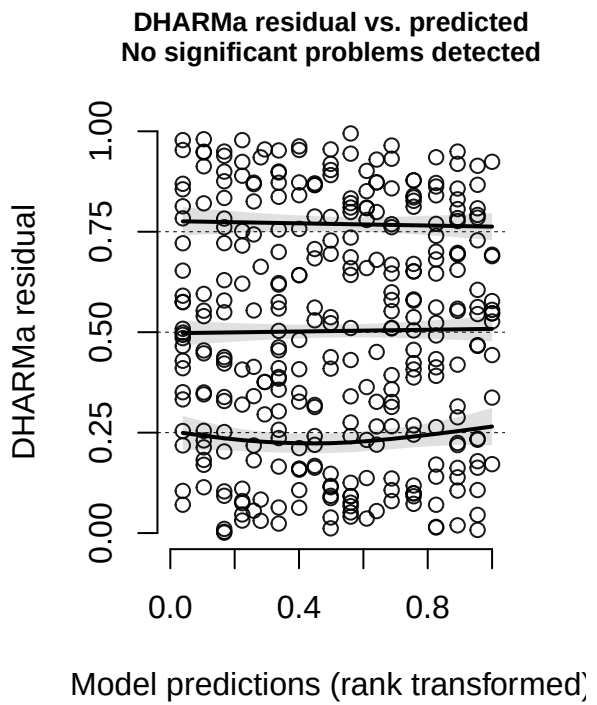
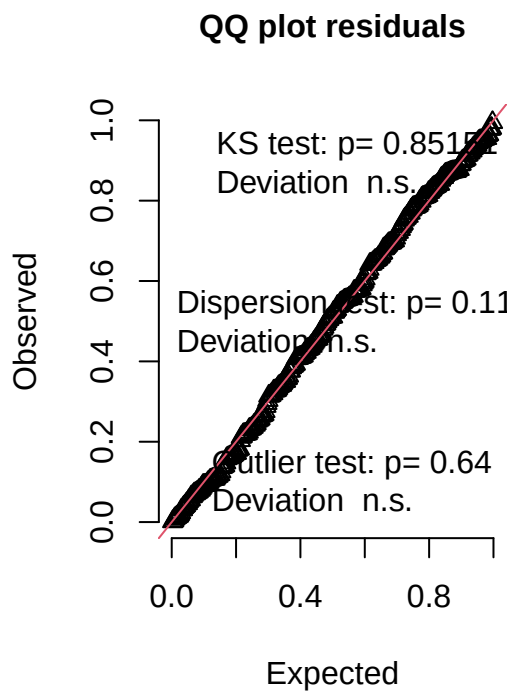
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.3931515	0.3090793	-1.2720084	0.2033701	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.3682646	0.3197436	-1.1517497	0.2494239	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.7728353	0.3458189	-2.2347975	0.0254306	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.7740098	0.3607445	-2.1455899	0.0319057	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.2208794	0.3203893	-0.6894095	0.4905656	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.2846858	0.3119506	-0.9125990	0.3614535	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-1.3655535	0.3936796	-3.4686923	0.0005230	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.8259524	0.3981860	-2.0742879	0.0380526	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.9155385	0.3877411	-2.3612110	0.0182154	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-1.3696233	0.4306481	-3.1803771	0.0014708	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-1.1076974	0.4324578	-2.5613998	0.0104251	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.6573342	0.4137352	-1.5887798	0.1121101	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-1.1217866	0.4748088	-2.3626069	0.0181469	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.6188482	0.4154817	-1.4894715	0.1363633	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.0391317	0.6658977	-1.5604975	0.1186424	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-1.0329659	0.6692427	-1.5434848	0.1227132	Poisson	BirdID+Terr

Territory only model is performing well on all selection criteria.

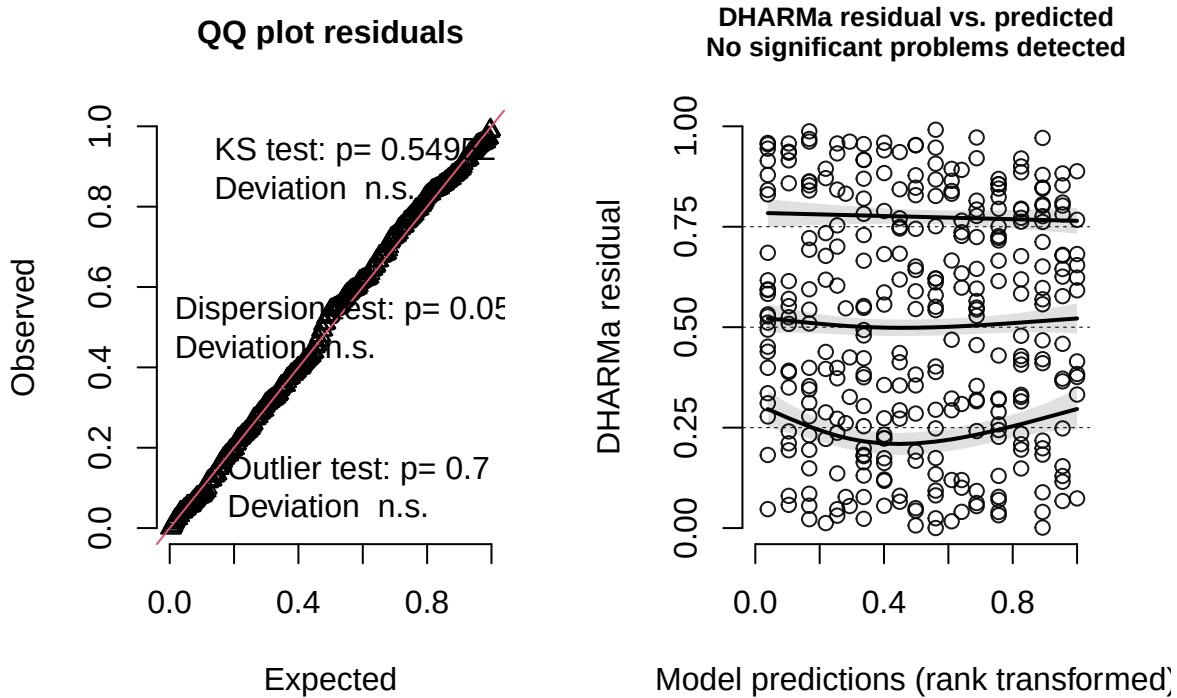
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics on the territory only model look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
327	1	-195.7668	435.5335	518.9126	252.0684	305	0.717	Binomial	BirdID+Terr
327	1	-197.3260	436.6519	516.2411	290.7631	306	0.648	Binomial	BirdID
327	1	-198.9401	439.8803	519.4694	319.5694	306	0.584	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.2072396	Binomial	BirdID
ran_pars	Terr	sd__(Intercept)	0.9648472	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.1652836	Binomial	BirdID+Terr
ran_pars	Terr	sd__(Intercept)	0.8564104	Binomial	BirdID+Terr

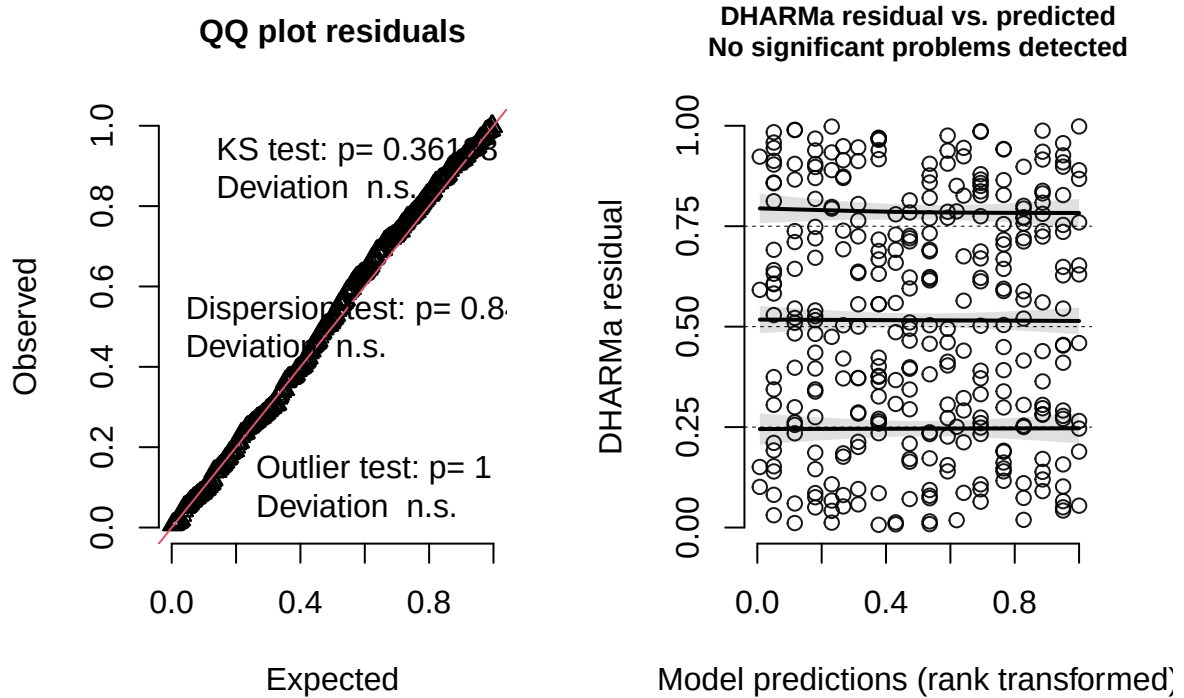
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	2.6134814	1.154019	2.2646789	0.0235324	Binomial	BirdID
fixed	factor(fac.yr)2	-1.1688912	1.280742	-0.9126673	0.3614175	Binomial	BirdID
fixed	factor(fac.yr)3	-3.6560011	1.249082	-2.9269496	0.0034230	Binomial	BirdID
fixed	factor(fac.yr)4	-2.4455623	1.235876	-1.9788088	0.0478375	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-1.5744585	1.265932	-1.2437144	0.2136047	Binomial	BirdID
fixed	factor(fac.yr)6	-1.1458911	1.298704	-0.8823343	0.3775961	Binomial	BirdID
fixed	factor(fac.yr)7	-2.2911917	1.257561	-1.8219331	0.0684651	Binomial	BirdID
fixed	factor(fac.yr)8	-2.3549013	1.280092	-1.8396342	0.0658220	Binomial	BirdID
fixed	factor(fac.yr)9	-1.1708646	1.334535	-0.8773577	0.3802924	Binomial	BirdID
fixed	factor(fac.yr)10	-1.4094830	1.287701	-1.0945731	0.2737037	Binomial	BirdID
fixed	factor(fac.yr)11	-3.0044168	1.260873	-2.3828078	0.0171812	Binomial	BirdID
fixed	factor(fac.yr)12	-2.2502103	1.330386	-1.6913961	0.0907612	Binomial	BirdID
fixed	factor(fac.yr)13	-2.7804124	1.298238	-2.1416811	0.0322192	Binomial	BirdID
fixed	factor(fac.yr)14	-3.5718066	1.314984	-2.7162354	0.0066029	Binomial	BirdID
fixed	factor(fac.yr)15	-2.8191942	1.328820	-2.1215772	0.0338733	Binomial	BirdID
fixed	factor(fac.yr)16	-2.4161576	1.388626	-1.7399632	0.0818655	Binomial	BirdID
fixed	factor(fac.yr)17	-2.9769812	1.374043	-2.1665846	0.0302665	Binomial	BirdID
fixed	factor(fac.yr)18	-2.0826838	1.396856	-1.4909799	0.1359668	Binomial	BirdID
fixed	factor(fac.yr)19	-2.2268271	1.686392	-1.3204683	0.1866787	Binomial	BirdID
fixed	factor(fac.yr)20	-3.8966428	1.871736	-2.0818333	0.0373577	Binomial	BirdID
fixed	(Intercept)	2.5763294	1.114062	2.3125546	0.0207471	Binomial	Terr
fixed	factor(fac.yr)2	-1.1713638	1.248655	-0.9381006	0.3481927	Binomial	Terr
fixed	factor(fac.yr)3	-3.3533172	1.207085	-2.7780297	0.0054690	Binomial	Terr
fixed	factor(fac.yr)4	-2.3508988	1.194414	-1.9682437	0.0490400	Binomial	Terr
fixed	factor(fac.yr)5	-1.4227362	1.212726	-1.1731722	0.2407268	Binomial	Terr
fixed	factor(fac.yr)6	-1.0617862	1.241374	-0.8553315	0.3923677	Binomial	Terr
fixed	factor(fac.yr)7	-2.2184508	1.205080	-1.8409166	0.0656338	Binomial	Terr
fixed	factor(fac.yr)8	-2.3453959	1.228092	-1.9097883	0.0561605	Binomial	Terr
fixed	factor(fac.yr)9	-0.9780474	1.275660	-0.7666989	0.4432605	Binomial	Terr
fixed	factor(fac.yr)10	-1.4759515	1.225570	-1.2042977	0.2284745	Binomial	Terr
fixed	factor(fac.yr)11	-3.1079328	1.208091	-2.5725989	0.0100938	Binomial	Terr
fixed	factor(fac.yr)12	-2.0078626	1.258947	-1.5948745	0.1107403	Binomial	Terr
fixed	factor(fac.yr)13	-2.7592708	1.235817	-2.2327506	0.0255654	Binomial	Terr
fixed	factor(fac.yr)14	-3.6541539	1.258333	-2.9039638	0.0036847	Binomial	Terr
fixed	factor(fac.yr)15	-2.7424453	1.269557	-2.1601590	0.0307604	Binomial	Terr
fixed	factor(fac.yr)16	-2.0591936	1.324294	-1.5549366	0.1199612	Binomial	Terr
fixed	factor(fac.yr)17	-2.8069614	1.316138	-2.1327258	0.0329472	Binomial	Terr
fixed	factor(fac.yr)18	-1.9725520	1.320667	-1.4936032	0.1352794	Binomial	Terr
fixed	factor(fac.yr)19	-2.8765829	1.589626	-1.8095977	0.0703582	Binomial	Terr
fixed	factor(fac.yr)20	-3.8033412	1.732406	-2.1954103	0.0281342	Binomial	Terr
fixed	(Intercept)	2.8471270	1.224258	2.3255937	0.0200402	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-1.2298421	1.353447	-0.9086736	0.3635224	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-3.9280589	1.323607	-2.9676923	0.0030004	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-2.5564663	1.308942	-1.9530788	0.0508103	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-1.6198880	1.334549	-1.2138090	0.2248207	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-1.1553025	1.371950	-0.8420880	0.3997387	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-2.4218215	1.328469	-1.8230169	0.0683008	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-2.5980789	1.354630	-1.9179253	0.0551205	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-1.2100040	1.413986	-0.8557396	0.3921418	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-1.6126375	1.361791	-1.1842036	0.2363325	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-3.3521193	1.339815	-2.5019266	0.0123520	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-2.3184276	1.401160	-1.6546486	0.0979958	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-2.9716018	1.373346	-2.1637679	0.0304822	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-3.9778849	1.399941	-2.8414656	0.0044907	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-3.1072501	1.409379	-2.2046938	0.0274756	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-2.4988453	1.474209	-1.6950413	0.0900676	Binomial	BirdID+Terr

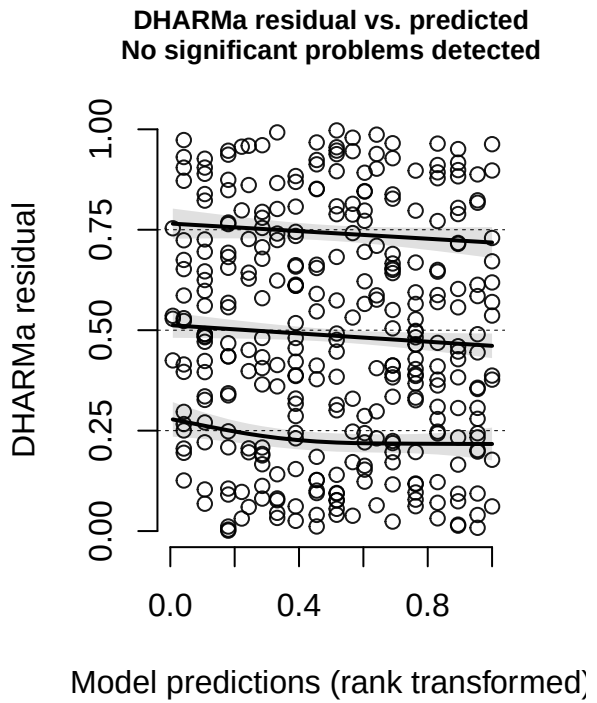
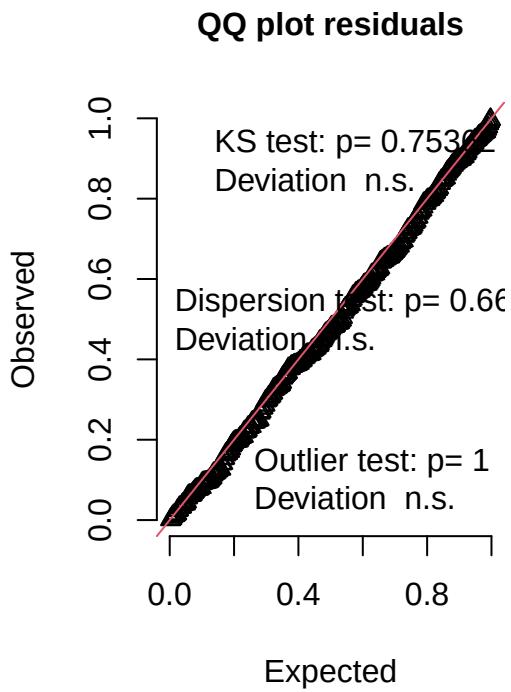
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-3.2560430	1.459653	-2.2306960	0.0257013	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-2.1658268	1.475041	-1.4683168	0.1420182	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-2.7358800	1.808510	-1.5127811	0.1303353	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-4.2365397	2.036531	-2.0802724	0.0375006	Binomial	BirdID+Terr

The territory only model is performing the worst on the model selection criteria.

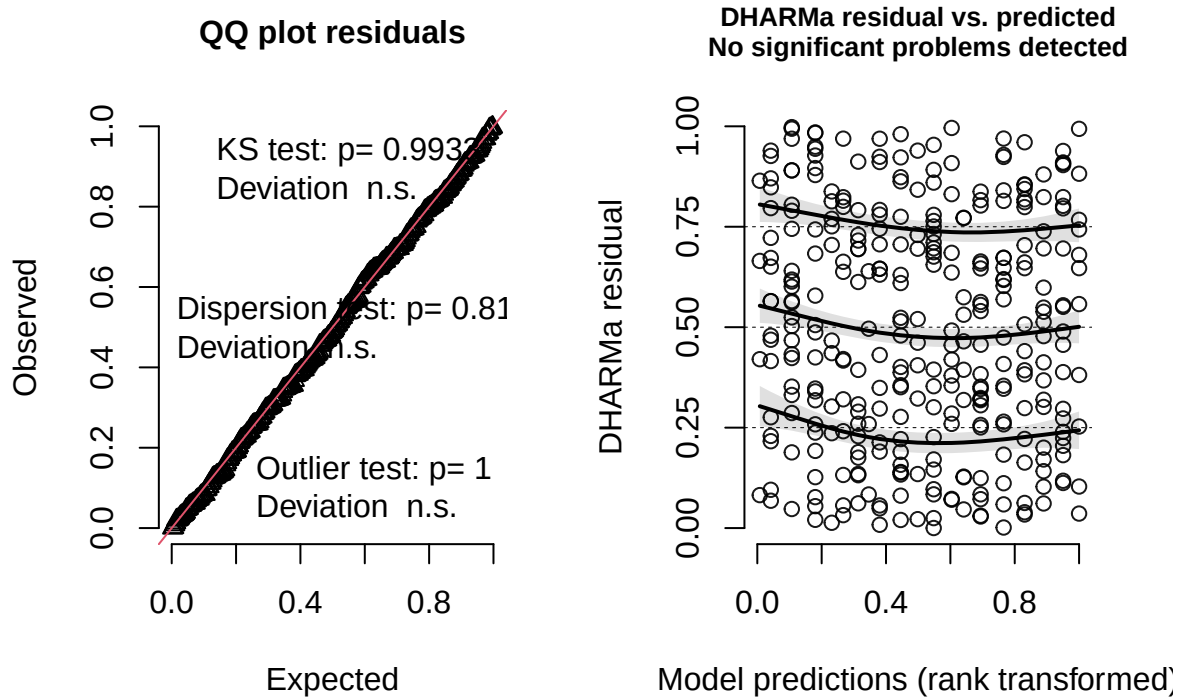
DHARMa residual



DHARMa residual



DHARMA residual



Diagnostics look good.

Probabilities used to generate simulation 14

Simulation 14
 Probability of mate fidelity: 0.746
 Probability of pair dispersal: 0.139
 Probability of single male dispersal: 0.528
 Probability of single female dispersal: 0.904

Simulation 15

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Family	RE
327	1	-423.0645	888.1290	967.7182	351.4939	306	Poisson	BirdID
327	1	-423.2117	888.4233	968.0125	363.1443	306	Poisson	Terr
327	1	-423.0645	890.1290	973.5081	351.4975	305	Poisson	BirdID+Terr

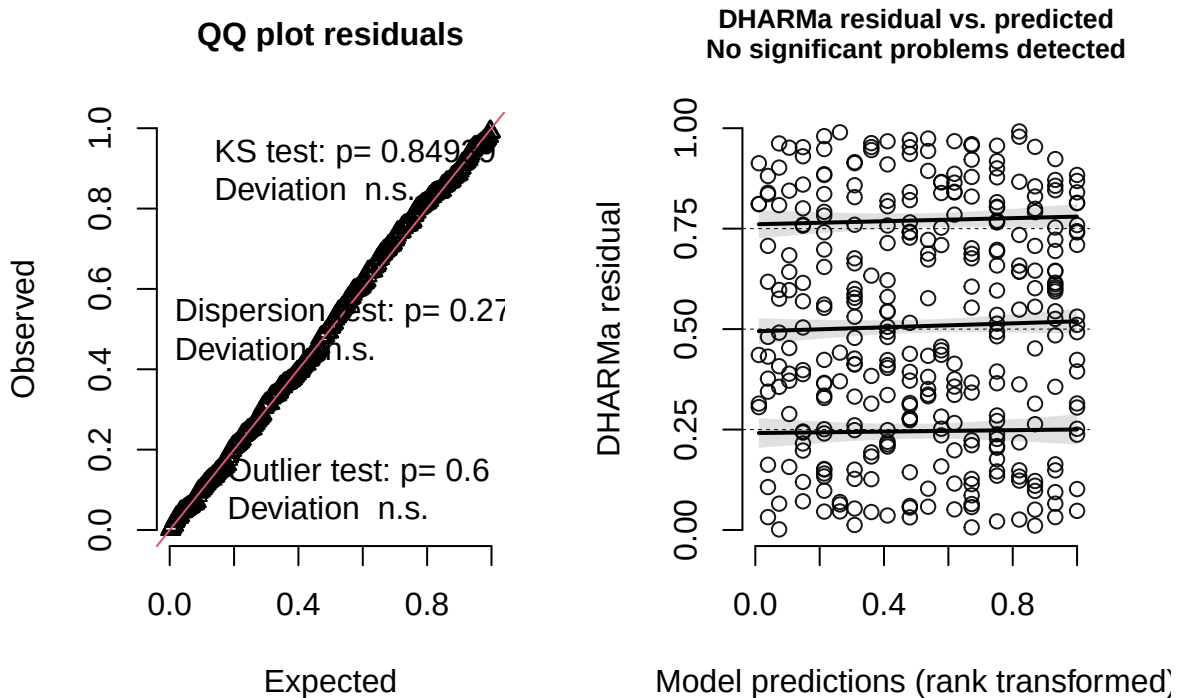
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.1316100	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.0000000	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.1315865	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.0003378	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.0704135	0.2810569	0.2505312	0.8021766	Poisson	BirdID
fixed	factor(fac.yr)2	0.4536873	0.3289402	1.3792393	0.1678210	Poisson	BirdID
fixed	factor(fac.yr)3	-0.2227954	0.3600115	-0.6188562	0.5360111	Poisson	BirdID
fixed	factor(fac.yr)4	-0.1781224	0.3601745	-0.4945446	0.6209216	Poisson	BirdID
fixed	factor(fac.yr)5	0.3364700	0.3290225	1.0226352	0.3064803	Poisson	BirdID
fixed	factor(fac.yr)6	0.1761056	0.3244780	0.5427352	0.5873122	Poisson	BirdID
fixed	factor(fac.yr)7	-0.3571355	0.3641085	-0.9808493	0.3266671	Poisson	BirdID
fixed	factor(fac.yr)8	-0.1245729	0.3574099	-0.3485436	0.7274320	Poisson	BirdID
fixed	factor(fac.yr)9	0.2973871	0.3468001	0.8575174	0.3911590	Poisson	BirdID
fixed	factor(fac.yr)10	0.1384519	0.3465886	0.3994704	0.6895466	Poisson	BirdID
fixed	factor(fac.yr)11	-0.3588183	0.3968213	-0.9042315	0.3658727	Poisson	BirdID
fixed	factor(fac.yr)12	0.0356010	0.3783235	0.0941021	0.9250281	Poisson	BirdID
fixed	factor(fac.yr)13	0.2623299	0.3613124	0.7260474	0.4678097	Poisson	BirdID
fixed	factor(fac.yr)14	-0.0243057	0.3838566	-0.0633198	0.9495118	Poisson	BirdID
fixed	factor(fac.yr)15	-0.4547942	0.4738614	-0.9597620	0.3371750	Poisson	BirdID
fixed	factor(fac.yr)16	-1.1910000	0.5764851	-2.0659683	0.0388315	Poisson	BirdID
fixed	factor(fac.yr)17	-0.1998925	0.4388215	-0.4555212	0.6487343	Poisson	BirdID
fixed	factor(fac.yr)18	-0.6004393	0.4985534	-1.2043630	0.2284493	Poisson	BirdID
fixed	factor(fac.yr)19	-1.8656540	1.0436155	-1.7876833	0.0738271	Poisson	BirdID
fixed	factor(fac.yr)20	-0.2548078	0.5323300	-0.4786651	0.6321769	Poisson	BirdID
fixed	(Intercept)	0.0800427	0.2773501	0.2885981	0.7728890	Poisson	Terr
fixed	factor(fac.yr)2	0.4505855	0.3260902	1.3817818	0.1670387	Poisson	Terr
fixed	factor(fac.yr)3	-0.2198047	0.3562627	-0.6169736	0.5372521	Poisson	Terr
fixed	factor(fac.yr)4	-0.1753529	0.3562627	-0.4922012	0.6225772	Poisson	Terr
fixed	factor(fac.yr)5	0.3398111	0.3247992	1.0462192	0.2954598	Poisson	Terr
fixed	factor(fac.yr)6	0.1823216	0.3202563	0.5692989	0.5691533	Poisson	Terr
fixed	factor(fac.yr)7	-0.3544796	0.3599370	-0.9848377	0.3247038	Poisson	Terr
fixed	factor(fac.yr)8	-0.1265627	0.3529053	-0.3586309	0.7198712	Poisson	Terr
fixed	factor(fac.yr)9	0.3056198	0.3419402	0.8937814	0.3714388	Poisson	Terr
fixed	factor(fac.yr)10	0.1431008	0.3419402	0.4184967	0.6755840	Poisson	Terr
fixed	factor(fac.yr)11	-0.3483067	0.3922323	-0.8880113	0.3745347	Poisson	Terr
fixed	factor(fac.yr)12	0.0534887	0.3733940	0.1432500	0.8860928	Poisson	Terr
fixed	factor(fac.yr)13	0.2766322	0.3562627	0.7764840	0.4374633	Poisson	Terr
fixed	factor(fac.yr)14	-0.0110498	0.3789324	-0.0291604	0.9767366	Poisson	Terr
fixed	factor(fac.yr)15	-0.4367177	0.4688072	-0.9315506	0.3515688	Poisson	Terr
fixed	factor(fac.yr)16	-1.1786550	0.5717719	-2.0614078	0.0392642	Poisson	Terr
fixed	factor(fac.yr)17	-0.1854032	0.4336291	-0.4275618	0.6689702	Poisson	Terr
fixed	factor(fac.yr)18	-0.5908683	0.4935481	-1.1971849	0.2312345	Poisson	Terr
fixed	factor(fac.yr)19	-1.8718022	1.0377490	-1.8037137	0.0712762	Poisson	Terr
fixed	factor(fac.yr)20	-0.2623643	0.5262348	-0.4985688	0.6180832	Poisson	Terr
fixed	(Intercept)	0.0701496	0.2810919	0.2495611	0.8029268	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	0.4539957	0.3289683	1.3800592	0.1675684	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.2224648	0.3600342	-0.6178991	0.5366419	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.1780396	0.3602143	-0.4942602	0.6211225	Poisson	BirdID+Terr

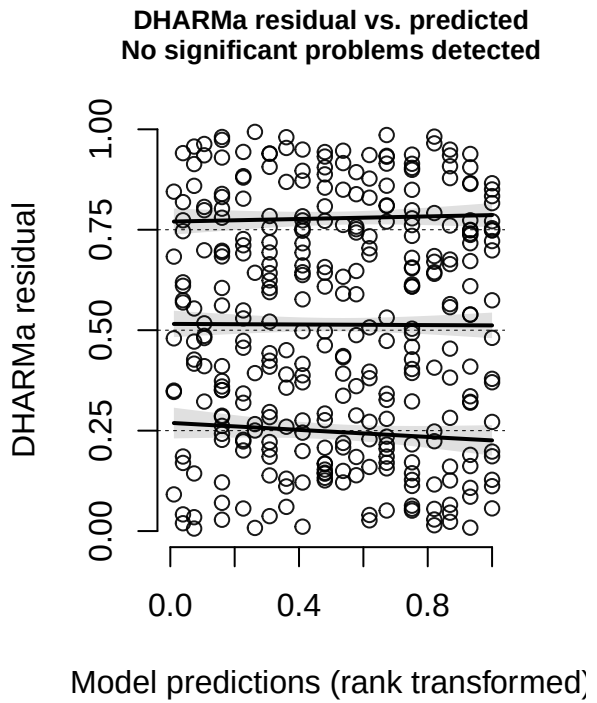
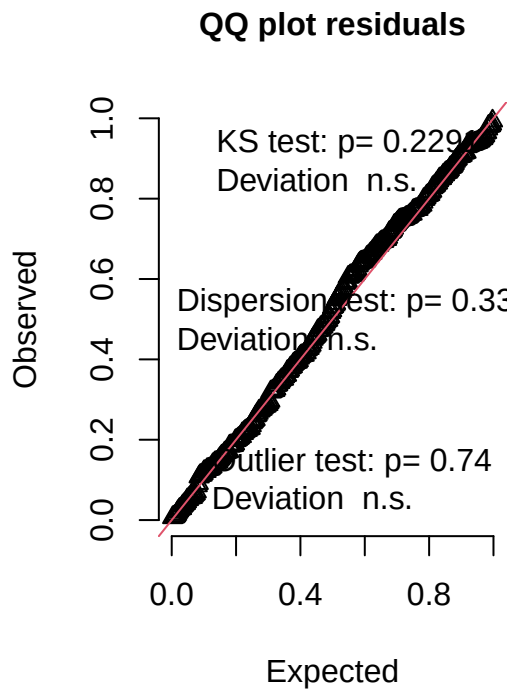
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.3366712	0.3290550	1.0231458	0.3062389	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	0.1764779	0.3245039	0.5438391	0.5865522	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.3568903	0.3641367	-0.9800998	0.3270369	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.1243306	0.3574386	-0.3478377	0.7279620	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	0.2976435	0.3468287	0.8581860	0.3907898	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	0.1387812	0.3466129	0.4003924	0.6888676	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.3584658	0.3968374	-0.9033065	0.3663632	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	0.0358263	0.3783526	0.0946902	0.9245609	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	0.2624971	0.3613461	0.7264424	0.4675676	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.0240640	0.3838840	-0.0626856	0.9500169	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.4544730	0.4738736	-0.9590596	0.3375287	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-1.1904546	0.5764408	-2.0651808	0.0389059	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.1998457	0.4388711	-0.4553630	0.6488481	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.6003458	0.4986011	-1.2040603	0.2285663	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.8651478	1.0435045	-1.7873883	0.0738747	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.2547974	0.5323942	-0.4785879	0.6322318	Poisson	BirdID+Terr

The models fitting only one random effect are performing better on AIC and BIC than the model fitted with both. Note that the band ID only model is performing slightly better than the territory only model.

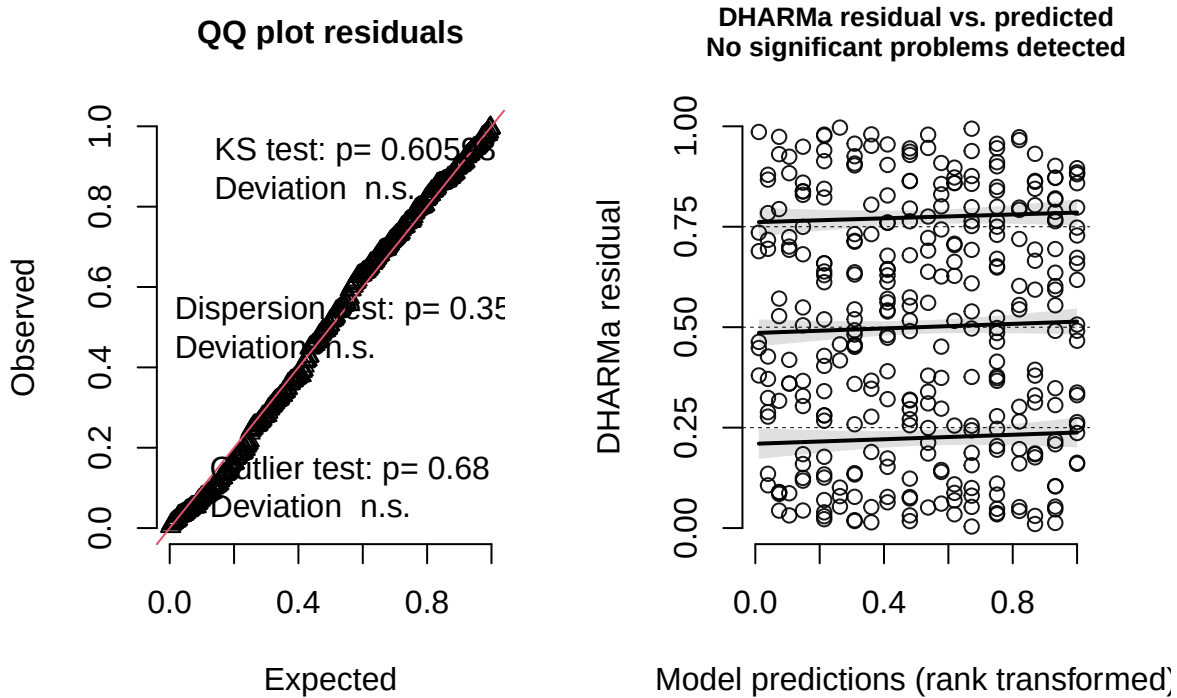
DHARMA residual



DHARMA residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
327	1	-199.0447	440.0895	519.6786	355.3955	306	0.379	Binomial	BirdID
327	1	-199.0344	442.0688	525.4479	351.2903	305	0.397	Binomial	BirdID+Terr
327	1	-200.3724	442.7448	522.3340	384.5064	306	0.329	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.6272431	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	0.3571618	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	0.6271382	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.1934513	Binomial	BirdID+Terr

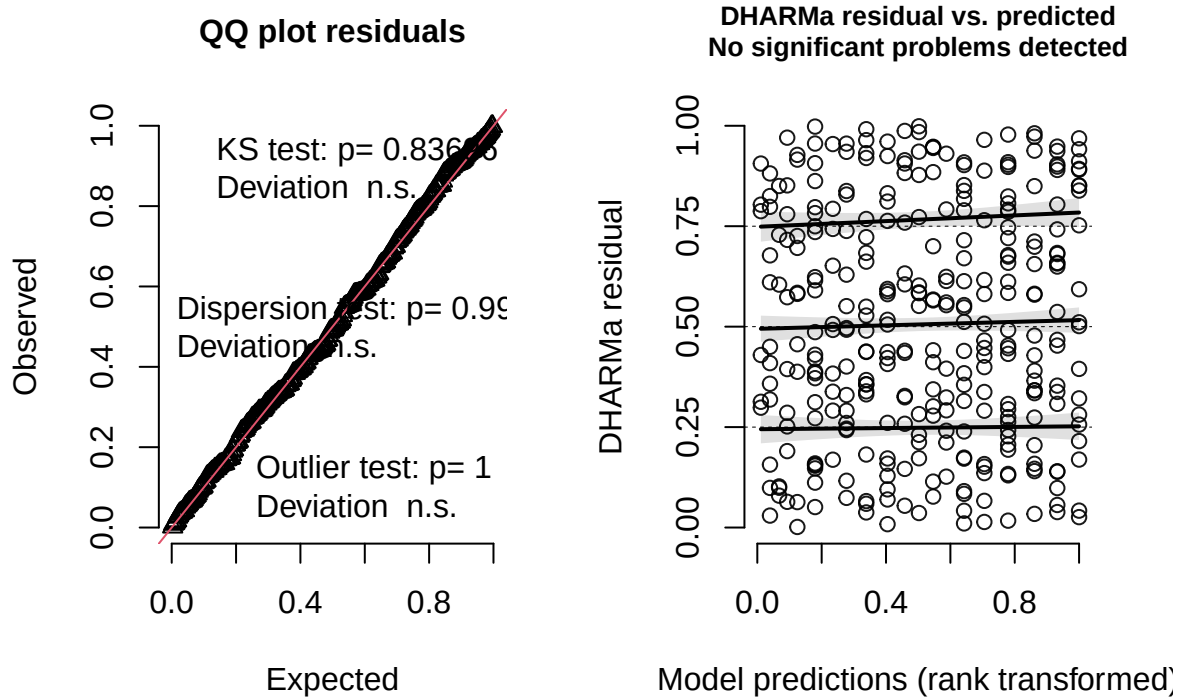
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.7530128	0.6453058	1.1669086	0.2432473	Binomial	BirdID
fixed	factor(fac.yr)2	1.5933834	1.0104034	1.5769775	0.1148007	Binomial	BirdID
fixed	factor(fac.yr)3	0.1027809	0.7958771	0.1291417	0.8972455	Binomial	BirdID
fixed	factor(fac.yr)4	0.2902423	0.8144223	0.3563781	0.7215574	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.5996969	0.8339110	0.7191378	0.4720560	Binomial	BirdID
fixed	factor(fac.yr)6	0.2874590	0.7776200	0.3696651	0.7116320	Binomial	BirdID
fixed	factor(fac.yr)7	-0.8292859	0.7693788	-1.0778643	0.2810943	Binomial	BirdID
fixed	factor(fac.yr)8	-0.5660450	0.7871660	-0.7190923	0.4720840	Binomial	BirdID
fixed	factor(fac.yr)9	0.1589388	0.8560572	0.1856638	0.8527084	Binomial	BirdID
fixed	factor(fac.yr)10	-0.3424862	0.8053686	-0.4252540	0.6706515	Binomial	BirdID
fixed	factor(fac.yr)11	-0.7183182	0.8250678	-0.8706172	0.3839632	Binomial	BirdID
fixed	factor(fac.yr)12	-0.2609040	0.8731745	-0.2987994	0.7650931	Binomial	BirdID
fixed	factor(fac.yr)13	-0.2322637	0.8729774	-0.2660592	0.7901936	Binomial	BirdID
fixed	factor(fac.yr)14	-0.1862769	0.8742237	-0.2130769	0.8312670	Binomial	BirdID
fixed	factor(fac.yr)15	-0.8227784	0.9287649	-0.8858845	0.3756797	Binomial	BirdID
fixed	factor(fac.yr)16	-1.5327724	0.9145511	-1.6759834	0.0937414	Binomial	BirdID
fixed	factor(fac.yr)17	-1.1596778	0.9401636	-1.2334852	0.2173948	Binomial	BirdID
fixed	factor(fac.yr)18	-1.1097954	0.9405960	-1.1798854	0.2380458	Binomial	BirdID
fixed	factor(fac.yr)19	-2.4343940	1.3270078	-1.8344987	0.0665800	Binomial	BirdID
fixed	factor(fac.yr)20	-1.4436945	1.1183536	-1.2909106	0.1967347	Binomial	BirdID
fixed	(Intercept)	0.7064034	0.6223827	1.1349984	0.2563760	Binomial	Terr
fixed	factor(fac.yr)2	1.5009033	0.9832669	1.5264455	0.1268990	Binomial	Terr
fixed	factor(fac.yr)3	0.1414714	0.7746763	0.1826201	0.8550961	Binomial	Terr
fixed	factor(fac.yr)4	0.3003949	0.7898628	0.3803127	0.7037133	Binomial	Terr
fixed	factor(fac.yr)5	0.6139432	0.8071708	0.7606112	0.4468893	Binomial	Terr
fixed	factor(fac.yr)6	0.3327959	0.7497602	0.4438697	0.6571368	Binomial	Terr
fixed	factor(fac.yr)7	-0.7806463	0.7413487	-1.0530083	0.2923372	Binomial	Terr
fixed	factor(fac.yr)8	-0.4992811	0.7581553	-0.6585473	0.5101865	Binomial	Terr
fixed	factor(fac.yr)9	0.2095758	0.8223740	0.2548424	0.7988448	Binomial	Terr
fixed	factor(fac.yr)10	-0.2499275	0.7749727	-0.3224984	0.7470751	Binomial	Terr
fixed	factor(fac.yr)11	-0.5472435	0.7937371	-0.6894519	0.4905389	Binomial	Terr
fixed	factor(fac.yr)12	-0.0798828	0.8396411	-0.0951392	0.9242043	Binomial	Terr
fixed	factor(fac.yr)13	-0.0598069	0.8387730	-0.0713029	0.9431567	Binomial	Terr
fixed	factor(fac.yr)14	-0.1195764	0.8393380	-0.1424652	0.8867126	Binomial	Terr
fixed	factor(fac.yr)15	-0.7240365	0.8947412	-0.8092133	0.4183925	Binomial	Terr
fixed	factor(fac.yr)16	-1.3896393	0.8796936	-1.5796855	0.1141789	Binomial	Terr
fixed	factor(fac.yr)17	-1.0822461	0.9019243	-1.1999301	0.2301665	Binomial	Terr
fixed	factor(fac.yr)18	-1.0587614	0.9019934	-1.1738018	0.2404744	Binomial	Terr
fixed	factor(fac.yr)19	-2.3315978	1.2807774	-1.8204551	0.0686897	Binomial	Terr
fixed	factor(fac.yr)20	-1.3246002	1.0755057	-1.2316068	0.2180960	Binomial	Terr
fixed	(Intercept)	0.7743864	0.6499161	1.1915175	0.2334505	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	1.5755596	1.0162702	1.5503353	0.1210611	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	0.0888523	0.8011122	0.1109112	0.9116868	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	0.2704404	0.8189830	0.3302149	0.7412376	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	0.5914730	0.8390509	0.7049310	0.4808532	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	0.2727195	0.7821305	0.3486880	0.7273235	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-0.8497736	0.7737542	-1.0982474	0.2720965	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-0.5769644	0.7918107	-0.7286645	0.4662069	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	0.1508302	0.8605869	0.1752644	0.8608719	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-0.3502804	0.8100560	-0.4324150	0.6654398	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-0.7282664	0.8299519	-0.8774802	0.3802259	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-0.2681074	0.8777837	-0.3054367	0.7600336	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-0.2353739	0.8772491	-0.2683091	0.7884614	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-0.2035391	0.8789369	-0.2315742	0.8168687	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-0.8530251	0.9341207	-0.9131851	0.3611452	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-1.5550383	0.9201091	-1.6900585	0.0910168	Binomial	BirdID+Terr

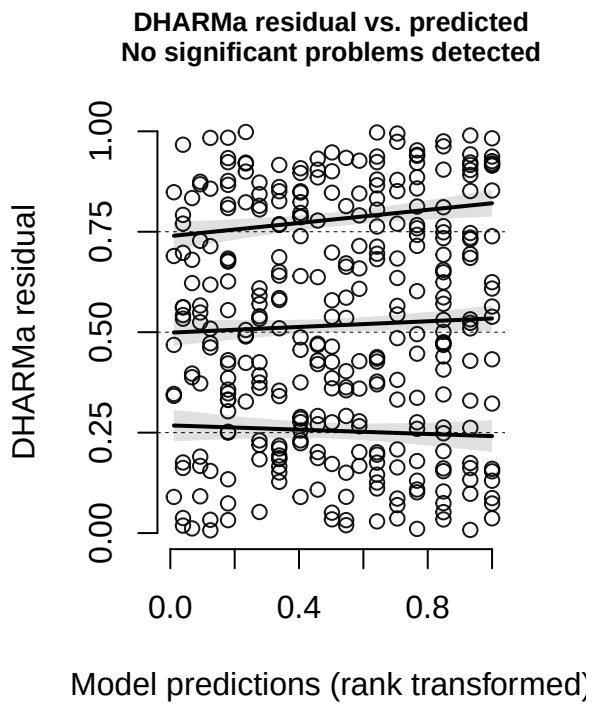
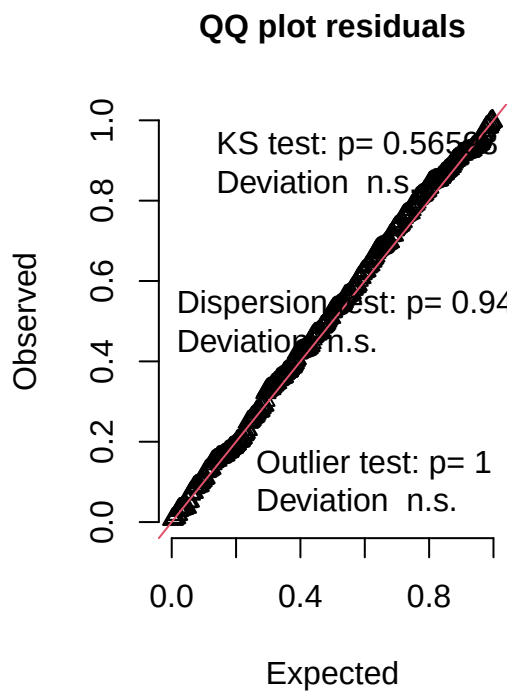
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-1.1803720	0.9447654	-1.2493811	0.2115257	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-1.1190637	0.9449675	-1.1842351	0.2363200	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-2.4781947	1.3388953	-1.8509250	0.0641803	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-1.4520091	1.1235729	-1.2923141	0.1962484	Binomial	BirdID+Terr

Similar to the previous batch of binomial models.

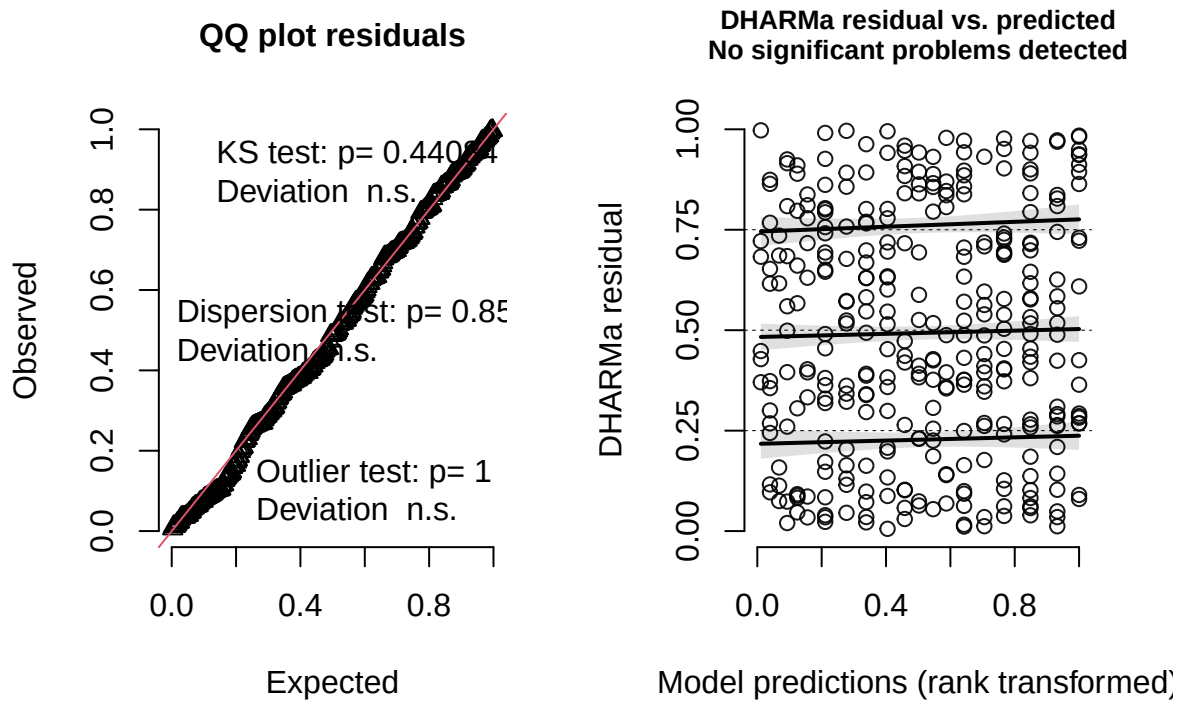
DHARMa residual



DHARMa residual



DHARMA residual



Diagnostics look good.

Probabilites used to generate simulation 15

Simulation 15
 Probability of mate fidelity: 0.77
 Probability of pair dispersal: 0.308
 Probability of single male dispersal: 0.371
 Probability of single female dispersal: 0.879

Simulation 16

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	REfamily	RE
311	1	-403.7941	849.5881	928.1238	283.6503	290	0.6367304	0.5224719	Poisson	BirdID
311	1	-403.3832	850.7665	933.0419	270.8160	289	0.6446159	0.5252362	Poisson	BirdID+Terr
311	1	-406.0567	854.1133	932.6490	311.4086	290	0.6246752	0.5463015	Poisson	Terr

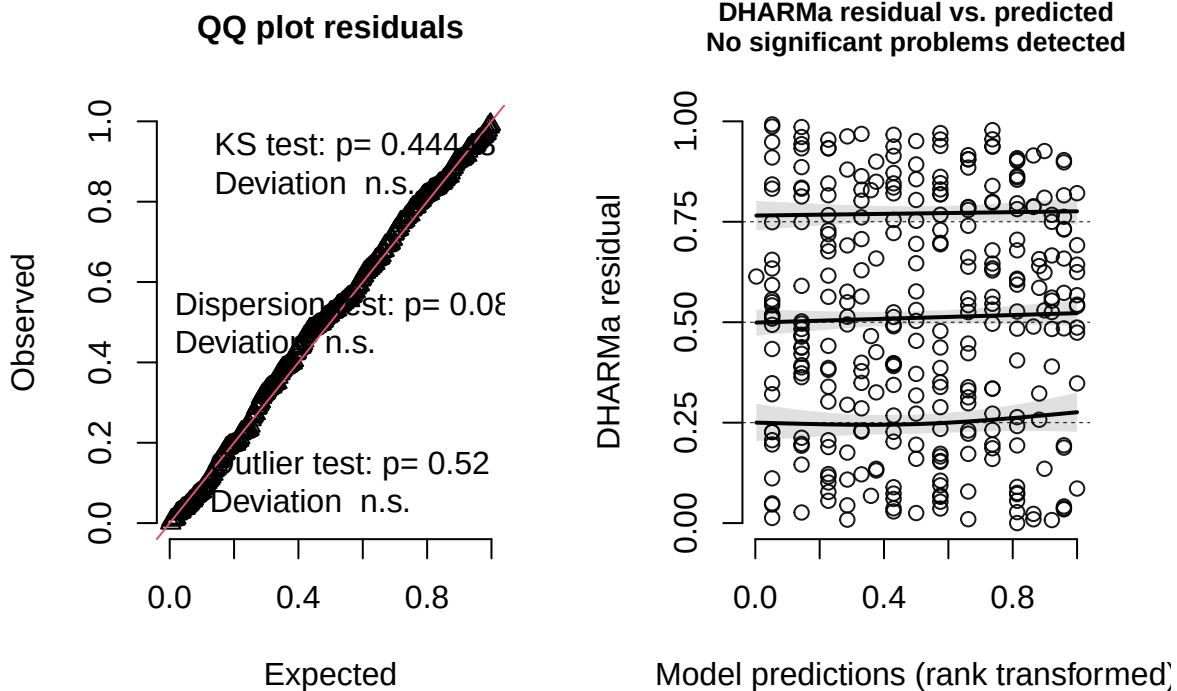
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.4502671	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.3669193	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.4038244	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.2319739	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.5264692	0.2263080	2.3263394	0.0200004	Poisson	BirdID
fixed	factor(fac.yr)2	-0.3857479	0.2822011	-1.3669255	0.1716487	Poisson	BirdID
fixed	factor(fac.yr)3	-1.4959541	0.3708462	-4.0338937	0.0000549	Poisson	BirdID
fixed	factor(fac.yr)4	-1.1649164	0.3540017	-3.2907084	0.0009994	Poisson	BirdID
fixed	factor(fac.yr)5	-0.4588111	0.2945643	-1.5575928	0.1193298	Poisson	BirdID
fixed	factor(fac.yr)6	-0.4046845	0.3025489	-1.3375839	0.1810321	Poisson	BirdID
fixed	factor(fac.yr)7	-1.1418333	0.3614832	-3.1587451	0.0015845	Poisson	BirdID
fixed	factor(fac.yr)8	-0.6432658	0.3145751	-2.0448719	0.0408675	Poisson	BirdID
fixed	factor(fac.yr)9	-0.3221085	0.2983143	-1.0797623	0.2802481	Poisson	BirdID
fixed	factor(fac.yr)10	-0.6277933	0.3478335	-1.8048673	0.0710954	Poisson	BirdID
fixed	factor(fac.yr)11	-0.8201542	0.4078031	-2.0111525	0.0443094	Poisson	BirdID
fixed	factor(fac.yr)12	-0.7563613	0.3761945	-2.0105590	0.0443721	Poisson	BirdID
fixed	factor(fac.yr)13	0.1971652	0.3330326	0.5920298	0.5538306	Poisson	BirdID
fixed	factor(fac.yr)14	-0.0380184	0.3674791	-0.1034572	0.9176001	Poisson	BirdID
fixed	factor(fac.yr)15	-0.6611220	0.4705563	-1.4049795	0.1600274	Poisson	BirdID
fixed	factor(fac.yr)16	-0.7450552	0.6657130	-1.1191837	0.2630618	Poisson	BirdID
fixed	factor(fac.yr)17	-0.0490822	0.4362046	-0.1125211	0.9104102	Poisson	BirdID
fixed	factor(fac.yr)18	-0.2470017	0.4598696	-0.5371124	0.5911899	Poisson	BirdID
fixed	factor(fac.yr)19	-0.1641237	0.4859086	-0.3377666	0.7355391	Poisson	BirdID
fixed	factor(fac.yr)20	-15.6725207	1945.0166419	-0.0080578	0.9935709	Poisson	BirdID
fixed	(Intercept)	0.6105920	0.2144985	2.8466025	0.0044189	Poisson	Terr
fixed	factor(fac.yr)2	-0.3638854	0.2862078	-1.2714028	0.2035854	Poisson	Terr
fixed	factor(fac.yr)3	-1.5618542	0.3659052	-4.2684663	0.0000197	Poisson	Terr
fixed	factor(fac.yr)4	-1.2129114	0.3406869	-3.5601941	0.0003706	Poisson	Terr
fixed	factor(fac.yr)5	-0.5318338	0.2824251	-1.8830967	0.0596873	Poisson	Terr
fixed	factor(fac.yr)6	-0.4518284	0.2915331	-1.5498358	0.1211809	Poisson	Terr
fixed	factor(fac.yr)7	-1.1697610	0.3489113	-3.3526030	0.0008006	Poisson	Terr
fixed	factor(fac.yr)8	-0.6594473	0.2978887	-2.2137375	0.0268468	Poisson	Terr
fixed	factor(fac.yr)9	-0.3943155	0.2836871	-1.3899664	0.1645391	Poisson	Terr
fixed	factor(fac.yr)10	-0.5989388	0.3329500	-1.7988852	0.0720368	Poisson	Terr
fixed	factor(fac.yr)11	-0.8076310	0.3954562	-2.0422768	0.0411241	Poisson	Terr
fixed	factor(fac.yr)12	-0.7933526	0.3613539	-2.1955002	0.0281277	Poisson	Terr
fixed	factor(fac.yr)13	0.0948096	0.3078354	0.3079878	0.7580916	Poisson	Terr
fixed	factor(fac.yr)14	-0.0258286	0.3450686	-0.0748506	0.9403336	Poisson	Terr
fixed	factor(fac.yr)15	-0.6589303	0.4568211	-1.4424253	0.1491824	Poisson	Terr
fixed	factor(fac.yr)16	-0.9241542	0.6510621	-1.4194562	0.1557661	Poisson	Terr
fixed	factor(fac.yr)17	0.1057014	0.4058223	0.2604622	0.7945072	Poisson	Terr
fixed	factor(fac.yr)18	-0.3492793	0.4322277	-0.8080910	0.4190382	Poisson	Terr
fixed	factor(fac.yr)19	-0.4439310	0.4539085	-0.9780187	0.3280651	Poisson	Terr
fixed	factor(fac.yr)20	-15.7724906	1920.8567873	-0.0082112	0.9934485	Poisson	Terr
fixed	(Intercept)	0.5394544	0.2297956	2.3475406	0.0188978	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.3914912	0.2888799	-1.3552043	0.1753525	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-1.5294832	0.3758039	-4.0698967	0.0000470	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-1.1749948	0.3568468	-3.2927146	0.0009923	Poisson	BirdID+Terr

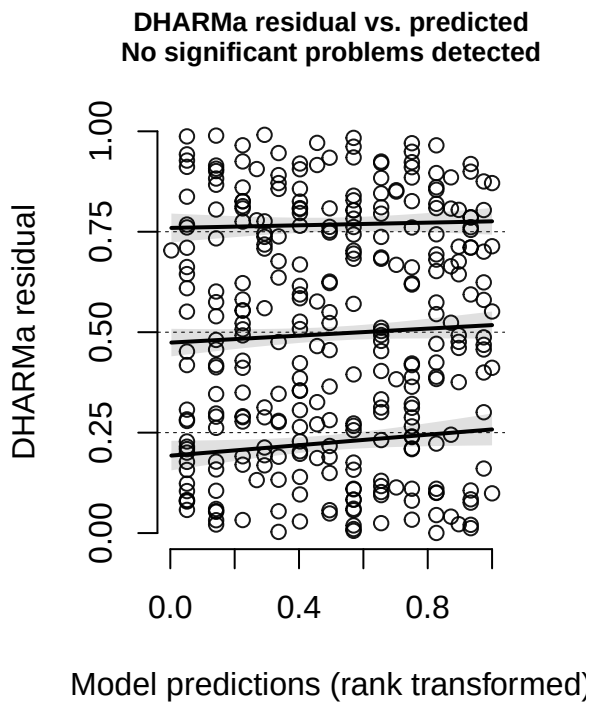
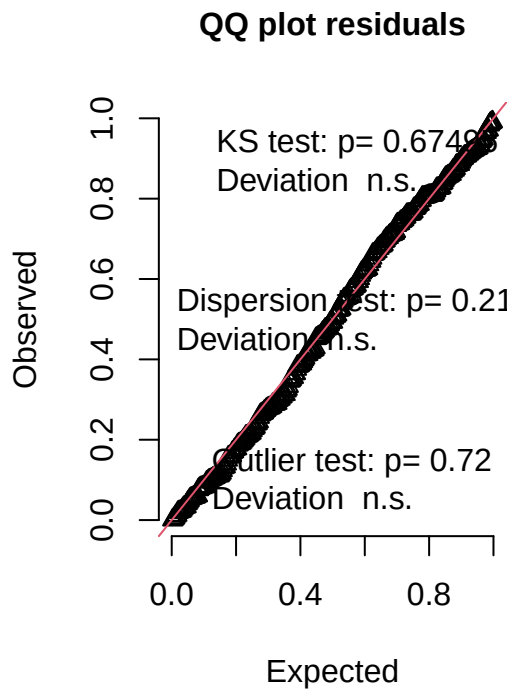
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.4924810	0.2985820	-1.6493996	0.0990658	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.4234925	0.3069161	-1.3798317	0.1676385	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-1.1547185	0.3648408	-3.1649928	0.0015509	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.6515738	0.3171949	-2.0541750	0.0399588	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.3494759	0.3018907	-1.1576239	0.2470175	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.6177528	0.3515170	-1.7573912	0.0788512	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.8463384	0.4130920	-2.0487890	0.0404828	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.7765967	0.3802568	-2.0422954	0.0411222	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	0.1580132	0.3357466	0.4706323	0.6379034	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.0322720	0.3714445	-0.0868824	0.9307650	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.6604202	0.4758092	-1.3879939	0.1651389	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.8454973	0.6768067	-1.2492448	0.2115756	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.0186661	0.4414159	-0.0422869	0.9662700	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.2496468	0.4630515	-0.5391339	0.5897945	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-0.2388373	0.4885318	-0.4888880	0.6249210	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-15.9445671	2200.8525660	-0.0072447	0.9942196	Poisson	BirdID+Terr

The territory only model is only performing best on marginal R^2 .

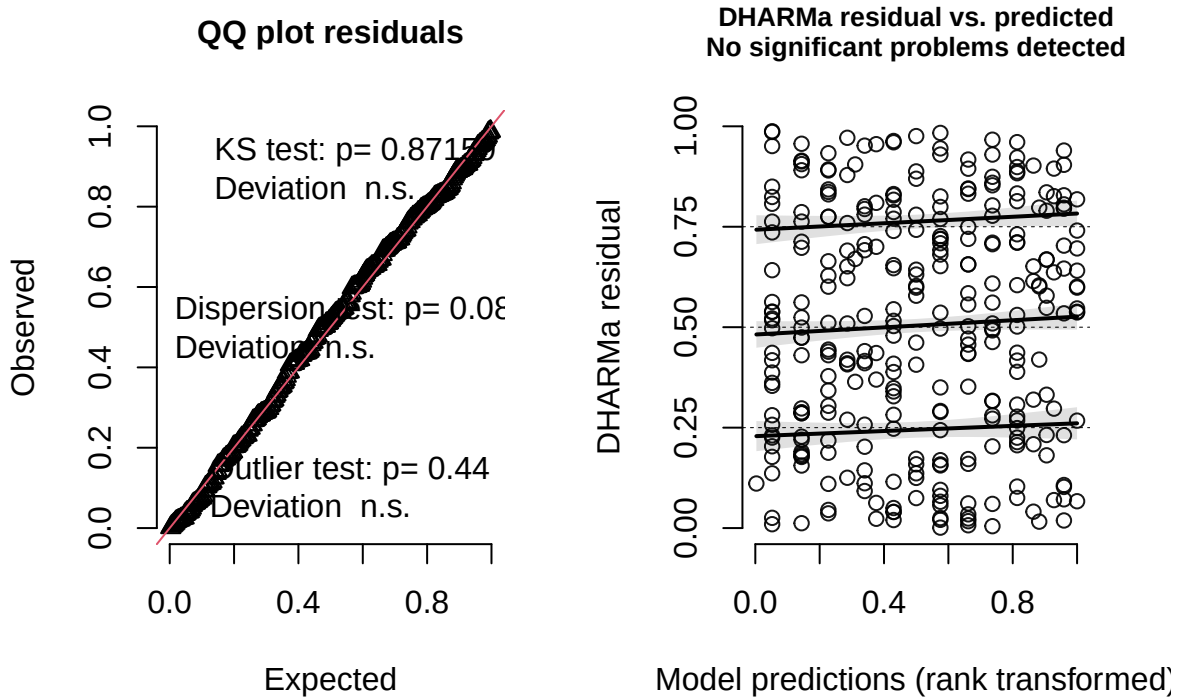
DHARMA residual



DHARMa residual



DHARMa residual



All diagnostic plots look reasonable.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
311	1	-175.4843	394.9687	477.2441	139.3563	289	0.915	Binomial	BirdID+Terr
311	1	-180.0777	402.1554	480.6910	227.2538	290	0.715	Binomial	BirdID
311	1	-181.9442	405.8885	484.4241	243.0007	290	0.686	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.652271	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.514835	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.898521	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.730740	Binomial	BirdID+Terr

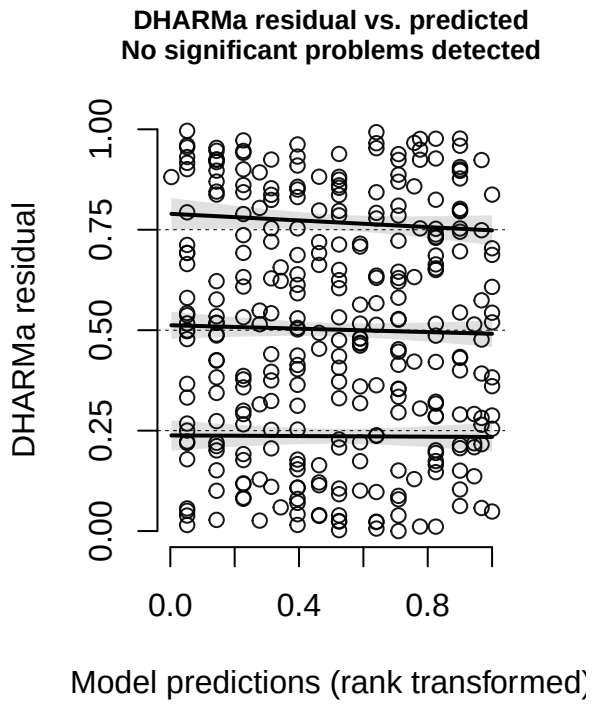
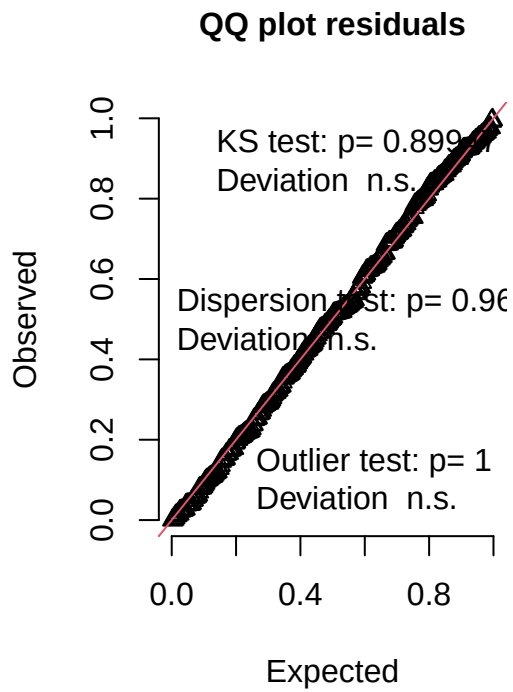
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.7036766	7.465835e-01	0.9425289	0.3459219	Binomial	BirdID
fixed	factor(fac.yr)2	0.5257773	8.924311e-01	0.5891517	0.5557595	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)3	-2.0556876	8.835699e-01	-2.3265705	0.0199881	Binomial	BirdID
fixed	factor(fac.yr)4	-1.4395370	9.039658e-01	-1.5924684	0.1112795	Binomial	BirdID
fixed	factor(fac.yr)5	-0.2985270	8.792116e-01	-0.3395395	0.7342034	Binomial	BirdID
fixed	factor(fac.yr)6	0.4940513	9.241929e-01	0.5345760	0.5929431	Binomial	BirdID
fixed	factor(fac.yr)7	-1.0919214	9.112468e-01	-1.1982719	0.2308112	Binomial	BirdID
fixed	factor(fac.yr)8	-0.1476135	9.102592e-01	-0.1621665	0.8711748	Binomial	BirdID
fixed	factor(fac.yr)9	0.2046466	9.283771e-01	0.2204348	0.8255326	Binomial	BirdID
fixed	factor(fac.yr)10	0.1392803	1.004070e+00	0.1387158	0.8896747	Binomial	BirdID
fixed	factor(fac.yr)11	-0.1601874	1.070202e+00	-0.1496797	0.8810173	Binomial	BirdID
fixed	factor(fac.yr)12	-0.5620342	1.012236e+00	-0.5552405	0.5787302	Binomial	BirdID
fixed	factor(fac.yr)13	2.6008997	1.639303e+00	1.5865885	0.1126059	Binomial	BirdID
fixed	factor(fac.yr)14	2.2881734	1.682335e+00	1.3601177	0.1737927	Binomial	BirdID
fixed	factor(fac.yr)15	-0.6379082	1.241739e+00	-0.5137215	0.6074467	Binomial	BirdID
fixed	factor(fac.yr)16	-0.5051113	1.655218e+00	-0.3051630	0.7602420	Binomial	BirdID
fixed	factor(fac.yr)17	0.3791328	1.578302e+00	0.2402156	0.8101631	Binomial	BirdID
fixed	factor(fac.yr)18	0.3833150	1.398312e+00	0.2741269	0.7839871	Binomial	BirdID
fixed	factor(fac.yr)19	1.5609429	1.696718e+00	0.9199776	0.3575845	Binomial	BirdID
fixed	factor(fac.yr)20	-	1.516604e+04	-0.0013159	0.9989500	Binomial	BirdID
		19.9573051					
fixed	(Intercept)	0.7282900	6.785328e-01	1.0733305	0.2831228	Binomial	Terr
fixed	factor(fac.yr)2	0.6713034	8.606007e-01	0.7800405	0.4353670	Binomial	Terr
fixed	factor(fac.yr)3	-2.0664505	8.448615e-01	-2.4459044	0.0144489	Binomial	Terr
fixed	factor(fac.yr)4	-1.1974407	8.151008e-01	-1.4690707	0.1418136	Binomial	Terr
fixed	factor(fac.yr)5	-0.3189056	8.144370e-01	-0.3915657	0.6953792	Binomial	Terr
fixed	factor(fac.yr)6	0.4748807	8.555759e-01	0.5550421	0.5788658	Binomial	Terr
fixed	factor(fac.yr)7	-0.8185635	8.432759e-01	-0.9706947	0.3317003	Binomial	Terr
fixed	factor(fac.yr)8	0.0906478	8.205503e-01	0.1104720	0.9120351	Binomial	Terr
fixed	factor(fac.yr)9	0.1686580	8.399154e-01	0.2008036	0.8408521	Binomial	Terr
fixed	factor(fac.yr)10	0.7487462	9.200571e-01	0.8138041	0.4157572	Binomial	Terr
fixed	factor(fac.yr)11	0.3587976	9.822878e-01	0.3652672	0.7149119	Binomial	Terr
fixed	factor(fac.yr)12	-0.4975012	9.248022e-01	-0.5379542	0.5906086	Binomial	Terr
fixed	factor(fac.yr)13	2.6004063	1.524489e+00	1.7057557	0.0880536	Binomial	Terr
fixed	factor(fac.yr)14	2.9021800	1.519969e+00	1.9093683	0.0562146	Binomial	Terr

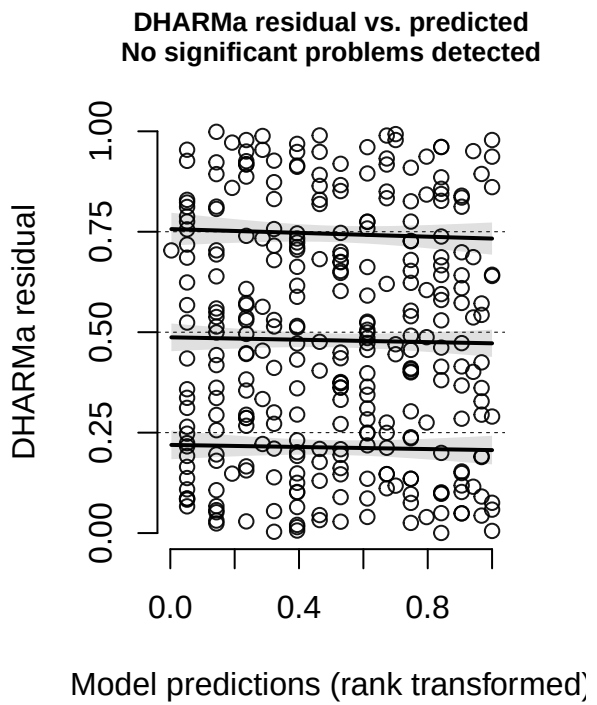
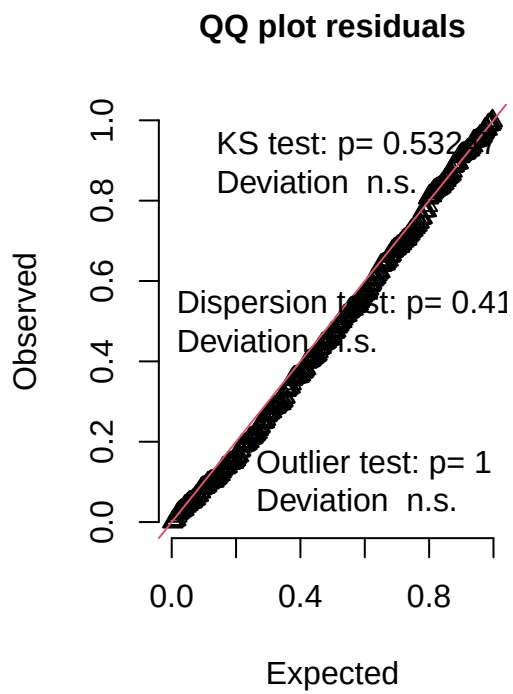
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)15	-0.6361795	1.134564e+00	-0.5607257	0.5749845	Binomial	Terr
fixed	factor(fac.yr)16	-0.9303382	1.591193e+00	-0.5846797	0.5587631	Binomial	Terr
fixed	factor(fac.yr)17	1.9525006	1.512969e+00	1.2905090	0.1968740	Binomial	Terr
fixed	factor(fac.yr)18	0.4821935	1.223839e+00	0.3940007	0.6935806	Binomial	Terr
fixed	factor(fac.yr)19	0.4004366	1.450727e+00	0.2760247	0.7825291	Binomial	Terr
fixed	factor(fac.yr)20	-	3.124358e+03	-0.0057059	0.9954474	Binomial	Terr
		17.8272350					
fixed	(Intercept)	1.1381156	9.317372e-01	1.2214985	0.2218973	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	0.4763511	1.075501e+00	0.4429109	0.6578302	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-3.1280558	1.116343e+00	-2.8020558	0.0050778	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-1.6903022	1.092559e+00	-1.5471038	0.1218382	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-0.5366485	1.092898e+00	-0.4910325	0.6234035	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	0.4659391	1.149008e+00	0.4055143	0.6850995	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-1.5250600	1.146055e+00	-1.3307045	0.1832863	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-0.0225568	1.127968e+00	-0.0199977	0.9840452	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	0.2181990	1.153633e+00	0.1891407	0.8499825	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	0.6351701	1.234638e+00	0.5144585	0.6069315	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-0.0370168	1.356357e+00	-0.0272913	0.9782274	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-0.9649806	1.249373e+00	-0.7723716	0.4398944	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	3.5894231	2.192909e+00	1.6368321	0.1016656	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	3.3504199	2.225411e+00	1.5055288	0.1321882	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-1.2802345	1.537741e+00	-0.8325421	0.4051030	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-1.9146506	2.014285e+00	-0.9505359	0.3418400	Binomial	BirdID+Terr
fixed	factor(fac.yr)17	0.7977937	1.900305e+00	0.4198241	0.6746140	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	0.8664098	1.710641e+00	0.5064826	0.6125179	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	1.0418339	2.010000e+00	0.5183252	0.6042314	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-	1.684220e+02	-0.0730569	0.9417608	Binomial	BirdID+Terr
		12.3043927					

Similar to the previous batch of binomial models.

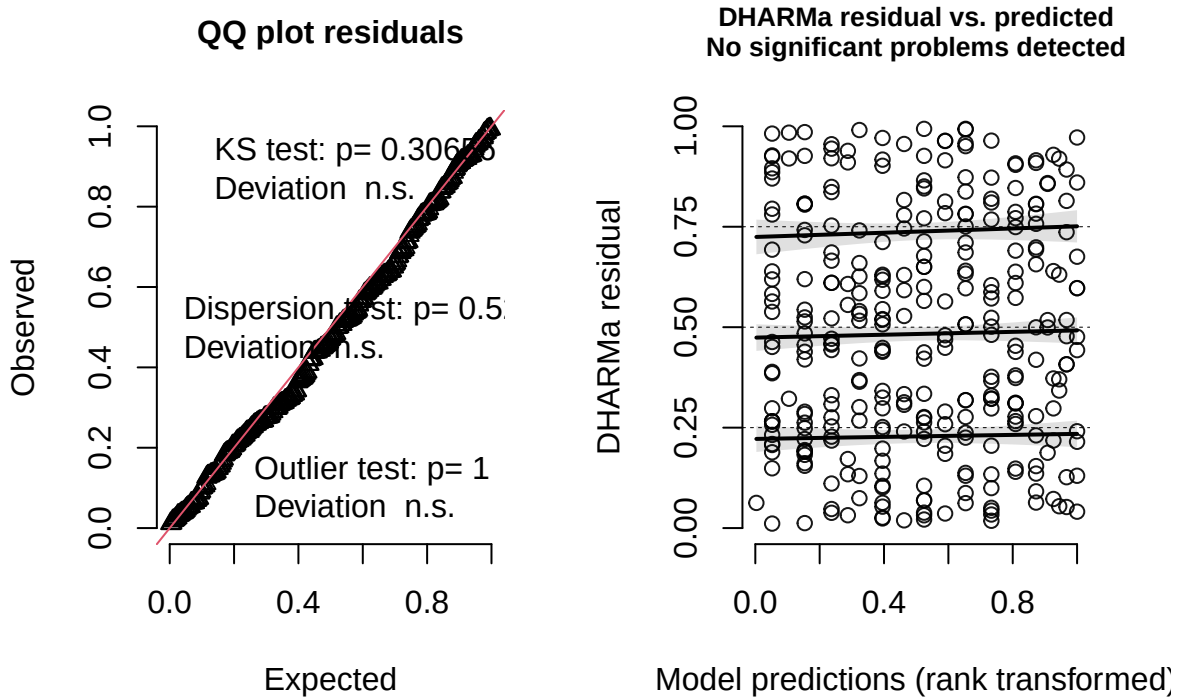
DHARMa residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Probabilites used to generate simulation 16

Simulation 16
 Probability of mate fidelity: 0.144
 Probability of pair dispersal: 0.192
 Probability of single male dispersal: 0.514
 Probability of single female dispersal: 0.219

Simulation 17

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
259	1	-360.9984	763.9967	838.6901	267.4453	238	0.2268677	0.1556790	Poisson	BirdID
259	1	-361.6970	765.3939	840.0873	274.0063	238	0.2142363	0.1581267	Poisson	Terr
259	1	-360.9960	765.9919	844.2421	266.2094	237	0.2292486	0.1557438	Poisson	BirdID+Terr

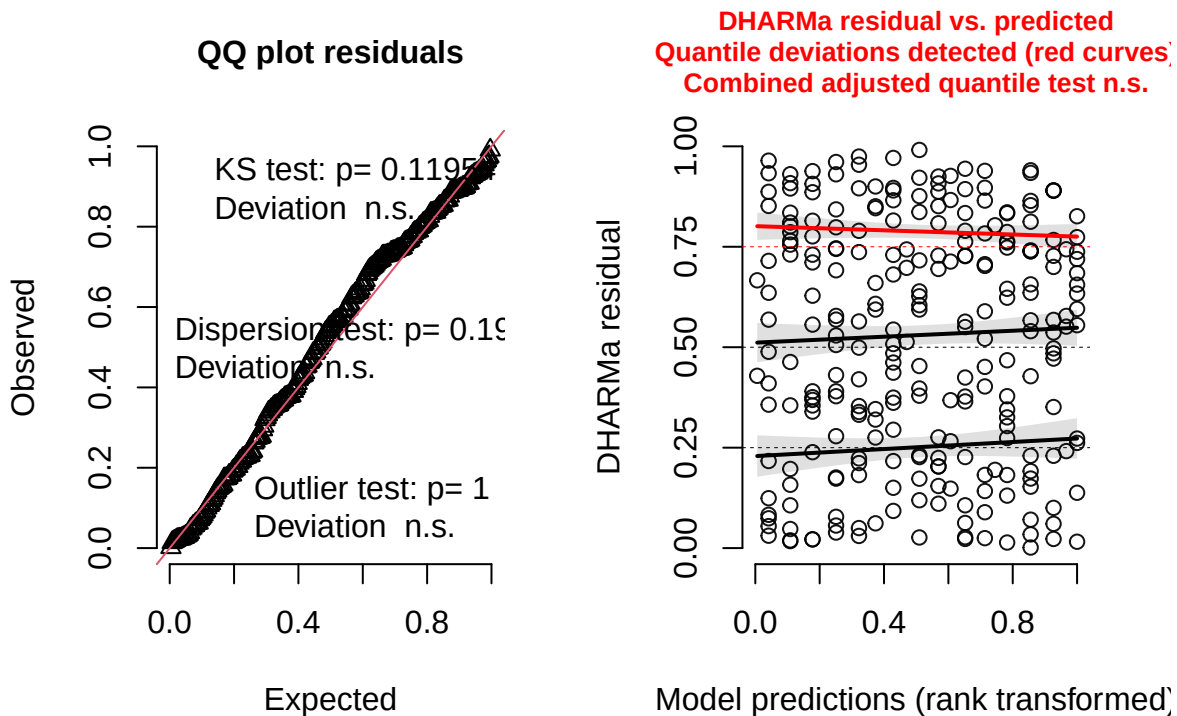
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd____(Intercept)	0.2330416	Poisson	BirdID
ran_pars	Terrs	sd____(Intercept)	0.2042373	Poisson	Terr
ran_pars	birdID	sd____(Intercept)	0.2289060	Poisson	BirdID+Terr
ran_pars	Terrs	sd____(Intercept)	0.0606568	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.2396209	0.2531626	0.9465099	0.3438886	Poisson	BirdID
fixed	factor(fac.yr)2	-0.4036023	0.3696642	-1.0918079	0.2749175	Poisson	BirdID
fixed	factor(fac.yr)3	-0.5437469	0.3878229	-1.4020497	0.1609004	Poisson	BirdID
fixed	factor(fac.yr)4	-0.4517869	0.3734294	-1.2098321	0.2263433	Poisson	BirdID
fixed	factor(fac.yr)5	0.5415779	0.3199717	1.6925810	0.0905353	Poisson	BirdID
fixed	factor(fac.yr)6	0.1286027	0.3276522	0.3924976	0.6946906	Poisson	BirdID
fixed	factor(fac.yr)7	-0.4274884	0.3552640	-1.2032981	0.2288610	Poisson	BirdID
fixed	factor(fac.yr)8	0.2097370	0.3229316	0.6494780	0.5160295	Poisson	BirdID
fixed	factor(fac.yr)9	0.2918011	0.3087475	0.9451126	0.3446014	Poisson	BirdID
fixed	factor(fac.yr)10	-0.1265835	0.3429647	-0.3690862	0.7120635	Poisson	BirdID
fixed	factor(fac.yr)11	-0.2851395	0.3511033	-0.8121242	0.4167204	Poisson	BirdID
fixed	factor(fac.yr)12	-0.4652414	0.3668573	-1.2681808	0.2047334	Poisson	BirdID
fixed	factor(fac.yr)13	0.3155464	0.3197036	0.9869967	0.3236442	Poisson	BirdID
fixed	factor(fac.yr)14	0.1583457	0.3371457	0.4696656	0.6385939	Poisson	BirdID
fixed	factor(fac.yr)15	-0.3953444	0.4258703	-0.9283211	0.3532410	Poisson	BirdID
fixed	factor(fac.yr)16	0.0514460	0.4465012	0.1152202	0.9082706	Poisson	BirdID
fixed	factor(fac.yr)17	0.4431237	0.4487191	0.9875304	0.3233827	Poisson	BirdID
fixed	factor(fac.yr)18	0.1931335	0.6496297	0.2972979	0.7662391	Poisson	BirdID
fixed	factor(fac.yr)19	-0.2737530	0.6487420	-0.4219752	0.6730431	Poisson	BirdID
fixed	factor(fac.yr)20	-0.9711091	1.0516501	-0.9234147	0.3557911	Poisson	BirdID
fixed	(Intercept)	0.2676473	0.2502513	1.0695141	0.2848381	Poisson	Terr
fixed	factor(fac.yr)2	-0.4253674	0.3690315	-1.1526589	0.2490504	Poisson	Terr
fixed	factor(fac.yr)3	-0.5525651	0.3844575	-1.4372593	0.1506443	Poisson	Terr
fixed	factor(fac.yr)4	-0.4754519	0.3697724	-1.2857962	0.1985142	Poisson	Terr
fixed	factor(fac.yr)5	0.5292930	0.3157864	1.6761110	0.0937165	Poisson	Terr
fixed	factor(fac.yr)6	0.1098050	0.3237518	0.3391641	0.7344861	Poisson	Terr
fixed	factor(fac.yr)7	-0.4488236	0.3516850	-1.2762091	0.2018816	Poisson	Terr
fixed	factor(fac.yr)8	0.1653760	0.3185921	0.5190837	0.6037024	Poisson	Terr
fixed	factor(fac.yr)9	0.2522824	0.3038634	0.8302493	0.4063978	Poisson	Terr
fixed	factor(fac.yr)10	-0.1375328	0.3388977	-0.4058241	0.6848719	Poisson	Terr
fixed	factor(fac.yr)11	-0.2907835	0.3461793	-0.8399794	0.4009199	Poisson	Terr
fixed	factor(fac.yr)12	-0.4688682	0.3626237	-1.2929883	0.1960151	Poisson	Terr
fixed	factor(fac.yr)13	0.3340118	0.3151649	1.0598003	0.2892355	Poisson	Terr
fixed	factor(fac.yr)14	0.1692755	0.3332647	0.5079312	0.6115016	Poisson	Terr
fixed	factor(fac.yr)15	-0.4060426	0.4233026	-0.9592253	0.3374452	Poisson	Terr
fixed	factor(fac.yr)16	-0.0382266	0.4421814	-0.0864500	0.9311087	Poisson	Terr
fixed	factor(fac.yr)17	0.3529490	0.4437557	0.7953678	0.4263996	Poisson	Terr
fixed	factor(fac.yr)18	0.0720597	0.6470094	0.1113735	0.9113202	Poisson	Terr
fixed	factor(fac.yr)19	-0.3873562	0.6424330	-0.6029518	0.5465407	Poisson	Terr
fixed	factor(fac.yr)20	-1.0092225	1.0483635	-0.9626646	0.3357159	Poisson	Terr
fixed	(Intercept)	0.2435150	0.2532013	0.9617447	0.3361779	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.4082541	0.3698925	-1.1037102	0.2697189	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.5484508	0.3880865	-1.4132180	0.1575916	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.4560792	0.3735922	-1.2207940	0.2221640	Poisson	BirdID+Terr

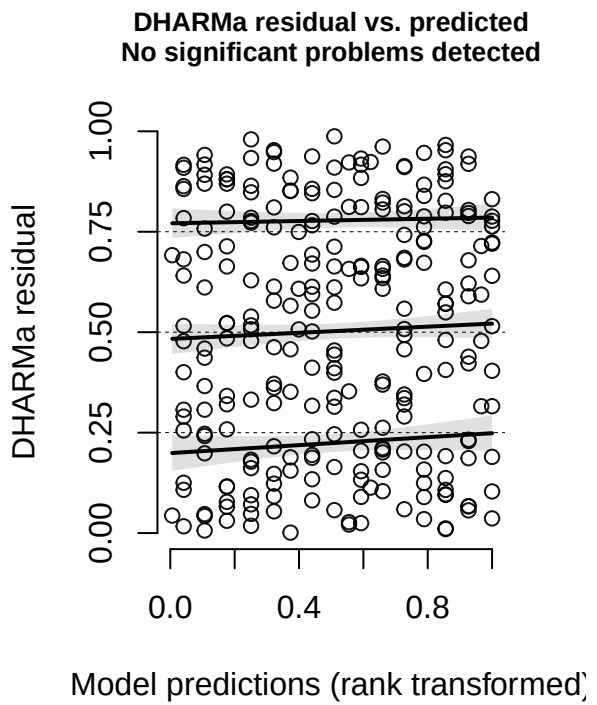
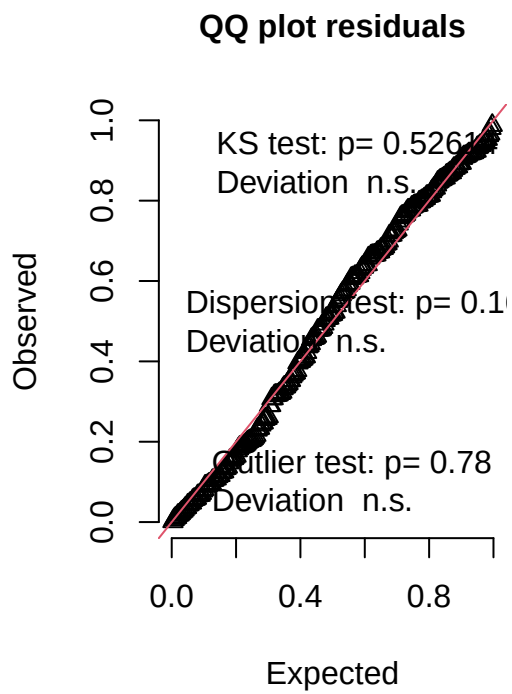
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.5372600	0.3201997	1.6778904	0.0933685	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	0.1242049	0.3278181	0.3788835	0.7047744	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.4326438	0.3554552	-1.2171540	0.2235456	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	0.2035132	0.3230553	0.6299639	0.5287182	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	0.2852989	0.3088084	0.9238701	0.3555540	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.1313115	0.3431136	-0.3827055	0.7019381	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.2897278	0.3512036	-0.8249568	0.4093961	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.4692114	0.3670199	-1.2784358	0.2010958	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	0.3142120	0.3198209	0.9824624	0.3258721	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	0.1552373	0.3373647	0.4601469	0.6454108	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.4019731	0.4264386	-0.9426284	0.3458710	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	0.0412964	0.4470860	0.0923678	0.9264058	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	0.4334614	0.4494745	0.9643739	0.3348585	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	0.1810020	0.6513429	0.2778905	0.7810964	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-0.2882846	0.6497408	-0.4436917	0.6572655	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.9780000	1.0525529	-0.9291695	0.3528013	Poisson	BirdID+Terr

The territory only model is performing best on conditional R^2 , and not by a significant margin.

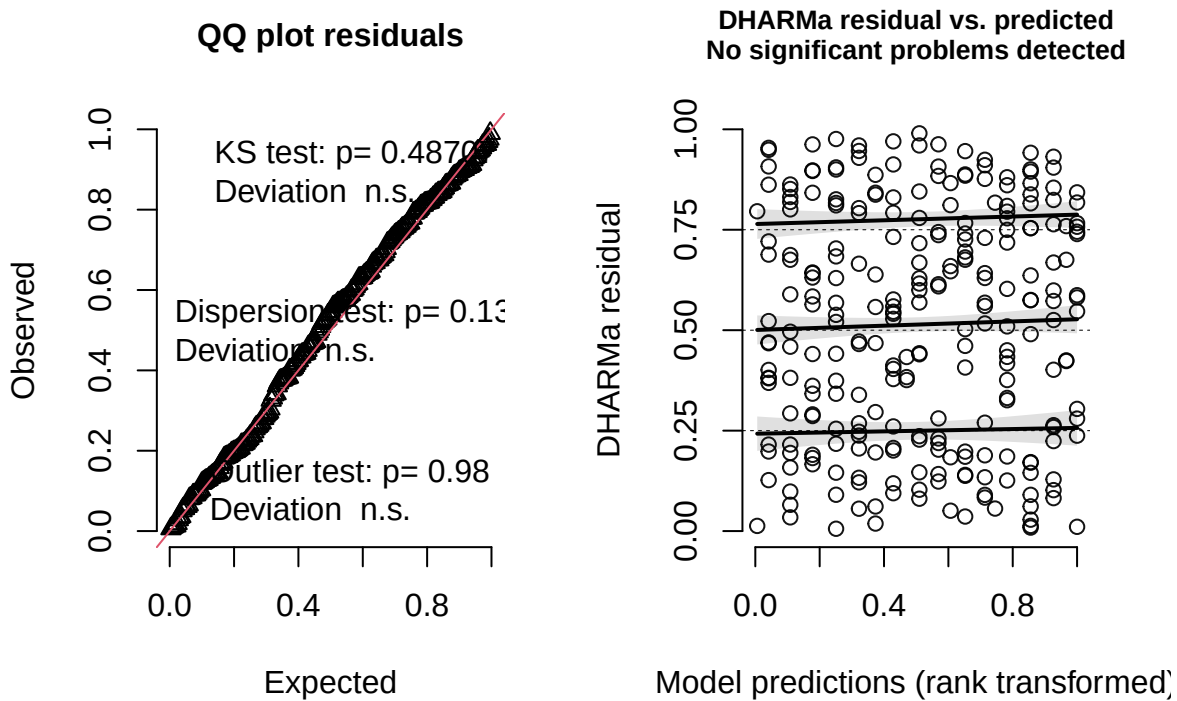
DHARMA residual



DHARMa residual



DHARMa residual



The band ID only model's diagnostics seem a bit concerning.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
259	1	-151.3283	344.6566	419.3500	258.8825	238	0.474	Binomial	BirdID
259	1	-151.3282	346.6565	424.9067	258.9059	237	0.474	Binomial	BirdID+Terr
259	1	-152.3330	346.6659	421.3593	277.3255	238	0.341	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.7497238	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	0.5535506	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	0.7493878	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.0006259	Binomial	BirdID+Terr

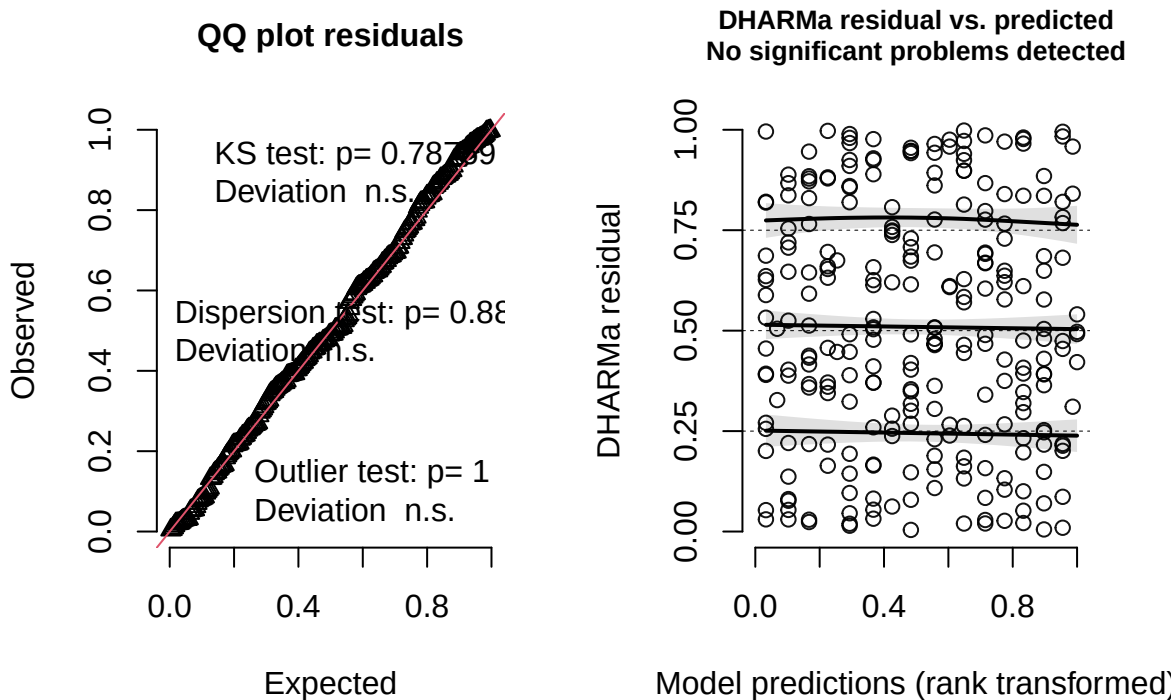
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.1914997	0.6005716	0.3188623	0.7498309	Binomial	BirdID
fixed	factor(fac.yr)2	-0.1735606	0.7860071	-0.2208130	0.8252381	Binomial	BirdID
fixed	factor(fac.yr)3	-0.0925820	0.7952989	-0.1164116	0.9073264	Binomial	BirdID
fixed	factor(fac.yr)4	-0.0279984	0.7909824	-0.0353970	0.9717632	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	2.5647828	1.2741391	2.0129534	0.0441195	Binomial	BirdID
fixed	factor(fac.yr)6	0.8783920	0.8280803	1.0607571	0.2888003	Binomial	BirdID
fixed	factor(fac.yr)7	0.2769205	0.7744524	0.3575695	0.7206655	Binomial	BirdID
fixed	factor(fac.yr)8	2.0864600	1.0108942	2.0639747	0.0390201	Binomial	BirdID
fixed	factor(fac.yr)9	0.4678576	0.7834839	0.5971503	0.5504071	Binomial	BirdID
fixed	factor(fac.yr)10	0.7260644	0.8304143	0.8743400	0.3819331	Binomial	BirdID
fixed	factor(fac.yr)11	0.5808358	0.8067827	0.7199409	0.4715614	Binomial	BirdID
fixed	factor(fac.yr)12	0.0828551	0.7881216	0.1051299	0.9162728	Binomial	BirdID
fixed	factor(fac.yr)13	1.5398911	0.9183450	1.6768112	0.0935794	Binomial	BirdID
fixed	factor(fac.yr)14	0.8805928	0.8750059	1.0063850	0.3142304	Binomial	BirdID
fixed	factor(fac.yr)15	0.3105555	0.9171580	0.3386063	0.7349064	Binomial	BirdID
fixed	factor(fac.yr)16	0.6739304	1.1009160	0.6121543	0.5404357	Binomial	BirdID
fixed	factor(fac.yr)17	19.5871724	9613.2832153	0.0020375	0.9983743	Binomial	BirdID
fixed	factor(fac.yr)18	0.0205522	1.5993514	0.0128503	0.9897472	Binomial	BirdID
fixed	factor(fac.yr)19	17.1855473	3284.8703316	0.0052317	0.9958257	Binomial	BirdID
fixed	factor(fac.yr)20	-0.1081744	1.6278040	-0.0664542	0.9470162	Binomial	BirdID
fixed	(Intercept)	0.2749902	0.5763531	0.4771210	0.6332759	Binomial	Terr
fixed	factor(fac.yr)2	-0.2888086	0.7693398	-0.3753979	0.7073646	Binomial	Terr
fixed	factor(fac.yr)3	-0.1874063	0.7661099	-0.2446206	0.8067502	Binomial	Terr
fixed	factor(fac.yr)4	-0.1571102	0.7626251	-0.2060123	0.8367813	Binomial	Terr
fixed	factor(fac.yr)5	2.3812697	1.2377208	1.9239151	0.0543652	Binomial	Terr
fixed	factor(fac.yr)6	0.7038421	0.7966923	0.8834553	0.3769903	Binomial	Terr
fixed	factor(fac.yr)7	0.1766806	0.7438943	0.2375077	0.8122630	Binomial	Terr
fixed	factor(fac.yr)8	1.8148598	0.9755899	1.8602690	0.0628475	Binomial	Terr
fixed	factor(fac.yr)9	0.3609566	0.7492388	0.4817645	0.6299733	Binomial	Terr
fixed	factor(fac.yr)10	0.6449236	0.7972899	0.8088948	0.4185757	Binomial	Terr
fixed	factor(fac.yr)11	0.4822398	0.7714883	0.6250773	0.5319203	Binomial	Terr
fixed	factor(fac.yr)12	-0.0380390	0.7536361	-0.0504740	0.9597447	Binomial	Terr
fixed	factor(fac.yr)13	1.3466510	0.8815321	1.5276256	0.1266055	Binomial	Terr
fixed	factor(fac.yr)14	0.7105672	0.8432075	0.8426955	0.3993987	Binomial	Terr
fixed	factor(fac.yr)15	0.1571926	0.8852299	0.1775727	0.8590586	Binomial	Terr
fixed	factor(fac.yr)16	0.3995249	1.0676420	0.3742124	0.7082463	Binomial	Terr
fixed	factor(fac.yr)17	17.2213839	3244.2836396	0.0053082	0.9957647	Binomial	Terr
fixed	factor(fac.yr)18	-0.4872080	1.5752808	-0.3092833	0.7571060	Binomial	Terr
fixed	factor(fac.yr)19	14.2095515	893.7309769	0.0158991	0.9873149	Binomial	Terr
fixed	factor(fac.yr)20	-0.3473242	1.5881745	-0.2186940	0.8268884	Binomial	Terr
fixed	(Intercept)	0.1938318	0.6005776	0.3227424	0.7468903	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-0.1770879	0.7860077	-0.2253005	0.8217455	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-0.0920764	0.7952886	-0.1157774	0.9078290	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-0.0273760	0.7909965	-0.0346096	0.9723911	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	2.5660134	1.2755060	2.0117612	0.0442451	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	0.8751068	0.8279925	1.0569018	0.2905564	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	0.2754338	0.7744595	0.3556464	0.7221054	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	2.0852057	1.0111392	2.0622341	0.0391855	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	0.4653277	0.7834616	0.5939381	0.5525535	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	0.7237236	0.8304013	0.8715347	0.3834623	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	0.5744851	0.8065570	0.7122685	0.4762986	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	0.0826324	0.7881412	0.1048447	0.9164990	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	1.5410126	0.9188612	1.6770896	0.0935250	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	0.8778475	0.8749625	1.0032973	0.3157175	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	0.3061632	0.9170443	0.3338587	0.7384862	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	0.6730049	1.1010739	0.6112259	0.5410501	Binomial	BirdID+Terr

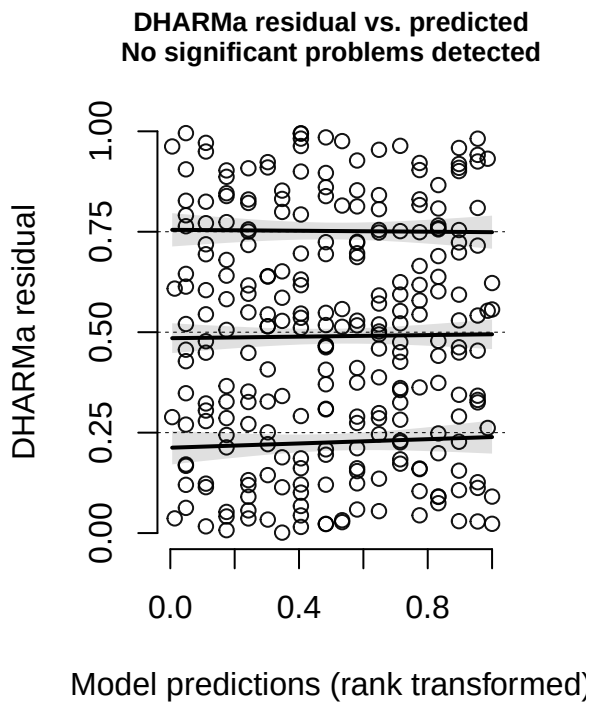
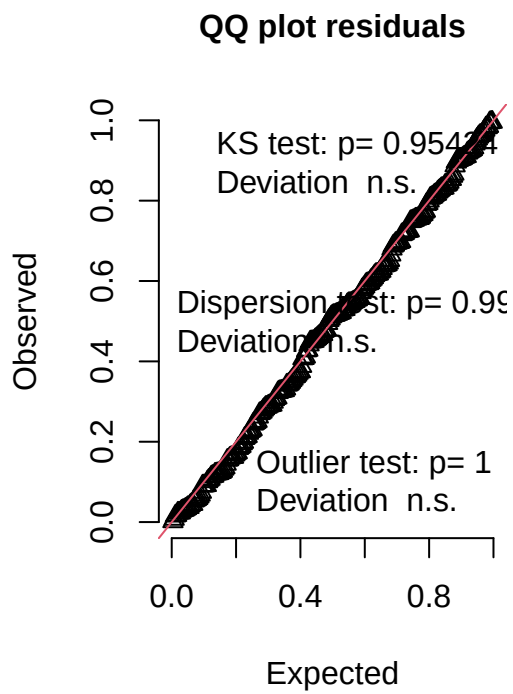
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	12.3390758	256.7620521	0.0480565	0.9616712	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	0.0194192	1.5992972	0.0121423	0.9903121	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	16.3853869	2204.7906680	0.0074317	0.9940704	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-0.1164524	1.6277527	-0.0715418	0.9429665	Binomial	BirdID+Terr

Similar to the previous batch of binomial models.

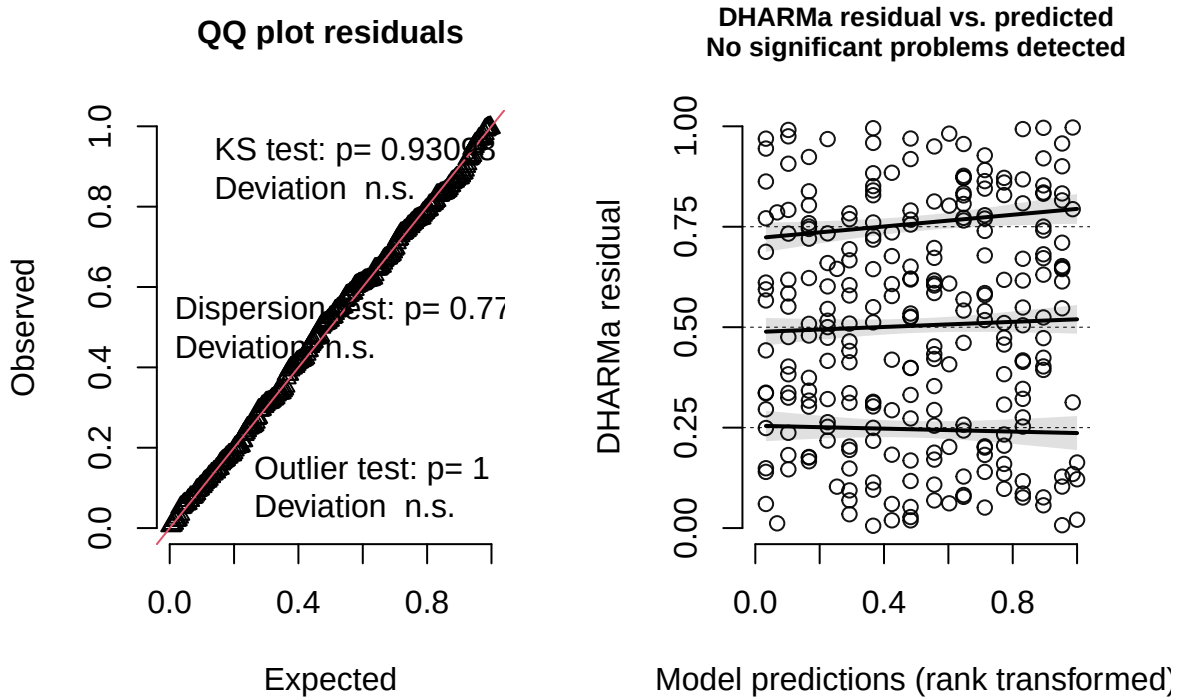
DHARMa residual



DHARMa residual



DHARMa residual



Diagnostics look reasonable.

Probabilites used to generate simulation 17

Simulation 17

Probability of mate fidelity: 0.184

Probability of pair dispersal: 0.526

Probability of single male dispersal: 0.223

Probability of single female dispersal: 0.978

Simulation 18

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
350	1	- 453.0785	948.1569	1029.174	311.0584	329	0.4092422	0.1932681	Poisson	Terr
350	1	- 452.4499	948.8997	1033.774	289.3140	328	0.4252632	0.1893273	Poisson	BirdID+Terr
350	1	- 459.7759	961.5518	1042.568	342.7290	329	0.3685965	0.1941239	Poisson	BirdID

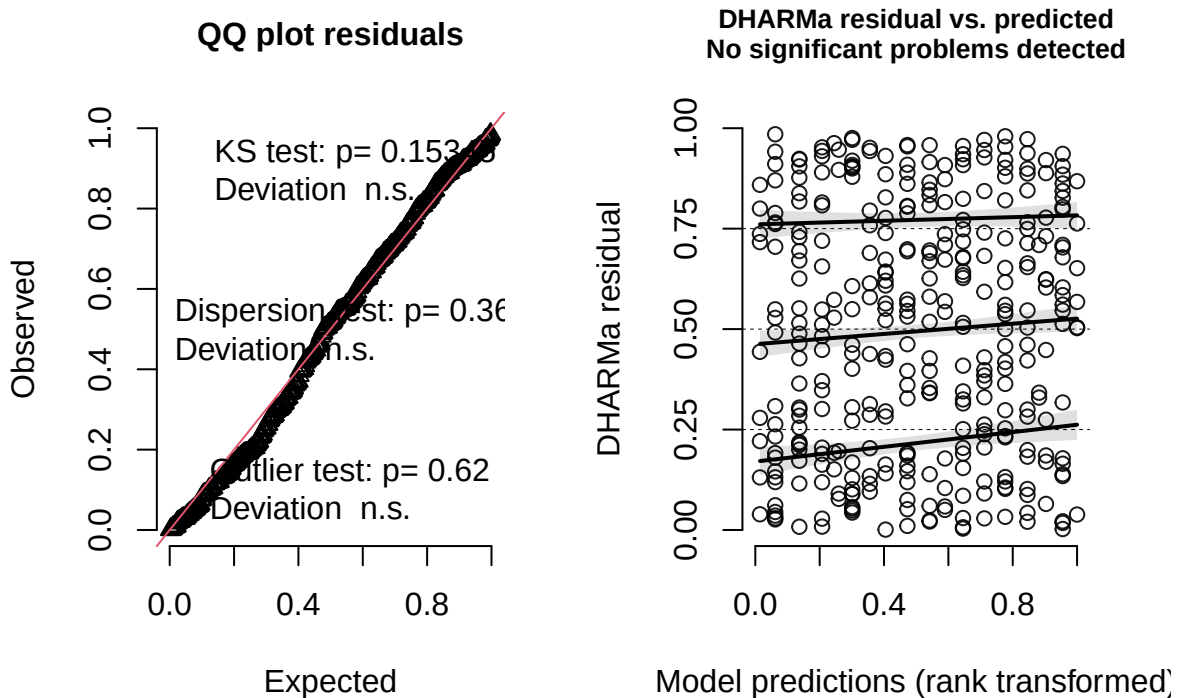
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.4266062	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.4915942	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.2428991	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.4615736	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.4859701	0.2579519	1.8839562	0.0595709	Poisson	BirdID
fixed	factor(fac.yr)2	-0.5279212	0.3332916	-1.5839618	0.1132024	Poisson	BirdID
fixed	factor(fac.yr)3	-0.5012192	0.3417975	-1.4664215	0.1425335	Poisson	BirdID
fixed	factor(fac.yr)4	-0.2885838	0.3189715	-0.9047323	0.3656072	Poisson	BirdID
fixed	factor(fac.yr)5	-0.2531485	0.3254867	-0.7777538	0.4367142	Poisson	BirdID
fixed	factor(fac.yr)6	-0.3802393	0.3213588	-1.1832235	0.2367206	Poisson	BirdID
fixed	factor(fac.yr)7	-1.3184696	0.3816482	-3.4546730	0.0005510	Poisson	BirdID
fixed	factor(fac.yr)8	-0.5028876	0.3257573	-1.5437494	0.1226490	Poisson	BirdID
fixed	factor(fac.yr)9	0.0103557	0.3089367	0.0335204	0.9732596	Poisson	BirdID
fixed	factor(fac.yr)10	-0.7904691	0.3526086	-2.2417747	0.0249759	Poisson	BirdID
fixed	factor(fac.yr)11	-1.3473359	0.4084134	-3.2989512	0.0009705	Poisson	BirdID
fixed	factor(fac.yr)12	-1.1737170	0.4095205	-2.8660761	0.0041559	Poisson	BirdID
fixed	factor(fac.yr)13	-0.3060193	0.3447675	-0.8876106	0.3747502	Poisson	BirdID
fixed	factor(fac.yr)14	-0.5674268	0.3937898	-1.4409383	0.1496021	Poisson	BirdID
fixed	factor(fac.yr)15	-0.3991310	0.3832088	-1.0415496	0.2976205	Poisson	BirdID
fixed	factor(fac.yr)16	-1.4978077	0.5794616	-2.5848266	0.0097428	Poisson	BirdID
fixed	factor(fac.yr)17	0.0973402	0.3884488	0.2505868	0.8021336	Poisson	BirdID
fixed	factor(fac.yr)18	-0.8871542	0.5873701	-1.5103835	0.1309456	Poisson	BirdID
fixed	factor(fac.yr)19	-1.0587305	0.6634280	-1.5958483	0.1105227	Poisson	BirdID
fixed	factor(fac.yr)20	-0.1986238	0.6148258	-0.3230571	0.7466520	Poisson	BirdID
fixed	(Intercept)	0.4731151	0.2559307	1.8486065	0.0645147	Poisson	Terr
fixed	factor(fac.yr)2	-0.5184511	0.3475968	-1.4915300	0.1358224	Poisson	Terr
fixed	factor(fac.yr)3	-0.6336501	0.3440271	-1.8418612	0.0654955	Poisson	Terr
fixed	factor(fac.yr)4	-0.2713828	0.3142760	-0.8635176	0.3878530	Poisson	Terr
fixed	factor(fac.yr)5	-0.2028180	0.3127100	-0.6485818	0.5166087	Poisson	Terr
fixed	factor(fac.yr)6	-0.3466071	0.3039751	-1.1402484	0.2541828	Poisson	Terr
fixed	factor(fac.yr)7	-1.2681845	0.3783143	-3.3521979	0.0008017	Poisson	Terr
fixed	factor(fac.yr)8	-0.4617538	0.3234132	-1.4277520	0.1533632	Poisson	Terr
fixed	factor(fac.yr)9	0.1152025	0.3058783	0.3766285	0.7064497	Poisson	Terr
fixed	factor(fac.yr)10	-0.7790667	0.3486637	-2.2344358	0.0254544	Poisson	Terr
fixed	factor(fac.yr)11	-1.3556953	0.3985108	-3.4019036	0.0006692	Poisson	Terr
fixed	factor(fac.yr)12	-1.2630293	0.4061327	-3.1098935	0.0018715	Poisson	Terr
fixed	factor(fac.yr)13	-0.4528685	0.3379107	-1.3402018	0.1801798	Poisson	Terr
fixed	factor(fac.yr)14	-0.6997796	0.3873014	-1.8068090	0.0707921	Poisson	Terr
fixed	factor(fac.yr)15	-0.4367315	0.3741647	-1.1672174	0.2431226	Poisson	Terr
fixed	factor(fac.yr)16	-1.4073664	0.5849985	-2.4057607	0.0161388	Poisson	Terr
fixed	factor(fac.yr)17	0.1602307	0.3918905	0.4088660	0.6826380	Poisson	Terr
fixed	factor(fac.yr)18	-1.0516973	0.5780362	-1.8194315	0.0688456	Poisson	Terr
fixed	factor(fac.yr)19	-1.2008389	0.6574304	-1.8265642	0.0677653	Poisson	Terr
fixed	factor(fac.yr)20	-0.2179917	0.5985525	-0.3641981	0.7157101	Poisson	Terr
fixed	(Intercept)	0.4644903	0.2654448	1.7498565	0.0801431	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.5289000	0.3523303	-1.5011482	0.1333172	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.6260964	0.3538828	-1.7692195	0.0768572	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.2784733	0.3249107	-0.8570763	0.3914027	Poisson	BirdID+Terr

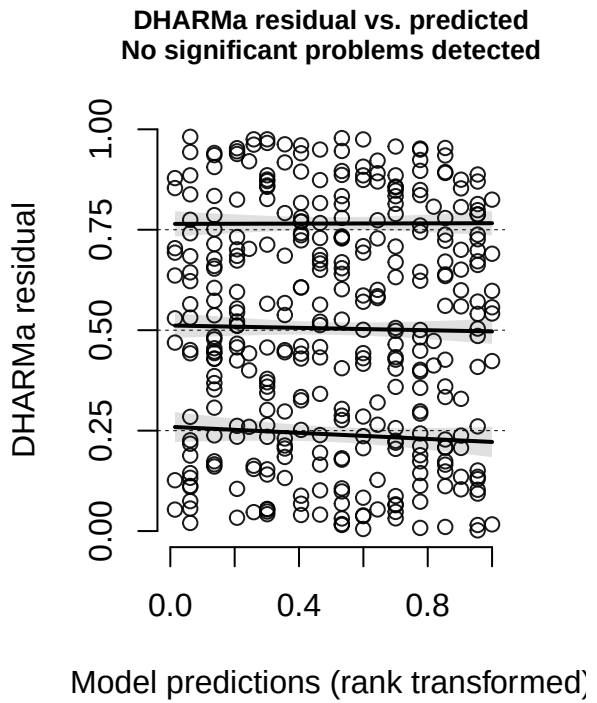
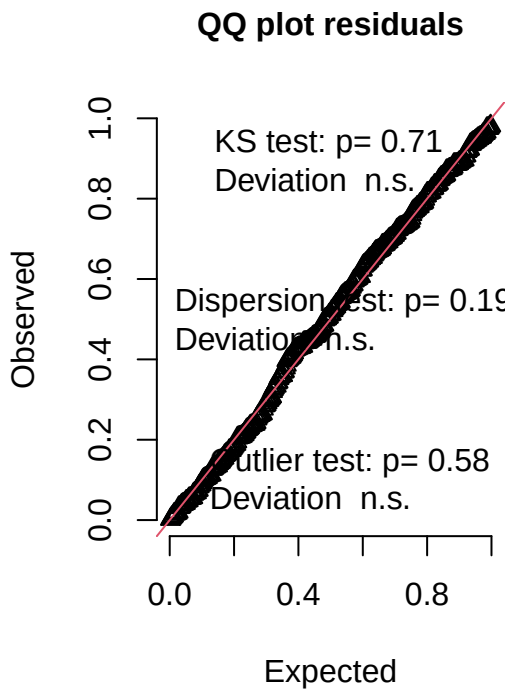
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.2178823	0.3253552	-0.6696752	0.5030649	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.3617415	0.3169214	-1.1414232	0.2536939	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-1.2944129	0.3882010	-3.3343884	0.0008549	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.4882265	0.3346129	-1.4590785	0.1445435	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	0.0835100	0.3177050	0.2628538	0.7926633	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.8081064	0.3595164	-2.2477592	0.0245915	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-1.3666846	0.4092088	-3.3398218	0.0008383	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-1.2681350	0.4163819	-3.0456057	0.0023221	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.4294650	0.3501787	-1.2264169	0.2200418	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.6769148	0.3993462	-1.6950577	0.0900645	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.4710456	0.3872701	-1.2163232	0.2238618	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-1.4806496	0.5939689	-2.4928066	0.0126738	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	0.1188308	0.4068204	0.2920964	0.7702129	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-1.0835405	0.5915756	-1.8316180	0.0670084	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.2217928	0.6707848	-1.8214379	0.0685403	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.1992301	0.6196468	-0.3215220	0.7478149	Poisson	BirdID+Terr

The AIC and BIC for the territory only model are the best, and the model produces R^2 s which are still competitive.

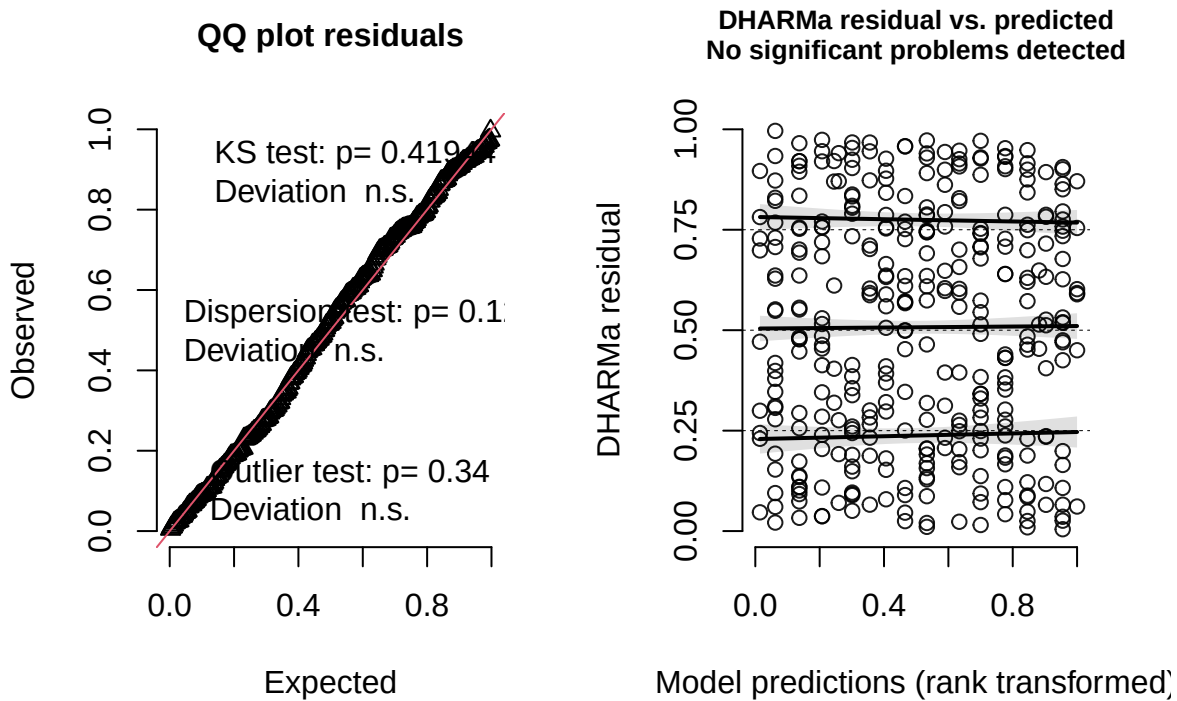
DHARMA residual



DHARMA residual



DHARMa residual



Mild concern for the residual vs predicted plot for the model fitting both REs.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
350	1	-203.0896	450.1792	535.0538	211.5483	328	0.805	Binomial	BirdID+Terr
350	1	-204.6461	451.2923	532.3089	259.5071	329	0.712	Binomial	BirdID
350	1	-215.4964	472.9927	554.0093	321.2230	329	0.584	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.763654	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.182732	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.813957	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.029431	Binomial	BirdID+Terr

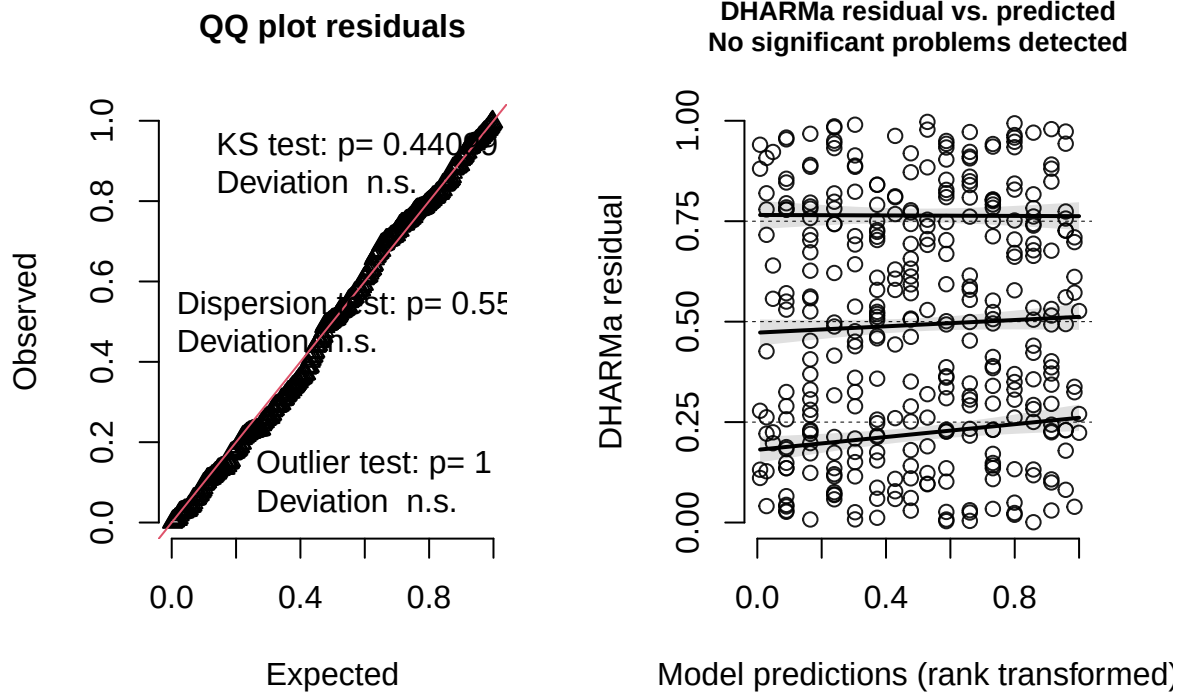
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	1.8867670	9.466426e-01	1.9931143	0.0462489	Binomial	BirdID
fixed	factor(fac.yr)2	-0.6322587	1.067283e+00	-0.5924003	0.5535825	Binomial	BirdID
fixed	factor(fac.yr)3	-0.6835651	1.068180e+00	-0.6399347	0.5222151	Binomial	BirdID
fixed	factor(fac.yr)4	-0.7192983	1.078312e+00	-0.6670594	0.5047342	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-1.0932683	1.111238e+00	-0.9838295	0.3251994	Binomial	BirdID
fixed	factor(fac.yr)6	-0.7939136	1.086348e+00	-0.7308096	0.4648955	Binomial	BirdID
fixed	factor(fac.yr)7	-2.7930491	1.072200e+00	-2.6049695	0.0091882	Binomial	BirdID
fixed	factor(fac.yr)8	-1.6251525	1.072293e+00	-1.5155866	0.1296239	Binomial	BirdID
fixed	factor(fac.yr)9	-0.7895620	1.120973e+00	-0.7043543	0.4812122	Binomial	BirdID
fixed	factor(fac.yr)10	-2.5937462	1.100210e+00	-2.3575018	0.0183984	Binomial	BirdID
fixed	factor(fac.yr)11	-2.9509259	1.110620e+00	-2.6570074	0.0078838	Binomial	BirdID
fixed	factor(fac.yr)12	-2.3365884	1.137082e+00	-2.0548997	0.0398887	Binomial	BirdID
fixed	factor(fac.yr)13	-0.5921960	1.142959e+00	-0.5181251	0.6043710	Binomial	BirdID
fixed	factor(fac.yr)14	-1.0392918	1.183393e+00	-0.8782304	0.3798187	Binomial	BirdID
fixed	factor(fac.yr)15	-1.2618261	1.266792e+00	-0.9960801	0.3192112	Binomial	BirdID
fixed	factor(fac.yr)16	-3.1053199	1.307157e+00	-2.3756283	0.0175191	Binomial	BirdID
fixed	factor(fac.yr)17	0.4093706	1.741486e+00	0.2350698	0.8141546	Binomial	BirdID
fixed	factor(fac.yr)18	-3.1387751	1.609685e+00	-1.9499310	0.0511843	Binomial	BirdID
fixed	factor(fac.yr)19	-2.9572082	1.637114e+00	-1.8063544	0.0708630	Binomial	BirdID
fixed	factor(fac.yr)20	64.0415866	3.874532e+07	0.0000017	0.9999987	Binomial	BirdID
fixed	(Intercept)	1.9894639	7.787487e-01	2.5546932	0.0106281	Binomial	Terr
fixed	factor(fac.yr)2	-0.7203447	9.477083e-01	-0.7600911	0.4472001	Binomial	Terr
fixed	factor(fac.yr)3	-1.1274760	9.395515e-01	-1.2000151	0.2301335	Binomial	Terr
fixed	factor(fac.yr)4	-0.9253343	8.945104e-01	-1.0344590	0.3009216	Binomial	Terr
fixed	factor(fac.yr)5	-1.1695743	9.159990e-01	-1.2768293	0.2016625	Binomial	Terr
fixed	factor(fac.yr)6	-0.9953543	8.883955e-01	-1.1203955	0.2625453	Binomial	Terr
fixed	factor(fac.yr)7	-2.5921882	8.924859e-01	-2.9044583	0.0036789	Binomial	Terr
fixed	factor(fac.yr)8	-1.6553816	8.908360e-01	-1.8582339	0.0631358	Binomial	Terr
fixed	factor(fac.yr)9	-0.7108267	9.254250e-01	-0.7681084	0.4424228	Binomial	Terr
fixed	factor(fac.yr)10	-2.4720410	9.093274e-01	-2.7185381	0.0065571	Binomial	Terr
fixed	factor(fac.yr)11	-2.8399774	9.163452e-01	-3.0992439	0.0019402	Binomial	Terr
fixed	factor(fac.yr)12	-2.5526195	9.308925e-01	-2.7421207	0.0061044	Binomial	Terr
fixed	factor(fac.yr)13	-1.2927766	9.440074e-01	-1.3694559	0.1708568	Binomial	Terr
fixed	factor(fac.yr)14	-1.7449575	9.866879e-01	-1.7684999	0.0769774	Binomial	Terr
fixed	factor(fac.yr)15	-1.1118508	1.053986e+00	-1.0549007	0.2914707	Binomial	Terr
fixed	factor(fac.yr)16	-2.4670897	1.098626e+00	-2.2456138	0.0247288	Binomial	Terr
fixed	factor(fac.yr)17	-0.1500201	1.439691e+00	-0.1042030	0.9170082	Binomial	Terr
fixed	factor(fac.yr)18	-2.7220873	1.308935e+00	-2.0796193	0.0375605	Binomial	Terr
fixed	factor(fac.yr)19	-2.7891484	1.317772e+00	-2.1165638	0.0342969	Binomial	Terr
fixed	factor(fac.yr)20	28.2957502	1.956457e+06	0.0000145	0.9999885	Binomial	Terr
fixed	(Intercept)	1.9823860	9.831191e-01	2.0164250	0.0437556	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-0.4681152	1.125792e+00	-0.4158095	0.6775494	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-0.7703466	1.133488e+00	-0.6796245	0.4967422	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-0.5932781	1.129862e+00	-0.5250889	0.5995214	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-1.0298511	1.161938e+00	-0.8863215	0.3754443	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-0.7293285	1.133592e+00	-0.6433786	0.5199785	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-2.9610166	1.126247e+00	-2.6291012	0.0085611	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-1.5945263	1.127165e+00	-1.4146343	0.1571757	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-0.5670454	1.181201e+00	-0.4800584	0.6311859	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-2.7488067	1.158606e+00	-2.3725116	0.0176676	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-3.1810948	1.165911e+00	-2.7284212	0.0063638	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-2.5661477	1.204549e+00	-2.1303796	0.0331403	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-0.6620058	1.186974e+00	-0.5577255	0.5770319	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-1.1803116	1.246686e+00	-0.9467594	0.3437614	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-1.1717621	1.366804e+00	-0.8573004	0.3912789	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-3.3965875	1.399373e+00	-2.4272210	0.0152150	Binomial	BirdID+Terr

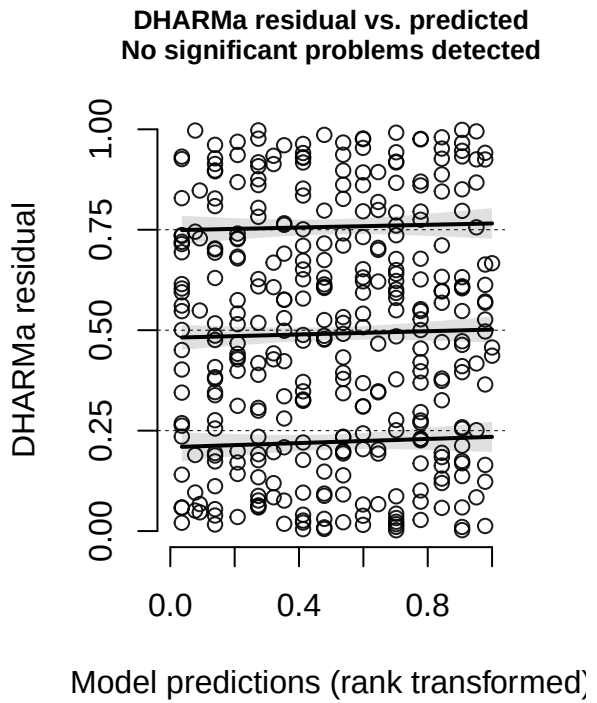
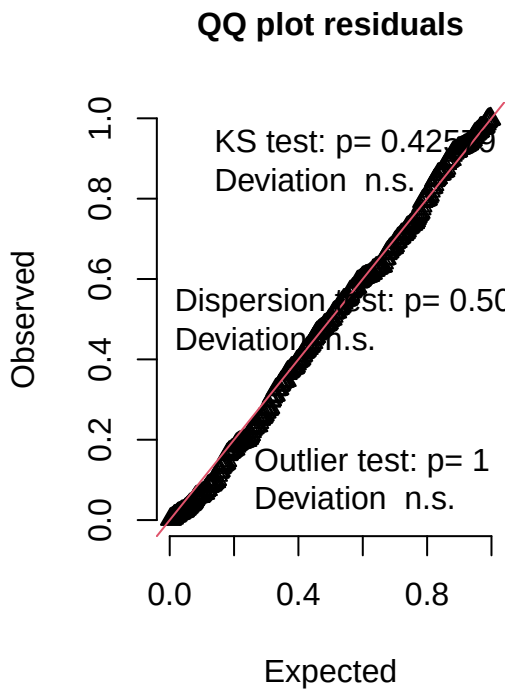
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	0.4997700	1.897526e+00	0.2633798	0.7922579	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-3.3654391	1.768958e+00	-1.9024976	0.0571061	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-3.3117004	1.813373e+00	-1.8262651	0.0678103	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	17.6414366	9.901547e+03	0.0017817	0.9985784	Binomial	BirdID+Terr

Similar as to the previous batch of binomial models.

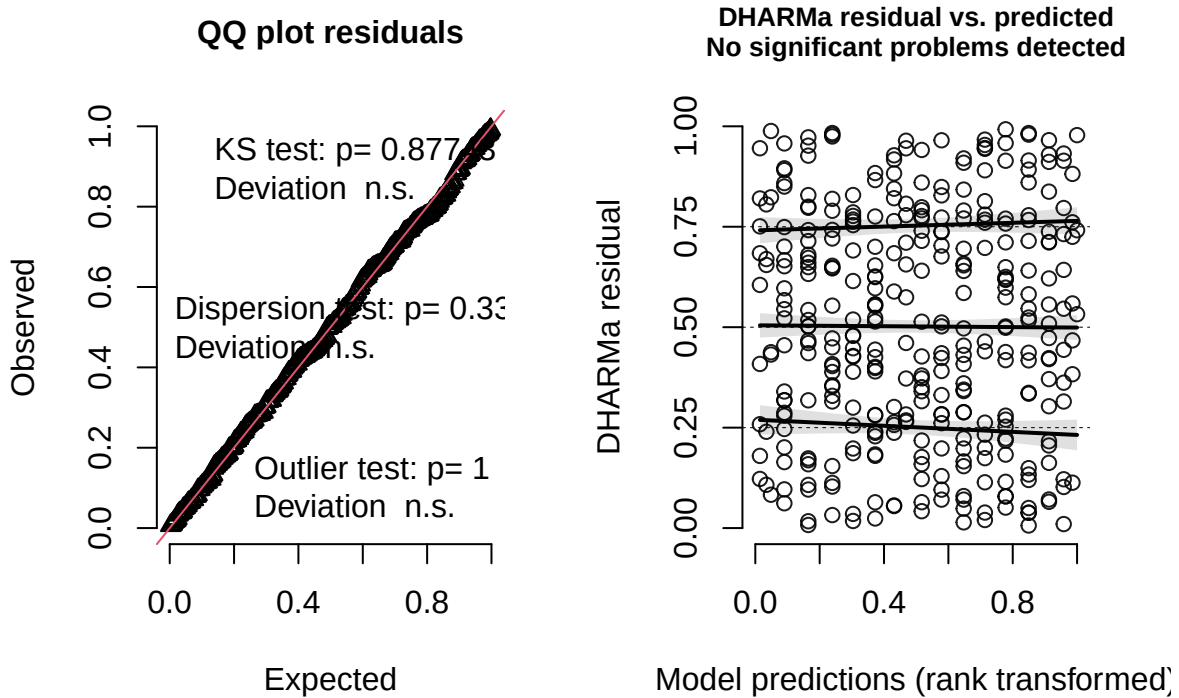
DHARMA residual



DHARMa residual



DHARMA residual



Diagnostics look good.

Probabilities used to generate simulation 18

Simulation 18
 Probability of mate fidelity: 0.871
 Probability of pair dispersal: 0.962
 Probability of single male dispersal: 0.662
 Probability of single female dispersal: 0.467

Simulation 19

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
311	1	-405.1043	852.2086	930.7443	275.3782	290	0.3170650	0.1881913	Poisson	BirdID
311	1	-405.0724	854.1449	936.4203	271.2072	289	0.3207556	0.1872916	Poisson	BirdID+Terr
311	1	-407.9129	857.8257	936.3614	301.5283	290	0.2850238	0.2084132	Poisson	Terr

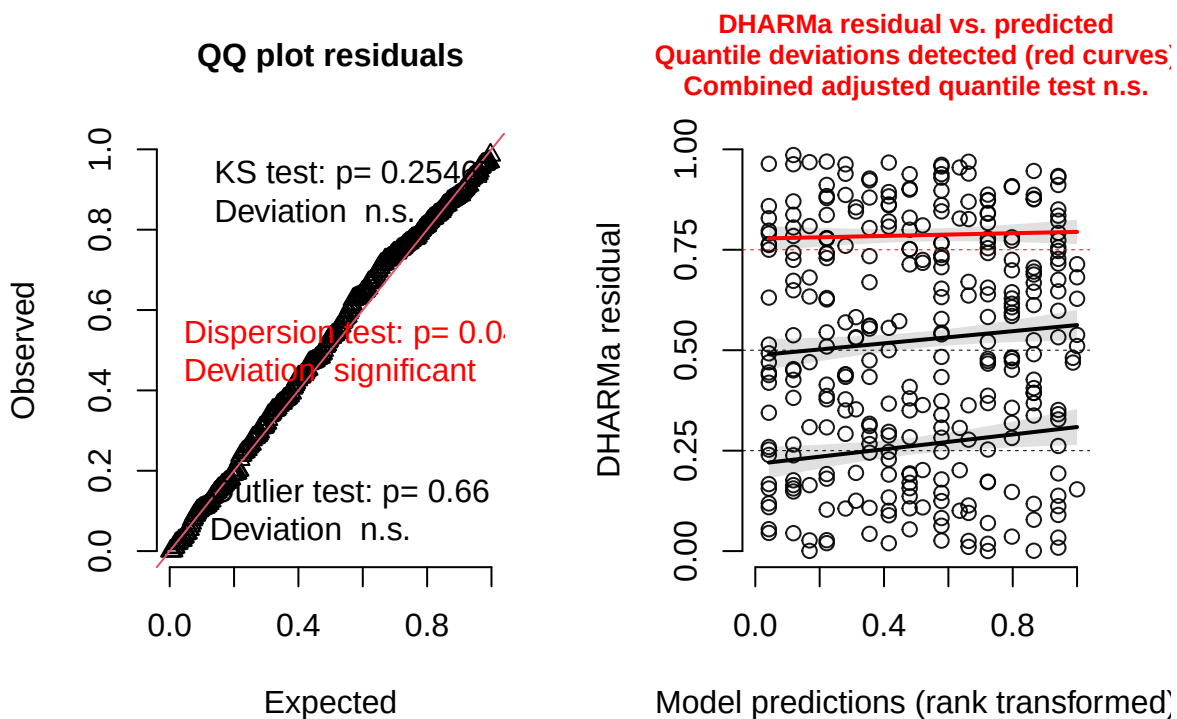
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.3484859	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.2593185	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.3389572	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.1074763	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.7422391	0.2985097	2.4864824	0.0129013	Poisson	BirdID
fixed	factor(fac.yr)2	-0.8626649	0.3804090	-2.2677299	0.0233457	Poisson	BirdID
fixed	factor(fac.yr)3	-1.0107837	0.3725540	-2.7131198	0.0066653	Poisson	BirdID
fixed	factor(fac.yr)4	-0.7028263	0.3432226	-2.0477271	0.0405867	Poisson	BirdID
fixed	factor(fac.yr)5	-0.1960216	0.3298454	-0.5942833	0.5523226	Poisson	BirdID
fixed	factor(fac.yr)6	-0.8002769	0.3817609	-2.0962778	0.0360576	Poisson	BirdID
fixed	factor(fac.yr)7	-1.2150706	0.3998398	-3.0388934	0.0023745	Poisson	BirdID
fixed	factor(fac.yr)8	-0.5525911	0.3533394	-1.5639103	0.1178386	Poisson	BirdID
fixed	factor(fac.yr)9	-0.3838657	0.3527416	-1.0882346	0.2764915	Poisson	BirdID
fixed	factor(fac.yr)10	-0.4444853	0.3555965	-1.2499708	0.2113102	Poisson	BirdID
fixed	factor(fac.yr)11	-1.8187096	0.4552348	-3.9951024	0.0000647	Poisson	BirdID
fixed	factor(fac.yr)12	-0.7932063	0.3808387	-2.0827882	0.0372705	Poisson	BirdID
fixed	factor(fac.yr)13	-0.5624622	0.4008229	-1.4032686	0.1605368	Poisson	BirdID
fixed	factor(fac.yr)14	-0.9470146	0.4464852	-2.1210436	0.0339181	Poisson	BirdID
fixed	factor(fac.yr)15	-1.0839390	0.5210366	-2.0803510	0.0374933	Poisson	BirdID
fixed	factor(fac.yr)16	-0.8807737	0.4986392	-1.7663547	0.0773364	Poisson	BirdID
fixed	factor(fac.yr)17	-0.7850963	0.5251509	-1.4949920	0.1349165	Poisson	BirdID
fixed	factor(fac.yr)18	-0.6975999	0.5589879	-1.2479695	0.2120422	Poisson	BirdID
fixed	factor(fac.yr)19	-0.7952708	1.1220407	-0.7087718	0.4784661	Poisson	BirdID
fixed	factor(fac.yr)20	-0.0972035	0.6400753	-0.1518626	0.8792953	Poisson	BirdID
fixed	(Intercept)	0.8255014	0.2875862	2.8704482	0.0040989	Poisson	Terr
fixed	factor(fac.yr)2	-0.8450389	0.3776357	-2.2377089	0.0252400	Poisson	Terr
fixed	factor(fac.yr)3	-1.0517618	0.3658981	-2.8744662	0.0040471	Poisson	Terr
fixed	factor(fac.yr)4	-0.7151942	0.3353417	-2.1327328	0.0329467	Poisson	Terr
fixed	factor(fac.yr)5	-0.2069117	0.3243923	-0.6378441	0.5235752	Poisson	Terr
fixed	factor(fac.yr)6	-0.8406564	0.3722625	-2.2582355	0.0239310	Poisson	Terr
fixed	factor(fac.yr)7	-1.2439130	0.3909845	-3.1814894	0.0014652	Poisson	Terr
fixed	factor(fac.yr)8	-0.6206472	0.3393281	-1.8290474	0.0673925	Poisson	Terr
fixed	factor(fac.yr)9	-0.4270623	0.3410610	-1.2521580	0.2105123	Poisson	Terr
fixed	factor(fac.yr)10	-0.4662313	0.3410743	-1.3669496	0.1716411	Poisson	Terr
fixed	factor(fac.yr)11	-1.8585082	0.4436955	-4.1887019	0.0000281	Poisson	Terr
fixed	factor(fac.yr)12	-0.7844299	0.3681126	-2.1309511	0.0330932	Poisson	Terr
fixed	factor(fac.yr)13	-0.5114062	0.3887753	-1.3154288	0.1883658	Poisson	Terr
fixed	factor(fac.yr)14	-0.9847063	0.4352098	-2.2626012	0.0236603	Poisson	Terr
fixed	factor(fac.yr)15	-1.1636667	0.5093298	-2.2847019	0.0223303	Poisson	Terr
fixed	factor(fac.yr)16	-0.9084647	0.4856668	-1.8705513	0.0614073	Poisson	Terr
fixed	factor(fac.yr)17	-0.7940134	0.5090831	-1.5596932	0.1188324	Poisson	Terr
fixed	factor(fac.yr)18	-0.8108084	0.5463953	-1.4839228	0.1378294	Poisson	Terr
fixed	factor(fac.yr)19	-0.8651788	1.0825684	-0.7991909	0.4241798	Poisson	Terr
fixed	factor(fac.yr)20	-0.1524729	0.6031494	-0.2527945	0.8004270	Poisson	Terr
fixed	(Intercept)	0.7465150	0.3004375	2.4847596	0.0129639	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.8666117	0.3832029	-2.2614958	0.0237286	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-1.0208604	0.3744787	-2.7260840	0.0064091	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.7109543	0.3453363	-2.0587304	0.0395201	Poisson	BirdID+Terr

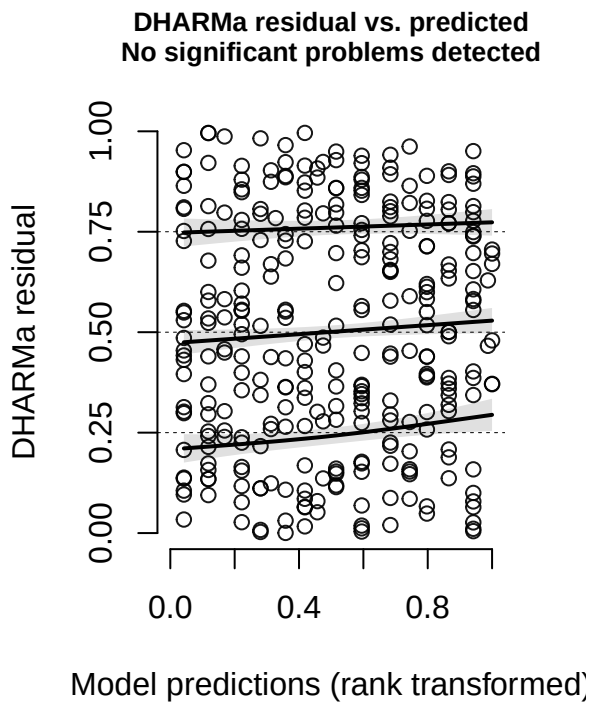
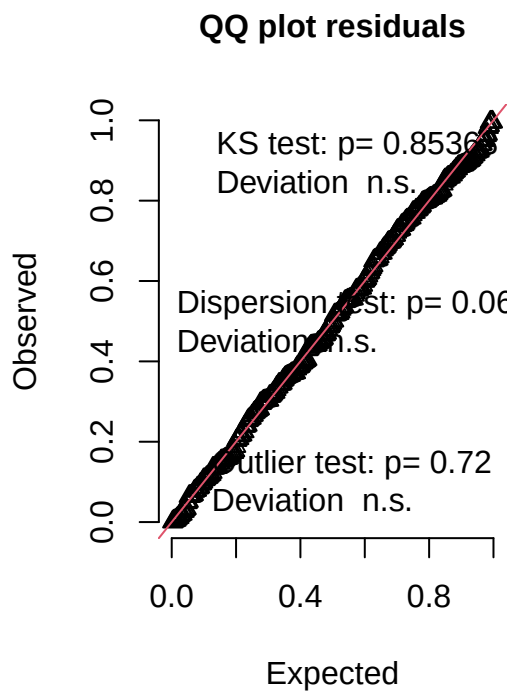
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.2040045	0.3324406	-0.6136571	0.5394420	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.8049344	0.3839452	-2.0964824	0.0360394	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-1.2229223	0.4021079	-3.0412790	0.0023558	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.5622742	0.3552186	-1.5828962	0.1134451	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.3926452	0.3549059	-1.1063360	0.2685811	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.4495465	0.3571375	-1.2587491	0.2081210	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-1.8253702	0.4568884	-3.9952209	0.0000646	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.7989883	0.3826513	-2.0880324	0.0367949	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.5621555	0.4034726	-1.3932928	0.1635313	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.9543950	0.4488634	-2.1262480	0.0334826	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-1.0855681	0.5234490	-2.0738754	0.0380909	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.8822013	0.5011490	-1.7603575	0.0783472	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.7887505	0.5274737	-1.4953361	0.1348268	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.7052174	0.5613983	-1.2561802	0.2090506	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-0.8050559	1.1259206	-0.7150201	0.4745966	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.1013773	0.6431364	-0.1576296	0.8747487	Poisson	BirdID+Terr

The territory model is not performing well on the selection criteria.

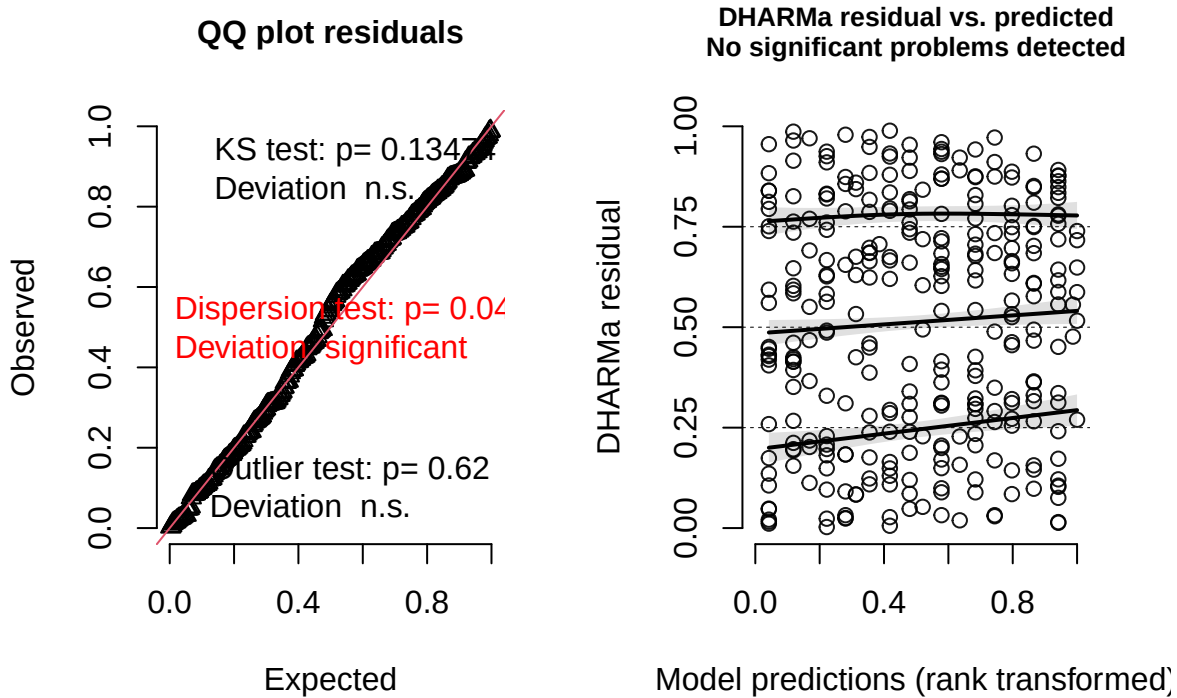
DHARMA residual



DHARMa residual



DHARMa residual



However, the diagnostics look best on the territory only model, and I would not trust the band ID only model.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
311	1	-170.4515	382.9030	461.4386	233.7354	290	0.677	Binomial	BirdID
311	1	-169.8182	383.6365	465.9119	209.9327	289	0.731	Binomial	BirdID+Terr
311	1	-176.9743	395.9486	474.4842	281.3419	290	0.572	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.4711091	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	0.9739511	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.4754769	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.6870788	Binomial	BirdID+Terr

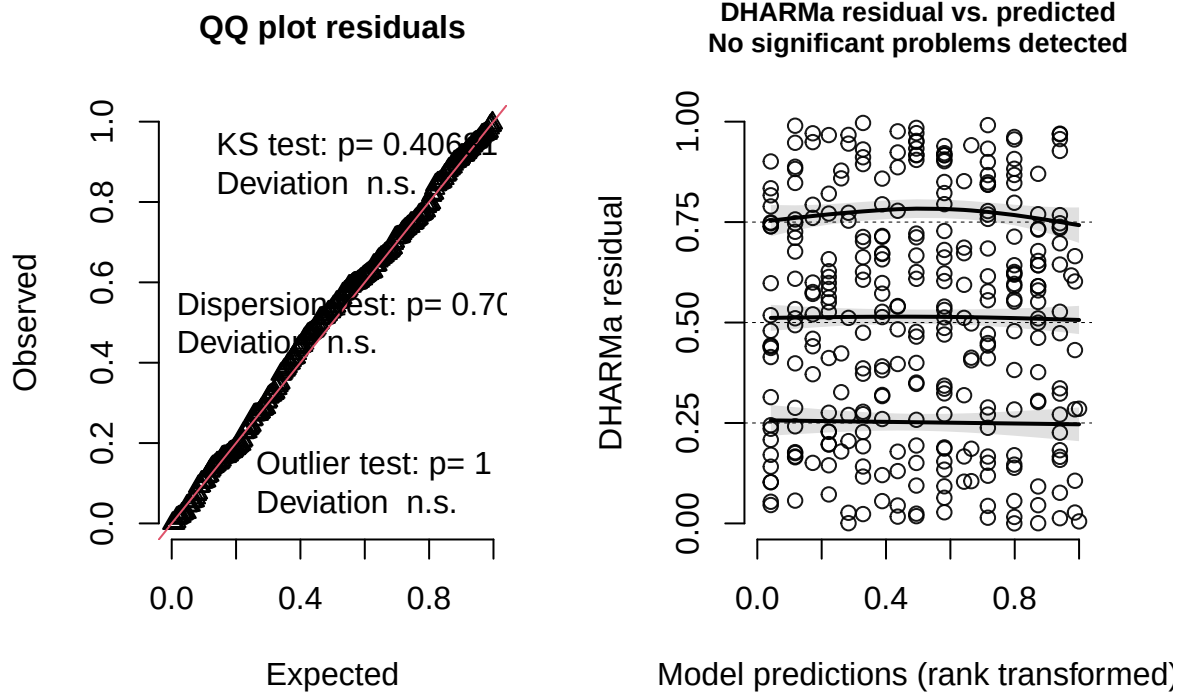
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	17.3288699	2675.295	0.0064774	0.9948318	Binomial	BirdID
fixed	factor(fac.yr)2	-17.0103475	2675.295	-0.0063583	0.9949268	Binomial	BirdID
fixed	factor(fac.yr)3	-16.7821928	2675.295	-0.0062730	0.9949949	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)4	-16.5573528	2675.295	-0.0061890	0.9950619	Binomial	BirdID
fixed	factor(fac.yr)5	-15.8631227	2675.295	-0.0059295	0.9952690	Binomial	BirdID
fixed	factor(fac.yr)6	-17.2078471	2675.295	-0.0064321	0.9948679	Binomial	BirdID
fixed	factor(fac.yr)7	-17.7870084	2675.295	-0.0066486	0.9946952	Binomial	BirdID
fixed	factor(fac.yr)8	-15.7840062	2675.295	-0.0058999	0.9952926	Binomial	BirdID
fixed	factor(fac.yr)9	-15.0811306	2675.295	-0.0056372	0.9955022	Binomial	BirdID
fixed	factor(fac.yr)10	-14.5247556	2675.295	-0.0054292	0.9956681	Binomial	BirdID
fixed	factor(fac.yr)11	-18.2640073	2675.295	-0.0068269	0.9945530	Binomial	BirdID
fixed	factor(fac.yr)12	-17.4050580	2675.295	-0.0065058	0.9948091	Binomial	BirdID
fixed	factor(fac.yr)13	-16.9859942	2675.295	-0.0063492	0.9949341	Binomial	BirdID
fixed	factor(fac.yr)14	-17.6844487	2675.295	-0.0066103	0.9947258	Binomial	BirdID
fixed	factor(fac.yr)15	-17.3047497	2675.295	-0.0064684	0.9948390	Binomial	BirdID
fixed	factor(fac.yr)16	-16.4564351	2675.295	-0.0061513	0.9950920	Binomial	BirdID
fixed	factor(fac.yr)17	-16.3571851	2675.295	-0.0061142	0.9951216	Binomial	BirdID
fixed	factor(fac.yr)18	-17.3695108	2675.295	-0.0064926	0.9948197	Binomial	BirdID
fixed	factor(fac.yr)19	-1.9934283	3424.780	-0.0005821	0.9995356	Binomial	BirdID
fixed	factor(fac.yr)20	0.3386936	5540.940	0.0000611	0.9999512	Binomial	BirdID
fixed	(Intercept)	16.6901069	1588.763	0.0105051	0.9916183	Binomial	Terr
fixed	factor(fac.yr)2	-16.1351196	1588.763	-0.0101558	0.9918970	Binomial	Terr
fixed	factor(fac.yr)3	-16.1385663	1588.763	-0.0101579	0.9918953	Binomial	Terr
fixed	factor(fac.yr)4	-15.8957525	1588.763	-0.0100051	0.9920172	Binomial	Terr
fixed	factor(fac.yr)5	-15.2492147	1588.763	-0.0095982	0.9923419	Binomial	Terr
fixed	factor(fac.yr)6	-16.4673500	1588.763	-0.0103649	0.9917302	Binomial	Terr
fixed	factor(fac.yr)7	-16.8688108	1588.763	-0.0106176	0.9915286	Binomial	Terr
fixed	factor(fac.yr)8	-15.3212217	1588.763	-0.0096435	0.9923057	Binomial	Terr
fixed	factor(fac.yr)9	-14.7270502	1588.763	-0.0092695	0.9926041	Binomial	Terr
fixed	factor(fac.yr)10	-14.0641841	1588.763	-0.0088523	0.9929370	Binomial	Terr
fixed	factor(fac.yr)11	-17.4819947	1588.763	-0.0110035	0.9912206	Binomial	Terr
fixed	factor(fac.yr)12	-16.3187969	1588.763	-0.0102714	0.9918048	Binomial	Terr
fixed	factor(fac.yr)13	-15.8799771	1588.763	-0.0099952	0.9920251	Binomial	Terr
fixed	factor(fac.yr)14	-16.5947628	1588.763	-0.0104451	0.9916662	Binomial	Terr
fixed	factor(fac.yr)15	-16.5375522	1588.763	-0.0104091	0.9916949	Binomial	Terr
fixed	factor(fac.yr)16	-15.5688755	1588.763	-0.0097994	0.9921814	Binomial	Terr
fixed	factor(fac.yr)17	-15.5819336	1588.763	-0.0098076	0.9921748	Binomial	Terr
fixed	factor(fac.yr)18	-16.9739650	1588.763	-0.0106838	0.9914758	Binomial	Terr
fixed	factor(fac.yr)19	-1.4201260	2868.218	-0.0004951	0.9996049	Binomial	Terr
fixed	factor(fac.yr)20	-1.0569716	2568.597	-0.0004115	0.9996717	Binomial	Terr
fixed	(Intercept)	17.2536426	2578.059	0.0066925	0.9946602	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-16.8758833	2578.059	-0.0065460	0.9947771	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-16.7195466	2578.059	-0.0064853	0.9948255	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-16.4362085	2578.059	-0.0063754	0.9949132	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-15.6634972	2578.059	-0.0060757	0.9951523	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-17.1286160	2578.059	-0.0066440	0.9946989	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-17.7078189	2578.059	-0.0068687	0.9945196	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-15.5648687	2578.059	-0.0060374	0.9951829	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-14.9574096	2578.059	-0.0058018	0.9953709	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-14.3869294	2578.060	-0.0055805	0.9955474	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-18.2445859	2578.059	-0.0070769	0.9943535	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-17.2567077	2578.059	-0.0066937	0.9946593	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-16.7259594	2578.059	-0.0064878	0.9948235	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-17.4679050	2578.059	-0.0067756	0.9945939	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-17.1132093	2578.059	-0.0066380	0.9947037	Binomial	BirdID+Terr

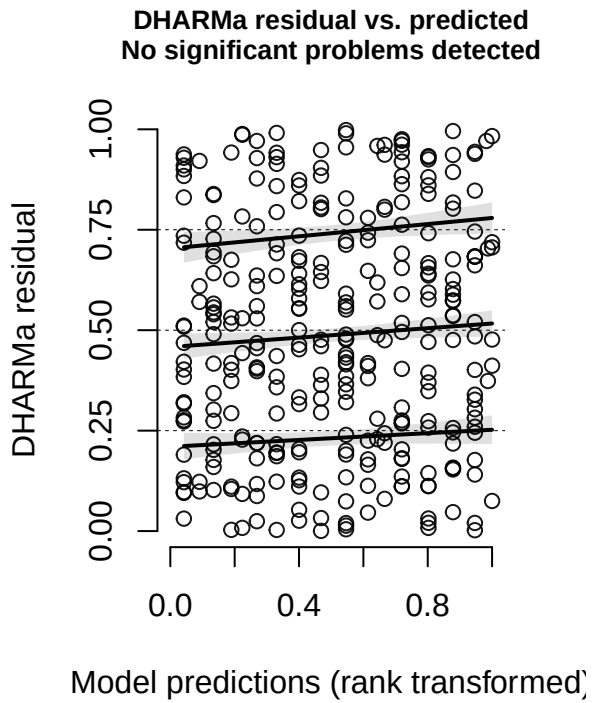
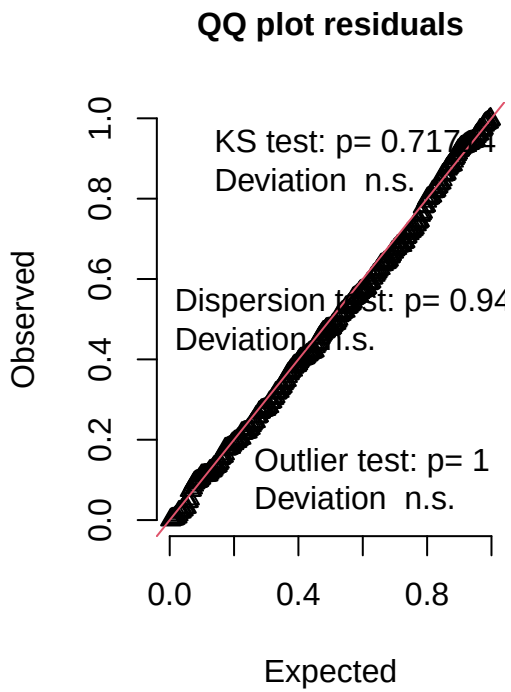
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)16	-16.2095007	2578.060	-0.0062875	0.9949833	Binomial	BirdID+Terr
fixed	factor(fac.yr)17	-16.2029060	2578.060	-0.0062849	0.9949854	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-17.3903061	2578.060	-0.0067455	0.9946179	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-3.9595481	2703.819	-0.0014644	0.9988316	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-0.8533865	3757.702	-0.0002271	0.9998188	Binomial	BirdID+Terr

Similar to the previous batch of binomial models.

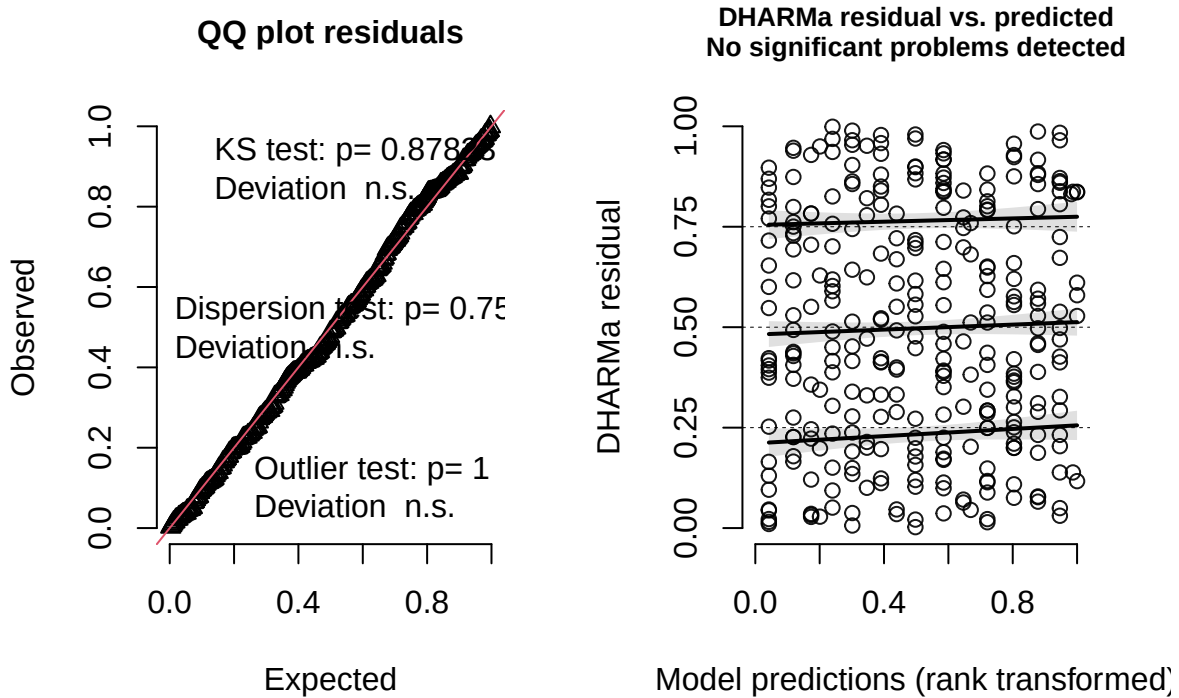
DHARMa residual



DHARMa residual



DHARMa residual



Diagnostics all look reasonable.

Probabilites used to generate simulation 19

Simulation 19
 Probability of mate fidelity: 0.29
 Probability of pair dispersal: 0.702
 Probability of single male dispersal: 0.396
 Probability of single female dispersal: 0.053

Simulation 20

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
305	1	-366.9052	775.8105	853.9370	278.0745	284	0.9121985	0.8898766	Poisson	Terr
305	1	-366.9054	777.8108	859.6577	278.1520	283	0.9086615	0.8859254	Poisson	BirdID+Terr
305	1	-370.8281	783.6562	861.7827	311.8688	284	0.9051581	0.8925994	Poisson	BirdID

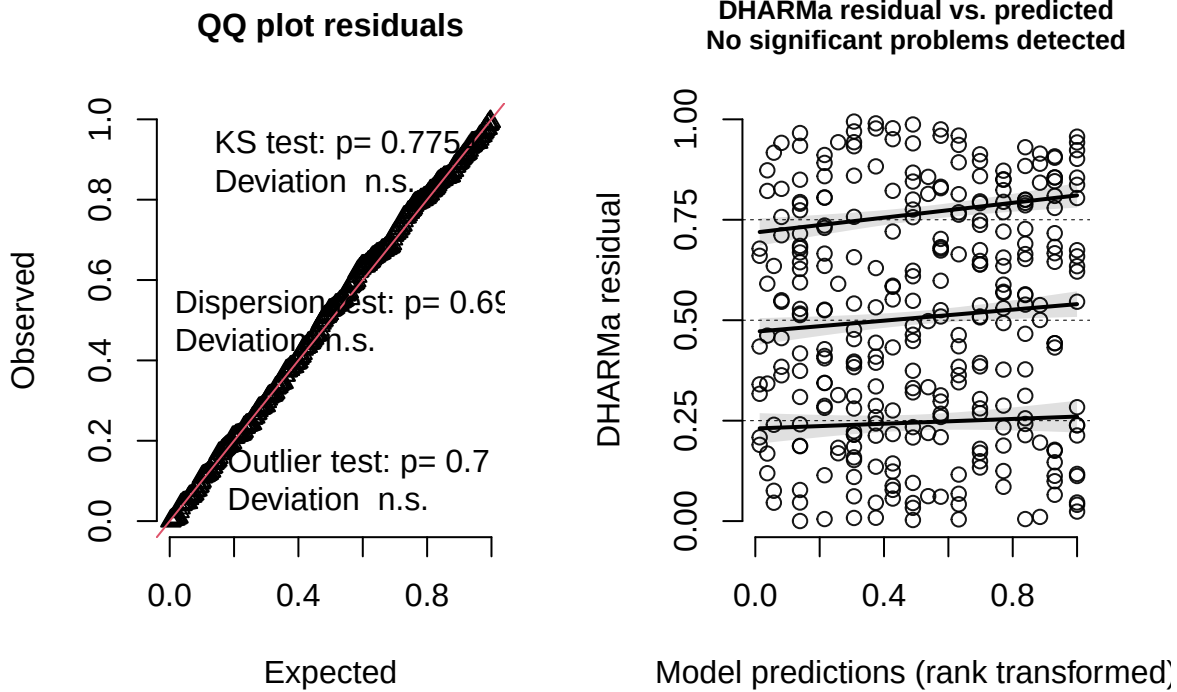
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.3052889	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.4156760	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.0019820	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.4152689	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	-0.3652362	0.3145726	-1.1610554	0.2456194	Poisson	BirdID
fixed	factor(fac.yr)2	0.4271655	0.3890905	1.0978565	0.2722671	Poisson	BirdID
fixed	factor(fac.yr)3	0.5262427	0.3725746	1.4124491	0.1578178	Poisson	BirdID
fixed	factor(fac.yr)4	0.6360704	0.3693725	1.7220297	0.0850641	Poisson	BirdID
fixed	factor(fac.yr)5	0.8805378	0.3621732	2.4312616	0.0150463	Poisson	BirdID
fixed	factor(fac.yr)6	0.4708752	0.3959158	1.1893316	0.2343092	Poisson	BirdID
fixed	factor(fac.yr)7	-0.2806384	0.4429132	-0.6336194	0.5263292	Poisson	BirdID
fixed	factor(fac.yr)8	0.5089855	0.3729522	1.3647472	0.1723325	Poisson	BirdID
fixed	factor(fac.yr)9	0.8954149	0.3604435	2.4842032	0.0129842	Poisson	BirdID
fixed	factor(fac.yr)10	0.0560531	0.4040465	0.1387293	0.8896640	Poisson	BirdID
fixed	factor(fac.yr)11	-0.6518422	0.4661101	-1.3984726	0.1619712	Poisson	BirdID
fixed	factor(fac.yr)12	-0.0487231	0.4089949	-0.1191288	0.9051733	Poisson	BirdID
fixed	factor(fac.yr)13	-0.0196295	0.4455323	-0.0440585	0.9648577	Poisson	BirdID
fixed	factor(fac.yr)14	-1.1281705	0.7917287	-1.4249457	0.1541729	Poisson	BirdID
fixed	factor(fac.yr)15	-16.5555329	1824.7398490	-0.0090728	0.9927610	Poisson	BirdID
fixed	factor(fac.yr)16	-1.6002042	1.0704865	-1.4948383	0.1349567	Poisson	BirdID
fixed	factor(fac.yr)17	-0.1842414	0.6793280	-0.2712113	0.7862285	Poisson	BirdID
fixed	factor(fac.yr)18	-1.2853207	1.0741849	-1.1965544	0.2314803	Poisson	BirdID
fixed	factor(fac.yr)19	0.1606700	0.5655533	0.2840935	0.7763388	Poisson	BirdID
fixed	factor(fac.yr)20	0.7836005	0.4536402	1.7273615	0.0841028	Poisson	BirdID
fixed	(Intercept)	-0.3454352	0.3237295	-1.0670489	0.2859497	Poisson	Terr
fixed	factor(fac.yr)2	0.3960589	0.4009277	0.9878561	0.3232231	Poisson	Terr
fixed	factor(fac.yr)3	0.4868676	0.3894675	1.2500853	0.2112684	Poisson	Terr
fixed	factor(fac.yr)4	0.7016998	0.3821116	1.8363738	0.0663024	Poisson	Terr
fixed	factor(fac.yr)5	0.8430286	0.3725959	2.2625815	0.0236615	Poisson	Terr
fixed	factor(fac.yr)6	0.4343926	0.4075428	1.0658822	0.2864769	Poisson	Terr
fixed	factor(fac.yr)7	-0.3997219	0.4514997	-0.8853204	0.3759838	Poisson	Terr
fixed	factor(fac.yr)8	0.5026774	0.3811000	1.3190171	0.1871634	Poisson	Terr
fixed	factor(fac.yr)9	0.8578110	0.3688726	2.3254939	0.0200456	Poisson	Terr
fixed	factor(fac.yr)10	0.0472951	0.4106254	0.1151782	0.9083039	Poisson	Terr
fixed	factor(fac.yr)11	-0.6405308	0.4752382	-1.3478100	0.1777195	Poisson	Terr
fixed	factor(fac.yr)12	-0.0988464	0.4161461	-0.2375282	0.8122471	Poisson	Terr
fixed	factor(fac.yr)13	-0.0278266	0.4539326	-0.0613013	0.9511193	Poisson	Terr
fixed	factor(fac.yr)14	-1.0811536	0.7971836	-1.3562165	0.1750303	Poisson	Terr
fixed	factor(fac.yr)15	-16.9117213	2227.4603404	-0.0075924	0.9939422	Poisson	Terr
fixed	factor(fac.yr)16	-1.7254161	1.0762034	-1.6032434	0.1088809	Poisson	Terr
fixed	factor(fac.yr)17	-0.2228539	0.7017201	-0.3175823	0.7508018	Poisson	Terr
fixed	factor(fac.yr)18	-1.3415234	1.0831595	-1.2385280	0.2155203	Poisson	Terr
fixed	factor(fac.yr)19	0.1727645	0.5776941	0.2990589	0.7648951	Poisson	Terr
fixed	factor(fac.yr)20	0.7965261	0.4704349	1.6931699	0.0904231	Poisson	Terr
fixed	(Intercept)	-0.3448632	0.3236297	-1.0656104	0.2865998	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	0.3956675	0.4008385	0.9870996	0.3235938	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	0.4863643	0.3893719	1.2490995	0.2116287	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	0.7008634	0.3820269	1.8345917	0.0665662	Poisson	BirdID+Terr

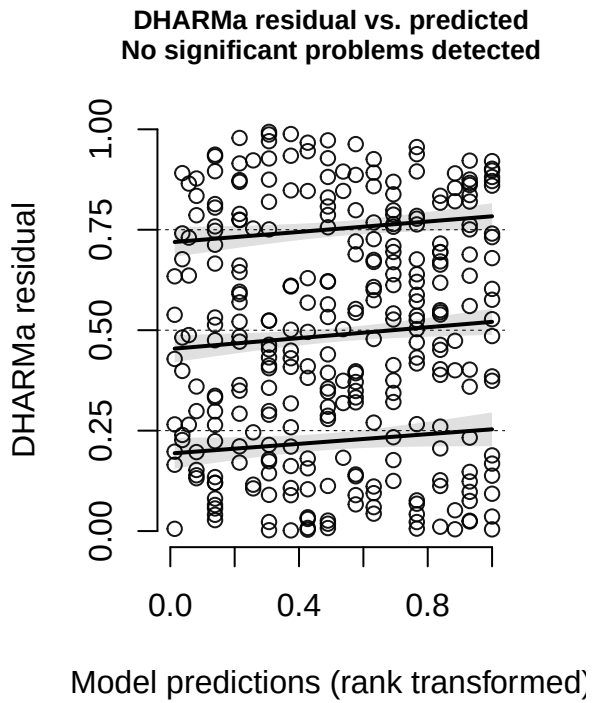
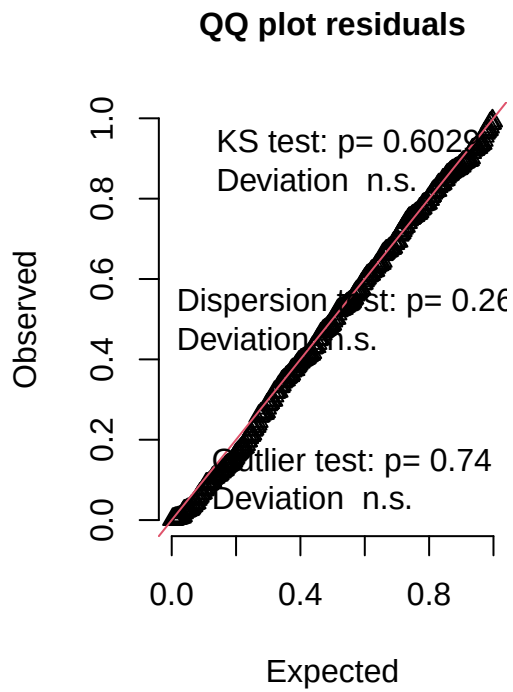
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.8423433	0.3725057	2.2612901	0.0237413	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	0.4314598	0.4075934	1.0585545	0.2898027	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.4004444	0.4514480	-0.8870223	0.3750669	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	0.5017983	0.3810222	1.3169793	0.1878456	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	0.8573287	0.3687740	2.3248075	0.0200823	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	0.0468197	0.4105425	0.1140435	0.9092033	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.6403109	0.4750636	-1.3478424	0.1777091	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.0976537	0.4159392	-0.2347788	0.8143804	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.0277846	0.4537918	-0.0612277	0.9511779	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-1.0844075	0.7979605	-1.3589738	0.1741549	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-16.6907727	1993.7940528	-0.0083714	0.9933207	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-1.7267593	1.0765786	-1.6039325	0.1087289	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.2251817	0.7020400	-0.3207533	0.7483973	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-1.3445504	1.0841904	-1.2401423	0.2149228	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	0.1702888	0.5779308	0.2946525	0.7682593	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	0.7973489	0.4702078	1.6957372	0.0899357	Poisson	BirdID+Terr

Territory only model is performing well on all criteria.

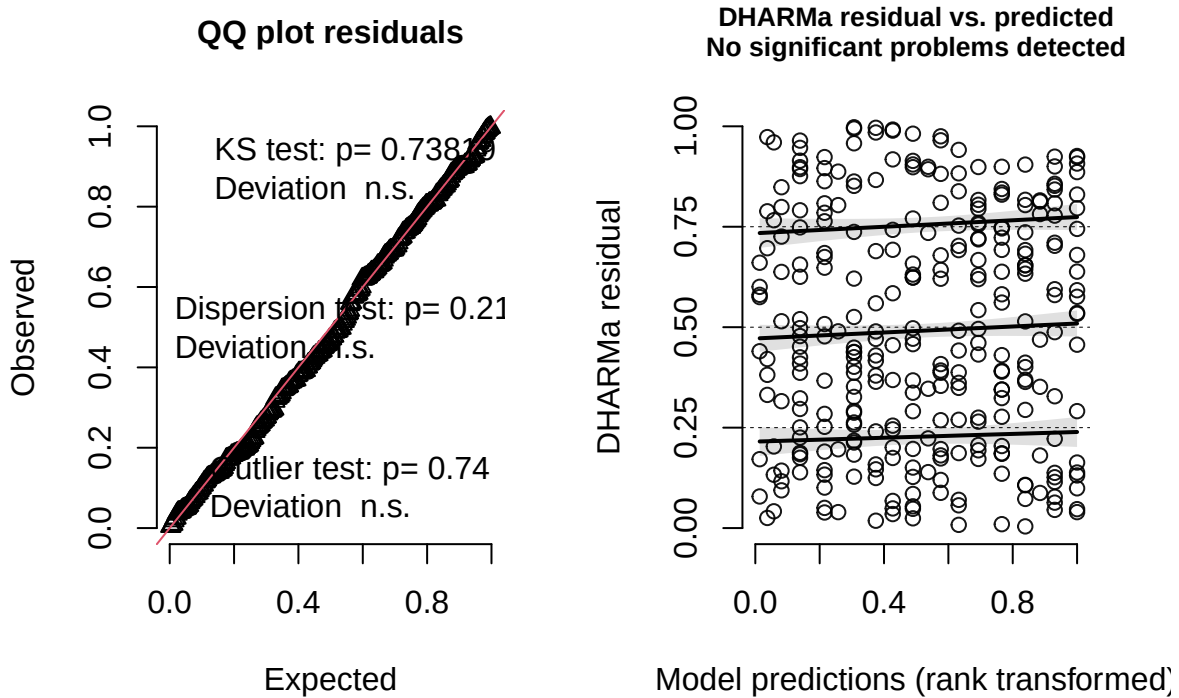
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics all look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
305	1	-170.4247	382.8494	460.9759	221.4504	284	0.750	Binomial	Terr
305	1	-169.9444	383.8888	465.7357	191.3722	283	0.789	Binomial	BirdID+Terr
305	1	-176.4720	394.9439	473.0705	260.6012	284	0.651	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.1619832	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.5791816	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	0.8011378	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.6173247	Binomial	BirdID+Terr

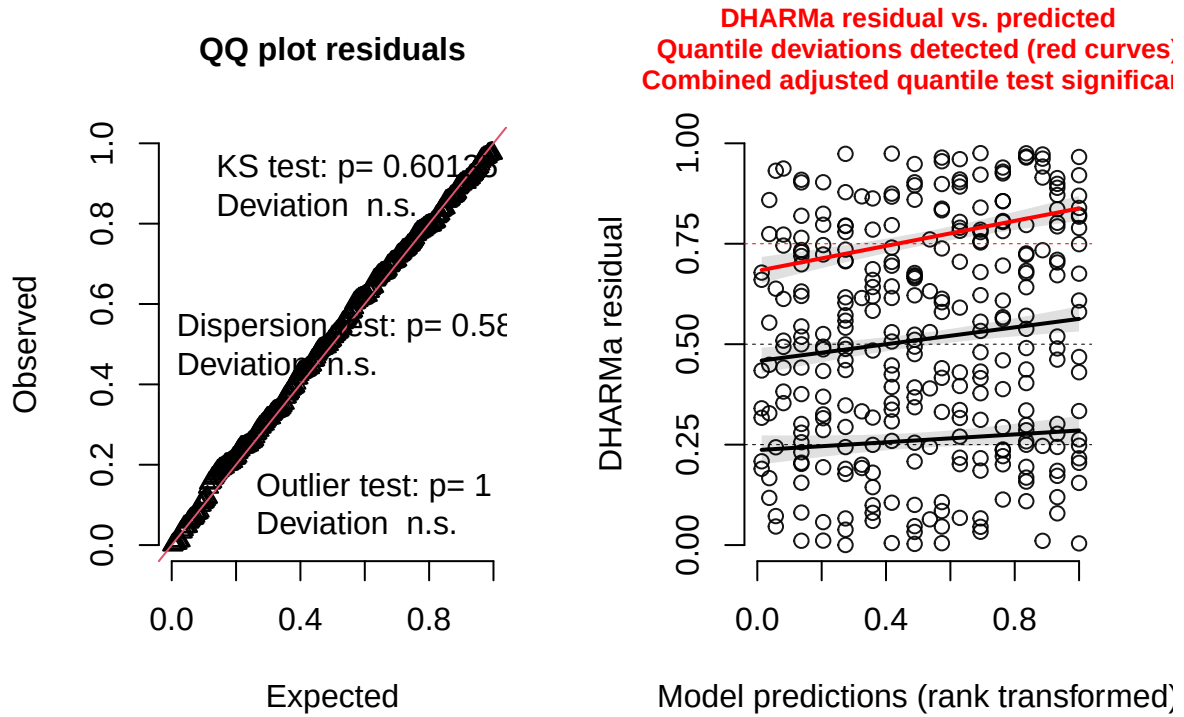
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	-0.4636017	6.238063e-01	-0.7431821	0.4573714	Binomial	BirdID
fixed	factor(fac.yr)2	1.2379863	8.219640e-01	1.5061319	0.1320333	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)3	1.7866814	8.333428e-01	2.1439933	0.0320334	Binomial	BirdID
fixed	factor(fac.yr)4	2.0752146	8.866392e-01	2.3405400	0.0192559	Binomial	BirdID
fixed	factor(fac.yr)5	1.5307988	8.361361e-01	1.8308009	0.0671303	Binomial	BirdID
fixed	factor(fac.yr)6	1.1630885	8.717014e-01	1.3342740	0.1821141	Binomial	BirdID
fixed	factor(fac.yr)7	0.0610519	8.174142e-01	0.0746891	0.9404621	Binomial	BirdID
fixed	factor(fac.yr)8	1.5982316	8.379696e-01	1.9072668	0.0564861	Binomial	BirdID
fixed	factor(fac.yr)9	1.4456838	8.431599e-01	1.7146022	0.0864182	Binomial	BirdID
fixed	factor(fac.yr)10	0.5141350	7.967278e-01	0.6453082	0.5187275	Binomial	BirdID
fixed	factor(fac.yr)11	-0.2440444	7.997598e-01	-0.3051471	0.7602542	Binomial	BirdID
fixed	factor(fac.yr)12	-0.4796146	8.203315e-01	-0.5846595	0.5587767	Binomial	BirdID
fixed	factor(fac.yr)13	-0.3842191	8.946139e-01	-0.4294804	0.6675737	Binomial	BirdID
fixed	factor(fac.yr)14	-0.7942542	1.125291e+00	-0.7058212	0.4802994	Binomial	BirdID
fixed	factor(fac.yr)15	-	1.517425e+04	-0.0013546	0.9989192	Binomial	BirdID
		20.5556286					
fixed	factor(fac.yr)16	-1.5145514	1.389412e+00	-1.0900666	0.2756838	Binomial	BirdID
fixed	factor(fac.yr)17	-0.1694368	1.217873e+00	-0.1391252	0.8893512	Binomial	BirdID
fixed	factor(fac.yr)18	-1.3298263	1.460669e+00	-0.9104228	0.3625996	Binomial	BirdID
fixed	factor(fac.yr)19	0.5276644	1.139864e+00	0.4629188	0.6434226	Binomial	BirdID
fixed	factor(fac.yr)20	1.7325352	1.174802e+00	1.4747468	0.1402806	Binomial	BirdID
fixed	(Intercept)	-0.0577704	6.845178e-01	-0.0843958	0.9327417	Binomial	Terr
fixed	factor(fac.yr)2	1.2021295	9.145304e-01	1.3144774	0.1886856	Binomial	Terr
fixed	factor(fac.yr)3	1.8403154	9.478845e-01	1.9414974	0.0521980	Binomial	Terr
fixed	factor(fac.yr)4	2.3996747	9.684426e-01	2.4778699	0.0132169	Binomial	Terr
fixed	factor(fac.yr)5	1.4472788	9.040238e-01	1.6009299	0.1093924	Binomial	Terr
fixed	factor(fac.yr)6	0.9835368	9.587038e-01	1.0259027	0.3049375	Binomial	Terr
fixed	factor(fac.yr)7	-0.5318068	8.832264e-01	-0.6021183	0.5470954	Binomial	Terr
fixed	factor(fac.yr)8	1.8168866	8.896407e-01	2.0422701	0.0411247	Binomial	Terr
fixed	factor(fac.yr)9	1.6309696	9.027177e-01	1.8067327	0.0708040	Binomial	Terr
fixed	factor(fac.yr)10	0.4231019	8.548620e-01	0.4949359	0.6206453	Binomial	Terr
fixed	factor(fac.yr)11	-0.4916964	8.680140e-01	-0.5664613	0.5710802	Binomial	Terr

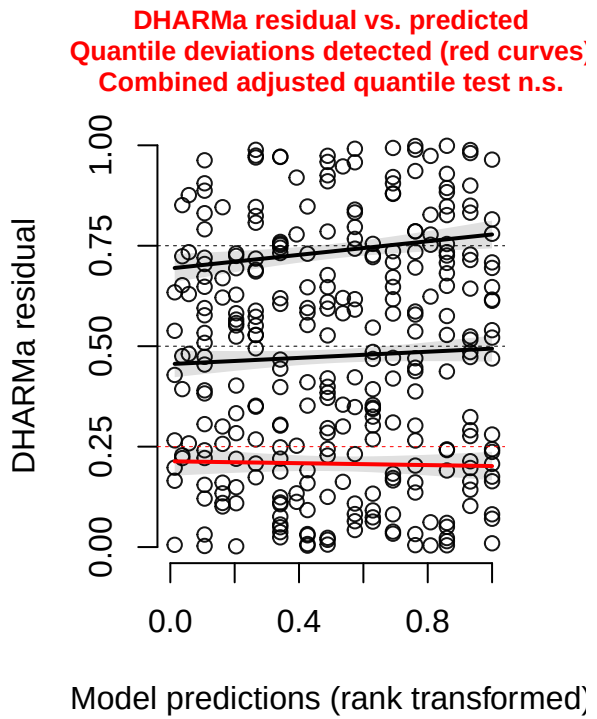
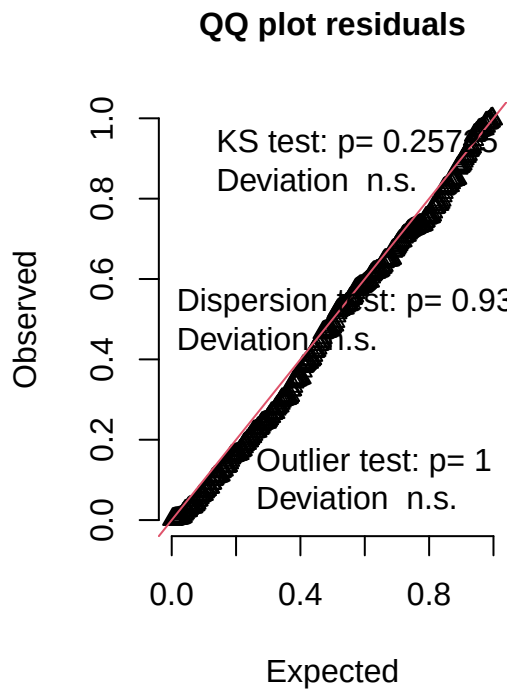
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)12	-0.8326379	8.795898e-01	-0.9466207	0.3438321	Binomial	Terr
fixed	factor(fac.yr)13	-0.6150160	9.599565e-01	-0.6406707	0.5217367	Binomial	Terr
fixed	factor(fac.yr)14	-0.7350886	1.146249e+00	-0.6412993	0.5213282	Binomial	Terr
fixed	factor(fac.yr)15	-28.4176199	7.112050e+05	-0.0000400	0.9999681	Binomial	Terr
fixed	factor(fac.yr)16	-2.3513600	1.448829e+00	-1.6229385	0.1046026	Binomial	Terr
fixed	factor(fac.yr)17	-0.4519094	1.338220e+00	-0.3376944	0.7355935	Binomial	Terr
fixed	factor(fac.yr)18	-2.0049719	1.589314e+00	-1.2615329	0.2071169	Binomial	Terr
fixed	factor(fac.yr)19	0.6130297	1.210235e+00	0.5065377	0.6124792	Binomial	Terr
fixed	factor(fac.yr)20	1.7816245	1.269831e+00	1.4030404	0.1606048	Binomial	Terr
fixed	(Intercept)	0.0521947	7.245116e-01	0.0720412	0.9425691	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	1.1283298	9.556826e-01	1.1806533	0.2377405	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	1.7951470	9.940915e-01	1.8058167	0.0709470	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	2.4119481	1.023786e+00	2.3559115	0.0184773	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	1.4460321	9.615055e-01	1.5039249	0.1326007	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	0.8832650	1.017799e+00	0.8678186	0.3854937	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-0.7740207	9.433570e-01	-0.8204960	0.4119334	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	1.7507403	9.434424e-01	1.8556939	0.0634972	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	1.5065608	9.626627e-01	1.5649935	0.1175844	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	0.3213819	9.080138e-01	0.3539394	0.7233843	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-0.6327991	9.240371e-01	-0.6848200	0.4934575	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-1.0075577	9.389341e-01	-1.0730867	0.2832322	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-0.7711449	1.026724e+00	-0.7510732	0.4526086	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-0.9879881	1.231415e+00	-0.8023191	0.4223684	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-26.0003236	1.944174e+05	-0.0001337	0.9998933	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-2.5856593	1.529285e+00	-1.6907634	0.0908820	Binomial	BirdID+Terr
fixed	factor(fac.yr)17	-0.8063209	1.409322e+00	-0.5721338	0.5672313	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-2.2551398	1.695477e+00	-1.3300919	0.1834880	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	0.5324530	1.301928e+00	0.4089727	0.6825597	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	1.8413569	1.378631e+00	1.3356417	0.1816664	Binomial	BirdID+Terr

Territory is performing well on all criteria.

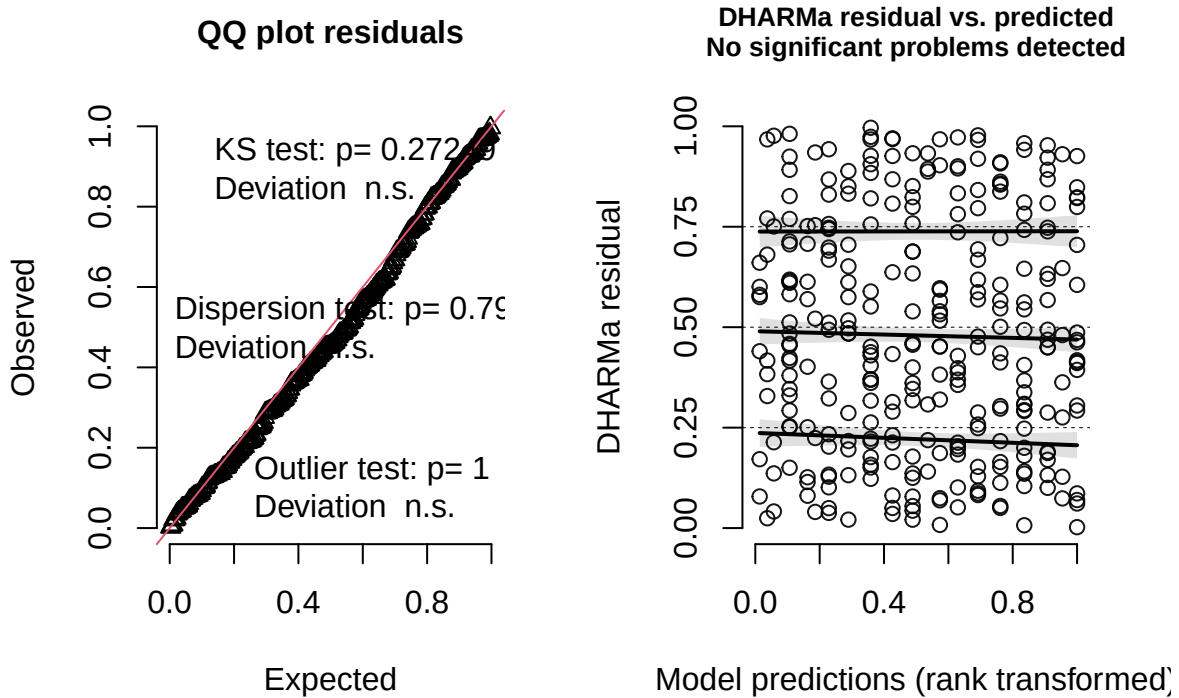
DHARMa residual



DHARMa residual



DHARMA residual



Mildly concerning diagnostics.

Probabilities used to generate simulation 20

Simulation 20
 Probability of mate fidelity: 0.338
 Probability of pair dispersal: 0.847
 Probability of single male dispersal: 0.058
 Probability of single female dispersal: 0.232

Simulation 21

Number of fledgling as the response (Poisson)

boundary (singular) fit: see `help('isSingular')`

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
323	1	-437.5264	917.0528	996.3835	278.8491	302	0.3012891	0.0953890	Poisson	Terr
323	1	-437.5263	919.0527	1002.1610	278.8003	301	0.3013928	0.0952893	Poisson	BirdID+Terr

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	condtional_R2	marginal_R2	Family	RE
323	1	-445.0315	932.0630	1011.3937	332.3511	302	0.2144401	0.1113532	Poisson	BirdID

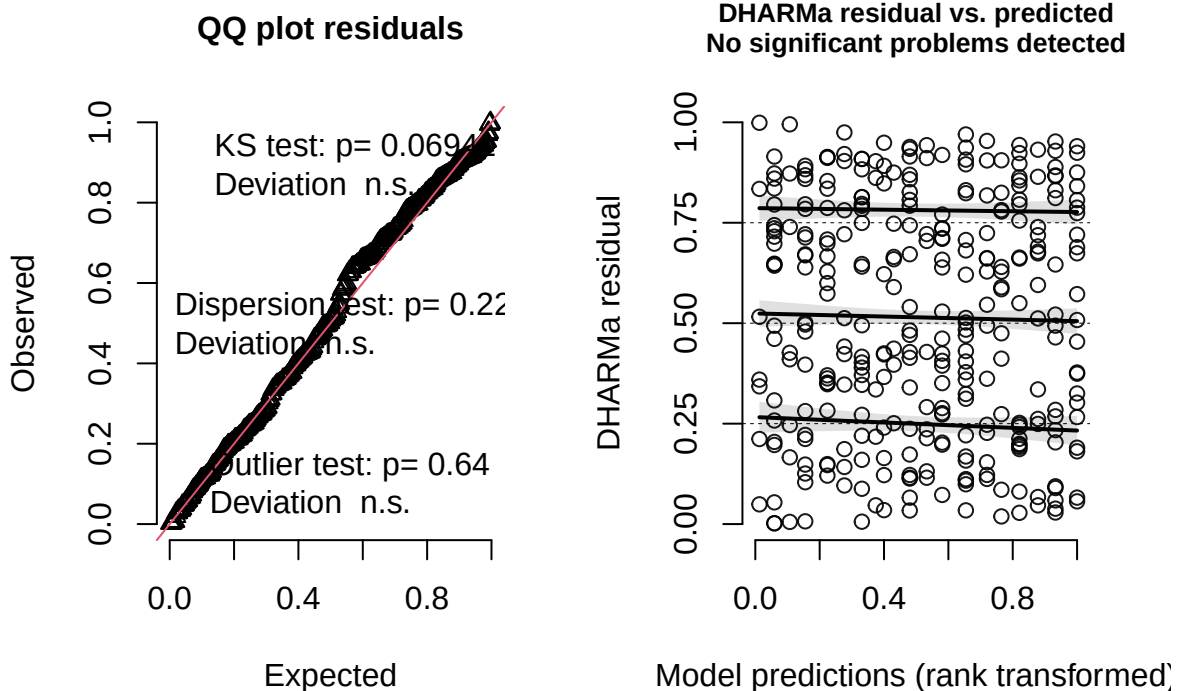
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd____(Intercept)	0.2837609	Poisson	BirdID
ran_pars	Terrs	sd____(Intercept)	0.4202343	Poisson	Terr
ran_pars	birdID	sd____(Intercept)	0.0001926	Poisson	BirdID+Terr
ran_pars	Terrs	sd____(Intercept)	0.4204720	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.1139088	0.3504836	0.3250048	0.7451774	Poisson	BirdID
fixed	factor(fac.yr)2	0.1656545	0.4146707	0.3994845	0.6895363	Poisson	BirdID
fixed	factor(fac.yr)3	-0.2972936	0.4243354	-0.7006101	0.4835464	Poisson	BirdID
fixed	factor(fac.yr)4	0.1562493	0.3903345	0.4002958	0.6889386	Poisson	BirdID
fixed	factor(fac.yr)5	0.2581609	0.3935049	0.6560550	0.5117887	Poisson	BirdID
fixed	factor(fac.yr)6	0.0534197	0.3998190	0.1336097	0.8937112	Poisson	BirdID
fixed	factor(fac.yr)7	-0.4152350	0.4300563	-0.9655365	0.3342762	Poisson	BirdID
fixed	factor(fac.yr)8	-0.2188833	0.4192816	-0.5220437	0.6016399	Poisson	BirdID
fixed	factor(fac.yr)9	0.2669733	0.3992383	0.6687067	0.5036826	Poisson	BirdID
fixed	factor(fac.yr)10	0.1005143	0.4087071	0.2459323	0.8057347	Poisson	BirdID
fixed	factor(fac.yr)11	-0.5582553	0.4406400	-1.2669194	0.2051841	Poisson	BirdID
fixed	factor(fac.yr)12	0.3826944	0.3911043	0.9784969	0.3278286	Poisson	BirdID
fixed	factor(fac.yr)13	0.2112109	0.4228931	0.4994429	0.6174674	Poisson	BirdID
fixed	factor(fac.yr)14	0.2609611	0.4175569	0.6249714	0.5319898	Poisson	BirdID
fixed	factor(fac.yr)15	-0.2826272	0.4706046	-0.6005620	0.5481318	Poisson	BirdID
fixed	factor(fac.yr)16	-0.0984857	0.4630526	-0.2126880	0.8315703	Poisson	BirdID
fixed	factor(fac.yr)17	0.0626702	0.4641018	0.1350355	0.8925838	Poisson	BirdID
fixed	factor(fac.yr)18	-0.4238588	0.5497285	-0.7710330	0.4406874	Poisson	BirdID
fixed	factor(fac.yr)19	-0.6683519	0.6244308	-1.0703378	0.2844673	Poisson	BirdID
fixed	factor(fac.yr)20	-0.1370922	0.5873041	-0.2334263	0.8154304	Poisson	BirdID
fixed	(Intercept)	-0.0108414	0.3579150	-0.0302904	0.9758354	Poisson	Terr
fixed	factor(fac.yr)2	0.1399113	0.4254350	0.3288664	0.7422567	Poisson	Terr
fixed	factor(fac.yr)3	-0.2182664	0.4325384	-0.5046175	0.6138275	Poisson	Terr
fixed	factor(fac.yr)4	0.2587663	0.3991623	0.6482733	0.5168082	Poisson	Terr
fixed	factor(fac.yr)5	0.4120198	0.4054873	1.0161101	0.3095769	Poisson	Terr
fixed	factor(fac.yr)6	0.1540592	0.4079505	0.3776419	0.7056967	Poisson	Terr
fixed	factor(fac.yr)7	-0.2810137	0.4337409	-0.6478838	0.5170601	Poisson	Terr
fixed	factor(fac.yr)8	-0.1335179	0.4248774	-0.3142503	0.7533309	Poisson	Terr
fixed	factor(fac.yr)9	0.2948579	0.4027383	0.7321328	0.4640875	Poisson	Terr
fixed	factor(fac.yr)10	0.2153935	0.4138234	0.5204962	0.6027178	Poisson	Terr
fixed	factor(fac.yr)11	-0.4230169	0.4446733	-0.9512980	0.3414531	Poisson	Terr
fixed	factor(fac.yr)12	0.4680276	0.3993007	1.1721181	0.2411496	Poisson	Terr
fixed	factor(fac.yr)13	0.3571469	0.4323480	0.8260636	0.4087680	Poisson	Terr
fixed	factor(fac.yr)14	0.3610849	0.4193406	0.8610778	0.3891952	Poisson	Terr
fixed	factor(fac.yr)15	-0.1022176	0.4773365	-0.2141416	0.8304366	Poisson	Terr
fixed	factor(fac.yr)16	0.0572325	0.4709742	0.1215195	0.9032796	Poisson	Terr
fixed	factor(fac.yr)17	0.1460298	0.4767553	0.3062993	0.7593768	Poisson	Terr
fixed	factor(fac.yr)18	-0.3199430	0.5619550	-0.5693392	0.5691259	Poisson	Terr

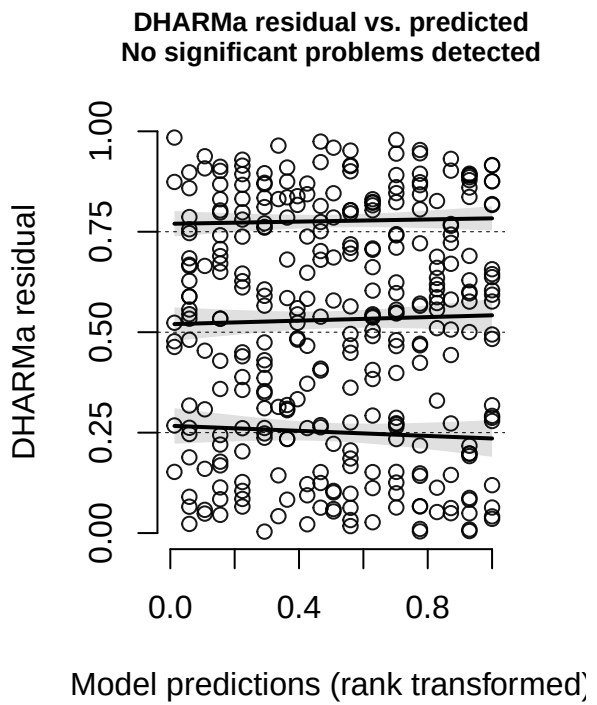
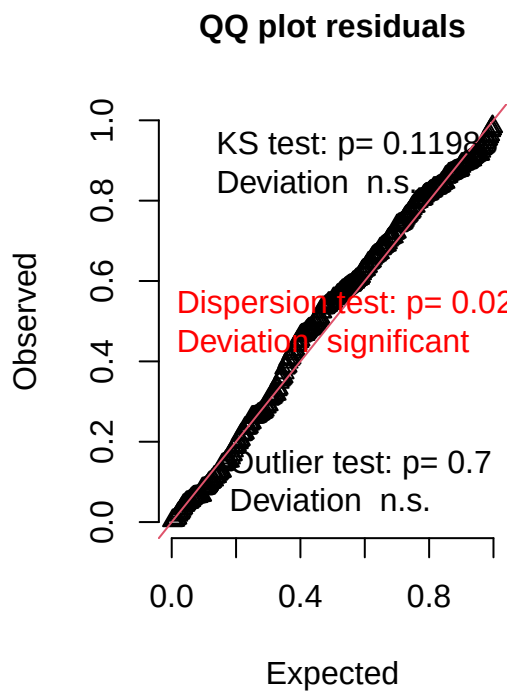
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)19	-0.5711984	0.6309137	-0.9053511	0.3652794	Poisson	Terr
fixed	factor(fac.yr)20	-0.1129667	0.6016125	-0.1877731	0.8510545	Poisson	Terr
fixed	(Intercept)	-0.0087552	0.3576172	-0.0244820	0.9804681	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	0.1375632	0.4251938	0.3235305	0.7462935	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.2204611	0.4322979	-0.5099750	0.6100690	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	0.2568399	0.3988897	0.6438872	0.5196486	Poisson	BirdID+Terr
fixed	factor(fac.yr)5	0.4096026	0.4052396	1.0107666	0.3121282	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	0.1517047	0.4076995	0.3720992	0.7098190	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.2818850	0.4334148	-0.6503816	0.5154458	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.1358049	0.4246369	-0.3198143	0.7491091	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	0.2925073	0.4024811	0.7267604	0.4673728	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	0.2126863	0.4135934	0.5142400	0.6070842	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.4245249	0.4443895	-0.9552992	0.3394264	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	0.4661059	0.3990305	1.1680959	0.2427681	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	0.3548166	0.4321191	0.8211083	0.4115846	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	0.3586455	0.4191016	0.8557483	0.3921370	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.1056776	0.4772366	-0.2214366	0.8247525	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	0.0557007	0.4706990	0.1183362	0.9058013	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	0.1432384	0.4765948	0.3005456	0.7637610	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.3235522	0.5619771	-0.5757392	0.5647915	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-0.5739783	0.6308649	-0.9098276	0.3629134	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.1146631	0.6013722	-0.1906691	0.8487848	Poisson	BirdID+Terr

Territory only model is performing well on AIC and BIC.

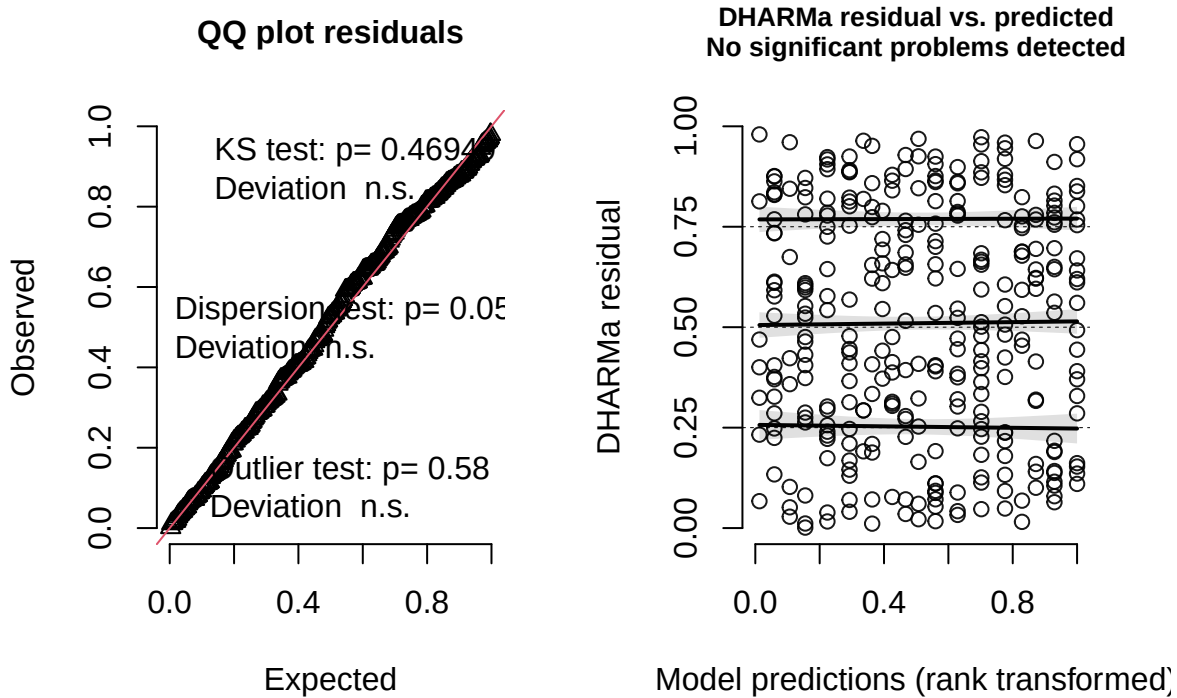
DHARMa residual



DHARMa residual



DHARMa residual



Dispersion test suggest a negative binomial model may be better for the territory only model.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
323	1	-186.4923	414.9847	494.3154	268.4642	302	0.531	Binomial	Terr
323	1	-185.7911	415.5821	498.6905	236.5497	301	0.689	Binomial	BirdID+Terr
323	1	-190.6548	423.3097	502.6404	287.3899	302	0.490	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.1653155	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.2961279	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	0.7547604	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.2917988	Binomial	BirdID+Terr

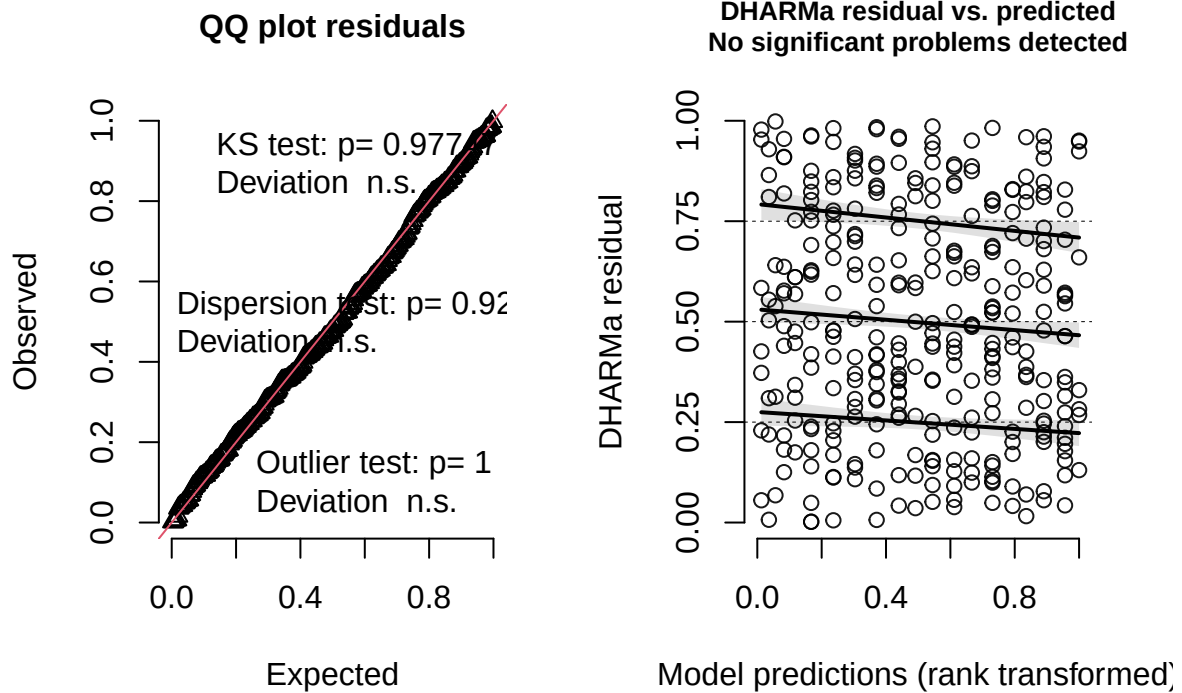
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	2.4575905	1.168167	2.1038002	0.0353959	Binomial	BirdID
fixed	factor(fac.yr)2	-1.2990687	1.287728	-1.0088065	0.3130674	Binomial	BirdID
fixed	factor(fac.yr)3	-1.9238091	1.253261	-1.5350425	0.1247734	Binomial	BirdID
fixed	factor(fac.yr)4	-1.0940946	1.258998	-0.8690198	0.3848363	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-1.8263886	1.255120	-1.4551508	0.1456275	Binomial	BirdID
fixed	factor(fac.yr)6	-1.7222500	1.262429	-1.3642351	0.1724936	Binomial	BirdID
fixed	factor(fac.yr)7	-2.0888962	1.265235	-1.6509949	0.0987396	Binomial	BirdID
fixed	factor(fac.yr)8	-1.8801378	1.265190	-1.4860517	0.1372654	Binomial	BirdID
fixed	factor(fac.yr)9	-1.6430129	1.287897	-1.2757332	0.2020498	Binomial	BirdID
fixed	factor(fac.yr)10	-0.5913410	1.341586	-0.4407775	0.6593741	Binomial	BirdID
fixed	factor(fac.yr)11	-2.1382136	1.271811	-1.6812349	0.0927173	Binomial	BirdID
fixed	factor(fac.yr)12	-0.7769364	1.332171	-0.5832107	0.5597514	Binomial	BirdID
fixed	factor(fac.yr)13	-0.8197277	1.458532	-0.5620223	0.5741009	Binomial	BirdID
fixed	factor(fac.yr)14	-0.8826246	1.399449	-0.6306942	0.5282405	Binomial	BirdID
fixed	factor(fac.yr)15	-2.6364322	1.353324	-1.9481157	0.0514011	Binomial	BirdID
fixed	factor(fac.yr)16	-1.7343118	1.398797	-1.2398598	0.2150273	Binomial	BirdID
fixed	factor(fac.yr)17	-2.1902109	1.395699	-1.5692571	0.1165880	Binomial	BirdID
fixed	factor(fac.yr)18	-2.8355851	1.454543	-1.9494683	0.0512395	Binomial	BirdID
fixed	factor(fac.yr)19	-3.1662345	1.527002	-2.0734970	0.0381261	Binomial	BirdID
fixed	factor(fac.yr)20	-2.6736028	1.600089	-1.6709087	0.0947397	Binomial	BirdID
fixed	(Intercept)	2.2308222	1.205912	1.8499043	0.0643273	Binomial	Terr
fixed	factor(fac.yr)2	-1.6979695	1.358239	-1.2501254	0.2112537	Binomial	Terr
fixed	factor(fac.yr)3	-1.6811111	1.312916	-1.2804407	0.2003902	Binomial	Terr
fixed	factor(fac.yr)4	-0.6680479	1.314064	-0.5083831	0.6111847	Binomial	Terr
fixed	factor(fac.yr)5	-1.2801583	1.310612	-0.9767640	0.3286860	Binomial	Terr
fixed	factor(fac.yr)6	-1.2260927	1.317395	-0.9306945	0.3520116	Binomial	Terr
fixed	factor(fac.yr)7	-1.4665545	1.293303	-1.1339607	0.2568110	Binomial	Terr
fixed	factor(fac.yr)8	-1.4403183	1.303493	-1.1049684	0.2691733	Binomial	Terr
fixed	factor(fac.yr)9	-1.6177807	1.321527	-1.2241757	0.2208860	Binomial	Terr
fixed	factor(fac.yr)10	-0.1476742	1.379767	-0.1070283	0.9147665	Binomial	Terr
fixed	factor(fac.yr)11	-1.5764549	1.297154	-1.2153182	0.2242447	Binomial	Terr
fixed	factor(fac.yr)12	-0.2294169	1.365005	-0.1680704	0.8665279	Binomial	Terr
fixed	factor(fac.yr)13	-0.0628866	1.514402	-0.0415257	0.9668768	Binomial	Terr
fixed	factor(fac.yr)14	-0.3351312	1.436338	-0.2333233	0.8155104	Binomial	Terr
fixed	factor(fac.yr)15	-2.2110208	1.367198	-1.6171910	0.1058371	Binomial	Terr
fixed	factor(fac.yr)16	-1.3272711	1.407077	-0.9432823	0.3455365	Binomial	Terr
fixed	factor(fac.yr)17	-2.1279679	1.429612	-1.4884932	0.1366209	Binomial	Terr
fixed	factor(fac.yr)18	-2.6828547	1.499251	-1.7894635	0.0735402	Binomial	Terr
fixed	factor(fac.yr)19	-3.0921580	1.553943	-1.9898788	0.0466043	Binomial	Terr
fixed	factor(fac.yr)20	-2.5693888	1.624839	-1.5813195	0.1138050	Binomial	Terr
fixed	(Intercept)	2.4677990	1.243781	1.9841108	0.0472435	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-1.7079116	1.397120	-1.2224512	0.2215370	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-1.8537124	1.347735	-1.3754284	0.1689987	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-0.7855636	1.353717	-0.5803012	0.5617115	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-1.4742927	1.350176	-1.0919261	0.2748656	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-1.4249376	1.358926	-1.0485761	0.2943733	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-1.7301053	1.335067	-1.2958943	0.1950119	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-1.6566019	1.343808	-1.2327670	0.2176627	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-1.7892994	1.369813	-1.3062365	0.1914721	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-0.2454383	1.431645	-0.1714379	0.8638794	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-1.8561880	1.342222	-1.3829220	0.1666888	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-0.3943699	1.414306	-0.2788434	0.7803650	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-0.3056347	1.577449	-0.1937525	0.8463697	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-0.4895434	1.494188	-0.3276318	0.7431901	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-2.5487487	1.420144	-1.7947114	0.0726997	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-1.5442935	1.473691	-1.0479087	0.2946807	Binomial	BirdID+Terr

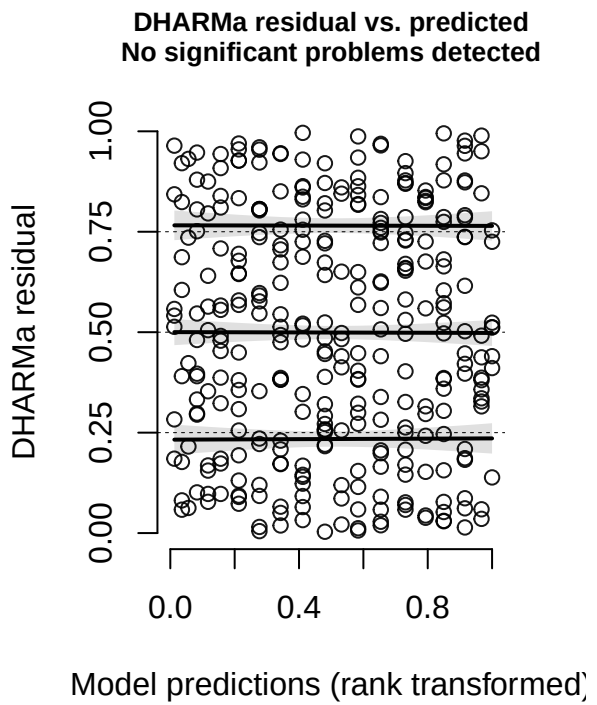
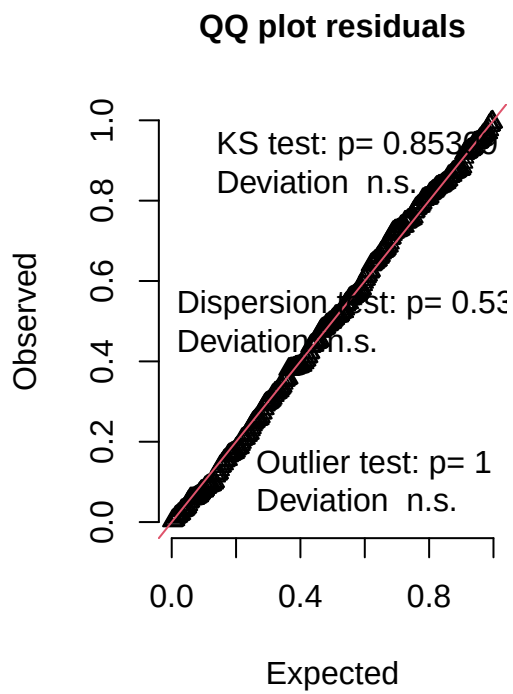
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-2.3078244	1.495891	-1.5427762	0.1228851	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-2.8583611	1.565484	-1.8258644	0.0678707	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-3.2485594	1.623498	-2.0009624	0.0453964	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-2.7470419	1.703087	-1.6129784	0.1067492	Binomial	BirdID+Terr

Territory only model is performing best on AIC and BIC.

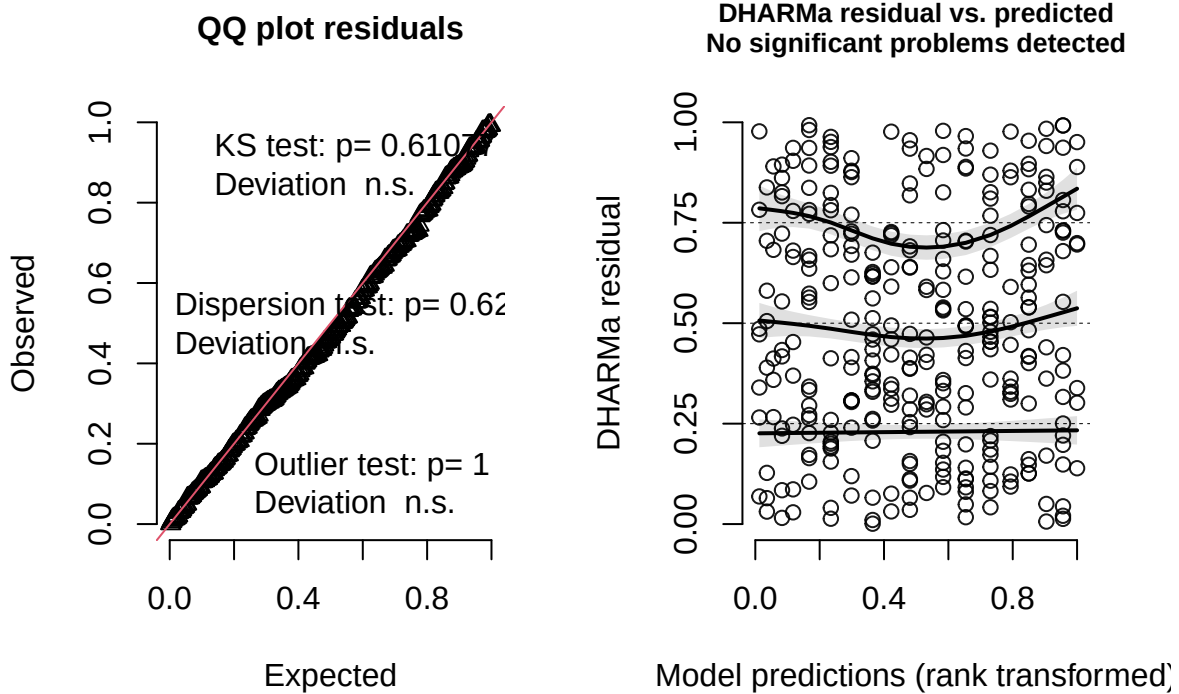
DHARMA residual



DHARMa residual



DHARMA residual



Diagnostics look good.

Probabilites used to generate simulation 21

Simulation 21
 Probability of mate fidelity: 0.084
 Probability of pair dispersal: 0.927
 Probability of single male dispersal: 0.437
 Probability of single female dispersal: 0.424

Simulation 22

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
363	1	-490.3527	1022.705	1104.488	336.7421	342	0.4117964	0.2098540	Poisson	Terr
363	1	-490.1991	1024.398	1110.075	326.1320	341	0.4166805	0.2081943	Poisson	BirdID+Terr
363	1	-497.7553	1037.511	1119.293	375.2772	342	0.3585550	0.2053308	Poisson	BirdID

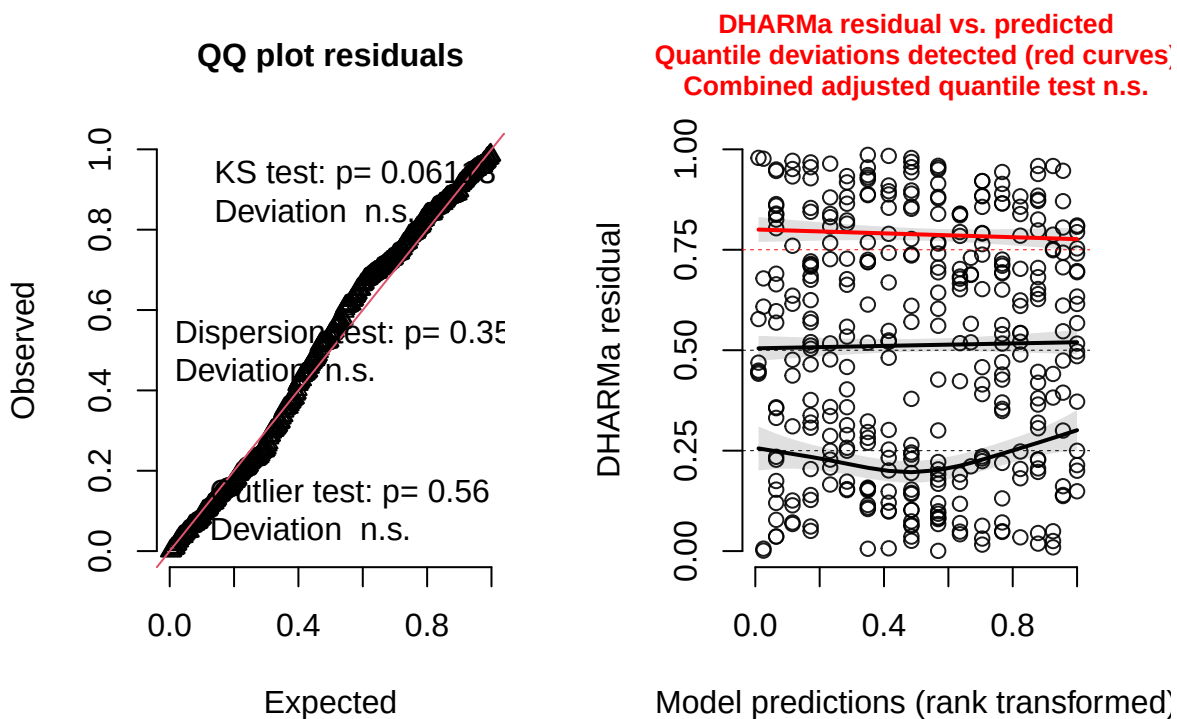
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.3888953	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.4546856	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.1595974	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.4386880	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.5531723	0.2276443	2.4299856	0.0150994	Poisson	BirdID
fixed	factor(fac.yr)2	-0.2358926	0.3040405	-0.7758592	0.4378321	Poisson	BirdID
fixed	factor(fac.yr)3	-1.2720297	0.3460998	-3.6753265	0.0002375	Poisson	BirdID
fixed	factor(fac.yr)4	-0.4930121	0.2897377	-1.7015806	0.0888340	Poisson	BirdID
fixed	factor(fac.yr)5	-0.4640028	0.2866183	-1.6188874	0.1054715	Poisson	BirdID
fixed	factor(fac.yr)6	-0.5280521	0.3023109	-1.7467185	0.0806862	Poisson	BirdID
fixed	factor(fac.yr)7	-1.3702010	0.3869377	-3.5411415	0.0003984	Poisson	BirdID
fixed	factor(fac.yr)8	-0.2072744	0.2889424	-0.7173554	0.4731548	Poisson	BirdID
fixed	factor(fac.yr)9	-0.2791021	0.2909604	-0.9592444	0.3374356	Poisson	BirdID
fixed	factor(fac.yr)10	-0.5112387	0.3153059	-1.6214057	0.1049307	Poisson	BirdID
fixed	factor(fac.yr)11	-0.6850043	0.3285837	-2.0847180	0.0370949	Poisson	BirdID
fixed	factor(fac.yr)12	-0.3063227	0.3093357	-0.9902599	0.3220471	Poisson	BirdID
fixed	factor(fac.yr)13	0.0369837	0.2934276	0.1260403	0.8997000	Poisson	BirdID
fixed	factor(fac.yr)14	-0.4057955	0.3232498	-1.2553621	0.2093474	Poisson	BirdID
fixed	factor(fac.yr)15	-0.8685981	0.3707734	-2.3426656	0.0191465	Poisson	BirdID
fixed	factor(fac.yr)16	-1.3030097	0.4535147	-2.8731366	0.0040642	Poisson	BirdID
fixed	factor(fac.yr)17	-0.1715391	0.3699786	-0.4636460	0.6429014	Poisson	BirdID
fixed	factor(fac.yr)18	-0.3999825	0.4504163	-0.8880285	0.3745254	Poisson	BirdID
fixed	factor(fac.yr)19	-1.6483552	0.7718831	-2.1354985	0.0327203	Poisson	BirdID
fixed	factor(fac.yr)20	-1.8041186	0.7753724	-2.3267767	0.0199772	Poisson	BirdID
fixed	(Intercept)	0.6200051	0.2297184	2.6989792	0.0069553	Poisson	Terr
fixed	factor(fac.yr)2	-0.2790510	0.3204394	-0.8708386	0.3838423	Poisson	Terr
fixed	factor(fac.yr)3	-1.2317528	0.3506637	-3.5126324	0.0004437	Poisson	Terr
fixed	factor(fac.yr)4	-0.5644878	0.2917365	-1.9349237	0.0529997	Poisson	Terr
fixed	factor(fac.yr)5	-0.4068184	0.2907627	-1.3991420	0.1617704	Poisson	Terr
fixed	factor(fac.yr)6	-0.5240783	0.3024231	-1.7329307	0.0831080	Poisson	Terr
fixed	factor(fac.yr)7	-1.4130981	0.3833932	-3.6857672	0.0002280	Poisson	Terr
fixed	factor(fac.yr)8	-0.2806964	0.2866369	-0.9792750	0.3274441	Poisson	Terr
fixed	factor(fac.yr)9	-0.3061427	0.2905202	-1.0537744	0.2919862	Poisson	Terr
fixed	factor(fac.yr)10	-0.5728106	0.3143571	-1.8221655	0.0684299	Poisson	Terr
fixed	factor(fac.yr)11	-0.7772733	0.3301259	-2.3544755	0.0185489	Poisson	Terr
fixed	factor(fac.yr)12	-0.4442827	0.3106007	-1.4303983	0.1526027	Poisson	Terr
fixed	factor(fac.yr)13	0.0833876	0.2914319	0.2861307	0.7747780	Poisson	Terr
fixed	factor(fac.yr)14	-0.4519321	0.3170832	-1.4252791	0.1540765	Poisson	Terr
fixed	factor(fac.yr)15	-0.9207108	0.3690094	-2.4950879	0.0125926	Poisson	Terr
fixed	factor(fac.yr)16	-1.3308844	0.4562178	-2.9172130	0.0035317	Poisson	Terr
fixed	factor(fac.yr)17	-0.2597680	0.3709737	-0.7002330	0.4837818	Poisson	Terr
fixed	factor(fac.yr)18	-0.5551555	0.4381871	-1.2669372	0.2051777	Poisson	Terr
fixed	factor(fac.yr)19	-1.8435266	0.7554521	-2.4402958	0.0146752	Poisson	Terr
fixed	factor(fac.yr)20	-2.0532362	0.7669394	-2.6771819	0.0074244	Poisson	Terr
fixed	(Intercept)	0.6093921	0.2331571	2.6136548	0.0089579	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.2808836	0.3231305	-0.8692572	0.3847065	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-1.2462615	0.3532841	-3.5276471	0.0004193	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.5612134	0.2952363	-1.9008959	0.0573157	Poisson	BirdID+Terr

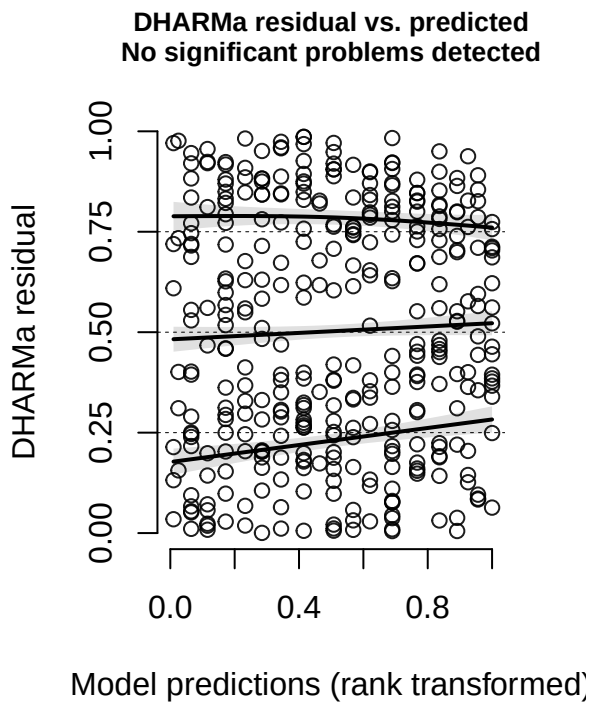
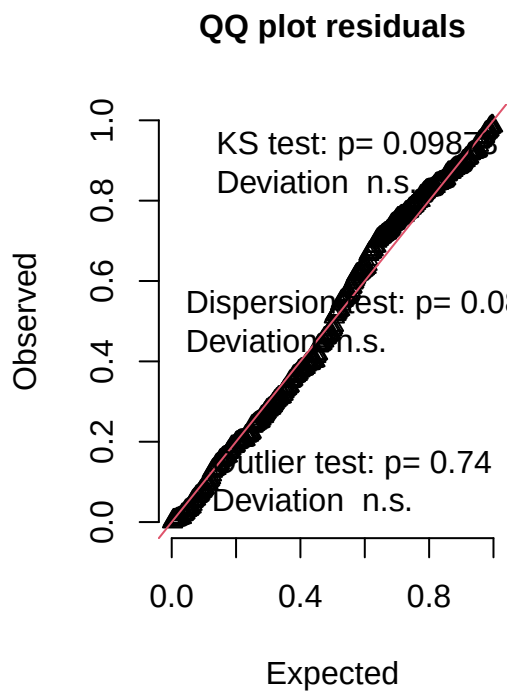
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.4152190	0.2943791	-1.4104907	0.1583949	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.5339112	0.3068076	-1.7402150	0.0818213	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-1.4200236	0.3872231	-3.6671972	0.0002452	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.2865268	0.2911401	-0.9841544	0.3250396	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.3038199	0.2946670	-1.0310619	0.3025118	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.5792681	0.3190423	-1.8156466	0.0694246	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.7850510	0.3343905	-2.3477072	0.0188894	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.4437615	0.3149629	-1.4089325	0.1588551	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	0.0708451	0.2961441	0.2392252	0.8109310	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.4543615	0.3224115	-1.4092596	0.1587584	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.9360458	0.3737774	-2.5042866	0.0122699	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-1.3520516	0.4603084	-2.9372734	0.0033111	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.2616023	0.3766259	-0.6945947	0.4873093	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.5503731	0.4457090	-1.2348260	0.2168953	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.8504501	0.7617622	-2.4291703	0.0151334	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-2.0647781	0.7727285	-2.6720617	0.0075387	Poisson	BirdID+Terr

Territory only model is performing well on all criteria.

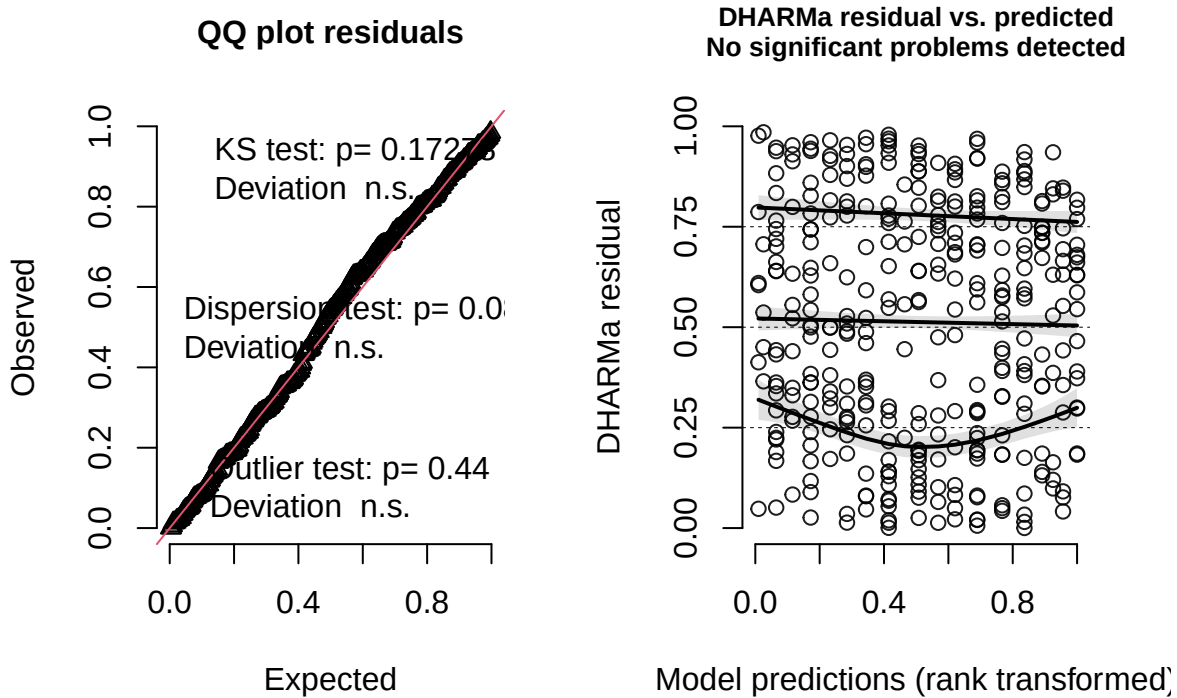
DHARMA residual



DHARMa residual



DHARMa residual



Territory only model diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
363	1	-203.7316	451.4631	537.1400	182.7792	341	0.885	Binomial	BirdID+Terr
363	1	-210.7421	463.4842	545.2667	308.7060	342	0.639	Binomial	Terr
363	1	-211.3103	464.6205	546.4030	305.1618	342	0.660	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.330063	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.270406	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.828019	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.587986	Binomial	BirdID+Terr

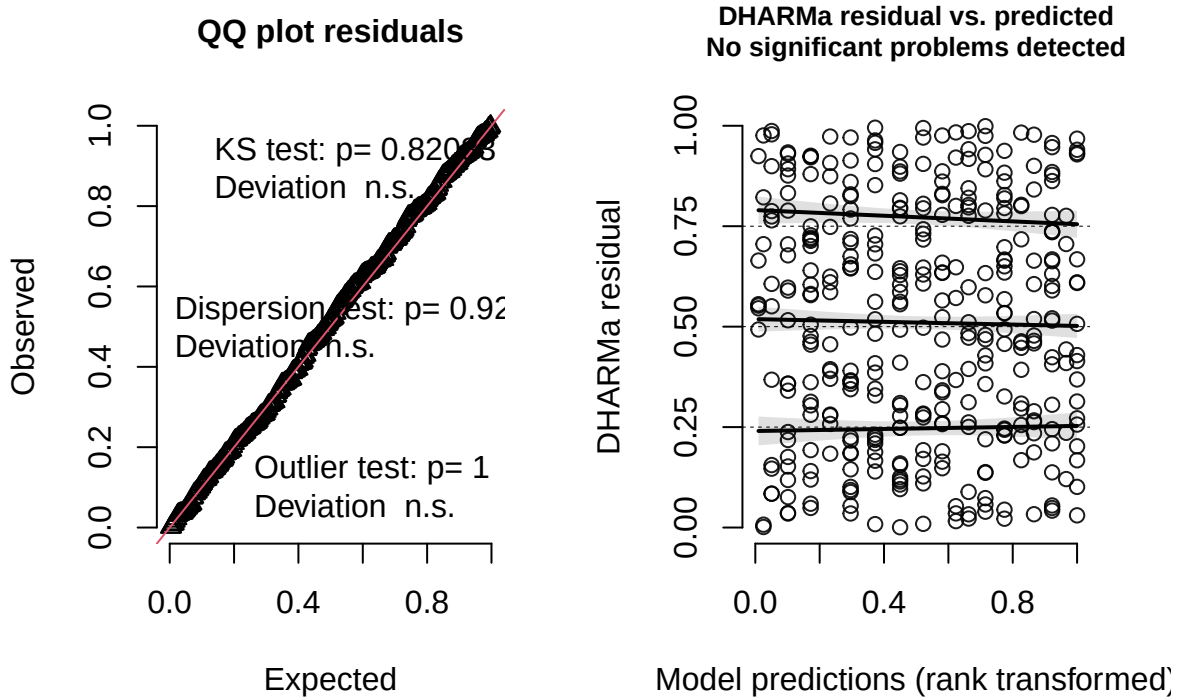
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	1.4703452	0.7902720	1.8605560	0.0628069	Binomial	BirdID
fixed	factor(fac.yr)2	0.0790532	1.0700825	0.0738758	0.9411092	Binomial	BirdID
fixed	factor(fac.yr)3	-1.9575659	0.8981133	-2.1796426	0.0292840	Binomial	BirdID
fixed	factor(fac.yr)4	-1.6659004	0.8961311	-1.8589918	0.0630283	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-1.3113218	0.8944163	-1.4661202	0.1426155	Binomial	BirdID
fixed	factor(fac.yr)6	-1.4789318	0.9257202	-1.5976014	0.1101317	Binomial	BirdID
fixed	factor(fac.yr)7	-2.0952705	0.9420397	-2.2241849	0.0261360	Binomial	BirdID
fixed	factor(fac.yr)8	-0.2921849	0.9728074	-0.3003522	0.7639085	Binomial	BirdID
fixed	factor(fac.yr)9	0.1328082	0.9898149	0.1341748	0.8932644	Binomial	BirdID
fixed	factor(fac.yr)10	-1.2257460	0.9475923	-1.2935373	0.1958253	Binomial	BirdID
fixed	factor(fac.yr)11	-1.1575984	0.9472289	-1.2220894	0.2216738	Binomial	BirdID
fixed	factor(fac.yr)12	-0.5539777	0.9841534	-0.5628977	0.5735046	Binomial	BirdID
fixed	factor(fac.yr)13	1.8306214	1.4557051	1.2575496	0.2085547	Binomial	BirdID
fixed	factor(fac.yr)14	-0.4565824	1.0022923	-0.4555381	0.6487221	Binomial	BirdID
fixed	factor(fac.yr)15	-1.6739152	1.0070172	-1.6622508	0.0964625	Binomial	BirdID
fixed	factor(fac.yr)16	-2.9721757	1.0759670	-2.7623298	0.0057390	Binomial	BirdID
fixed	factor(fac.yr)17	-0.6089676	1.1864337	-0.5132757	0.6077585	Binomial	BirdID
fixed	factor(fac.yr)18	0.4975805	1.6300985	0.3052457	0.7601790	Binomial	BirdID
fixed	factor(fac.yr)19	-3.8143644	1.5740978	-2.4232068	0.0153842	Binomial	BirdID
fixed	factor(fac.yr)20	-3.9568258	1.5619124	-2.5333211	0.0112987	Binomial	BirdID
fixed	(Intercept)	1.8019408	0.7831847	2.3007865	0.0214037	Binomial	Terr
fixed	factor(fac.yr)2	0.0302143	1.0824022	0.0279141	0.9777307	Binomial	Terr
fixed	factor(fac.yr)3	-1.7833228	0.8949397	-1.9926737	0.0462972	Binomial	Terr
fixed	factor(fac.yr)4	-1.7065918	0.8887569	-1.9202009	0.0548325	Binomial	Terr
fixed	factor(fac.yr)5	-1.1751262	0.8787763	-1.3372302	0.1811475	Binomial	Terr
fixed	factor(fac.yr)6	-1.4572192	0.9098169	-1.6016621	0.1092304	Binomial	Terr
fixed	factor(fac.yr)7	-2.1838901	0.9094944	-2.4012134	0.0163408	Binomial	Terr
fixed	factor(fac.yr)8	-0.3504096	0.9483047	-0.3695117	0.7117464	Binomial	Terr
fixed	factor(fac.yr)9	-0.0266961	0.9646369	-0.0276747	0.9779216	Binomial	Terr
fixed	factor(fac.yr)10	-1.3596151	0.9308118	-1.4606768	0.1441042	Binomial	Terr
fixed	factor(fac.yr)11	-1.4237606	0.9268686	-1.5360975	0.1245145	Binomial	Terr
fixed	factor(fac.yr)12	-1.0233096	0.9634964	-1.0620793	0.2881997	Binomial	Terr
fixed	factor(fac.yr)13	2.0234118	1.4240788	1.4208566	0.1553585	Binomial	Terr
fixed	factor(fac.yr)14	-0.5825581	0.9693996	-0.6009473	0.5478751	Binomial	Terr
fixed	factor(fac.yr)15	-1.7035425	0.9805344	-1.7373612	0.0823234	Binomial	Terr
fixed	factor(fac.yr)16	-2.8151385	1.0673675	-2.6374594	0.0083530	Binomial	Terr
fixed	factor(fac.yr)17	-0.8349931	1.1448672	-0.7293362	0.4657961	Binomial	Terr
fixed	factor(fac.yr)18	0.1889733	1.4744723	0.1281633	0.8980197	Binomial	Terr
fixed	factor(fac.yr)19	-3.9319775	1.5249743	-2.5783893	0.0099262	Binomial	Terr
fixed	factor(fac.yr)20	-4.3914118	1.5124628	-2.9034841	0.0036904	Binomial	Terr
fixed	(Intercept)	2.4434671	1.0836897	2.2547663	0.0241480	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-0.2991723	1.3981483	-0.2139775	0.8305646	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-2.8757125	1.2149133	-2.3670105	0.0179324	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-2.5327708	1.2062380	-2.0997272	0.0357528	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-1.5980560	1.1958106	-1.3363789	0.1814255	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-2.0710893	1.2447481	-1.6638622	0.0961400	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-3.1908495	1.2449188	-2.5630985	0.0103743	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-0.6957900	1.3110175	-0.5307252	0.5956093	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-0.1894821	1.3179682	-0.1437683	0.8856834	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-2.1244434	1.2674763	-1.6761208	0.0937145	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-2.0932076	1.2680867	-1.6506818	0.0988036	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-1.3598519	1.3013919	-1.0449212	0.2960594	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	2.2605469	1.9455465	1.1619084	0.2452727	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-0.7554283	1.3395263	-0.5639518	0.5727869	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-2.7769878	1.3625414	-2.0380942	0.0415405	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-4.5153615	1.4510231	-3.1118467	0.0018592	Binomial	BirdID+Terr

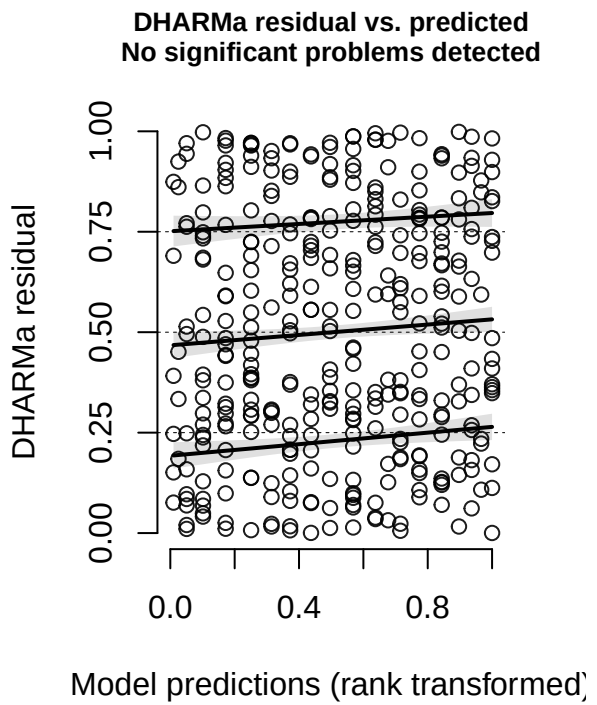
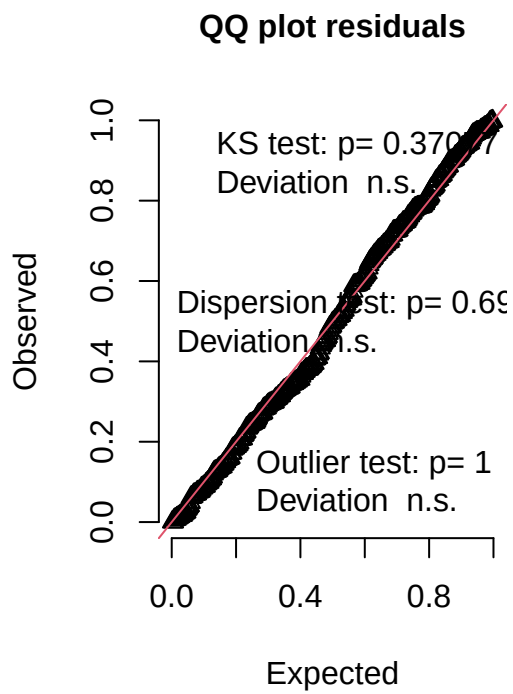
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-1.1797881	1.6194433	-0.7285146	0.4662986	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	0.0064919	2.2112989	0.0029358	0.9976576	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-5.7667371	2.0001415	-2.8831645	0.0039370	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-6.5898731	2.0564750	-3.2044509	0.0013532	Binomial	BirdID+Terr

The model fitted with both REs is best optimizing the selection criteria.

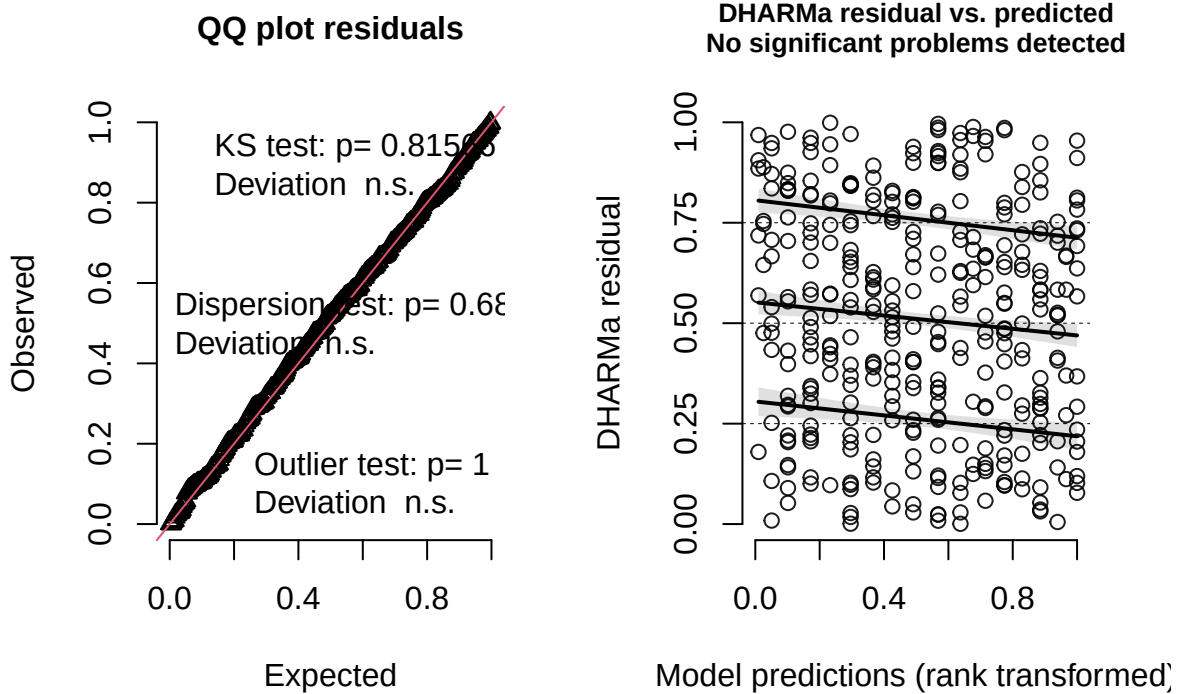
DHARMA residual



DHARMA residual



DHARMa residual



Diagnostics look good.

Probabilites used to generate simulation 22

Simulation 22
 Probability of mate fidelity: 0.725
 Probability of pair dispersal: 0.504
 Probability of single male dispersal: 0.731
 Probability of single female dispersal: 0.999

Simulation 23

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
300	1	-408.6014	859.2028	936.9822	333.4774	279	0.3157006	0.2233828	Poisson	BirdID
300	1	-409.2554	860.5107	938.2902	343.6564	279	0.2933053	0.2194405	Poisson	Terr
300	1	-408.4782	860.9565	942.4397	325.7227	278	0.3220571	0.2174303	Poisson	BirdID+Terr

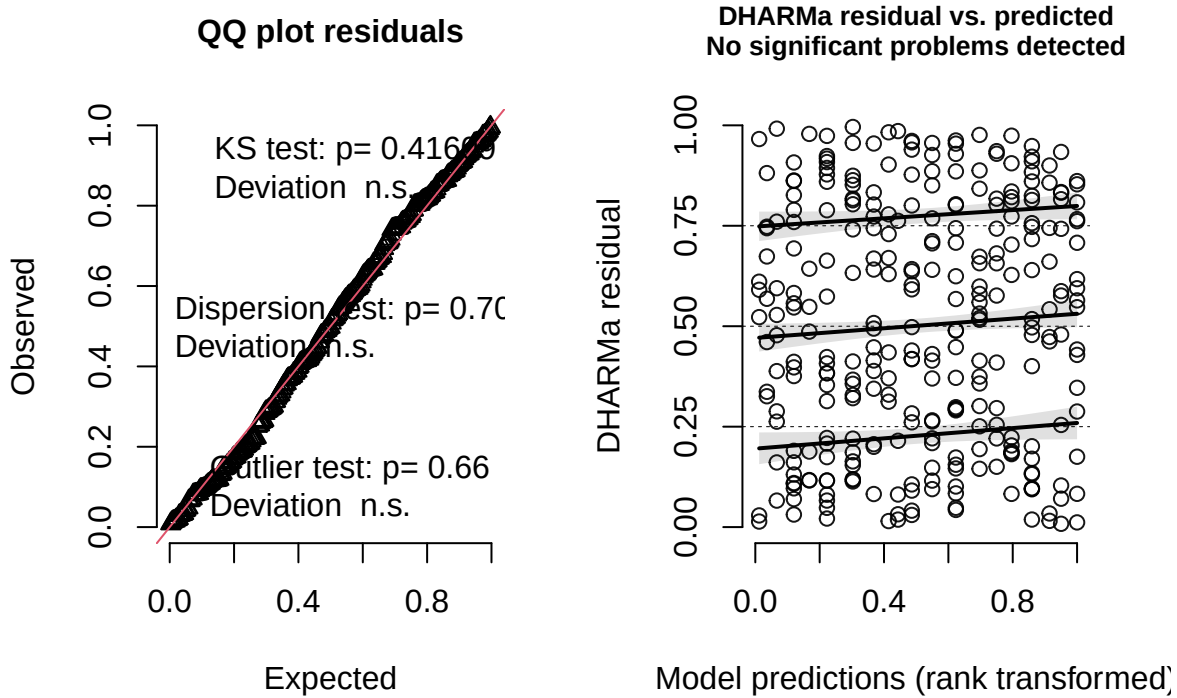
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.2904601	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.2540539	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.2645897	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.1600294	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.6426141	0.2225107	2.8880147	0.0038768	Poisson	BirdID
fixed	factor(fac.yr)2	-0.4830358	0.3082881	-1.5668328	0.1171538	Poisson	BirdID
fixed	factor(fac.yr)3	-1.0793110	0.3517717	-3.0682148	0.0021534	Poisson	BirdID
fixed	factor(fac.yr)4	-0.5573142	0.3178789	-1.7532281	0.0795629	Poisson	BirdID
fixed	factor(fac.yr)5	0.1383075	0.2838840	0.4871971	0.6261186	Poisson	BirdID
fixed	factor(fac.yr)6	-0.4970814	0.2963291	-1.6774638	0.0934518	Poisson	BirdID
fixed	factor(fac.yr)7	-0.9981407	0.3307627	-3.0176943	0.0025471	Poisson	BirdID
fixed	factor(fac.yr)8	-0.9188461	0.3361382	-2.7335364	0.0062658	Poisson	BirdID
fixed	factor(fac.yr)9	-0.1944023	0.2833718	-0.6860326	0.4926925	Poisson	BirdID
fixed	factor(fac.yr)10	-0.5552113	0.3237372	-1.7150065	0.0863440	Poisson	BirdID
fixed	factor(fac.yr)11	-0.9034300	0.3643950	-2.4792605	0.0131655	Poisson	BirdID
fixed	factor(fac.yr)12	-0.3257516	0.3261502	-0.9987778	0.3179024	Poisson	BirdID
fixed	factor(fac.yr)13	-0.2877554	0.3183021	-0.9040324	0.3659782	Poisson	BirdID
fixed	factor(fac.yr)14	-0.8976971	0.4127795	-2.1747617	0.0296480	Poisson	BirdID
fixed	factor(fac.yr)15	-1.3880929	0.5122246	-2.7099301	0.0067297	Poisson	BirdID
fixed	factor(fac.yr)16	-1.0585226	0.5141260	-2.0588779	0.0395059	Poisson	BirdID
fixed	factor(fac.yr)17	-0.0329152	0.3387738	-0.0971599	0.9225994	Poisson	BirdID
fixed	factor(fac.yr)18	-1.6930711	0.6330222	-2.6745840	0.0074822	Poisson	BirdID
fixed	factor(fac.yr)19	-1.7354630	0.7609485	-2.2806575	0.0225687	Poisson	BirdID
fixed	factor(fac.yr)20	-0.6900285	0.4849948	-1.4227546	0.1548073	Poisson	BirdID
fixed	(Intercept)	0.6054848	0.2176323	2.7821454	0.0054001	Poisson	Terr
fixed	factor(fac.yr)2	-0.4307869	0.3063414	-1.4062314	0.1596554	Poisson	Terr
fixed	factor(fac.yr)3	-1.0454374	0.3485947	-2.9990055	0.0027086	Poisson	Terr
fixed	factor(fac.yr)4	-0.5530764	0.3139452	-1.7616975	0.0781204	Poisson	Terr
fixed	factor(fac.yr)5	0.1559971	0.2773922	0.5623703	0.5738637	Poisson	Terr
fixed	factor(fac.yr)6	-0.4958893	0.2893760	-1.7136503	0.0865930	Poisson	Terr
fixed	factor(fac.yr)7	-0.9298593	0.3244276	-2.8661535	0.0041549	Poisson	Terr
fixed	factor(fac.yr)8	-0.8855888	0.3306941	-2.6779700	0.0074070	Poisson	Terr
fixed	factor(fac.yr)9	-0.1435848	0.2769886	-0.5183779	0.6041946	Poisson	Terr
fixed	factor(fac.yr)10	-0.4933028	0.3173540	-1.5544244	0.1200832	Poisson	Terr
fixed	factor(fac.yr)11	-0.8386571	0.3581902	-2.3413736	0.0192129	Poisson	Terr
fixed	factor(fac.yr)12	-0.2110182	0.3180638	-0.6634461	0.5070449	Poisson	Terr
fixed	factor(fac.yr)13	-0.1928562	0.3124657	-0.6172075	0.5370979	Poisson	Terr
fixed	factor(fac.yr)14	-0.7830560	0.4072086	-1.9229851	0.0544819	Poisson	Terr
fixed	factor(fac.yr)15	-1.2385830	0.5077029	-2.4395821	0.0147043	Poisson	Terr
fixed	factor(fac.yr)16	-0.9110462	0.5094306	-1.7883618	0.0737177	Poisson	Terr
fixed	factor(fac.yr)17	0.0030092	0.3322629	0.0090566	0.9927740	Poisson	Terr
fixed	factor(fac.yr)18	-1.6044442	0.6284823	-2.5528867	0.0106834	Poisson	Terr
fixed	factor(fac.yr)19	-1.6910753	0.7545476	-2.2411776	0.0250146	Poisson	Terr
fixed	factor(fac.yr)20	-0.5992329	0.4748945	-1.2618231	0.2070124	Poisson	Terr
fixed	(Intercept)	0.6185459	0.2245667	2.7543968	0.0058800	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.4622977	0.3102907	-1.4898855	0.1362543	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-1.0633237	0.3539488	-3.0041734	0.0026630	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.5490877	0.3206164	-1.7126000	0.0867861	Poisson	BirdID+Terr

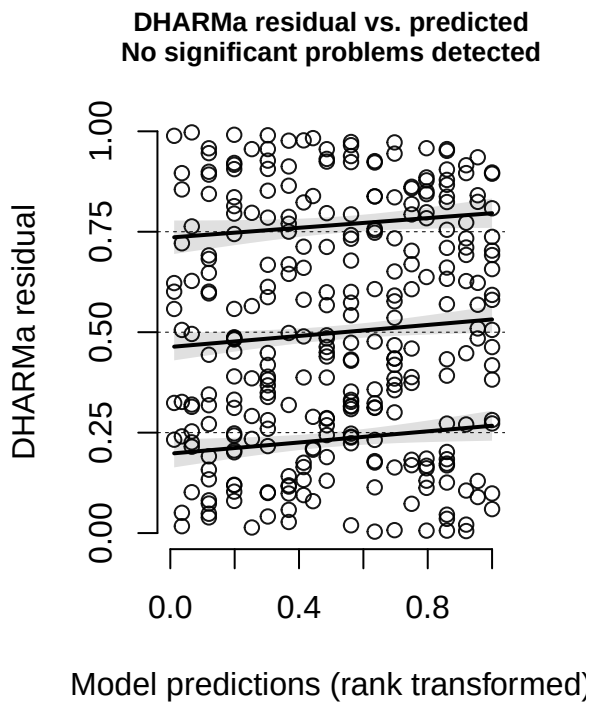
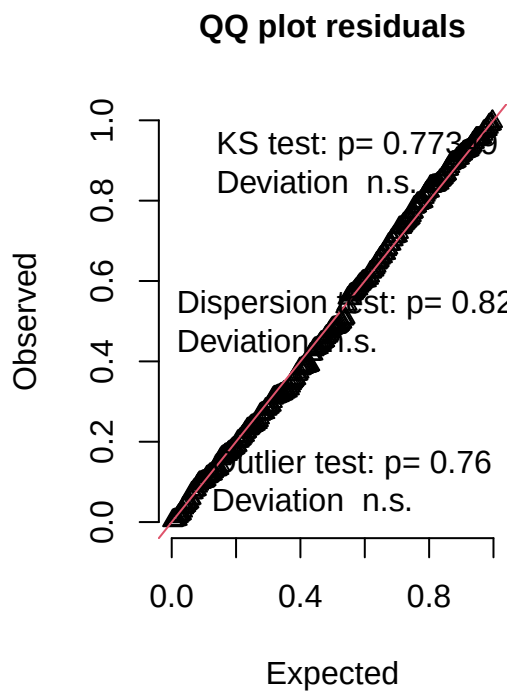
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.1517679	0.2865482	0.5296417	0.5963603	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.4917598	0.2982774	-1.6486657	0.0992161	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.9691918	0.3326019	-2.9139692	0.0035687	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.9047978	0.3384142	-2.6736402	0.0075033	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.1741246	0.2857781	-0.6093000	0.5423256	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.5268259	0.3260486	-1.6157897	0.1061398	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.8806688	0.3665983	-2.4022722	0.0162936	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.2952130	0.3283972	-0.8989511	0.3686787	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.2628687	0.3214505	-0.8177578	0.4134955	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.8584594	0.4156470	-2.0653569	0.0388892	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-1.3332447	0.5153630	-2.5870011	0.0096815	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-1.0085477	0.5179231	-1.9472922	0.0514997	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.0267205	0.3425194	-0.0780118	0.9378187	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-1.6645459	0.6357058	-2.6184217	0.0088338	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.7095984	0.7644157	-2.2364774	0.0253205	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.6541073	0.4874575	-1.3418756	0.1796363	Poisson	BirdID+Terr

The band ID only model seems to be performing well on all criteria.

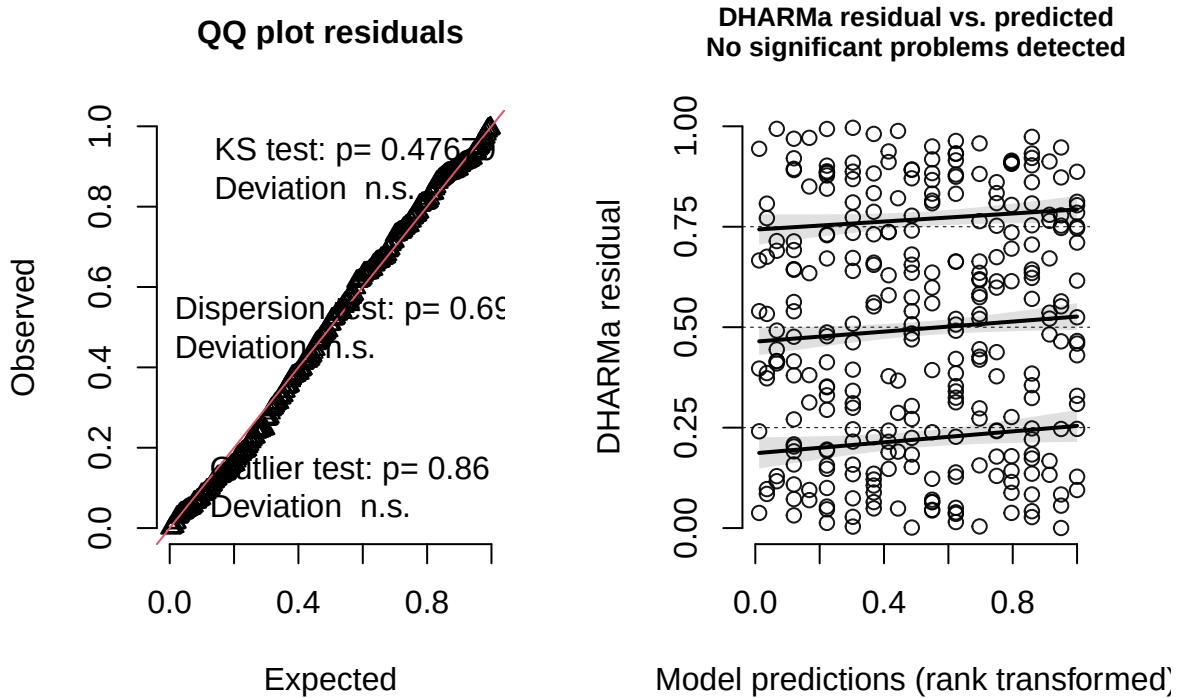
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
300	1	-181.9042	405.8084	483.5879	296.0913	279	0.565	Binomial	BirdID
300	1	-181.9040	407.8080	489.2912	296.2723	278	0.557	Binomial	BirdID+Terr
300	1	-184.8186	411.6372	489.4166	347.4284	279	0.346	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.9019941	Binomial	BirdID
ran_pars	Terr	sd__(Intercept)	0.4414269	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	0.8992788	Binomial	BirdID+Terr
ran_pars	Terr	sd__(Intercept)	0.0366122	Binomial	BirdID+Terr

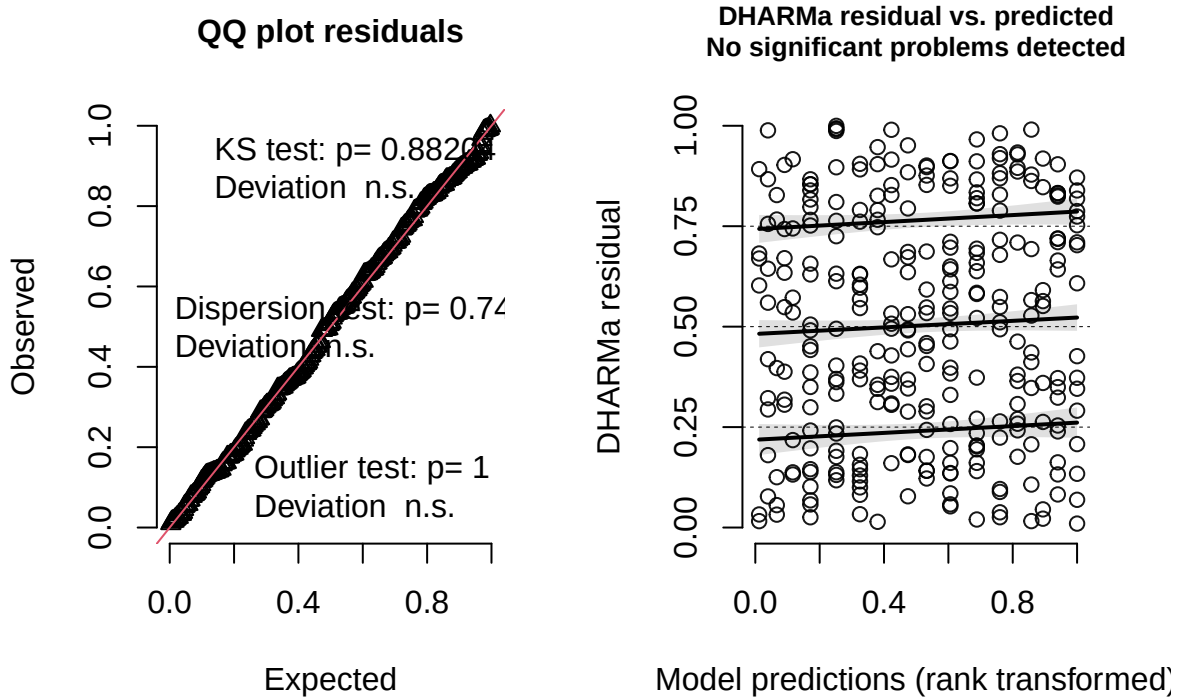
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	1.2544233	0.7354231	1.7057165	0.0880609	Binomial	BirdID
fixed	factor(fac.yr)2	0.5059053	0.9936140	0.5091568	0.6106423	Binomial	BirdID
fixed	factor(fac.yr)3	-1.3465623	0.8648132	-1.5570557	0.1194573	Binomial	BirdID
fixed	factor(fac.yr)4	-0.3188549	0.9084469	-0.3509891	0.7255965	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	1.1798932	1.1143676	1.0588007	0.2896905	Binomial	BirdID
fixed	factor(fac.yr)6	-0.4417184	0.8679523	-0.5089202	0.6108082	Binomial	BirdID
fixed	factor(fac.yr)7	-1.6168951	0.8635413	-1.8724005	0.0611512	Binomial	BirdID
fixed	factor(fac.yr)8	-1.3942852	0.8757318	-1.5921372	0.1113539	Binomial	BirdID
fixed	factor(fac.yr)9	-0.3900955	0.8827209	-0.4419240	0.6585442	Binomial	BirdID
fixed	factor(fac.yr)10	-0.5658138	0.9119976	-0.6204115	0.5349869	Binomial	BirdID
fixed	factor(fac.yr)11	-0.9145288	0.9240374	-0.9897097	0.3223160	Binomial	BirdID
fixed	factor(fac.yr)12	-0.0866832	1.0324362	-0.0839598	0.9330884	Binomial	BirdID
fixed	factor(fac.yr)13	-1.0671108	0.9491312	-1.1243028	0.2608846	Binomial	BirdID
fixed	factor(fac.yr)14	-1.0906066	1.0038720	-1.0864001	0.2773020	Binomial	BirdID
fixed	factor(fac.yr)15	-2.3328317	1.0727959	-2.1745345	0.0296650	Binomial	BirdID
fixed	factor(fac.yr)16	-1.6783939	1.1285161	-1.4872573	0.1369469	Binomial	BirdID
fixed	factor(fac.yr)17	0.1009110	1.1658916	0.0865527	0.9310271	Binomial	BirdID
fixed	factor(fac.yr)18	-1.9632050	1.1019766	-1.7815305	0.0748258	Binomial	BirdID
fixed	factor(fac.yr)19	-3.0823677	1.4627292	-2.1072716	0.0350940	Binomial	BirdID
fixed	factor(fac.yr)20	-2.1256814	1.2317157	-1.7257890	0.0843854	Binomial	BirdID
fixed	(Intercept)	1.1038523	0.6855077	1.6102698	0.1073390	Binomial	Terr
fixed	factor(fac.yr)2	0.5929096	0.9420437	0.6293865	0.5290960	Binomial	Terr
fixed	factor(fac.yr)3	-1.1988042	0.8159743	-1.4691690	0.1417870	Binomial	Terr
fixed	factor(fac.yr)4	-0.2987221	0.8516614	-0.3507521	0.7257743	Binomial	Terr
fixed	factor(fac.yr)5	0.9791208	1.0352252	0.9458047	0.3442482	Binomial	Terr
fixed	factor(fac.yr)6	-0.5059403	0.8063498	-0.6274451	0.5303675	Binomial	Terr
fixed	factor(fac.yr)7	-1.4829176	0.8012009	-1.8508687	0.0641884	Binomial	Terr
fixed	factor(fac.yr)8	-1.2663025	0.8135685	-1.5564794	0.1195941	Binomial	Terr
fixed	factor(fac.yr)9	-0.4046105	0.8192008	-0.4939088	0.6213706	Binomial	Terr
fixed	factor(fac.yr)10	-0.5896262	0.8458328	-0.6970955	0.4857431	Binomial	Terr
fixed	factor(fac.yr)11	-0.8405097	0.8559585	-0.9819515	0.3261237	Binomial	Terr
fixed	factor(fac.yr)12	0.1505512	0.9607950	0.1566944	0.8754857	Binomial	Terr
fixed	factor(fac.yr)13	-0.8542003	0.8808696	-0.9697239	0.3321841	Binomial	Terr
fixed	factor(fac.yr)14	-0.8596361	0.9247972	-0.9295401	0.3526093	Binomial	Terr
fixed	factor(fac.yr)15	-1.8726378	0.9864684	-1.8983251	0.0576533	Binomial	Terr
fixed	factor(fac.yr)16	-1.3297820	1.0403413	-1.2782170	0.2011729	Binomial	Terr
fixed	factor(fac.yr)17	0.1507857	1.0728691	0.1405443	0.8882299	Binomial	Terr
fixed	factor(fac.yr)18	-1.6650324	1.0150651	-1.6403208	0.1009385	Binomial	Terr
fixed	factor(fac.yr)19	-2.7459677	1.3234473	-2.0748599	0.0379995	Binomial	Terr
fixed	factor(fac.yr)20	-1.8012054	1.1194927	-1.6089479	0.1076277	Binomial	Terr
fixed	(Intercept)	1.2642683	0.7367068	1.7161080	0.0861423	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	0.5072605	0.9961909	0.5092001	0.6106120	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-1.3570340	0.8658962	-1.5672018	0.1170675	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-0.3299898	0.9094024	-0.3628645	0.7167061	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	1.1608813	1.1139891	1.0420940	0.2973681	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-0.4620521	0.8687503	-0.5318584	0.5948241	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-1.6417221	0.8647671	-1.8984557	0.0576361	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-1.4112019	0.8767287	-1.6096222	0.1074804	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-0.4200315	0.8831131	-0.4756259	0.6343409	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-0.5787144	0.9128858	-0.6339395	0.5261203	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-0.9307822	0.9248290	-1.0064372	0.3142053	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-0.0882167	1.0342454	-0.0852957	0.9320263	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-1.0806302	0.9498854	-1.1376427	0.2552697	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-1.0801297	1.0048563	-1.0749096	0.2824152	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-2.3245726	1.0721889	-2.1680625	0.0301539	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-1.6880495	1.1293210	-1.4947473	0.1349804	Binomial	BirdID+Terr

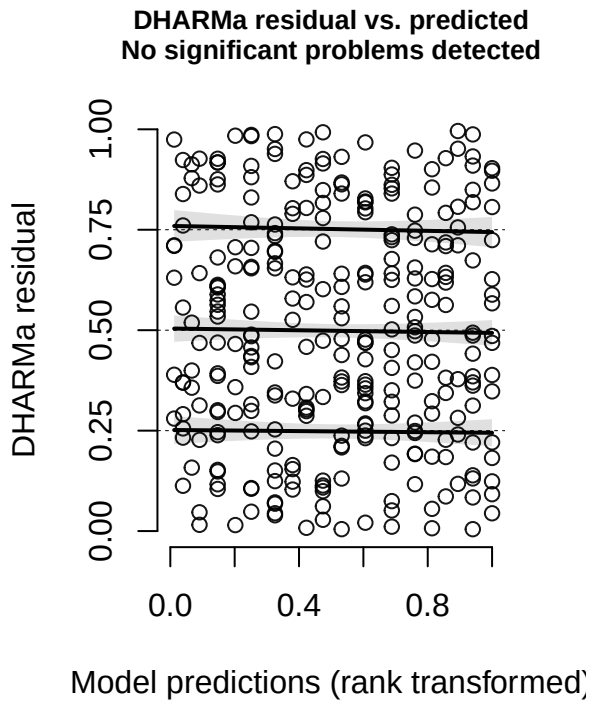
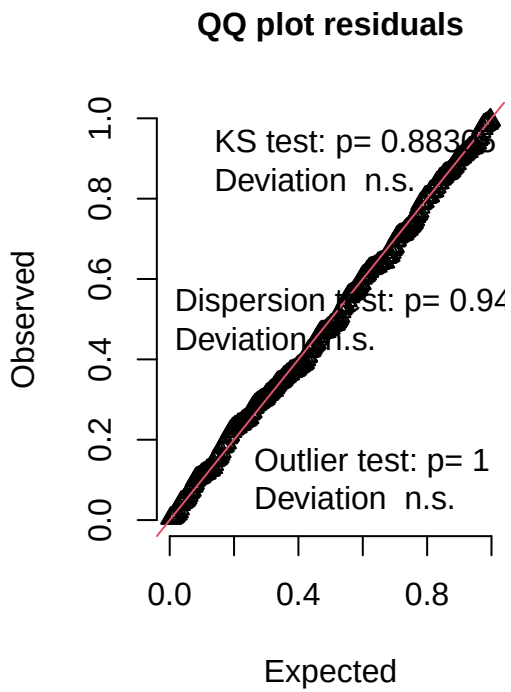
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	0.1033206	1.1680213	0.0884578	0.9295128	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-1.9670026	1.1023666	-1.7843452	0.0743676	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-3.0579939	1.4534794	-2.1039128	0.0353861	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-2.1194294	1.2306144	-1.7222531	0.0850237	Binomial	BirdID+Terr

Band ID only model is performing the best on selection criteria.

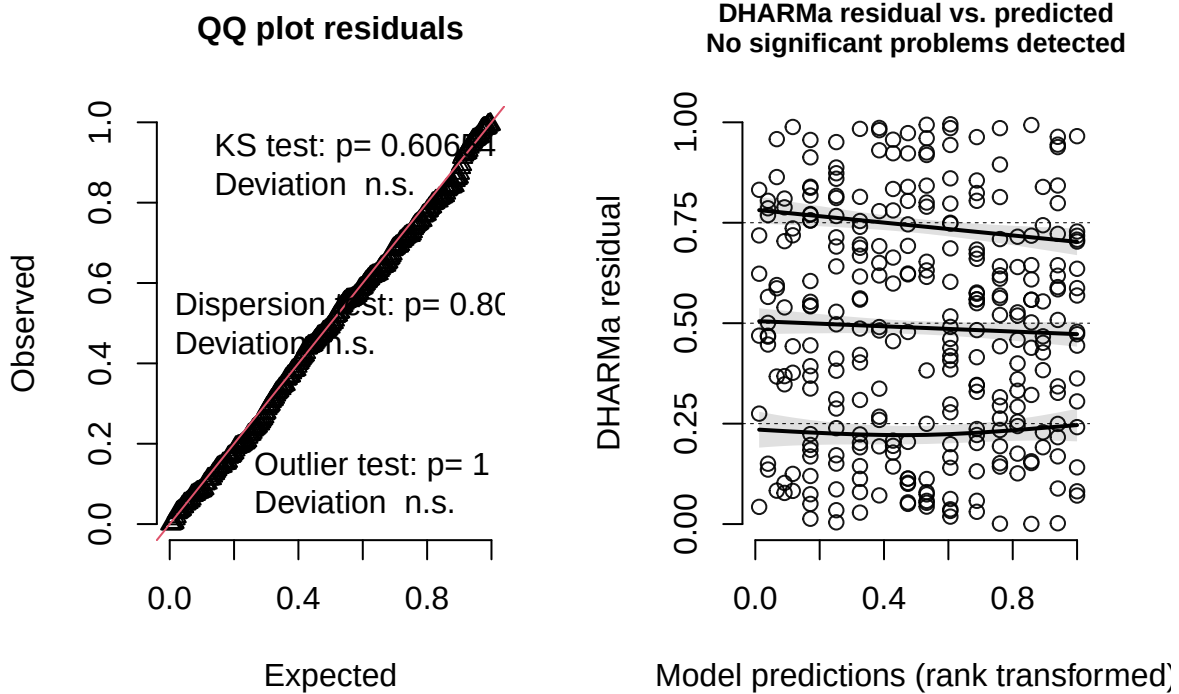
DHARMa residual



DHARMa residual



DHARMA residual



Probabilites used to generate simulation 23

Simulation 23

Probability of mate fidelity: 0.213

Probability of pair dispersal: 0.242

Probability of single male dispersal: 0.356

Probability of single female dispersal: 0.714

Simulation 24

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
314	1	-437.2100	916.4199	995.1572	311.8398	293	0.3146184	0.1974984	Poisson	Terr
314	1	-436.9425	917.8849	1000.3716	299.7577	292	0.3275126	0.1951470	Poisson	BirdID+Terr
314	1	-438.0380	918.0760	996.8133	325.1154	293	0.2936035	0.2046759	Poisson	BirdID

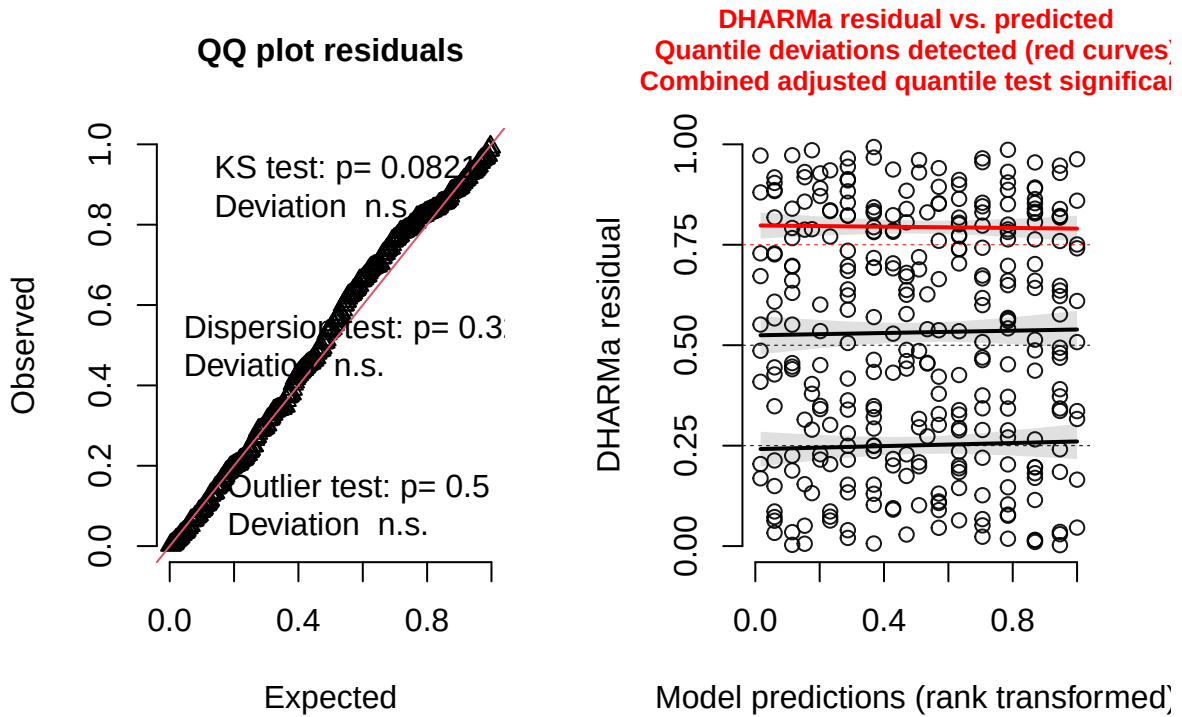
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.2741573	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.3158257	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.1864736	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.2840747	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.5114094	0.2528286	2.0227512	0.0430988	Poisson	BirdID
fixed	factor(fac.yr)2	-0.2992381	0.3759546	-0.7959422	0.4260656	Poisson	BirdID
fixed	factor(fac.yr)3	-1.0711029	0.3970409	-2.6977145	0.0069817	Poisson	BirdID
fixed	factor(fac.yr)4	-0.1482540	0.3077044	-0.4818066	0.6299433	Poisson	BirdID
fixed	factor(fac.yr)5	-0.0377653	0.3013916	-0.1253030	0.9002837	Poisson	BirdID
fixed	factor(fac.yr)6	-0.0879661	0.3000208	-0.2932002	0.7693691	Poisson	BirdID
fixed	factor(fac.yr)7	-0.4906611	0.3251768	-1.5089058	0.1313229	Poisson	BirdID
fixed	factor(fac.yr)8	-0.5360757	0.3252168	-1.6483644	0.0992779	Poisson	BirdID
fixed	factor(fac.yr)9	-0.0323027	0.3047907	-0.1059831	0.9155958	Poisson	BirdID
fixed	factor(fac.yr)10	-0.1808674	0.3128204	-0.5781828	0.5631407	Poisson	BirdID
fixed	factor(fac.yr)11	-0.8607642	0.3914176	-2.1990944	0.0278712	Poisson	BirdID
fixed	factor(fac.yr)12	-0.4179217	0.3639637	-1.1482511	0.2508649	Poisson	BirdID
fixed	factor(fac.yr)13	-0.2218853	0.3387371	-0.6550368	0.5124440	Poisson	BirdID
fixed	factor(fac.yr)14	-0.4744591	0.3846286	-1.2335512	0.2173702	Poisson	BirdID
fixed	factor(fac.yr)15	-0.6759183	0.4292023	-1.5748244	0.1152970	Poisson	BirdID
fixed	factor(fac.yr)16	-2.0787271	0.7644061	-2.7194015	0.0065400	Poisson	BirdID
fixed	factor(fac.yr)17	-0.5558020	0.4308534	-1.2900027	0.1970497	Poisson	BirdID
fixed	factor(fac.yr)18	-0.2578172	0.4512045	-0.5713978	0.5677300	Poisson	BirdID
fixed	factor(fac.yr)19	-0.7978435	0.4936922	-1.6160750	0.1060781	Poisson	BirdID
fixed	factor(fac.yr)20	-0.6987917	0.5291702	-1.3205423	0.1866540	Poisson	BirdID
fixed	(Intercept)	0.5723852	0.2554675	2.2405400	0.0250559	Poisson	Terr
fixed	factor(fac.yr)2	-0.3902374	0.3865308	-1.0095895	0.3126920	Poisson	Terr
fixed	factor(fac.yr)3	-1.1354585	0.4031338	-2.8165796	0.0048538	Poisson	Terr
fixed	factor(fac.yr)4	-0.2060573	0.3134176	-0.6574529	0.5108897	Poisson	Terr
fixed	factor(fac.yr)5	-0.1006725	0.3039668	-0.3311957	0.7404967	Poisson	Terr
fixed	factor(fac.yr)6	-0.1599190	0.3009079	-0.5314548	0.5951037	Poisson	Terr
fixed	factor(fac.yr)7	-0.5719218	0.3259070	-1.7548617	0.0792830	Poisson	Terr
fixed	factor(fac.yr)8	-0.5778171	0.3266597	-1.7688656	0.0769163	Poisson	Terr
fixed	factor(fac.yr)9	-0.0632448	0.3067142	-0.2062009	0.8366340	Poisson	Terr
fixed	factor(fac.yr)10	-0.2421847	0.3124132	-0.7752064	0.4382177	Poisson	Terr
fixed	factor(fac.yr)11	-0.8622110	0.3945555	-2.1852716	0.0288689	Poisson	Terr
fixed	factor(fac.yr)12	-0.4252717	0.3630128	-1.1715058	0.2413955	Poisson	Terr
fixed	factor(fac.yr)13	-0.2911780	0.3390427	-0.8588239	0.3904377	Poisson	Terr
fixed	factor(fac.yr)14	-0.5022478	0.3880866	-1.2941643	0.1956087	Poisson	Terr
fixed	factor(fac.yr)15	-0.6422640	0.4331725	-1.4826983	0.1381546	Poisson	Terr
fixed	factor(fac.yr)16	-2.0988533	0.7638669	-2.7476687	0.0060021	Poisson	Terr
fixed	factor(fac.yr)17	-0.5612894	0.4318188	-1.2998264	0.1936605	Poisson	Terr
fixed	factor(fac.yr)18	-0.2772142	0.4489514	-0.6174705	0.5369244	Poisson	Terr
fixed	factor(fac.yr)19	-0.8769914	0.4939567	-1.7754418	0.0758250	Poisson	Terr
fixed	factor(fac.yr)20	-0.7634120	0.5251497	-1.4537035	0.1460285	Poisson	Terr
fixed	(Intercept)	0.5371464	0.2602732	2.0637790	0.0390387	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.3655321	0.3888623	-0.9400039	0.3472156	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-1.1140141	0.4066194	-2.7396970	0.0061496	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.1940581	0.3173050	-0.6115821	0.5408143	Poisson	BirdID+Terr

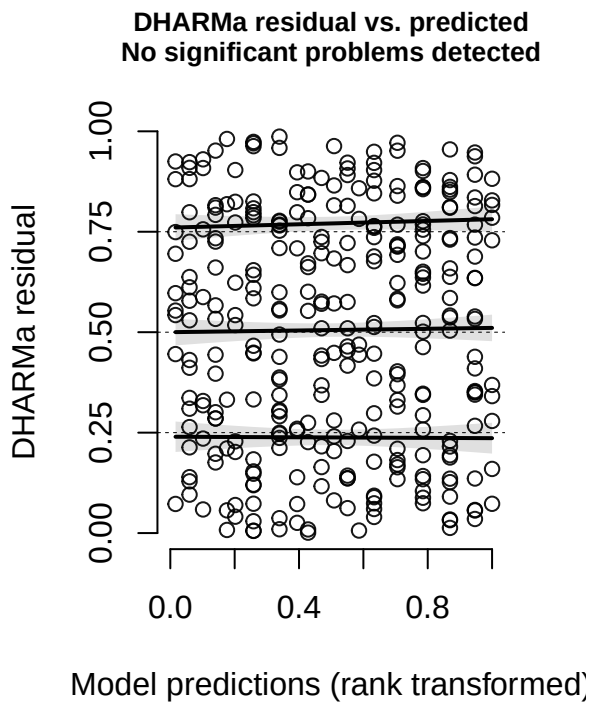
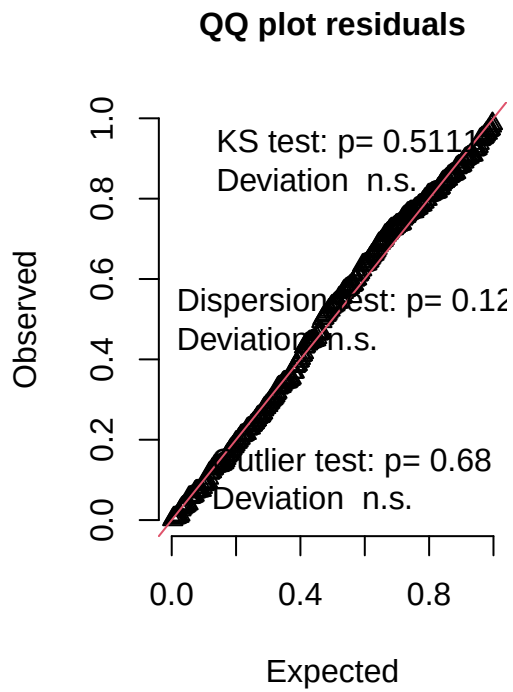
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.0724017	0.3093464	-0.2340475	0.8149481	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.1310233	0.3068310	-0.4270212	0.6693639	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.5374881	0.3316013	-1.6208867	0.1050419	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.5639415	0.3322920	-1.6971262	0.0896728	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.0440267	0.3125644	-0.1408563	0.8879834	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.2151795	0.3185288	-0.6755418	0.4993316	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.8572254	0.3998355	-2.1439453	0.0320373	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.4101097	0.3695275	-1.1098218	0.2670758	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.2610173	0.3455015	-0.7554736	0.4499648	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.4943679	0.3941732	-1.2541895	0.2097732	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.6391823	0.4388262	-1.4565727	0.1452344	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-2.0930926	0.7698113	-2.7189683	0.0065486	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.5497965	0.4387546	-1.2530842	0.2101751	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.2608523	0.4573114	-0.5704040	0.5684037	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-0.8396326	0.5007464	-1.6767620	0.0935890	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.7364600	0.5330555	-1.3815821	0.1671001	Poisson	BirdID+Terr

Territory only model is performing well on all criteria, with it being the best on AIC and BIC.

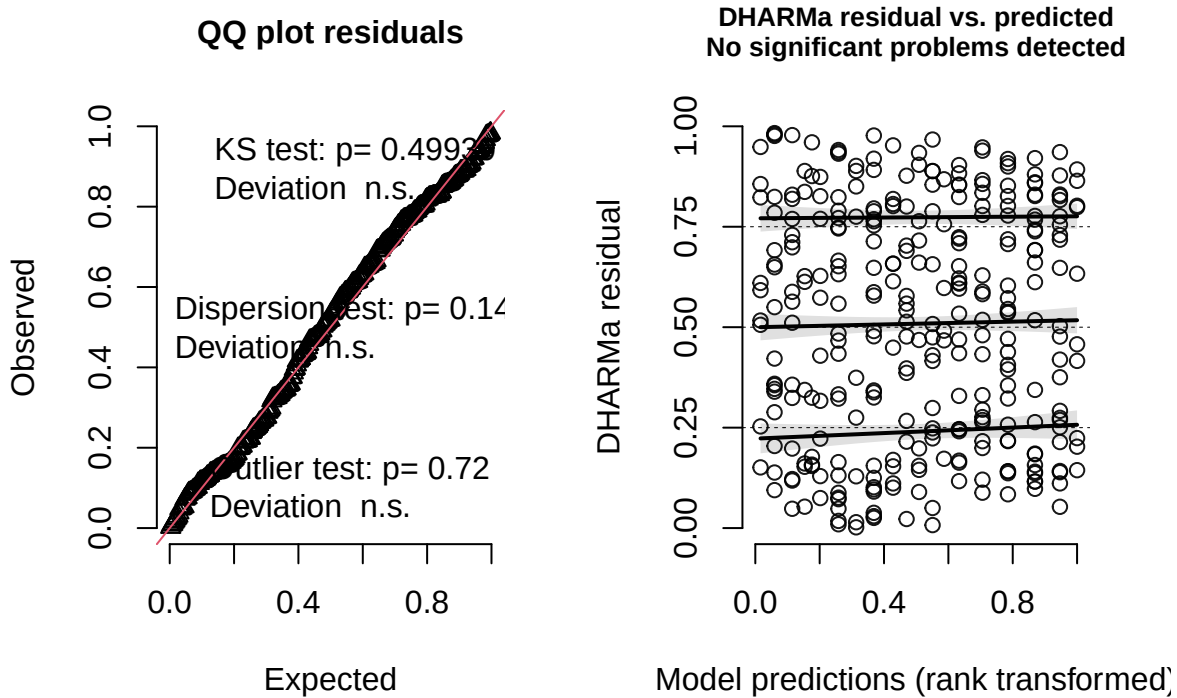
DHARMA residual



DHARMa residual



DHARMa residual



Territory only model diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
314	1	-174.7935	393.5869	476.0736	159.8155	292	0.834	Binomial	BirdID+Terr
314	1	-177.3564	396.7128	475.4501	224.9718	293	0.712	Binomial	BirdID
314	1	-182.8095	407.6190	486.3562	275.1015	293	0.548	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.666761	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.144381	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.942218	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.259375	Binomial	BirdID+Terr

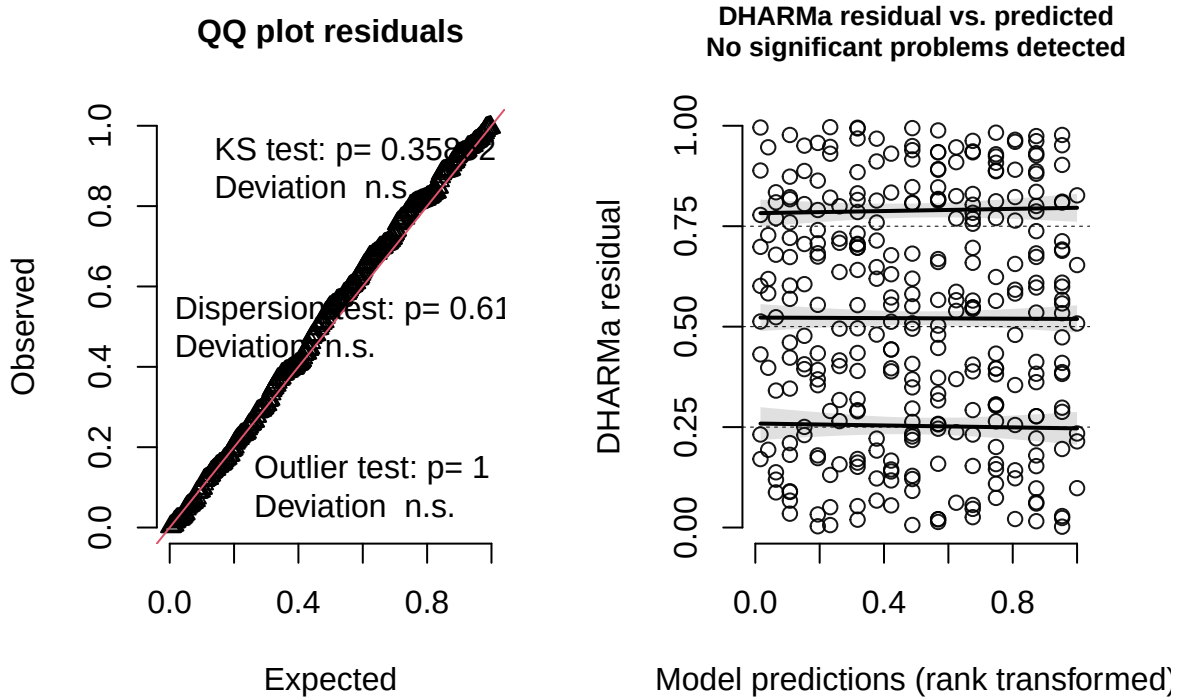
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	1.4070062	1.032624	1.3625538	0.1730232	Binomial	BirdID
fixed	factor(fac.yr)2	-0.8303625	1.270212	-0.6537196	0.5132925	Binomial	BirdID
fixed	factor(fac.yr)3	-1.8598375	1.180970	-1.5748389	0.1152936	Binomial	BirdID
fixed	factor(fac.yr)4	0.1320437	1.214866	0.1086900	0.9134484	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.1278613	1.168540	-0.1094198	0.9128696	Binomial	BirdID
fixed	factor(fac.yr)6	0.5480997	1.187241	0.4616582	0.6443265	Binomial	BirdID
fixed	factor(fac.yr)7	-0.3314160	1.159029	-0.2859429	0.7749219	Binomial	BirdID
fixed	factor(fac.yr)8	-0.7915191	1.160047	-0.6823166	0.4950388	Binomial	BirdID
fixed	factor(fac.yr)9	0.8687654	1.255248	0.6921066	0.4888704	Binomial	BirdID
fixed	factor(fac.yr)10	0.1823132	1.215047	0.1500462	0.8807282	Binomial	BirdID
fixed	factor(fac.yr)11	-1.1138695	1.227451	-0.9074659	0.3641605	Binomial	BirdID
fixed	factor(fac.yr)12	0.3394965	1.310220	0.2591141	0.7955472	Binomial	BirdID
fixed	factor(fac.yr)13	-0.3635005	1.248392	-0.2911750	0.7709175	Binomial	BirdID
fixed	factor(fac.yr)14	-0.5195230	1.326245	-0.3917247	0.6952617	Binomial	BirdID
fixed	factor(fac.yr)15	-1.1189437	1.364196	-0.8202223	0.4120894	Binomial	BirdID
fixed	factor(fac.yr)16	-3.9269867	1.500054	-2.6178961	0.0088474	Binomial	BirdID
fixed	factor(fac.yr)17	-1.9740171	1.419581	-1.3905628	0.1643580	Binomial	BirdID
fixed	factor(fac.yr)18	1.3056574	1.844074	0.7080286	0.4789275	Binomial	BirdID
fixed	factor(fac.yr)19	-0.9642884	1.434193	-0.6723560	0.5013571	Binomial	BirdID
fixed	factor(fac.yr)20	-2.3301904	1.582191	-1.4727619	0.1408152	Binomial	BirdID
fixed	(Intercept)	1.8362273	0.906338	2.0259851	0.0427663	Binomial	Terr
fixed	factor(fac.yr)2	-1.4915498	1.150761	-1.2961419	0.1949266	Binomial	Terr
fixed	factor(fac.yr)3	-2.1537271	1.055405	-2.0406652	0.0412841	Binomial	Terr
fixed	factor(fac.yr)4	-0.2095012	1.075266	-0.1948366	0.8455209	Binomial	Terr
fixed	factor(fac.yr)5	-0.8165999	1.030263	-0.7926134	0.4280031	Binomial	Terr
fixed	factor(fac.yr)6	-0.3076151	1.043711	-0.2947320	0.7681986	Binomial	Terr
fixed	factor(fac.yr)7	-1.1373698	1.006720	-1.1297777	0.2585699	Binomial	Terr
fixed	factor(fac.yr)8	-1.4182730	1.008103	-1.4068726	0.1594652	Binomial	Terr
fixed	factor(fac.yr)9	0.1707076	1.099257	0.1552936	0.8765898	Binomial	Terr
fixed	factor(fac.yr)10	-0.3361633	1.043227	-0.3222340	0.7472755	Binomial	Terr
fixed	factor(fac.yr)11	-1.2703665	1.079345	-1.1769793	0.2392038	Binomial	Terr
fixed	factor(fac.yr)12	0.0103841	1.156202	0.0089812	0.9928341	Binomial	Terr
fixed	factor(fac.yr)13	-0.7754266	1.087786	-0.7128484	0.4759395	Binomial	Terr
fixed	factor(fac.yr)14	-0.6494346	1.140814	-0.5692729	0.5691709	Binomial	Terr
fixed	factor(fac.yr)15	-0.9611382	1.174340	-0.8184499	0.4131003	Binomial	Terr
fixed	factor(fac.yr)16	-3.1108229	1.282179	-2.4262011	0.0152578	Binomial	Terr
fixed	factor(fac.yr)17	-2.1038563	1.197179	-1.7573446	0.0788591	Binomial	Terr
fixed	factor(fac.yr)18	0.3097020	1.534198	0.2018657	0.8400217	Binomial	Terr
fixed	factor(fac.yr)19	-1.9138689	1.212294	-1.5787164	0.1144011	Binomial	Terr
fixed	factor(fac.yr)20	-2.8882205	1.334930	-2.1635744	0.0304970	Binomial	Terr
fixed	(Intercept)	2.0970675	1.293250	1.6215480	0.1049002	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-1.5994082	1.515040	-1.0556873	0.2911111	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-2.6096166	1.446956	-1.8035215	0.0713064	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-0.1730508	1.489990	-0.1161422	0.9075398	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-0.4989840	1.443607	-0.3456508	0.7296052	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	0.2623183	1.450559	0.1808395	0.8564936	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-0.5557110	1.417703	-0.3919799	0.6950731	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-1.3862189	1.427954	-0.9707725	0.3316616	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	0.8783518	1.534132	0.5725399	0.5669563	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	0.0092805	1.487249	0.0062400	0.9950212	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-1.5878977	1.518573	-1.0456510	0.2957222	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	0.3748697	1.569284	0.2388794	0.8111991	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-0.6847731	1.532797	-0.4467474	0.6550575	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-0.6135556	1.625876	-0.3773693	0.7058992	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-1.1720933	1.694977	-0.6915099	0.4892452	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-4.8421678	1.843004	-2.6273228	0.0086060	Binomial	BirdID+Terr

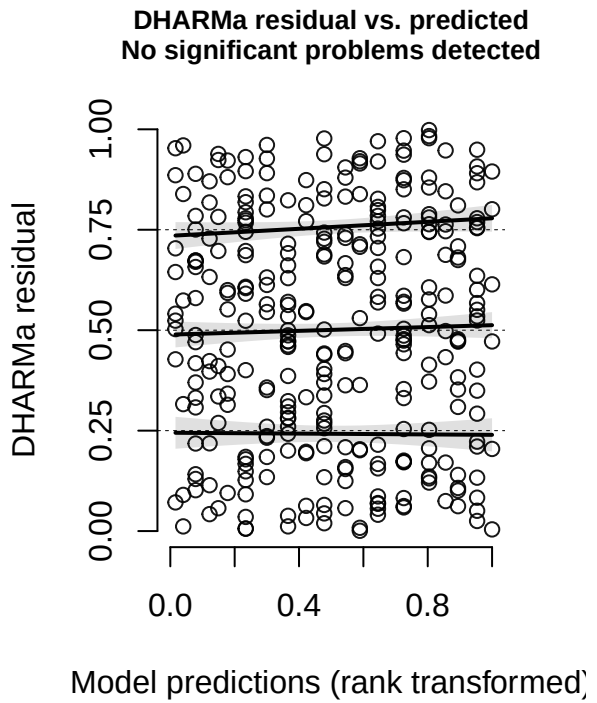
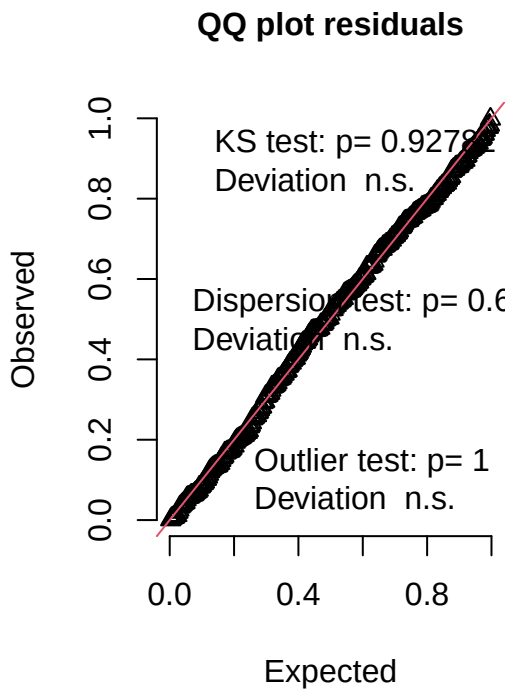
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-2.7410887	1.767546	-1.5507874	0.1209526	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	1.2280857	2.182699	0.5626455	0.5736763	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-1.9363462	1.750882	-1.1059258	0.2687586	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-3.6312148	1.944648	-1.8672867	0.0618616	Binomial	BirdID+Terr

Territory only model is performing the worst on the selection criteria.

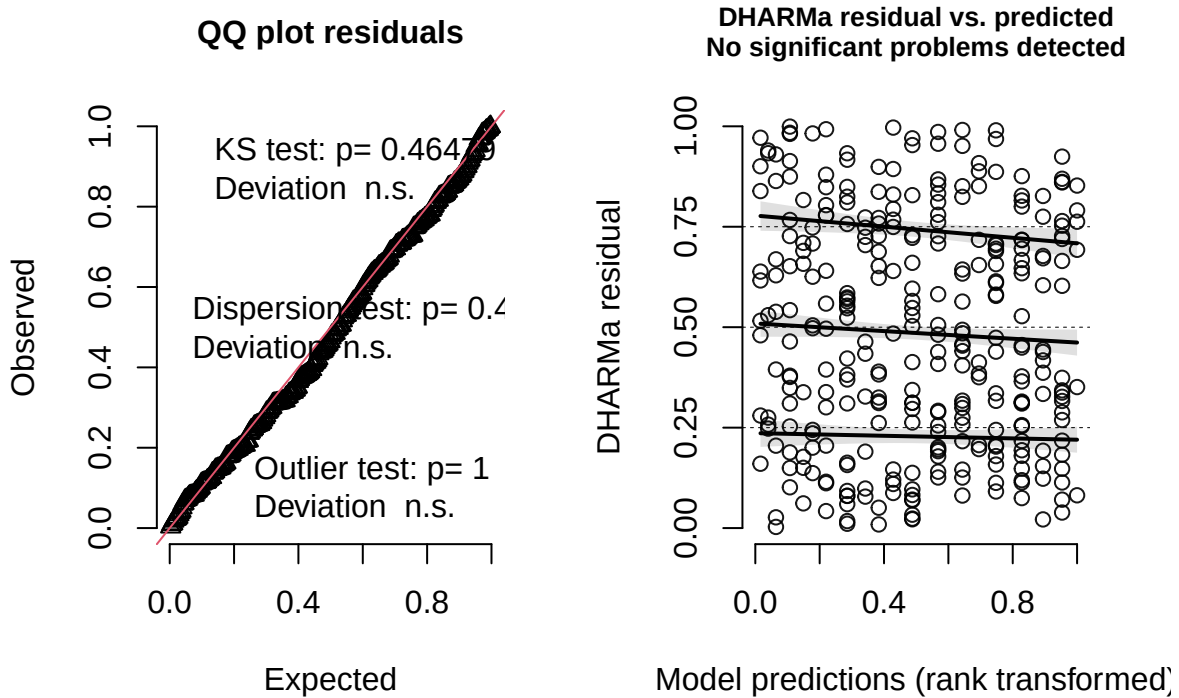
DHARMA residual



DHARMa residual



DHARMA residual



Diagnostics look good.

Probabilites used to generate simulation 24

Simulation 24
 Probability of mate fidelity: 0.985
 Probability of pair dispersal: 0.654
 Probability of single male dispersal: 0.597
 Probability of single female dispersal: 0.502

Simulation 25

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
411	1	-505.8495	1053.699	1138.090	380.3998	390	0.3774217	0.1020718	Poisson	BirdID
411	1	-504.9357	1053.871	1142.281	355.2114	389	0.3928088	0.1039704	Poisson	BirdID+Terr
411	1	-511.6206	1065.241	1149.632	435.1326	390	0.2981021	0.1322127	Poisson	Terr

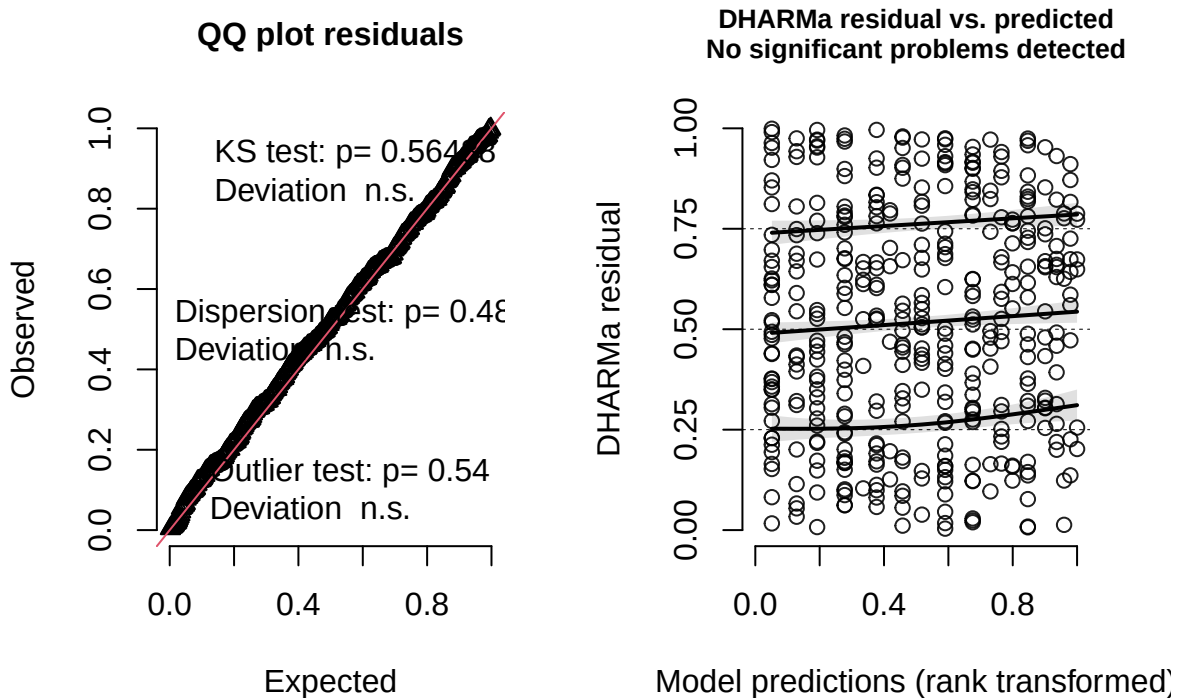
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.5799399	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.4129893	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.5308821	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.2769862	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.3914379	0.2531096	1.5465152	0.1219802	Poisson	BirdID
fixed	factor(fac.yr)2	-0.4159133	0.3178747	-1.3084190	0.1907312	Poisson	BirdID
fixed	factor(fac.yr)3	-1.0968713	0.3620931	-3.0292523	0.0024516	Poisson	BirdID
fixed	factor(fac.yr)4	-0.7643253	0.3296392	-2.3186723	0.0204128	Poisson	BirdID
fixed	factor(fac.yr)5	-0.3663102	0.2926699	-1.2516155	0.2107100	Poisson	BirdID
fixed	factor(fac.yr)6	-0.6584180	0.3075915	-2.1405600	0.0323095	Poisson	BirdID
fixed	factor(fac.yr)7	-1.2991486	0.3433057	-3.7842327	0.0001542	Poisson	BirdID
fixed	factor(fac.yr)8	-1.0281213	0.3293574	-3.1215976	0.0017987	Poisson	BirdID
fixed	factor(fac.yr)9	-0.7035890	0.3218853	-2.1858372	0.0288275	Poisson	BirdID
fixed	factor(fac.yr)10	-1.0831274	0.3587600	-3.0190867	0.0025354	Poisson	BirdID
fixed	factor(fac.yr)11	-0.9778485	0.3614376	-2.7054423	0.0068213	Poisson	BirdID
fixed	factor(fac.yr)12	-0.7994878	0.3582760	-2.2314856	0.0256490	Poisson	BirdID
fixed	factor(fac.yr)13	-0.2819122	0.3625035	-0.7776812	0.4367570	Poisson	BirdID
fixed	factor(fac.yr)14	-0.2590324	0.3660154	-0.7077090	0.4791260	Poisson	BirdID
fixed	factor(fac.yr)15	-0.5927012	0.4301353	-1.3779414	0.1682214	Poisson	BirdID
fixed	factor(fac.yr)16	-0.3697832	0.4089851	-0.9041485	0.3659167	Poisson	BirdID
fixed	factor(fac.yr)17	0.0081319	0.4446343	0.0182889	0.9854084	Poisson	BirdID
fixed	factor(fac.yr)18	-0.0014193	0.4955011	-0.0028644	0.9977145	Poisson	BirdID
fixed	factor(fac.yr)19	-0.9711148	0.6049336	-1.6053246	0.1084224	Poisson	BirdID
fixed	factor(fac.yr)20	-1.0132418	0.6119837	-1.6556679	0.0977891	Poisson	BirdID
fixed	(Intercept)	0.5634013	0.2371892	2.3753247	0.0175335	Poisson	Terr
fixed	factor(fac.yr)2	-0.3041498	0.3247640	-0.9365257	0.3490026	Poisson	Terr
fixed	factor(fac.yr)3	-1.0719777	0.3628742	-2.9541308	0.0031355	Poisson	Terr
fixed	factor(fac.yr)4	-0.7663889	0.3304015	-2.3195688	0.0203642	Poisson	Terr
fixed	factor(fac.yr)5	-0.3251348	0.2929784	-1.1097570	0.2671038	Poisson	Terr
fixed	factor(fac.yr)6	-0.6186785	0.2969355	-2.0835449	0.0372016	Poisson	Terr
fixed	factor(fac.yr)7	-1.3788494	0.3293747	-4.1862634	0.0000284	Poisson	Terr
fixed	factor(fac.yr)8	-1.0587618	0.3139068	-3.3728541	0.0007439	Poisson	Terr
fixed	factor(fac.yr)9	-0.7534948	0.3029378	-2.4872923	0.0128720	Poisson	Terr
fixed	factor(fac.yr)10	-1.0703524	0.3395846	-3.1519459	0.0016219	Poisson	Terr
fixed	factor(fac.yr)11	-1.0046082	0.3398619	-2.9559311	0.0031173	Poisson	Terr
fixed	factor(fac.yr)12	-0.8804114	0.3365195	-2.6162267	0.0088908	Poisson	Terr
fixed	factor(fac.yr)13	-0.3380198	0.3381702	-0.9995553	0.3175258	Poisson	Terr
fixed	factor(fac.yr)14	-0.3472260	0.3440655	-1.0091857	0.3128856	Poisson	Terr
fixed	factor(fac.yr)15	-0.4995427	0.4048901	-1.2337734	0.2172873	Poisson	Terr
fixed	factor(fac.yr)16	-0.4985136	0.3885396	-1.2830447	0.1994764	Poisson	Terr
fixed	factor(fac.yr)17	-0.0944667	0.4042184	-0.2337022	0.8152161	Poisson	Terr
fixed	factor(fac.yr)18	-0.2495304	0.4563615	-0.5467823	0.5845283	Poisson	Terr
fixed	factor(fac.yr)19	-1.2970640	0.5691538	-2.2789341	0.0226710	Poisson	Terr
fixed	factor(fac.yr)20	-1.1565586	0.5773798	-2.0031160	0.0451648	Poisson	Terr
fixed	(Intercept)	0.3876270	0.2613413	1.4832216	0.1380156	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.3539038	0.3309412	-1.0693859	0.2848958	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-1.0754059	0.3720393	-2.8905711	0.0038454	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.7576179	0.3416177	-2.2177360	0.0265728	Poisson	BirdID+Terr

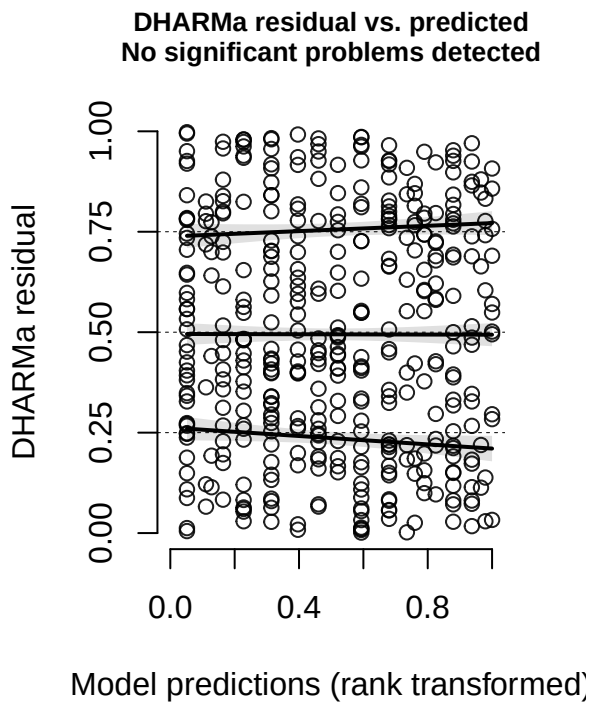
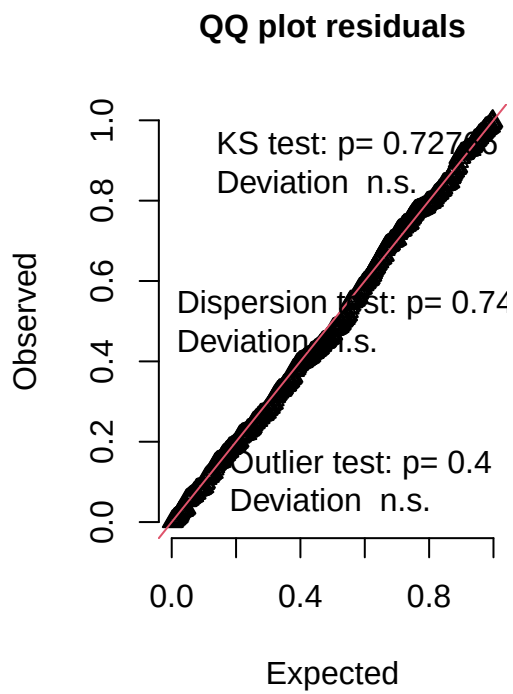
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.3520807	0.3046042	-1.1558630	0.2477372	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.6238879	0.3171018	-1.9674686	0.0491292	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-1.3240475	0.3514401	-3.7674907	0.0001649	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-1.0190732	0.3365811	-3.0277194	0.0024641	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.6913918	0.3289066	-2.1020917	0.0355452	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-1.0670799	0.3666688	-2.9102012	0.0036120	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.9772145	0.3700570	-2.6407135	0.0082732	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.8060459	0.3663931	-2.1999486	0.0278105	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.2690612	0.3717020	-0.7238627	0.4691500	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.2562480	0.3748790	-0.6835485	0.4942603	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.5388687	0.4398096	-1.2252319	0.2204879	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.3971979	0.4211841	-0.9430505	0.3456551	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	0.0073752	0.4498038	0.0163965	0.9869181	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.0959062	0.5081831	-0.1887236	0.8503094	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.0611212	0.6096965	-1.7404089	0.0817872	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-1.0398516	0.6208374	-1.6749179	0.0939503	Poisson	BirdID+Terr

The territory only model has the best marginal R^2 , and that is it.

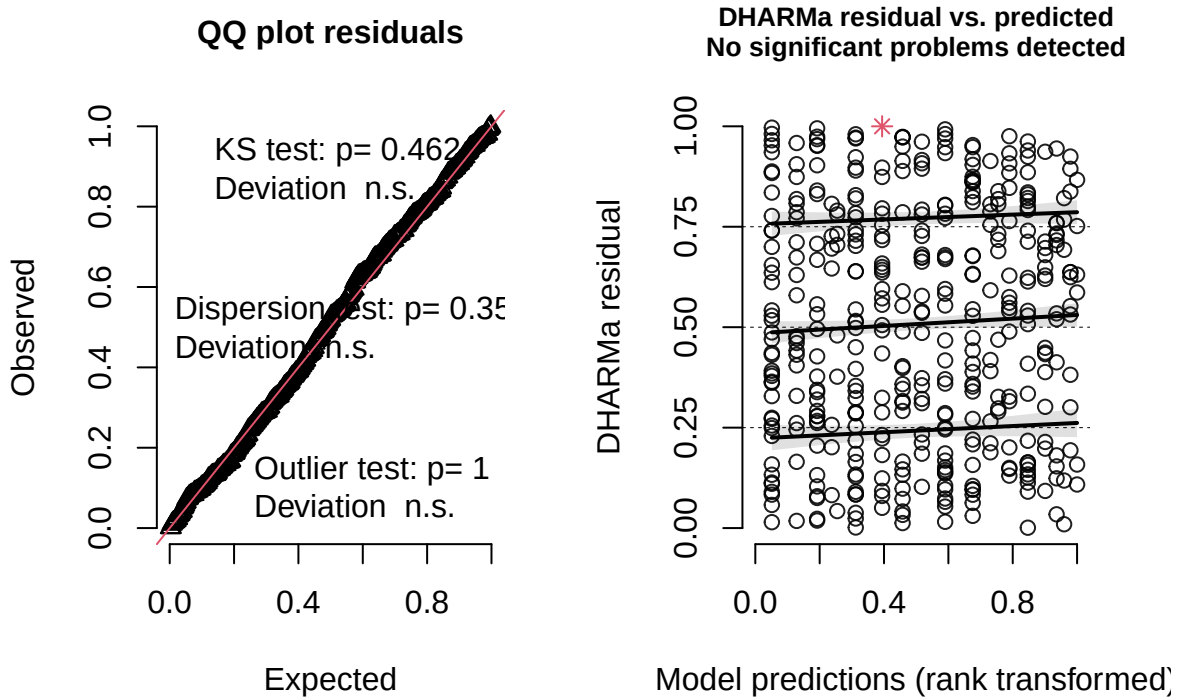
DHARMa residual



DHARMa residual



DHARMa residual



Diagnostics look reasonable.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
411	1	-245.7587	535.5174	623.9265	295.0886	389	0.771	Binomial	BirdID+Terr
411	1	-247.5407	537.0814	621.4718	346.1900	390	0.708	Binomial	BirdID
411	1	-257.9876	557.9751	642.3656	421.7846	390	0.543	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.4843071	Binomial	BirdID
ran_pars	Terr	sd__(Intercept)	0.9487810	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.5091726	Binomial	BirdID+Terr
ran_pars	Terr	sd__(Intercept)	0.8852716	Binomial	BirdID+Terr

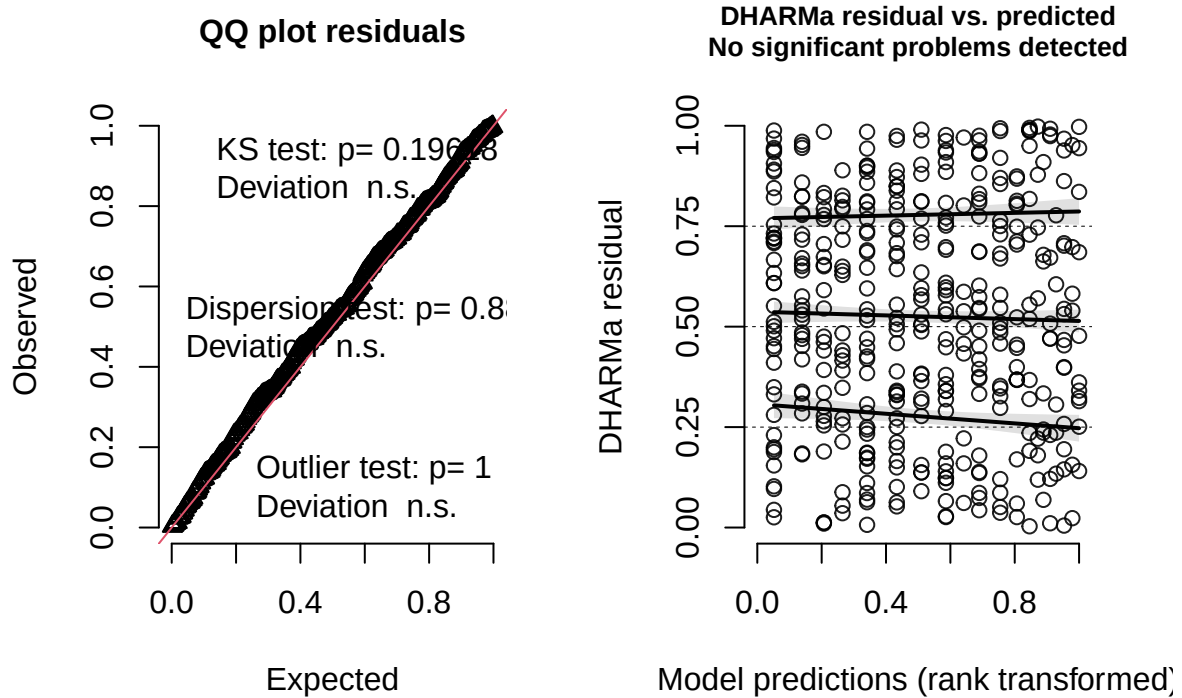
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	2.1643389	1.151242	1.8800037	0.0601076	Binomial	BirdID
fixed	factor(fac.yr)2	-1.5195662	1.246308	-1.2192539	0.2227478	Binomial	BirdID
fixed	factor(fac.yr)3	-2.7972942	1.233578	-2.2676271	0.0233519	Binomial	BirdID
fixed	factor(fac.yr)4	-1.5819193	1.235410	-1.2804811	0.2003760	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-1.6765422	1.228402	-1.3648155	0.1723110	Binomial	BirdID
fixed	factor(fac.yr)6	-2.1362970	1.217776	-1.7542606	0.0793859	Binomial	BirdID
fixed	factor(fac.yr)7	-3.6185652	1.220881	-2.9638976	0.0030377	Binomial	BirdID
fixed	factor(fac.yr)8	-2.7684817	1.216740	-2.2753267	0.0228863	Binomial	BirdID
fixed	factor(fac.yr)9	-2.5470461	1.225695	-2.0780421	0.0377055	Binomial	BirdID
fixed	factor(fac.yr)10	-3.2871176	1.250427	-2.6287970	0.0085687	Binomial	BirdID
fixed	factor(fac.yr)11	-2.3664708	1.248471	-1.8954957	0.0580268	Binomial	BirdID
fixed	factor(fac.yr)12	-3.0046441	1.261834	-2.3811725	0.0172576	Binomial	BirdID
fixed	factor(fac.yr)13	-1.0416922	1.363741	-0.7638493	0.4449571	Binomial	BirdID
fixed	factor(fac.yr)14	-1.3243411	1.355020	-0.9773590	0.3283915	Binomial	BirdID
fixed	factor(fac.yr)15	-1.9312939	1.382688	-1.3967677	0.1624834	Binomial	BirdID
fixed	factor(fac.yr)16	-1.1427709	1.425241	-0.8018090	0.4226634	Binomial	BirdID
fixed	factor(fac.yr)17	-0.9338858	1.555234	-0.6004793	0.5481868	Binomial	BirdID
fixed	factor(fac.yr)18	-1.0618350	1.667485	-0.6367883	0.5242627	Binomial	BirdID
fixed	factor(fac.yr)19	-1.1690393	1.514913	-0.7716873	0.4402997	Binomial	BirdID
fixed	factor(fac.yr)20	-1.1645625	1.533049	-0.7596384	0.4474708	Binomial	BirdID
fixed	(Intercept)	2.0843157	1.097673	1.8988499	0.0575842	Binomial	Terr
fixed	factor(fac.yr)2	-1.0965632	1.220056	-0.8987809	0.3687694	Binomial	Terr
fixed	factor(fac.yr)3	-2.2475659	1.190974	-1.8871667	0.0591379	Binomial	Terr
fixed	factor(fac.yr)4	-1.1522819	1.195618	-0.9637539	0.3351693	Binomial	Terr
fixed	factor(fac.yr)5	-1.2874929	1.178043	-1.0929084	0.2744340	Binomial	Terr
fixed	factor(fac.yr)6	-1.7326024	1.157609	-1.4967083	0.1344692	Binomial	Terr
fixed	factor(fac.yr)7	-3.1261997	1.155856	-2.7046613	0.0068374	Binomial	Terr
fixed	factor(fac.yr)8	-2.3748160	1.148781	-2.0672483	0.0387108	Binomial	Terr
fixed	factor(fac.yr)9	-2.3574628	1.155911	-2.0394843	0.0414017	Binomial	Terr
fixed	factor(fac.yr)10	-2.6588320	1.173938	-2.2648838	0.0235198	Binomial	Terr
fixed	factor(fac.yr)11	-1.9916575	1.171393	-1.7002473	0.0890844	Binomial	Terr
fixed	factor(fac.yr)12	-2.6343747	1.185191	-2.2227425	0.0262332	Binomial	Terr
fixed	factor(fac.yr)13	-0.8350556	1.274514	-0.6551952	0.5123421	Binomial	Terr
fixed	factor(fac.yr)14	-0.9107248	1.281103	-0.7108914	0.4771515	Binomial	Terr
fixed	factor(fac.yr)15	-1.3212356	1.289897	-1.0242953	0.3056958	Binomial	Terr
fixed	factor(fac.yr)16	-1.1612419	1.345721	-0.8629147	0.3881844	Binomial	Terr
fixed	factor(fac.yr)17	-1.0499407	1.446244	-0.7259775	0.4678526	Binomial	Terr
fixed	factor(fac.yr)18	-1.6622758	1.508732	-1.1017701	0.2705616	Binomial	Terr
fixed	factor(fac.yr)19	-2.0112708	1.387733	-1.4493209	0.1472480	Binomial	Terr
fixed	factor(fac.yr)20	-1.6948580	1.388191	-1.2209112	0.2221196	Binomial	Terr
fixed	(Intercept)	2.1090253	1.235887	1.7064876	0.0879173	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-1.1636289	1.339254	-0.8688632	0.3849219	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-2.7186705	1.317868	-2.0629314	0.0391191	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-1.3739586	1.326113	-1.0360792	0.3001652	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-1.5419560	1.316443	-1.1713050	0.2414762	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-2.0016446	1.300827	-1.5387477	0.1238659	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-3.7144386	1.306196	-2.8437065	0.0044592	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-2.6843924	1.299972	-2.0649618	0.0389266	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-2.5053719	1.311222	-1.9107154	0.0560412	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-3.1817221	1.336993	-2.3797588	0.0173240	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-2.2114439	1.334990	-1.6565249	0.0976156	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-2.9543776	1.350296	-2.1879479	0.0286734	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-0.7726587	1.465447	-0.5272512	0.5980192	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-0.9492948	1.462337	-0.6491629	0.5162331	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-1.6063033	1.479932	-1.0853900	0.2777490	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-0.9565690	1.538707	-0.6216705	0.5341586	Binomial	BirdID+Terr

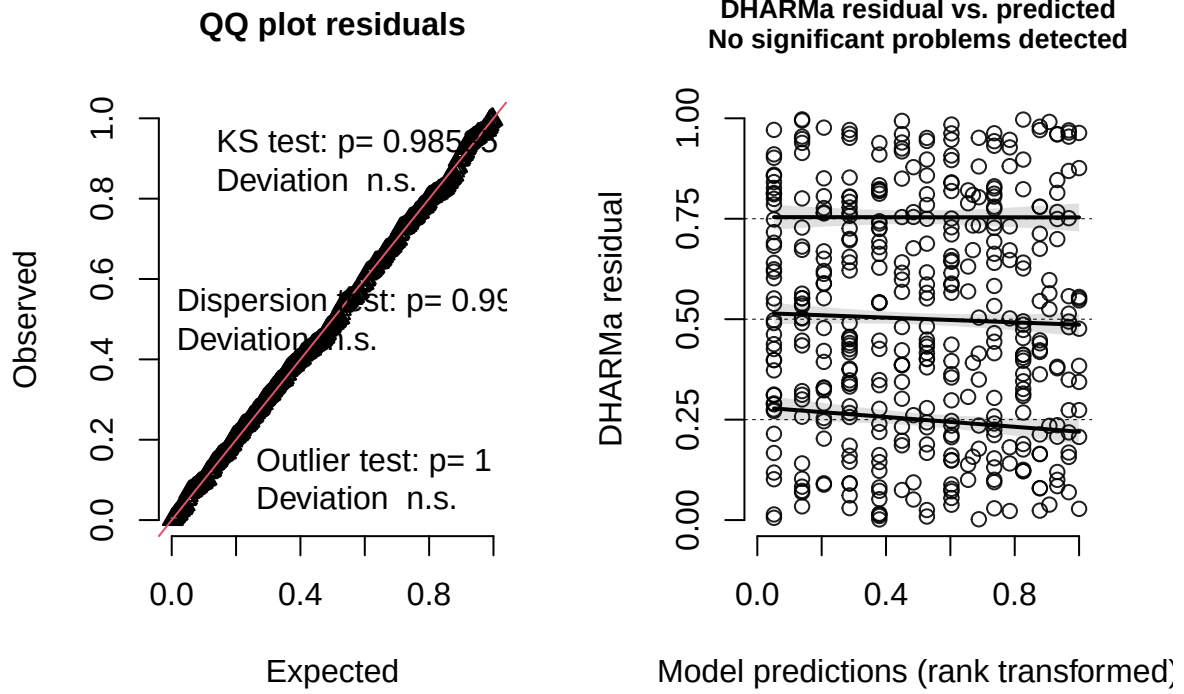
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-0.6701459	1.686637	-0.3973268	0.6911265	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-1.1045741	1.828409	-0.6041177	0.5457654	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-1.3172444	1.643485	-0.8014948	0.4228452	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-1.0550255	1.646546	-0.6407506	0.5216847	Binomial	BirdID+Terr

Similar to the previous batch of models.

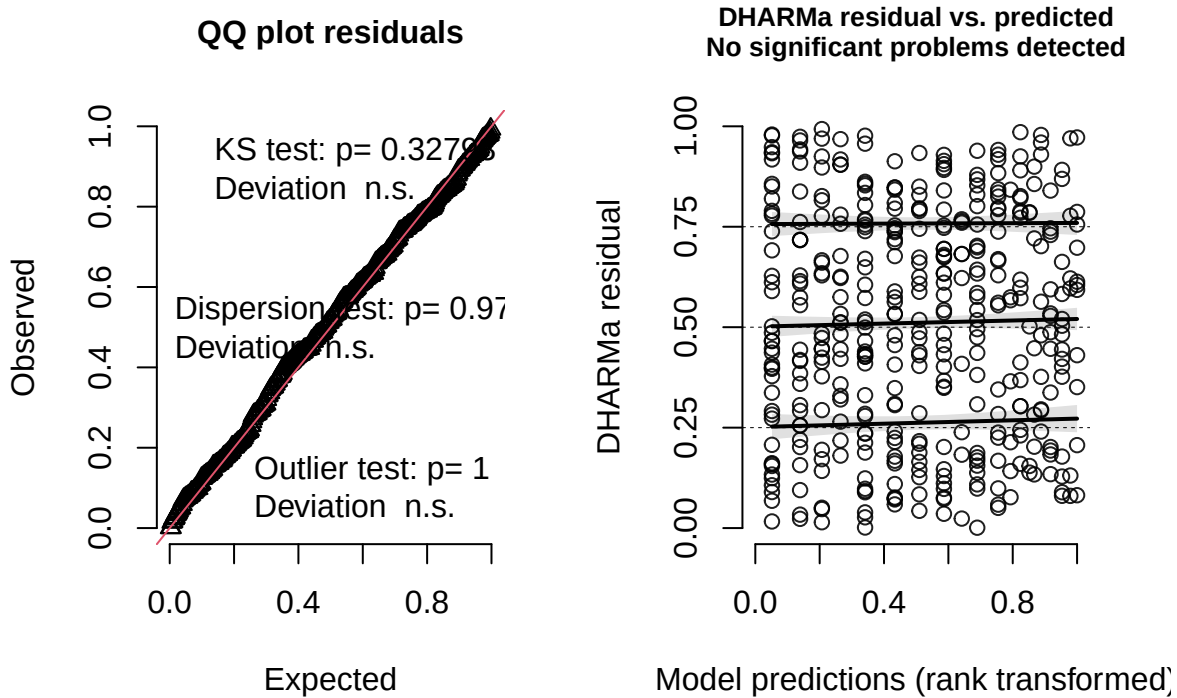
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Probabilites used to generate simulation 25

Simulation 25
 Probability of mate fidelity: 0.859
 Probability of pair dispersal: 0
 Probability of single male dispersal: 0.11
 Probability of single female dispersal: 0.833

Simulation 26

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	REfamily	RE
330	1	-466.9020	975.8040	1055.585	327.9644	309	0.3096390	0.1091107	Poisson	Terr
330	1	-466.8837	977.7674	1061.347	325.5428	308	0.3117266	0.1088383	Poisson	BirdID+Terr
330	1	-471.6875	985.3749	1065.156	360.4063	309	0.2534239	0.1171110	Poisson	BirdID

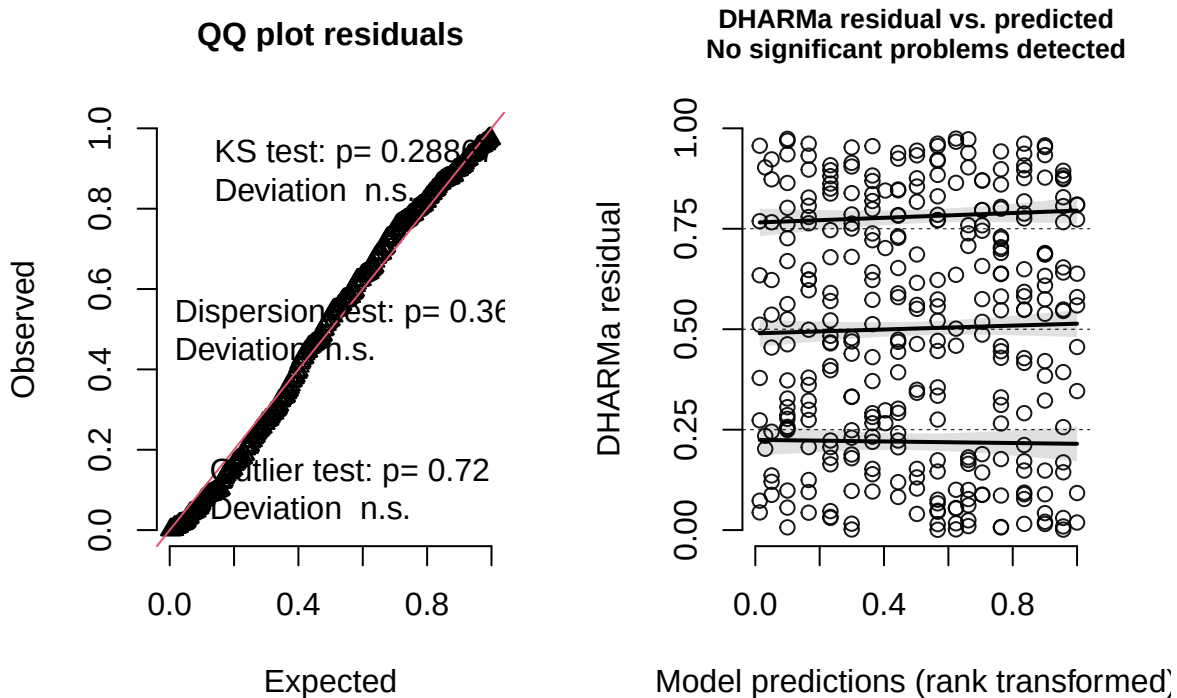
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.3266750	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.4115622	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.0757749	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.4075267	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.1879551	0.2610584	0.7199736	0.4715412	Poisson	BirdID
fixed	factor(fac.yr)2	-0.0611593	0.3502065	-0.1746379	0.8613641	Poisson	BirdID
fixed	factor(fac.yr)3	-0.4389202	0.3554037	-1.2349905	0.2168340	Poisson	BirdID
fixed	factor(fac.yr)4	0.1547366	0.3145408	0.4919444	0.6227587	Poisson	BirdID
fixed	factor(fac.yr)5	0.3130849	0.3220723	0.9720953	0.3310032	Poisson	BirdID
fixed	factor(fac.yr)6	-0.0178370	0.3231214	-0.0552021	0.9559774	Poisson	BirdID
fixed	factor(fac.yr)7	-0.0716581	0.3357188	-0.2134468	0.8309785	Poisson	BirdID
fixed	factor(fac.yr)8	-0.1249022	0.3389483	-0.3684994	0.7125009	Poisson	BirdID
fixed	factor(fac.yr)9	0.2389019	0.3146868	0.7591736	0.4477487	Poisson	BirdID
fixed	factor(fac.yr)10	-0.4209495	0.3549942	-1.1857928	0.2357041	Poisson	BirdID
fixed	factor(fac.yr)11	-0.4253276	0.3594666	-1.1832186	0.2367225	Poisson	BirdID
fixed	factor(fac.yr)12	-0.1286921	0.3422762	-0.3759890	0.7069250	Poisson	BirdID
fixed	factor(fac.yr)13	0.3411643	0.3227365	1.0570987	0.2904665	Poisson	BirdID
fixed	factor(fac.yr)14	0.0722858	0.3580908	0.2018643	0.8400228	Poisson	BirdID
fixed	factor(fac.yr)15	-0.5574127	0.4718135	-1.1814258	0.2374336	Poisson	BirdID
fixed	factor(fac.yr)16	-0.0032427	0.3858824	-0.0084033	0.9932952	Poisson	BirdID
fixed	factor(fac.yr)17	0.6961898	0.3436629	2.0257928	0.0427860	Poisson	BirdID
fixed	factor(fac.yr)18	-0.6466149	0.5316951	-1.2161385	0.2239322	Poisson	BirdID
fixed	factor(fac.yr)19	-0.1227076	0.5859104	-0.2094307	0.8341120	Poisson	BirdID
fixed	factor(fac.yr)20	-0.6304460	0.7779262	-0.8104188	0.4176995	Poisson	BirdID
fixed	(Intercept)	0.0430419	0.2642797	0.1628650	0.8706248	Poisson	Terr
fixed	factor(fac.yr)2	0.0076088	0.3527447	0.0215703	0.9827907	Poisson	Terr
fixed	factor(fac.yr)3	-0.2949508	0.3600255	-0.8192497	0.4126440	Poisson	Terr
fixed	factor(fac.yr)4	0.3347253	0.3167769	1.0566596	0.2906670	Poisson	Terr
fixed	factor(fac.yr)5	0.4338923	0.3248006	1.3358731	0.1815908	Poisson	Terr
fixed	factor(fac.yr)6	0.0456821	0.3221233	0.1418156	0.8872256	Poisson	Terr
fixed	factor(fac.yr)7	0.0254525	0.3387347	0.0751400	0.9401033	Poisson	Terr
fixed	factor(fac.yr)8	0.0509684	0.3411834	0.1493872	0.8812481	Poisson	Terr
fixed	factor(fac.yr)9	0.3933677	0.3159665	1.2449662	0.2131442	Poisson	Terr
fixed	factor(fac.yr)10	-0.1985093	0.3551215	-0.5589896	0.5761688	Poisson	Terr
fixed	factor(fac.yr)11	-0.1600826	0.3605245	-0.4440270	0.6570231	Poisson	Terr
fixed	factor(fac.yr)12	0.0362602	0.3415085	0.1061765	0.9154423	Poisson	Terr
fixed	factor(fac.yr)13	0.4982805	0.3239739	1.5380269	0.1240420	Poisson	Terr
fixed	factor(fac.yr)14	0.1846923	0.3589680	0.5145092	0.6068960	Poisson	Terr
fixed	factor(fac.yr)15	-0.4171036	0.4780547	-0.8725017	0.3829347	Poisson	Terr
fixed	factor(fac.yr)16	0.0870594	0.3889577	0.2238274	0.8228916	Poisson	Terr
fixed	factor(fac.yr)17	0.7617928	0.3451136	2.2073681	0.0272884	Poisson	Terr
fixed	factor(fac.yr)18	-0.7767546	0.5337987	-1.4551452	0.1456291	Poisson	Terr
fixed	factor(fac.yr)19	-0.2374157	0.5907989	-0.4018554	0.6877904	Poisson	Terr
fixed	factor(fac.yr)20	-0.5192450	0.7921897	-0.6554554	0.5121746	Poisson	Terr
fixed	(Intercept)	0.0439288	0.2650637	0.1657294	0.8683699	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	0.0056837	0.3536800	0.0160701	0.9871784	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.2990166	0.3609205	-0.8284833	0.4073968	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	0.3310002	0.3177461	1.0417130	0.2975448	Poisson	BirdID+Terr

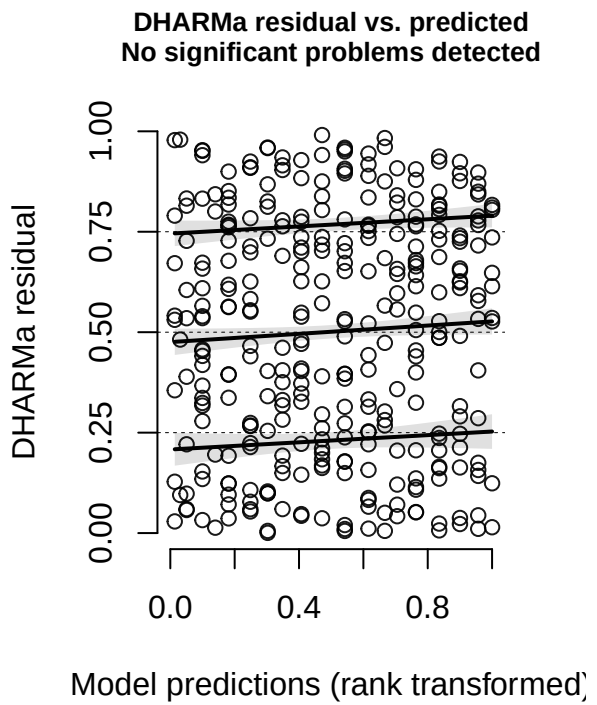
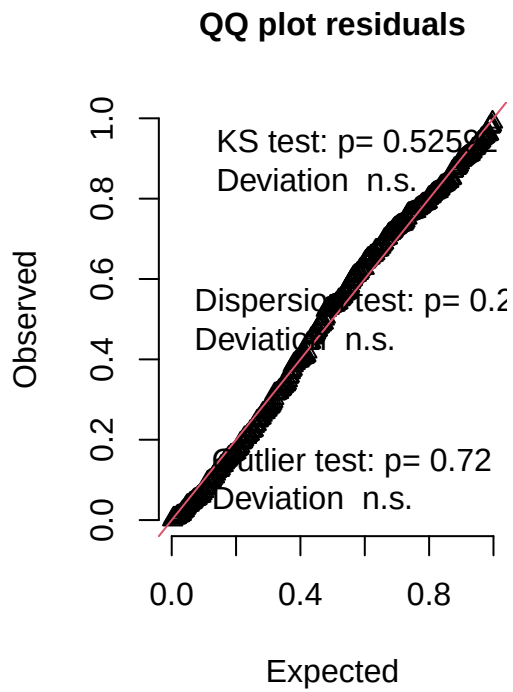
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.4304910	0.3258909	1.3209665	0.1865125	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	0.0436097	0.3232333	0.1349172	0.8926773	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	0.0235212	0.3397996	0.0692208	0.9448138	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	0.0476153	0.3422216	0.1391359	0.8893428	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	0.3868679	0.3171385	1.2198704	0.2225140	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.2030199	0.3561302	-0.5700720	0.5686288	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.1693467	0.3616423	-0.4682714	0.6395906	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	0.0280096	0.3427280	0.0817255	0.9348650	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	0.4922947	0.3251756	1.5139349	0.1300423	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	0.1818395	0.3601233	0.5049368	0.6136032	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.4198151	0.4791441	-0.8761771	0.3809338	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	0.0850140	0.3900056	0.2179814	0.8274436	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	0.7598524	0.3465620	2.1925435	0.0283403	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.7804473	0.5361138	-1.4557492	0.1454620	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-0.2337219	0.5924517	-0.3944996	0.6932122	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.5272408	0.7945845	-0.6635428	0.5069830	Poisson	BirdID+Terr

The territory only model has the best AIC and BIC, and competitive R^2 s.

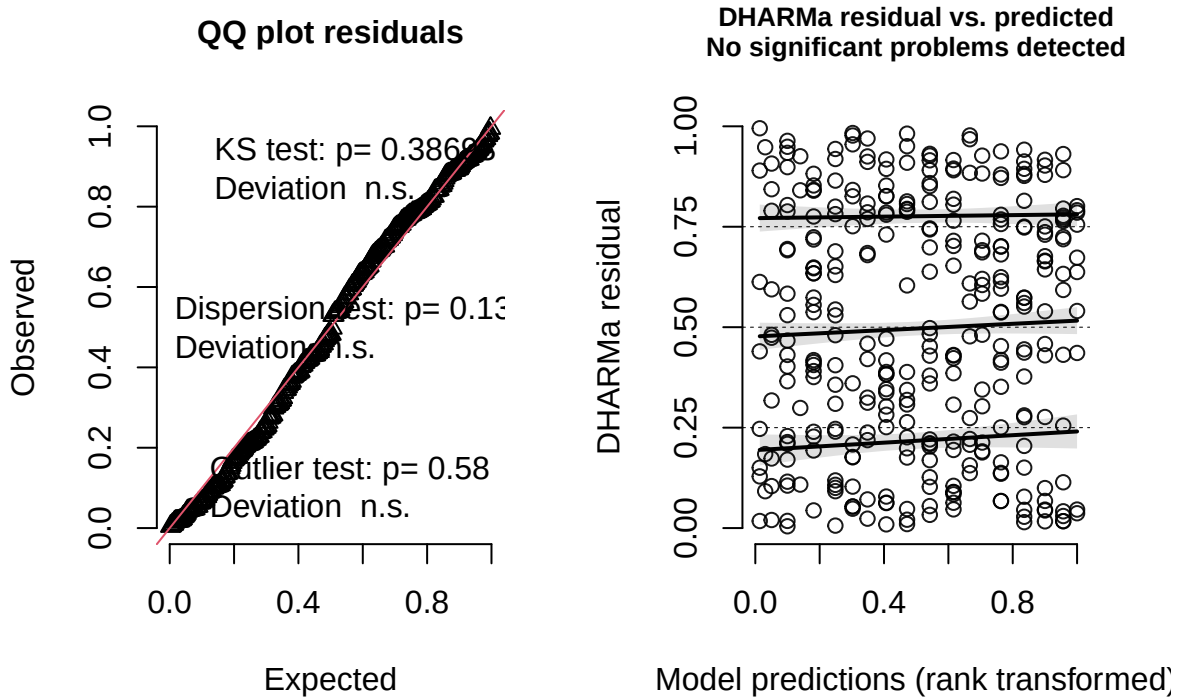
DHARMA residual



DHARMA residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
330	1	-196.4733	436.9466	520.5266	249.8174	308	0.707	Binomial	BirdID+Terr
330	1	-198.1837	438.3673	518.1483	294.9617	309	0.607	Binomial	Terr
330	1	-201.9418	445.8835	525.6645	313.8654	309	0.512	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.0819723	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.2345024	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	0.8937299	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.2132570	Binomial	BirdID+Terr

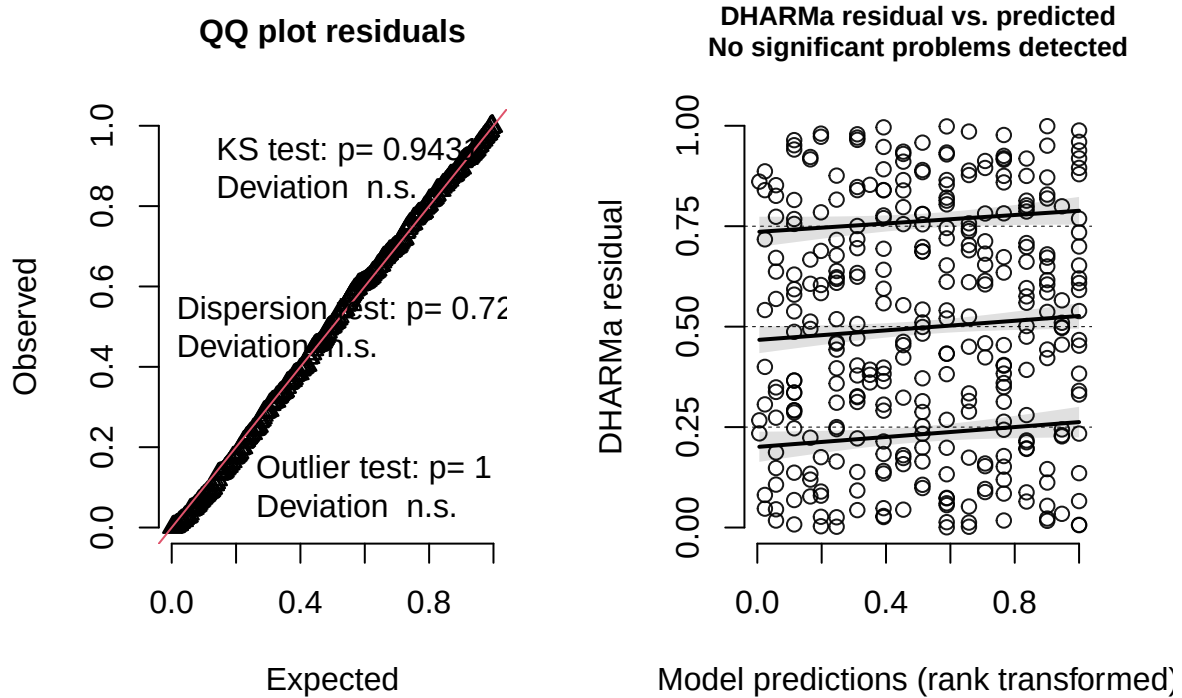
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	-0.1577573	0.6294432	-0.2506299	0.8021003	Binomial	BirdID
fixed	factor(fac.yr)2	0.8072426	0.8290509	0.9736949	0.3302081	Binomial	BirdID
fixed	factor(fac.yr)3	0.1820808	0.7768099	0.2343956	0.8146779	Binomial	BirdID
fixed	factor(fac.yr)4	1.7778384	0.8446898	2.1047234	0.0353154	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	1.7085471	0.8595834	1.9876454	0.0468509	Binomial	BirdID
fixed	factor(fac.yr)6	1.2183372	0.7856232	1.5507908	0.1209518	Binomial	BirdID
fixed	factor(fac.yr)7	1.3770889	0.8152920	1.6890745	0.0912052	Binomial	BirdID
fixed	factor(fac.yr)8	0.9972708	0.8087614	1.2330840	0.2175444	Binomial	BirdID
fixed	factor(fac.yr)9	1.3116201	0.8317925	1.5768598	0.1148278	Binomial	BirdID
fixed	factor(fac.yr)10	0.7266923	0.8072894	0.9001633	0.3680334	Binomial	BirdID
fixed	factor(fac.yr)11	0.3690570	0.8147111	0.4529912	0.6505550	Binomial	BirdID
fixed	factor(fac.yr)12	0.3306390	0.8182438	0.4040837	0.6861512	Binomial	BirdID
fixed	factor(fac.yr)13	1.2631386	0.8795479	1.4361226	0.1509675	Binomial	BirdID
fixed	factor(fac.yr)14	1.2883768	0.9313326	1.3833691	0.1665517	Binomial	BirdID
fixed	factor(fac.yr)15	0.2213091	0.9725313	0.2275599	0.8199884	Binomial	BirdID
fixed	factor(fac.yr)16	0.2469191	0.9383675	0.2631369	0.7924451	Binomial	BirdID
fixed	factor(fac.yr)17	1.7638858	1.1020201	1.6005930	0.1094671	Binomial	BirdID
fixed	factor(fac.yr)18	-0.2694682	1.0418344	-0.2586478	0.7959070	Binomial	BirdID
fixed	factor(fac.yr)19	0.5675341	1.2876923	0.4407373	0.6594032	Binomial	BirdID
fixed	factor(fac.yr)20	-0.5982979	1.4995575	-0.3989830	0.6899057	Binomial	BirdID
fixed	(Intercept)	-0.4871229	0.6362679	-0.7655941	0.4439179	Binomial	Terr
fixed	factor(fac.yr)2	1.2876185	0.8641958	1.4899616	0.1362343	Binomial	Terr
fixed	factor(fac.yr)3	0.6997842	0.7951158	0.8801035	0.3788032	Binomial	Terr
fixed	factor(fac.yr)4	2.2896657	0.8644778	2.6486113	0.0080823	Binomial	Terr
fixed	factor(fac.yr)5	1.9159625	0.8808589	2.1751073	0.0296221	Binomial	Terr
fixed	factor(fac.yr)6	1.3770238	0.7957197	1.7305388	0.0835341	Binomial	Terr
fixed	factor(fac.yr)7	1.6436286	0.8282769	1.9843950	0.0472118	Binomial	Terr
fixed	factor(fac.yr)8	1.6150364	0.8079355	1.9989670	0.0456119	Binomial	Terr
fixed	factor(fac.yr)9	1.9034069	0.8367433	2.2747801	0.0229191	Binomial	Terr
fixed	factor(fac.yr)10	1.3621129	0.8093802	1.6829085	0.0923928	Binomial	Terr
fixed	factor(fac.yr)11	1.3317692	0.8084507	1.6473103	0.0994943	Binomial	Terr
fixed	factor(fac.yr)12	0.9210841	0.8164834	1.1281113	0.2592729	Binomial	Terr
fixed	factor(fac.yr)13	1.7974707	0.8785585	2.0459318	0.0407631	Binomial	Terr
fixed	factor(fac.yr)14	1.6902458	0.9220324	1.8331739	0.0667767	Binomial	Terr
fixed	factor(fac.yr)15	0.7754275	0.9626499	0.8055135	0.4205234	Binomial	Terr
fixed	factor(fac.yr)16	0.8194288	0.9403159	0.8714399	0.3835140	Binomial	Terr
fixed	factor(fac.yr)17	2.3461657	1.1106955	2.1123394	0.0346574	Binomial	Terr
fixed	factor(fac.yr)18	-0.5276477	1.0613672	-0.4971396	0.6190906	Binomial	Terr
fixed	factor(fac.yr)19	0.1462805	1.2958556	0.1128833	0.9101230	Binomial	Terr
fixed	factor(fac.yr)20	-0.8549638	1.6046419	-0.5328066	0.5941675	Binomial	Terr
fixed	(Intercept)	-0.1707806	0.6868040	-0.2486599	0.8036238	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	0.9955845	0.8998105	1.1064380	0.2685370	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	0.3784869	0.8457143	0.4475352	0.6544887	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	2.1406256	0.9262745	2.3110057	0.0208325	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	1.8492797	0.9430714	1.9609117	0.0498893	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	1.2956347	0.8588010	1.5086554	0.1313869	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	1.5086817	0.8879800	1.6990041	0.0893184	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	1.4198310	0.8755474	1.6216494	0.1048784	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	1.5900023	0.9032743	1.7602652	0.0783629	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	1.0606420	0.8773586	1.2089036	0.2266999	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	0.9432968	0.8803707	1.0714768	0.2839551	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	0.5390739	0.8906276	0.6052742	0.5449968	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	1.5428072	0.9558265	1.6141080	0.1065040	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	1.4808610	0.9993208	1.4818676	0.1383755	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	0.5184759	1.0467993	0.4952964	0.6203909	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	0.4693350	1.0273288	0.4568498	0.6477790	Binomial	BirdID+Terr

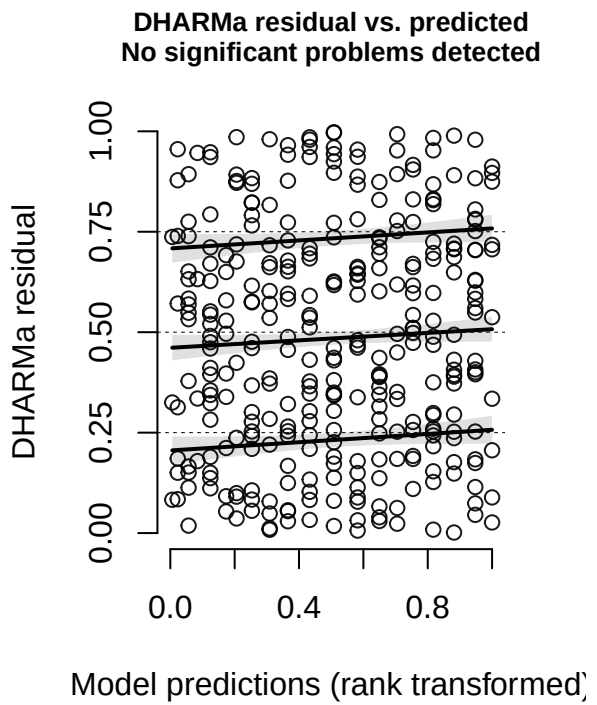
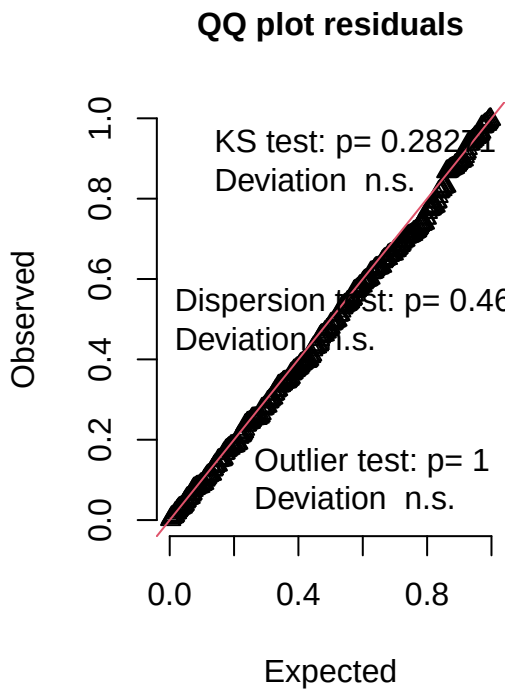
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	2.0830370	1.1857755	1.7566875	0.0789711	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-0.8194133	1.1602093	-0.7062633	0.4800244	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	0.1042077	1.4048700	0.0741760	0.9408703	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-1.2173437	1.7039909	-0.7144074	0.4749753	Binomial	BirdID+Terr

The model fitting both random effects is performing the best on the criteria.

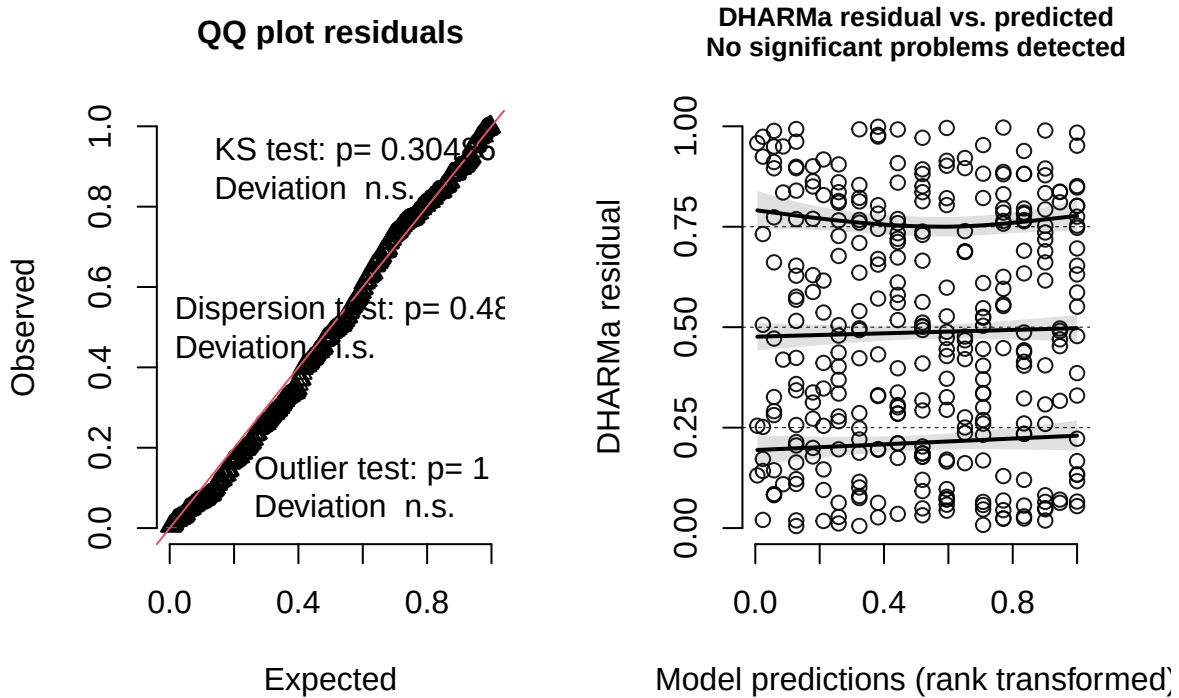
DHARMA residual



DHARMA residual



DHARMa residual



Diagnostics look good.

Probabilites used to generate simulation 26

Simulation 26
 Probability of mate fidelity: 0.8
 Probability of pair dispersal: 0.473
 Probability of single male dispersal: 0.334
 Probability of single female dispersal: 0.066

Simulation 27

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
351	1	-440.0704	922.1409	1003.217	323.3283	330	0.4166312	0.2126581	Poisson	Terr
351	1	-439.8822	923.7644	1008.702	313.1159	329	0.4223471	0.2097993	Poisson	BirdID+Terr
351	1	-446.7504	935.5009	1016.577	385.4069	330	0.3365282	0.2505706	Poisson	BirdID

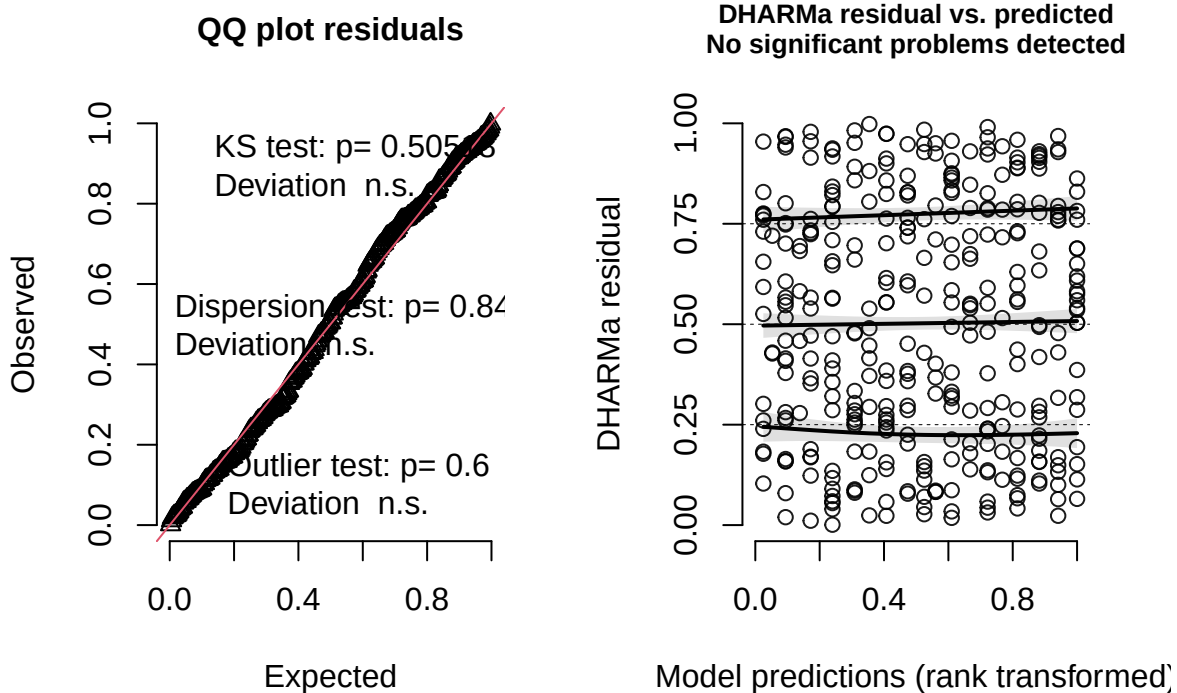
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.3005440	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.4838863	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.1676313	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.4686051	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	-0.2053556	0.3030789	-0.6775648	0.4980477	Poisson	BirdID
fixed	factor(fac.yr)2	0.2150167	0.3940038	0.5457225	0.5852567	Poisson	BirdID
fixed	factor(fac.yr)3	-0.2082131	0.4231370	-0.4920702	0.6226697	Poisson	BirdID
fixed	factor(fac.yr)4	0.9080089	0.3399446	2.6710496	0.0075614	Poisson	BirdID
fixed	factor(fac.yr)5	0.5391211	0.3532248	1.5262831	0.1269394	Poisson	BirdID
fixed	factor(fac.yr)6	0.2744962	0.3687472	0.7444021	0.4566332	Poisson	BirdID
fixed	factor(fac.yr)7	-0.1967257	0.3814220	-0.5157692	0.6060156	Poisson	BirdID
fixed	factor(fac.yr)8	-0.0681362	0.3824788	-0.1781439	0.8586100	Poisson	BirdID
fixed	factor(fac.yr)9	0.4141160	0.3581336	1.1563171	0.2475515	Poisson	BirdID
fixed	factor(fac.yr)10	0.1939637	0.3668792	0.5286854	0.5970237	Poisson	BirdID
fixed	factor(fac.yr)11	-0.8827160	0.4571780	-1.9307929	0.0535087	Poisson	BirdID
fixed	factor(fac.yr)12	-0.0586350	0.3948980	-0.1484813	0.8819629	Poisson	BirdID
fixed	factor(fac.yr)13	0.7523876	0.3613893	2.0819311	0.0373488	Poisson	BirdID
fixed	factor(fac.yr)14	-0.1937880	0.4180376	-0.4635660	0.6429587	Poisson	BirdID
fixed	factor(fac.yr)15	-1.1999869	0.5945357	-2.0183595	0.0435538	Poisson	BirdID
fixed	factor(fac.yr)16	-0.1708854	0.4761105	-0.3589197	0.7196552	Poisson	BirdID
fixed	factor(fac.yr)17	0.3923009	0.4406958	0.8901852	0.3733664	Poisson	BirdID
fixed	factor(fac.yr)18	0.0723256	0.4631738	0.1561523	0.8759130	Poisson	BirdID
fixed	factor(fac.yr)19	-0.3181740	0.6714750	-0.4738434	0.6356116	Poisson	BirdID
fixed	factor(fac.yr)20	-0.8980064	1.0746702	-0.8356112	0.4033737	Poisson	BirdID
fixed	(Intercept)	-0.3333242	0.3131899	-1.0642879	0.2871983	Poisson	Terr
fixed	factor(fac.yr)2	0.2939901	0.4119797	0.7136034	0.4754725	Poisson	Terr
fixed	factor(fac.yr)3	-0.1376655	0.4361135	-0.3156645	0.7522572	Poisson	Terr
fixed	factor(fac.yr)4	0.9291981	0.3516023	2.6427531	0.0082235	Poisson	Terr
fixed	factor(fac.yr)5	0.6885288	0.3631366	1.8960603	0.0579521	Poisson	Terr
fixed	factor(fac.yr)6	0.4994117	0.3799893	1.3142782	0.1887526	Poisson	Terr
fixed	factor(fac.yr)7	-0.0050811	0.3943108	-0.0128860	0.9897188	Poisson	Terr
fixed	factor(fac.yr)8	0.0612325	0.3918697	0.1562572	0.8758303	Poisson	Terr
fixed	factor(fac.yr)9	0.5767254	0.3647912	1.5809743	0.1138839	Poisson	Terr
fixed	factor(fac.yr)10	0.3031883	0.3731907	0.8124219	0.4165496	Poisson	Terr
fixed	factor(fac.yr)11	-0.7778952	0.4681339	-1.6616938	0.0965742	Poisson	Terr
fixed	factor(fac.yr)12	-0.0921204	0.4025047	-0.2288678	0.8189717	Poisson	Terr
fixed	factor(fac.yr)13	0.6698169	0.3698850	1.8108788	0.0701596	Poisson	Terr
fixed	factor(fac.yr)14	-0.1171354	0.4262263	-0.2748198	0.7834547	Poisson	Terr
fixed	factor(fac.yr)15	-1.0038053	0.6015326	-1.6687462	0.0951677	Poisson	Terr
fixed	factor(fac.yr)16	-0.1109642	0.4811765	-0.2306103	0.8176176	Poisson	Terr
fixed	factor(fac.yr)17	0.4740595	0.4545130	1.0430054	0.2969458	Poisson	Terr
fixed	factor(fac.yr)18	0.1702423	0.4710865	0.3613823	0.7178136	Poisson	Terr
fixed	factor(fac.yr)19	-0.1739495	0.6934734	-0.2508380	0.8019394	Poisson	Terr
fixed	factor(fac.yr)20	-0.7152689	1.0721951	-0.6671070	0.5047038	Poisson	Terr
fixed	(Intercept)	-0.3389402	0.3161413	-1.0721164	0.2836678	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	0.2943613	0.4140044	0.7110102	0.4770779	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.1348185	0.4395548	-0.3067160	0.7590595	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	0.9326961	0.3551712	2.6260463	0.0086383	Poisson	BirdID+Terr

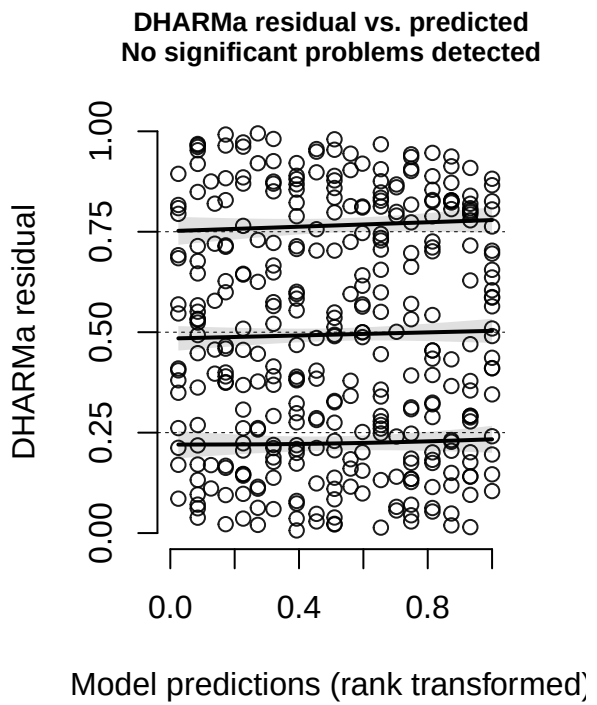
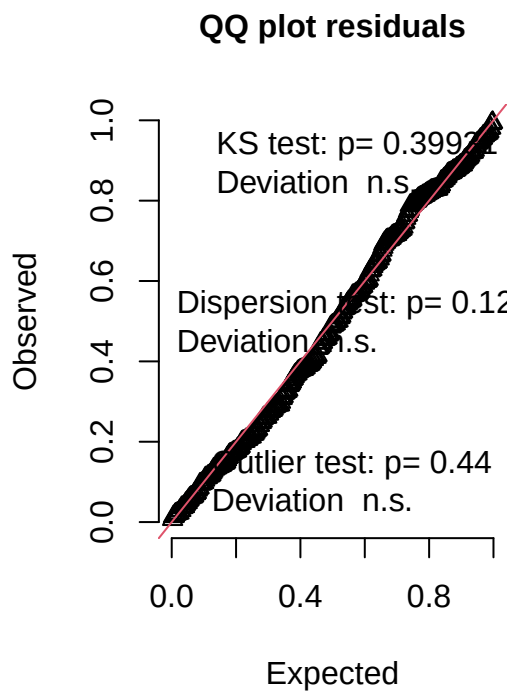
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.6757967	0.3668827	1.8419966	0.0654756	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	0.4836701	0.3839928	1.2595812	0.2078205	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.0197546	0.3978784	-0.0496498	0.9604015	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	0.0462179	0.3958879	0.1167449	0.9070622	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	0.5535282	0.3690905	1.4997089	0.1336898	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	0.2881850	0.3775190	0.7633657	0.4452453	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.7859026	0.4717520	-1.6659233	0.0957287	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.0965816	0.4066951	-0.2374792	0.8122851	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	0.6826147	0.3751601	1.8195291	0.0688307	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.1246241	0.4303415	-0.2895935	0.7721272	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-1.0126071	0.6058524	-1.6713760	0.0946474	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.0950455	0.4859520	-0.1955863	0.8449340	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	0.4824934	0.4599913	1.0489185	0.2942156	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	0.1576383	0.4764836	0.3308368	0.7407677	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-0.1747225	0.6982717	-0.2502214	0.8024162	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.7228423	1.0809833	-0.6686895	0.5036935	Poisson	BirdID+Terr

Similar response as to the last batch of Poisson models.

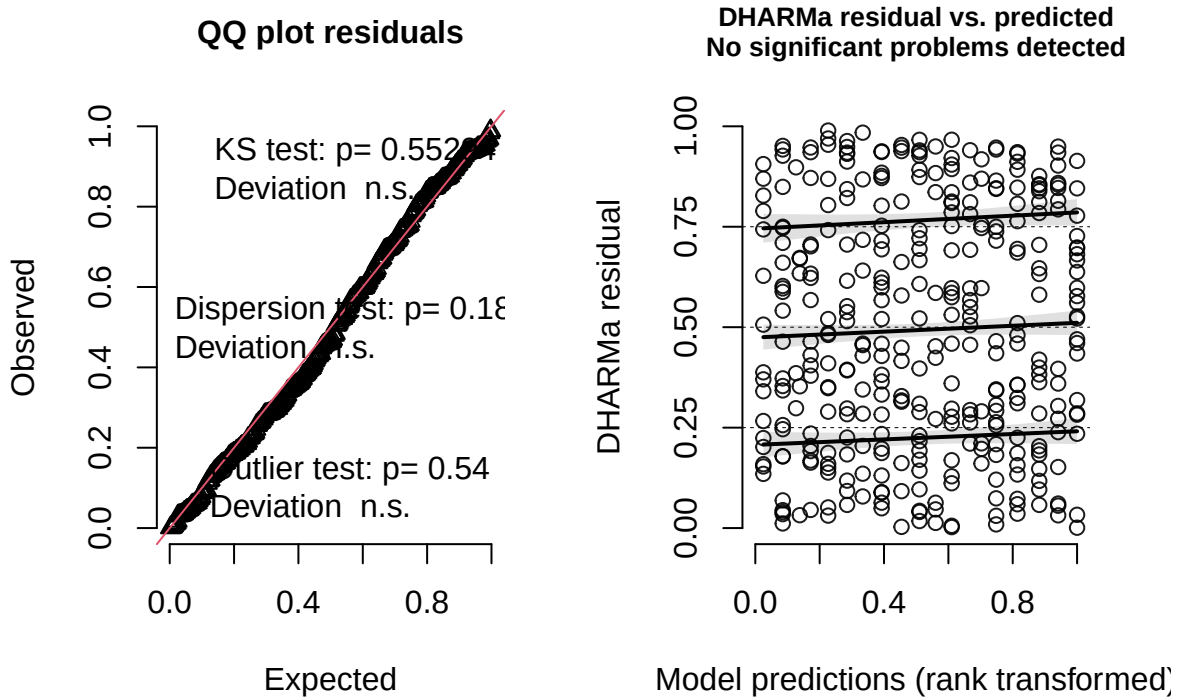
DHARMA residual



DHARMa residual



DHARMA residual



Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
351	1	-207.7336	459.4671	544.4044	248.8754	329	0.787	Binomial	BirdID+Terr
351	1	-211.6228	465.2456	546.3221	325.8684	330	0.626	Binomial	Terr
351	1	-214.2015	470.4029	551.4794	347.7095	330	0.563	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.9354364	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.1155798	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.0625009	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.2466950	Binomial	BirdID+Terr

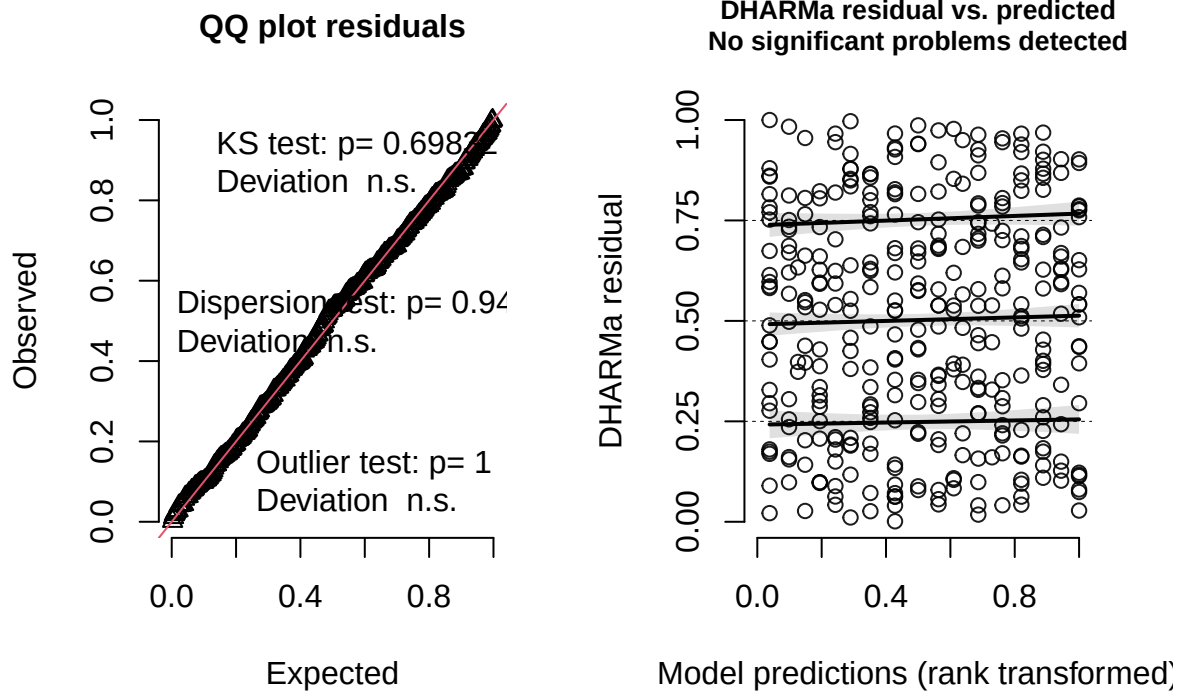
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	-0.3467688	0.5995641	-0.5783683	0.5630155	Binomial	BirdID
fixed	factor(fac.yr)2	1.4933290	0.8563709	1.7437876	0.0811961	Binomial	BirdID
fixed	factor(fac.yr)3	0.1308589	0.7861673	0.1664517	0.8678015	Binomial	BirdID
fixed	factor(fac.yr)4	3.1060197	1.0275808	3.0226525	0.0025057	Binomial	BirdID
fixed	factor(fac.yr)5	1.3220770	0.7786141	1.6979873	0.0895102	Binomial	BirdID
fixed	factor(fac.yr)6	0.4805316	0.7681545	0.6255664	0.5315994	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)7	0.2843024	0.7325519	0.3880986	0.6979431	Binomial	BirdID
fixed	factor(fac.yr)8	0.2289439	0.7552659	0.3031303	0.7617906	Binomial	BirdID
fixed	factor(fac.yr)9	1.2624807	0.7863707	1.6054523	0.1083943	Binomial	BirdID
fixed	factor(fac.yr)10	0.8309050	0.7636318	1.0880964	0.2765526	Binomial	BirdID
fixed	factor(fac.yr)11	-1.0587919	0.7977055	-1.3272968	0.1844105	Binomial	BirdID
fixed	factor(fac.yr)12	0.4708189	0.7725145	0.6094628	0.5422177	Binomial	BirdID
fixed	factor(fac.yr)13	1.2200520	0.8253635	1.4781997	0.1393543	Binomial	BirdID
fixed	factor(fac.yr)14	-0.0837452	0.7995982	-0.1047341	0.9165868	Binomial	BirdID
fixed	factor(fac.yr)15	-0.8727356	0.8798834	-0.9918765	0.3212578	Binomial	BirdID
fixed	factor(fac.yr)16	-0.1834212	0.9026240	-0.2032089	0.8389718	Binomial	BirdID
fixed	factor(fac.yr)17	0.7927883	0.9584200	0.8271825	0.4081336	Binomial	BirdID
fixed	factor(fac.yr)18	0.5379948	0.9270595	0.5803239	0.5616962	Binomial	BirdID
fixed	factor(fac.yr)19	1.0872224	1.1962923	0.9088267	0.3634416	Binomial	BirdID
fixed	factor(fac.yr)20	-0.2607149	1.5152018	-0.1720661	0.8633855	Binomial	BirdID
fixed	(Intercept)	-0.5392803	0.6240664	-0.8641393	0.3875114	Binomial	Terr
fixed	factor(fac.yr)2	1.9262258	0.9208552	2.0917794	0.0364583	Binomial	Terr
fixed	factor(fac.yr)3	0.4205864	0.8176337	0.5143947	0.6069761	Binomial	Terr
fixed	factor(fac.yr)4	3.2995108	1.0562675	3.1237455	0.0017856	Binomial	Terr
fixed	factor(fac.yr)5	1.7965604	0.8005343	2.2442017	0.0248194	Binomial	Terr
fixed	factor(fac.yr)6	1.0596100	0.7896626	1.3418516	0.1796441	Binomial	Terr
fixed	factor(fac.yr)7	0.7861343	0.7581704	1.0368834	0.2997902	Binomial	Terr
fixed	factor(fac.yr)8	0.6906194	0.7757454	0.8902655	0.3733234	Binomial	Terr
fixed	factor(fac.yr)9	1.6484847	0.8027514	2.0535433	0.0400199	Binomial	Terr
fixed	factor(fac.yr)10	1.1648882	0.7779722	1.4973391	0.1343050	Binomial	Terr
fixed	factor(fac.yr)11	-0.7345649	0.8248012	-0.8905963	0.3731458	Binomial	Terr
fixed	factor(fac.yr)12	0.4709829	0.7811466	0.6029380	0.5465499	Binomial	Terr
fixed	factor(fac.yr)13	1.1527028	0.8381175	1.3753475	0.1690237	Binomial	Terr
fixed	factor(fac.yr)14	-0.1453175	0.8160394	-0.1780766	0.8586629	Binomial	Terr
fixed	factor(fac.yr)15	-0.5594199	0.9027098	-0.6197117	0.5354476	Binomial	Terr
fixed	factor(fac.yr)16	-0.0435096	0.9179873	-0.0473968	0.9621970	Binomial	Terr
fixed	factor(fac.yr)17	1.2101719	0.9899436	1.2224656	0.2215316	Binomial	Terr
fixed	factor(fac.yr)18	0.5339905	0.9449334	0.5651092	0.5719995	Binomial	Terr
fixed	factor(fac.yr)19	1.3286597	1.2145593	1.0939438	0.2739796	Binomial	Terr
fixed	factor(fac.yr)20	-0.1094726	1.5073928	-0.0726238	0.9421055	Binomial	Terr
fixed	(Intercept)	-0.4595420	0.7117662	-0.6456361	0.5185150	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	2.0884237	1.0158629	2.0558126	0.0398006	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	0.3223466	0.9240586	0.3488378	0.7272111	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	3.6643639	1.1951322	3.0660742	0.0021689	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	1.8819371	0.9121037	2.0632930	0.0390848	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	0.9804380	0.9037550	1.0848493	0.2779884	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	0.6370146	0.8660531	0.7355375	0.4620122	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	0.5398931	0.8900839	0.6065642	0.5441401	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	1.7021538	0.9195590	1.8510544	0.0641617	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	1.0639124	0.8870650	1.1993624	0.2303870	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-1.1821292	0.9480044	-1.2469659	0.2124100	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	0.3178213	0.8907979	0.3567827	0.7212545	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	1.3420073	0.9519382	1.4097630	0.1586097	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-0.2225568	0.9310808	-0.2390307	0.8110818	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-0.8879055	1.0304962	-0.8616291	0.3888917	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	0.0179305	1.0399652	0.0172414	0.9862440	Binomial	BirdID+Terr
fixed	factor(fac.yr)17	1.5035185	1.1357897	1.3237648	0.1855812	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	0.5482199	1.0954735	0.5004411	0.6167645	Binomial	BirdID+Terr

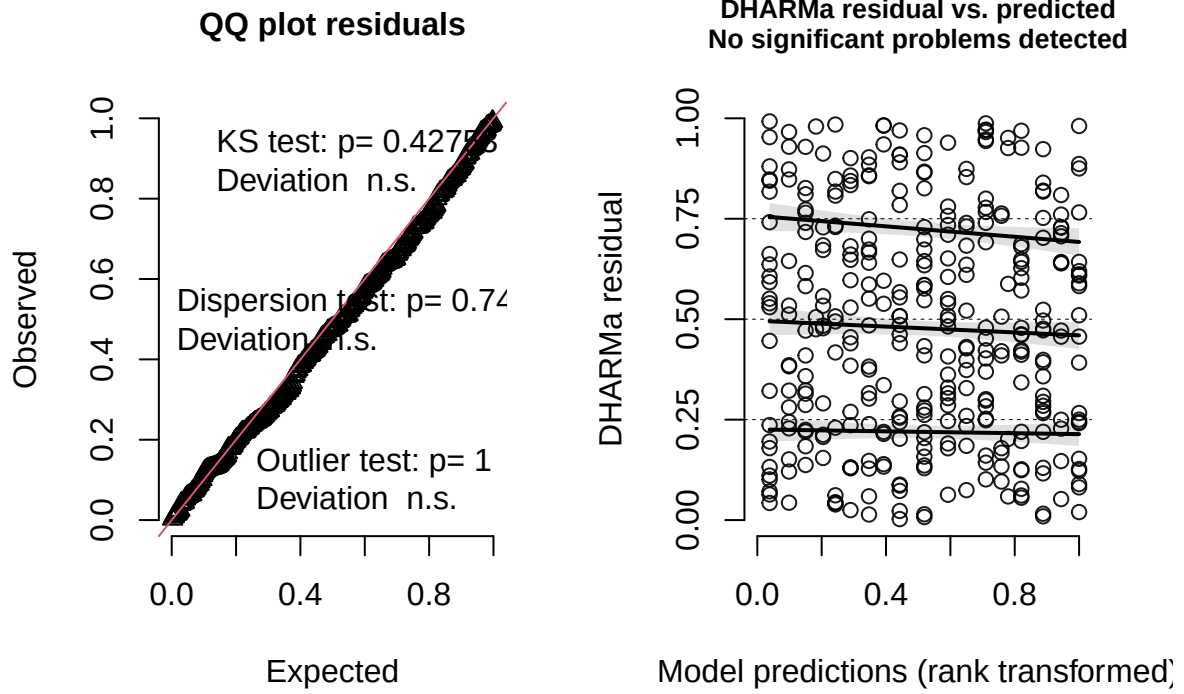
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)19	1.5328293	1.3947976	1.0989618	0.2717847	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-0.2096582	1.7941069	-0.1168594	0.9069715	Binomial	BirdID+Terr

Similar response as to the last batch of binomial models.

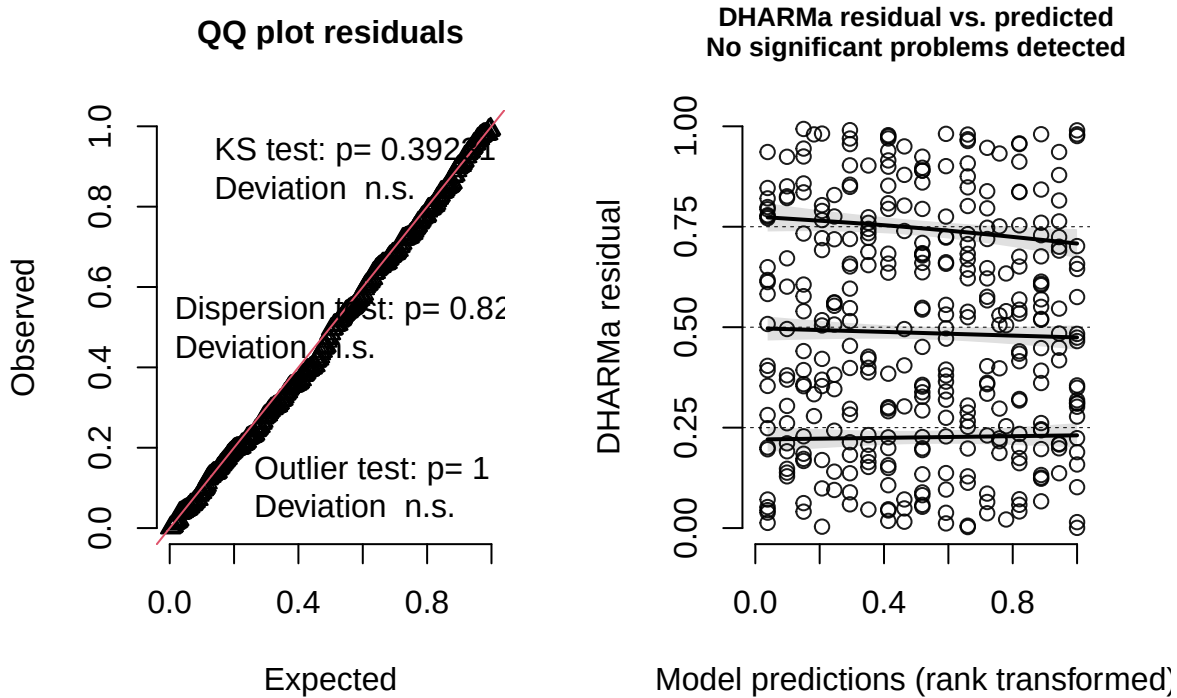
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Probabilites used to generate simulation 27

Simulation 27

Probability of mate fidelity: 0.406
 Probability of pair dispersal: 0.901
 Probability of single male dispersal: 0.121
 Probability of single female dispersal: 0.13

Simulation 28

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	REfamily	RE
318	1	-416.0608	874.1216	953.1246	343.9688	297	0.2365774	0.1609617	Poisson	Terr
318	1	-416.2402	874.4804	953.4835	347.8331	297	0.2367536	0.1688667	Poisson	BirdID
318	1	-415.7603	875.5207	958.2858	328.5713	296	0.2617361	0.1584547	Poisson	BirdID+Terr

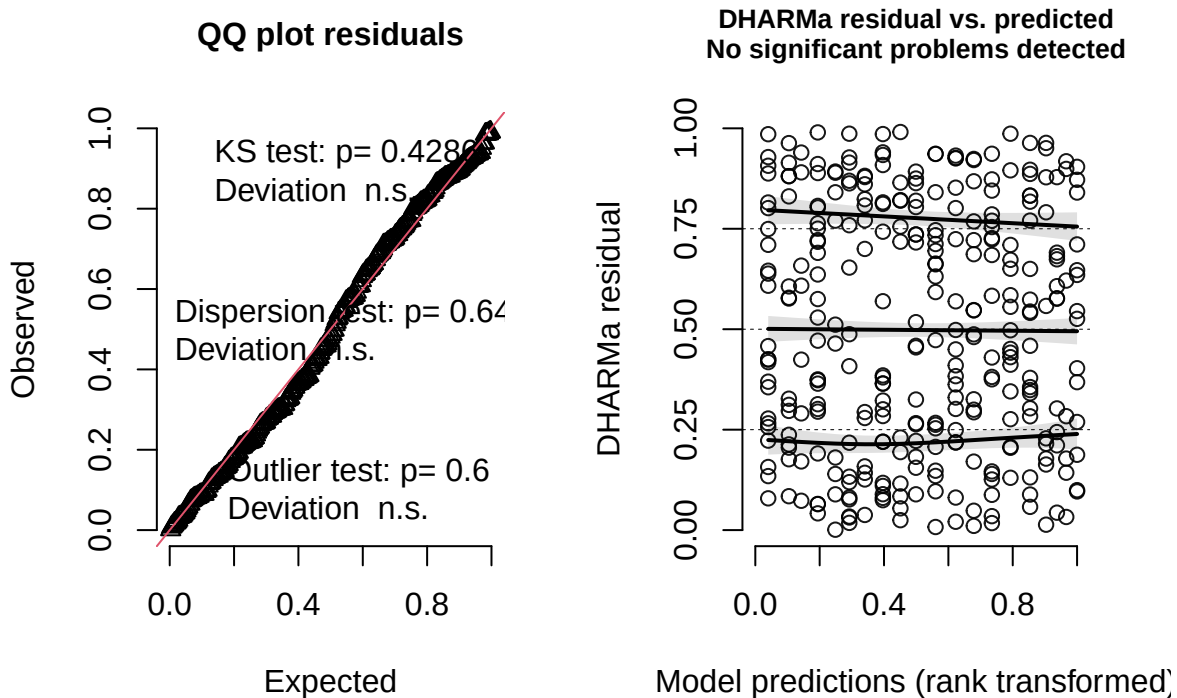
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.2452007	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.2565784	Poisson	Terr
ran_pars	Terrs	sd__(Intercept)	0.2277617	Poisson	BirdID+Terr
ran_pars	birdID	sd__(Intercept)	0.2035807	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.2948829	0.2705306	1.0900166	0.2757058	Poisson	BirdID
fixed	factor(fac.yr)2	-0.1529460	0.3551178	-0.4306907	0.6666933	Poisson	BirdID
fixed	factor(fac.yr)3	-0.5037860	0.3832630	-1.3144656	0.1886896	Poisson	BirdID
fixed	factor(fac.yr)4	-0.1586344	0.3478076	-0.4560983	0.6483193	Poisson	BirdID
fixed	factor(fac.yr)5	-0.1599785	0.3447103	-0.4640956	0.6425793	Poisson	BirdID
fixed	factor(fac.yr)6	-0.1109779	0.3516072	-0.3156303	0.7522831	Poisson	BirdID
fixed	factor(fac.yr)7	-0.6719682	0.3668806	-1.8315718	0.0670152	Poisson	BirdID
fixed	factor(fac.yr)8	-0.0917659	0.3360989	-0.2730323	0.7848284	Poisson	BirdID
fixed	factor(fac.yr)9	-0.2172196	0.3521447	-0.6168475	0.5373353	Poisson	BirdID
fixed	factor(fac.yr)10	-0.1397599	0.3440540	-0.4062151	0.6845846	Poisson	BirdID
fixed	factor(fac.yr)11	-1.2305068	0.4210343	-2.9225810	0.0034714	Poisson	BirdID
fixed	factor(fac.yr)12	-0.2808171	0.3463207	-0.8108585	0.4174469	Poisson	BirdID
fixed	factor(fac.yr)13	0.2710908	0.3405459	0.7960479	0.4260042	Poisson	BirdID
fixed	factor(fac.yr)14	-0.3385205	0.3943203	-0.8584911	0.3906213	Poisson	BirdID
fixed	factor(fac.yr)15	-1.0202129	0.4514811	-2.2597022	0.0238397	Poisson	BirdID
fixed	factor(fac.yr)16	-0.0337584	0.3958311	-0.0852848	0.9320349	Poisson	BirdID
fixed	factor(fac.yr)17	-0.2390778	0.3949692	-0.6053075	0.5449747	Poisson	BirdID
fixed	factor(fac.yr)18	-0.5464122	0.4537219	-1.2042888	0.2284780	Poisson	BirdID
fixed	factor(fac.yr)19	-1.0201289	0.5782394	-1.7641984	0.0776986	Poisson	BirdID
fixed	factor(fac.yr)20	0.0610414	0.4269629	0.1429665	0.8863166	Poisson	BirdID
fixed	(Intercept)	0.2913252	0.2709477	1.0752082	0.2822815	Poisson	Terr
fixed	factor(fac.yr)2	-0.1405036	0.3590156	-0.3913578	0.6955328	Poisson	Terr
fixed	factor(fac.yr)3	-0.5051684	0.3858517	-1.3092296	0.1904566	Poisson	Terr
fixed	factor(fac.yr)4	-0.1907807	0.3470767	-0.5496789	0.5825396	Poisson	Terr
fixed	factor(fac.yr)5	-0.1642776	0.3459363	-0.4748782	0.6348738	Poisson	Terr
fixed	factor(fac.yr)6	-0.0782529	0.3537364	-0.2212180	0.8249227	Poisson	Terr
fixed	factor(fac.yr)7	-0.6725500	0.3672674	-1.8312269	0.0670667	Poisson	Terr
fixed	factor(fac.yr)8	-0.0715679	0.3346313	-0.2138711	0.8306476	Poisson	Terr
fixed	factor(fac.yr)9	-0.1484949	0.3519585	-0.4219102	0.6730906	Poisson	Terr
fixed	factor(fac.yr)10	-0.0877917	0.3432659	-0.2557542	0.7981407	Poisson	Terr
fixed	factor(fac.yr)11	-1.1602005	0.4198404	-2.7634324	0.0057197	Poisson	Terr
fixed	factor(fac.yr)12	-0.2536986	0.3452983	-0.7347230	0.4625082	Poisson	Terr
fixed	factor(fac.yr)13	0.2817607	0.3393193	0.8303703	0.4063295	Poisson	Terr
fixed	factor(fac.yr)14	-0.3553582	0.3950579	-0.8995091	0.3683816	Poisson	Terr
fixed	factor(fac.yr)15	-0.9892627	0.4515084	-2.1910175	0.0284505	Poisson	Terr
fixed	factor(fac.yr)16	-0.0718800	0.3965049	-0.1812839	0.8561447	Poisson	Terr
fixed	factor(fac.yr)17	-0.2346577	0.3955431	-0.5932543	0.5530110	Poisson	Terr
fixed	factor(fac.yr)18	-0.5327411	0.4546453	-1.1717731	0.2412882	Poisson	Terr
fixed	factor(fac.yr)19	-1.0103873	0.5776393	-1.7491664	0.0802623	Poisson	Terr
fixed	factor(fac.yr)20	-0.0669723	0.4236509	-0.1580837	0.8743908	Poisson	Terr
fixed	(Intercept)	0.2885094	0.2763872	1.0438593	0.2965505	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.1518265	0.3625448	-0.4187800	0.6753769	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.5070347	0.3900651	-1.2998719	0.1936449	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.1894089	0.3529190	-0.5366922	0.5914802	Poisson	BirdID+Terr

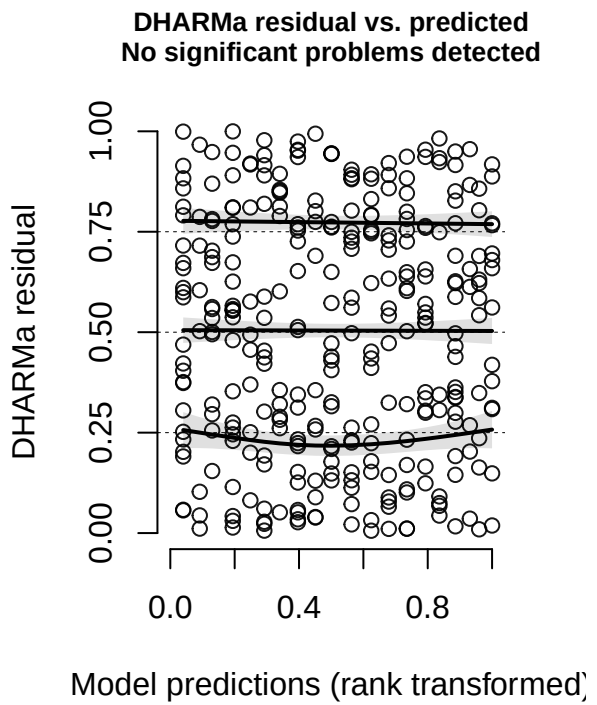
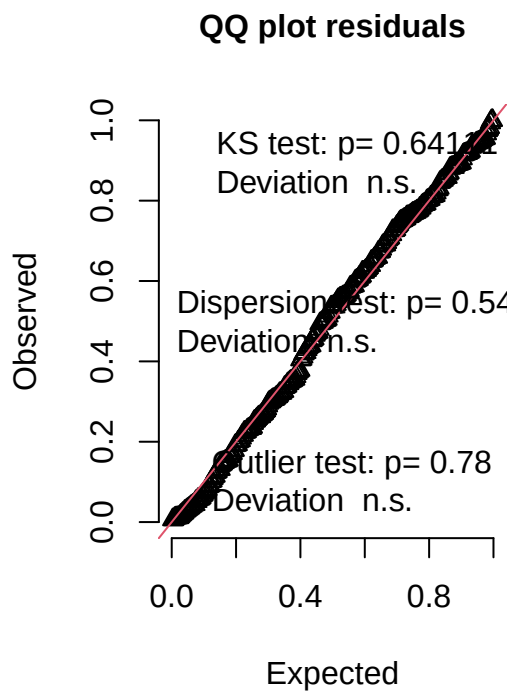
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.1625017	0.3513976	-0.4624441	0.6437629	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.0964438	0.3596462	-0.2681630	0.7885739	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.6882009	0.3728784	-1.8456440	0.0649439	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.0960808	0.3414022	-0.2814300	0.7783806	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.1963608	0.3582664	-0.5480860	0.5836328	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.1320687	0.3502901	-0.3770267	0.7061537	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-1.2035729	0.4258226	-2.8264656	0.0047065	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.2736529	0.3522501	-0.7768709	0.4372349	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	0.2569207	0.3472859	0.7397959	0.4594238	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.3677466	0.4021323	-0.9144915	0.3604586	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-1.0064200	0.4577014	-2.1988571	0.0278881	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.0604359	0.4041943	-0.1495218	0.8811419	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.2435028	0.4028762	-0.6044110	0.5455704	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.5417714	0.4617785	-1.1732279	0.2407044	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.0217559	0.5852363	-1.7458860	0.0808308	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.0395938	0.4334934	-0.0913365	0.9272252	Poisson	BirdID+Terr

Similar to the previous Poisson models.

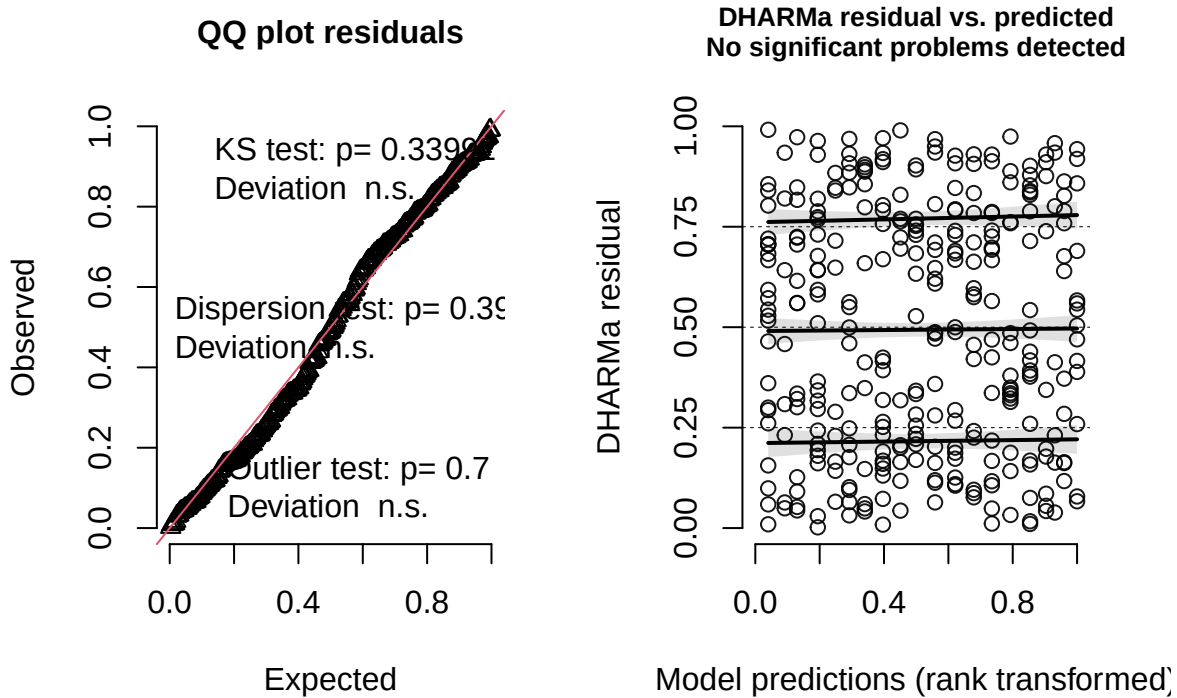
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
318	1	-198.5825	439.1650	518.1680	314.5513	297	0.537	Binomial	Terr
318	1	-197.6817	439.3634	522.1286	286.9513	296	0.620	Binomial	BirdID+Terr
318	1	-198.9883	439.9766	518.9797	324.3836	297	0.511	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.9411482	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.0111627	Binomial	Terr
ran_pars	Terrs	sd__(Intercept)	0.8913520	Binomial	BirdID+Terr
ran_pars	birdID	sd__(Intercept)	0.7446943	Binomial	BirdID+Terr

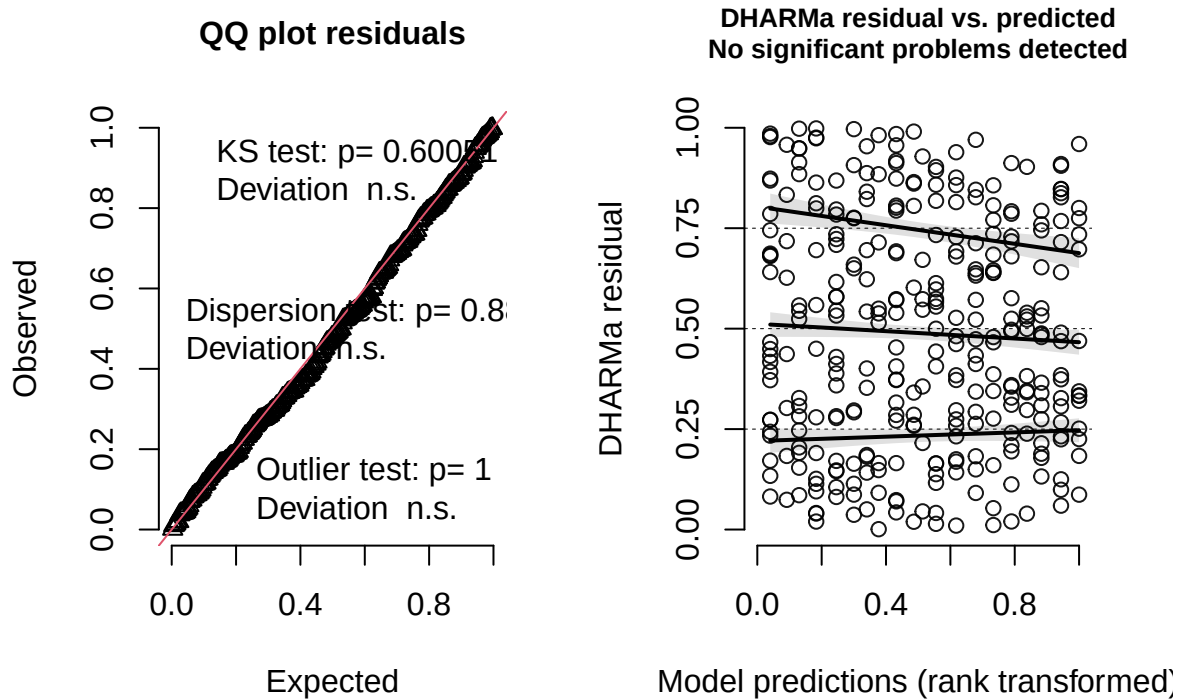
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	1.1636532	0.7499073	1.5517295	0.1207270	Binomial	BirdID
fixed	factor(fac.yr)2	-0.3119619	0.9250051	-0.3372542	0.7359253	Binomial	BirdID
fixed	factor(fac.yr)3	-1.2848987	0.9076999	-1.4155545	0.1569060	Binomial	BirdID
fixed	factor(fac.yr)4	-0.1530707	0.9165715	-0.1670036	0.8673672	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.5287342	0.8942718	-0.5912455	0.5543559	Binomial	BirdID
fixed	factor(fac.yr)6	0.0833888	0.9785347	0.0852180	0.9320881	Binomial	BirdID
fixed	factor(fac.yr)7	-1.0175888	0.8729638	-1.1656712	0.2437474	Binomial	BirdID
fixed	factor(fac.yr)8	0.1435008	0.9440697	0.1520023	0.8791851	Binomial	BirdID
fixed	factor(fac.yr)9	-0.5260532	0.9360906	-0.5619682	0.5741377	Binomial	BirdID
fixed	factor(fac.yr)10	-0.5320221	0.9301782	-0.5719571	0.5673510	Binomial	BirdID
fixed	factor(fac.yr)11	-2.1458719	0.8919633	-2.4057849	0.0161378	Binomial	BirdID
fixed	factor(fac.yr)12	-1.1692441	0.8871369	-1.3179974	0.1875045	Binomial	BirdID
fixed	factor(fac.yr)13	0.7048928	1.1353986	0.6208329	0.5347096	Binomial	BirdID
fixed	factor(fac.yr)14	-1.0880885	0.9733092	-1.1179269	0.2635982	Binomial	BirdID
fixed	factor(fac.yr)15	-1.4881965	0.9382631	-1.5861186	0.1127124	Binomial	BirdID
fixed	factor(fac.yr)16	-0.9737854	1.0217717	-0.9530362	0.3405717	Binomial	BirdID
fixed	factor(fac.yr)17	-1.1018018	0.9870835	-1.1162194	0.2643282	Binomial	BirdID
fixed	factor(fac.yr)18	-1.0841520	1.0275853	-1.0550482	0.2914033	Binomial	BirdID
fixed	factor(fac.yr)19	-1.8319561	1.1031562	-1.6606497	0.0967838	Binomial	BirdID
fixed	factor(fac.yr)20	-0.6218257	1.1262160	-0.5521371	0.5808544	Binomial	BirdID
fixed	(Intercept)	1.1506354	0.7465092	1.5413547	0.1232305	Binomial	Terr
fixed	factor(fac.yr)2	-0.1192007	0.9375553	-0.1271399	0.8988297	Binomial	Terr
fixed	factor(fac.yr)3	-1.1291328	0.9164199	-1.2321129	0.2179069	Binomial	Terr
fixed	factor(fac.yr)4	-0.2164300	0.9075563	-0.2384756	0.8115122	Binomial	Terr
fixed	factor(fac.yr)5	-0.3039617	0.8951497	-0.3395652	0.7341840	Binomial	Terr
fixed	factor(fac.yr)6	0.4869405	0.9828972	0.4954134	0.6203083	Binomial	Terr
fixed	factor(fac.yr)7	-0.8465439	0.8668656	-0.9765573	0.3287884	Binomial	Terr
fixed	factor(fac.yr)8	0.3074997	0.9322759	0.3298376	0.7415226	Binomial	Terr
fixed	factor(fac.yr)9	-0.1444546	0.9287339	-0.1555393	0.8763962	Binomial	Terr
fixed	factor(fac.yr)10	-0.1792332	0.9198375	-0.1948531	0.8455079	Binomial	Terr
fixed	factor(fac.yr)11	-1.7565327	0.8766446	-2.0036999	0.0451022	Binomial	Terr
fixed	factor(fac.yr)12	-1.0549173	0.8786593	-1.2005988	0.2299069	Binomial	Terr
fixed	factor(fac.yr)13	0.8782777	1.1294891	0.7775884	0.4368117	Binomial	Terr
fixed	factor(fac.yr)14	-1.0605673	0.9739343	-1.0889515	0.2761753	Binomial	Terr
fixed	factor(fac.yr)15	-1.3552310	0.9364802	-1.4471539	0.1478538	Binomial	Terr
fixed	factor(fac.yr)16	-1.1790691	1.0255540	-1.1496899	0.2502716	Binomial	Terr
fixed	factor(fac.yr)17	-1.0543521	0.9801900	-1.0756610	0.2820789	Binomial	Terr
fixed	factor(fac.yr)18	-1.0568099	1.0280684	-1.0279568	0.3039702	Binomial	Terr
fixed	factor(fac.yr)19	-1.6396216	1.0996046	-1.4911011	0.1359350	Binomial	Terr
fixed	factor(fac.yr)20	-1.3894578	1.1139643	-1.2473091	0.2122842	Binomial	Terr
fixed	(Intercept)	1.2762547	0.7803541	1.6354816	0.1019481	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-0.2122470	0.9647070	-0.2200118	0.8258619	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-1.2929419	0.9462905	-1.3663267	0.1718365	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-0.2421386	0.9449103	-0.2562556	0.7977535	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-0.3913041	0.9296960	-0.4208947	0.6738320	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	0.2822340	1.0236501	0.2757133	0.7827682	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-1.0351467	0.9068351	-1.1414939	0.2536644	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	0.2380242	0.9782398	0.2433188	0.8077584	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-0.3880985	0.9747979	-0.3981323	0.6905327	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-0.4645457	0.9641880	-0.4818000	0.6299480	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-2.1085950	0.9200846	-2.2917404	0.0219206	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-1.2390264	0.9213976	-1.3447251	0.1787140	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	0.7630913	1.1869131	0.6429210	0.5202754	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-1.2087874	1.0174686	-1.1880341	0.2348200	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-1.5209994	0.9791626	-1.5533676	0.1203353	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-1.1456436	1.0728065	-1.0678940	0.2855683	Binomial	BirdID+Terr

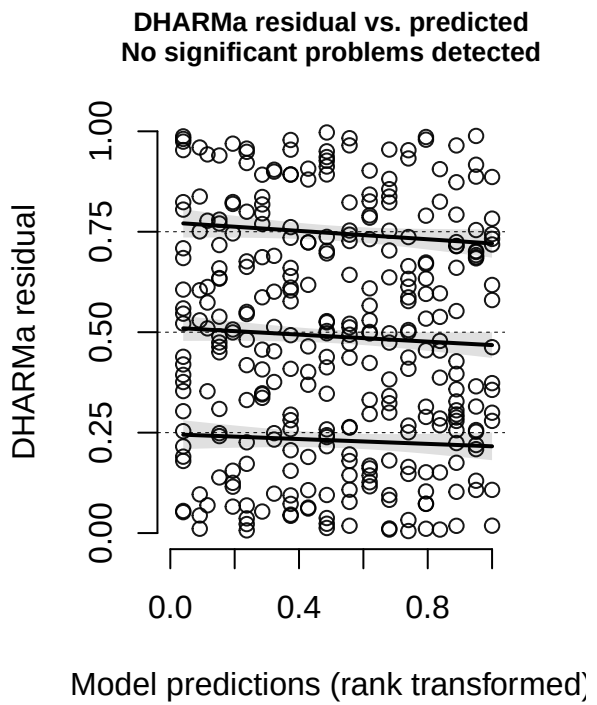
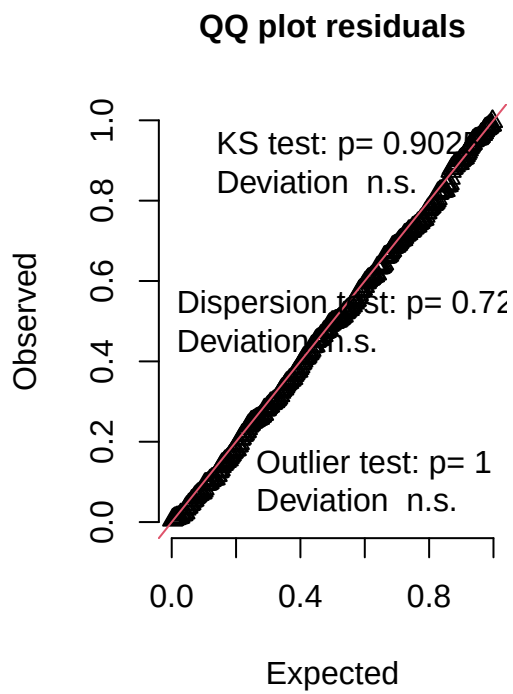
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-1.1173557	1.0288790	-1.0859933	0.2774820	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-1.0990732	1.0715915	-1.0256457	0.3050586	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-1.8110742	1.1532725	-1.5703783	0.1163271	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-1.2196653	1.1694234	-1.0429630	0.2969654	Binomial	BirdID+Terr

The territory only model is performing best on AIC and BIC.

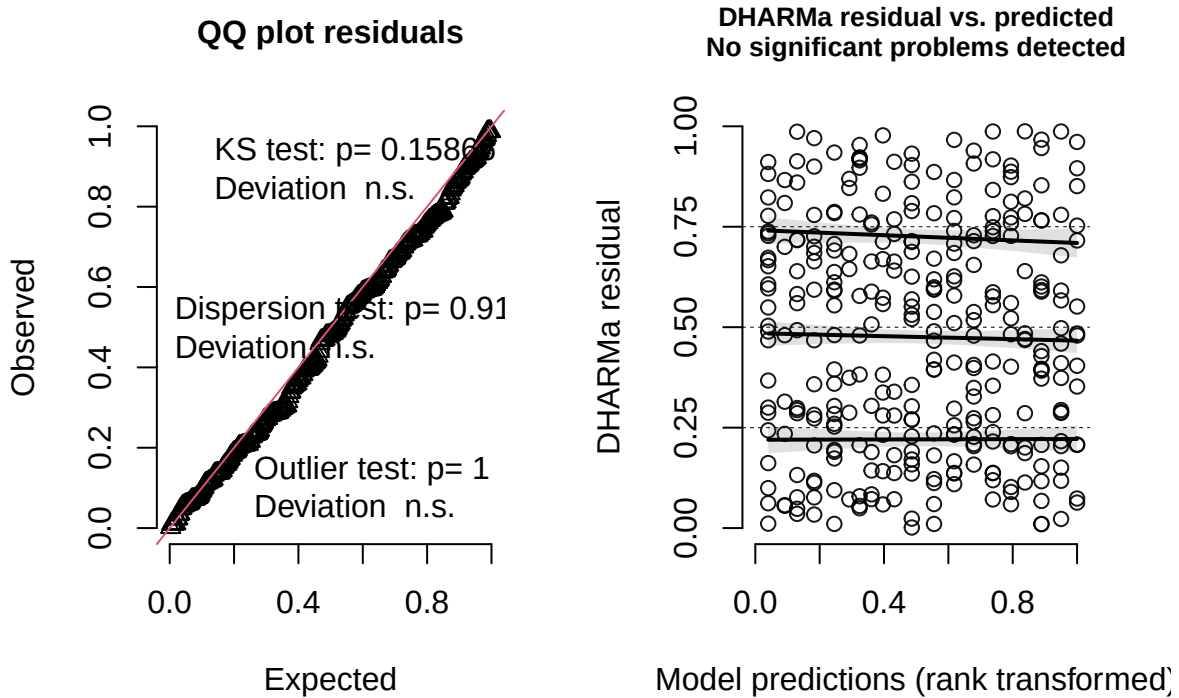
DHARMA residual



DHARMa residual



DHARMA residual



Diagnostics look good.

Probabilities used to generate simulation 28

Simulation 28
 Probability of mate fidelity: 0.465
 Probability of pair dispersal: 0.162
 Probability of single male dispersal: 0.755
 Probability of single female dispersal: 0.928

Simulation 29

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
330	1	-443.0576	928.1151	1007.896	337.7799	309	0.2929493	0.2184603	Poisson	BirdID
330	1	-443.5922	929.1844	1008.965	356.8043	309	0.2736671	0.2345858	Poisson	Terr
330	1	-442.9821	929.9641	1013.544	331.1632	308	0.3015468	0.2175067	Poisson	BirdID+Terr

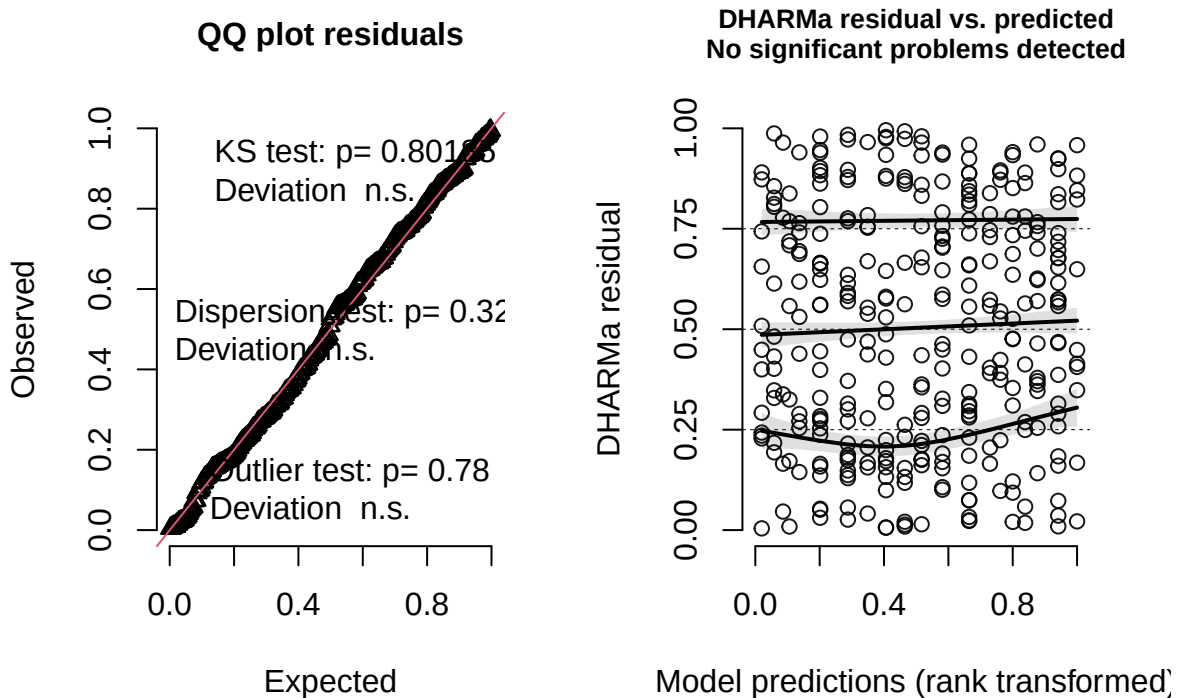
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.2551386	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.1821023	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.2400529	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.1285586	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.6875498	0.2235766	3.0752320	0.0021034	Poisson	BirdID
fixed	factor(fac.yr)2	-0.6467534	0.3337516	-1.9378284	0.0526442	Poisson	BirdID
fixed	factor(fac.yr)3	-0.8850177	0.3097682	-2.8570318	0.0042762	Poisson	BirdID
fixed	factor(fac.yr)4	-0.5265381	0.2966186	-1.7751349	0.0758756	Poisson	BirdID
fixed	factor(fac.yr)5	-0.0243554	0.2656695	-0.0916757	0.9269557	Poisson	BirdID
fixed	factor(fac.yr)6	-0.4969848	0.2805594	-1.7714070	0.0764931	Poisson	BirdID
fixed	factor(fac.yr)7	-0.8927921	0.3047241	-2.9298372	0.0033914	Poisson	BirdID
fixed	factor(fac.yr)8	-0.7444494	0.3145465	-2.3667388	0.0179456	Poisson	BirdID
fixed	factor(fac.yr)9	-0.6006004	0.3189951	-1.8827888	0.0597290	Poisson	BirdID
fixed	factor(fac.yr)10	-0.8400492	0.3731240	-2.2513943	0.0243606	Poisson	BirdID
fixed	factor(fac.yr)11	-2.0649217	0.6288054	-3.2838802	0.0010239	Poisson	BirdID
fixed	factor(fac.yr)12	-1.3157619	0.4472053	-2.9421874	0.0032590	Poisson	BirdID
fixed	factor(fac.yr)13	-0.3155368	0.3134592	-1.0066281	0.3141135	Poisson	BirdID
fixed	factor(fac.yr)14	-0.2175486	0.3051869	-0.7128374	0.4759464	Poisson	BirdID
fixed	factor(fac.yr)15	-1.0500825	0.4096376	-2.5634428	0.0103640	Poisson	BirdID
fixed	factor(fac.yr)16	-0.4405922	0.3506799	-1.2563943	0.2089730	Poisson	BirdID
fixed	factor(fac.yr)17	-0.2998905	0.3512102	-0.8538777	0.3931728	Poisson	BirdID
fixed	factor(fac.yr)18	-0.3811335	0.3581403	-1.0642016	0.2872374	Poisson	BirdID
fixed	factor(fac.yr)19	-1.1858630	0.5103024	-2.3238436	0.0201339	Poisson	BirdID
fixed	factor(fac.yr)20	-1.2277250	0.6329161	-1.9397912	0.0524051	Poisson	BirdID
fixed	(Intercept)	0.7422217	0.2165045	3.4282045	0.0006076	Poisson	Terr
fixed	factor(fac.yr)2	-0.6766842	0.3343481	-2.0238911	0.0429814	Poisson	Terr
fixed	factor(fac.yr)3	-0.9271640	0.3063532	-3.0264541	0.0024744	Poisson	Terr
fixed	factor(fac.yr)4	-0.5574923	0.2895743	-1.9252133	0.0542027	Poisson	Terr
fixed	factor(fac.yr)5	-0.0569799	0.2577896	-0.2210326	0.8250670	Poisson	Terr
fixed	factor(fac.yr)6	-0.5448497	0.2723849	-2.0002934	0.0454686	Poisson	Terr
fixed	factor(fac.yr)7	-0.9294277	0.2969590	-3.1298189	0.0017491	Poisson	Terr
fixed	factor(fac.yr)8	-0.8040216	0.3060607	-2.6270003	0.0086141	Poisson	Terr
fixed	factor(fac.yr)9	-0.6725940	0.3095570	-2.1727633	0.0297981	Poisson	Terr
fixed	factor(fac.yr)10	-0.8983022	0.3635934	-2.4706231	0.0134878	Poisson	Terr
fixed	factor(fac.yr)11	-2.1329672	0.6219746	-3.4293475	0.0006050	Poisson	Terr
fixed	factor(fac.yr)12	-1.3715551	0.4396380	-3.1197376	0.0018101	Poisson	Terr
fixed	factor(fac.yr)13	-0.3110746	0.3045933	-1.0212785	0.3071225	Poisson	Terr
fixed	factor(fac.yr)14	-0.1975428	0.2951215	-0.6693609	0.5032653	Poisson	Terr
fixed	factor(fac.yr)15	-1.0318955	0.4017922	-2.5682316	0.0102219	Poisson	Terr
fixed	factor(fac.yr)16	-0.4283954	0.3407476	-1.2572221	0.2086732	Poisson	Terr
fixed	factor(fac.yr)17	-0.3715302	0.3423814	-1.0851357	0.2778616	Poisson	Terr
fixed	factor(fac.yr)18	-0.4172734	0.3482245	-1.1982885	0.2308047	Poisson	Terr
fixed	factor(fac.yr)19	-1.2095394	0.5020452	-2.4092241	0.0159865	Poisson	Terr
fixed	factor(fac.yr)20	-1.2179503	0.6239784	-1.9519111	0.0509488	Poisson	Terr
fixed	(Intercept)	0.6957827	0.2257067	3.0826848	0.0020514	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.6533671	0.3371684	-1.9378062	0.0526469	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.9004872	0.3123620	-2.8828317	0.0039412	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.5374563	0.2988242	-1.7985703	0.0720867	Poisson	BirdID+Terr

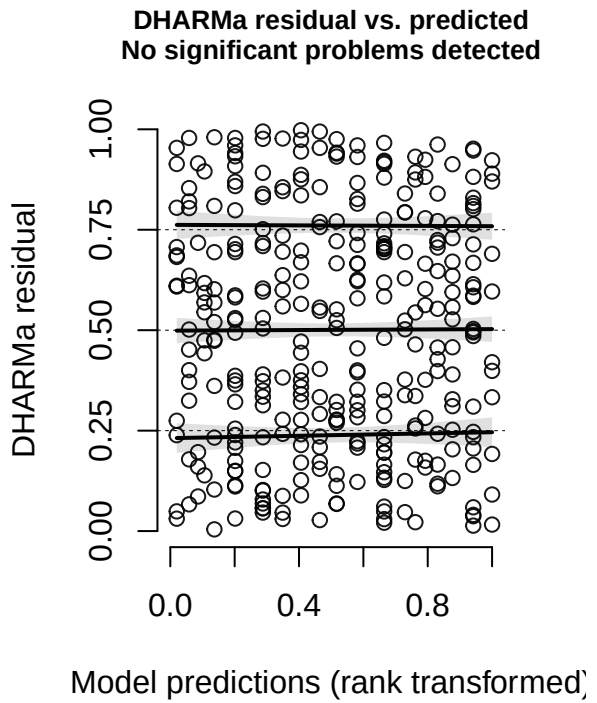
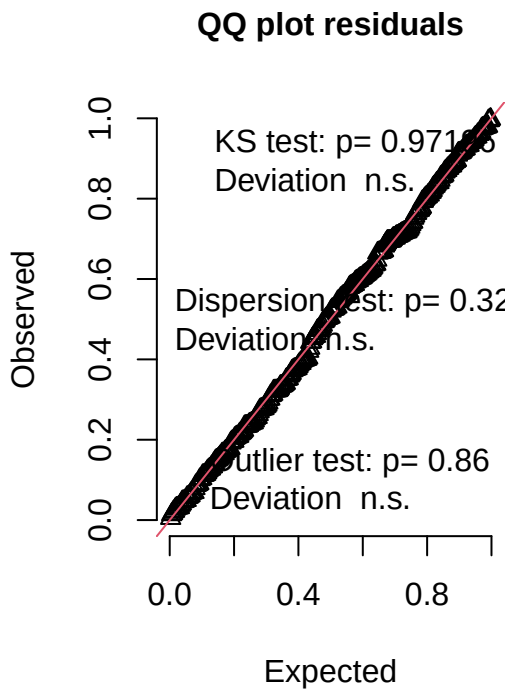
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.0339534	0.2681242	-0.1266329	0.8992309	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.5077465	0.2826457	-1.7964064	0.0724299	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.9015335	0.3068051	-2.9384571	0.0032985	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.7611115	0.3165659	-2.4042752	0.0162046	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.6233909	0.3207920	-1.9432868	0.0519815	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.8489990	0.3747485	-2.2655164	0.0234810	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-2.0790735	0.6308105	-3.2958764	0.0009812	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-1.3296308	0.4492583	-2.9596135	0.0030803	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.3147029	0.3160439	-0.9957569	0.3193683	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.2180867	0.3073885	-0.7094823	0.4780253	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-1.0522523	0.4119531	-2.5543015	0.0106401	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.4407260	0.3527847	-1.2492777	0.2115635	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.3226356	0.3542280	-0.9108133	0.3623938	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.3911847	0.3604516	-1.0852628	0.2778053	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.1878055	0.5124745	-2.3177848	0.0204610	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-1.2150605	0.6355720	-1.9117589	0.0559071	Poisson	BirdID+Terr

The territory only model has the best marginal R^2 .

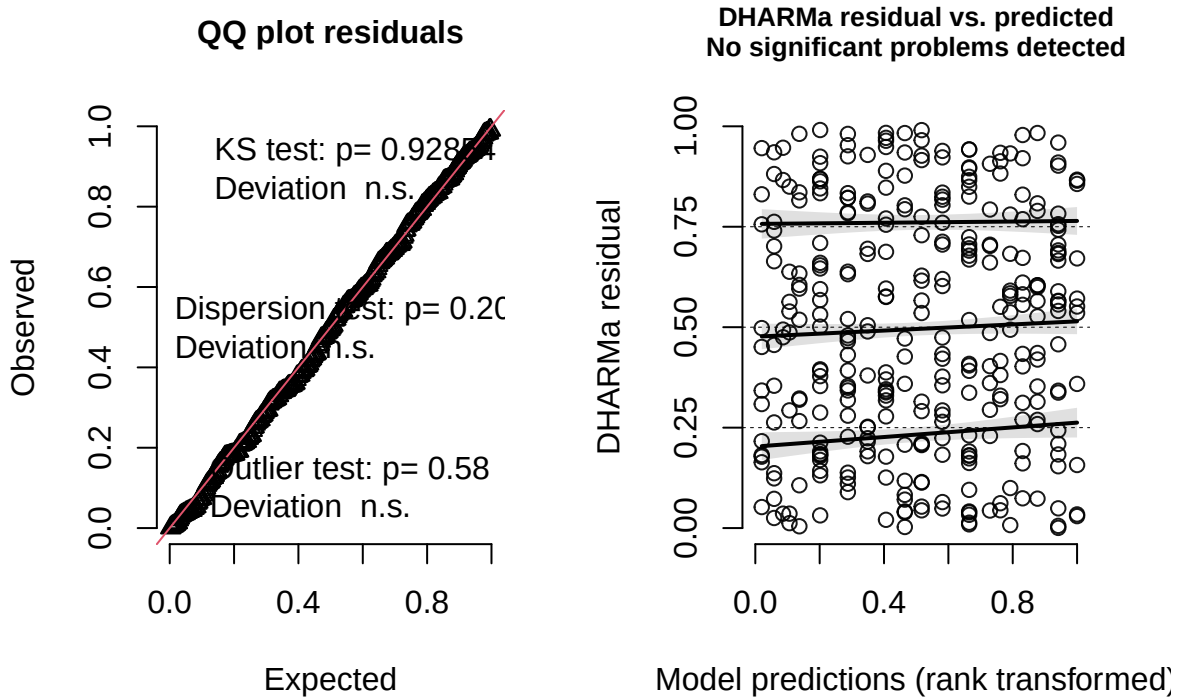
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
330	1	-193.6018	429.2036	508.9846	306.0495	309	0.555	Binomial	BirdID
330	1	-193.5545	431.1091	514.6891	298.1519	308	0.578	Binomial	BirdID+Terr
330	1	-196.1356	434.2713	514.0522	356.4981	309	0.402	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.0041476	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	0.5722407	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.0030152	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.3125047	Binomial	BirdID+Terr

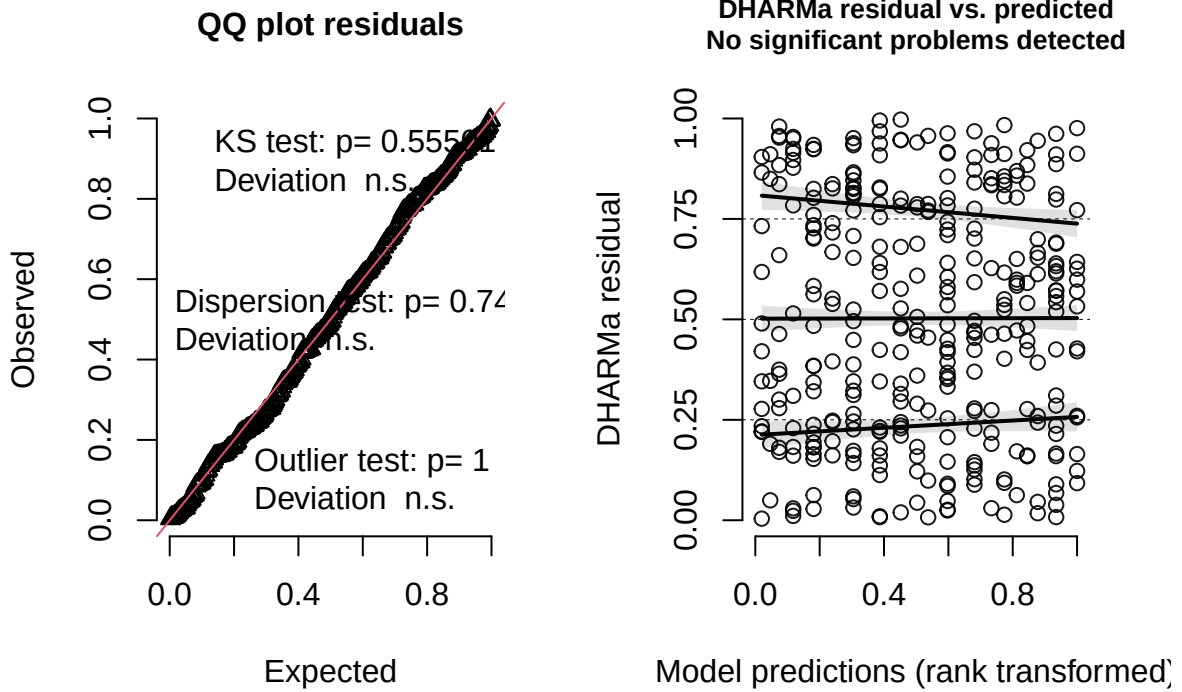
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	1.4885306	0.8740319	1.7030621	0.0885565	Binomial	BirdID
fixed	factor(fac.yr)2	-1.5129443	1.0097555	-1.4983273	0.1340482	Binomial	BirdID
fixed	factor(fac.yr)3	-1.4899669	0.9629002	-1.5473741	0.1217731	Binomial	BirdID
fixed	factor(fac.yr)4	-0.2311534	1.0202879	-0.2265570	0.8207682	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.5489724	1.0688843	0.5135938	0.6075360	Binomial	BirdID
fixed	factor(fac.yr)6	-0.7718760	0.9673037	-0.7979666	0.4248899	Binomial	BirdID
fixed	factor(fac.yr)7	-1.2502831	0.9675920	-1.2921593	0.1963020	Binomial	BirdID
fixed	factor(fac.yr)8	-1.2439474	0.9922324	-1.2536854	0.2099564	Binomial	BirdID
fixed	factor(fac.yr)9	-1.2201680	1.0161755	-1.2007454	0.2298500	Binomial	BirdID
fixed	factor(fac.yr)10	-0.9365326	1.0717641	-0.8738234	0.3822144	Binomial	BirdID
fixed	factor(fac.yr)11	-2.5985898	1.1529533	-2.2538552	0.0242053	Binomial	BirdID
fixed	factor(fac.yr)12	-1.5975470	1.0778323	-1.4821852	0.1382910	Binomial	BirdID
fixed	factor(fac.yr)13	0.0374840	1.1472728	0.0326723	0.9739359	Binomial	BirdID
fixed	factor(fac.yr)14	1.3520736	1.4733006	0.9177174	0.3587668	Binomial	BirdID
fixed	factor(fac.yr)15	-1.3662152	1.0918102	-1.2513304	0.2108140	Binomial	BirdID
fixed	factor(fac.yr)16	0.0902002	1.2364040	0.0729536	0.9418430	Binomial	BirdID
fixed	factor(fac.yr)17	0.0910307	1.2410959	0.0733471	0.9415300	Binomial	BirdID
fixed	factor(fac.yr)18	0.0616036	1.2434509	0.0495425	0.9604870	Binomial	BirdID
fixed	factor(fac.yr)19	-0.8708487	1.1944441	-0.7290829	0.4659510	Binomial	BirdID
fixed	factor(fac.yr)20	-1.9909101	1.3490130	-1.4758272	0.1399903	Binomial	BirdID
fixed	(Intercept)	1.5913567	0.8135797	1.9559935	0.0504659	Binomial	Terr
fixed	factor(fac.yr)2	-1.3839770	0.9755046	-1.4187294	0.1559779	Binomial	Terr
fixed	factor(fac.yr)3	-1.5605873	0.9095048	-1.7158649	0.0861868	Binomial	Terr
fixed	factor(fac.yr)4	-0.4520340	0.9514122	-0.4751190	0.6347022	Binomial	Terr
fixed	factor(fac.yr)5	0.3361639	0.9931979	0.3384662	0.7350119	Binomial	Terr
fixed	factor(fac.yr)6	-1.0465919	0.8980957	-1.1653456	0.2438791	Binomial	Terr
fixed	factor(fac.yr)7	-1.3624166	0.8976359	-1.5177831	0.1290691	Binomial	Terr
fixed	factor(fac.yr)8	-1.4716588	0.9197143	-1.6001261	0.1095706	Binomial	Terr
fixed	factor(fac.yr)9	-1.5067490	0.9396750	-1.6034789	0.1088290	Binomial	Terr
fixed	factor(fac.yr)10	-1.2729671	0.9842659	-1.2933163	0.1959017	Binomial	Terr
fixed	factor(fac.yr)11	-2.6907902	1.0697576	-2.5153271	0.0118922	Binomial	Terr
fixed	factor(fac.yr)12	-1.6757507	0.9961454	-1.6822351	0.0925233	Binomial	Terr
fixed	factor(fac.yr)13	0.0015636	1.0571320	0.0014791	0.9988198	Binomial	Terr
fixed	factor(fac.yr)14	1.2106009	1.3575676	0.8917426	0.3725309	Binomial	Terr
fixed	factor(fac.yr)15	-1.2041804	1.0144287	-1.1870527	0.2352068	Binomial	Terr
fixed	factor(fac.yr)16	0.1003196	1.1517676	0.0871006	0.9305916	Binomial	Terr
fixed	factor(fac.yr)17	-0.2231150	1.1580811	-0.1926592	0.8472259	Binomial	Terr
fixed	factor(fac.yr)18	-0.0776402	1.1588693	-0.0669965	0.9465845	Binomial	Terr
fixed	factor(fac.yr)19	-0.9347174	1.1090618	-0.8428001	0.3993402	Binomial	Terr
fixed	factor(fac.yr)20	-1.8518515	1.2515207	-1.4796810	0.1389584	Binomial	Terr
fixed	(Intercept)	1.5692755	0.8947587	1.7538533	0.0794556	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-1.5836820	1.0311922	-1.5357777	0.1245929	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-1.5680368	0.9836417	-1.5941139	0.1109105	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-0.2995828	1.0398811	-0.2880933	0.7732753	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	0.5120150	1.0904870	0.4695287	0.6386918	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-0.8432032	0.9874919	-0.8538837	0.3931694	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-1.3136961	0.9880619	-1.3295685	0.1836605	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-1.3322602	1.0121735	-1.3162369	0.1880945	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-1.3157343	1.0361481	-1.2698323	0.2041444	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-1.0091431	1.0918143	-0.9242809	0.3553401	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-2.6708307	1.1707545	-2.2812901	0.0225313	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-1.6806862	1.0986987	-1.5297062	0.1260895	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	0.0130045	1.1677185	0.0111366	0.9911144	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	1.3155531	1.4939852	0.8805664	0.3785525	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-1.4331685	1.1121271	-1.2886733	0.1975117	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	0.0693828	1.2573999	0.0551796	0.9559954	Binomial	BirdID+Terr

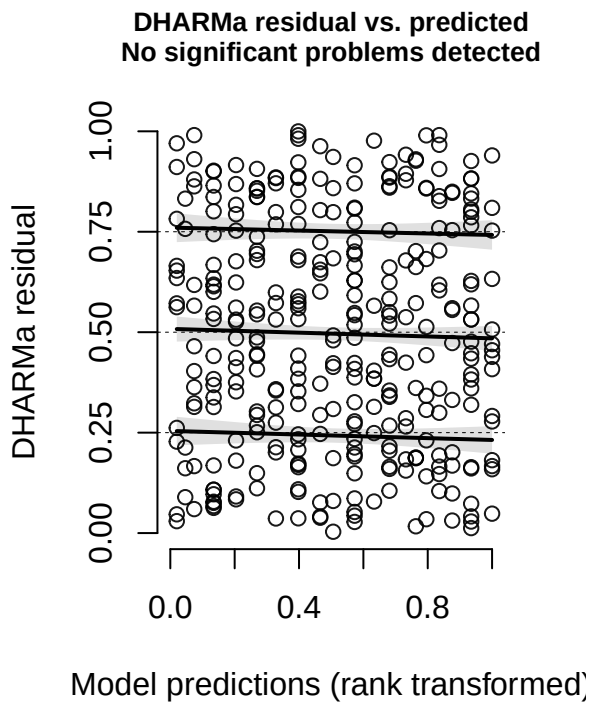
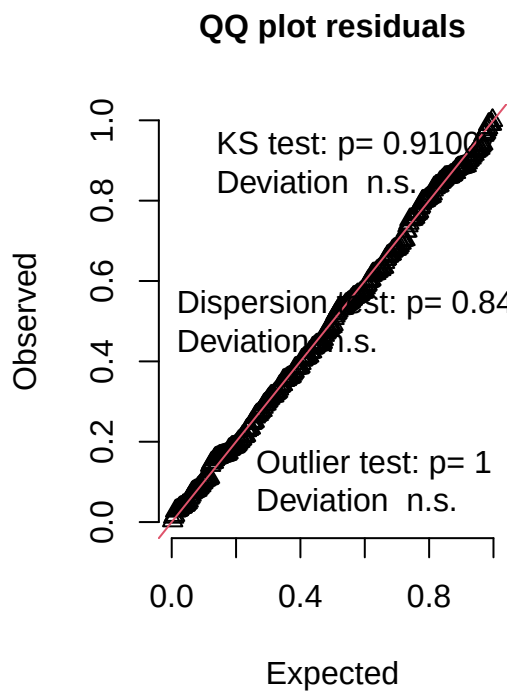
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	0.0141455	1.2627003	0.0112026	0.9910618	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	0.0424384	1.2680754	0.0334668	0.9733024	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-0.9223165	1.2139942	-0.7597371	0.4474117	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-2.0649168	1.3728129	-1.5041502	0.1325427	Binomial	BirdID+Terr

The territory only model is performing the worst on the selection criteria, when compared to the other 2 models.

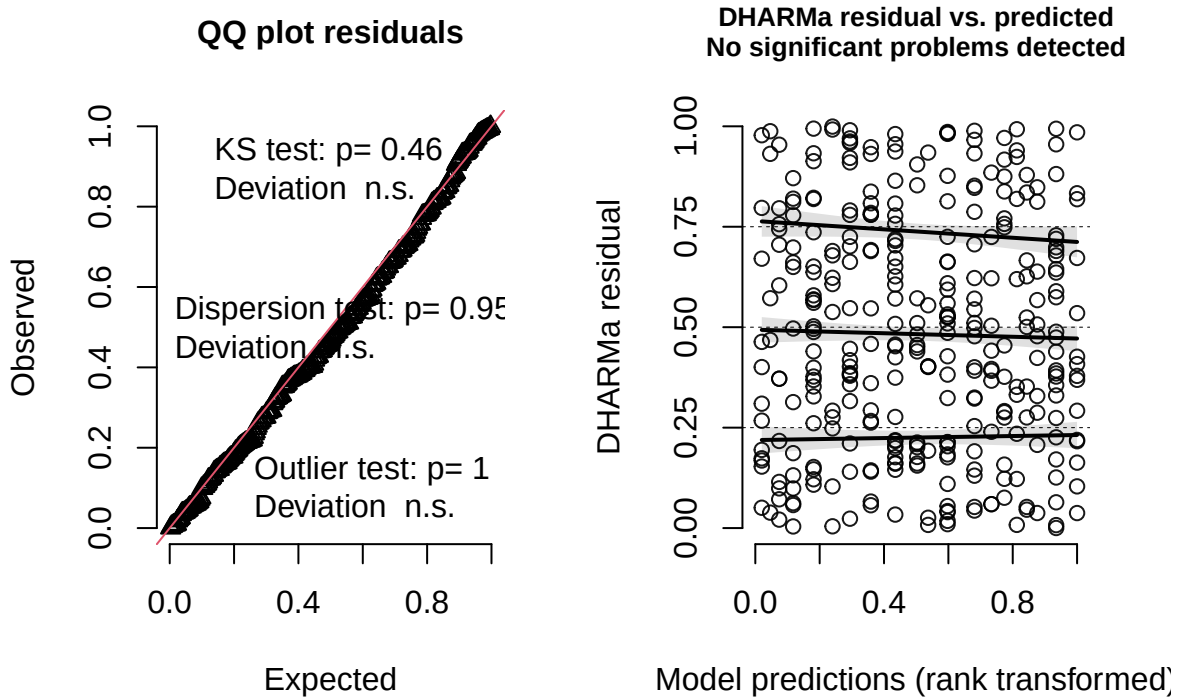
DHARMa residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Probabilities used to generate simulation 29

Simulation 29
 Probability of mate fidelity: 0.521
 Probability of pair dispersal: 0.35
 Probability of single male dispersal: 0.095
 Probability of single female dispersal: 0.665

Simulation 30

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
315	1	-421.3554	884.7107	963.5148	295.0288	294	0.3678685	0.2776095	Poisson	BirdID
315	1	-420.9919	885.9837	968.5403	278.8668	293	0.3831141	0.2709260	Poisson	BirdID+Terr
315	1	-422.3591	886.7182	965.5222	303.2032	294	0.3636571	0.2878284	Poisson	Terr

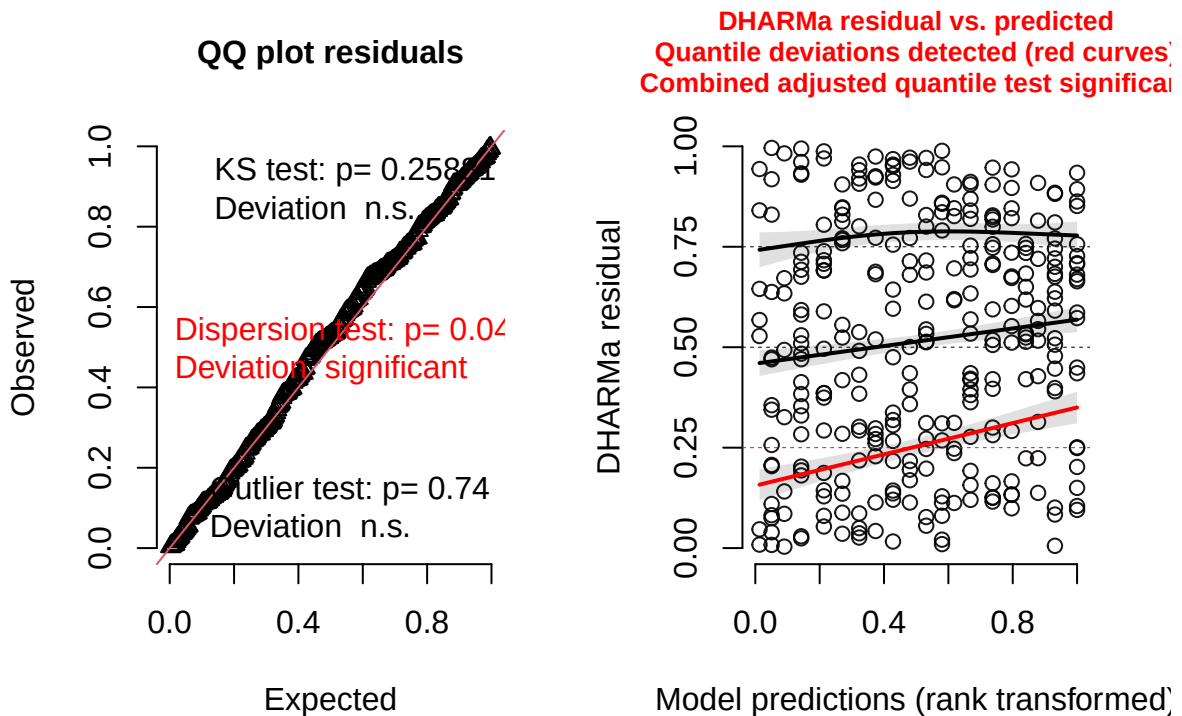
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.2951344	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.2635552	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.2676362	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.1944875	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.5888249	0.2367017	2.4876245	0.0128599	Poisson	BirdID
fixed	factor(fac.yr)2	-0.1427691	0.3254502	-0.4386817	0.6608922	Poisson	BirdID
fixed	factor(fac.yr)3	-0.8456242	0.3689010	-2.2922794	0.0218895	Poisson	BirdID
fixed	factor(fac.yr)4	-0.4882961	0.3308385	-1.4759347	0.1399614	Poisson	BirdID
fixed	factor(fac.yr)5	0.0373616	0.2879365	0.1297565	0.8967591	Poisson	BirdID
fixed	factor(fac.yr)6	-0.6733129	0.3398698	-1.9810905	0.0475811	Poisson	BirdID
fixed	factor(fac.yr)7	-1.1341920	0.3555130	-3.1902973	0.0014213	Poisson	BirdID
fixed	factor(fac.yr)8	-0.1886628	0.2960423	-0.6372833	0.5239403	Poisson	BirdID
fixed	factor(fac.yr)9	0.1816932	0.2823765	0.6434433	0.5199365	Poisson	BirdID
fixed	factor(fac.yr)10	-0.3139198	0.3054967	-1.0275719	0.3041513	Poisson	BirdID
fixed	factor(fac.yr)11	-0.9177597	0.3571311	-2.5698121	0.0101754	Poisson	BirdID
fixed	factor(fac.yr)12	-0.7538005	0.3655933	-2.0618550	0.0392215	Poisson	BirdID
fixed	factor(fac.yr)13	-0.1653757	0.3186904	-0.5189228	0.6038146	Poisson	BirdID
fixed	factor(fac.yr)14	-0.7809804	0.3654865	-2.1368242	0.0326123	Poisson	BirdID
fixed	factor(fac.yr)15	-1.7892318	0.5151347	-3.4733280	0.0005140	Poisson	BirdID
fixed	factor(fac.yr)16	-0.5766516	0.3596280	-1.6034666	0.1088317	Poisson	BirdID
fixed	factor(fac.yr)17	-0.4852895	0.3604302	-1.3464175	0.1781679	Poisson	BirdID
fixed	factor(fac.yr)18	-0.4619507	0.3849494	-1.2000296	0.2301278	Poisson	BirdID
fixed	factor(fac.yr)19	-1.1392632	0.5184283	-2.1975329	0.0279824	Poisson	BirdID
fixed	factor(fac.yr)20	-1.9130061	0.7611232	-2.5133985	0.0119574	Poisson	BirdID
fixed	(Intercept)	0.6134779	0.2332747	2.6298514	0.0085422	Poisson	Terr
fixed	factor(fac.yr)2	-0.1364991	0.3293141	-0.4144951	0.6785115	Poisson	Terr
fixed	factor(fac.yr)3	-0.9029880	0.3672904	-2.4585123	0.0139514	Poisson	Terr
fixed	factor(fac.yr)4	-0.5010389	0.3266709	-1.5337724	0.1250856	Poisson	Terr
fixed	factor(fac.yr)5	0.0344270	0.2845266	0.1209974	0.9036931	Poisson	Terr
fixed	factor(fac.yr)6	-0.6539325	0.3381756	-1.9337070	0.0531492	Poisson	Terr
fixed	factor(fac.yr)7	-1.1206412	0.3525907	-3.1783059	0.0014814	Poisson	Terr
fixed	factor(fac.yr)8	-0.1331037	0.2939226	-0.4528530	0.6506546	Poisson	Terr
fixed	factor(fac.yr)9	0.2205067	0.2786445	0.7913550	0.4287369	Poisson	Terr
fixed	factor(fac.yr)10	-0.3084682	0.3011403	-1.0243338	0.3056776	Poisson	Terr
fixed	factor(fac.yr)11	-0.8766155	0.3543739	-2.4737024	0.0133721	Poisson	Terr
fixed	factor(fac.yr)12	-0.7280623	0.3599394	-2.0227356	0.0431004	Poisson	Terr
fixed	factor(fac.yr)13	-0.1331050	0.3150541	-0.4224830	0.6726725	Poisson	Terr
fixed	factor(fac.yr)14	-0.7416453	0.3603177	-2.0583092	0.0395605	Poisson	Terr
fixed	factor(fac.yr)15	-1.7699162	0.5100073	-3.4703744	0.0005197	Poisson	Terr
fixed	factor(fac.yr)16	-0.5045109	0.3519467	-1.4334865	0.1517189	Poisson	Terr
fixed	factor(fac.yr)17	-0.4518189	0.3556952	-1.2702419	0.2039985	Poisson	Terr
fixed	factor(fac.yr)18	-0.4230201	0.3780196	-1.1190428	0.2631219	Poisson	Terr
fixed	factor(fac.yr)19	-1.0175735	0.5149936	-1.9758953	0.0481666	Poisson	Terr
fixed	factor(fac.yr)20	-1.8198528	0.7570677	-2.4038177	0.0162249	Poisson	Terr
fixed	(Intercept)	0.5798267	0.2417570	2.3983859	0.0164675	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.1392302	0.3322015	-0.4191138	0.6751329	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.8715312	0.3742257	-2.3288918	0.0198648	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.5003412	0.3360557	-1.4888638	0.1365232	Poisson	BirdID+Terr

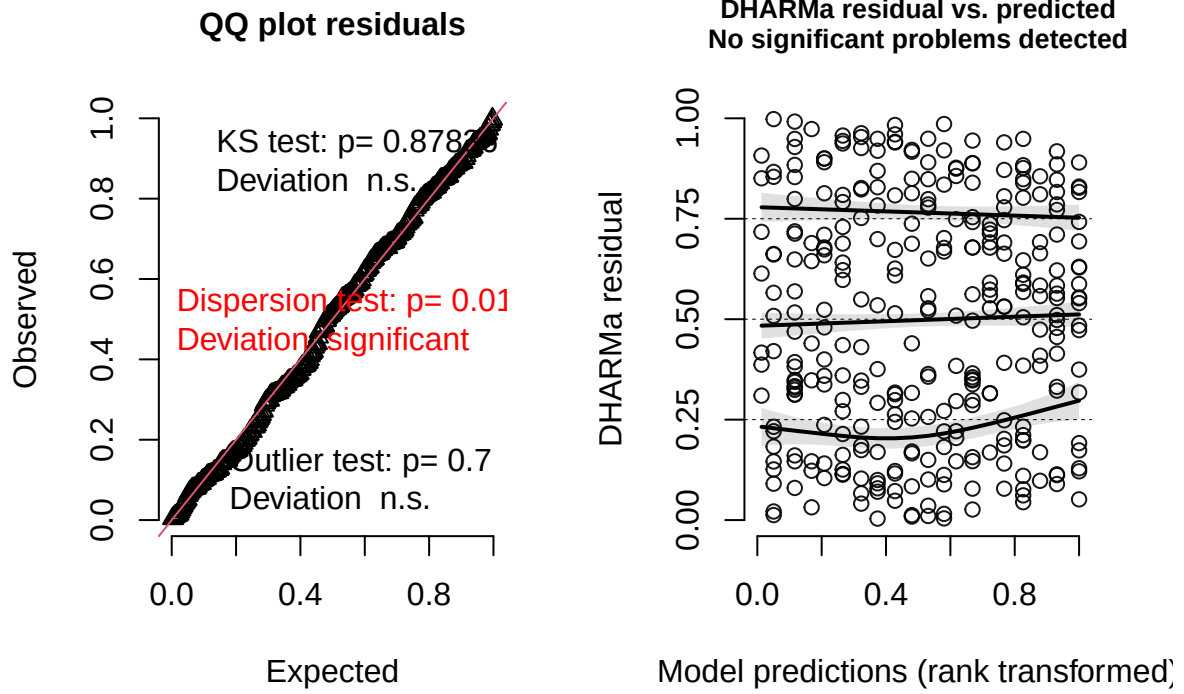
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.0318052	0.2941670	0.1081195	0.9139009	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.6731801	0.3464043	-1.9433362	0.0519756	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-1.1349349	0.3608138	-3.1454862	0.0016581	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.1789695	0.3027142	-0.5912159	0.5543758	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	0.1900329	0.2887590	0.6581021	0.5104725	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.3193150	0.3113572	-1.0255582	0.3050999	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.9153238	0.3632355	-2.5199186	0.0117382	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.7509956	0.3704633	-2.0271798	0.0426440	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.1588901	0.3254915	-0.4881544	0.6254405	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.7635325	0.3706163	-2.0601697	0.0393823	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-1.7754209	0.5192725	-3.4190541	0.0006284	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.5468728	0.3640344	-1.5022558	0.1330310	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.4697592	0.3667381	-1.2809119	0.2002246	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.4503297	0.3902745	-1.1538794	0.2485496	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.0907244	0.5246119	-2.0791073	0.0376075	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-1.8770432	0.7679563	-2.4442059	0.0145171	Poisson	BirdID+Terr

Similar to the previous batch of Poisson models.

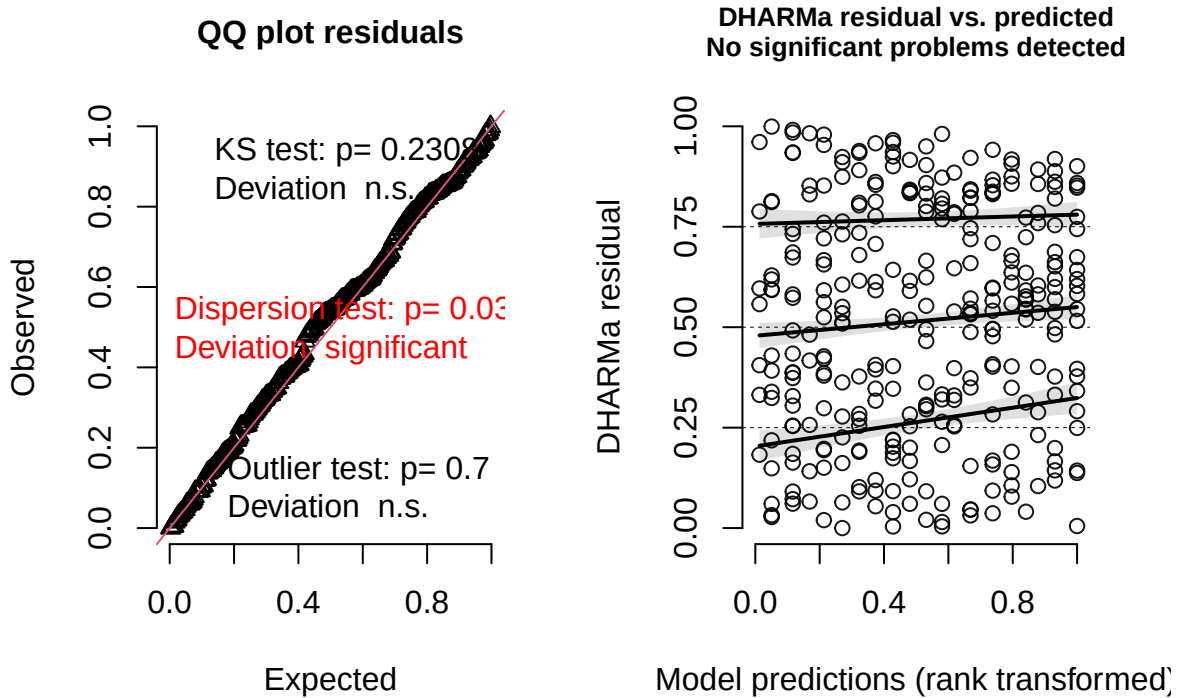
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics suggest that perhaps a negative binomial distribution should have been used for the response. Because of this, I will disregard these models.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
315	1	-159.0838	362.1676	444.7242	145.2940	293	0.832	Binomial	BirdID+Terr
315	1	-160.2965	362.5930	441.3970	190.4288	294	0.768	Binomial	BirdID
315	1	-171.5123	385.0246	463.8286	277.8025	294	0.603	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.9511535	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	0.9333263	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	2.0918108	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.1411091	Binomial	BirdID+Terr

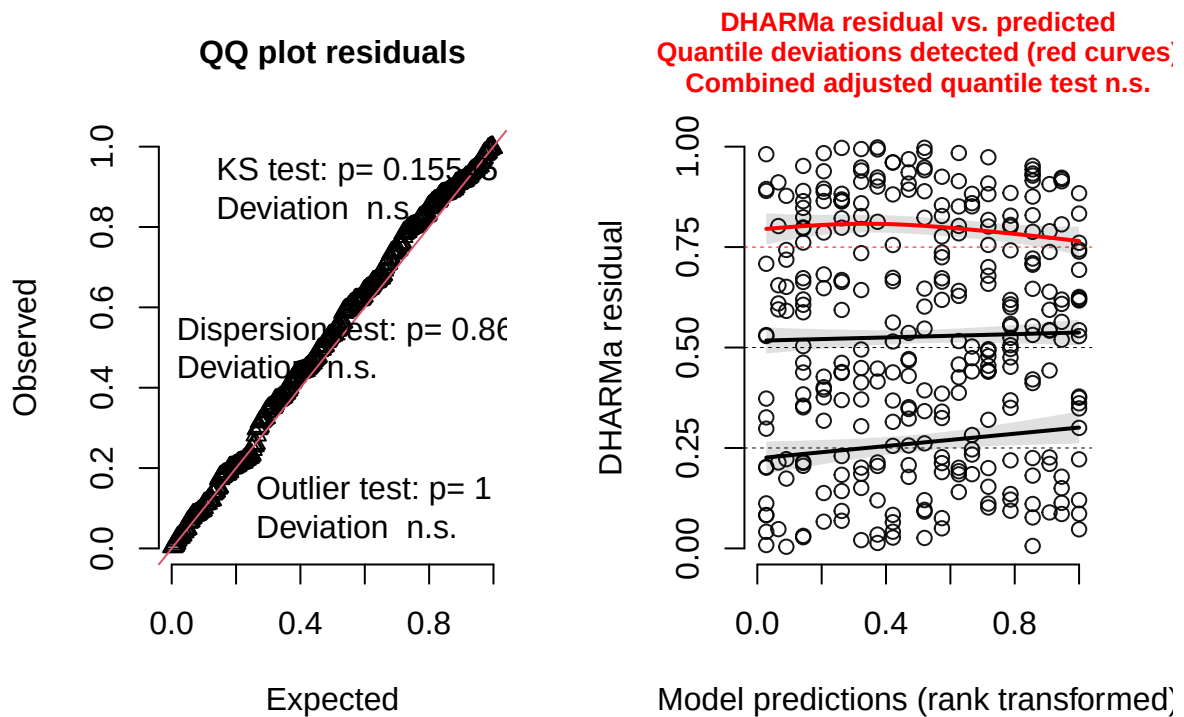
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	3.0160570	1.334906	2.2593775	0.0238599	Binomial	BirdID
fixed	factor(fac.yr)2	0.1860939	1.866461	0.0997042	0.9205792	Binomial	BirdID
fixed	factor(fac.yr)3	-2.0289778	1.463900	-1.3860088	0.1657442	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)4	-1.7502781	1.484187	-1.1792838	0.2382852	Binomial	BirdID
fixed	factor(fac.yr)5	-0.3909568	1.589139	-0.2460179	0.8056683	Binomial	BirdID
fixed	factor(fac.yr)6	-3.2720964	1.477731	-2.2142707	0.0268102	Binomial	BirdID
fixed	factor(fac.yr)7	-3.7987606	1.451844	-2.6165079	0.0088834	Binomial	BirdID
fixed	factor(fac.yr)8	-1.3760380	1.528024	-0.9005340	0.3678361	Binomial	BirdID
fixed	factor(fac.yr)9	0.8853623	1.978509	0.4474897	0.6545216	Binomial	BirdID
fixed	factor(fac.yr)10	-1.0134734	1.558780	-0.6501711	0.5155817	Binomial	BirdID
fixed	factor(fac.yr)11	-3.5451498	1.477655	-2.3991725	0.0164322	Binomial	BirdID
fixed	factor(fac.yr)12	-3.6299003	1.496209	-2.4260643	0.0152636	Binomial	BirdID
fixed	factor(fac.yr)13	-1.7409377	1.595669	-1.0910394	0.2752555	Binomial	BirdID
fixed	factor(fac.yr)14	-2.9730064	1.531986	-1.9406219	0.0523042	Binomial	BirdID
fixed	factor(fac.yr)15	-5.1215892	1.599127	-3.2027411	0.0013613	Binomial	BirdID
fixed	factor(fac.yr)16	-2.4002332	1.558005	-1.5405809	0.1234188	Binomial	BirdID
fixed	factor(fac.yr)17	-3.0213941	1.568947	-1.9257465	0.0541360	Binomial	BirdID
fixed	factor(fac.yr)18	-1.5638152	1.686859	-0.9270575	0.3538967	Binomial	BirdID
fixed	factor(fac.yr)19	-4.1762131	1.663576	-2.5103830	0.0120600	Binomial	BirdID
fixed	factor(fac.yr)20	-4.8518811	1.794725	-2.7034127	0.0068631	Binomial	BirdID
fixed	(Intercept)	2.5802944	1.152397	2.2390677	0.0251515	Binomial	Terr
fixed	factor(fac.yr)2	0.2002985	1.647839	0.1215522	0.9032537	Binomial	Terr
fixed	factor(fac.yr)3	-2.0709231	1.282550	-1.6146923	0.1063774	Binomial	Terr
fixed	factor(fac.yr)4	-1.5434438	1.288243	-1.1980999	0.2308781	Binomial	Terr
fixed	factor(fac.yr)5	-0.5629415	1.345052	-0.4185278	0.6755613	Binomial	Terr
fixed	factor(fac.yr)6	-2.5875147	1.263705	-2.0475627	0.0406029	Binomial	Terr
fixed	factor(fac.yr)7	-3.0553633	1.238529	-2.4669288	0.0136277	Binomial	Terr
fixed	factor(fac.yr)8	-0.8991440	1.307906	-0.6874683	0.4917876	Binomial	Terr
fixed	factor(fac.yr)9	0.7715993	1.644504	0.4691987	0.6389276	Binomial	Terr
fixed	factor(fac.yr)10	-0.9707803	1.305891	-0.7433853	0.4572484	Binomial	Terr
fixed	factor(fac.yr)11	-2.3017489	1.259915	-1.8269077	0.0677137	Binomial	Terr
fixed	factor(fac.yr)12	-2.4793116	1.282819	-1.9327050	0.0532725	Binomial	Terr
fixed	factor(fac.yr)13	-1.2392123	1.326133	-0.9344554	0.3500691	Binomial	Terr
fixed	factor(fac.yr)14	-2.3138818	1.270231	-1.8216235	0.0685121	Binomial	Terr
fixed	factor(fac.yr)15	-3.9594227	1.349163	-2.9347260	0.0033384	Binomial	Terr
fixed	factor(fac.yr)16	-1.9446714	1.292042	-1.5051147	0.1322946	Binomial	Terr
fixed	factor(fac.yr)17	-2.5428552	1.303708	-1.9504789	0.0511191	Binomial	Terr
fixed	factor(fac.yr)18	-1.3614194	1.375685	-0.9896305	0.3223548	Binomial	Terr
fixed	factor(fac.yr)19	-2.7762347	1.405826	-1.9748062	0.0482901	Binomial	Terr
fixed	factor(fac.yr)20	-3.2962067	1.475822	-2.2334713	0.0255179	Binomial	Terr
fixed	(Intercept)	3.5887949	1.701076	2.1097208	0.0348824	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	0.0049859	2.248203	0.0022177	0.9982305	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-2.6096751	1.828380	-1.4273155	0.1534890	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-2.3034994	1.860155	-1.2383375	0.2155910	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-0.5384843	1.964689	-0.2740812	0.7840222	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-3.8674852	1.850858	-2.0895640	0.0366570	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-4.5892725	1.820858	-2.5203908	0.0117225	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-1.6853503	1.894985	-0.8893741	0.3738020	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	0.8280396	2.365657	0.3500253	0.7263197	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-1.1907622	1.950685	-0.6104328	0.5415752	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-4.0996009	1.850179	-2.2157862	0.0267061	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-4.3478513	1.860468	-2.3369663	0.0194409	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-2.0067697	1.959440	-1.0241549	0.3057621	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-3.5020041	1.901076	-1.8421171	0.0654580	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-5.7771990	1.969260	-2.9336907	0.0033496	Binomial	BirdID+Terr

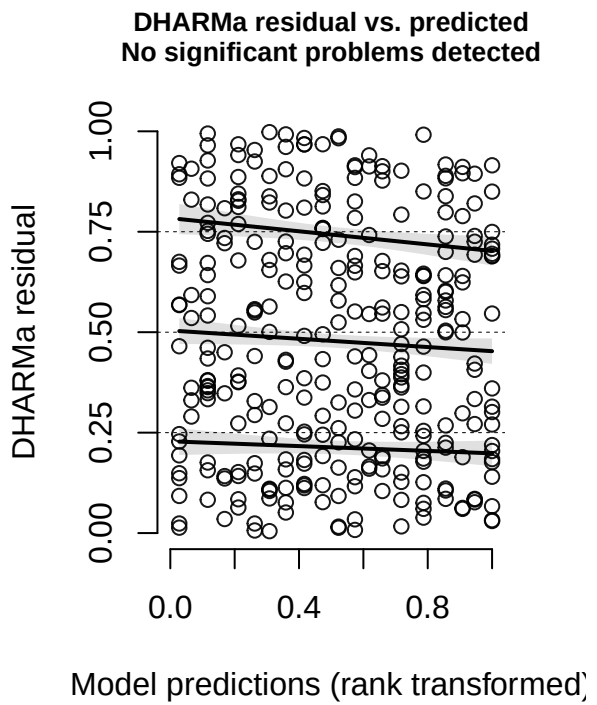
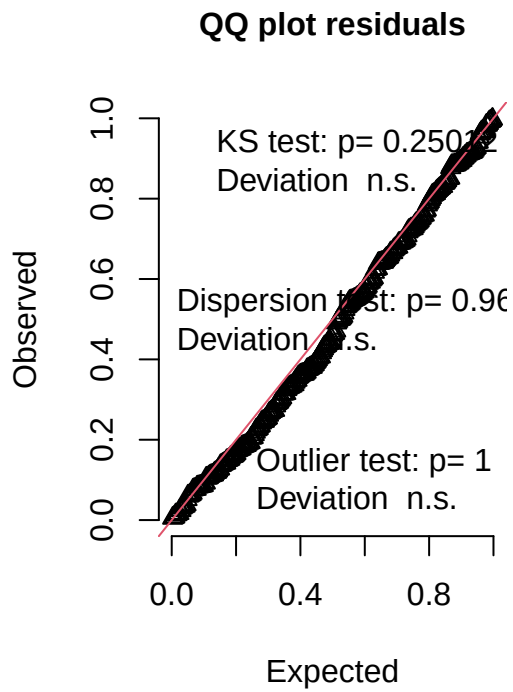
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)16	-2.7083802	1.923789	-1.4078361	0.1591796	Binomial	BirdID+Terr
fixed	factor(fac.yr)17	-3.3511161	1.953634	-1.7153242	0.0862858	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-1.9656603	2.056162	-0.9559854	0.3390796	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-4.4518400	2.038491	-2.1838896	0.0289704	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-5.4381663	2.203330	-2.4681582	0.0135810	Binomial	BirdID+Terr

Similar to the previous batch of binomial models.

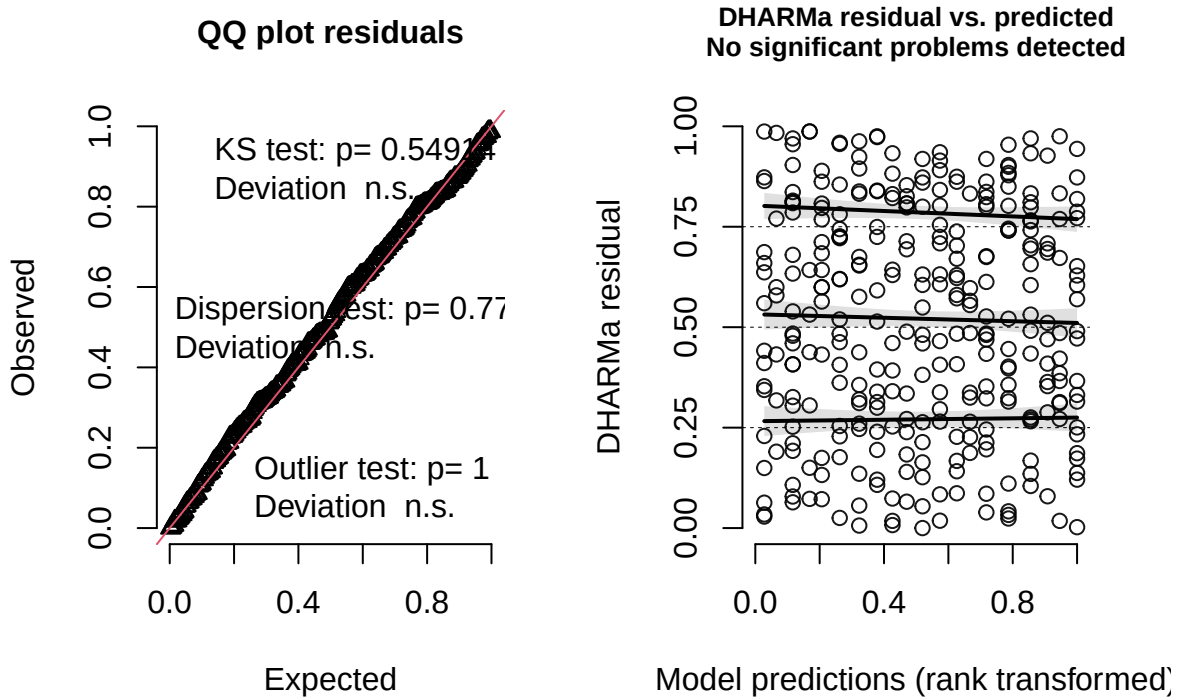
DHARMa residual



DHARMa residual



DHARMa residual



Mild concern coming from the band ID only model.

Probabilities used to generate simulation 30

Simulation 30
 Probability of mate fidelity: 0.693
 Probability of pair dispersal: 0.433
 Probability of single male dispersal: 0.076
 Probability of single female dispersal: 0.009

Simulation 31

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
344	1	- 411.7287	867.4574	951.9516	243.2245	322	0.4566654	0.1676806	Poisson	BirdID+Terr
344	1	- 415.0705	872.1410	952.7944	290.1512	323	0.4085458	0.1809608	Poisson	BirdID
344	1	- 417.2902	876.5803	957.2338	302.3452	323	0.3864540	0.1693632	Poisson	Terr

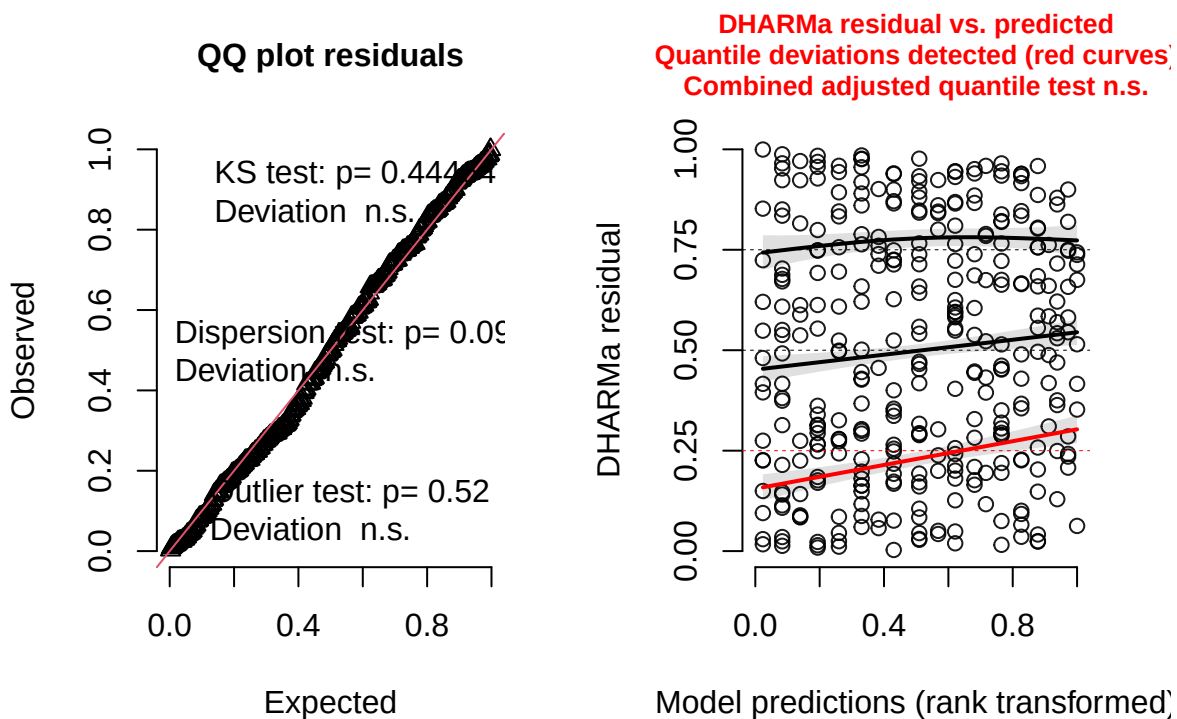
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.5239380	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.4923688	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.4604485	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.3974281	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.6599652	0.3082946	2.1406967	0.0322985	Poisson	BirdID
fixed	factor(fac.yr)2	-0.1989963	0.3738127	-0.5323422	0.5944890	Poisson	BirdID
fixed	factor(fac.yr)3	-1.1812181	0.4185848	-2.8219325	0.0047735	Poisson	BirdID
fixed	factor(fac.yr)4	-0.4338983	0.3864629	-1.1227424	0.2615469	Poisson	BirdID
fixed	factor(fac.yr)5	-0.0757132	0.3974879	-0.1904793	0.8489335	Poisson	BirdID
fixed	factor(fac.yr)6	-0.7295654	0.3811301	-1.9142165	0.0555925	Poisson	BirdID
fixed	factor(fac.yr)7	-0.8372515	0.3780630	-2.2145820	0.0267888	Poisson	BirdID
fixed	factor(fac.yr)8	-0.9268103	0.3751939	-2.4702168	0.0135031	Poisson	BirdID
fixed	factor(fac.yr)9	-1.1758981	0.3900919	-3.0144132	0.0025748	Poisson	BirdID
fixed	factor(fac.yr)10	-1.4011853	0.4221729	-3.3189848	0.0009035	Poisson	BirdID
fixed	factor(fac.yr)11	-1.5738535	0.4365224	-3.6054355	0.0003116	Poisson	BirdID
fixed	factor(fac.yr)12	-1.0598307	0.4025211	-2.6329819	0.0084639	Poisson	BirdID
fixed	factor(fac.yr)13	-0.6938170	0.3981293	-1.7426926	0.0813873	Poisson	BirdID
fixed	factor(fac.yr)14	-0.7924084	0.4108991	-1.9284742	0.0537962	Poisson	BirdID
fixed	factor(fac.yr)15	-2.1088825	0.6083181	-3.4667428	0.0005268	Poisson	BirdID
fixed	factor(fac.yr)16	-1.4714213	0.5353231	-2.7486603	0.0059839	Poisson	BirdID
fixed	factor(fac.yr)17	-0.8477283	0.4695532	-1.8053936	0.0710131	Poisson	BirdID
fixed	factor(fac.yr)18	-0.7672974	0.4715991	-1.6270123	0.1037345	Poisson	BirdID
fixed	factor(fac.yr)19	-1.1379251	0.5399863	-2.1073221	0.0350897	Poisson	BirdID
fixed	factor(fac.yr)20	-0.2438175	0.4808041	-0.5071037	0.6120821	Poisson	BirdID
fixed	(Intercept)	0.6734383	0.2919190	2.3069359	0.0210584	Poisson	Terr
fixed	factor(fac.yr)2	-0.2157585	0.3788051	-0.5695764	0.5689650	Poisson	Terr
fixed	factor(fac.yr)3	-1.3002307	0.4073579	-3.1918633	0.0014136	Poisson	Terr
fixed	factor(fac.yr)4	-0.5840535	0.3619643	-1.6135666	0.1066215	Poisson	Terr
fixed	factor(fac.yr)5	-0.1668225	0.3804701	-0.4384641	0.6610499	Poisson	Terr
fixed	factor(fac.yr)6	-0.7041728	0.3627620	-1.9411426	0.0522410	Poisson	Terr
fixed	factor(fac.yr)7	-0.7532799	0.3559954	-2.1159821	0.0343463	Poisson	Terr
fixed	factor(fac.yr)8	-0.8263538	0.3534890	-2.3377071	0.0194024	Poisson	Terr
fixed	factor(fac.yr)9	-0.9865576	0.3677420	-2.6827437	0.0073021	Poisson	Terr
fixed	factor(fac.yr)10	-1.3123075	0.4003545	-3.2778640	0.0010460	Poisson	Terr
fixed	factor(fac.yr)11	-1.4313508	0.4204197	-3.4045764	0.0006627	Poisson	Terr
fixed	factor(fac.yr)12	-0.9445161	0.3734407	-2.5292265	0.0114314	Poisson	Terr
fixed	factor(fac.yr)13	-0.6245717	0.3805036	-1.6414347	0.1007072	Poisson	Terr
fixed	factor(fac.yr)14	-0.5313436	0.3962114	-1.3410610	0.1799006	Poisson	Terr
fixed	factor(fac.yr)15	-1.9879414	0.5934253	-3.3499435	0.0008083	Poisson	Terr
fixed	factor(fac.yr)16	-1.3264819	0.5204691	-2.5486276	0.0108148	Poisson	Terr
fixed	factor(fac.yr)17	-0.7518181	0.4535567	-1.6576055	0.0973971	Poisson	Terr
fixed	factor(fac.yr)18	-0.6176173	0.4546718	-1.3583806	0.1743429	Poisson	Terr
fixed	factor(fac.yr)19	-0.9847064	0.5302774	-1.8569647	0.0633162	Poisson	Terr
fixed	factor(fac.yr)20	-0.0695225	0.4537109	-0.1532308	0.8782163	Poisson	Terr
fixed	(Intercept)	0.6657714	0.3243962	2.0523404	0.0401366	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.2419599	0.3992885	-0.6059775	0.5445297	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-1.2832148	0.4339654	-2.9569518	0.0031070	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.5327630	0.4011045	-1.3282398	0.1840989	Poisson	BirdID+Terr

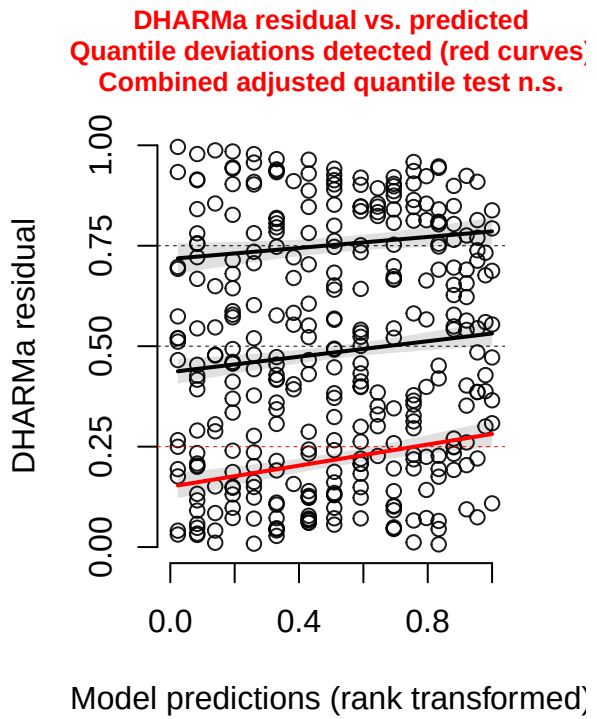
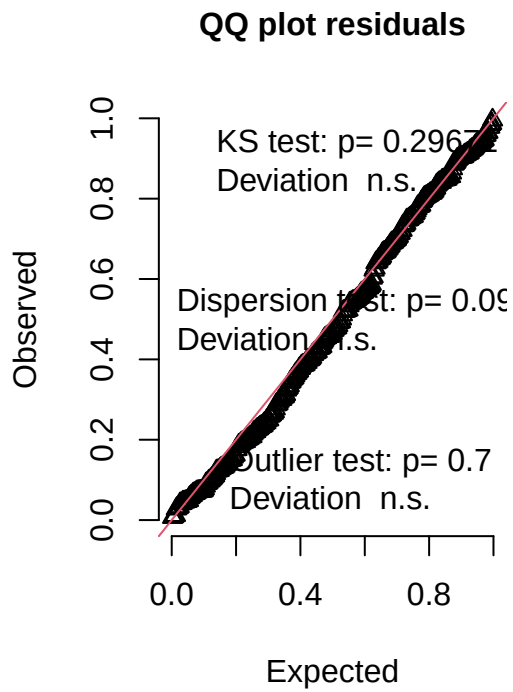
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.1754577	0.4194592	-0.4182951	0.6757313	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.7467851	0.3966491	-1.8827349	0.0597363	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.8512777	0.3921125	-2.1710038	0.0299309	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.9331395	0.3889299	-2.3992483	0.0164288	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-1.1619988	0.4037399	-2.8780873	0.0040009	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-1.4427502	0.4364464	-3.3056755	0.0009475	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-1.5542299	0.4523722	-3.4357327	0.0005910	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-1.0443923	0.4110460	-2.5408163	0.0110594	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.7137816	0.4166469	-1.7131573	0.0866836	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.7713694	0.4327207	-1.7826033	0.0746509	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-2.1458662	0.6276885	-3.4186801	0.0006293	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-1.4459593	0.5544669	-2.6078369	0.0091116	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.8496772	0.4919014	-1.7273323	0.0841080	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.7266628	0.4935327	-1.4723703	0.1409209	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.0848695	0.5643188	-1.9224408	0.0545503	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.1080121	0.5015972	-0.2153363	0.8295051	Poisson	BirdID+Terr

the territory only model is not the best on any criteria.

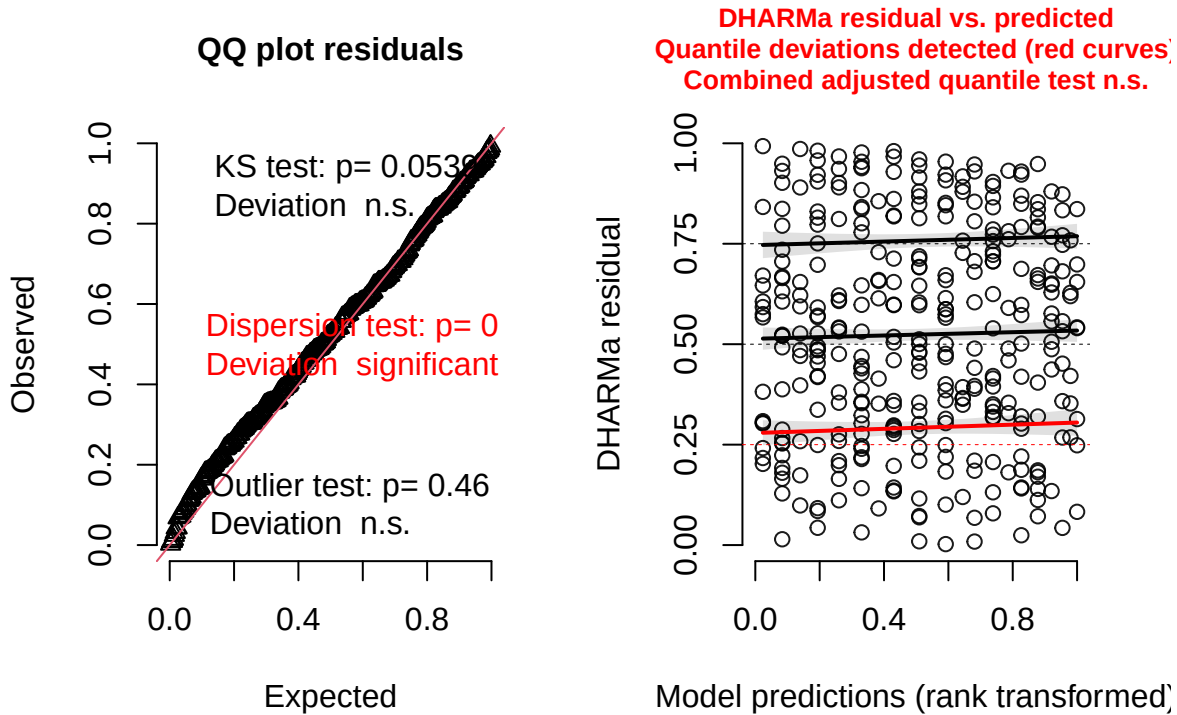
DHARMa residual



DHARMa residual



DHARMa residual



I would not trust these models.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
344	1	-185.5148	415.0296	499.5237	141.5905	322	0.901	Binomial	BirdID+Terr
344	1	-190.0297	422.0594	502.7129	241.7957	323	0.749	Binomial	BirdID
344	1	-199.5078	441.0155	521.6690	296.9708	323	0.662	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.763558	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.230017	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	2.352458	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.746597	Binomial	BirdID+Terr

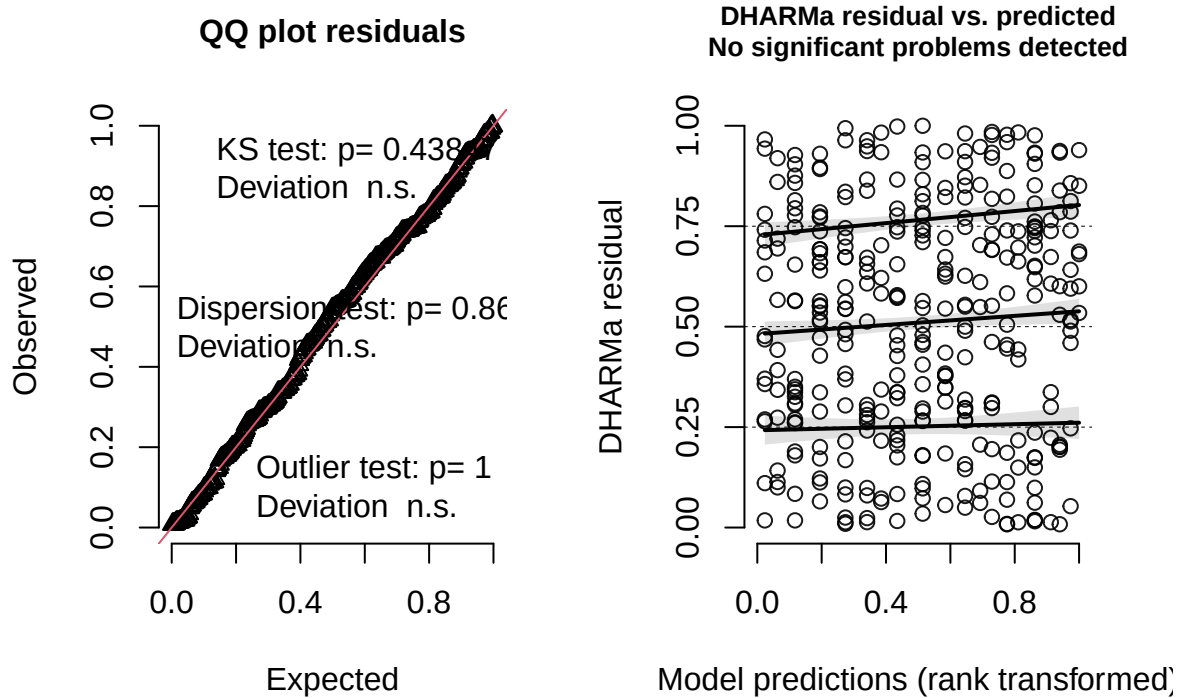
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	2.4812548	1.346011	1.8434139	0.0652686	Binomial	BirdID
fixed	factor(fac.yr)2	0.6945505	1.803470	0.3851190	0.7001493	Binomial	BirdID
fixed	factor(fac.yr)3	-3.1611057	1.435203	-2.2025492	0.0276265	Binomial	BirdID
fixed	factor(fac.yr)4	-1.5249900	1.467299	-1.0393176	0.2986571	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.3229990	1.849876	0.1746057	0.8613895	Binomial	BirdID
fixed	factor(fac.yr)6	-1.6438159	1.456468	-1.1286319	0.2590531	Binomial	BirdID
fixed	factor(fac.yr)7	-1.4023329	1.457179	-0.9623615	0.3358681	Binomial	BirdID
fixed	factor(fac.yr)8	-2.0169394	1.436737	-1.4038330	0.1603686	Binomial	BirdID
fixed	factor(fac.yr)9	-3.3362926	1.446792	-2.3059928	0.0211110	Binomial	BirdID
fixed	factor(fac.yr)10	-3.3703293	1.462952	-2.3037871	0.0212346	Binomial	BirdID
fixed	factor(fac.yr)11	-3.3076792	1.458493	-2.2678742	0.0233369	Binomial	BirdID
fixed	factor(fac.yr)12	-2.4031966	1.456403	-1.6500901	0.0989245	Binomial	BirdID
fixed	factor(fac.yr)13	-1.9902066	1.493075	-1.3329584	0.1825454	Binomial	BirdID
fixed	factor(fac.yr)14	-1.5235842	1.547025	-0.9848481	0.3246987	Binomial	BirdID
fixed	factor(fac.yr)15	-5.1619003	1.699831	-3.0367136	0.0023917	Binomial	BirdID
fixed	factor(fac.yr)16	-3.9612902	1.630371	-2.4296867	0.0151119	Binomial	BirdID
fixed	factor(fac.yr)17	-2.5479518	1.595557	-1.5969040	0.1102871	Binomial	BirdID
fixed	factor(fac.yr)18	-1.4906026	1.658966	-0.8985133	0.3689120	Binomial	BirdID
fixed	factor(fac.yr)19	-1.6013663	1.667143	-0.9605450	0.3367810	Binomial	BirdID
fixed	factor(fac.yr)20	20.7937566	45444.423636	0.0004576	0.9996349	Binomial	BirdID
fixed	(Intercept)	2.5797245	1.205279	2.1403548	0.0323261	Binomial	Terr
fixed	factor(fac.yr)2	0.3371833	1.701450	0.1981740	0.8429089	Binomial	Terr
fixed	factor(fac.yr)3	-3.2491731	1.305393	-2.4890385	0.0128089	Binomial	Terr
fixed	factor(fac.yr)4	-1.9858535	1.333454	-1.4892548	0.1364203	Binomial	Terr
fixed	factor(fac.yr)5	0.1089454	1.713922	0.0635650	0.9493166	Binomial	Terr
fixed	factor(fac.yr)6	-2.0620725	1.285727	-1.6038186	0.1087540	Binomial	Terr
fixed	factor(fac.yr)7	-1.7098106	1.283147	-1.3325137	0.1826915	Binomial	Terr
fixed	factor(fac.yr)8	-1.9949012	1.259591	-1.5837684	0.1132464	Binomial	Terr
fixed	factor(fac.yr)9	-2.8417206	1.275285	-2.2283027	0.0258603	Binomial	Terr
fixed	factor(fac.yr)10	-3.0298159	1.287662	-2.3529590	0.0186247	Binomial	Terr
fixed	factor(fac.yr)11	-2.8932336	1.293945	-2.2359788	0.0253532	Binomial	Terr
fixed	factor(fac.yr)12	-2.3521997	1.283518	-1.8326190	0.0668593	Binomial	Terr
fixed	factor(fac.yr)13	-1.9981199	1.326798	-1.5059718	0.1320744	Binomial	Terr
fixed	factor(fac.yr)14	-1.1737542	1.366712	-0.8588160	0.3904420	Binomial	Terr
fixed	factor(fac.yr)15	-4.5300077	1.494102	-3.0319270	0.0024300	Binomial	Terr
fixed	factor(fac.yr)16	-3.5684781	1.427060	-2.5005811	0.0123990	Binomial	Terr
fixed	factor(fac.yr)17	-2.3191610	1.429111	-1.6227998	0.1046322	Binomial	Terr
fixed	factor(fac.yr)18	-1.3638265	1.481127	-0.9208029	0.3571533	Binomial	Terr
fixed	factor(fac.yr)19	-1.4946052	1.492576	-1.0013597	0.3166529	Binomial	Terr
fixed	factor(fac.yr)20	16.4928187	3535.927167	0.0046644	0.9962784	Binomial	Terr
fixed	(Intercept)	3.5002945	1.655178	2.1147541	0.0344509	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	0.3756645	2.087838	0.1799299	0.8572076	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-4.6305581	1.748038	-2.6490031	0.0080730	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-2.2583403	1.833095	-1.2319823	0.2179557	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	0.5878873	2.439932	0.2409442	0.8095984	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-2.2119798	1.805699	-1.2249993	0.2205755	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-1.8105027	1.796207	-1.0079590	0.3134742	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-2.4775164	1.747693	-1.4175924	0.1563098	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-4.2141642	1.771410	-2.3789885	0.0173602	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-4.3603295	1.795545	-2.4284157	0.0151650	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-3.8837320	1.804833	-2.1518513	0.0314091	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-2.9558529	1.785853	-1.6551491	0.0978943	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-2.6635928	1.868594	-1.4254526	0.1540264	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-1.4765419	1.927427	-0.7660690	0.4436352	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-6.8640278	2.194966	-3.1271679	0.0017650	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-5.3429940	2.051112	-2.6049261	0.0091894	Binomial	BirdID+Terr

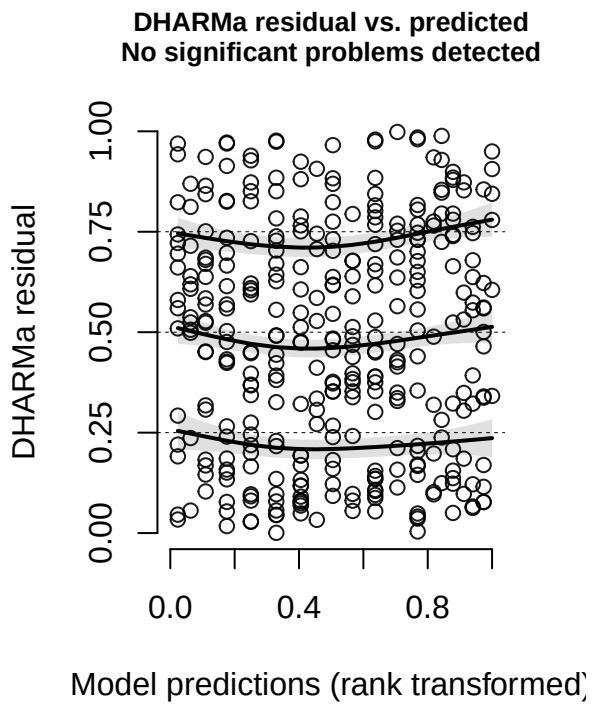
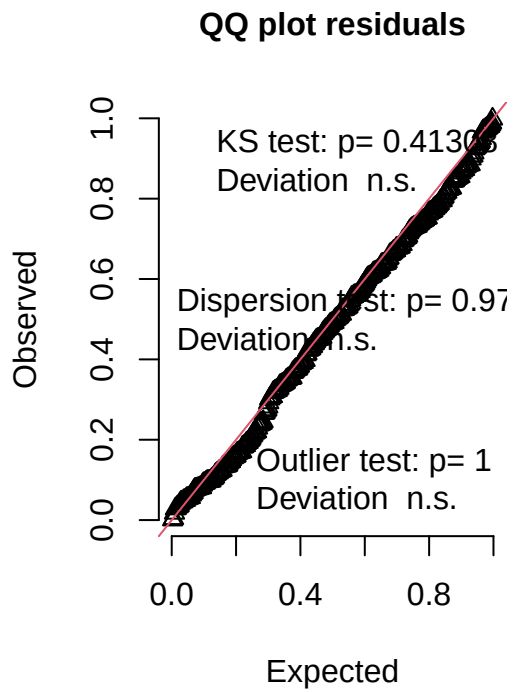
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-3.4605653	2.032659	-1.7024818	0.0886651	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-1.7254320	2.122288	-0.8130056	0.4162149	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-2.0224326	2.109947	-0.9585228	0.3377992	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	15.4685726	2419.387551	0.0063936	0.9948987	Binomial	BirdID+Terr

Similar to the previous batch of binomial models.

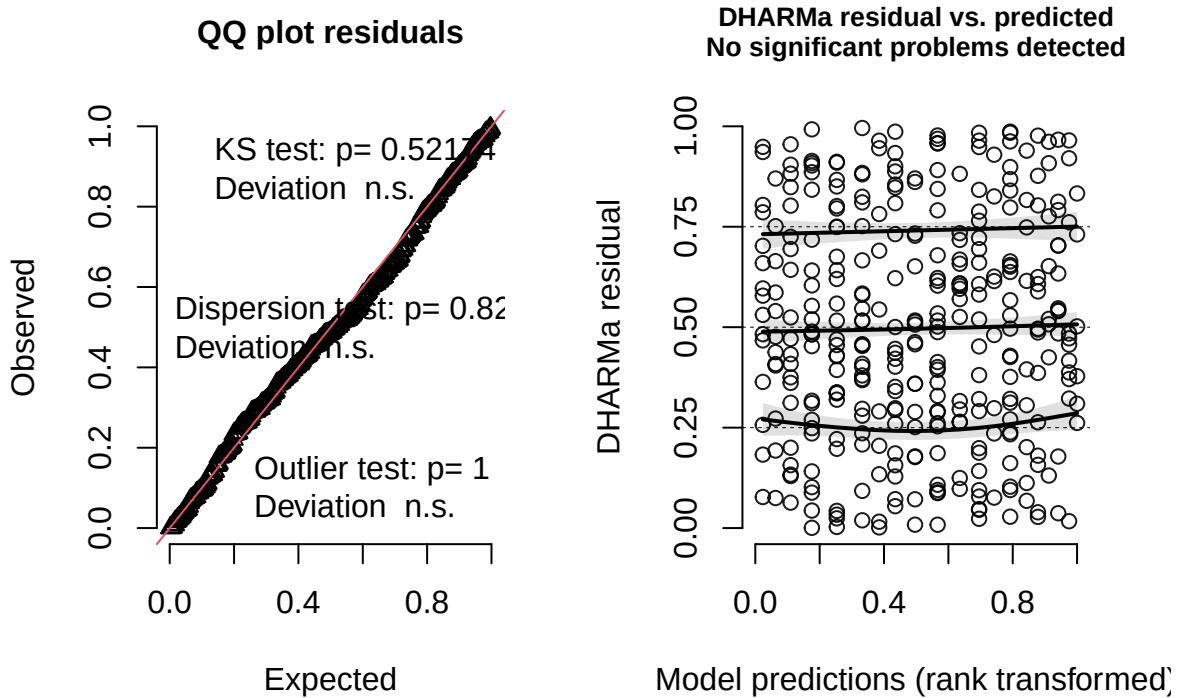
DHARMa residual



DHARMa residual



DHARMA residual



Diagnostics look good.

Probabilities used to generate simulation 31

Simulation 31
 Probability of mate fidelity: 0.912
 Probability of pair dispersal: 0.618
 Probability of single male dispersal: 0.542
 Probability of single female dispersal: 0.893

Simulation 32

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
319	1	-391.1253	824.2506	903.3196	300.9555	298	0.8880899	0.8608370	Poisson	Terr
319	1	-391.0613	826.1227	908.9569	293.2507	297	0.8933020	0.8657136	Poisson	BirdID+Terr
319	1	-394.2216	830.4433	909.5123	312.3365	298	0.8954263	0.8727497	Poisson	BirdID

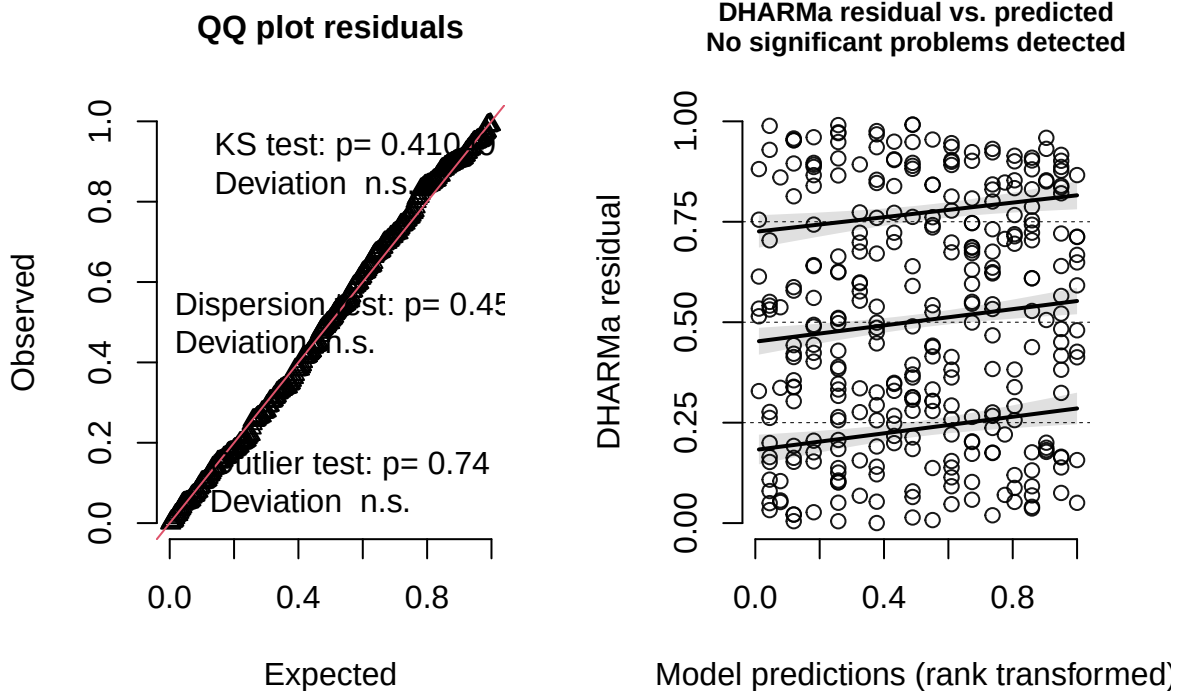
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.3876407	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.4091594	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.1597371	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.3917721	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.7221736	0.2323356	3.1083213	0.0018815	Poisson	BirdID
fixed	factor(fac.yr)2	-0.9150771	0.3208891	-2.8516932	0.0043487	Poisson	BirdID
fixed	factor(fac.yr)3	-1.5434353	0.4117569	-3.7484137	0.0001780	Poisson	BirdID
fixed	factor(fac.yr)4	-0.5379380	0.3128148	-1.7196694	0.0854925	Poisson	BirdID
fixed	factor(fac.yr)5	-0.2181254	0.2969247	-0.7346153	0.4625738	Poisson	BirdID
fixed	factor(fac.yr)6	-1.2104103	0.3360506	-3.6018687	0.0003159	Poisson	BirdID
fixed	factor(fac.yr)7	-1.6039428	0.4351908	-3.6856081	0.0002282	Poisson	BirdID
fixed	factor(fac.yr)8	-0.5419650	0.3178938	-1.7048619	0.0882202	Poisson	BirdID
fixed	factor(fac.yr)9	-0.4246970	0.3112390	-1.3645366	0.1723987	Poisson	BirdID
fixed	factor(fac.yr)10	-0.7016268	0.3268345	-2.1467341	0.0318145	Poisson	BirdID
fixed	factor(fac.yr)11	-1.1130033	0.3659982	-3.0410075	0.0023579	Poisson	BirdID
fixed	factor(fac.yr)12	-0.8804657	0.3602136	-2.4442876	0.0145139	Poisson	BirdID
fixed	factor(fac.yr)13	-0.4880839	0.3298343	-1.4797853	0.1389306	Poisson	BirdID
fixed	factor(fac.yr)14	-0.9300392	0.3767158	-2.4688087	0.0135564	Poisson	BirdID
fixed	factor(fac.yr)15	-1.1501834	0.4080848	-2.8184909	0.0048250	Poisson	BirdID
fixed	factor(fac.yr)16	-2.8690510	0.7620936	-3.7646963	0.0001668	Poisson	BirdID
fixed	factor(fac.yr)17	-0.2961784	0.3699337	-0.8006256	0.4233484	Poisson	BirdID
fixed	factor(fac.yr)18	-2.4634503	1.0623806	-2.3188021	0.0204058	Poisson	BirdID
fixed	factor(fac.yr)19	-18.0005340	2367.8149506	-0.0076022	0.9939344	Poisson	BirdID
fixed	factor(fac.yr)20	-0.4972322	0.5482580	-0.9069311	0.3644432	Poisson	BirdID
fixed	(Intercept)	0.8342302	0.2319886	3.5959958	0.0003232	Poisson	Terr
fixed	factor(fac.yr)2	-0.9500227	0.3321138	-2.8605332	0.0042293	Poisson	Terr
fixed	factor(fac.yr)3	-1.5920342	0.4165451	-3.8219976	0.0001324	Poisson	Terr
fixed	factor(fac.yr)4	-0.6969966	0.3146213	-2.2153512	0.0267360	Poisson	Terr
fixed	factor(fac.yr)5	-0.3520177	0.2908382	-1.2103557	0.2261424	Poisson	Terr
fixed	factor(fac.yr)6	-1.3836522	0.3342460	-4.1396227	0.0000348	Poisson	Terr
fixed	factor(fac.yr)7	-1.8205144	0.4342034	-4.1927687	0.0000276	Poisson	Terr
fixed	factor(fac.yr)8	-0.7022605	0.3147043	-2.2314930	0.0256485	Poisson	Terr
fixed	factor(fac.yr)9	-0.5282496	0.3085361	-1.7121159	0.0868753	Poisson	Terr
fixed	factor(fac.yr)10	-0.8462104	0.3246424	-2.6065924	0.0091448	Poisson	Terr
fixed	factor(fac.yr)11	-1.2718140	0.3560819	-3.5716896	0.0003547	Poisson	Terr
fixed	factor(fac.yr)12	-0.9835581	0.3561811	-2.7613986	0.0057554	Poisson	Terr
fixed	factor(fac.yr)13	-0.5087306	0.3268787	-1.5563281	0.1196301	Poisson	Terr
fixed	factor(fac.yr)14	-0.9657324	0.3764265	-2.5655270	0.0103019	Poisson	Terr
fixed	factor(fac.yr)15	-1.2100608	0.4021537	-3.0089509	0.0026215	Poisson	Terr
fixed	factor(fac.yr)16	-2.8492390	0.7628095	-3.7351906	0.0001876	Poisson	Terr
fixed	factor(fac.yr)17	-0.4004232	0.3704946	-1.0807803	0.2797949	Poisson	Terr
fixed	factor(fac.yr)18	-2.6989906	1.0563355	-2.5550505	0.0106172	Poisson	Terr
fixed	factor(fac.yr)19	-17.3128716	1538.9345521	-0.0112499	0.9910241	Poisson	Terr
fixed	factor(fac.yr)20	-0.8674630	0.5312663	-1.6328215	0.1025065	Poisson	Terr
fixed	(Intercept)	0.8198099	0.2351883	3.4857591	0.0004907	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.9583590	0.3337618	-2.8713860	0.0040868	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-1.5880743	0.4189571	-3.7905415	0.0001503	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.6871285	0.3185022	-2.1573743	0.0309765	Poisson	BirdID+Terr

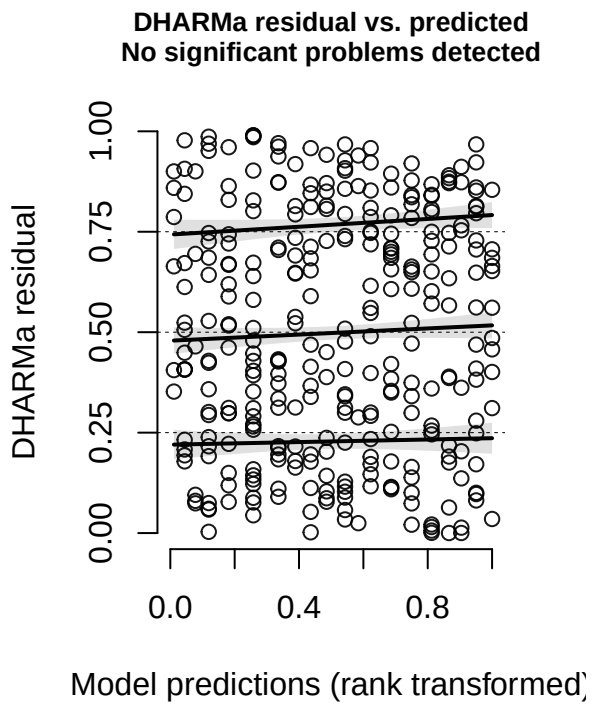
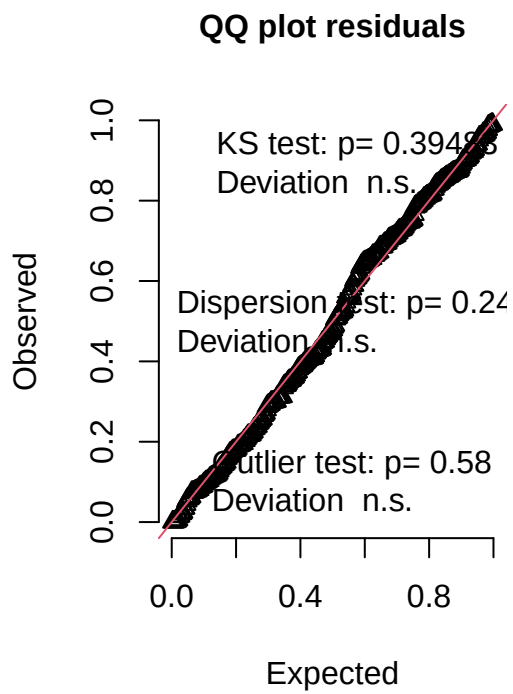
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.3473784	0.2957711	-1.1744842	0.2402011	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-1.3720660	0.3377216	-4.0627130	0.0000485	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-1.7928912	0.4375465	-4.0976014	0.0000417	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.6842301	0.3188963	-2.1456193	0.0319034	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.5234726	0.3126830	-1.6741319	0.0941047	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.8374170	0.3288520	-2.5464857	0.0108814	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-1.2616633	0.3606543	-3.4982624	0.0004683	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.9774365	0.3605915	-2.7106479	0.0067152	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.5150868	0.3313454	-1.5545313	0.1200577	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.9645469	0.3804226	-2.5354620	0.0112299	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-1.2215144	0.4069712	-3.0014761	0.0026867	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-2.8680191	0.7666567	-3.7409434	0.0001833	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.4071320	0.3759869	-1.0828355	0.2788814	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-2.6667820	1.0596244	-2.5167239	0.0118452	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-17.8461616	2032.4596440	-0.0087806	0.9929942	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.8319413	0.5395338	-1.5419633	0.1230825	Poisson	BirdID+Terr

The territory only model is performing the best on AIC and BIC, and still well on the R^2 s.

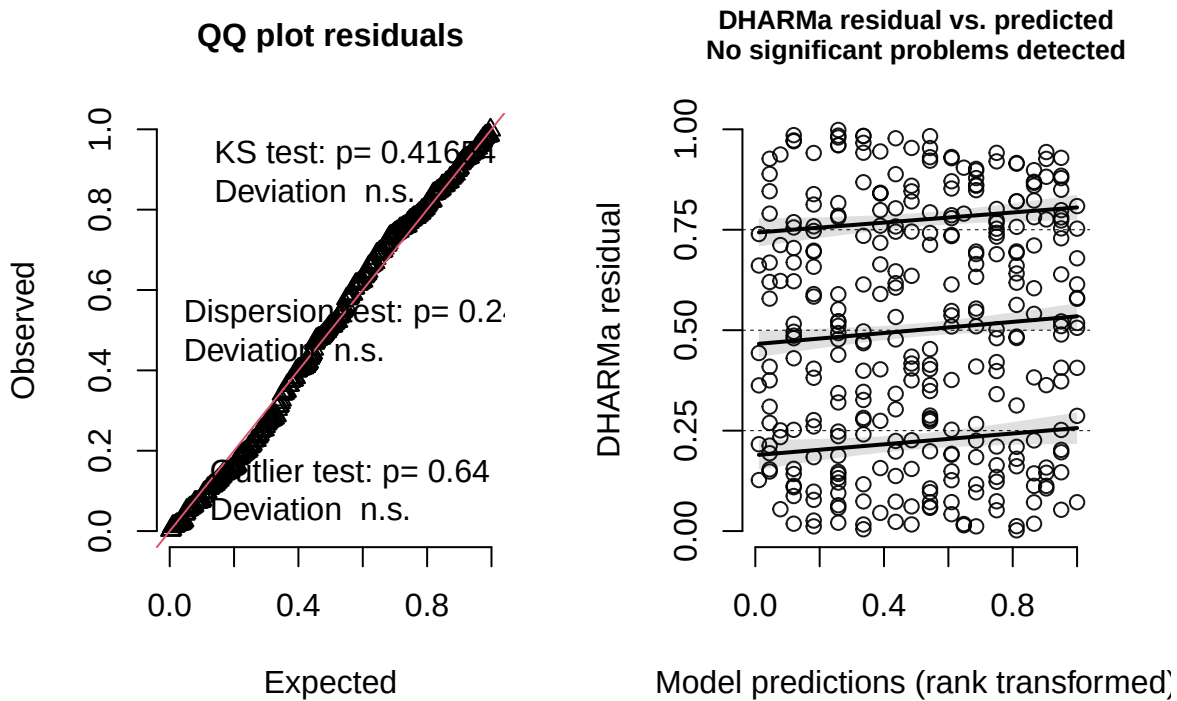
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
319	1	-181.6250	407.2500	490.0842	217.5132	297	0.768	Binomial	BirdID+Terr
319	1	-183.0847	408.1694	487.2384	257.5462	298	0.692	Binomial	BirdID
319	1	-185.4875	412.9751	492.0441	294.7032	298	0.541	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.3547546	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	0.9880499	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.3196221	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.9393147	Binomial	BirdID+Terr

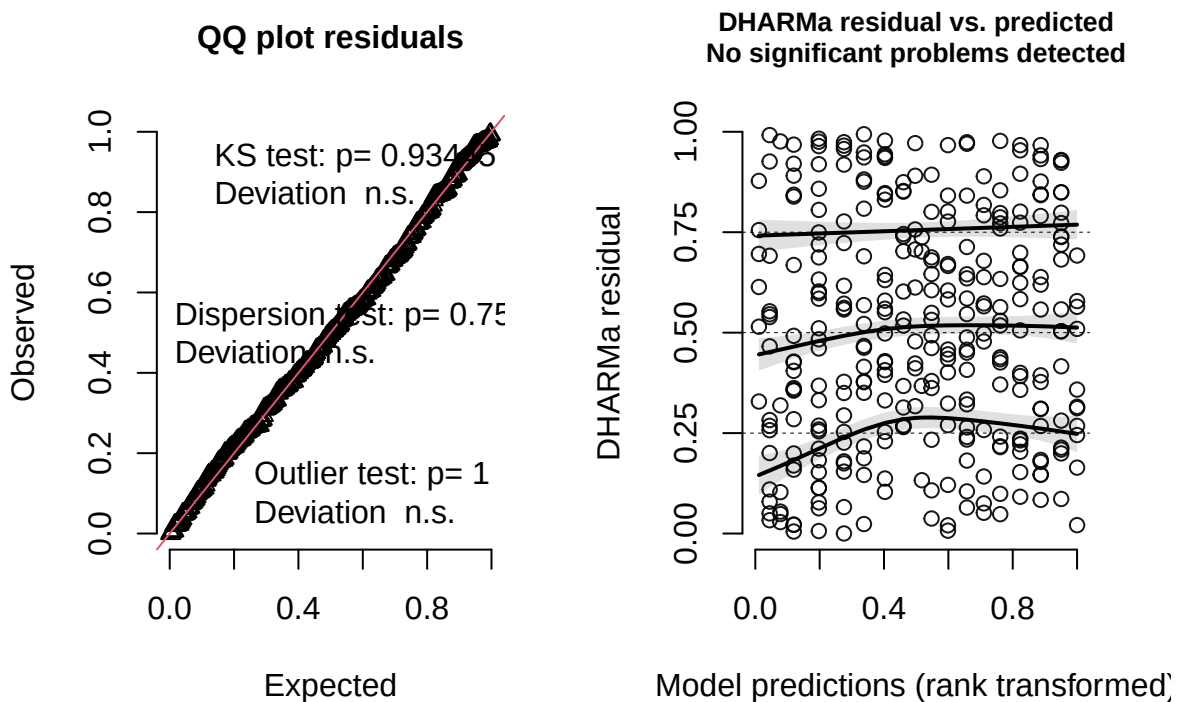
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	2.4121233	1.227976	1.9643074	0.0494944	Binomial	BirdID
fixed	factor(fac.yr)2	-2.8171020	1.309655	-2.1510256	0.0314742	Binomial	BirdID
fixed	factor(fac.yr)3	-2.8501002	1.336340	-2.1327655	0.0329440	Binomial	BirdID
fixed	factor(fac.yr)4	-1.1407359	1.370389	-0.8324178	0.4051732	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.8079908	1.410087	-0.5730080	0.5666393	Binomial	BirdID
fixed	factor(fac.yr)6	-3.3637088	1.318222	-2.5517005	0.0107199	Binomial	BirdID
fixed	factor(fac.yr)7	-3.3822359	1.374625	-2.4604797	0.0138751	Binomial	BirdID
fixed	factor(fac.yr)8	-1.1507669	1.361971	-0.8449274	0.3981514	Binomial	BirdID
fixed	factor(fac.yr)9	-1.4878966	1.363907	-1.0909078	0.2753135	Binomial	BirdID
fixed	factor(fac.yr)10	-1.7857811	1.347941	-1.3248209	0.1852306	Binomial	BirdID
fixed	factor(fac.yr)11	-3.0388698	1.356671	-2.2399459	0.0250944	Binomial	BirdID
fixed	factor(fac.yr)12	-1.8229419	1.370035	-1.3305804	0.1833271	Binomial	BirdID
fixed	factor(fac.yr)13	-1.9097893	1.398741	-1.3653627	0.1721391	Binomial	BirdID
fixed	factor(fac.yr)14	-2.6511353	1.392680	-1.9036211	0.0569595	Binomial	BirdID
fixed	factor(fac.yr)15	-1.6802749	1.437137	-1.1691822	0.2423302	Binomial	BirdID
fixed	factor(fac.yr)16	-5.1350628	1.545213	-3.3232078	0.0008899	Binomial	BirdID
fixed	factor(fac.yr)17	-2.3361014	1.484977	-1.5731565	0.1156826	Binomial	BirdID
fixed	factor(fac.yr)18	-4.3359902	1.854977	-2.3374897	0.0194137	Binomial	BirdID
fixed	factor(fac.yr)19	-11.2682426	29.083866	-0.3874396	0.6984308	Binomial	BirdID
fixed	factor(fac.yr)20	-1.9997258	1.726841	-1.1580256	0.2468536	Binomial	BirdID
fixed	(Intercept)	2.7126856	1.136347	2.3871991	0.0169773	Binomial	Terr
fixed	factor(fac.yr)2	-2.7778316	1.229656	-2.2590314	0.0238814	Binomial	Terr
fixed	factor(fac.yr)3	-2.9641970	1.240309	-2.3898849	0.0168537	Binomial	Terr
fixed	factor(fac.yr)4	-1.5579004	1.259353	-1.2370639	0.2160634	Binomial	Terr
fixed	factor(fac.yr)5	-0.9975817	1.295073	-0.7702899	0.4411280	Binomial	Terr
fixed	factor(fac.yr)6	-3.5794055	1.216036	-2.9435038	0.0032452	Binomial	Terr
fixed	factor(fac.yr)7	-4.0502677	1.272087	-3.1839557	0.0014528	Binomial	Terr
fixed	factor(fac.yr)8	-1.7015628	1.255980	-1.3547692	0.1754911	Binomial	Terr
fixed	factor(fac.yr)9	-1.8178810	1.254114	-1.4495339	0.1471885	Binomial	Terr
fixed	factor(fac.yr)10	-2.2229975	1.241735	-1.7902355	0.0734161	Binomial	Terr
fixed	factor(fac.yr)11	-3.4370872	1.238403	-2.7754189	0.0055131	Binomial	Terr
fixed	factor(fac.yr)12	-2.2073686	1.252698	-1.7620913	0.0780539	Binomial	Terr
fixed	factor(fac.yr)13	-1.7650537	1.285214	-1.3733539	0.1696423	Binomial	Terr
fixed	factor(fac.yr)14	-2.5772041	1.288250	-2.0005468	0.0454413	Binomial	Terr
fixed	factor(fac.yr)15	-1.7109281	1.320188	-1.2959726	0.1949849	Binomial	Terr
fixed	factor(fac.yr)16	-4.8594655	1.430637	-3.3967153	0.0006820	Binomial	Terr
fixed	factor(fac.yr)17	-2.2577151	1.366158	-1.6526015	0.0984120	Binomial	Terr
fixed	factor(fac.yr)18	-4.4995178	1.677720	-2.6819242	0.0073200	Binomial	Terr
fixed	factor(fac.yr)19	-12.2216815	49.643412	-0.2461894	0.8055356	Binomial	Terr
fixed	factor(fac.yr)20	-3.0143051	1.611100	-1.8709606	0.0613505	Binomial	Terr
fixed	(Intercept)	2.7431352	1.306862	2.0990248	0.0358147	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-3.0746509	1.402307	-2.1925669	0.0283386	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-3.0645217	1.420813	-2.1568790	0.0310151	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-1.3691379	1.464790	-0.9346989	0.3499435	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-0.8519916	1.512219	-0.5634049	0.5731592	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-3.8461185	1.405214	-2.7370337	0.0061996	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-3.9991349	1.468619	-2.7230574	0.0064681	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-1.4013418	1.448758	-0.9672710	0.3334086	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-1.7328137	1.440791	-1.2026820	0.2290994	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-2.0180688	1.433613	-1.4076801	0.1592258	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-3.4995772	1.435685	-2.4375664	0.0147865	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-2.0647639	1.451813	-1.4221969	0.1549691	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-2.0046443	1.489781	-1.3455968	0.1784326	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-2.8331258	1.491237	-1.8998500	0.0574528	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-1.8934572	1.533931	-1.2343827	0.2170603	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-5.7316235	1.670898	-3.4302646	0.0006030	Binomial	BirdID+Terr

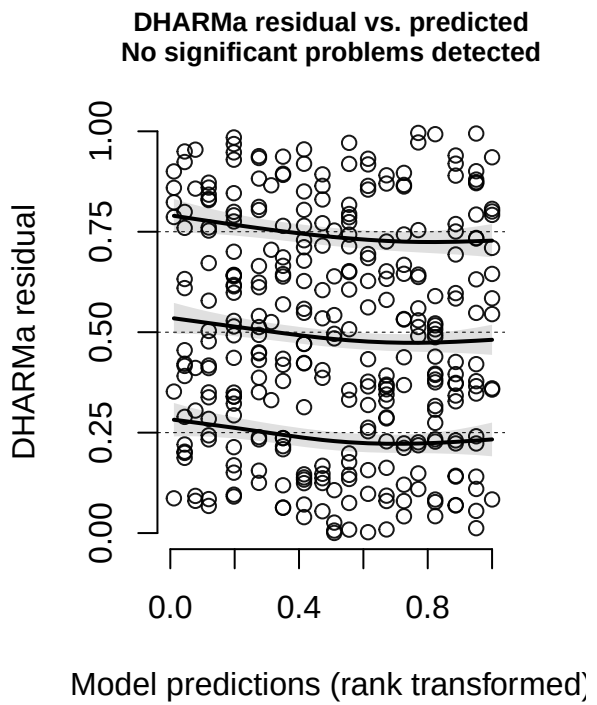
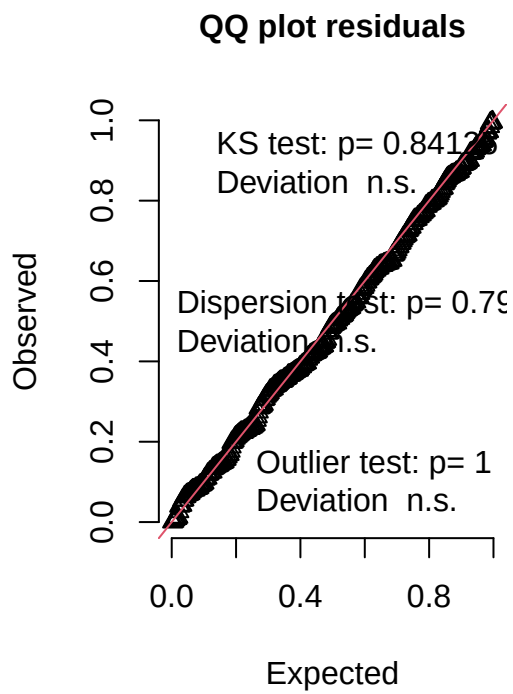
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-2.5242923	1.586804	-1.5908025	0.1116540	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-4.7987367	2.005683	-2.3925693	0.0167309	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-12.0108997	36.733772	-0.3269716	0.7436894	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-2.6531600	1.852080	-1.4325303	0.1519921	Binomial	BirdID+Terr

Similar to most other binomial models in this report.

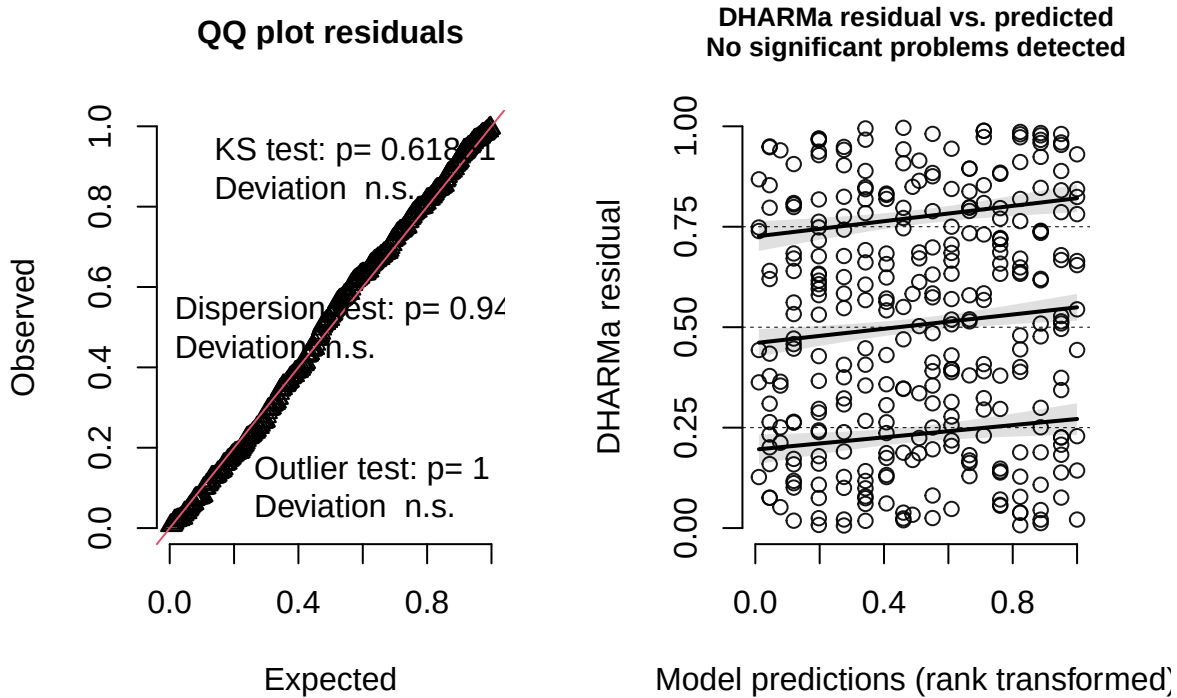
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Probabilities used to generate simulation 32

Simulation 32
 Probability of mate fidelity: 0.272
 Probability of pair dispersal: 1
 Probability of single male dispersal: 0.417
 Probability of single female dispersal: 0.092

Simulation 33

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
326	1	-455.2397	952.4793	1032.004	325.3513	305	0.8563363	0.8273982	Poisson	BirdID
326	1	-455.2397	954.4793	1037.791	325.3517	304	0.8409607	0.8088554	Poisson	BirdID+Terr
326	1	-459.0381	960.0761	1039.601	367.4779	305	0.8215248	0.8074398	Poisson	Terr

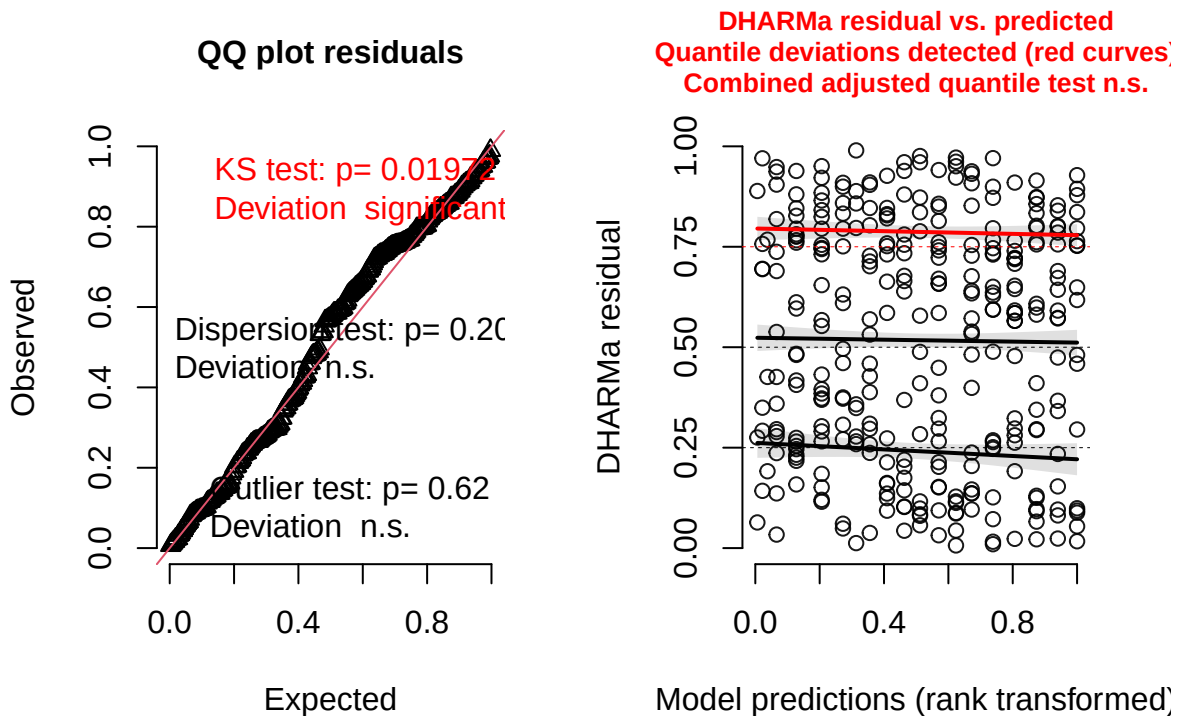
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.3423324	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.2141720	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.3423129	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.0025076	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.1816826	0.2552349	0.7118250	0.4765731	Poisson	BirdID
fixed	factor(fac.yr)2	-0.0495608	0.3376041	-0.1468016	0.8832886	Poisson	BirdID
fixed	factor(fac.yr)3	-0.0922971	0.3432541	-0.2688886	0.7880154	Poisson	BirdID
fixed	factor(fac.yr)4	-0.0793284	0.3462438	-0.2291113	0.8187824	Poisson	BirdID
fixed	factor(fac.yr)5	0.4704661	0.3060511	1.5372143	0.1242408	Poisson	BirdID
fixed	factor(fac.yr)6	0.1930870	0.3184244	0.6063826	0.5442607	Poisson	BirdID
fixed	factor(fac.yr)7	-0.5397863	0.3597805	-1.5003213	0.1335312	Poisson	BirdID
fixed	factor(fac.yr)8	0.0561400	0.3215890	0.1745708	0.8614169	Poisson	BirdID
fixed	factor(fac.yr)9	0.2356924	0.3142166	0.7500954	0.4531973	Poisson	BirdID
fixed	factor(fac.yr)10	0.1756893	0.3168505	0.5544865	0.5792460	Poisson	BirdID
fixed	factor(fac.yr)11	-0.5258300	0.3505599	-1.4999721	0.1336216	Poisson	BirdID
fixed	factor(fac.yr)12	0.1653773	0.3225402	0.5127340	0.6081374	Poisson	BirdID
fixed	factor(fac.yr)13	0.5204897	0.3118944	1.6688012	0.0951568	Poisson	BirdID
fixed	factor(fac.yr)14	-0.2883634	0.3675182	-0.7846235	0.4326743	Poisson	BirdID
fixed	factor(fac.yr)15	-0.6829484	0.4171082	-1.6373411	0.1015592	Poisson	BirdID
fixed	factor(fac.yr)16	-0.1011210	0.3984468	-0.2537879	0.7996594	Poisson	BirdID
fixed	factor(fac.yr)17	0.0671346	0.3757573	0.1786647	0.8582010	Poisson	BirdID
fixed	factor(fac.yr)18	-1.4775960	0.7677783	-1.9245086	0.0542909	Poisson	BirdID
fixed	factor(fac.yr)19	-18.7314731	6248.3627808	-0.0029978	0.9976081	Poisson	BirdID
fixed	factor(fac.yr)20	-1.2754727	1.0735548	-1.1880834	0.2348005	Poisson	BirdID
fixed	(Intercept)	0.2412743	0.2437049	0.9900264	0.3221612	Poisson	Terr
fixed	factor(fac.yr)2	-0.0289338	0.3292600	-0.0878752	0.9299759	Poisson	Terr
fixed	factor(fac.yr)3	-0.1033122	0.3342984	-0.3090419	0.7572897	Poisson	Terr
fixed	factor(fac.yr)4	-0.0653104	0.3340263	-0.1955246	0.8449822	Poisson	Terr
fixed	factor(fac.yr)5	0.4175351	0.2940086	1.4201460	0.1555652	Poisson	Terr
fixed	factor(fac.yr)6	0.1431972	0.3043103	0.4705630	0.6379529	Poisson	Terr
fixed	factor(fac.yr)7	-0.6194074	0.3453752	-1.7934333	0.0729037	Poisson	Terr
fixed	factor(fac.yr)8	0.0147224	0.3069256	0.0479673	0.9617423	Poisson	Terr
fixed	factor(fac.yr)9	0.1972351	0.2991672	0.6592805	0.5097157	Poisson	Terr
fixed	factor(fac.yr)10	0.1283968	0.3008065	0.4268420	0.6694944	Poisson	Terr
fixed	factor(fac.yr)11	-0.5843313	0.3371505	-1.7331467	0.0830696	Poisson	Terr
fixed	factor(fac.yr)12	0.1459087	0.3076076	0.4743340	0.6352618	Poisson	Terr
fixed	factor(fac.yr)13	0.4766191	0.2966522	1.6066593	0.1081291	Poisson	Terr
fixed	factor(fac.yr)14	-0.2700201	0.3534860	-0.7638778	0.4449401	Poisson	Terr
fixed	factor(fac.yr)15	-0.6336143	0.4041267	-1.5678605	0.1169137	Poisson	Terr
fixed	factor(fac.yr)16	-0.0481967	0.3835311	-0.1256658	0.8999965	Poisson	Terr
fixed	factor(fac.yr)17	0.0356629	0.3607807	0.0988493	0.9212579	Poisson	Terr
fixed	factor(fac.yr)18	-1.4789411	0.7565289	-1.9549036	0.0505944	Poisson	Terr
fixed	factor(fac.yr)19	-16.5212417	2070.2379810	-0.0079804	0.9936327	Poisson	Terr
fixed	factor(fac.yr)20	-1.3240976	1.0421634	-1.2705279	0.2038967	Poisson	Terr
fixed	(Intercept)	0.1817201	0.2552308	0.7119835	0.4764750	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.0497926	0.3376154	-0.1474831	0.8827507	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.0920388	0.3432302	-0.2681546	0.7885803	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.0794635	0.3462469	-0.2294995	0.8184807	Poisson	BirdID+Terr

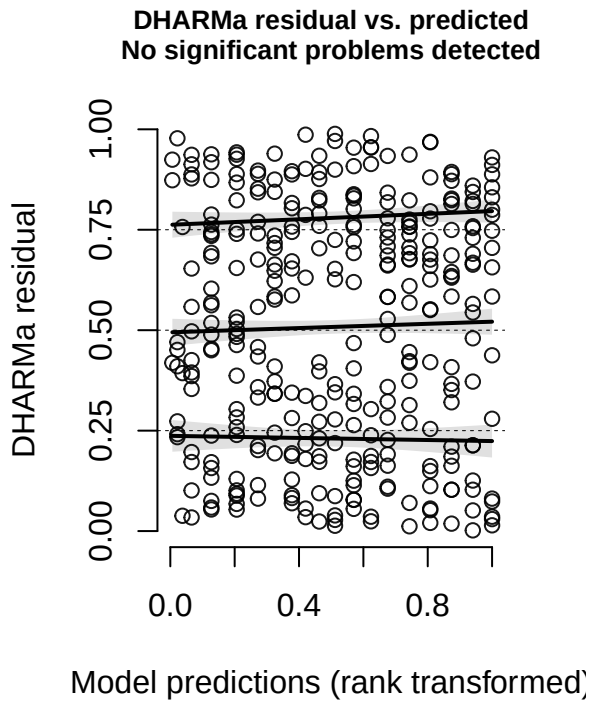
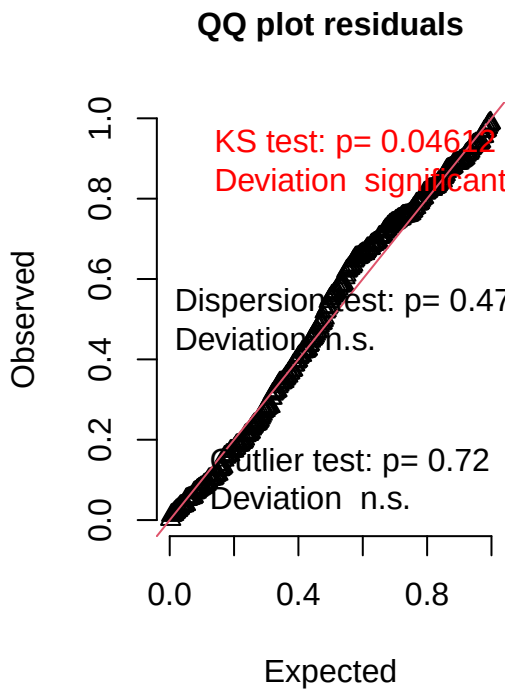
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.4704430	0.3060459	1.5371648	0.1242529	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	0.1925980	0.3184409	0.6048156	0.5453015	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.5403602	0.3598187	-1.5017569	0.1331599	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	0.0563599	0.3215721	0.1752637	0.8608724	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	0.2358681	0.3142038	0.7506850	0.4528422	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	0.1754842	0.3168548	0.5538317	0.5796940	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.5249586	0.3504880	-1.4977933	0.1341870	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	0.1654807	0.3225290	0.5130721	0.6079009	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	0.5198261	0.3119149	1.6665637	0.0956012	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.2885521	0.3675253	-0.7851216	0.4323823	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.6826195	0.4170600	-1.6367418	0.1016844	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.1014441	0.3984727	-0.2545823	0.7990458	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	0.0673251	0.3757327	0.1791834	0.8577937	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-1.4773669	0.7676892	-1.9244335	0.0543003	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-17.5255262	3418.9364456	-0.0051260	0.9959100	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-1.2754541	1.0735317	-1.1880917	0.2347973	Poisson	BirdID+Terr

Territory only model is performing the worst on all criteria.

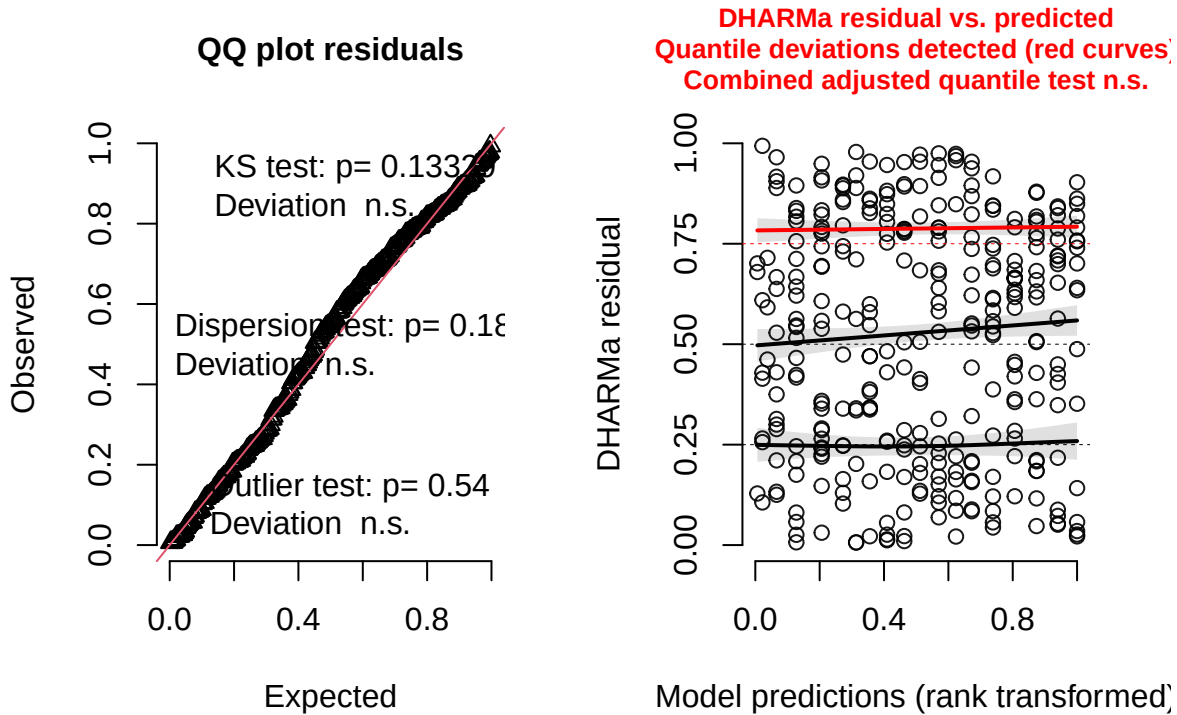
DHARMA residual



DHARMa residual



DHARMa residual



None of the diagnostics here are ideal.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
326	1	-190.8659	423.7318	503.2566	312.4264	305	0.513	Binomial	BirdID
326	1	-190.2115	424.4230	507.7348	276.5290	304	0.606	Binomial	BirdID+Terr
326	1	-193.5077	429.0155	508.5403	341.3999	305	0.380	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.9075948	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	0.6697953	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	0.9550311	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.6484579	Binomial	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.3107310	6.082077e-01	0.5108963	0.6094237	Binomial	BirdID
fixed	factor(fac.yr)2	0.0759255	7.960550e-01	0.0953772	0.9240152	Binomial	BirdID

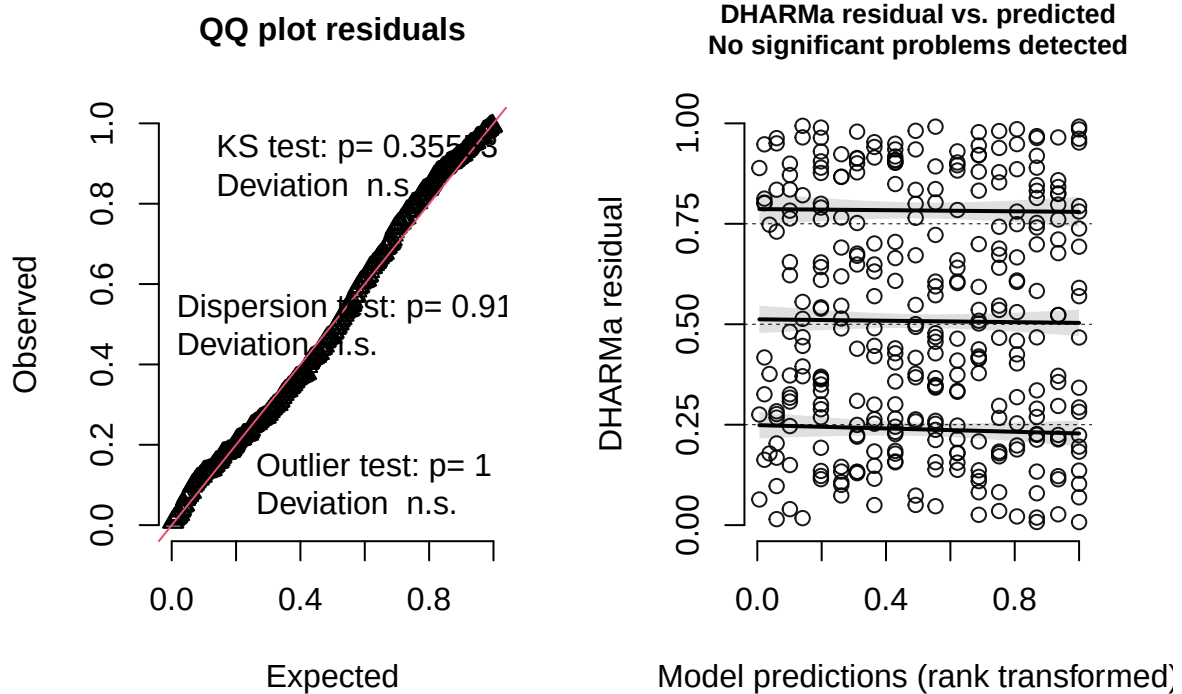
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)3	1.0475612	8.658923e-01	1.2098054	0.2263536	Binomial	BirdID
fixed	factor(fac.yr)4	0.0498602	8.120547e-01	0.0614001	0.9510406	Binomial	BirdID
fixed	factor(fac.yr)5	1.3022483	8.689145e-01	1.4987070	0.1339497	Binomial	BirdID
fixed	factor(fac.yr)6	2.2210546	1.027869e+00	2.1608342	0.0307081	Binomial	BirdID
fixed	factor(fac.yr)7	0.1509461	7.602849e-01	0.1985389	0.8426235	Binomial	BirdID
fixed	factor(fac.yr)8	0.4838130	7.772610e-01	0.6224588	0.5336402	Binomial	BirdID
fixed	factor(fac.yr)9	0.8671860	8.071479e-01	1.0743830	0.2826511	Binomial	BirdID
fixed	factor(fac.yr)10	1.0863660	8.304022e-01	1.3082408	0.1907916	Binomial	BirdID
fixed	factor(fac.yr)11	-0.1007468	7.490784e-01	-0.1344943	0.8930117	Binomial	BirdID
fixed	factor(fac.yr)12	0.7376839	8.168447e-01	0.9030896	0.3664784	Binomial	BirdID
fixed	factor(fac.yr)13	0.9364773	8.457494e-01	1.1072751	0.2681750	Binomial	BirdID
fixed	factor(fac.yr)14	0.2175965	8.374400e-01	0.2598353	0.7949908	Binomial	BirdID
fixed	factor(fac.yr)15	-0.8157878	8.467155e-01	-0.9634733	0.3353101	Binomial	BirdID
fixed	factor(fac.yr)16	-0.2784014	9.324476e-01	-0.2985706	0.7652677	Binomial	BirdID
fixed	factor(fac.yr)17	-0.9459147	9.217403e-01	-1.0262270	0.3047846	Binomial	BirdID
fixed	factor(fac.yr)18	-2.2994276	1.338052e+00	-1.7184888	0.0857075	Binomial	BirdID
fixed	factor(fac.yr)19	-	2.048662e+04	-0.0010284	0.9991794	Binomial	BirdID
fixed	factor(fac.yr)20	21.0685811	1.495096e+00	-0.6770599	0.4983680	Binomial	BirdID
fixed	(Intercept)	-1.0122696	5.708987e-01	0.5012237	0.6162137	Binomial	Terr
fixed	factor(fac.yr)2	0.2861479	7.623251e-01	0.2363792	0.8131385	Binomial	Terr
fixed	factor(fac.yr)3	0.1801978	8.280211e-01	1.2344738	0.2170264	Binomial	Terr
fixed	factor(fac.yr)4	1.0221703	7.648858e-01	0.2476030	0.8044416	Binomial	Terr
fixed	factor(fac.yr)5	0.1893880	8.221518e-01	1.4320221	0.1521375	Binomial	Terr
fixed	factor(fac.yr)6	1.1773396	9.769875e-01	2.1646378	0.0304154	Binomial	Terr
fixed	factor(fac.yr)7	2.1148241	7.089829e-01	0.0803257	0.9359782	Binomial	Terr
fixed	factor(fac.yr)8	0.0569496	7.293449e-01	0.6589311	0.5099400	Binomial	Terr
fixed	factor(fac.yr)9	0.4805880	7.580820e-01	1.0740177	0.2828147	Binomial	Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)10	0.9487793	7.767095e-01	1.2215368	0.2218828	Binomial	Terr
fixed	factor(fac.yr)11	-0.1367734	7.022492e-01	-0.1947648	0.8455770	Binomial	Terr
fixed	factor(fac.yr)12	0.6624886	7.665372e-01	0.8642615	0.3874442	Binomial	Terr
fixed	factor(fac.yr)13	0.7611683	7.967671e-01	0.9553209	0.3394154	Binomial	Terr
fixed	factor(fac.yr)14	0.3269469	7.889233e-01	0.4144217	0.6785653	Binomial	Terr
fixed	factor(fac.yr)15	-0.6113101	7.965166e-01	-0.7674794	0.4427965	Binomial	Terr
fixed	factor(fac.yr)16	-0.1626364	8.769530e-01	-0.1854562	0.8528712	Binomial	Terr
fixed	factor(fac.yr)17	-0.9764909	8.727610e-01	-1.1188526	0.2632030	Binomial	Terr
fixed	factor(fac.yr)18	-2.1090020	1.284869e+00	-1.6414144	0.1007114	Binomial	Terr
fixed	factor(fac.yr)19	-	5.891485e+03	-0.0031388	0.9974956	Binomial	Terr
fixed	factor(fac.yr)20	18.4924836	1.408373e+00	-0.6169076	0.5372957	Binomial	Terr
fixed	(Intercept)	-0.8688359	6.486733e-01	0.5691353	0.5692643	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	0.1215158	8.395412e-01	0.1447407	0.8849156	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	1.1256138	9.162901e-01	1.2284470	0.2192792	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	0.1066440	8.586122e-01	0.1242051	0.9011529	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	1.3476290	9.187083e-01	1.4668736	0.1424105	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	2.3073715	1.089338e+00	2.1181421	0.0341630	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	0.0981940	7.987051e-01	0.1229415	0.9021534	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	0.5646973	8.205322e-01	0.6882087	0.4913214	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	0.9334954	8.486236e-01	1.1000110	0.2713273	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	1.0453512	8.744942e-01	1.1953781	0.2319393	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-0.1757033	7.939391e-01	-0.2213058	0.8248543	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	0.7532290	8.622914e-01	0.8735203	0.3823796	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	0.9559758	8.985559e-01	1.0639024	0.2873729	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	0.2490539	8.930695e-01	0.2788741	0.7803415	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-0.8173409	8.981963e-01	-0.9099802	0.3628329	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-0.2222612	9.837088e-01	-0.2259420	0.8212465	Binomial	BirdID+Terr

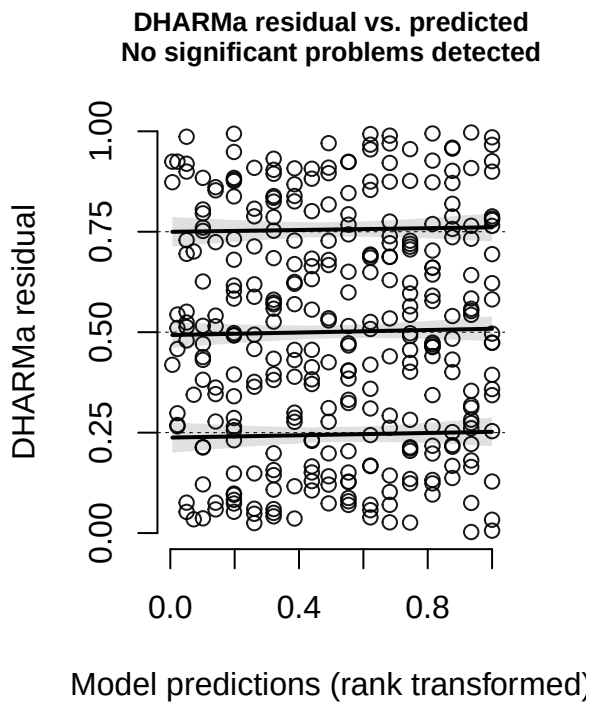
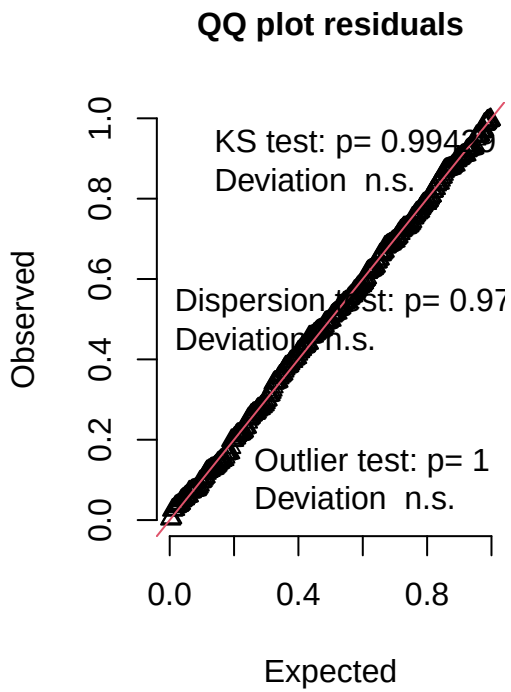
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-1.1361547	9.767675e-01	-1.1631781	0.2447572	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-2.3845168	1.420619e+00	-1.6785059	0.0932484	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-31.2399736	3.453177e+06	-0.0000090	0.9999928	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-1.0105953	1.586407e+00	-0.6370342	0.5241025	Binomial	BirdID+Terr

Similar to most other binomial models.

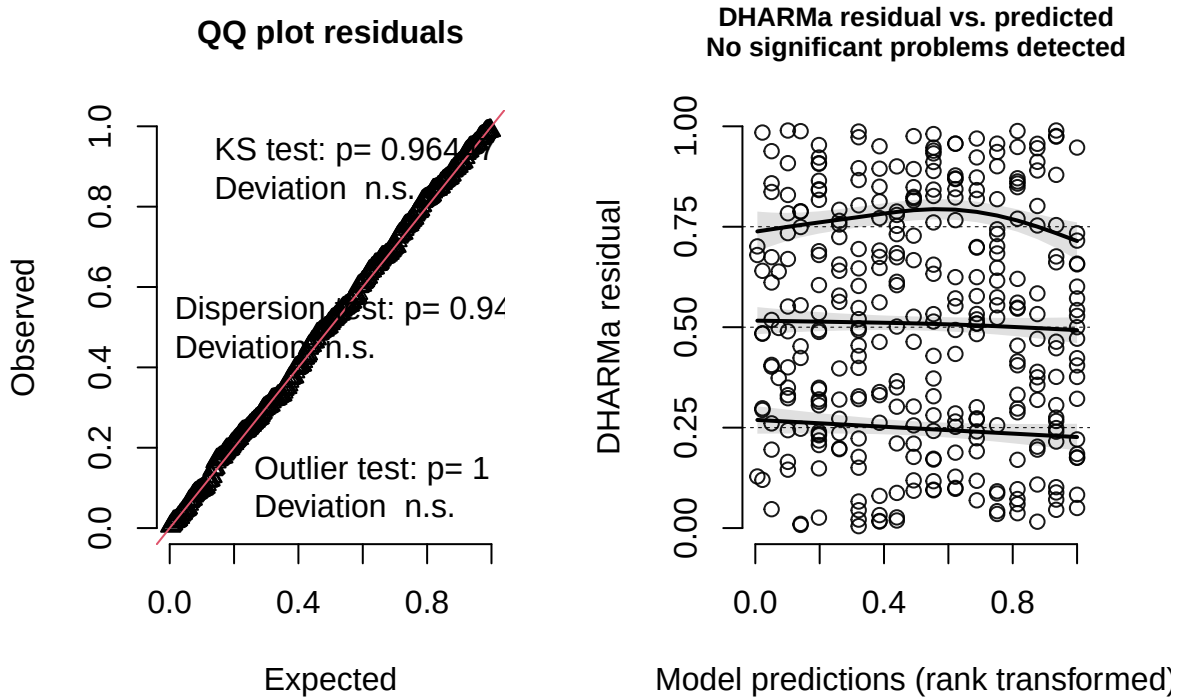
DHARMa residual



DHARMa residual



DHARMA residual



Diagnostics look good.

Probabilities used to generate simulation 33

Simulation 33
 Probability of mate fidelity: 0.127
 Probability of pair dispersal: 0.485
 Probability of single male dispersal: 0.875
 Probability of single female dispersal: 0.448

Simulation 34

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
321	1	-439.3525	920.7051	999.9053	313.8326	300	0.3750773	0.1494943	Poisson	Terr
321	1	-439.2771	922.5541	1005.5258	306.9031	299	0.3781020	0.1483459	Poisson	BirdID+Terr
321	1	-444.5907	931.1813	1010.3816	344.9388	300	0.3387239	0.1701340	Poisson	BirdID

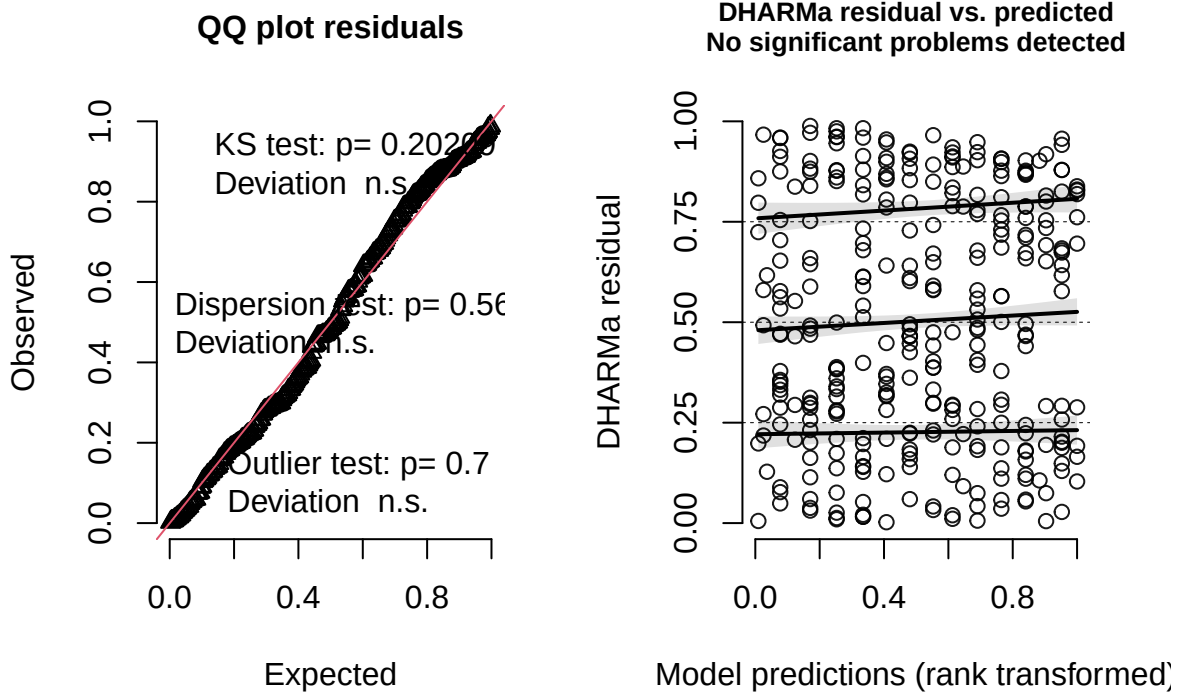
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.3984241	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.4722414	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.1437714	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.4566270	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.4445151	0.2661752	1.6700095	0.0949175	Poisson	BirdID
fixed	factor(fac.yr)2	-0.4133621	0.3263054	-1.2667951	0.2052286	Poisson	BirdID
fixed	factor(fac.yr)3	-0.9640002	0.3647612	-2.6428252	0.0082217	Poisson	BirdID
fixed	factor(fac.yr)4	-0.6068960	0.3362640	-1.8048200	0.0711029	Poisson	BirdID
fixed	factor(fac.yr)5	-0.5298978	0.3406028	-1.5557646	0.1197641	Poisson	BirdID
fixed	factor(fac.yr)6	-0.1196500	0.3220740	-0.3714985	0.7102663	Poisson	BirdID
fixed	factor(fac.yr)7	-0.2271731	0.3316631	-0.6849514	0.4933746	Poisson	BirdID
fixed	factor(fac.yr)8	-0.3267753	0.3325671	-0.9825847	0.3258119	Poisson	BirdID
fixed	factor(fac.yr)9	-0.0389358	0.3296529	-0.1181114	0.9059794	Poisson	BirdID
fixed	factor(fac.yr)10	-0.1631592	0.3275437	-0.4981296	0.6183927	Poisson	BirdID
fixed	factor(fac.yr)11	-0.6683626	0.3533353	-1.8915819	0.0585467	Poisson	BirdID
fixed	factor(fac.yr)12	-0.3118646	0.3458080	-0.9018434	0.3671401	Poisson	BirdID
fixed	factor(fac.yr)13	-0.2587730	0.3537197	-0.7315764	0.4644271	Poisson	BirdID
fixed	factor(fac.yr)14	-0.0686435	0.3735208	-0.1837742	0.8541906	Poisson	BirdID
fixed	factor(fac.yr)15	-2.0574904	1.0598037	-1.9413881	0.0522112	Poisson	BirdID
fixed	factor(fac.yr)16	-2.0685671	1.0655431	-1.9413265	0.0522187	Poisson	BirdID
fixed	factor(fac.yr)17	-0.2397221	0.5549650	-0.4319589	0.6657713	Poisson	BirdID
fixed	factor(fac.yr)18	-0.9620011	0.6649526	-1.4467214	0.1479750	Poisson	BirdID
fixed	factor(fac.yr)19	-1.1502391	1.0934129	-1.0519714	0.2928127	Poisson	BirdID
fixed	factor(fac.yr)20	-0.1067381	0.7038275	-0.1516537	0.8794601	Poisson	BirdID
fixed	(Intercept)	0.6662791	0.2735064	2.4360642	0.0148480	Poisson	Terr
fixed	factor(fac.yr)2	-0.5409908	0.3393506	-1.5941946	0.1108925	Poisson	Terr
fixed	factor(fac.yr)3	-1.1626382	0.3780094	-3.0756859	0.0021002	Poisson	Terr
fixed	factor(fac.yr)4	-0.8129735	0.3443397	-2.3609636	0.0182275	Poisson	Terr
fixed	factor(fac.yr)5	-0.6990151	0.3490239	-2.0027711	0.0452019	Poisson	Terr
fixed	factor(fac.yr)6	-0.4308680	0.3271874	-1.3168843	0.1878774	Poisson	Terr
fixed	factor(fac.yr)7	-0.5681507	0.3401247	-1.6704189	0.0948365	Poisson	Terr
fixed	factor(fac.yr)8	-0.6993200	0.3345424	-2.0903774	0.0365839	Poisson	Terr
fixed	factor(fac.yr)9	-0.3307506	0.3350070	-0.9872945	0.3234983	Poisson	Terr
fixed	factor(fac.yr)10	-0.4647490	0.3282888	-1.4156710	0.1568719	Poisson	Terr
fixed	factor(fac.yr)11	-0.8714812	0.3575898	-2.4370974	0.0148057	Poisson	Terr
fixed	factor(fac.yr)12	-0.6234786	0.3506605	-1.7780122	0.0754019	Poisson	Terr
fixed	factor(fac.yr)13	-0.4903500	0.3595574	-1.3637601	0.1726431	Poisson	Terr
fixed	factor(fac.yr)14	-0.3754669	0.3791229	-0.9903568	0.3219998	Poisson	Terr
fixed	factor(fac.yr)15	-2.3316762	1.0685840	-2.1820243	0.0291077	Poisson	Terr
fixed	factor(fac.yr)16	-2.1102732	1.0711367	-1.9701250	0.0488240	Poisson	Terr
fixed	factor(fac.yr)17	-0.4306227	0.5589699	-0.7703863	0.4410708	Poisson	Terr
fixed	factor(fac.yr)18	-1.1676269	0.6653599	-1.7548802	0.0792798	Poisson	Terr
fixed	factor(fac.yr)19	-1.8841830	1.0728462	-1.7562470	0.0790462	Poisson	Terr
fixed	factor(fac.yr)20	-0.7523494	0.7097512	-1.0600185	0.2891362	Poisson	Terr
fixed	(Intercept)	0.6495956	0.2750854	2.3614323	0.0182045	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.5394490	0.3400407	-1.5864247	0.1126430	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-1.1588062	0.3789543	-3.0579046	0.0022289	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.8074721	0.3460459	-2.3334247	0.0196259	Poisson	BirdID+Terr

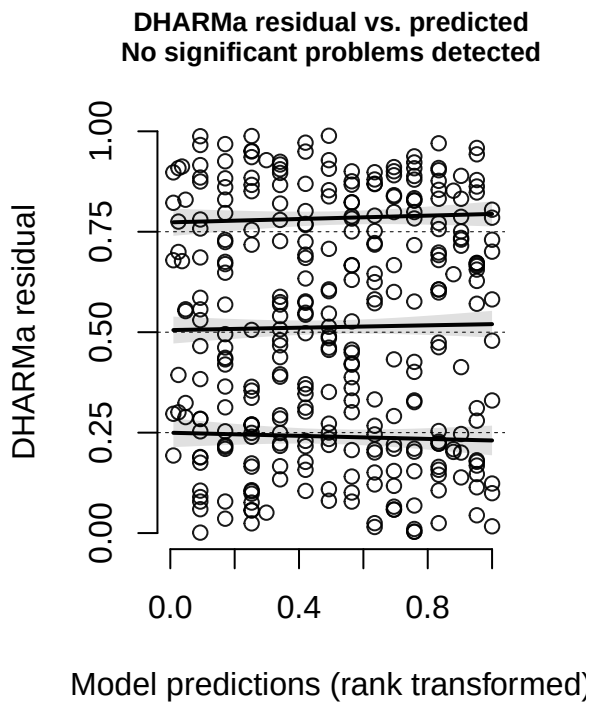
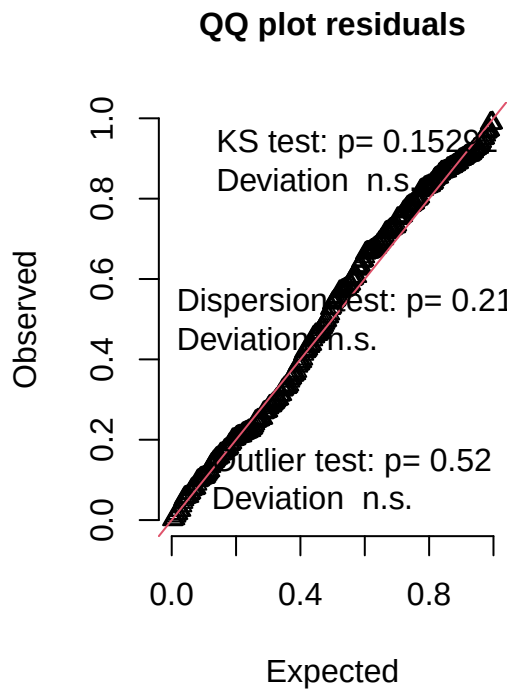
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.6859179	0.3507671	-1.9554798	0.0505265	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.4081312	0.3295030	-1.2386269	0.2154837	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.5426649	0.3425405	-1.5842358	0.1131401	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.6825534	0.3373224	-2.0234453	0.0430273	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.3206940	0.3377652	-0.9494584	0.3423875	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.4558567	0.3310431	-1.3770313	0.1685026	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.8654365	0.3600481	-2.4036691	0.0162315	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.6024031	0.3530640	-1.7062150	0.0879681	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.4865049	0.3622041	-1.3431789	0.1792141	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.3683552	0.3826005	-0.9627672	0.3356644	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-2.3130894	1.0743682	-2.1529764	0.0313205	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-2.1154936	1.0762110	-1.9656867	0.0493348	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.4314890	0.5649817	-0.7637220	0.4450329	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-1.1551800	0.6704087	-1.7230983	0.0848708	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.8279017	1.0808629	-1.6911503	0.0908081	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.7023705	0.7159817	-0.9809895	0.3265979	Poisson	BirdID+Terr

Territory only model has the best AIC and BIC.

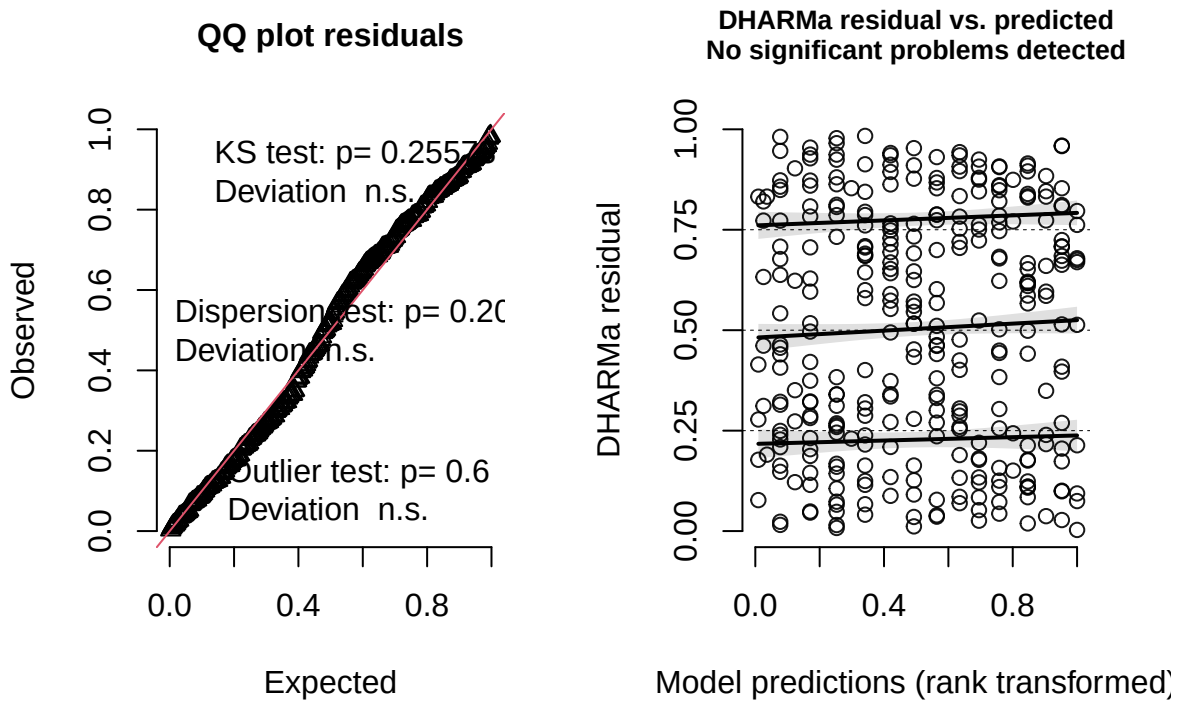
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
321	1	-195.7104	435.4207	518.3924	267.0278	299	0.663	Binomial	BirdID+Terr
321	1	-197.3287	436.6574	515.8577	309.6779	300	0.550	Binomial	Terr
321	1	-197.3602	436.7204	515.9207	303.9841	300	0.555	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.1084906	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.0444731	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	0.9337985	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.9070190	Binomial	BirdID+Terr

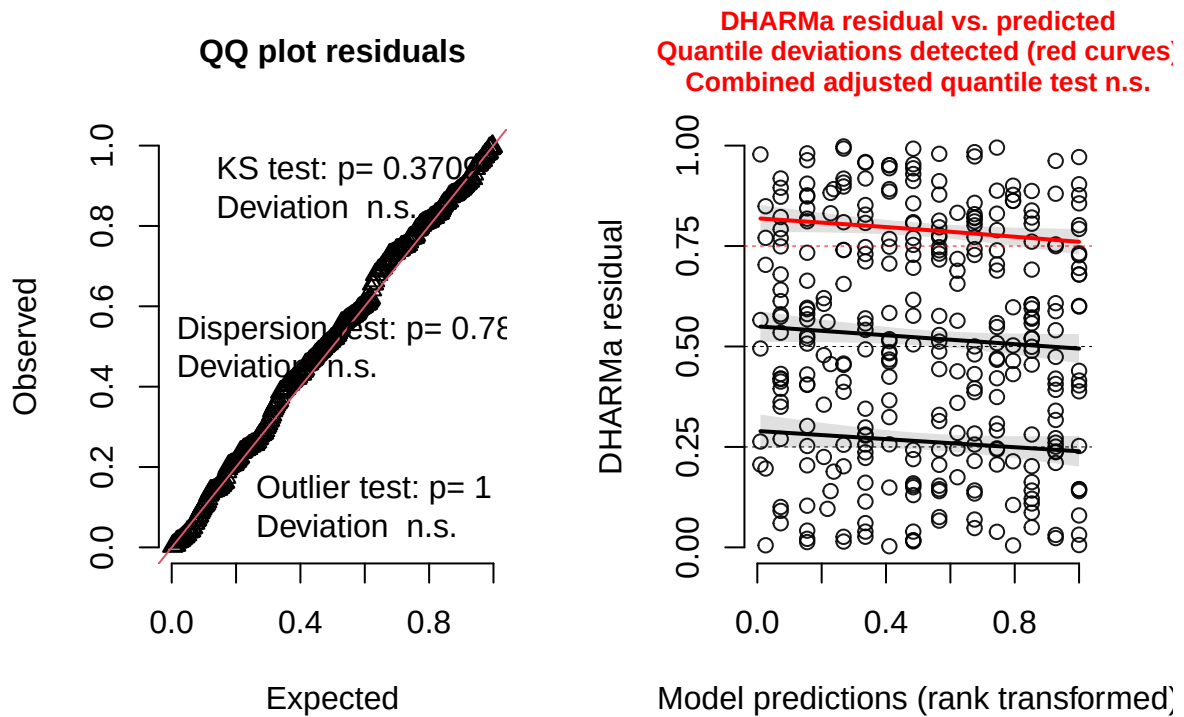
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.8913193	0.8162751	1.0919350	0.2748617	Binomial	BirdID
fixed	factor(fac.yr)2	-0.6527295	0.9295347	-0.7022110	0.4825476	Binomial	BirdID
fixed	factor(fac.yr)3	-1.6484708	0.9229768	-1.7860370	0.0740933	Binomial	BirdID
fixed	factor(fac.yr)4	-1.5836959	0.9218377	-1.7179770	0.0858008	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.3335821	0.9269978	-0.3598521	0.7189577	Binomial	BirdID
fixed	factor(fac.yr)6	-0.0379341	0.9358830	-0.0405330	0.9676682	Binomial	BirdID
fixed	factor(fac.yr)7	0.6291844	0.9681909	0.6498558	0.5157854	Binomial	BirdID
fixed	factor(fac.yr)8	0.4400882	0.9592711	0.4587735	0.6463968	Binomial	BirdID
fixed	factor(fac.yr)9	0.0889201	0.9858649	0.0901950	0.9281323	Binomial	BirdID
fixed	factor(fac.yr)10	0.0836204	0.9655270	0.0866059	0.9309847	Binomial	BirdID
fixed	factor(fac.yr)11	-0.6974098	0.9389109	-0.7427860	0.4576113	Binomial	BirdID
fixed	factor(fac.yr)12	0.4637580	0.9896706	0.4685983	0.6393568	Binomial	BirdID
fixed	factor(fac.yr)13	-0.7303040	0.9948472	-0.7340866	0.4628959	Binomial	BirdID
fixed	factor(fac.yr)14	0.2081281	1.1330597	0.1836868	0.8542591	Binomial	BirdID
fixed	factor(fac.yr)15	-2.4939694	1.5485031	-1.6105679	0.1072739	Binomial	BirdID
fixed	factor(fac.yr)16	-2.4731538	1.5223106	-1.6246052	0.1042467	Binomial	BirdID
fixed	factor(fac.yr)17	-0.8016079	1.4105009	-0.5683143	0.5698215	Binomial	BirdID
fixed	factor(fac.yr)18	-1.2811585	1.3353613	-0.9594097	0.3373524	Binomial	BirdID
fixed	factor(fac.yr)19	-0.8450061	1.8176101	-0.4648995	0.6420034	Binomial	BirdID
fixed	factor(fac.yr)20	-0.7517248	1.8170805	-0.4136992	0.6790944	Binomial	BirdID
fixed	(Intercept)	1.3094462	0.7691555	1.7024467	0.0886717	Binomial	Terr
fixed	factor(fac.yr)2	-0.9131066	0.8974244	-1.0174747	0.3089277	Binomial	Terr
fixed	factor(fac.yr)3	-1.9336459	0.8989269	-2.1510601	0.0314715	Binomial	Terr
fixed	factor(fac.yr)4	-1.8832611	0.8862295	-2.1250265	0.0335844	Binomial	Terr
fixed	factor(fac.yr)5	-0.5317642	0.8899891	-0.5974952	0.5501768	Binomial	Terr
fixed	factor(fac.yr)6	-0.5507816	0.8902693	-0.6186685	0.5361347	Binomial	Terr
fixed	factor(fac.yr)7	0.0365642	0.9352462	0.0390958	0.9688140	Binomial	Terr
fixed	factor(fac.yr)8	-0.2047048	0.9082849	-0.2253752	0.8216874	Binomial	Terr
fixed	factor(fac.yr)9	-0.2335089	0.9395558	-0.2485312	0.8037234	Binomial	Terr
fixed	factor(fac.yr)10	-0.4293211	0.9098980	-0.4718344	0.6370450	Binomial	Terr
fixed	factor(fac.yr)11	-0.9931403	0.8839733	-1.1234959	0.2612269	Binomial	Terr
fixed	factor(fac.yr)12	-0.0408596	0.9334732	-0.0437716	0.9650865	Binomial	Terr
fixed	factor(fac.yr)13	-1.1145950	0.9398819	-1.1858884	0.2356664	Binomial	Terr
fixed	factor(fac.yr)14	-0.2783693	1.0837250	-0.2568634	0.7972842	Binomial	Terr
fixed	factor(fac.yr)15	-3.1107522	1.5354158	-2.0259999	0.0427648	Binomial	Terr
fixed	factor(fac.yr)16	-2.6331799	1.4786813	-1.7807623	0.0749513	Binomial	Terr
fixed	factor(fac.yr)17	-0.9559563	1.3540417	-0.7060021	0.4801868	Binomial	Terr
fixed	factor(fac.yr)18	-1.4966807	1.2626919	-1.1853095	0.2358951	Binomial	Terr
fixed	factor(fac.yr)19	-2.1167032	1.7849771	-1.1858434	0.2356842	Binomial	Terr
fixed	factor(fac.yr)20	-1.5863976	1.7455908	-0.9088027	0.3634543	Binomial	Terr
fixed	(Intercept)	1.2338911	0.8424028	1.4647281	0.1429951	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-0.8984631	0.9586908	-0.9371772	0.3486674	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-2.0572244	0.9602690	-2.1423418	0.0321660	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-1.9493925	0.9554293	-2.0403315	0.0413173	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-0.4485504	0.9618865	-0.4663236	0.6409839	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-0.3293535	0.9691744	-0.3398290	0.7339853	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	0.3302741	1.0105751	0.3268179	0.7438056	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	0.1177465	0.9946414	0.1183808	0.9057659	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-0.1334096	1.0205485	-0.1307234	0.8959941	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-0.3355476	0.9932473	-0.3378289	0.7354921	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-0.9604255	0.9681538	-0.9920175	0.3211890	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	0.2128304	1.0199390	0.2086697	0.8347061	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-1.0539544	1.0324382	-1.0208401	0.3073302	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-0.1391231	1.1817017	-0.1177312	0.9062807	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-3.0678077	1.6653140	-1.8421797	0.0654489	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-2.7967537	1.6188707	-1.7275955	0.0840608	Binomial	BirdID+Terr

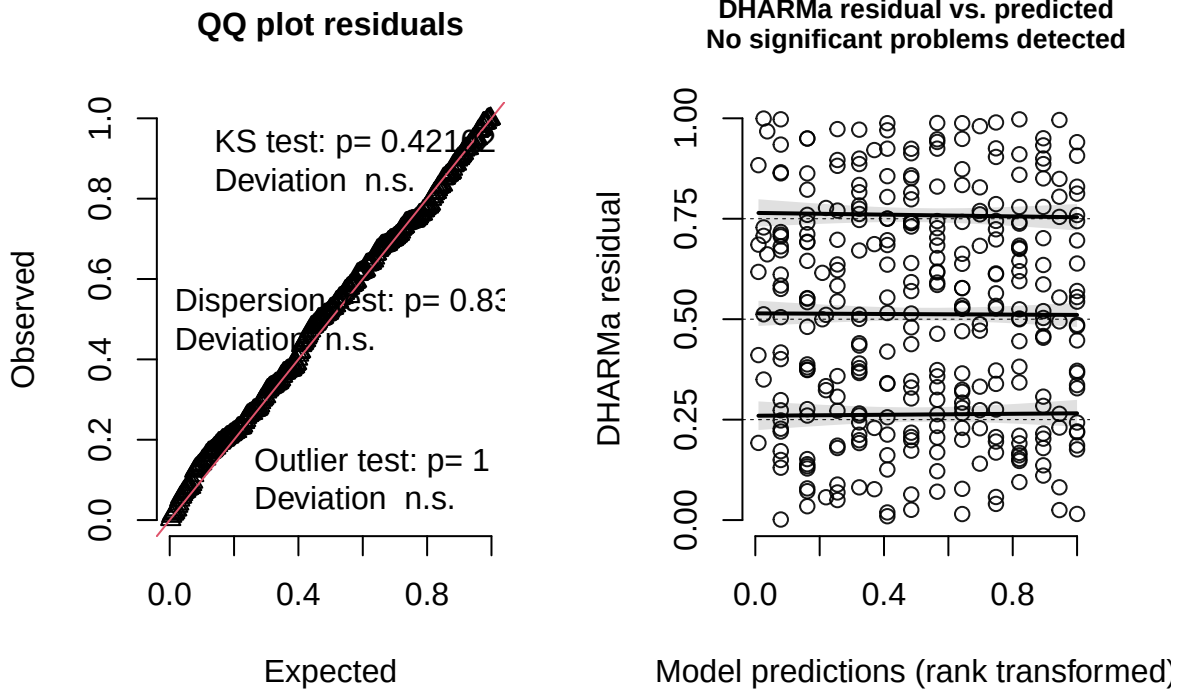
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-0.8953566	1.4871745	-0.6020522	0.5471394	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-1.4449659	1.3815879	-1.0458733	0.2956195	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-1.8291790	1.9613965	-0.9325901	0.3510316	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-1.2622590	1.8804538	-0.6712523	0.5020598	Binomial	BirdID+Terr

Similar results to other binomial models.

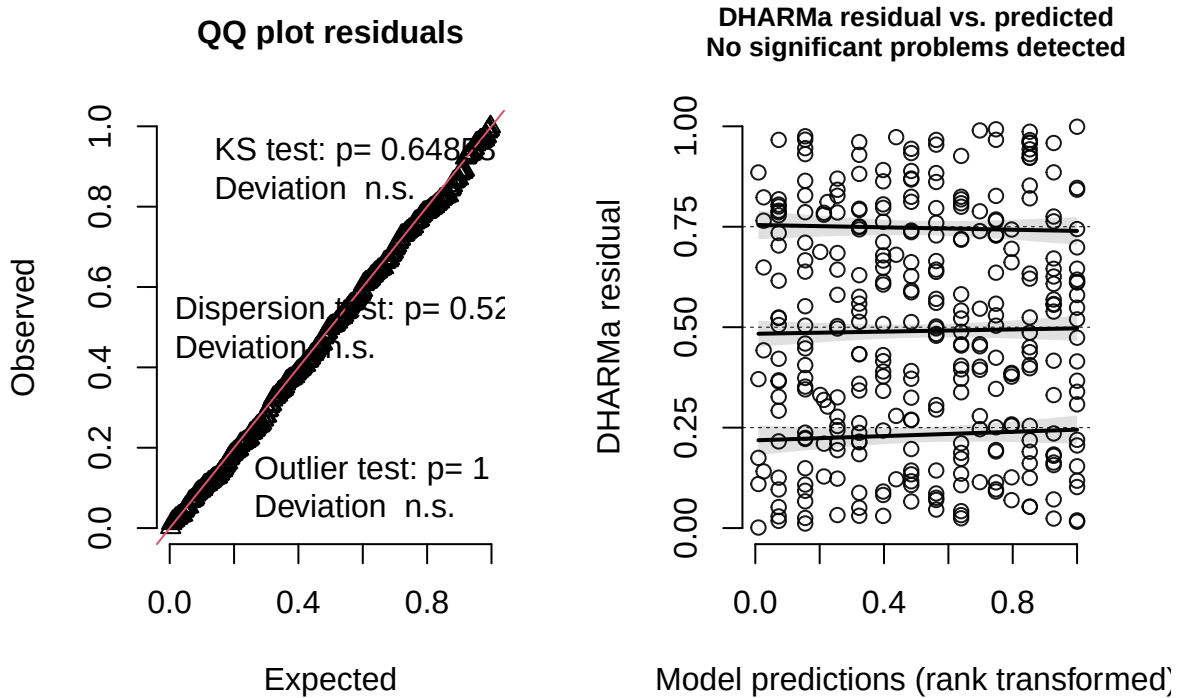
DHARMA residual



DHARMa residual



DHARMa residual



The band ID only model diagnostics are mildly concerning.

Probabilites used to generate simulation 34

Simulation 34
 Probability of mate fidelity: 0.044
 Probability of pair dispersal: 0.379
 Probability of single male dispersal: 0.579
 Probability of single female dispersal: 0.587

Simulation 35

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	REfamily	RE
329	1	-429.0916	900.1831	979.9003	316.1398	308	0.3063098	0.1541205	Poisson	BirdID
329	1	-428.8318	901.6635	985.1768	303.5295	307	0.3204188	0.1539955	Poisson	BirdID+Terr
329	1	-430.9298	903.8596	983.5769	344.5323	308	0.2645456	0.1716451	Poisson	Terr

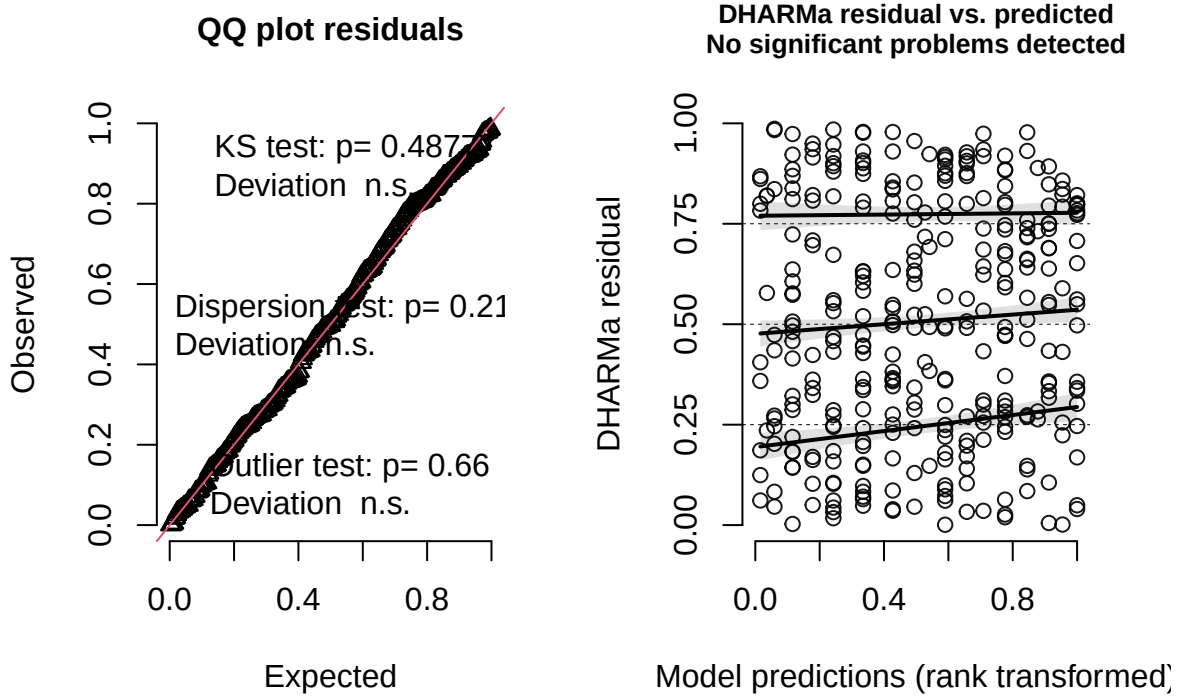
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.3856322	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.2875916	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.3568683	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.1945807	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.5097223	0.2719390	1.8743992	0.0608754	Poisson	BirdID
fixed	factor(fac.yr)2	-0.3036421	0.3345428	-0.9076331	0.3640721	Poisson	BirdID
fixed	factor(fac.yr)3	-1.0216599	0.3704721	-2.7577242	0.0058205	Poisson	BirdID
fixed	factor(fac.yr)4	-0.8330373	0.3524304	-2.3636933	0.0180938	Poisson	BirdID
fixed	factor(fac.yr)5	-0.4668641	0.3328527	-1.4026144	0.1607319	Poisson	BirdID
fixed	factor(fac.yr)6	-0.9009304	0.3440970	-2.6182457	0.0088383	Poisson	BirdID
fixed	factor(fac.yr)7	-0.9131409	0.3613450	-2.5270612	0.0115021	Poisson	BirdID
fixed	factor(fac.yr)8	-0.3462380	0.3270464	-1.0586817	0.2897448	Poisson	BirdID
fixed	factor(fac.yr)9	0.0874390	0.3247574	0.2692440	0.7877419	Poisson	BirdID
fixed	factor(fac.yr)10	-0.0955385	0.3400410	-0.2809618	0.7787397	Poisson	BirdID
fixed	factor(fac.yr)11	-0.7201063	0.3800205	-1.8949143	0.0581038	Poisson	BirdID
fixed	factor(fac.yr)12	-0.3718455	0.3582854	-1.0378473	0.2993412	Poisson	BirdID
fixed	factor(fac.yr)13	-0.3850568	0.3556662	-1.0826354	0.2789703	Poisson	BirdID
fixed	factor(fac.yr)14	-0.9605903	0.4196148	-2.2892193	0.0220666	Poisson	BirdID
fixed	factor(fac.yr)15	-1.4518273	0.5880839	-2.4687416	0.0135589	Poisson	BirdID
fixed	factor(fac.yr)16	-1.1872242	0.5087393	-2.3336595	0.0196136	Poisson	BirdID
fixed	factor(fac.yr)17	-0.5424503	0.5159771	-1.0513069	0.2931177	Poisson	BirdID
fixed	factor(fac.yr)18	-0.1277085	0.5000623	-0.2553851	0.7984257	Poisson	BirdID
fixed	factor(fac.yr)19	-0.6816526	0.6008976	-1.1343906	0.2566307	Poisson	BirdID
fixed	factor(fac.yr)20	-1.3226163	0.7939328	-1.6659046	0.0957324	Poisson	BirdID
fixed	(Intercept)	0.6101227	0.2598737	2.3477666	0.0188864	Poisson	Terr
fixed	factor(fac.yr)2	-0.3205080	0.3327251	-0.9632817	0.3354062	Poisson	Terr
fixed	factor(fac.yr)3	-1.1049038	0.3659536	-3.0192458	0.0025340	Poisson	Terr
fixed	factor(fac.yr)4	-0.8510760	0.3431634	-2.4800895	0.0131349	Poisson	Terr
fixed	factor(fac.yr)5	-0.5478839	0.3208421	-1.7076432	0.0877026	Poisson	Terr
fixed	factor(fac.yr)6	-0.9584823	0.3320884	-2.8862265	0.0038989	Poisson	Terr
fixed	factor(fac.yr)7	-0.9547238	0.3478930	-2.7443028	0.0060640	Poisson	Terr
fixed	factor(fac.yr)8	-0.4192606	0.3127539	-1.3405449	0.1800683	Poisson	Terr
fixed	factor(fac.yr)9	0.0481926	0.3127570	0.1540897	0.8775390	Poisson	Terr
fixed	factor(fac.yr)10	-0.1743564	0.3272596	-0.5327770	0.5941879	Poisson	Terr
fixed	factor(fac.yr)11	-0.7052835	0.3653915	-1.9302130	0.0535805	Poisson	Terr
fixed	factor(fac.yr)12	-0.3045215	0.3401686	-0.8952076	0.3706761	Poisson	Terr
fixed	factor(fac.yr)13	-0.2959856	0.3358279	-0.8813609	0.3781225	Poisson	Terr
fixed	factor(fac.yr)14	-0.9100781	0.4061545	-2.2407190	0.0250443	Poisson	Terr
fixed	factor(fac.yr)15	-1.3785005	0.5750414	-2.3972196	0.0165200	Poisson	Terr
fixed	factor(fac.yr)16	-1.2318963	0.4936811	-2.4953279	0.0125841	Poisson	Terr
fixed	factor(fac.yr)17	-0.5875464	0.5026368	-1.1689285	0.2424324	Poisson	Terr
fixed	factor(fac.yr)18	-0.0419380	0.4793290	-0.0874931	0.9302796	Poisson	Terr
fixed	factor(fac.yr)19	-0.6459074	0.5866010	-1.1011017	0.2708524	Poisson	Terr
fixed	factor(fac.yr)20	-1.3691925	0.7713226	-1.7751230	0.0758776	Poisson	Terr
fixed	(Intercept)	0.5186269	0.2760613	1.8786660	0.0602901	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.3208265	0.3400780	-0.9433911	0.3454809	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-1.0469307	0.3760956	-2.7836828	0.0053746	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.8420068	0.3571182	-2.3577820	0.0183845	Poisson	BirdID+Terr

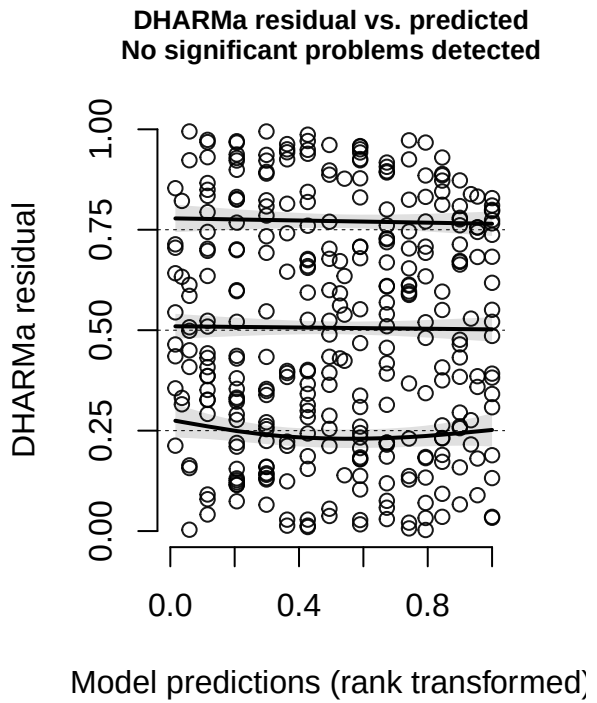
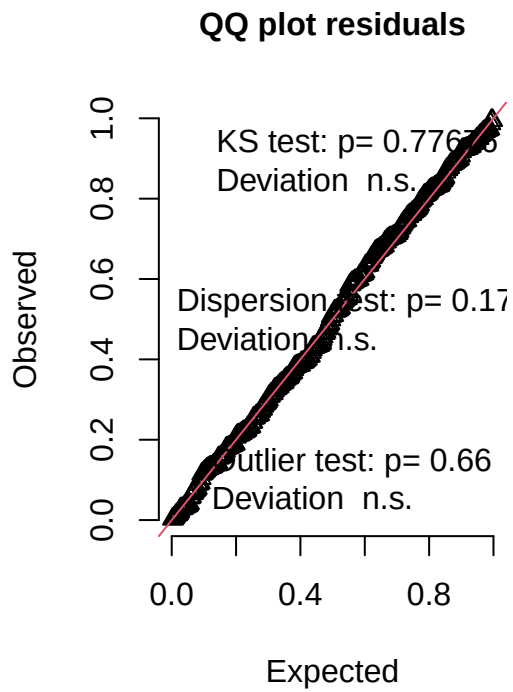
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.4937624	0.3363127	-1.4681645	0.1420595	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.9164445	0.3478319	-2.6347341	0.0084203	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.9232096	0.3646465	-2.5317932	0.0113481	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.3646168	0.3305781	-1.1029670	0.2700415	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	0.1067540	0.3298961	0.3235990	0.7462417	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.1022942	0.3447239	-0.2967423	0.7666633	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.7255247	0.3837266	-1.8907335	0.0586599	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.3666545	0.3620515	-1.0127136	0.3111970	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.3665687	0.3583615	-1.0229018	0.3063543	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.9485097	0.4241972	-2.2360113	0.0253510	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-1.4216145	0.5922937	-2.4001853	0.0163868	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-1.1940880	0.5137335	-2.3243335	0.0201076	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.5866836	0.5236106	-1.1204578	0.2625187	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.1076327	0.5085124	-0.2116619	0.8323708	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-0.6871376	0.6093419	-1.1276717	0.2594586	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-1.3406828	0.7984604	-1.6790848	0.0931355	Poisson	BirdID+Terr

The territory only model did not perform too well, comparatively.

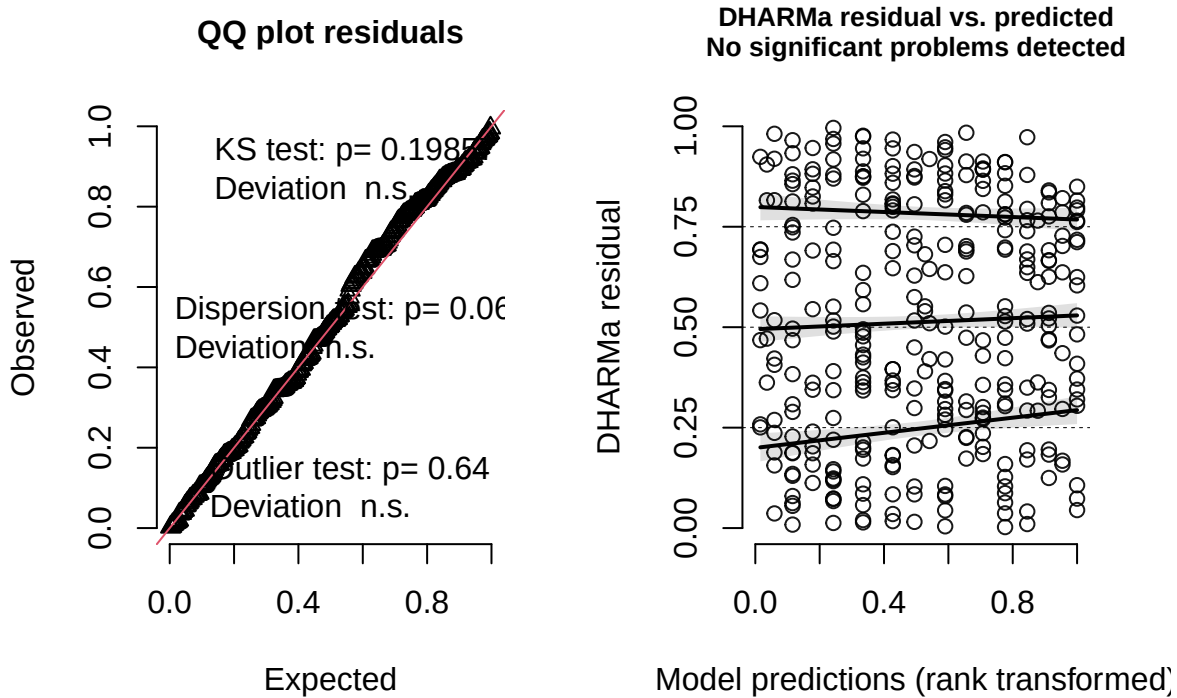
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
329	1	-188.3808	418.7615	498.4787	265.8076	308	0.643	Binomial	BirdID
329	1	-188.3777	420.7554	504.2687	265.7091	307	0.643	Binomial	BirdID+Terr
329	1	-197.6775	437.3550	517.0722	356.6995	308	0.493	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.3606892	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	0.5924633	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.3582172	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.0598523	Binomial	BirdID+Terr

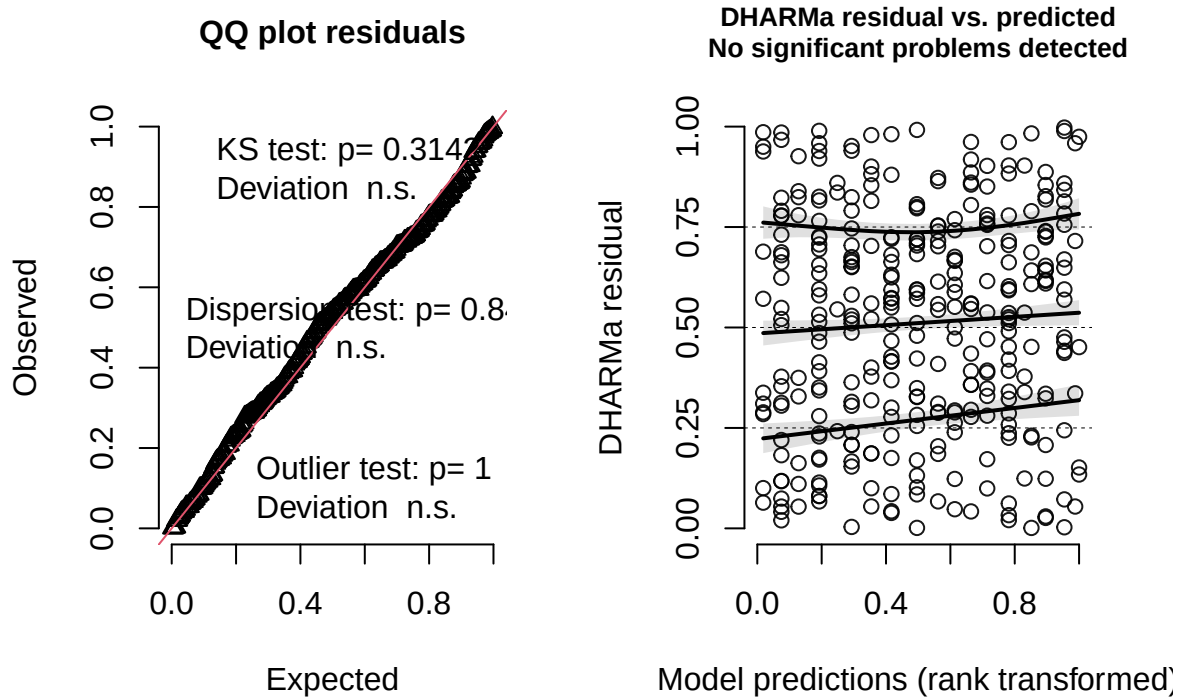
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	2.1097770	1.258586e+00	1.6763069	0.0936781	Binomial	BirdID
fixed	factor(fac.yr)2	-1.3741530	1.366526e+00	-1.0055814	0.3146170	Binomial	BirdID
fixed	factor(fac.yr)3	-2.5547156	1.324919e+00	-1.9282055	0.0538296	Binomial	BirdID
fixed	factor(fac.yr)4	-2.2653723	1.329067e+00	-1.7044827	0.0882909	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-2.1489050	1.331297e+00	-1.6141441	0.1064962	Binomial	BirdID
fixed	factor(fac.yr)6	-2.6076485	1.319189e+00	-1.9767063	0.0480748	Binomial	BirdID
fixed	factor(fac.yr)7	-2.6917967	1.338575e+00	-2.0109421	0.0443316	Binomial	BirdID
fixed	factor(fac.yr)8	-1.0216679	1.350833e+00	-0.7563241	0.4494549	Binomial	BirdID
fixed	factor(fac.yr)9	0.2941535	1.458208e+00	0.2017225	0.8401336	Binomial	BirdID
fixed	factor(fac.yr)10	0.5811208	1.549860e+00	0.3749506	0.7076972	Binomial	BirdID
fixed	factor(fac.yr)11	-1.6677451	1.396105e+00	-1.1945697	0.2322552	Binomial	BirdID
fixed	factor(fac.yr)12	-1.3813578	1.425148e+00	-0.9692733	0.3324088	Binomial	BirdID
fixed	factor(fac.yr)13	-1.9647306	1.403623e+00	-1.3997561	0.1615864	Binomial	BirdID
fixed	factor(fac.yr)14	-2.4717934	1.423585e+00	-1.7363159	0.0825080	Binomial	BirdID
fixed	factor(fac.yr)15	-2.6107601	1.546867e+00	-1.6877731	0.0914548	Binomial	BirdID
fixed	factor(fac.yr)16	-3.5258585	1.534345e+00	-2.2979568	0.0215642	Binomial	BirdID
fixed	factor(fac.yr)17	-0.7084953	1.918124e+00	-0.3693689	0.7118527	Binomial	BirdID
fixed	factor(fac.yr)18	34.4257044	3.355443e+07	0.0000010	0.9999992	Binomial	BirdID
fixed	factor(fac.yr)19	10.4414785	3.734702e+02	0.0279580	0.9776957	Binomial	BirdID
fixed	factor(fac.yr)20	-2.6062761	1.752176e+00	-1.4874514	0.1368956	Binomial	BirdID
fixed	(Intercept)	2.3337572	1.112438e+00	2.0978755	0.0359161	Binomial	Terr
fixed	factor(fac.yr)2	-1.4184826	1.239488e+00	-1.1444099	0.2524537	Binomial	Terr
fixed	factor(fac.yr)3	-2.5881008	1.188367e+00	-2.1778638	0.0294162	Binomial	Terr
fixed	factor(fac.yr)4	-2.2937800	1.183426e+00	-1.9382532	0.0525923	Binomial	Terr
fixed	factor(fac.yr)5	-2.2858353	1.180245e+00	-1.9367464	0.0527764	Binomial	Terr
fixed	factor(fac.yr)6	-2.6445323	1.168630e+00	-2.2629339	0.0236398	Binomial	Terr
fixed	factor(fac.yr)7	-2.6343418	1.182098e+00	-2.2285302	0.0258452	Binomial	Terr
fixed	factor(fac.yr)8	-1.4138597	1.191028e+00	-1.1870915	0.2351915	Binomial	Terr
fixed	factor(fac.yr)9	-0.3430637	1.291820e+00	-0.2655662	0.7905733	Binomial	Terr
fixed	factor(fac.yr)10	-0.1057664	1.363330e+00	-0.0775794	0.9381626	Binomial	Terr
fixed	factor(fac.yr)11	-1.6474040	1.228550e+00	-1.3409336	0.1799420	Binomial	Terr
fixed	factor(fac.yr)12	-1.3822050	1.242341e+00	-1.1125814	0.2658883	Binomial	Terr
fixed	factor(fac.yr)13	-1.8355278	1.219524e+00	-1.5051180	0.1322938	Binomial	Terr
fixed	factor(fac.yr)14	-2.3416648	1.235998e+00	-1.8945535	0.0581516	Binomial	Terr
fixed	factor(fac.yr)15	-2.4398101	1.313295e+00	-1.8577782	0.0632005	Binomial	Terr
fixed	factor(fac.yr)16	-3.3773828	1.318389e+00	-2.5617506	0.0104146	Binomial	Terr
fixed	factor(fac.yr)17	-0.9998094	1.619476e+00	-0.6173659	0.5369934	Binomial	Terr
fixed	factor(fac.yr)18	24.2515890	2.935909e+05	0.0000826	0.9999341	Binomial	Terr
fixed	factor(fac.yr)19	16.4347307	5.914835e+03	0.0027786	0.9977830	Binomial	Terr
fixed	factor(fac.yr)20	-2.3950644	1.518746e+00	-1.5770014	0.1147952	Binomial	Terr
fixed	(Intercept)	2.1459263	1.273333e+00	1.6852832	0.0919339	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-1.4171549	1.379842e+00	-1.0270412	0.3044011	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-2.5871872	1.338977e+00	-1.9322122	0.0533333	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-2.3045354	1.343055e+00	-1.7158903	0.0861821	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-2.1934999	1.345241e+00	-1.6305623	0.1029827	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-2.6455404	1.333267e+00	-1.9842541	0.0472275	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-2.7392016	1.352511e+00	-2.0252706	0.0428396	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-1.0546649	1.364620e+00	-0.7728637	0.4396030	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	0.2627030	1.471428e+00	0.1785360	0.8583020	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	0.5274883	1.558965e+00	0.3383580	0.7350934	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-1.6967813	1.409506e+00	-1.2038126	0.2286620	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-1.4174629	1.438043e+00	-0.9856886	0.3242859	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-1.9952842	1.416799e+00	-1.4083039	0.1590411	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-2.5061086	1.436519e+00	-1.7445700	0.0810597	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-2.6508464	1.558680e+00	-1.7006991	0.0889995	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-3.5557034	1.545804e+00	-2.3002293	0.0214352	Binomial	BirdID+Terr

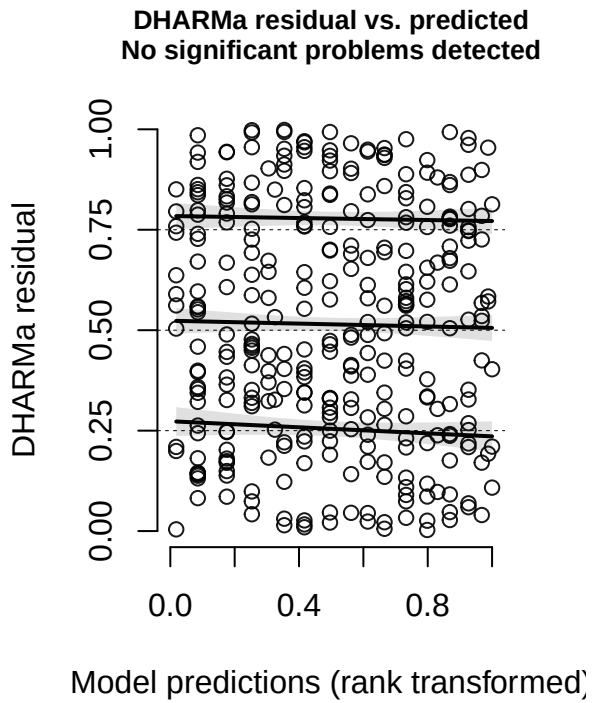
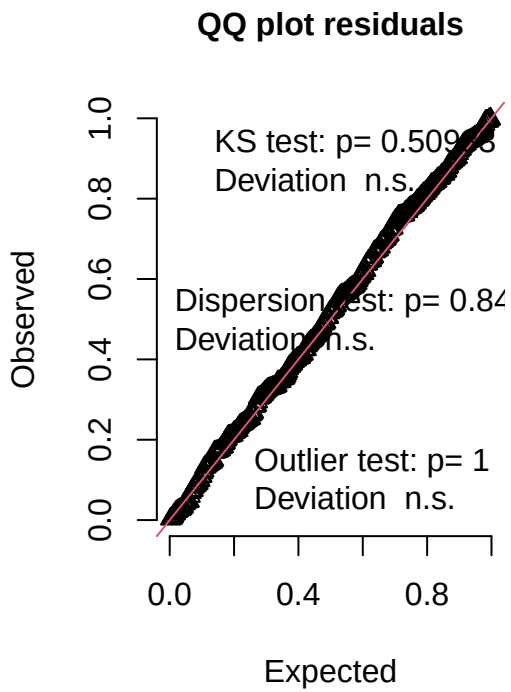
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-0.7680900	1.922999e+00	-0.3994229	0.6895816	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	35.1404762	3.355443e+07	0.0000010	0.9999992	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	23.3172782	2.374990e+05	0.0000982	0.9999217	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-2.6458301	1.762481e+00	-1.5011963	0.1333048	Binomial	BirdID+Terr

Similar to other binomial models.

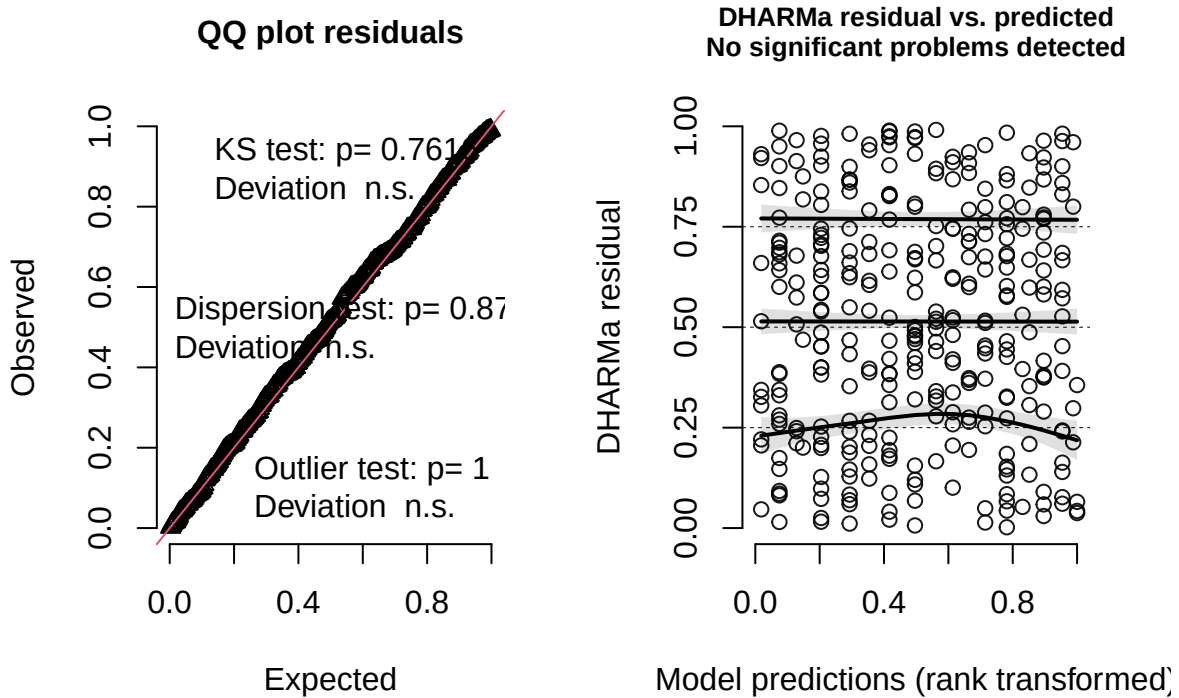
DHARMa residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Probabilites used to generate simulation 35

Simulation 35
 Probability of mate fidelity: 0.584
 Probability of pair dispersal: 0.839
 Probability of single male dispersal: 0.931
 Probability of single female dispersal: 0.654

Simulation 36

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
369	1	-474.9290	993.8580	1079.896	295.0319	347	0.9348906	0.9050156	Poisson	BirdID+Terr
369	1	-477.5558	997.1115	1079.238	330.3635	348	0.9236923	0.8929489	Poisson	BirdID
369	1	-478.7182	999.4364	1081.563	343.9280	348	0.9290591	0.9048004	Poisson	Terr

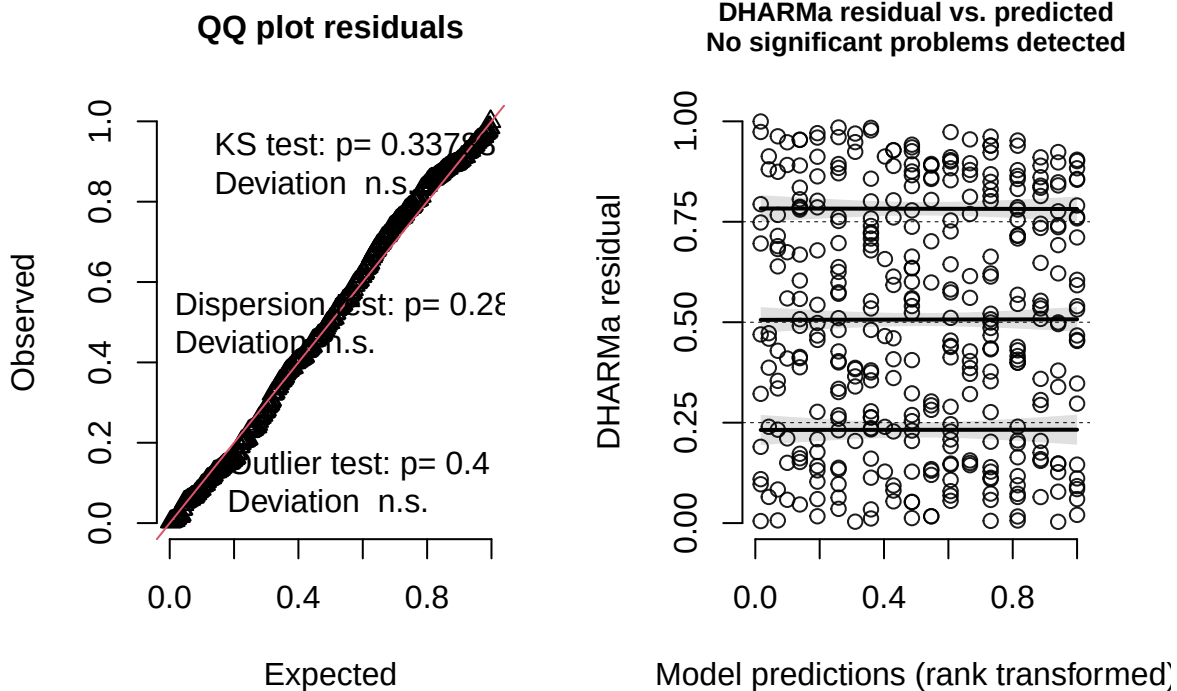
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.5156138	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.4653689	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.4109007	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.3583775	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	-0.0161116	0.2650683	-0.0607828	0.9515322	Poisson	BirdID
fixed	factor(fac.yr)2	0.3451858	0.3162795	1.0913949	0.2750992	Poisson	BirdID
fixed	factor(fac.yr)3	-0.2123919	0.3387814	-0.6269289	0.5307058	Poisson	BirdID
fixed	factor(fac.yr)4	0.0144483	0.3246686	0.0445016	0.9645046	Poisson	BirdID
fixed	factor(fac.yr)5	0.1570851	0.3095902	0.5073969	0.6118764	Poisson	BirdID
fixed	factor(fac.yr)6	0.1047168	0.3148827	0.3325580	0.7394679	Poisson	BirdID
fixed	factor(fac.yr)7	-0.3941822	0.3512010	-1.1223834	0.2616995	Poisson	BirdID
fixed	factor(fac.yr)8	-0.0192036	0.3302879	-0.0581421	0.9536355	Poisson	BirdID
fixed	factor(fac.yr)9	0.5258048	0.3153550	1.6673422	0.0954464	Poisson	BirdID
fixed	factor(fac.yr)10	0.3340197	0.3275211	1.0198416	0.3078036	Poisson	BirdID
fixed	factor(fac.yr)11	-0.6366226	0.4053762	-1.5704487	0.1163108	Poisson	BirdID
fixed	factor(fac.yr)12	-0.3965541	0.3839529	-1.0328195	0.3016884	Poisson	BirdID
fixed	factor(fac.yr)13	0.0908998	0.3583637	0.2536525	0.7997640	Poisson	BirdID
fixed	factor(fac.yr)14	-0.0549834	0.3876667	-0.1418316	0.8872130	Poisson	BirdID
fixed	factor(fac.yr)15	-1.2742213	0.5903098	-2.1585636	0.0308840	Poisson	BirdID
fixed	factor(fac.yr)16	-16.2329033	988.5377210	-0.0164211	0.9868984	Poisson	BirdID
fixed	factor(fac.yr)17	-0.7374384	0.5503141	-1.3400318	0.1802350	Poisson	BirdID
fixed	factor(fac.yr)18	-0.3065512	0.4752431	-0.6450407	0.5189008	Poisson	BirdID
fixed	factor(fac.yr)19	-1.5187671	0.7865105	-1.9310195	0.0534806	Poisson	BirdID
fixed	factor(fac.yr)20	-0.1518596	0.5736915	-0.2647060	0.7912360	Poisson	BirdID
fixed	(Intercept)	0.0534265	0.2572173	0.2077095	0.8354558	Poisson	Terr
fixed	factor(fac.yr)2	0.3217335	0.3140341	1.0245179	0.3055907	Poisson	Terr
fixed	factor(fac.yr)3	-0.2845166	0.3413169	-0.8335850	0.4045149	Poisson	Terr
fixed	factor(fac.yr)4	-0.0402068	0.3207601	-0.1253486	0.9002476	Poisson	Terr
fixed	factor(fac.yr)5	0.1988589	0.3031793	0.6559120	0.5118807	Poisson	Terr
fixed	factor(fac.yr)6	0.1255323	0.3107957	0.4039063	0.6862816	Poisson	Terr
fixed	factor(fac.yr)7	-0.3967032	0.3438497	-1.1537111	0.2486186	Poisson	Terr
fixed	factor(fac.yr)8	-0.0292205	0.3212804	-0.0909501	0.9275322	Poisson	Terr
fixed	factor(fac.yr)9	0.4623529	0.3053460	1.5141932	0.1299769	Poisson	Terr
fixed	factor(fac.yr)10	0.4477697	0.3120231	1.4350532	0.1512719	Poisson	Terr
fixed	factor(fac.yr)11	-0.5920844	0.3963903	-1.4936904	0.1352566	Poisson	Terr
fixed	factor(fac.yr)12	-0.3455956	0.3739361	-0.9242103	0.3553768	Poisson	Terr
fixed	factor(fac.yr)13	0.2534801	0.3425592	0.7399600	0.4593243	Poisson	Terr
fixed	factor(fac.yr)14	-0.0544205	0.3650837	-0.1490630	0.8815039	Poisson	Terr
fixed	factor(fac.yr)15	-1.2185061	0.5755439	-2.1171382	0.0342481	Poisson	Terr
fixed	factor(fac.yr)16	-16.6039235	1211.2850985	-0.0137077	0.9890632	Poisson	Terr
fixed	factor(fac.yr)17	-0.5859707	0.5330442	-1.0992909	0.2716412	Poisson	Terr
fixed	factor(fac.yr)18	-0.2753793	0.4556475	-0.6043692	0.5455982	Poisson	Terr
fixed	factor(fac.yr)19	-1.3327760	0.7729449	-1.7242832	0.0846567	Poisson	Terr
fixed	factor(fac.yr)20	-0.0684141	0.5505628	-0.1242621	0.9011077	Poisson	Terr
fixed	(Intercept)	-0.0198828	0.2706967	-0.0734506	0.9414476	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	0.3180219	0.3210124	0.9906841	0.3218399	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.2410861	0.3499107	-0.6889933	0.4908275	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.0224420	0.3331908	-0.0673547	0.9462993	Poisson	BirdID+Terr

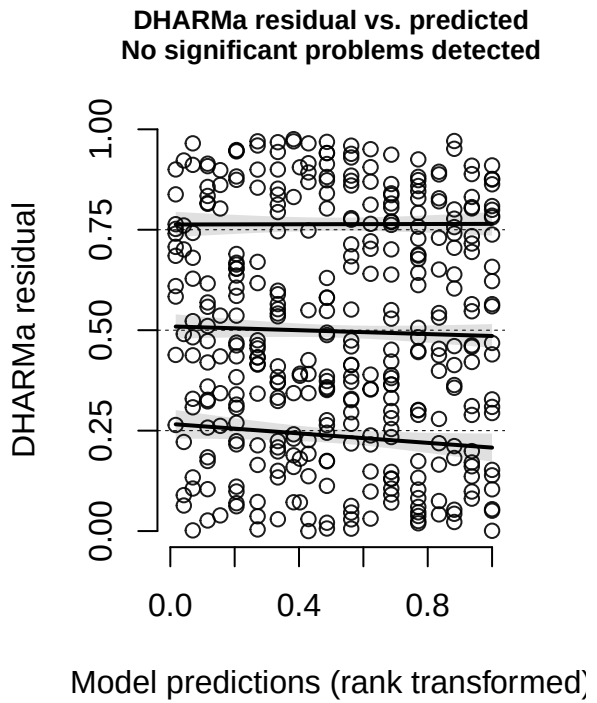
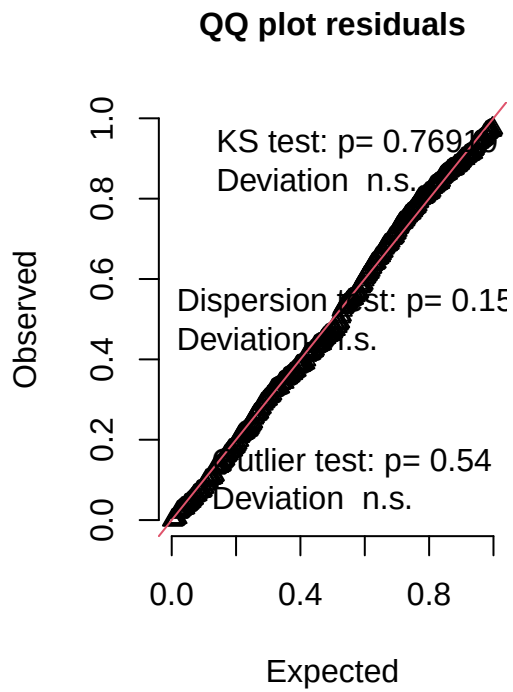
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.2075397	0.3165840	0.6555595	0.5121076	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	0.1295668	0.3238684	0.4000601	0.6891123	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.3663076	0.3578465	-1.0236444	0.3060033	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.0101086	0.3376930	-0.0299342	0.9761195	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	0.5189774	0.3217384	1.6130417	0.1067354	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	0.4085202	0.3321915	1.2297732	0.2187821	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.5912646	0.4127564	-1.4324782	0.1520070	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.3568185	0.3896386	-0.9157676	0.3597888	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	0.1685507	0.3646181	0.4622664	0.6438903	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.0729317	0.3900695	-0.1869711	0.8516833	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-1.2120653	0.5968139	-2.0308934	0.0422658	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-17.5395494	1963.6807803	-0.0089320	0.9928734	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.6724911	0.5575510	-1.2061515	0.2277591	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.3058756	0.4831015	-0.6331498	0.5266358	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.4496973	0.7923442	-1.8296308	0.0673052	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.1025609	0.5797968	-0.1768912	0.8595939	Poisson	BirdID+Terr

The territory only model seems to be the weakest on the criteria.

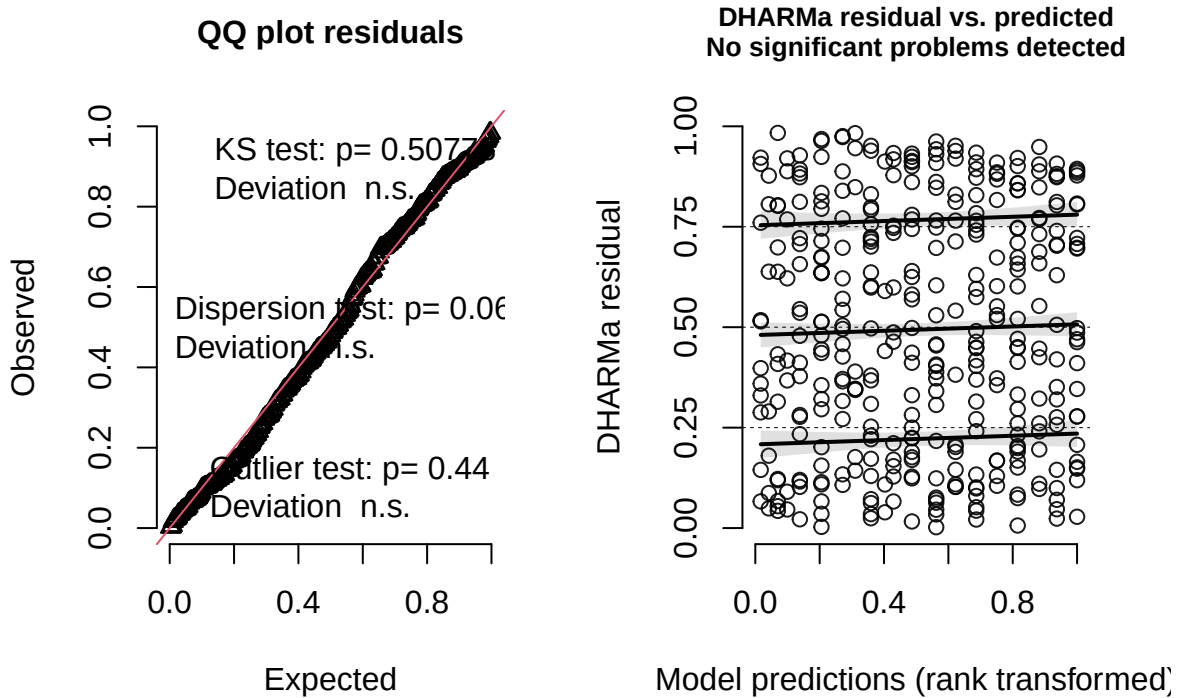
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
369	1	-213.5096	471.0192	557.0567	267.8000	347	0.748	Binomial	BirdID+Terr
369	1	-215.5742	473.1484	555.2751	310.4050	348	0.668	Binomial	BirdID
369	1	-218.6050	479.2099	561.3366	331.2375	348	0.554	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.3477179	Binomial	BirdID
ran_pars	Terr	sd__(Intercept)	1.1822321	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.2013054	Binomial	BirdID+Terr
ran_pars	Terr	sd__(Intercept)	0.9675987	Binomial	BirdID+Terr

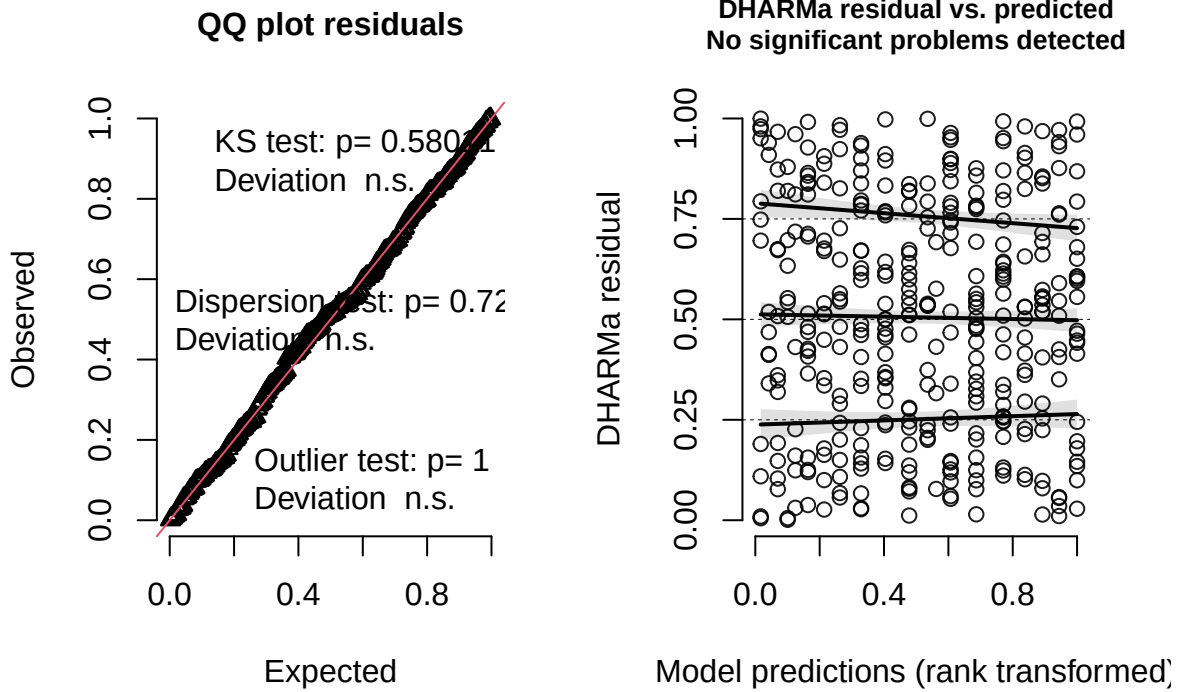
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	-0.1862377	6.066624e-01	-0.3069873	0.7588530	Binomial	BirdID
fixed	factor(fac.yr)2	1.1089308	8.180794e-01	1.3555295	0.1752489	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)3	0.4468346	7.365318e-01	0.6066739	0.5440673	Binomial	BirdID
fixed	factor(fac.yr)4	0.6048310	7.477954e-01	0.8088189	0.4186193	Binomial	BirdID
fixed	factor(fac.yr)5	1.0516976	7.431460e-01	1.4151966	0.1570109	Binomial	BirdID
fixed	factor(fac.yr)6	1.0894514	7.505526e-01	1.4515325	0.1466316	Binomial	BirdID
fixed	factor(fac.yr)7	0.1332518	7.466231e-01	0.1784727	0.8583518	Binomial	BirdID
fixed	factor(fac.yr)8	1.0214955	7.778568e-01	1.3132180	0.1891095	Binomial	BirdID
fixed	factor(fac.yr)9	1.6391300	8.267107e-01	1.9827130	0.0473995	Binomial	BirdID
fixed	factor(fac.yr)10	1.4811958	8.660399e-01	1.7103090	0.0872088	Binomial	BirdID
fixed	factor(fac.yr)11	-0.0264917	8.237430e-01	-0.0321601	0.9743444	Binomial	BirdID
fixed	factor(fac.yr)12	0.0506570	8.255942e-01	0.0613583	0.9510739	Binomial	BirdID
fixed	factor(fac.yr)13	1.5771468	9.421186e-01	1.6740427	0.0941222	Binomial	BirdID
fixed	factor(fac.yr)14	0.6132369	9.050475e-01	0.6775743	0.4980416	Binomial	BirdID
fixed	factor(fac.yr)15	-1.1564917	1.009506e+00	-1.1456011	0.2519602	Binomial	BirdID
fixed	factor(fac.yr)16	-	6.290304e+05	-0.0000461	0.9999632	Binomial	BirdID
		28.9939237					
fixed	factor(fac.yr)17	-0.7703219	1.084811e+00	-0.7100975	0.4776437	Binomial	BirdID
fixed	factor(fac.yr)18	-0.8137831	1.066097e+00	-0.7633295	0.4452669	Binomial	BirdID
fixed	factor(fac.yr)19	-1.2515797	1.195673e+00	-1.0467573	0.2952115	Binomial	BirdID
fixed	factor(fac.yr)20	0.8152770	1.300488e+00	0.6269008	0.5307243	Binomial	BirdID
fixed	(Intercept)	-0.0292299	5.694784e-01	-0.0513275	0.9590645	Binomial	Terr
fixed	factor(fac.yr)2	1.1716259	7.958561e-01	1.4721580	0.1409782	Binomial	Terr
fixed	factor(fac.yr)3	0.4583770	7.251685e-01	0.6320972	0.5273234	Binomial	Terr
fixed	factor(fac.yr)4	0.6434481	7.214587e-01	0.8918709	0.3724621	Binomial	Terr
fixed	factor(fac.yr)5	1.2043647	7.087864e-01	1.6991927	0.0892829	Binomial	Terr
fixed	factor(fac.yr)6	1.1445216	7.131351e-01	1.6049156	0.1085124	Binomial	Terr
fixed	factor(fac.yr)7	0.1768187	7.091069e-01	0.2493541	0.8030869	Binomial	Terr
fixed	factor(fac.yr)8	0.8220826	7.306089e-01	1.1252020	0.2605035	Binomial	Terr
fixed	factor(fac.yr)9	1.5121437	7.722808e-01	1.9580232	0.0502273	Binomial	Terr
fixed	factor(fac.yr)10	1.8112693	8.079408e-01	2.2418343	0.0249721	Binomial	Terr

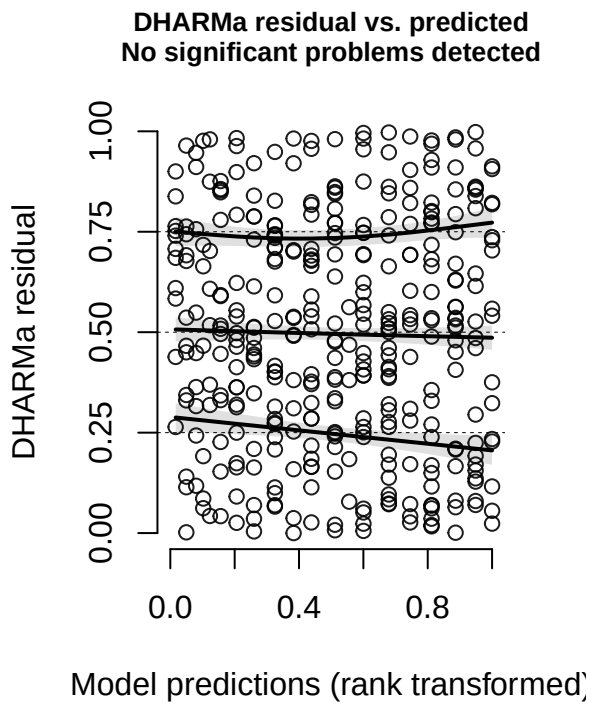
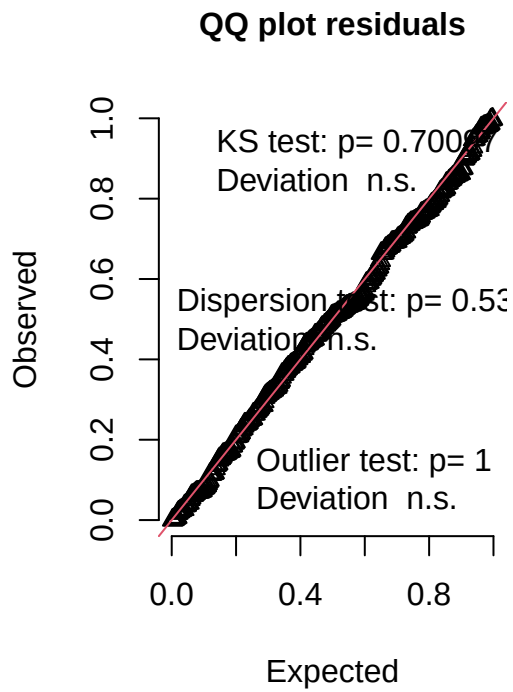
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)11	0.0572064	7.606130e-01	0.0752109	0.9400469	Binomial	Terr
fixed	factor(fac.yr)12	0.1620076	7.670365e-01	0.2112124	0.8327215	Binomial	Terr
fixed	factor(fac.yr)13	1.8138272	8.702695e-01	2.0842132	0.0371408	Binomial	Terr
fixed	factor(fac.yr)14	0.3585147	8.178268e-01	0.4383748	0.6611146	Binomial	Terr
fixed	factor(fac.yr)15	-1.3341689	9.404824e-01	-1.4186006	0.1560155	Binomial	Terr
fixed	factor(fac.yr)16	-28.1765913	4.070178e+05	-0.0000692	0.9999448	Binomial	Terr
fixed	factor(fac.yr)17	-0.6201864	1.034762e+00	-0.5993516	0.5489385	Binomial	Terr
fixed	factor(fac.yr)18	-1.0344961	9.922223e-01	-1.0426052	0.2971312	Binomial	Terr
fixed	factor(fac.yr)19	-1.0313101	1.093119e+00	-0.9434563	0.3454475	Binomial	Terr
fixed	factor(fac.yr)20	0.6690445	1.211137e+00	0.5524103	0.5806673	Binomial	Terr
fixed	(Intercept)	-0.1427540	6.345832e-01	-0.2249571	0.8220126	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	1.1694090	8.497039e-01	1.3762547	0.1687428	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	0.5012078	7.799907e-01	0.6425818	0.5204955	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	0.6236676	7.842651e-01	0.7952254	0.4264824	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	1.2965594	7.747543e-01	1.6735105	0.0942268	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	1.2814759	7.898507e-01	1.6224279	0.1047118	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	0.1832923	7.834147e-01	0.2339659	0.8150114	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	1.0594030	8.207310e-01	1.2908042	0.1967716	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	1.8443618	8.650099e-01	2.1321857	0.0329916	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	1.8542870	9.071799e-01	2.0440124	0.0409523	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	0.0784539	8.600065e-01	0.0912248	0.9273140	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	0.1524256	8.568638e-01	0.1778878	0.8588111	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	1.9819070	9.891269e-01	2.0036933	0.0451029	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	0.6110786	9.280979e-01	0.6584205	0.5102680	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-1.2793305	1.063023e+00	-1.2034831	0.2287894	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-27.2277501	2.767680e+05	-0.0000984	0.9999215	Binomial	BirdID+Terr
fixed	factor(fac.yr)17	-0.8424783	1.151134e+00	-0.7318683	0.4642490	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-1.0286706	1.116682e+00	-0.9211848	0.3569540	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-1.4127166	1.256065e+00	-1.1247161	0.2607094	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	0.8310420	1.398980e+00	0.5940343	0.5524892	Binomial	BirdID+Terr

Similar to most other binomial models.

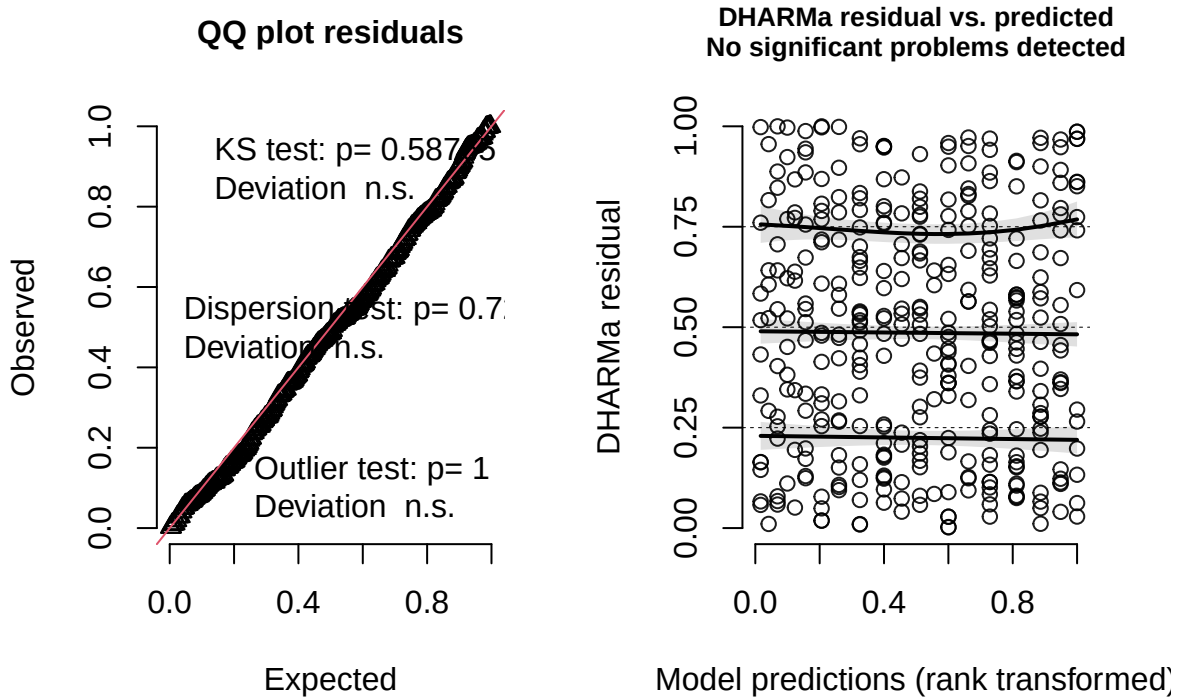
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Probabilites used to generate simulation 36

Simulation 36
 Probability of mate fidelity: 0.168
 Probability of pair dispersal: 0.759
 Probability of single male dispersal: 0.159
 Probability of single female dispersal: 0.782

Simulation 37

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
354	1	-454.6499	951.2998	1032.555	338.2725	333	0.3089073	0.1824884	Poisson	Terr
354	1	-454.6209	953.2417	1038.366	333.6227	332	0.3084341	0.1784212	Poisson	BirdID+Terr
354	1	-456.8980	955.7961	1037.051	351.7194	333	0.2866533	0.1818841	Poisson	BirdID

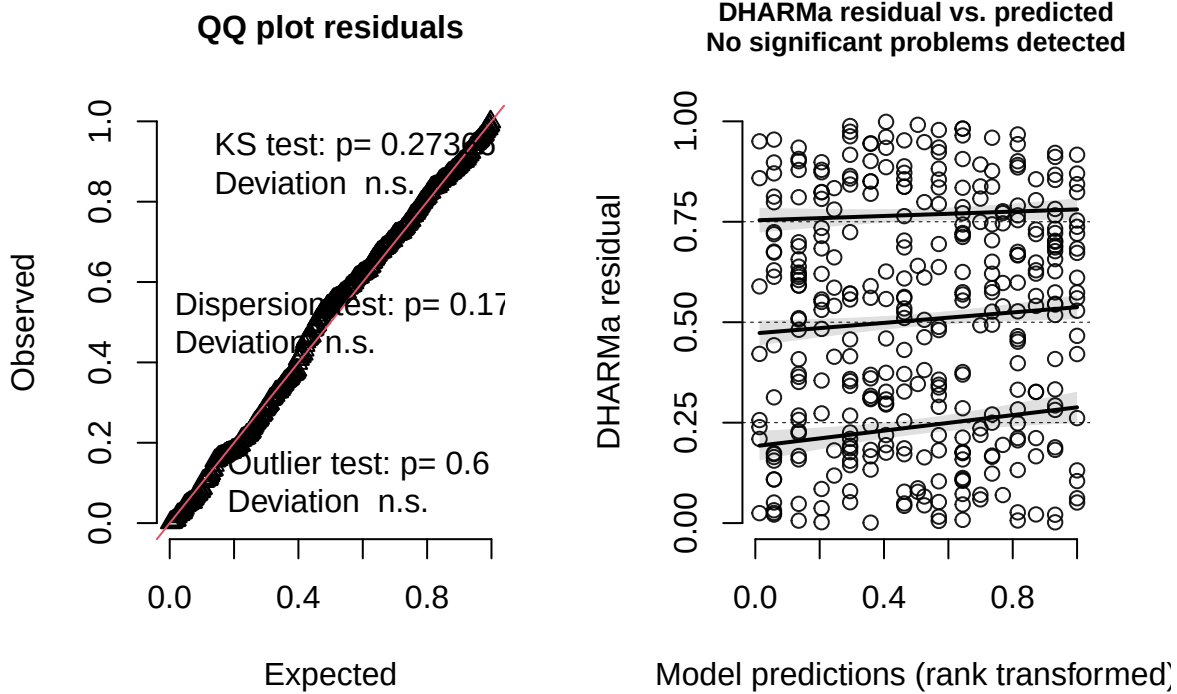
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.3161710	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.3446117	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.1143107	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.3335929	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	-0.1739229	0.2933982	-0.5927880	0.5533230	Poisson	BirdID
fixed	factor(fac.yr)2	-0.1385728	0.3976581	-0.3484721	0.7274856	Poisson	BirdID
fixed	factor(fac.yr)3	-0.2787572	0.4097151	-0.6803685	0.4962712	Poisson	BirdID
fixed	factor(fac.yr)4	-0.2688313	0.3821921	-0.7033932	0.4818107	Poisson	BirdID
fixed	factor(fac.yr)5	0.7873902	0.3394772	2.3194203	0.0203723	Poisson	BirdID
fixed	factor(fac.yr)6	0.3980579	0.3450505	1.1536222	0.2486551	Poisson	BirdID
fixed	factor(fac.yr)7	-0.3570632	0.3773997	-0.9461140	0.3440904	Poisson	BirdID
fixed	factor(fac.yr)8	0.1886243	0.3479166	0.5421537	0.5877126	Poisson	BirdID
fixed	factor(fac.yr)9	0.7255235	0.3304492	2.1955677	0.0281229	Poisson	BirdID
fixed	factor(fac.yr)10	0.0857114	0.3633066	0.2359204	0.8134945	Poisson	BirdID
fixed	factor(fac.yr)11	-0.3584200	0.4036679	-0.8879083	0.3745901	Poisson	BirdID
fixed	factor(fac.yr)12	0.1539602	0.3696977	0.4164488	0.6770816	Poisson	BirdID
fixed	factor(fac.yr)13	0.5366770	0.3748628	1.4316624	0.1522405	Poisson	BirdID
fixed	factor(fac.yr)14	0.2939308	0.3810906	0.7712886	0.4405359	Poisson	BirdID
fixed	factor(fac.yr)15	-0.2689237	0.5489768	-0.4898636	0.6242304	Poisson	BirdID
fixed	factor(fac.yr)16	0.3663864	0.4595312	0.7973049	0.4252740	Poisson	BirdID
fixed	factor(fac.yr)17	0.1161466	0.4575572	0.2538407	0.7996186	Poisson	BirdID
fixed	factor(fac.yr)18	0.2167300	0.4465859	0.4853041	0.6274607	Poisson	BirdID
fixed	factor(fac.yr)19	-1.2852347	0.7833958	-1.6405944	0.1008816	Poisson	BirdID
fixed	factor(fac.yr)20	0.0962213	0.5578873	0.1724744	0.8630646	Poisson	BirdID
fixed	(Intercept)	-0.1453185	0.2930087	-0.4959527	0.6199278	Poisson	Terr
fixed	factor(fac.yr)2	-0.1386395	0.4009381	-0.3457877	0.7295023	Poisson	Terr
fixed	factor(fac.yr)3	-0.2444268	0.4086202	-0.5981761	0.5497225	Poisson	Terr
fixed	factor(fac.yr)4	-0.3000738	0.3812246	-0.7871314	0.4312049	Poisson	Terr
fixed	factor(fac.yr)5	0.7829956	0.3405584	2.2991520	0.0214963	Poisson	Terr
fixed	factor(fac.yr)6	0.3984587	0.3446035	1.1562817	0.2475659	Poisson	Terr
fixed	factor(fac.yr)7	-0.3837713	0.3758849	-1.0209808	0.3072635	Poisson	Terr
fixed	factor(fac.yr)8	0.1573130	0.3440859	0.4571912	0.6475336	Poisson	Terr
fixed	factor(fac.yr)9	0.7204996	0.3270181	2.2032404	0.0275778	Poisson	Terr
fixed	factor(fac.yr)10	0.2110287	0.3616362	0.5835387	0.5595307	Poisson	Terr
fixed	factor(fac.yr)11	-0.3501867	0.4001010	-0.8752458	0.3814402	Poisson	Terr
fixed	factor(fac.yr)12	0.1231432	0.3680428	0.3345893	0.7379349	Poisson	Terr
fixed	factor(fac.yr)13	0.5481537	0.3756093	1.4593721	0.1444627	Poisson	Terr
fixed	factor(fac.yr)14	0.3597506	0.3790720	0.9490297	0.3426055	Poisson	Terr
fixed	factor(fac.yr)15	-0.2407846	0.5489377	-0.4386374	0.6609243	Poisson	Terr
fixed	factor(fac.yr)16	0.4242605	0.4601233	0.9220583	0.3564982	Poisson	Terr
fixed	factor(fac.yr)17	0.1346371	0.4587616	0.2934795	0.7691556	Poisson	Terr
fixed	factor(fac.yr)18	0.1456300	0.4414982	0.3298541	0.7415102	Poisson	Terr
fixed	factor(fac.yr)19	-1.1918052	0.7834674	-1.5211931	0.1282114	Poisson	Terr
fixed	factor(fac.yr)20	0.1303465	0.5501632	0.2369233	0.8127163	Poisson	Terr
fixed	(Intercept)	-0.1512549	0.2943352	-0.5138865	0.6073314	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.1428093	0.4020399	-0.3552119	0.7224309	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.2481415	0.4105191	-0.6044578	0.5455393	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.2999579	0.3828574	-0.7834717	0.4333502	Poisson	BirdID+Terr

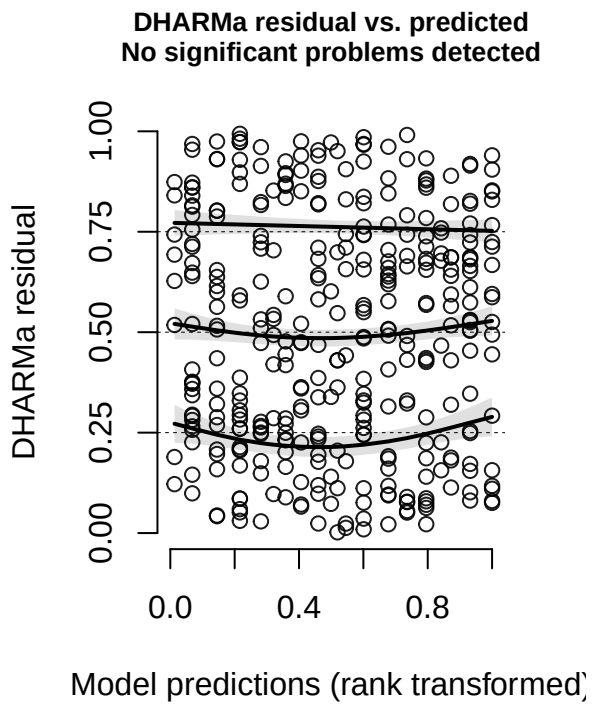
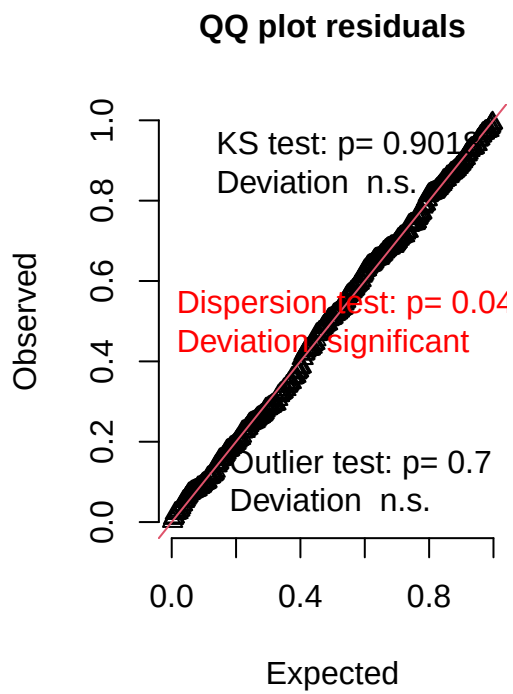
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.7788495	0.3422900	2.2754082	0.0228815	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	0.3963758	0.3462985	1.1446072	0.2523719	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.3872208	0.3775868	-1.0255147	0.3051204	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	0.1532140	0.3459983	0.4428172	0.6578980	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	0.7175098	0.3289524	2.1811968	0.0291689	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	0.1990278	0.3634720	0.5475739	0.5839845	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.3508025	0.4020244	-0.8725900	0.3828866	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	0.1261715	0.3698758	0.3411187	0.7330142	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	0.5450497	0.3775727	1.4435621	0.1488622	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	0.3503635	0.3811424	0.9192456	0.3579671	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.2364495	0.5509132	-0.4291956	0.6677809	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	0.4195158	0.4627423	0.9065861	0.3646257	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	0.1299025	0.4611991	0.2816626	0.7782022	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	0.1467934	0.4442446	0.3304337	0.7410723	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.1936858	0.7840091	-1.5225408	0.1278736	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	0.1216837	0.5538464	0.2197065	0.8260997	Poisson	BirdID+Terr

Territory only model is performing the best on all criteria.

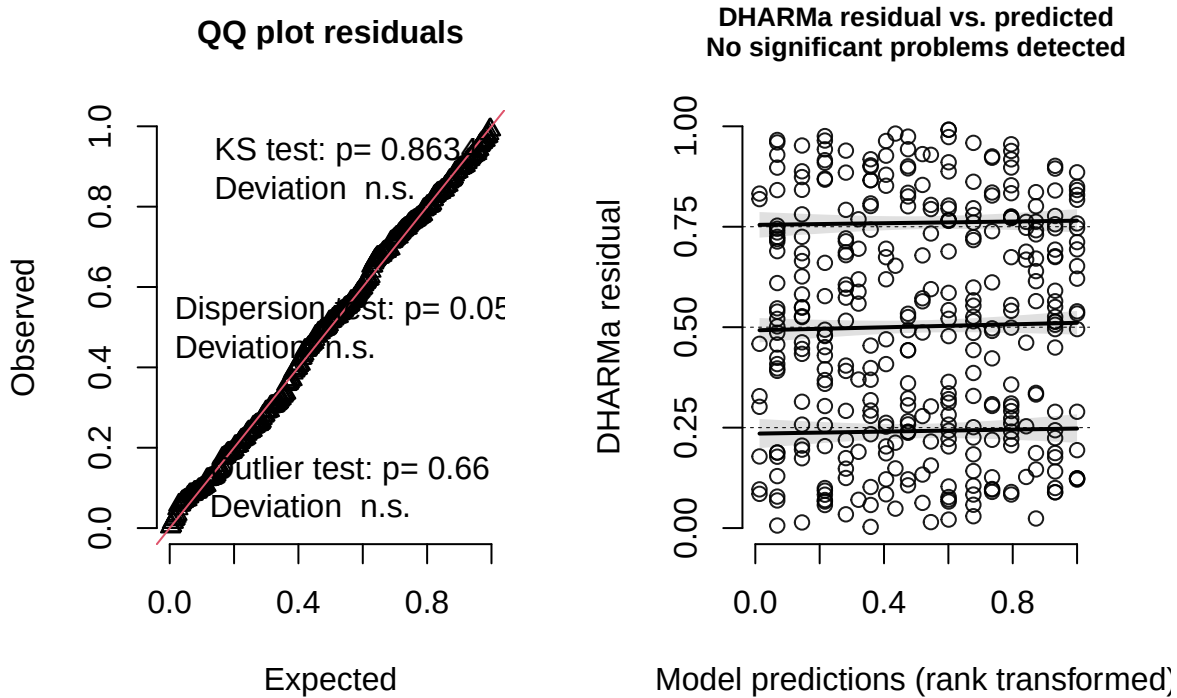
DHARMa residual



DHARMa residual



DHARMa residual



The low p value on the dispersion test is mildly alarming and suggests that perhaps a negative binomial response would be a better fit.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
354	1	-211.3427	466.6854	551.8099	283.6815	332	0.696	Binomial	BirdID+Terr
354	1	-213.2595	468.5191	549.7743	323.0075	333	0.595	Binomial	BirdID
354	1	-214.2017	470.4034	551.6586	336.4659	333	0.581	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.1495355	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.0400446	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.0021279	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.8795409	Binomial	BirdID+Terr

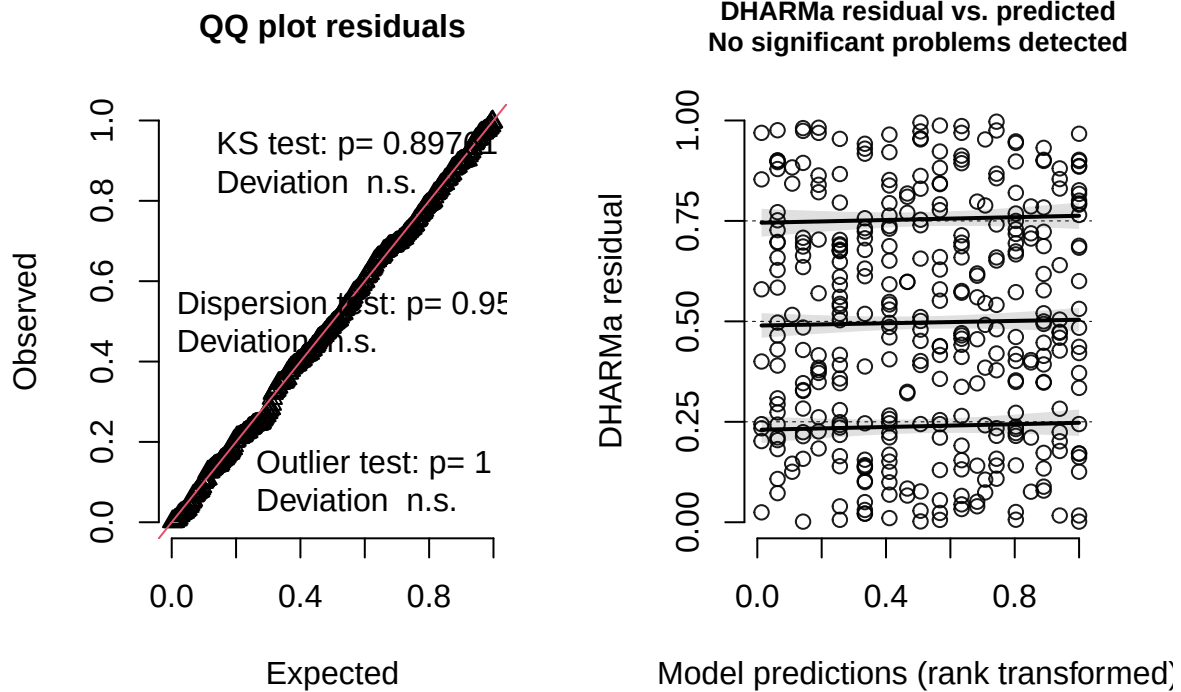
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	-0.1780950	0.6016771	-0.2959976	0.7672319	Binomial	BirdID
fixed	factor(fac.yr)2	-0.2001526	0.7591817	-0.2636426	0.7920554	Binomial	BirdID
fixed	factor(fac.yr)3	0.3697745	0.7859999	0.4704511	0.6380327	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)4	-0.8075325	0.7622400	-1.0594202	0.2894084	Binomial	BirdID
fixed	factor(fac.yr)5	1.7342146	0.8829276	1.9641639	0.0495111	Binomial	BirdID
fixed	factor(fac.yr)6	1.2482995	0.7862765	1.5876089	0.1123748	Binomial	BirdID
fixed	factor(fac.yr)7	0.1273126	0.7325301	0.1737985	0.8620238	Binomial	BirdID
fixed	factor(fac.yr)8	0.3017758	0.7433021	0.4059935	0.6847474	Binomial	BirdID
fixed	factor(fac.yr)9	2.2164527	0.8770895	2.5270543	0.0115024	Binomial	BirdID
fixed	factor(fac.yr)10	0.9145481	0.7851575	1.1647958	0.2441016	Binomial	BirdID
fixed	factor(fac.yr)11	0.1301695	0.7764821	0.1676401	0.8668664	Binomial	BirdID
fixed	factor(fac.yr)12	0.5814807	0.7882915	0.7376467	0.4607292	Binomial	BirdID
fixed	factor(fac.yr)13	1.9354055	0.9901699	1.9546197	0.0506280	Binomial	BirdID
fixed	factor(fac.yr)14	1.1902593	0.8896587	1.3378831	0.1809345	Binomial	BirdID
fixed	factor(fac.yr)15	1.1835949	1.0912135	1.0846594	0.2780726	Binomial	BirdID
fixed	factor(fac.yr)16	1.3048118	1.2042760	1.0834824	0.2785944	Binomial	BirdID
fixed	factor(fac.yr)17	1.0422920	1.0610618	0.9823104	0.3259470	Binomial	BirdID
fixed	factor(fac.yr)18	0.3450295	1.0112517	0.3411905	0.7329602	Binomial	BirdID
fixed	factor(fac.yr)19	-1.3765641	1.1542158	-1.1926402	0.2330103	Binomial	BirdID
fixed	factor(fac.yr)20	-0.3576592	1.2312028	-0.2904958	0.7714370	Binomial	BirdID
fixed	(Intercept)	-0.0518797	0.5901352	-0.0879116	0.9299469	Binomial	Terr
fixed	factor(fac.yr)2	-0.0977189	0.7758463	-0.1259514	0.8997704	Binomial	Terr
fixed	factor(fac.yr)3	0.5235983	0.7714959	0.6786794	0.4973410	Binomial	Terr
fixed	factor(fac.yr)4	-0.8578092	0.7421412	-1.1558571	0.2477396	Binomial	Terr
fixed	factor(fac.yr)5	1.7825641	0.8701309	2.0486161	0.0404997	Binomial	Terr
fixed	factor(fac.yr)6	1.1416435	0.7674010	1.4876752	0.1368366	Binomial	Terr
fixed	factor(fac.yr)7	-0.0342306	0.7068897	-0.0484242	0.9613782	Binomial	Terr
fixed	factor(fac.yr)8	0.2427466	0.7084767	0.3426317	0.7318755	Binomial	Terr
fixed	factor(fac.yr)9	2.1777305	0.8419447	2.5865483	0.0096943	Binomial	Terr
fixed	factor(fac.yr)10	1.2913211	0.7606036	1.6977582	0.0895534	Binomial	Terr
fixed	factor(fac.yr)11	0.2721249	0.7427872	0.3663565	0.7140991	Binomial	Terr
fixed	factor(fac.yr)12	0.4829602	0.7631002	0.6328922	0.5268040	Binomial	Terr
fixed	factor(fac.yr)13	1.9480882	0.9603289	2.0285636	0.0425028	Binomial	Terr
fixed	factor(fac.yr)14	1.4547444	0.8490224	1.7134346	0.0866326	Binomial	Terr
fixed	factor(fac.yr)15	1.0528035	1.0278040	1.0243232	0.3056826	Binomial	Terr
fixed	factor(fac.yr)16	1.3959559	1.1340569	1.2309398	0.2183454	Binomial	Terr
fixed	factor(fac.yr)17	1.1933187	0.9969951	1.1969154	0.2313395	Binomial	Terr
fixed	factor(fac.yr)18	0.3114437	0.9571610	0.3253828	0.7448914	Binomial	Terr
fixed	factor(fac.yr)19	-0.9307792	1.0943003	-0.8505702	0.3950081	Binomial	Terr
fixed	factor(fac.yr)20	-0.5560461	1.1882677	-0.4679469	0.6398226	Binomial	Terr
fixed	(Intercept)	0.0374708	0.6378690	0.0587437	0.9531563	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-0.4101069	0.8053283	-0.5092418	0.6105827	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	0.3260112	0.8219152	0.3966482	0.6916269	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-1.1420586	0.8013920	-1.4250936	0.1541302	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	1.6399913	0.9300746	1.7632901	0.0778516	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	1.1342326	0.8251079	1.3746476	0.1692407	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-0.1529234	0.7667925	-0.1994326	0.8419243	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	0.0841705	0.7737555	0.1087817	0.9133756	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	2.1835941	0.9154640	2.3852322	0.0170683	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	1.0601639	0.8247939	1.2853682	0.1986636	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	0.0963801	0.8047893	0.1197582	0.9046747	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	0.4687189	0.8335063	0.5623460	0.5738803	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	2.0296339	1.0601098	1.9145507	0.0555498	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	1.3396027	0.9384720	1.4274296	0.1534561	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	1.1449916	1.1556727	0.9907577	0.3218039	Binomial	BirdID+Terr

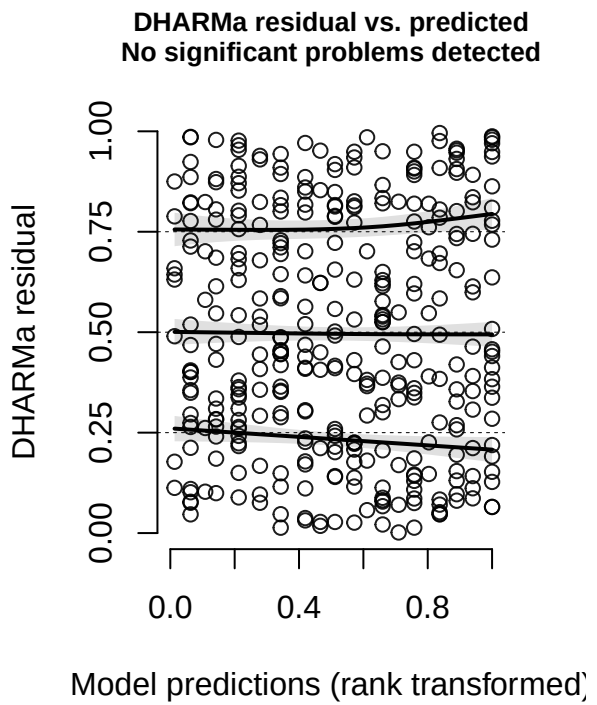
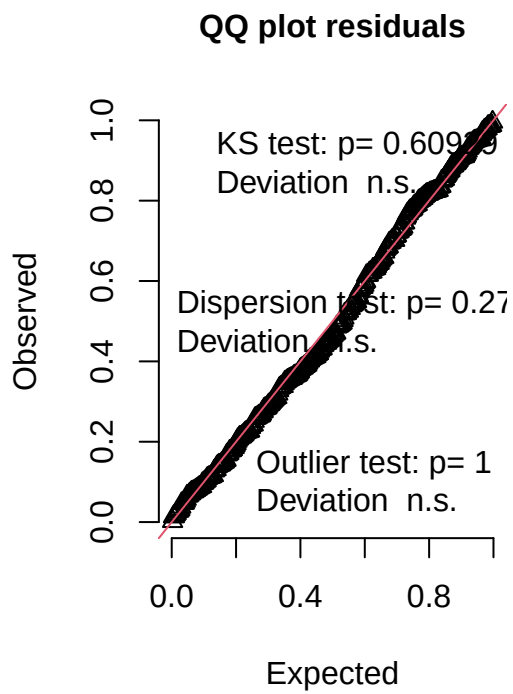
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)16	1.3414459	1.2890056	1.0406828	0.2980228	Binomial	BirdID+Terr
fixed	factor(fac.yr)17	1.0170958	1.1136170	0.9133264	0.3610709	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	0.1053898	1.0665431	0.0988144	0.9212856	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-1.4345013	1.2163220	-1.1793763	0.2382484	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-0.6646733	1.2971592	-0.5124068	0.6083663	Binomial	BirdID+Terr

Similar to the other binomial models.

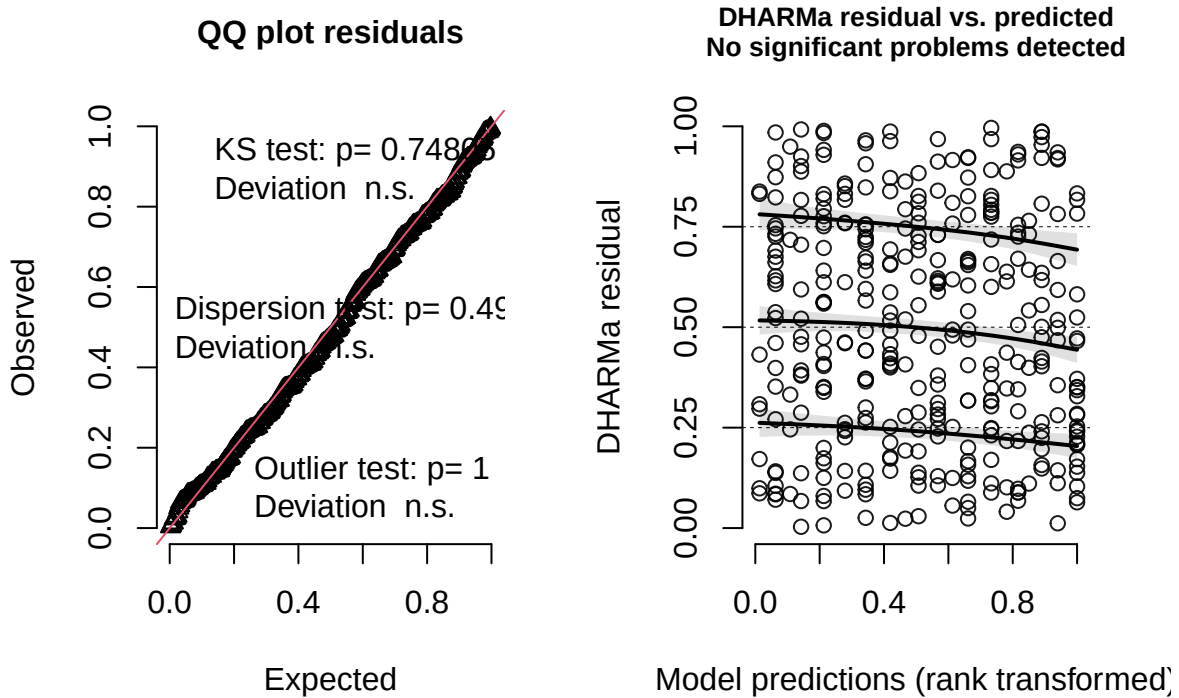
DHARMa residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Probabilities used to generate simulation 37

Simulation 37
 Probability of mate fidelity: 0.518
 Probability of pair dispersal: 0.274
 Probability of single male dispersal: 0.831
 Probability of single female dispersal: 0.493

Simulation 38

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Family	RE
327	1	-459.4471	960.8942	1040.483	313.0399	306	Poisson	Terr
327	1	-459.4471	962.8942	1046.273	313.1002	305	Poisson	BirdID+Terr
327	1	-463.1592	968.3184	1047.908	380.7317	306	Poisson	BirdID

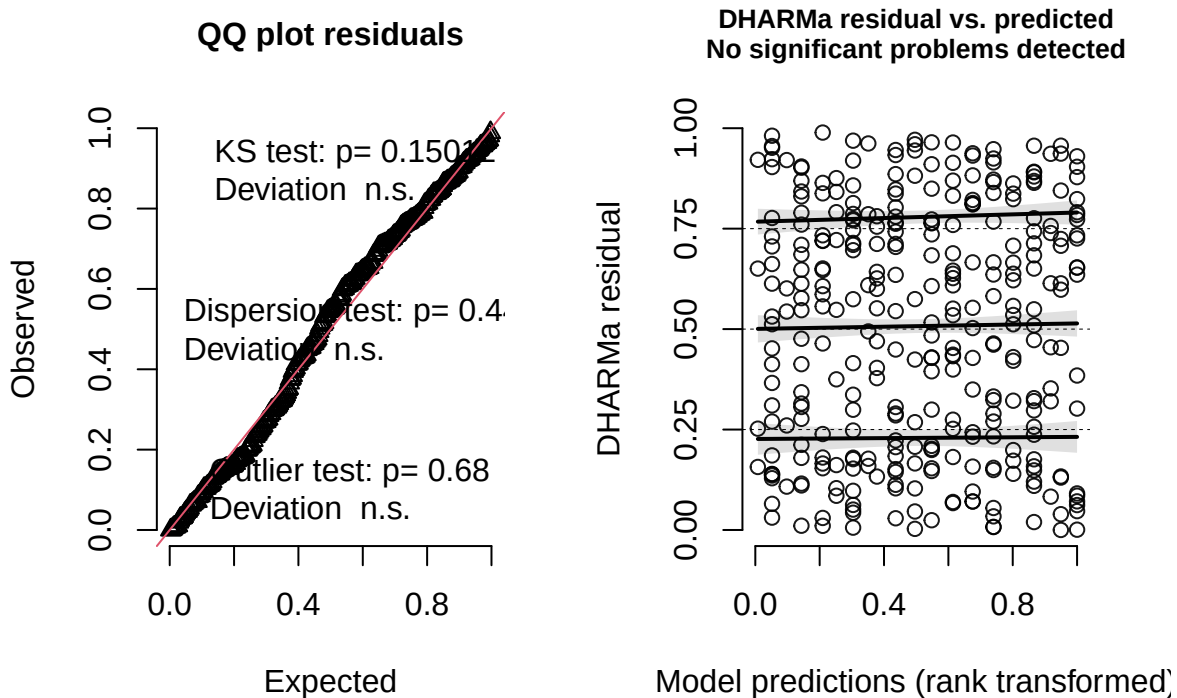
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.0000000	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.3163433	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.0002333	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.3160785	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.5108256	0.2236068	2.2844816	0.0223432	Poisson	BirdID
fixed	factor(fac.yr)2	-0.2595112	0.2950204	-0.8796381	0.3790554	Poisson	BirdID
fixed	factor(fac.yr)3	-0.5978370	0.3089572	-1.9350157	0.0529884	Poisson	BirdID
fixed	factor(fac.yr)4	-0.1199593	0.2818009	-0.4256881	0.6703351	Poisson	BirdID
fixed	factor(fac.yr)5	0.0769610	0.2788867	0.2759581	0.7825802	Poisson	BirdID
fixed	factor(fac.yr)6	-0.4700036	0.3000000	-1.5666788	0.1171898	Poisson	BirdID
fixed	factor(fac.yr)7	-0.7444405	0.3203616	-2.3237504	0.0201389	Poisson	BirdID
fixed	factor(fac.yr)8	-0.3349350	0.2868067	-1.1678074	0.2428845	Poisson	BirdID
fixed	factor(fac.yr)9	-0.0921153	0.2762531	-0.3334452	0.7387982	Poisson	BirdID
fixed	factor(fac.yr)10	-0.2231436	0.2927700	-0.7621803	0.4459524	Poisson	BirdID
fixed	factor(fac.yr)11	-0.5679840	0.3298841	-1.7217684	0.0851115	Poisson	BirdID
fixed	factor(fac.yr)12	-0.1823216	0.3000000	-0.6077385	0.5433609	Poisson	BirdID
fixed	factor(fac.yr)13	-0.1053605	0.3027650	-0.3479943	0.7278444	Poisson	BirdID
fixed	factor(fac.yr)14	-0.2876821	0.3415650	-0.8422469	0.3996498	Poisson	BirdID
fixed	factor(fac.yr)15	-0.3437715	0.3562627	-0.9649385	0.3345756	Poisson	BirdID
fixed	factor(fac.yr)16	-0.0253178	0.3562627	-0.0710650	0.9433460	Poisson	BirdID
fixed	factor(fac.yr)17	-0.5108256	0.4013865	-1.2726528	0.2031413	Poisson	BirdID
fixed	factor(fac.yr)18	-0.3566749	0.4391550	-0.8121846	0.4166857	Poisson	BirdID
fixed	factor(fac.yr)19	-0.7985077	0.6191392	-1.2897063	0.1971527	Poisson	BirdID
fixed	factor(fac.yr)20	-0.7339692	0.5477226	-1.3400382	0.1802329	Poisson	BirdID
fixed	(Intercept)	0.5001581	0.2407389	2.0775958	0.0377466	Poisson	Terr
fixed	factor(fac.yr)2	-0.2993739	0.3094349	-0.9674858	0.3333012	Poisson	Terr
fixed	factor(fac.yr)3	-0.6363350	0.3241163	-1.9632923	0.0496122	Poisson	Terr
fixed	factor(fac.yr)4	-0.1483735	0.3012268	-0.4925641	0.6223206	Poisson	Terr
fixed	factor(fac.yr)5	0.0770641	0.2968217	0.2596311	0.7951484	Poisson	Terr
fixed	factor(fac.yr)6	-0.5052060	0.3180116	-1.5886404	0.1121416	Poisson	Terr
fixed	factor(fac.yr)7	-0.7734051	0.3382755	-2.2863175	0.0222357	Poisson	Terr
fixed	factor(fac.yr)8	-0.3715581	0.3024802	-1.2283716	0.2193075	Poisson	Terr
fixed	factor(fac.yr)9	-0.1195224	0.2941134	-0.4063820	0.6844619	Poisson	Terr
fixed	factor(fac.yr)10	-0.2277822	0.3104460	-0.7337257	0.4631160	Poisson	Terr
fixed	factor(fac.yr)11	-0.6178754	0.3487399	-1.7717373	0.0764382	Poisson	Terr
fixed	factor(fac.yr)12	-0.2040829	0.3183771	-0.6410100	0.5215162	Poisson	Terr
fixed	factor(fac.yr)13	-0.1118304	0.3245312	-0.3445905	0.7304023	Poisson	Terr
fixed	factor(fac.yr)14	-0.3147632	0.3611594	-0.8715354	0.3834619	Poisson	Terr
fixed	factor(fac.yr)15	-0.4052126	0.3751442	-1.0801514	0.2800748	Poisson	Terr
fixed	factor(fac.yr)16	-0.0900662	0.3820649	-0.2357354	0.8136380	Poisson	Terr
fixed	factor(fac.yr)17	-0.5249740	0.4250439	-1.2351055	0.2167912	Poisson	Terr
fixed	factor(fac.yr)18	-0.3516436	0.4623465	-0.7605630	0.4469181	Poisson	Terr
fixed	factor(fac.yr)19	-0.9045220	0.6468612	-1.3983247	0.1620156	Poisson	Terr
fixed	factor(fac.yr)20	-0.7294727	0.5671829	-1.2861330	0.1983966	Poisson	Terr
fixed	(Intercept)	0.5005515	0.2406786	2.0797504	0.0375484	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.3007601	0.3094489	-0.9719216	0.3310895	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.6372003	0.3241050	-1.9660304	0.0492951	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.1484357	0.3011589	-0.4928818	0.6220961	Poisson	BirdID+Terr

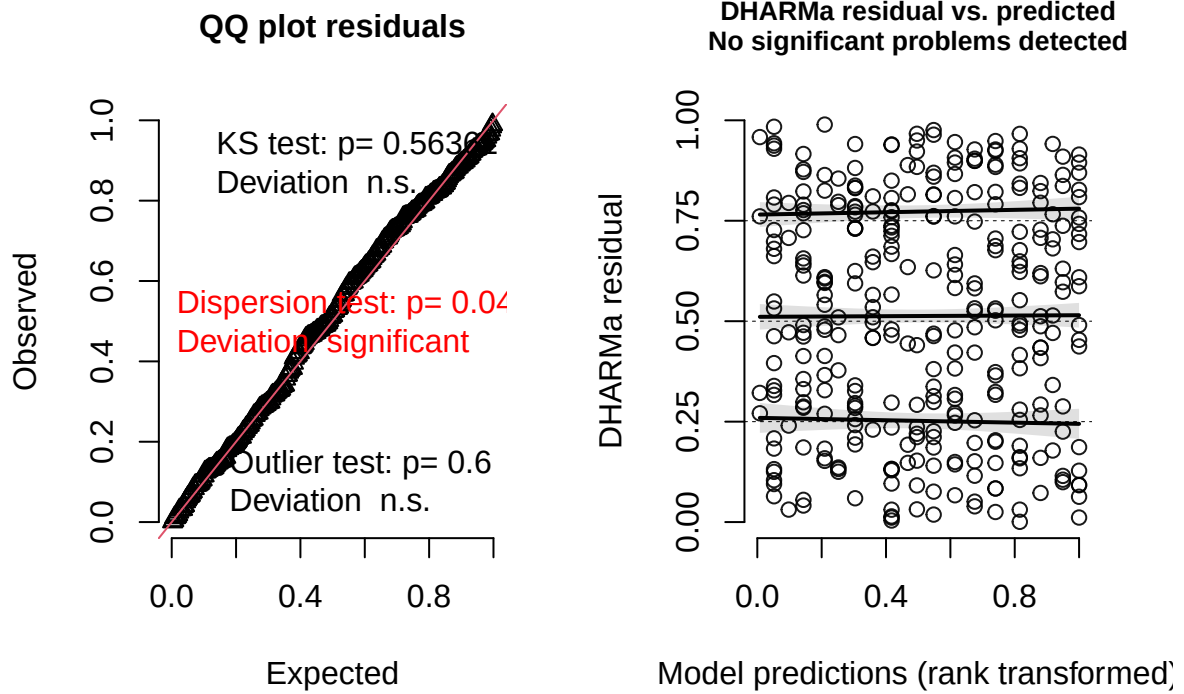
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.0766409	0.2967697	0.2582503	0.7962137	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.5053345	0.3179505	-1.5893495	0.1119815	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.7735370	0.3382138	-2.2871240	0.0221886	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.3710224	0.3023879	-1.2269750	0.2198320	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.1194163	0.2940409	-0.4061215	0.6846533	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.2278028	0.3103767	-0.7339558	0.4629757	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.6187803	0.3487376	-1.7743435	0.0760064	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.2045746	0.3183373	-0.6426348	0.5204611	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.1123242	0.3244848	-0.3461616	0.7292213	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.3143058	0.3610401	-0.8705563	0.3839965	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.4071632	0.3752608	-1.0850140	0.2779155	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.0909071	0.3820578	-0.2379407	0.8119271	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.5235608	0.4247770	-1.2325546	0.2177420	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.3508267	0.4621281	-0.7591547	0.4477601	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-0.9062469	0.6471914	-1.4002765	0.1614305	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.7329619	0.5678014	-1.2908773	0.1967462	Poisson	BirdID+Terr

Territory only model is performing best on AIC and BIC. Note that we don't include R^2 s because some variance components were too close to 0, leading to numerical instability.

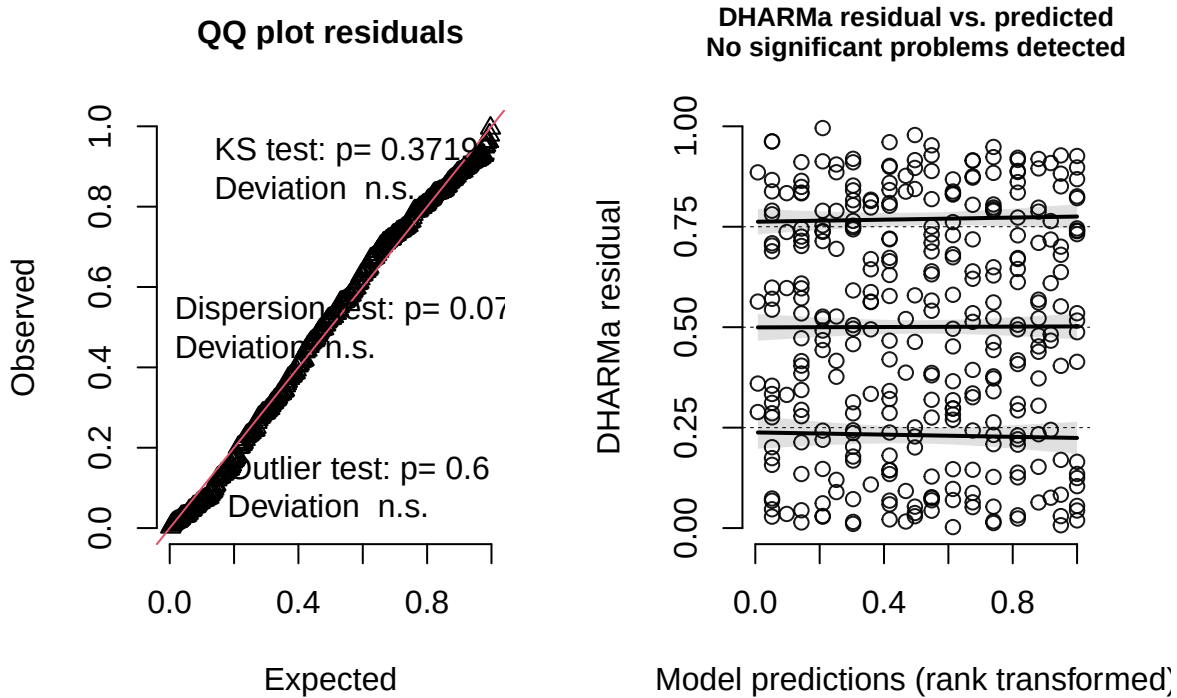
DHARMA residual



DHARMa residual



DHARMa residual



Similar to the previous batch of diagnostic plots for Poisson response.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
327	1	-190.8865	423.773	503.3621	308.5376	306	0.384	Binomial	Terr
327	1	-190.8865	425.773	509.1521	308.5214	305	0.384	Binomial	BirdID+Terr
327	1	-194.8235	431.647	511.2362	357.4784	306	0.208	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.5390149	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	0.9504437	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	0.0007031	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.9506756	Binomial	BirdID+Terr

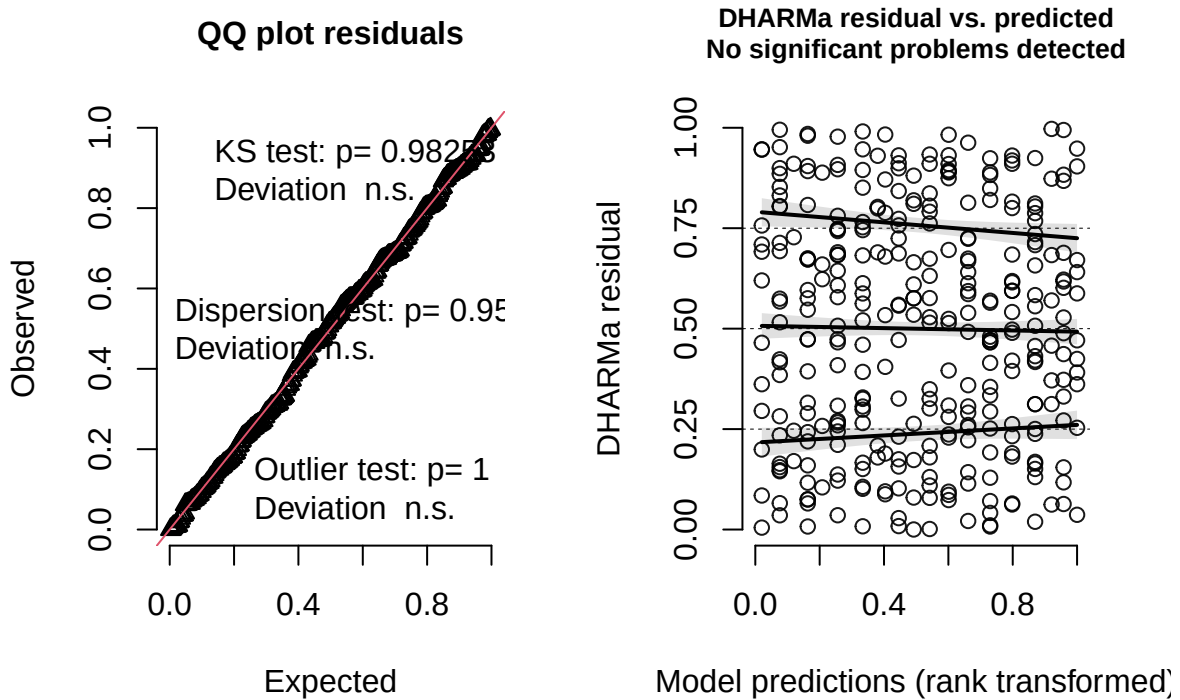
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.7889693	0.6383951	1.2358636	0.2165093	Binomial	BirdID
fixed	factor(fac.yr)2	0.2290669	0.8008617	0.2860256	0.7748585	Binomial	BirdID
fixed	factor(fac.yr)3	-0.2017720	0.7643917	-0.2639642	0.7918075	Binomial	BirdID
fixed	factor(fac.yr)4	0.3514653	0.7997127	0.4394894	0.6603070	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.1274650	0.8078412	0.1577848	0.8746264	Binomial	BirdID
fixed	factor(fac.yr)6	-0.4305125	0.7668237	-0.5614231	0.5745091	Binomial	BirdID
fixed	factor(fac.yr)7	-0.8049483	0.7661122	-1.0506926	0.2933998	Binomial	BirdID
fixed	factor(fac.yr)8	-0.2854336	0.7635429	-0.3738278	0.7085324	Binomial	BirdID
fixed	factor(fac.yr)9	0.4655463	0.8039225	0.5790935	0.5625261	Binomial	BirdID
fixed	factor(fac.yr)10	0.4582034	0.8330011	0.5500634	0.5822759	Binomial	BirdID
fixed	factor(fac.yr)11	-0.0891248	0.8234419	-0.1082344	0.9138097	Binomial	BirdID
fixed	factor(fac.yr)12	0.2221054	0.8414671	0.2639502	0.7918183	Binomial	BirdID
fixed	factor(fac.yr)13	1.2494226	1.0174051	1.2280483	0.2194288	Binomial	BirdID
fixed	factor(fac.yr)14	-0.8163766	0.8743674	-0.9336769	0.3504706	Binomial	BirdID
fixed	factor(fac.yr)15	1.5433832	1.2691459	1.2160802	0.2239544	Binomial	BirdID
fixed	factor(fac.yr)16	1.2004612	1.2856827	0.9337150	0.3504509	Binomial	BirdID
fixed	factor(fac.yr)17	-0.1529143	0.9725359	-0.1572325	0.8750616	Binomial	BirdID
fixed	factor(fac.yr)18	-0.1900925	1.1005072	-0.1727317	0.8628623	Binomial	BirdID
fixed	factor(fac.yr)19	-0.7794351	1.2169459	-0.6404846	0.5218576	Binomial	BirdID
fixed	factor(fac.yr)20	-0.4280731	1.1427801	-0.3745892	0.7079660	Binomial	BirdID
fixed	(Intercept)	1.0535789	0.6832431	1.5420265	0.1230671	Binomial	Terr
fixed	factor(fac.yr)2	0.1577917	0.8541927	0.1847261	0.8534439	Binomial	Terr
fixed	factor(fac.yr)3	-0.3628051	0.8134404	-0.4460132	0.6555877	Binomial	Terr
fixed	factor(fac.yr)4	0.2535841	0.8603249	0.2947539	0.7681819	Binomial	Terr
fixed	factor(fac.yr)5	0.1620206	0.8555193	0.1893827	0.8497929	Binomial	Terr
fixed	factor(fac.yr)6	-0.6875148	0.8138064	-0.8448138	0.3982148	Binomial	Terr
fixed	factor(fac.yr)7	-0.9721344	0.8152870	-1.1923830	0.2331111	Binomial	Terr
fixed	factor(fac.yr)8	-0.3939072	0.8101317	-0.4862261	0.6268069	Binomial	Terr
fixed	factor(fac.yr)9	0.3612990	0.8518055	0.4241567	0.6714515	Binomial	Terr
fixed	factor(fac.yr)10	0.4598835	0.8782788	0.5236190	0.6005436	Binomial	Terr
fixed	factor(fac.yr)11	-0.0485200	0.8822447	-0.0549961	0.9561416	Binomial	Terr
fixed	factor(fac.yr)12	0.2376152	0.8975911	0.2647255	0.7912210	Binomial	Terr
fixed	factor(fac.yr)13	1.2483697	1.0919357	1.1432630	0.2529294	Binomial	Terr
fixed	factor(fac.yr)14	-0.9894773	0.9189320	-1.0767687	0.2815836	Binomial	Terr
fixed	factor(fac.yr)15	1.5651149	1.3326767	1.1744145	0.2402290	Binomial	Terr
fixed	factor(fac.yr)16	0.9185613	1.3704181	0.6702781	0.5026805	Binomial	Terr
fixed	factor(fac.yr)17	-0.2305846	1.0279962	-0.2243049	0.8225201	Binomial	Terr
fixed	factor(fac.yr)18	-0.0342166	1.1495196	-0.0297660	0.9762537	Binomial	Terr
fixed	factor(fac.yr)19	-1.0984261	1.2836756	-0.8556883	0.3921702	Binomial	Terr
fixed	factor(fac.yr)20	-0.4221323	1.1948841	-0.3532830	0.7238763	Binomial	Terr
fixed	(Intercept)	1.0452936	0.6825441	1.5314667	0.1256541	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	0.1658822	0.8536473	0.1943217	0.8459240	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-0.3544194	0.8128720	-0.4360088	0.6628303	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	0.2633667	0.8598694	0.3062869	0.7593862	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	0.1706514	0.8549969	0.1995930	0.8417989	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-0.6780357	0.8132432	-0.8337428	0.4044259	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-0.9635799	0.8147084	-1.1827298	0.2369163	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-0.3851148	0.8095542	-0.4757122	0.6342794	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	0.3686206	0.8511841	0.4330680	0.6649654	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	0.4681977	0.8777452	0.5334095	0.5937501	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-0.0393852	0.8817564	-0.0446668	0.9643729	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	0.2461703	0.8970962	0.2744079	0.7837712	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	1.2587888	1.0919202	1.1528212	0.2489838	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-0.9801302	0.9184125	-1.0672005	0.2858813	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	1.5751399	1.3329771	1.1816706	0.2373364	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	0.9248932	1.3693604	0.6754199	0.4994091	Binomial	BirdID+Terr

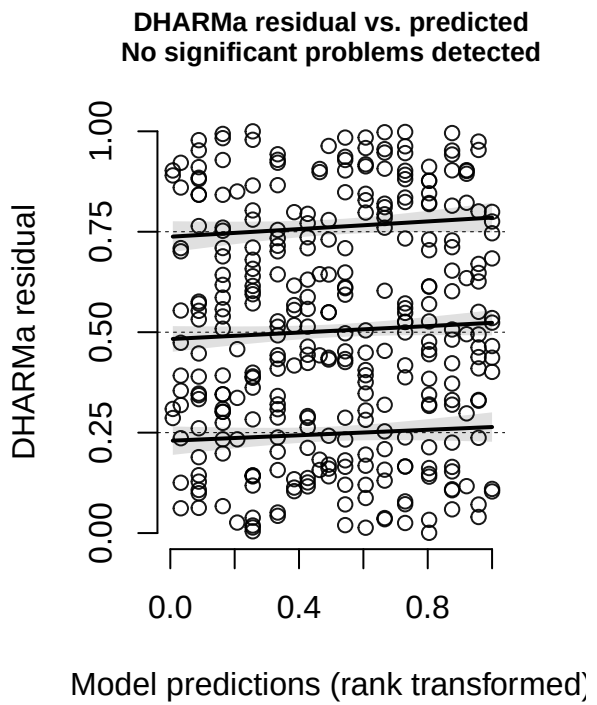
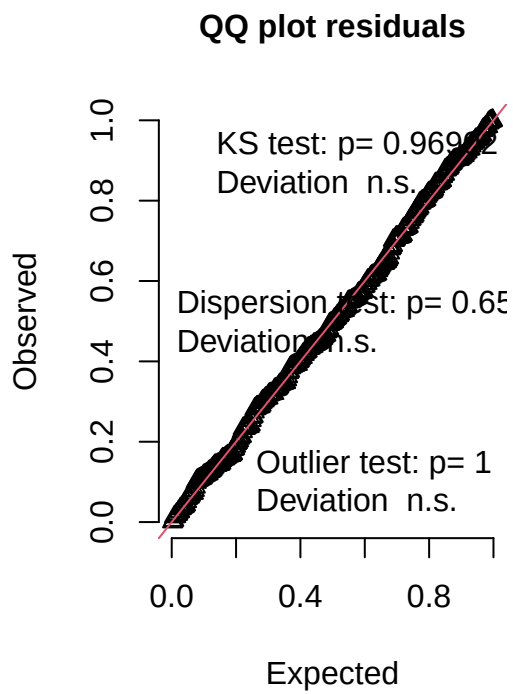
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-0.2218989	1.0275697	-0.2159454	0.8290303	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-0.0278658	1.1488960	-0.0242544	0.9806497	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-1.0905168	1.2833157	-0.8497650	0.3954557	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-0.4131977	1.1945579	-0.3459001	0.7294178	Binomial	BirdID+Terr

Territory only model seems to perform the best on all criteria.

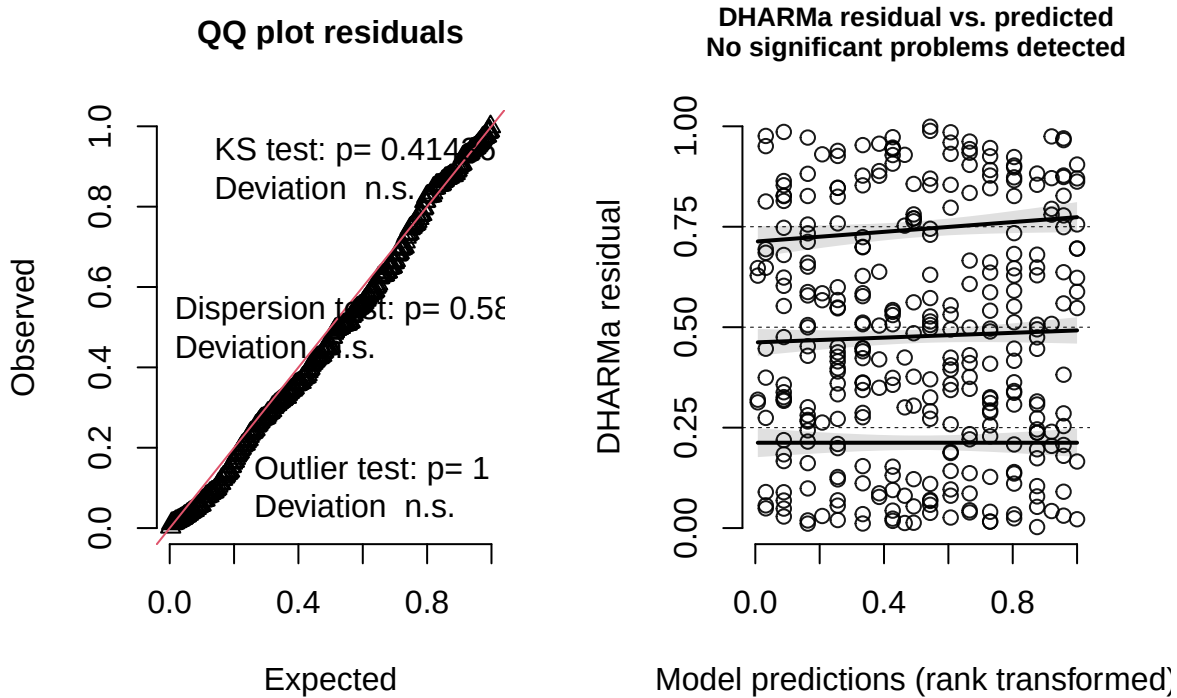
DHARMA residual



DHARMa residual



DHARMA residual



Diagnostics look good.

Probabilities used to generate simulation 38

Simulation 38
 Probability of mate fidelity: 0.804
 Probability of pair dispersal: 0.412
 Probability of single male dispersal: 0.214
 Probability of single female dispersal: 0.535

Simulation 39

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
351	1	-457.5215	957.0431	1038.12	327.8389	330	0.9173559	0.8968878	Poisson	Terr
351	1	-457.5213	959.0426	1043.98	327.7850	329	0.9131281	0.8917485	Poisson	BirdID+Terr
351	1	-463.7619	969.5238	1050.60	370.7297	330	0.9123249	0.9005900	Poisson	BirdID

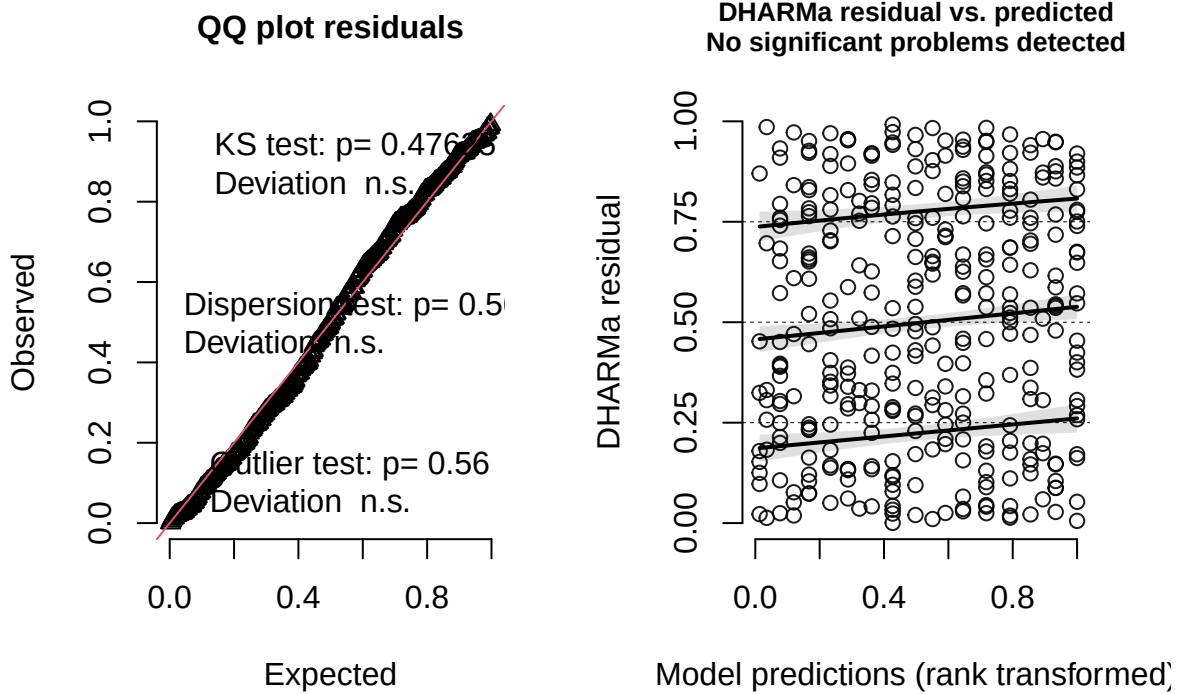
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.2964066	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.4014959	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.0045184	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.4016899	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	-0.1634528	0.2799059	-0.5839563	0.5592497	Poisson	BirdID
fixed	factor(fac.yr)2	-0.0319281	0.3652641	-0.0874109	0.9303449	Poisson	BirdID
fixed	factor(fac.yr)3	-0.1169573	0.3627790	-0.3223927	0.7471552	Poisson	BirdID
fixed	factor(fac.yr)4	0.2765038	0.3384702	0.8169221	0.4139729	Poisson	BirdID
fixed	factor(fac.yr)5	0.6314101	0.3336936	1.8921855	0.0584663	Poisson	BirdID
fixed	factor(fac.yr)6	0.3571430	0.3345165	1.0676395	0.2856832	Poisson	BirdID
fixed	factor(fac.yr)7	0.1491417	0.3535347	0.4218587	0.6731282	Poisson	BirdID
fixed	factor(fac.yr)8	0.2296589	0.3443008	0.6670297	0.5047531	Poisson	BirdID
fixed	factor(fac.yr)9	0.8110547	0.3175284	2.5542746	0.0106409	Poisson	BirdID
fixed	factor(fac.yr)10	0.1341987	0.3469900	0.3867511	0.6989405	Poisson	BirdID
fixed	factor(fac.yr)11	-0.4926096	0.4180188	-1.1784389	0.2386217	Poisson	BirdID
fixed	factor(fac.yr)12	0.1255682	0.3730283	0.3366185	0.7364045	Poisson	BirdID
fixed	factor(fac.yr)13	0.3722804	0.3562016	1.0451397	0.2959584	Poisson	BirdID
fixed	factor(fac.yr)14	0.1683534	0.3895575	0.4321657	0.6656210	Poisson	BirdID
fixed	factor(fac.yr)15	0.1730329	0.3960455	0.4369016	0.6621827	Poisson	BirdID
fixed	factor(fac.yr)16	0.4304450	0.4243276	1.0144167	0.3103840	Poisson	BirdID
fixed	factor(fac.yr)17	0.0865892	0.4648628	0.1862683	0.8522344	Poisson	BirdID
fixed	factor(fac.yr)18	-0.1647528	0.5096014	-0.3232974	0.7464700	Poisson	BirdID
fixed	factor(fac.yr)19	-17.0525386	1897.4543754	-0.0089871	0.9928295	Poisson	BirdID
fixed	factor(fac.yr)20	-1.1570411	0.7778944	-1.4874013	0.1369089	Poisson	BirdID
fixed	(Intercept)	-0.1758457	0.2866625	-0.6134243	0.5395958	Poisson	Terr
fixed	factor(fac.yr)2	-0.1057394	0.3749518	-0.2820079	0.7779375	Poisson	Terr
fixed	factor(fac.yr)3	-0.0964390	0.3718628	-0.2593402	0.7953727	Poisson	Terr
fixed	factor(fac.yr)4	0.2296253	0.3467324	0.6622551	0.5078077	Poisson	Terr
fixed	factor(fac.yr)5	0.6059523	0.3426752	1.7682994	0.0770109	Poisson	Terr
fixed	factor(fac.yr)6	0.3472183	0.3441677	1.0088635	0.3130401	Poisson	Terr
fixed	factor(fac.yr)7	0.1335381	0.3589690	0.3720046	0.7098894	Poisson	Terr
fixed	factor(fac.yr)8	0.1527821	0.3495678	0.4370601	0.6620678	Poisson	Terr
fixed	factor(fac.yr)9	0.7820155	0.3214439	2.4328211	0.0149817	Poisson	Terr
fixed	factor(fac.yr)10	0.0848416	0.3525748	0.2406342	0.8098386	Poisson	Terr
fixed	factor(fac.yr)11	-0.5795393	0.4223317	-1.3722374	0.1699896	Poisson	Terr
fixed	factor(fac.yr)12	0.1132815	0.3797395	0.2983138	0.7654636	Poisson	Terr
fixed	factor(fac.yr)13	0.3771000	0.3617931	1.0423084	0.2972687	Poisson	Terr
fixed	factor(fac.yr)14	0.2989379	0.3974289	0.7521795	0.4519431	Poisson	Terr
fixed	factor(fac.yr)15	0.2022006	0.4042397	0.5001998	0.6169344	Poisson	Terr
fixed	factor(fac.yr)16	0.5331210	0.4357650	1.2234139	0.2211734	Poisson	Terr
fixed	factor(fac.yr)17	0.1812327	0.4762978	0.3805030	0.7035721	Poisson	Terr
fixed	factor(fac.yr)18	-0.1398231	0.5170280	-0.2704362	0.7868247	Poisson	Terr
fixed	factor(fac.yr)19	-17.4689628	2322.4714213	-0.0075217	0.9939986	Poisson	Terr
fixed	factor(fac.yr)20	-1.2419932	0.7881006	-1.5759324	0.1150414	Poisson	Terr
fixed	(Intercept)	-0.1750297	0.2865763	-0.6107613	0.5413576	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.1075399	0.3749539	-0.2868082	0.7742592	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.0970282	0.3717821	-0.2609814	0.7941069	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	0.2293000	0.3466401	0.6614930	0.5082962	Poisson	BirdID+Terr

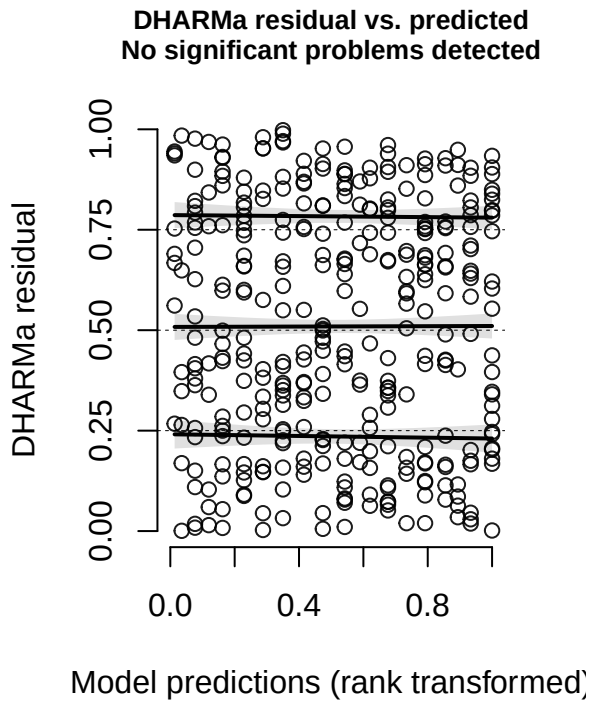
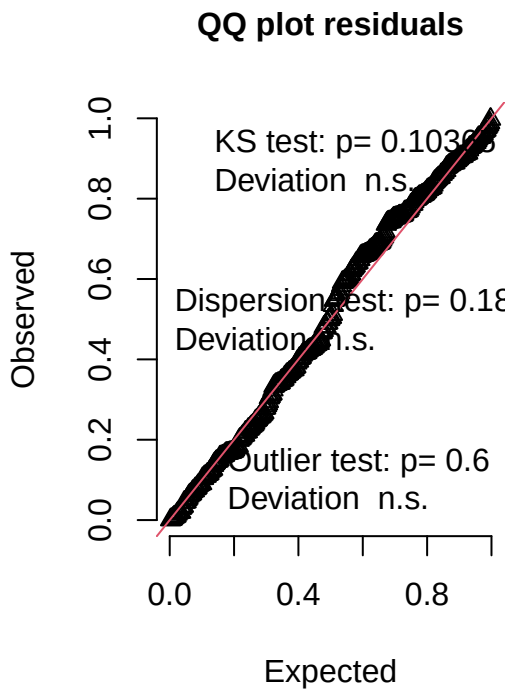
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.6049013	0.3426139	1.7655479	0.0774717	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	0.3459363	0.3441188	1.0052818	0.3147612	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	0.1332910	0.3588677	0.3714210	0.7103240	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	0.1523183	0.3494795	0.4358434	0.6629504	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	0.7811660	0.3213690	2.4307444	0.0150678	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	0.0838340	0.3525163	0.2378158	0.8120239	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.5794881	0.4221802	-1.3726084	0.1698741	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	0.1113676	0.3797585	0.2932590	0.7693242	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	0.3764365	0.3617193	1.0406867	0.2980210	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	0.2976192	0.3974146	0.7488884	0.4539245	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	0.2022499	0.4041081	0.5004847	0.6167338	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	0.5303534	0.4359125	1.2166510	0.2237370	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	0.1821138	0.4760449	0.3825560	0.7020490	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.1396062	0.5168334	-0.2701183	0.7870692	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-17.0415154	1874.6008536	-0.0090907	0.9927467	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-1.2534630	0.7914544	-1.5837462	0.1132515	Poisson	BirdID+Terr

The territory model has the best AIC and BIC values, and good R^2 values

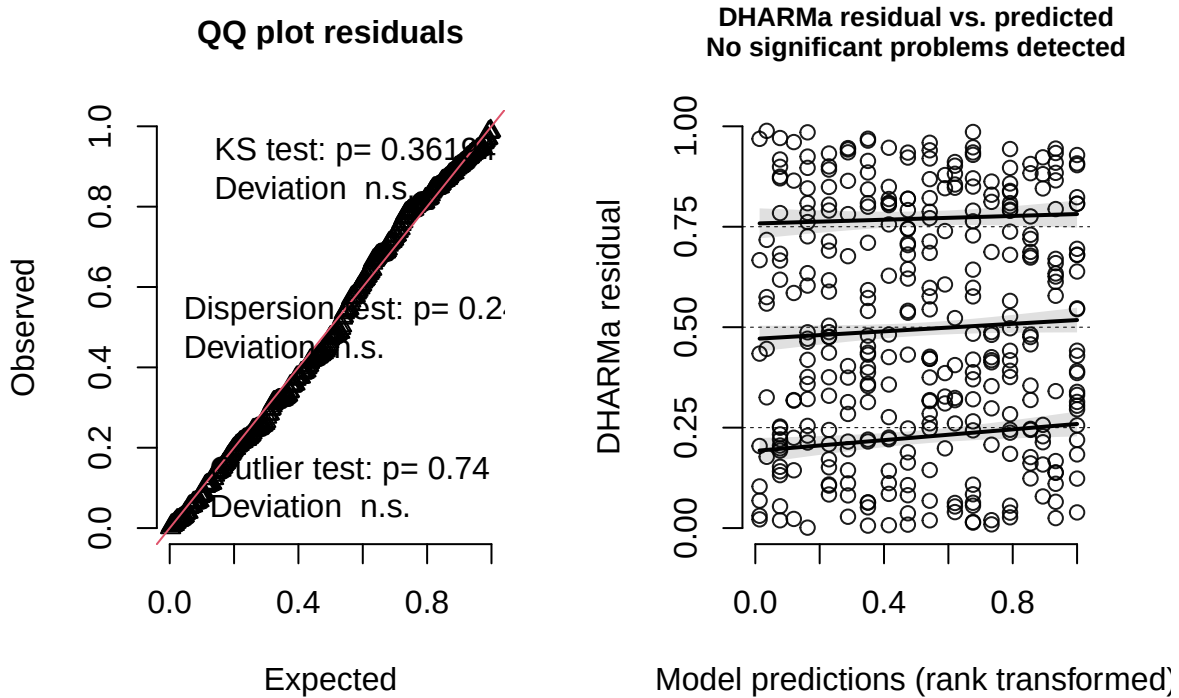
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
351	1	-203.8798	451.7595	536.6968	269.4076	329	0.699	Binomial	BirdID+Terr
351	1	-206.5905	455.1810	536.2575	327.2868	330	0.591	Binomial	Terr
351	1	-207.0136	456.0273	537.1038	331.7575	330	0.558	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.9866148	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.0122648	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	0.9677080	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.9941441	Binomial	BirdID+Terr

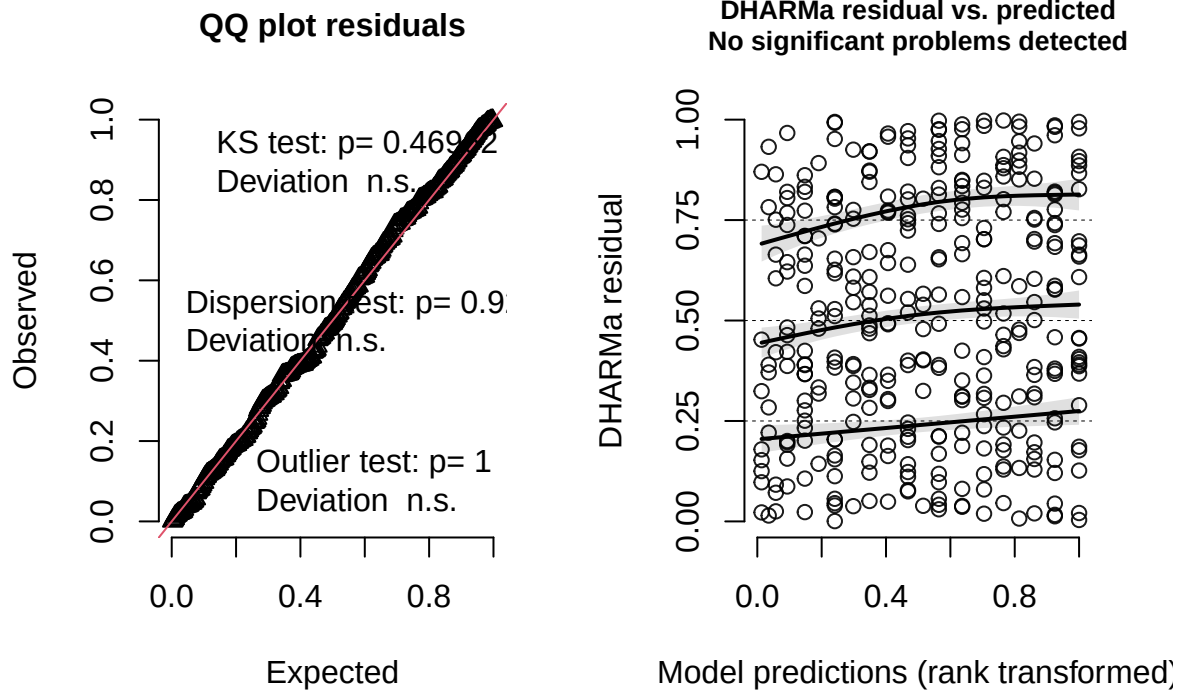
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	-0.5254661	5.899398e-01	-0.8907114	0.3730840	Binomial	BirdID
fixed	factor(fac.yr)2	0.6702302	7.324198e-01	0.9150902	0.3601443	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)3	0.9488118	7.301432e-01	1.2994873	0.1937768	Binomial	BirdID
fixed	factor(fac.yr)4	1.0765358	7.361865e-01	1.4623140	0.1436552	Binomial	BirdID
fixed	factor(fac.yr)5	1.4001157	7.981454e-01	1.7542113	0.0793943	Binomial	BirdID
fixed	factor(fac.yr)6	1.6843843	7.662374e-01	2.1982538	0.0279310	Binomial	BirdID
fixed	factor(fac.yr)7	1.3865962	7.733239e-01	1.7930342	0.0729675	Binomial	BirdID
fixed	factor(fac.yr)8	1.2273395	7.610205e-01	1.6127549	0.1067978	Binomial	BirdID
fixed	factor(fac.yr)9	3.2719117	1.019483e+00	3.2093817	0.0013302	Binomial	BirdID
fixed	factor(fac.yr)10	0.4801496	7.350679e-01	0.6532044	0.5136245	Binomial	BirdID
fixed	factor(fac.yr)11	0.1272442	7.726752e-01	0.1646801	0.8691958	Binomial	BirdID
fixed	factor(fac.yr)12	0.6896539	7.971992e-01	0.8650961	0.3869860	Binomial	BirdID
fixed	factor(fac.yr)13	1.5982872	8.412102e-01	1.8999855	0.0574350	Binomial	BirdID
fixed	factor(fac.yr)14	1.4895660	8.986860e-01	1.6574932	0.0974198	Binomial	BirdID
fixed	factor(fac.yr)15	0.4823487	8.626883e-01	0.5591228	0.5760779	Binomial	BirdID
fixed	factor(fac.yr)16	0.4137665	1.007383e+00	0.4107343	0.6812674	Binomial	BirdID
fixed	factor(fac.yr)17	1.0248342	1.016956e+00	1.0077471	0.3135759	Binomial	BirdID
fixed	factor(fac.yr)18	-0.0983808	1.021358e+00	-0.0963235	0.9232636	Binomial	BirdID
fixed	factor(fac.yr)19	-	2.372657e+07	-0.0000020	0.9999984	Binomial	BirdID
		47.0457334					
fixed	factor(fac.yr)20	-1.6894901	1.369331e+00	-1.2338071	0.2172748	Binomial	BirdID
fixed	(Intercept)	-0.4792065	5.801534e-01	-0.8259997	0.4088043	Binomial	Terr
fixed	factor(fac.yr)2	0.5474846	7.416423e-01	0.7382057	0.4603895	Binomial	Terr
fixed	factor(fac.yr)3	1.0013959	7.318357e-01	1.3683344	0.1712074	Binomial	Terr
fixed	factor(fac.yr)4	0.9531098	7.307395e-01	1.3043086	0.1921284	Binomial	Terr
fixed	factor(fac.yr)5	1.3103461	7.954392e-01	1.6473240	0.0994915	Binomial	Terr
fixed	factor(fac.yr)6	1.5830490	7.628080e-01	2.0752915	0.0379595	Binomial	Terr
fixed	factor(fac.yr)7	1.3615157	7.644119e-01	1.7811282	0.0748915	Binomial	Terr
fixed	factor(fac.yr)8	1.1468614	7.446868e-01	1.5400587	0.1235461	Binomial	Terr
fixed	factor(fac.yr)9	3.3568770	1.011140e+00	3.3198934	0.0009005	Binomial	Terr
fixed	factor(fac.yr)10	0.4630323	7.218251e-01	0.6414744	0.5212145	Binomial	Terr
fixed	factor(fac.yr)11	0.0688952	7.591368e-01	0.0907547	0.9276875	Binomial	Terr

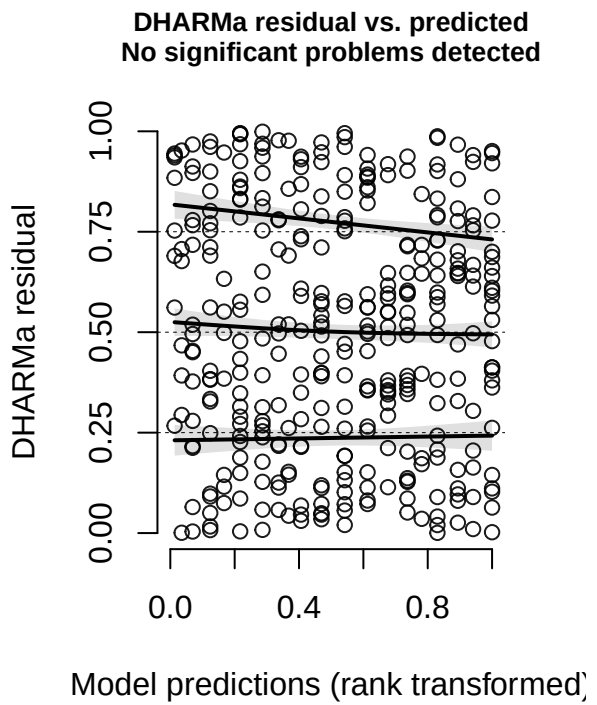
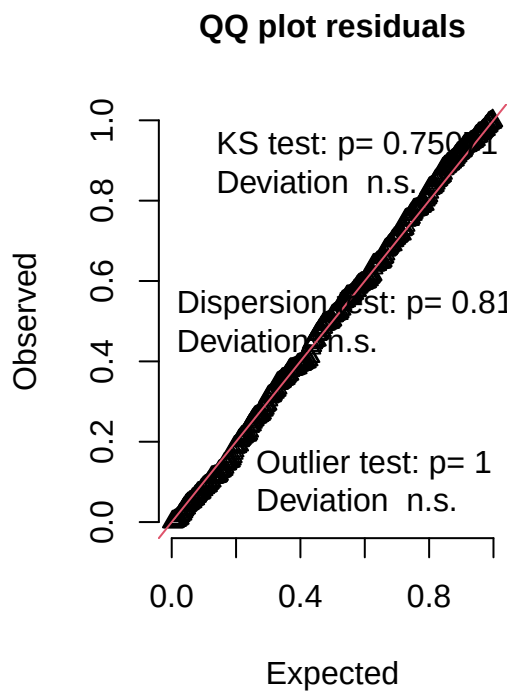
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)12	0.9272796	7.849162e-01	1.1813739	0.2374542	Binomial	Terr
fixed	factor(fac.yr)13	1.7444337	8.220560e-01	2.1220376	0.0338346	Binomial	Terr
fixed	factor(fac.yr)14	1.7450516	8.785245e-01	1.9863437	0.0469952	Binomial	Terr
fixed	factor(fac.yr)15	0.7813735	8.444373e-01	0.9253185	0.3548002	Binomial	Terr
fixed	factor(fac.yr)16	0.8625005	9.815467e-01	0.8787157	0.3795554	Binomial	Terr
fixed	factor(fac.yr)17	1.3786350	9.846692e-01	1.4000996	0.1614835	Binomial	Terr
fixed	factor(fac.yr)18	0.0877227	9.982276e-01	0.0878784	0.9299733	Binomial	Terr
fixed	factor(fac.yr)19	-38.5520244	2.372657e+07	-0.0000016	0.9999987	Binomial	Terr
fixed	factor(fac.yr)20	-1.7814372	1.388655e+00	-1.2828507	0.1995444	Binomial	Terr
fixed	(Intercept)	-0.4992218	6.364641e-01	-0.7843676	0.4328244	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	0.6222605	7.853898e-01	0.7922950	0.4281887	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	1.0953900	7.917467e-01	1.3835106	0.1665084	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	1.1064436	7.955613e-01	1.3907710	0.1642949	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	1.4442745	8.590480e-01	1.6812500	0.0927144	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	1.7810313	8.266174e-01	2.1546017	0.0311930	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	1.4703907	8.307047e-01	1.7700523	0.0767184	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	1.1760234	8.191587e-01	1.4356479	0.1511026	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	3.6398926	1.114838e+00	3.2649510	0.0010948	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	0.3982203	7.942305e-01	0.5013914	0.6160957	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-0.0564524	8.359143e-01	-0.0675338	0.9461568	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	0.7382591	8.628851e-01	0.8555705	0.3922354	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	1.7440995	9.083589e-01	1.9200554	0.0548509	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	1.7289011	9.695853e-01	1.7831346	0.0745644	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	0.5215310	9.322206e-01	0.5594502	0.5758545	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	0.6400732	1.095810e+00	0.5841097	0.5591465	Binomial	BirdID+Terr
fixed	factor(fac.yr)17	1.3488582	1.093873e+00	1.2331029	0.2175374	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-0.0011533	1.102959e+00	-0.0010457	0.9991657	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-50.4317877	2.372657e+07	-0.0000021	0.9999983	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-2.0377890	1.517894e+00	-1.3425110	0.1794304	Binomial	BirdID+Terr

The territory only model is performing the best on BIC, but not by much.

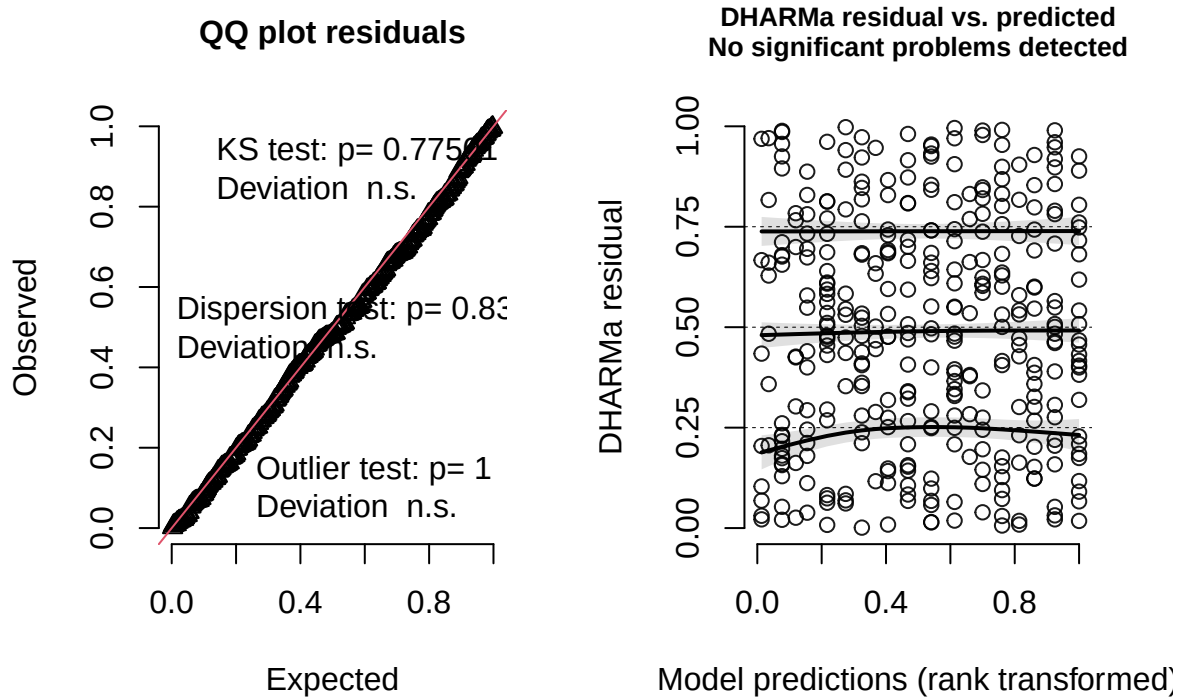
DHARMA residual



DHARMa residual



DHARMA residual



Diagnostics look good.

Probabilities used to generate simulation 39

Simulation 39
 Probability of mate fidelity: 0.232
 Probability of pair dispersal: 0.802
 Probability of single male dispersal: 0.279
 Probability of single female dispersal: 0.338

Simulation 40

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Family	RE
353	1	-481.8545	1005.709	1086.905	286.4179	332	Poisson	Terr
353	1	-481.8543	1007.709	1092.771	286.5018	331	Poisson	BirdID+Terr
353	1	-498.5572	1039.114	1120.310	389.9528	332	Poisson	BirdID

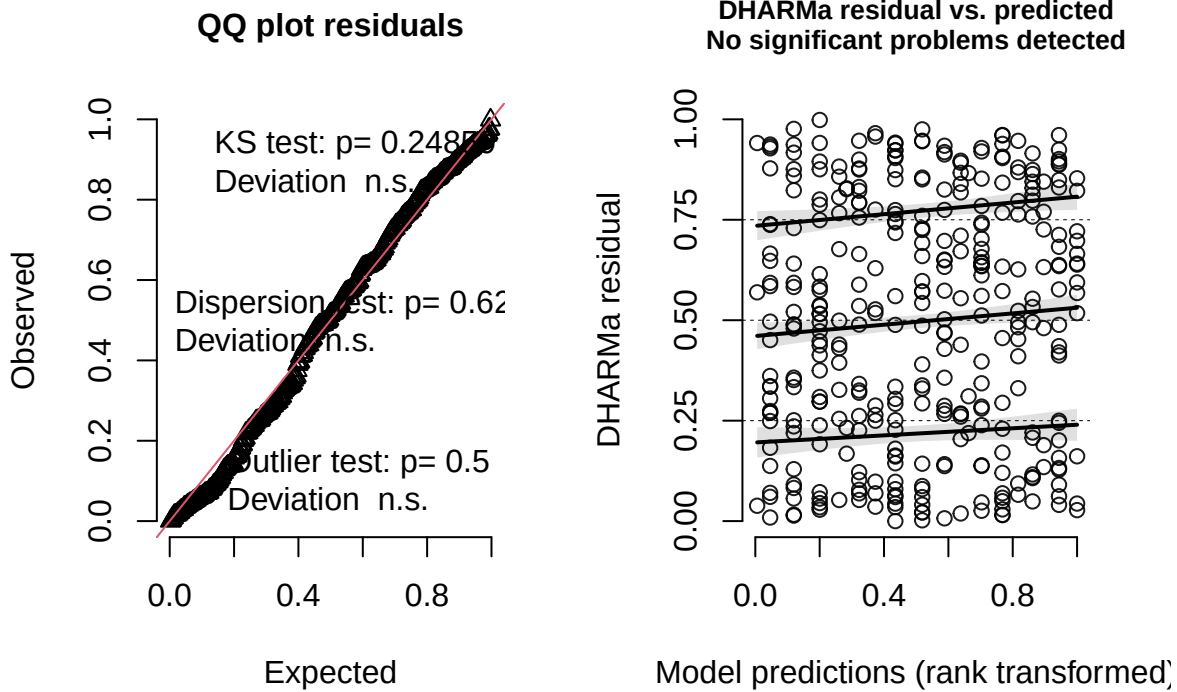
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.3414039	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.5715837	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.0000272	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.5711787	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.8174816	0.2027792	4.0313869	0.0000554	Poisson	BirdID
fixed	factor(fac.yr)2	-0.3788517	0.2921097	-1.2969502	0.1946483	Poisson	BirdID
fixed	factor(fac.yr)3	-1.6258960	0.3562261	-4.5642254	0.0000050	Poisson	BirdID
fixed	factor(fac.yr)4	-0.7848013	0.2893767	-2.7120408	0.0066870	Poisson	BirdID
fixed	factor(fac.yr)5	-0.3816785	0.2787556	-1.3692226	0.1709297	Poisson	BirdID
fixed	factor(fac.yr)6	-0.7721530	0.2717534	-2.8413736	0.0044920	Poisson	BirdID
fixed	factor(fac.yr)7	-1.1890804	0.3005993	-3.9556993	0.0000763	Poisson	BirdID
fixed	factor(fac.yr)8	-0.7006542	0.2755751	-2.5425164	0.0110057	Poisson	BirdID
fixed	factor(fac.yr)9	-0.2919098	0.2565249	-1.1379395	0.2551458	Poisson	BirdID
fixed	factor(fac.yr)10	-0.4519713	0.2653980	-1.7029945	0.0885691	Poisson	BirdID
fixed	factor(fac.yr)11	-1.2598566	0.3177519	-3.9649068	0.0000734	Poisson	BirdID
fixed	factor(fac.yr)12	-0.6830608	0.2902215	-2.3535844	0.0185934	Poisson	BirdID
fixed	factor(fac.yr)13	-0.3599428	0.2844689	-1.2653152	0.2057584	Poisson	BirdID
fixed	factor(fac.yr)14	-0.5272479	0.3134528	-1.6820648	0.0925563	Poisson	BirdID
fixed	factor(fac.yr)15	-0.9793353	0.3880321	-2.5238512	0.0116077	Poisson	BirdID
fixed	factor(fac.yr)16	-0.7814542	0.3663464	-2.1331020	0.0329164	Poisson	BirdID
fixed	factor(fac.yr)17	-0.3150190	0.3810867	-0.8266334	0.4084449	Poisson	BirdID
fixed	factor(fac.yr)18	-0.4877104	0.5639942	-0.8647436	0.3871795	Poisson	BirdID
fixed	factor(fac.yr)19	-2.0095412	1.0467381	-1.9198128	0.0548816	Poisson	BirdID
fixed	factor(fac.yr)20	-0.8782177	0.5649296	-1.5545614	0.1200506	Poisson	BirdID
fixed	(Intercept)	0.6511979	0.2232090	2.9174358	0.0035292	Poisson	Terr
fixed	factor(fac.yr)2	-0.2994646	0.3193799	-0.9376440	0.3484274	Poisson	Terr
fixed	factor(fac.yr)3	-1.5654301	0.3694497	-4.2371938	0.0000226	Poisson	Terr
fixed	factor(fac.yr)4	-0.6283612	0.3115723	-2.0167427	0.0437224	Poisson	Terr
fixed	factor(fac.yr)5	-0.3349734	0.2927108	-1.1443837	0.2524646	Poisson	Terr
fixed	factor(fac.yr)6	-0.6719333	0.2867054	-2.3436365	0.0190968	Poisson	Terr
fixed	factor(fac.yr)7	-1.0841142	0.3131062	-3.4624487	0.0005353	Poisson	Terr
fixed	factor(fac.yr)8	-0.6874429	0.2881550	-2.3856708	0.0170480	Poisson	Terr
fixed	factor(fac.yr)9	-0.1155173	0.2720029	-0.4246913	0.6710617	Poisson	Terr
fixed	factor(fac.yr)10	-0.2008243	0.2840203	-0.7070774	0.4795184	Poisson	Terr
fixed	factor(fac.yr)11	-0.9889962	0.3363987	-2.9399525	0.0032826	Poisson	Terr
fixed	factor(fac.yr)12	-0.3748088	0.3088372	-1.2136127	0.2248956	Poisson	Terr
fixed	factor(fac.yr)13	-0.2589607	0.3063106	-0.8454189	0.3978770	Poisson	Terr
fixed	factor(fac.yr)14	-0.4626525	0.3233389	-1.4308596	0.1524705	Poisson	Terr
fixed	factor(fac.yr)15	-0.8223866	0.4073024	-2.0191058	0.0434762	Poisson	Terr
fixed	factor(fac.yr)16	-0.7926168	0.3883973	-2.0407371	0.0412770	Poisson	Terr
fixed	factor(fac.yr)17	-0.4004444	0.4281539	-0.9352815	0.3496432	Poisson	Terr
fixed	factor(fac.yr)18	-0.3943197	0.5866111	-0.6721995	0.5014567	Poisson	Terr
fixed	factor(fac.yr)19	-1.7835426	1.0896278	-1.6368366	0.1016646	Poisson	Terr
fixed	factor(fac.yr)20	-0.3750521	0.5980861	-0.6270871	0.5306021	Poisson	Terr
fixed	(Intercept)	0.6518360	0.2231412	2.9211820	0.0034871	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.3014163	0.3193925	-0.9437175	0.3453140	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-1.5665383	0.3694502	-4.2401882	0.0000223	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.6284316	0.3114744	-2.0176024	0.0436327	Poisson	BirdID+Terr

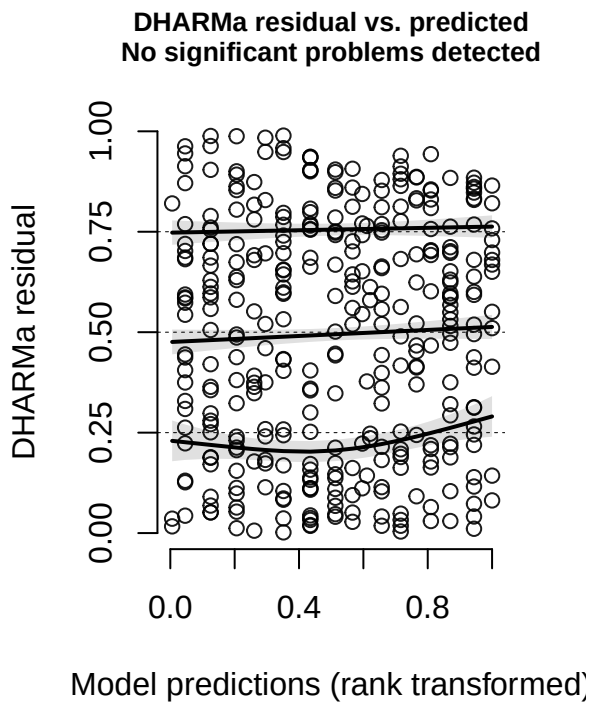
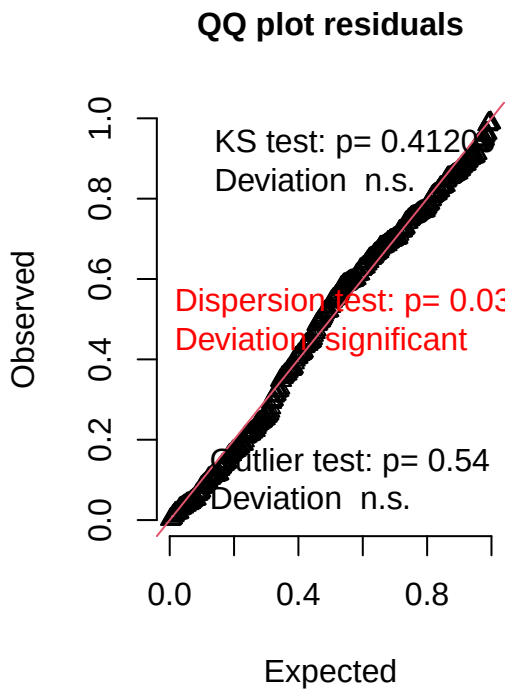
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.3353709	0.2926528	-1.1459683	0.2518083	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.6714572	0.2865937	-2.3428888	0.0191351	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-1.0846314	0.3130427	-3.4648030	0.0005306	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.6882149	0.2881098	-2.3887246	0.0169070	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.1166854	0.2719593	-0.4290544	0.6678836	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.2017255	0.2839641	-0.7103909	0.4774618	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.9934142	0.3366146	-2.9511917	0.0031655	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.3764709	0.3088130	-1.2190901	0.2228100	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.2604137	0.3062825	-0.8502403	0.3951915	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.4638063	0.3233282	-1.4344753	0.1514367	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.8255017	0.4074805	-2.0258680	0.0427783	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.7931061	0.3883161	-2.0424240	0.0411095	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.4037233	0.4282865	-0.9426477	0.3458611	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.3967197	0.5867884	-0.6760866	0.4989857	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.7826116	1.0885309	-1.6376306	0.1014988	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.3793672	0.5986217	-0.6337344	0.5262542	Poisson	BirdID+Terr

Territory only model has the best AIC and BIC.

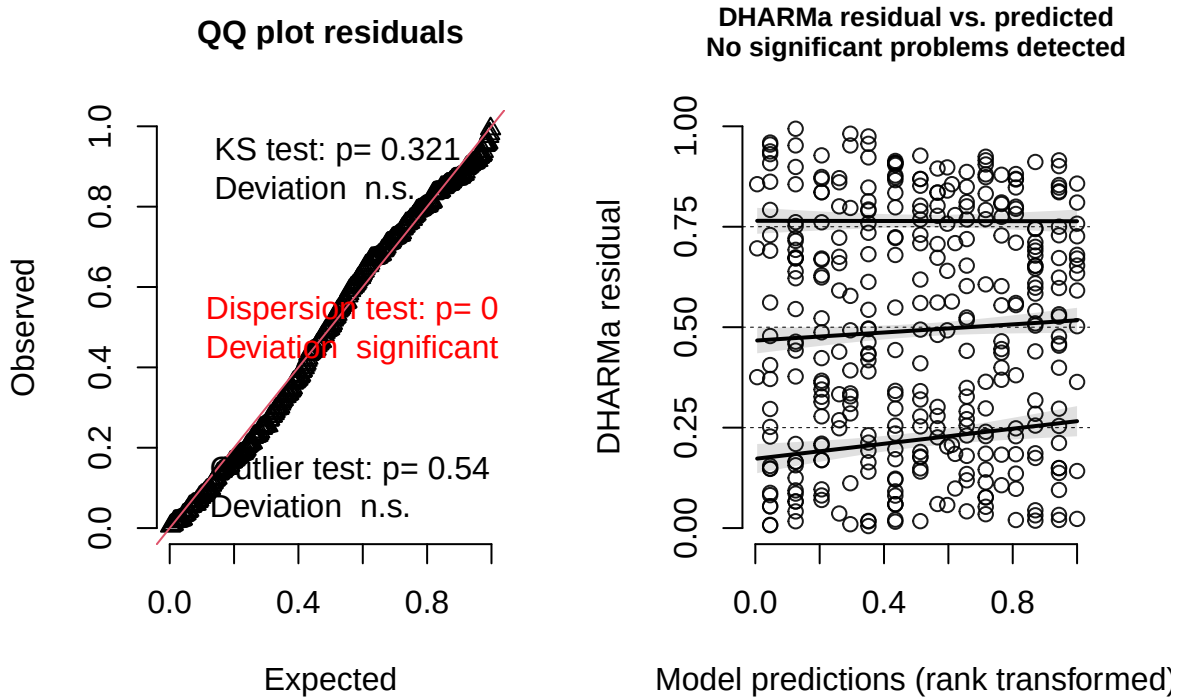
DHARMA residual



DHARMa residual



DHARMa residual



Similar response to other models with significant dispersion tests.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
353	1	-199.4354	440.8708	522.0667	259.3226	332	0.663	Binomial	Terr
353	1	-199.3051	442.6102	527.6725	248.0851	331	0.718	Binomial	BirdID+Terr
353	1	-217.4382	476.8765	558.0723	369.4754	332	0.491	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.8225099	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.6803494	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	0.3741964	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.7278301	Binomial	BirdID+Terr

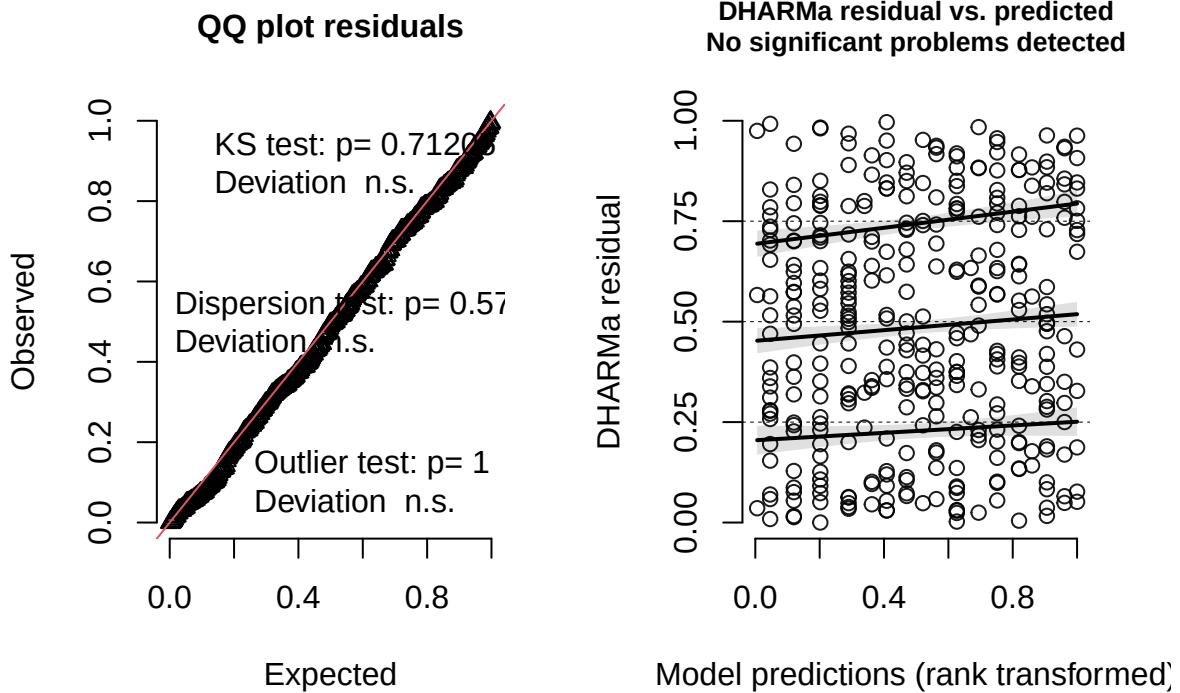
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	1.8705896	0.8326496	2.2465507	0.0246688	Binomial	BirdID
fixed	factor(fac.yr)2	0.0573513	1.1682648	0.0490910	0.9608468	Binomial	BirdID
fixed	factor(fac.yr)3	-2.5510787	0.9386854	-2.7177142	0.0065735	Binomial	BirdID
fixed	factor(fac.yr)4	-1.5436451	0.9420386	-1.6386219	0.1012920	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-1.5646955	0.9621473	-1.6262536	0.1038957	Binomial	BirdID
fixed	factor(fac.yr)6	-1.9731595	0.9182580	-2.1488072	0.0316497	Binomial	BirdID
fixed	factor(fac.yr)7	-1.9119187	0.9214085	-2.0749959	0.0379869	Binomial	BirdID
fixed	factor(fac.yr)8	-1.0800876	0.9446547	-1.1433677	0.2528860	Binomial	BirdID
fixed	factor(fac.yr)9	-0.7044723	0.9626127	-0.7318336	0.4642702	Binomial	BirdID
fixed	factor(fac.yr)10	-0.2115707	0.9976167	-0.2120761	0.8320476	Binomial	BirdID
fixed	factor(fac.yr)11	-1.9940781	0.9370289	-2.1280860	0.0333300	Binomial	BirdID
fixed	factor(fac.yr)12	-0.7038653	0.9868463	-0.7132471	0.4756929	Binomial	BirdID
fixed	factor(fac.yr)13	-1.0862102	0.9974063	-1.0890348	0.2761385	Binomial	BirdID
fixed	factor(fac.yr)14	-0.7521799	1.0535232	-0.7139662	0.4752481	Binomial	BirdID
fixed	factor(fac.yr)15	-1.2902989	1.0491199	-1.2298870	0.2187394	Binomial	BirdID
fixed	factor(fac.yr)16	-1.6408773	1.0438743	-1.5719107	0.1159713	Binomial	BirdID
fixed	factor(fac.yr)17	-0.5159897	1.2344548	-0.4179900	0.6759545	Binomial	BirdID
fixed	factor(fac.yr)18	-0.8757067	1.5562863	-0.5626900	0.5736460	Binomial	BirdID
fixed	factor(fac.yr)19	-2.8454056	1.5688607	-1.8136764	0.0697276	Binomial	BirdID
fixed	factor(fac.yr)20	-1.8337151	1.3630734	-1.3452798	0.1785349	Binomial	BirdID
fixed	(Intercept)	1.6345453	0.9635054	1.6964568	0.0897994	Binomial	Terr
fixed	factor(fac.yr)2	0.6253041	1.3544954	0.4616510	0.6443316	Binomial	Terr
fixed	factor(fac.yr)3	-2.7710461	1.0802318	-2.5652328	0.0103107	Binomial	Terr
fixed	factor(fac.yr)4	-0.8537081	1.0858713	-0.7861964	0.4317524	Binomial	Terr
fixed	factor(fac.yr)5	-1.2191134	1.1002300	-1.1080532	0.2678388	Binomial	Terr
fixed	factor(fac.yr)6	-1.5435038	1.0442522	-1.4780948	0.1393824	Binomial	Terr
fixed	factor(fac.yr)7	-1.5700003	1.0531166	-1.4908134	0.1360105	Binomial	Terr
fixed	factor(fac.yr)8	-0.6216006	1.1003664	-0.5649033	0.5721395	Binomial	Terr
fixed	factor(fac.yr)9	0.2394972	1.1268665	0.2125338	0.8316906	Binomial	Terr
fixed	factor(fac.yr)10	0.8602947	1.1682937	0.7363686	0.4615064	Binomial	Terr
fixed	factor(fac.yr)11	-1.4628332	1.0748686	-1.3609415	0.1735322	Binomial	Terr
fixed	factor(fac.yr)12	0.4063675	1.1453231	0.3548060	0.7227350	Binomial	Terr
fixed	factor(fac.yr)13	-0.5661842	1.2016203	-0.4711840	0.6375094	Binomial	Terr
fixed	factor(fac.yr)14	0.2064467	1.2571864	0.1642133	0.8695633	Binomial	Terr
fixed	factor(fac.yr)15	-0.2023701	1.2769178	-0.1584832	0.8740760	Binomial	Terr
fixed	factor(fac.yr)16	-1.6146259	1.2246316	-1.3184585	0.1873502	Binomial	Terr
fixed	factor(fac.yr)17	-0.0915809	1.5403369	-0.0594551	0.9525896	Binomial	Terr
fixed	factor(fac.yr)18	0.0435609	2.0636459	0.0211087	0.9831589	Binomial	Terr
fixed	factor(fac.yr)19	-2.4313636	1.8379321	-1.3228800	0.1858753	Binomial	Terr
fixed	factor(fac.yr)20	-0.3292002	1.6026387	-0.2054114	0.8372507	Binomial	Terr
fixed	(Intercept)	1.8032902	1.0141476	1.7781339	0.0753819	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	0.5240026	1.4067440	0.3724932	0.7095257	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-2.9972629	1.1297841	-2.6529519	0.0079791	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-0.9945128	1.1340764	-0.8769363	0.3805212	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-1.3906416	1.1486435	-1.2106816	0.2260175	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-1.6955894	1.0928773	-1.5514911	0.1207840	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-1.7526307	1.1039968	-1.5875324	0.1123921	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-0.8120971	1.1496114	-0.7064101	0.4799331	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	0.0864311	1.1762657	0.0734793	0.9414247	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	0.6878205	1.2154570	0.5658946	0.5714655	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-1.6671704	1.1256100	-1.4811262	0.1385730	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	0.2683522	1.1926745	0.2250004	0.8219790	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-0.6483809	1.2543444	-0.5169082	0.6052203	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	0.1478264	1.3082388	0.1129965	0.9100333	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-0.2832875	1.3288819	-0.2131774	0.8311886	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-1.7940562	1.2746156	-1.4075272	0.1592711	Binomial	BirdID+Terr

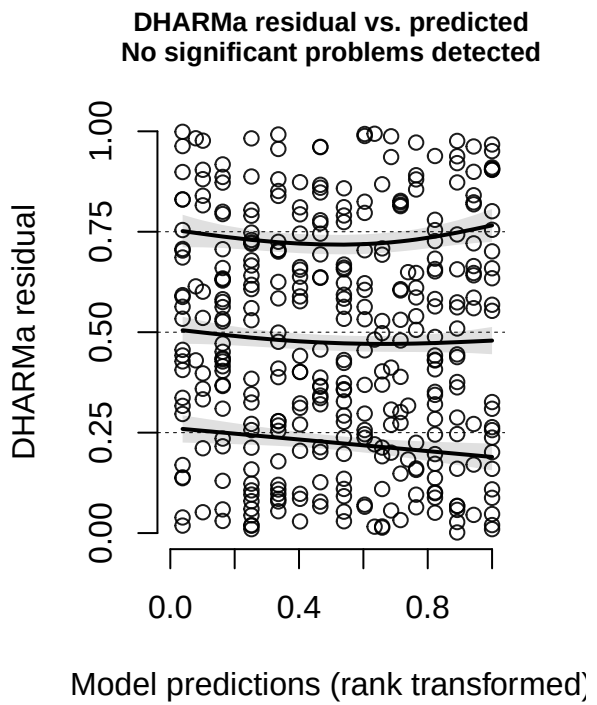
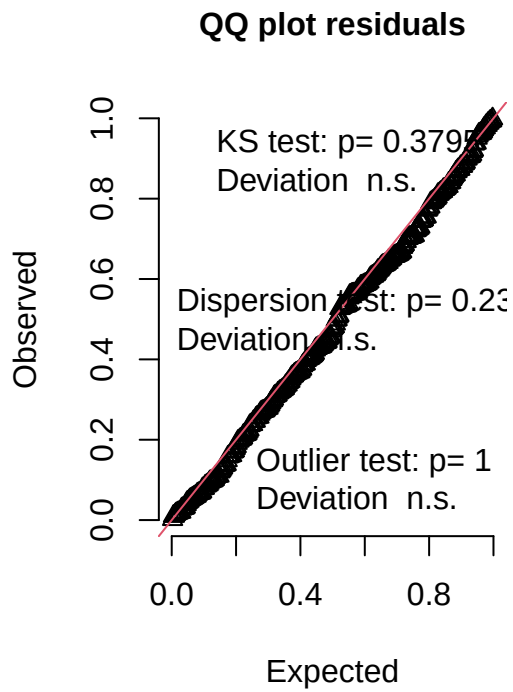
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-0.1374099	1.6019130	-0.0857786	0.9316424	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-0.0349424	2.1241318	-0.0164502	0.9868752	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-2.6092382	1.8887984	-1.3814276	0.1671475	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-0.4924687	1.6737028	-0.2942390	0.7685753	Binomial	BirdID+Terr

Territory only model is performing well on AIC and BIC.

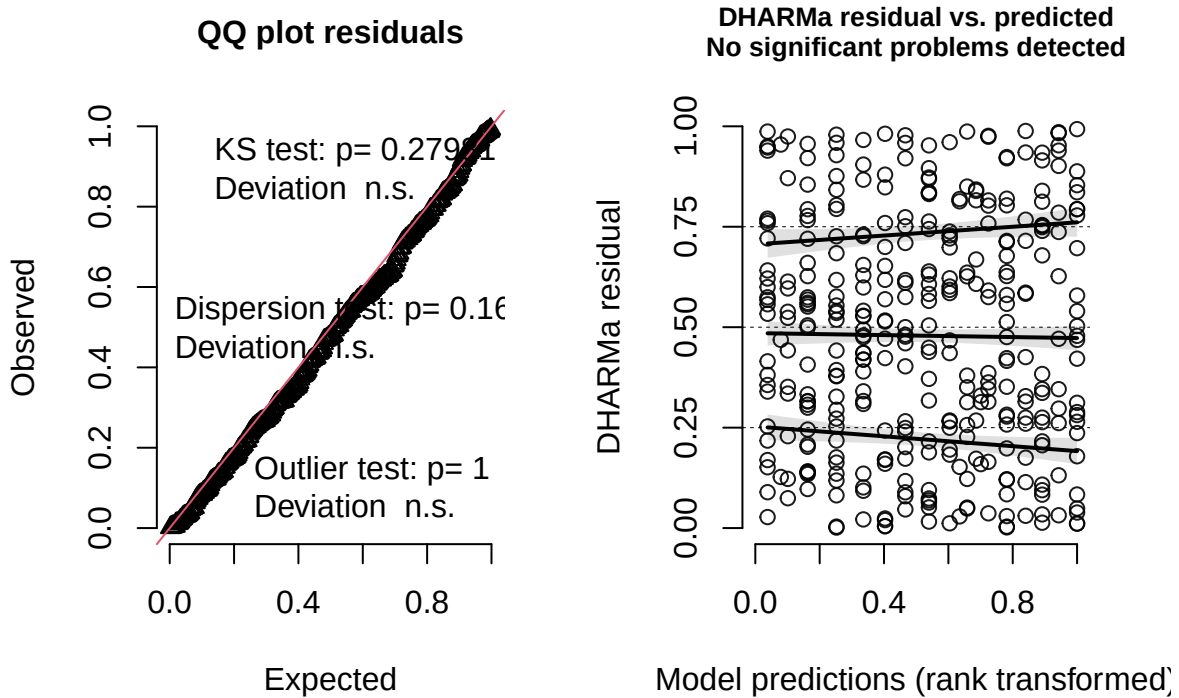
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Probabilities used to generate simulation 40

Simulation 40
 Probability of mate fidelity: 0.253
 Probability of pair dispersal: 0.736
 Probability of single male dispersal: 0.631
 Probability of single female dispersal: 0.401

Simulation 41

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
348	1	-457.4271	956.8543	1037.751	340.4038	327	0.9192461	0.9044776	Poisson	BirdID
348	1	-456.6835	957.3669	1042.115	318.4799	326	0.9296454	0.9135031	Poisson	BirdID+Terr
348	1	-457.9929	957.9859	1038.882	345.6307	327	0.9267220	0.9141238	Poisson	Terr

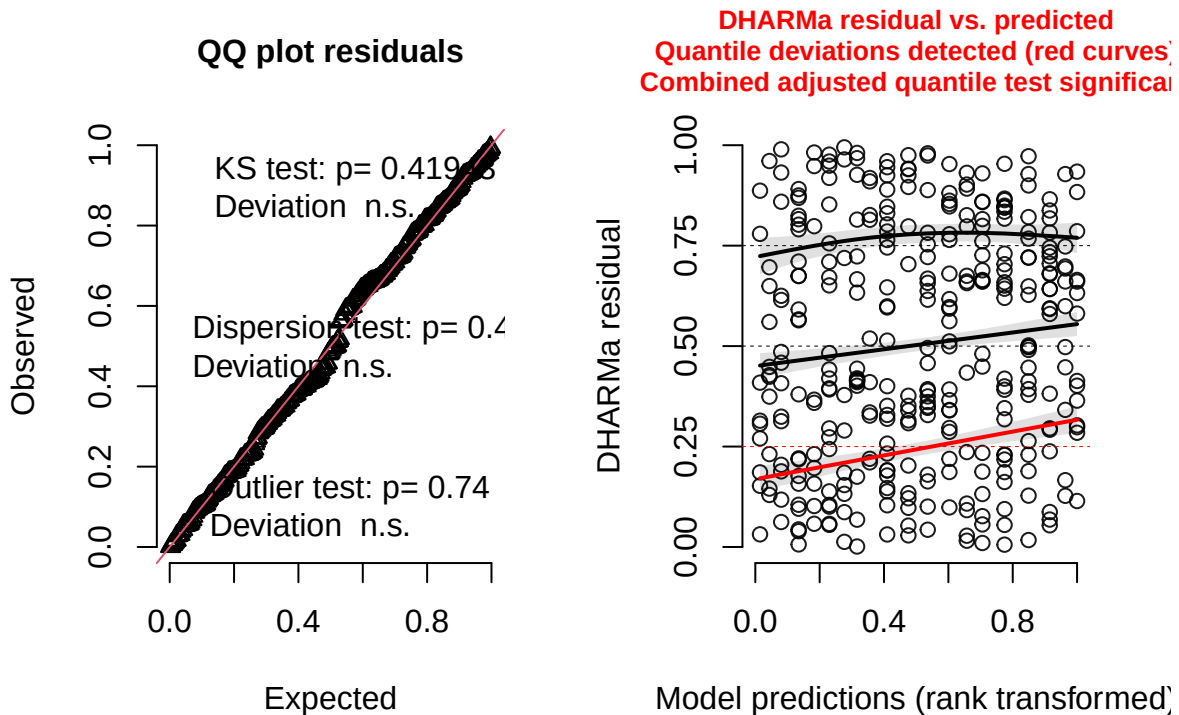
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.3470388	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.3288450	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.2906965	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.2549400	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.4706009	0.2606525	1.8054721	0.0710008	Poisson	BirdID
fixed	factor(fac.yr)2	0.0348401	0.3296812	0.1056781	0.9158378	Poisson	BirdID
fixed	factor(fac.yr)3	-0.6841541	0.3754278	-1.8223320	0.0684046	Poisson	BirdID
fixed	factor(fac.yr)4	-0.3347124	0.3277436	-1.0212629	0.3071299	Poisson	BirdID
fixed	factor(fac.yr)5	-0.0541435	0.3161214	-0.1712743	0.8640081	Poisson	BirdID
fixed	factor(fac.yr)6	-0.1540114	0.3198147	-0.4815645	0.6301154	Poisson	BirdID
fixed	factor(fac.yr)7	-1.0523804	0.3814148	-2.7591491	0.0057952	Poisson	BirdID
fixed	factor(fac.yr)8	-0.5735665	0.3311973	-1.7317973	0.0833097	Poisson	BirdID
fixed	factor(fac.yr)9	-0.1850435	0.3096209	-0.5976453	0.5500766	Poisson	BirdID
fixed	factor(fac.yr)10	-0.5288060	0.3390687	-1.5595835	0.1188583	Poisson	BirdID
fixed	factor(fac.yr)11	-0.8299282	0.3642956	-2.2781725	0.0227163	Poisson	BirdID
fixed	factor(fac.yr)12	-0.1881055	0.3290859	-0.5715999	0.5675930	Poisson	BirdID
fixed	factor(fac.yr)13	-0.3285761	0.3585839	-0.9163157	0.3595014	Poisson	BirdID
fixed	factor(fac.yr)14	-0.5438238	0.3571657	-1.5226094	0.1278565	Poisson	BirdID
fixed	factor(fac.yr)15	-1.1254042	0.4516026	-2.4920234	0.0127018	Poisson	BirdID
fixed	factor(fac.yr)16	-0.5796164	0.4253213	-1.3627731	0.1729540	Poisson	BirdID
fixed	factor(fac.yr)17	-0.7450311	0.4557304	-1.6348067	0.1020896	Poisson	BirdID
fixed	factor(fac.yr)18	-1.0007164	0.5009348	-1.9976981	0.0457494	Poisson	BirdID
fixed	factor(fac.yr)19	-1.5380480	0.5800382	-2.6516320	0.0080104	Poisson	BirdID
fixed	factor(fac.yr)20	-17.4272434	1567.5637231	-0.0111174	0.9911298	Poisson	BirdID
fixed	(Intercept)	0.3996788	0.2544754	1.5705993	0.1162758	Poisson	Terr
fixed	factor(fac.yr)2	0.1332636	0.3302374	0.4035388	0.6865519	Poisson	Terr
fixed	factor(fac.yr)3	-0.5862786	0.3686745	-1.5902337	0.1117821	Poisson	Terr
fixed	factor(fac.yr)4	-0.2004917	0.3220387	-0.6225701	0.5335671	Poisson	Terr
fixed	factor(fac.yr)5	0.1173127	0.3061391	0.3832007	0.7015710	Poisson	Terr
fixed	factor(fac.yr)6	-0.0221700	0.3106843	-0.0713586	0.9431123	Poisson	Terr
fixed	factor(fac.yr)7	-0.9209179	0.3763191	-2.4471732	0.0143982	Poisson	Terr
fixed	factor(fac.yr)8	-0.3980884	0.3219166	-1.2366197	0.2162283	Poisson	Terr
fixed	factor(fac.yr)9	-0.0091333	0.2991089	-0.0305352	0.9756402	Poisson	Terr
fixed	factor(fac.yr)10	-0.4246080	0.3305885	-1.2844004	0.1990019	Poisson	Terr
fixed	factor(fac.yr)11	-0.6890034	0.3588680	-1.9199352	0.0548661	Poisson	Terr
fixed	factor(fac.yr)12	-0.0479474	0.3210743	-0.1493341	0.8812900	Poisson	Terr
fixed	factor(fac.yr)13	-0.1144729	0.3518832	-0.3253151	0.7449426	Poisson	Terr
fixed	factor(fac.yr)14	-0.4289045	0.3481068	-1.2321060	0.2179095	Poisson	Terr
fixed	factor(fac.yr)15	-0.9817430	0.4463060	-2.1997083	0.0278276	Poisson	Terr
fixed	factor(fac.yr)16	-0.4122997	0.4177996	-0.9868360	0.3237230	Poisson	Terr
fixed	factor(fac.yr)17	-0.6102831	0.4479820	-1.3622938	0.1731052	Poisson	Terr
fixed	factor(fac.yr)18	-0.9007655	0.4943877	-1.8219822	0.0684577	Poisson	Terr
fixed	factor(fac.yr)19	-1.5014122	0.5732814	-2.6189796	0.0088193	Poisson	Terr
fixed	factor(fac.yr)20	-17.8259373	2093.6045256	-0.0085145	0.9932065	Poisson	Terr
fixed	(Intercept)	0.4187985	0.2651534	1.5794574	0.1142312	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	0.0820071	0.3370138	0.2433345	0.8077462	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.6423063	0.3802054	-1.6893665	0.0911492	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.2706166	0.3338065	-0.8106988	0.4175386	Poisson	BirdID+Terr

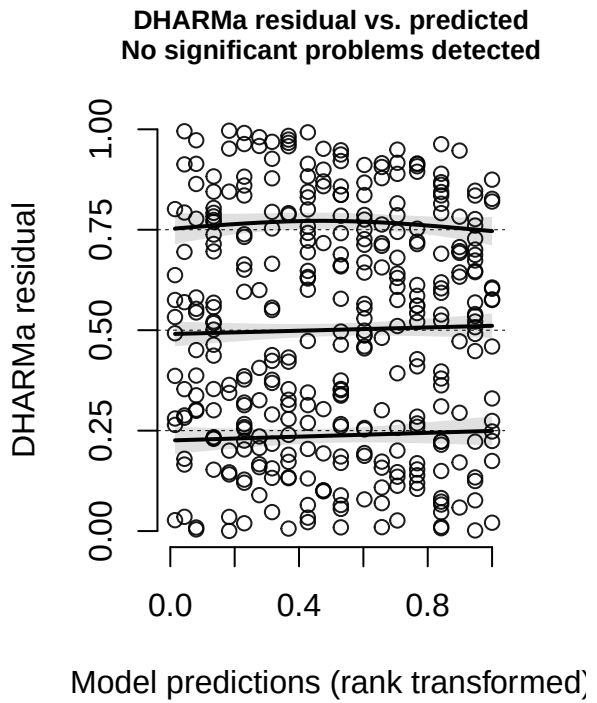
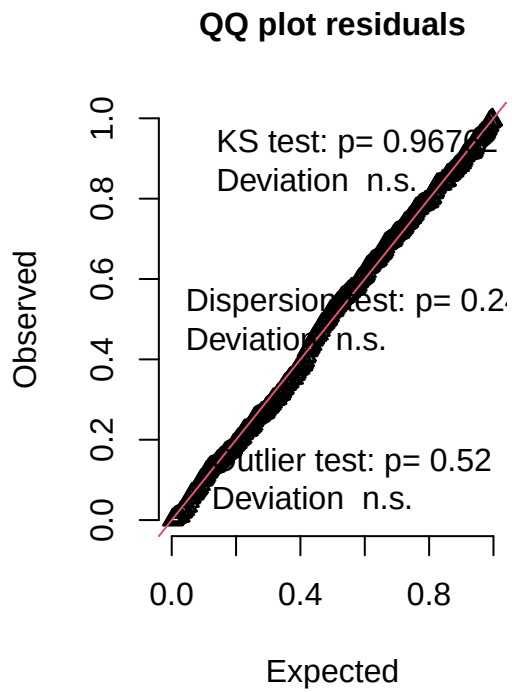
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.0281200	0.3200360	0.0878653	0.9299838	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.0860085	0.3242634	-0.2652428	0.7908224	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.9892434	0.3871352	-2.5552918	0.0106099	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.4953286	0.3347867	-1.4795347	0.1389975	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.1064472	0.3126274	-0.3404922	0.7334859	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.4910640	0.3432599	-1.4305892	0.1525480	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.7755472	0.3698553	-2.0968935	0.0360030	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.1448245	0.3339785	-0.4336342	0.6645541	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.2483990	0.3647615	-0.6809903	0.4958776	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.5143218	0.3617685	-1.4216876	0.1551170	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-1.0636248	0.4576321	-2.3241920	0.0201152	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.5080126	0.4314868	-1.1773537	0.2390544	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.6856162	0.4622463	-1.4832269	0.1380141	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.9461232	0.5075789	-1.8639925	0.0623228	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.5375806	0.5858464	-2.6245455	0.0086765	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-18.5895770	2942.9805328	-0.0063166	0.9949601	Poisson	BirdID+Terr

The band ID only and territory models are performing similarly on the criteria, but the band ID only model is performing marginally better.

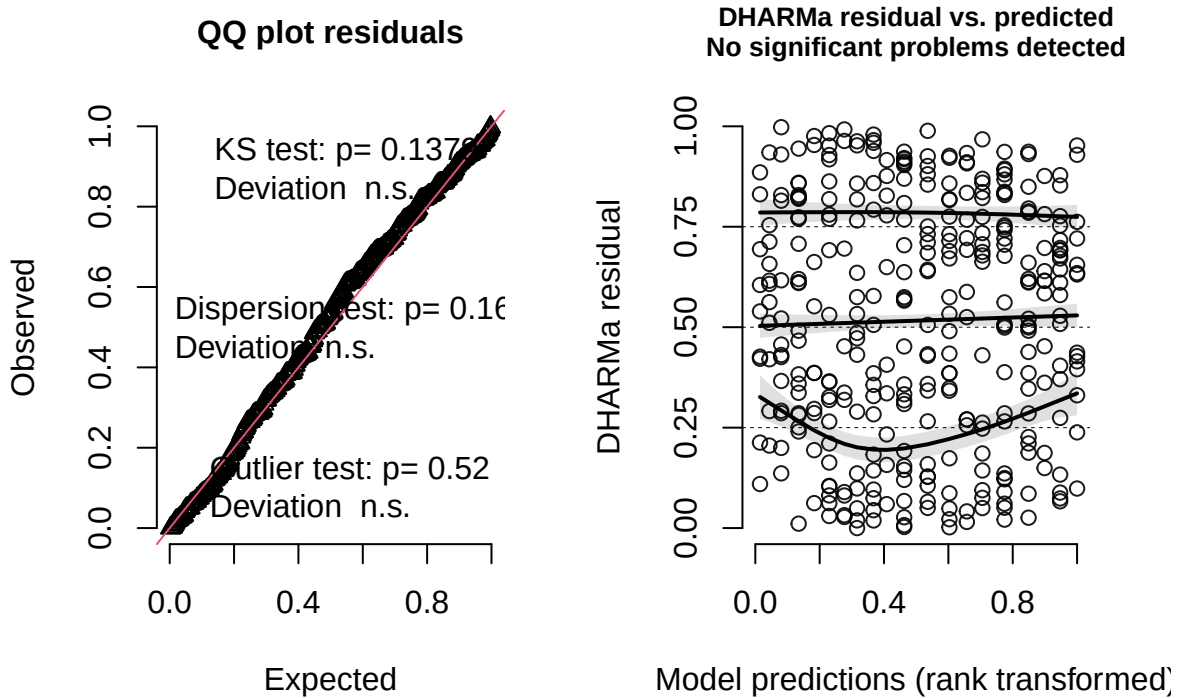
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics for the band ID only model are mildly concerning.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
348	1	-193.9609	431.9218	516.6702	237.4178	326	0.733	Binomial	BirdID+Terr
348	1	-195.1203	432.2406	513.1369	285.1107	327	0.616	Binomial	BirdID
348	1	-200.2286	442.4572	523.3534	325.6853	327	0.563	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.2607182	Binomial	BirdID
ran_pars	Terr	sd__(Intercept)	0.9281646	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.3051138	Binomial	BirdID+Terr
ran_pars	Terr	sd__(Intercept)	0.8923807	Binomial	BirdID+Terr

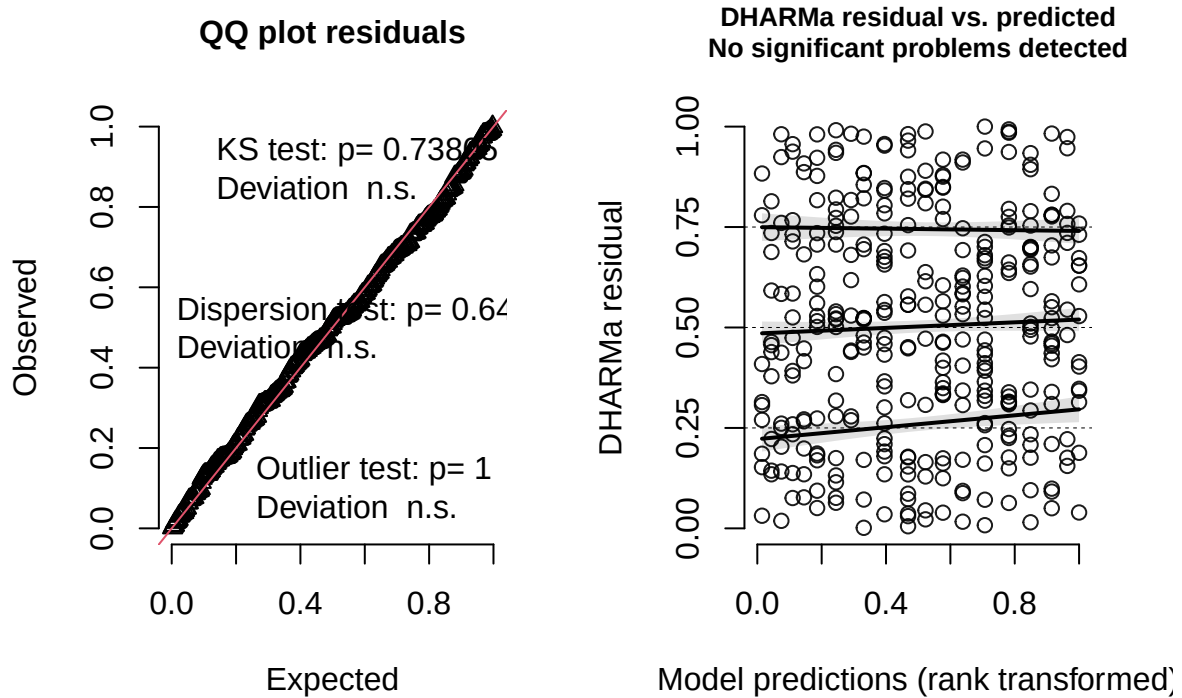
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	1.7378742	0.9121944	1.9051577	0.0567596	Binomial	BirdID
fixed	factor(fac.yr)2	1.4555693	1.5628886	0.9313327	0.3516815	Binomial	BirdID
fixed	factor(fac.yr)3	-1.7357488	1.0743750	-1.6155894	0.1061831	Binomial	BirdID
fixed	factor(fac.yr)4	-0.2308810	1.1046509	-0.2090082	0.8344419	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.0491073	1.1241117	-0.0436854	0.9651552	Binomial	BirdID
fixed	factor(fac.yr)6	-0.1869924	1.0978273	-0.1703295	0.8647510	Binomial	BirdID
fixed	factor(fac.yr)7	-1.5131905	1.0419889	-1.4522136	0.1464422	Binomial	BirdID
fixed	factor(fac.yr)8	-1.6140636	1.0249137	-1.5748287	0.1152960	Binomial	BirdID
fixed	factor(fac.yr)9	-0.7587377	1.0424506	-0.7278404	0.4667113	Binomial	BirdID
fixed	factor(fac.yr)10	-1.1111042	1.0528726	-1.0553074	0.2912848	Binomial	BirdID
fixed	factor(fac.yr)11	-2.2334239	1.0490443	-2.1290083	0.0332536	Binomial	BirdID
fixed	factor(fac.yr)12	-1.1099067	1.0923112	-1.0161086	0.3095777	Binomial	BirdID
fixed	factor(fac.yr)13	-1.4147657	1.1310673	-1.2508237	0.2109988	Binomial	BirdID
fixed	factor(fac.yr)14	-2.2957455	1.0746260	-2.1363204	0.0326533	Binomial	BirdID
fixed	factor(fac.yr)15	-2.7828521	1.1290111	-2.4648580	0.0137068	Binomial	BirdID
fixed	factor(fac.yr)16	-1.9625998	1.1923506	-1.6459921	0.0997654	Binomial	BirdID
fixed	factor(fac.yr)17	-2.8729824	1.2406168	-2.3157693	0.0205709	Binomial	BirdID
fixed	factor(fac.yr)18	-2.6865856	1.2365634	-2.1726226	0.0298087	Binomial	BirdID
fixed	factor(fac.yr)19	-3.0051052	1.2280909	-2.4469729	0.0144062	Binomial	BirdID
fixed	factor(fac.yr)20	-10.7929076	23.5431333	-0.4584312	0.6466427	Binomial	BirdID
fixed	(Intercept)	1.5055873	0.8499206	1.7714446	0.0764868	Binomial	Terr
fixed	factor(fac.yr)2	1.3924045	1.4403163	0.9667353	0.3336764	Binomial	Terr
fixed	factor(fac.yr)3	-1.4062674	0.9993591	-1.4071693	0.1593772	Binomial	Terr
fixed	factor(fac.yr)4	-0.0267312	1.0184495	-0.0262470	0.9790603	Binomial	Terr
fixed	factor(fac.yr)5	0.3031897	1.0411372	0.2912101	0.7708906	Binomial	Terr
fixed	factor(fac.yr)6	0.1482427	1.0118779	0.1465025	0.8835247	Binomial	Terr
fixed	factor(fac.yr)7	-1.1019805	0.9695973	-1.1365343	0.2557330	Binomial	Terr
fixed	factor(fac.yr)8	-1.0353699	0.9476038	-1.0926189	0.2745612	Binomial	Terr
fixed	factor(fac.yr)9	-0.2994538	0.9588518	-0.3123046	0.7548091	Binomial	Terr
fixed	factor(fac.yr)10	-0.7946522	0.9735871	-0.8162107	0.4143797	Binomial	Terr
fixed	factor(fac.yr)11	-1.6485979	0.9745177	-1.6917064	0.0907020	Binomial	Terr
fixed	factor(fac.yr)12	-0.6298238	1.0026585	-0.6281539	0.5299032	Binomial	Terr
fixed	factor(fac.yr)13	-0.7245130	1.0485449	-0.6909699	0.4895845	Binomial	Terr
fixed	factor(fac.yr)14	-1.8284677	0.9870425	-1.8524712	0.0639582	Binomial	Terr
fixed	factor(fac.yr)15	-2.1251986	1.0533470	-2.0175674	0.0436363	Binomial	Terr
fixed	factor(fac.yr)16	-1.3275864	1.0930892	-1.2145270	0.2245465	Binomial	Terr
fixed	factor(fac.yr)17	-2.2189414	1.1401919	-1.9461122	0.0516413	Binomial	Terr
fixed	factor(fac.yr)18	-2.2528397	1.1397974	-1.9765264	0.0480952	Binomial	Terr
fixed	factor(fac.yr)19	-2.5781997	1.1365296	-2.2684845	0.0232997	Binomial	Terr
fixed	factor(fac.yr)20	-11.6587525	56.1815642	-0.2075192	0.8356044	Binomial	Terr
fixed	(Intercept)	1.8373998	0.9954400	1.8458167	0.0649188	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	1.6200517	1.7123562	0.9460950	0.3441001	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-1.8346862	1.1708931	-1.5669118	0.1171353	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-0.1312970	1.2078462	-0.1087034	0.9134377	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	0.1776205	1.2172463	0.1459199	0.8839846	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-0.0056731	1.1874199	-0.0047777	0.9961880	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-1.5231318	1.1341247	-1.3430021	0.1792713	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-1.5041611	1.1144895	-1.3496413	0.1771311	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-0.5787557	1.1278146	-0.5131657	0.6078354	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-1.1216732	1.1445518	-0.9800109	0.3270807	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-2.3889287	1.1423914	-2.0911647	0.0365133	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-1.1107241	1.1871599	-0.9356145	0.3494717	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-1.3251611	1.2390950	-1.0694588	0.2848630	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-2.4854974	1.1651265	-2.1332426	0.0329048	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-2.9725389	1.2321143	-2.4125513	0.0158413	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-1.9269630	1.2836612	-1.5011461	0.1333178	Binomial	BirdID+Terr

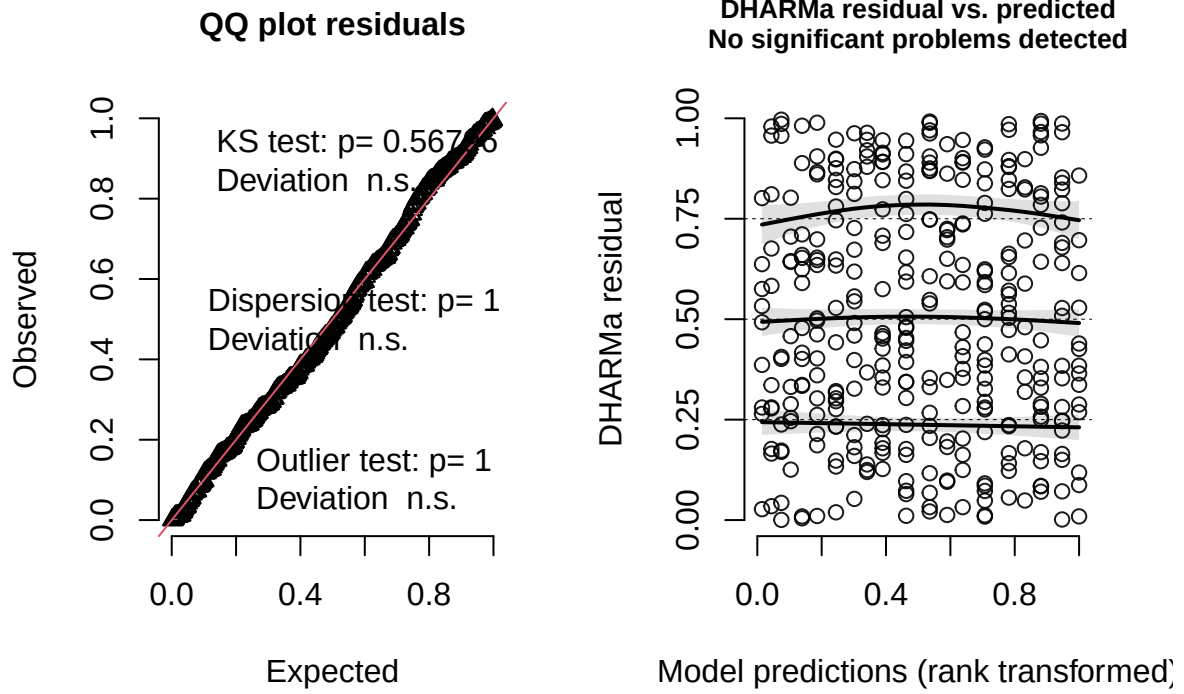
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-2.9226918	1.3350315	-2.1892306	0.0285801	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-2.7384581	1.3399103	-2.0437622	0.0409770	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-3.2901510	1.3482175	-2.4403710	0.0146722	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-11.9784844	36.9257641	-0.3243937	0.7456400	Binomial	BirdID+Terr

Similar to other binomial models

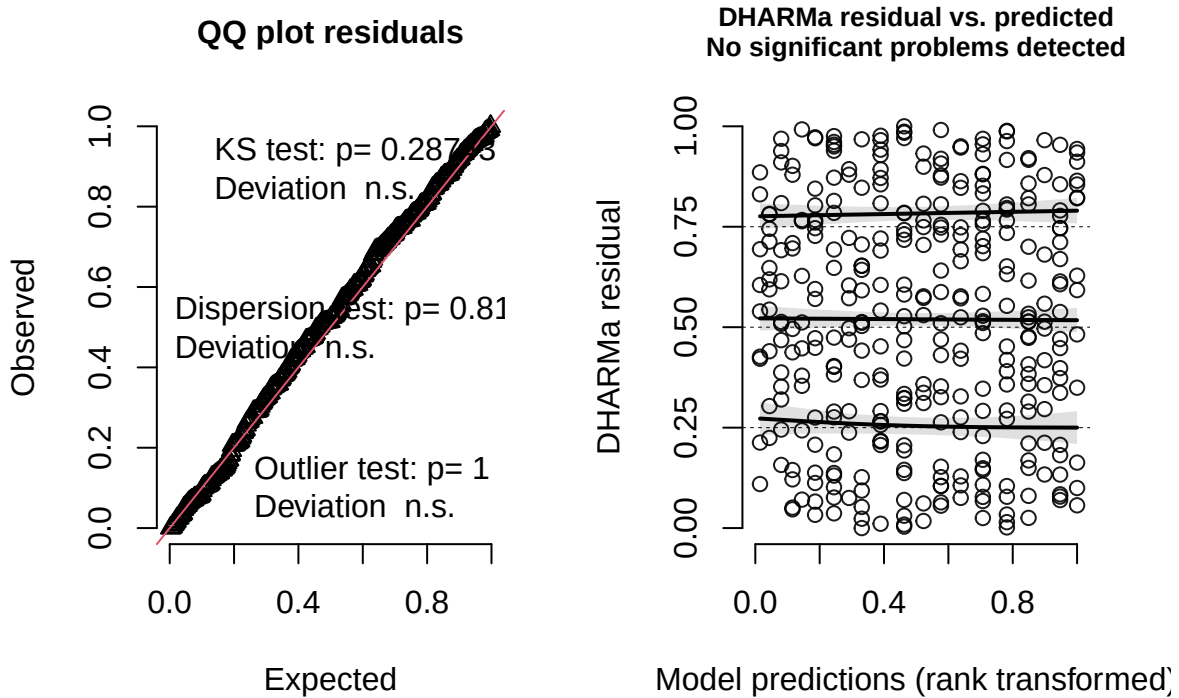
DHARMa residual



DHARMa residual



DHARMA residual



Diagnostics look good.

Probabilities used to generate simulation 41

Simulation 41
 Probability of mate fidelity: 0.558
 Probability of pair dispersal: 0.294
 Probability of single male dispersal: 0.963
 Probability of single female dispersal: 0.754

Simulation 42

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
329	1	-453.2977	948.5954	1028.313	306.2803	308	0.8858221	0.8739806	Poisson	Terr
329	1	-453.7703	949.5406	1029.258	314.8273	308	0.8861117	0.8766141	Poisson	BirdID
329	1	-453.0300	950.0600	1033.573	294.0753	307	0.8811849	0.8657183	Poisson	BirdID+Terr

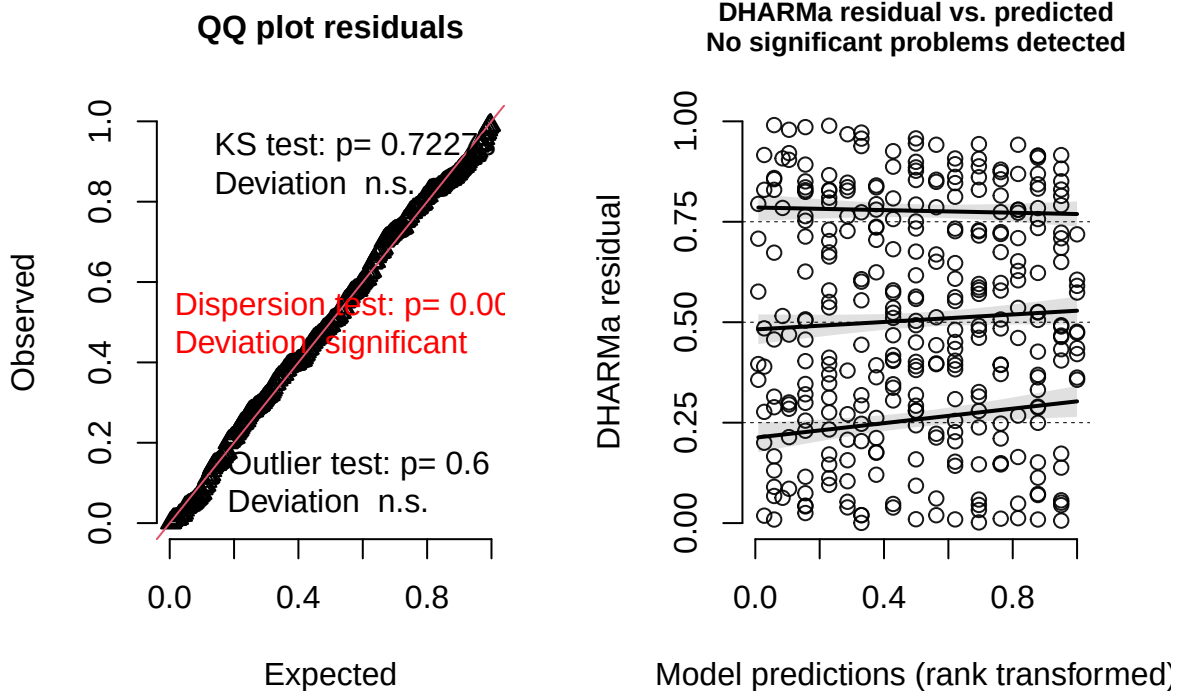
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.2193154	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.2421110	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.1671937	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.2152406	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.8958181	0.2120582	4.2243978	0.0000240	Poisson	BirdID
fixed	factor(fac.yr)2	-0.7289008	0.3217586	-2.2653652	0.0234903	Poisson	BirdID
fixed	factor(fac.yr)3	-0.9383475	0.3367889	-2.7861591	0.0053337	Poisson	BirdID
fixed	factor(fac.yr)4	-0.7468998	0.2994509	-2.4942312	0.0126230	Poisson	BirdID
fixed	factor(fac.yr)5	-0.1490646	0.2587398	-0.5761178	0.5645356	Poisson	BirdID
fixed	factor(fac.yr)6	-0.7338342	0.2765293	-2.6537301	0.0079607	Poisson	BirdID
fixed	factor(fac.yr)7	-0.9860070	0.3001954	-3.2845503	0.0010215	Poisson	BirdID
fixed	factor(fac.yr)8	-0.5738020	0.2753584	-2.0838368	0.0371750	Poisson	BirdID
fixed	factor(fac.yr)9	-0.3054014	0.2607849	-1.1710854	0.2415645	Poisson	BirdID
fixed	factor(fac.yr)10	-0.4990490	0.2836262	-1.7595305	0.0784874	Poisson	BirdID
fixed	factor(fac.yr)11	-0.9618548	0.3009490	-3.1960725	0.0013931	Poisson	BirdID
fixed	factor(fac.yr)12	-0.5761775	0.2768160	-2.0814462	0.0373931	Poisson	BirdID
fixed	factor(fac.yr)13	-0.3311822	0.2896474	-1.1433977	0.2528735	Poisson	BirdID
fixed	factor(fac.yr)14	-0.8816172	0.3332998	-2.6451177	0.0081663	Poisson	BirdID
fixed	factor(fac.yr)15	-1.5665412	0.4400598	-3.5598373	0.0003711	Poisson	BirdID
fixed	factor(fac.yr)16	-0.9536451	0.3559838	-2.6789002	0.0073864	Poisson	BirdID
fixed	factor(fac.yr)17	-1.0754524	0.4211355	-2.5536967	0.0106586	Poisson	BirdID
fixed	factor(fac.yr)18	-1.8001004	0.6251648	-2.8794014	0.0039843	Poisson	BirdID
fixed	factor(fac.yr)19	-17.6403316	1880.9377658	-0.0093785	0.9925172	Poisson	BirdID
fixed	factor(fac.yr)20	-1.2722279	0.6284034	-2.0245402	0.0429146	Poisson	BirdID
fixed	(Intercept)	0.8958093	0.2136954	4.1919916	0.0000277	Poisson	Terr
fixed	factor(fac.yr)2	-0.7135167	0.3254407	-2.1924630	0.0283461	Poisson	Terr
fixed	factor(fac.yr)3	-0.9308087	0.3421394	-2.7205537	0.0065173	Poisson	Terr
fixed	factor(fac.yr)4	-0.7372125	0.3015657	-2.4446166	0.0145006	Poisson	Terr
fixed	factor(fac.yr)5	-0.1347146	0.2621274	-0.5139280	0.6073023	Poisson	Terr
fixed	factor(fac.yr)6	-0.7116801	0.2792316	-2.5487086	0.0108123	Poisson	Terr
fixed	factor(fac.yr)7	-0.9692282	0.3009581	-3.2204753	0.0012798	Poisson	Terr
fixed	factor(fac.yr)8	-0.5827344	0.2765097	-2.1074641	0.0350774	Poisson	Terr
fixed	factor(fac.yr)9	-0.3160567	0.2617254	-1.2075890	0.2272054	Poisson	Terr
fixed	factor(fac.yr)10	-0.5263197	0.2838180	-1.8544271	0.0636781	Poisson	Terr
fixed	factor(fac.yr)11	-0.9749077	0.3023079	-3.2248829	0.0012602	Poisson	Terr
fixed	factor(fac.yr)12	-0.5921806	0.2773716	-2.1349718	0.0327633	Poisson	Terr
fixed	factor(fac.yr)13	-0.3254688	0.2907476	-1.1194207	0.2629607	Poisson	Terr
fixed	factor(fac.yr)14	-0.8473244	0.3343937	-2.5339126	0.0112797	Poisson	Terr
fixed	factor(fac.yr)15	-1.5411738	0.4404058	-3.4994406	0.0004662	Poisson	Terr
fixed	factor(fac.yr)16	-0.8730937	0.3555636	-2.4555205	0.0140681	Poisson	Terr
fixed	factor(fac.yr)17	-1.0201717	0.4223183	-2.4156464	0.0157073	Poisson	Terr
fixed	factor(fac.yr)18	-1.7396058	0.6261929	-2.7780669	0.0054683	Poisson	Terr
fixed	factor(fac.yr)19	-17.4232240	1789.0809325	-0.0097386	0.9922298	Poisson	Terr
fixed	factor(fac.yr)20	-1.2679096	0.6268412	-2.0226968	0.0431044	Poisson	Terr
fixed	(Intercept)	0.8944738	0.2179958	4.1031703	0.0000408	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.7241948	0.3273522	-2.2122801	0.0269473	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.9495867	0.3450588	-2.7519562	0.0059240	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.7527879	0.3052308	-2.4662904	0.0136521	Poisson	BirdID+Terr

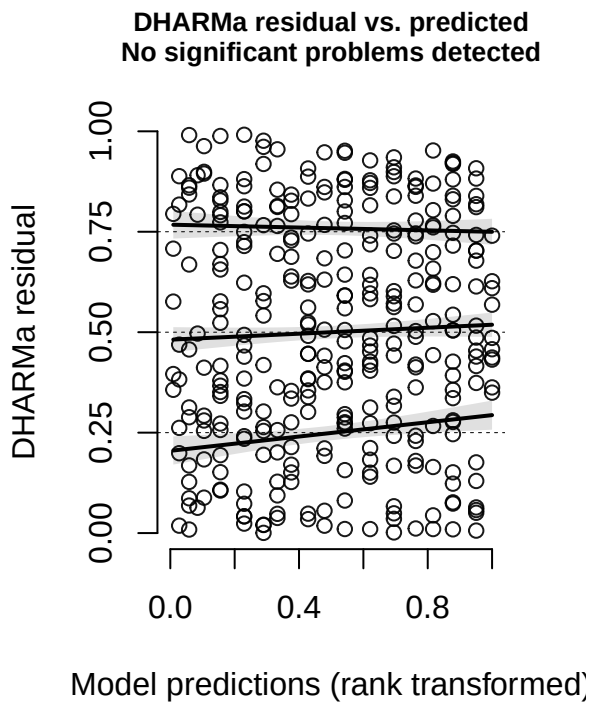
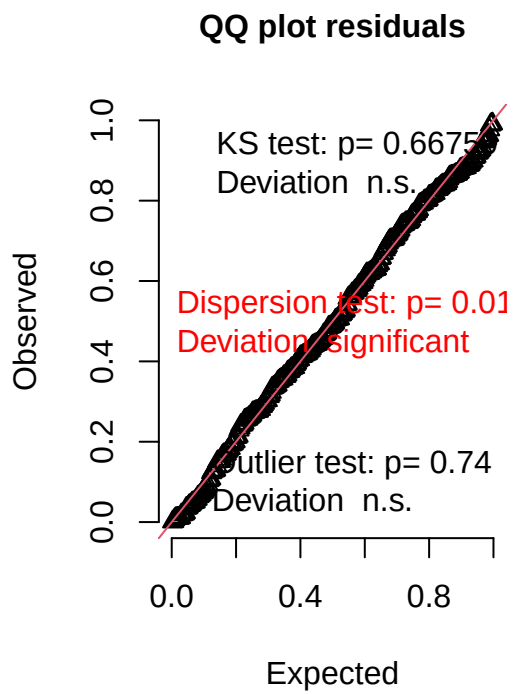
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.1514219	0.2661900	-0.5688489	0.5694587	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.7290326	0.2830040	-2.5760506	0.0099936	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.9926583	0.3052518	-3.2519331	0.0011462	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.5913366	0.2808265	-2.1057007	0.0352304	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.3228703	0.2662997	-1.2124323	0.2253469	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.5226130	0.2887416	-1.8099680	0.0703007	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.9778645	0.3065934	-3.1894504	0.0014254	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.5977027	0.2823678	-2.1167521	0.0342809	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.3481330	0.2962727	-1.1750426	0.2399777	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.8748157	0.3394709	-2.5769975	0.0099663	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-1.5663107	0.4452622	-3.5177268	0.0004353	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.9129147	0.3610410	-2.5285624	0.0114531	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-1.0521042	0.4279025	-2.4587474	0.0139423	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-1.7727834	0.6311552	-2.8087916	0.0049728	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-17.0909578	1481.6564010	-0.0115350	0.9907966	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-1.3042395	0.6337493	-2.0579739	0.0395926	Poisson	BirdID+Terr

Territory only model is performing well on all criteria.

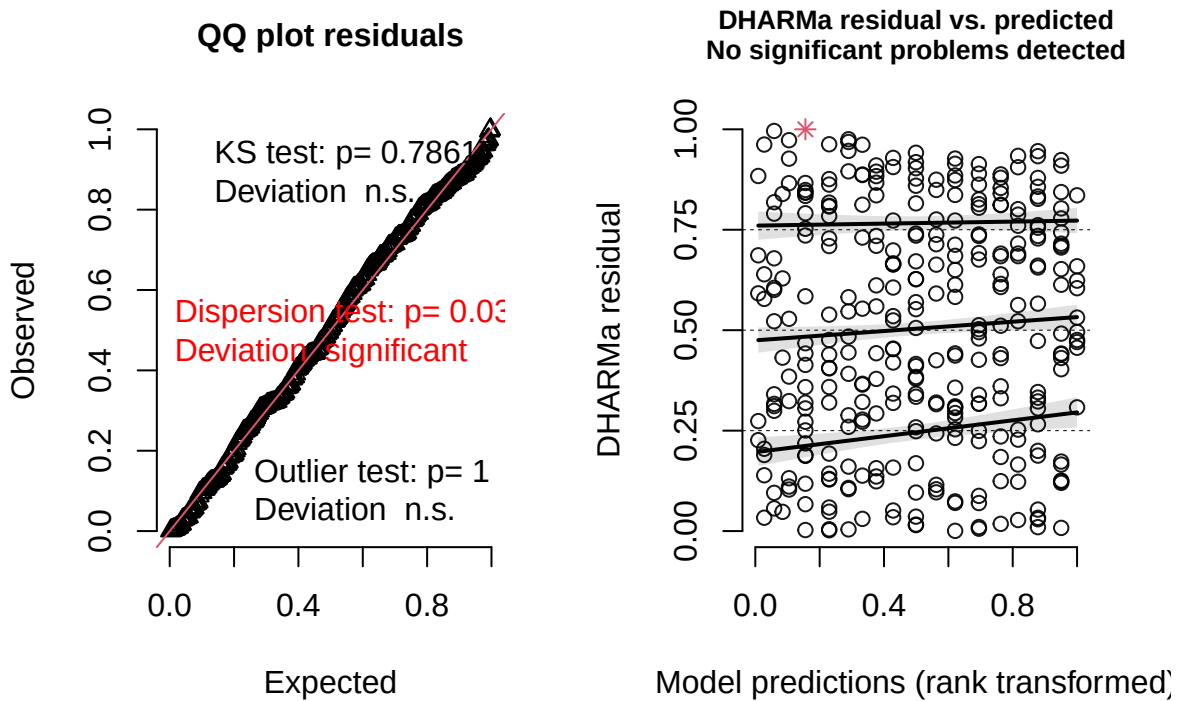
DHARMA residual



DHARMa residual



DHARMa residual



Similar response for other models who had significant dispersion tests.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
329	1	-173.3582	390.7163	474.2296	220.1482	307	0.625	Binomial	BirdID+Terr
329	1	-174.4176	390.8351	470.5523	261.2698	308	0.510	Binomial	BirdID
329	1	-177.5544	397.1089	476.8261	290.3995	308	0.440	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.1758980	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	0.8956844	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.2257135	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.8219306	Binomial	BirdID+Terr

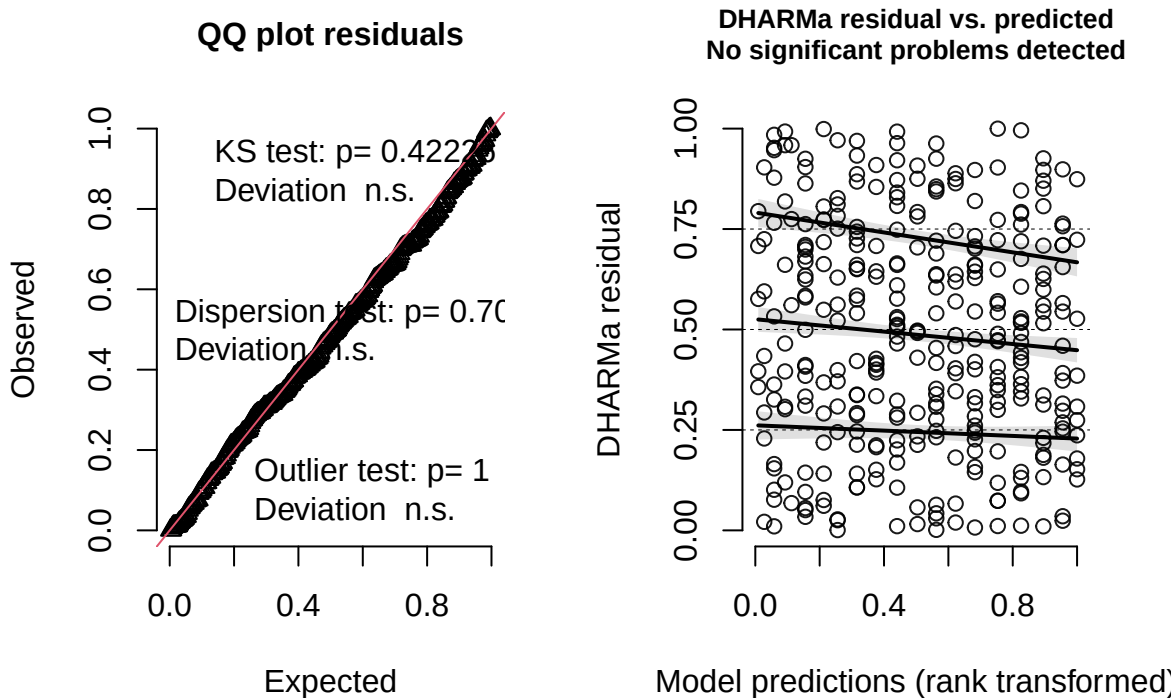
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	17.61263	2097.711	0.0083961	0.9933009	Binomial	BirdID
fixed	factor(fac.yr)2	-16.51974	2097.711	-0.0078751	0.9937166	Binomial	BirdID
fixed	factor(fac.yr)3	-17.61208	2097.711	-0.0083959	0.9933012	Binomial	BirdID
fixed	factor(fac.yr)4	-15.77979	2097.711	-0.0075224	0.9939981	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-15.97553	2097.711	-0.0076157	0.9939236	Binomial	BirdID
fixed	factor(fac.yr)6	-16.54975	2097.711	-0.0078894	0.9937052	Binomial	BirdID
fixed	factor(fac.yr)7	-17.66102	2097.711	-0.0084192	0.9932825	Binomial	BirdID
fixed	factor(fac.yr)8	-16.11098	2097.711	-0.0076803	0.9938721	Binomial	BirdID
fixed	factor(fac.yr)9	-15.92277	2097.711	-0.0075905	0.9939437	Binomial	BirdID
fixed	factor(fac.yr)10	-15.84310	2097.711	-0.0075526	0.9939740	Binomial	BirdID
fixed	factor(fac.yr)11	-17.04596	2097.711	-0.0081260	0.9935165	Binomial	BirdID
fixed	factor(fac.yr)12	-15.98344	2097.711	-0.0076195	0.9939206	Binomial	BirdID
fixed	factor(fac.yr)13	-16.10456	2097.711	-0.0076772	0.9938745	Binomial	BirdID
fixed	factor(fac.yr)14	-16.92987	2097.711	-0.0080706	0.9935606	Binomial	BirdID
fixed	factor(fac.yr)15	-18.50073	2097.711	-0.0088195	0.9929632	Binomial	BirdID
fixed	factor(fac.yr)16	-17.64407	2097.711	-0.0084111	0.9932890	Binomial	BirdID
fixed	factor(fac.yr)17	-18.11655	2097.711	-0.0086363	0.9931093	Binomial	BirdID
fixed	factor(fac.yr)18	-19.08187	2097.711	-0.0090965	0.9927421	Binomial	BirdID
fixed	factor(fac.yr)19	-37.38588	6064.751	-0.0061645	0.9950815	Binomial	BirdID
fixed	factor(fac.yr)20	-17.96755	2097.711	-0.0085653	0.9931660	Binomial	BirdID
fixed	(Intercept)	17.60467	2139.127	0.0082298	0.9934336	Binomial	Terr
fixed	factor(fac.yr)2	-16.44826	2139.127	-0.0076892	0.9938649	Binomial	Terr
fixed	factor(fac.yr)3	-17.50284	2139.127	-0.0081822	0.9934716	Binomial	Terr
fixed	factor(fac.yr)4	-15.72296	2139.127	-0.0073502	0.9941355	Binomial	Terr
fixed	factor(fac.yr)5	-15.79819	2139.127	-0.0073853	0.9941074	Binomial	Terr
fixed	factor(fac.yr)6	-16.29001	2139.127	-0.0076153	0.9939240	Binomial	Terr
fixed	factor(fac.yr)7	-17.31175	2139.127	-0.0080929	0.9935429	Binomial	Terr
fixed	factor(fac.yr)8	-16.07832	2139.127	-0.0075163	0.9940029	Binomial	Terr
fixed	factor(fac.yr)9	-15.92710	2139.127	-0.0074456	0.9940593	Binomial	Terr
fixed	factor(fac.yr)10	-16.02393	2139.127	-0.0074909	0.9940232	Binomial	Terr
fixed	factor(fac.yr)11	-17.02099	2139.127	-0.0079570	0.9936513	Binomial	Terr
fixed	factor(fac.yr)12	-16.09494	2139.127	-0.0075241	0.9939967	Binomial	Terr
fixed	factor(fac.yr)13	-16.12729	2139.127	-0.0075392	0.9939847	Binomial	Terr
fixed	factor(fac.yr)14	-16.63280	2139.127	-0.0077755	0.9937961	Binomial	Terr
fixed	factor(fac.yr)15	-18.17283	2139.127	-0.0084954	0.9932217	Binomial	Terr
fixed	factor(fac.yr)16	-17.24522	2139.127	-0.0080618	0.9935677	Binomial	Terr
fixed	factor(fac.yr)17	-17.57862	2139.127	-0.0082177	0.9934433	Binomial	Terr
fixed	factor(fac.yr)18	-18.52697	2139.127	-0.0086610	0.9930896	Binomial	Terr
fixed	factor(fac.yr)19	-37.46603	10862.164	-0.0034492	0.9972479	Binomial	Terr
fixed	factor(fac.yr)20	-17.56774	2139.127	-0.0082126	0.9934474	Binomial	Terr
fixed	(Intercept)	17.66887	2079.119	0.0084983	0.9932195	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-16.38013	2079.119	-0.0078784	0.9937140	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-17.68341	2079.119	-0.0085052	0.9932139	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-15.67403	2079.119	-0.0075388	0.9939850	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-15.81637	2079.119	-0.0076072	0.9939304	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-16.40573	2079.119	-0.0078907	0.9937042	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-17.66700	2079.119	-0.0084974	0.9932202	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-16.04814	2079.119	-0.0077187	0.9938414	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-15.77109	2079.119	-0.0075855	0.9939477	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-15.69676	2079.119	-0.0075497	0.9939763	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-16.99859	2079.119	-0.0081759	0.9934767	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-15.88844	2079.119	-0.0076419	0.9939027	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-16.00394	2079.119	-0.0076975	0.9938584	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-16.70182	2079.119	-0.0080331	0.9935906	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-18.65249	2079.119	-0.0089713	0.9928420	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-17.57279	2079.119	-0.0084520	0.9932563	Binomial	BirdID+Terr

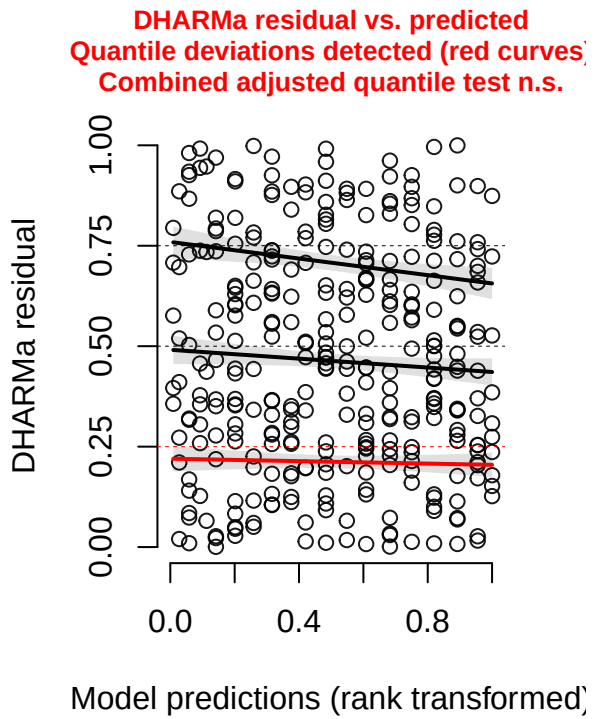
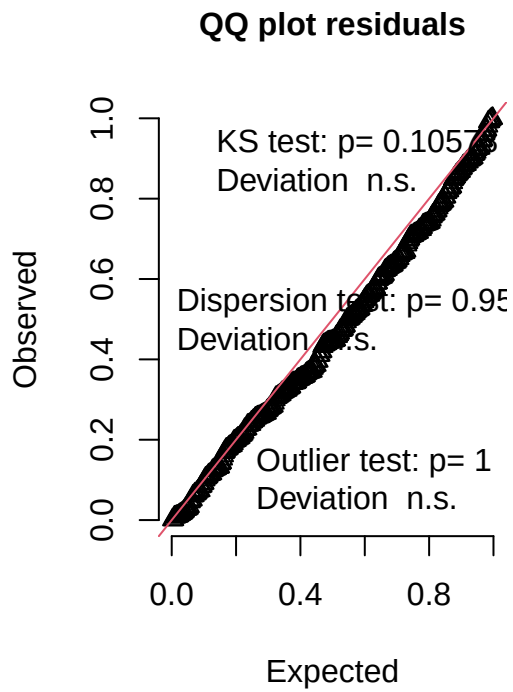
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-18.01309	2079.119	-0.0086638	0.9930874	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-19.28058	2079.119	-0.0092734	0.9926010	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-51.32619	6053320.192	-0.0000085	0.9999932	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-18.10158	2079.119	-0.0087064	0.9930534	Binomial	BirdID+Terr

Similar to the other binomial models.

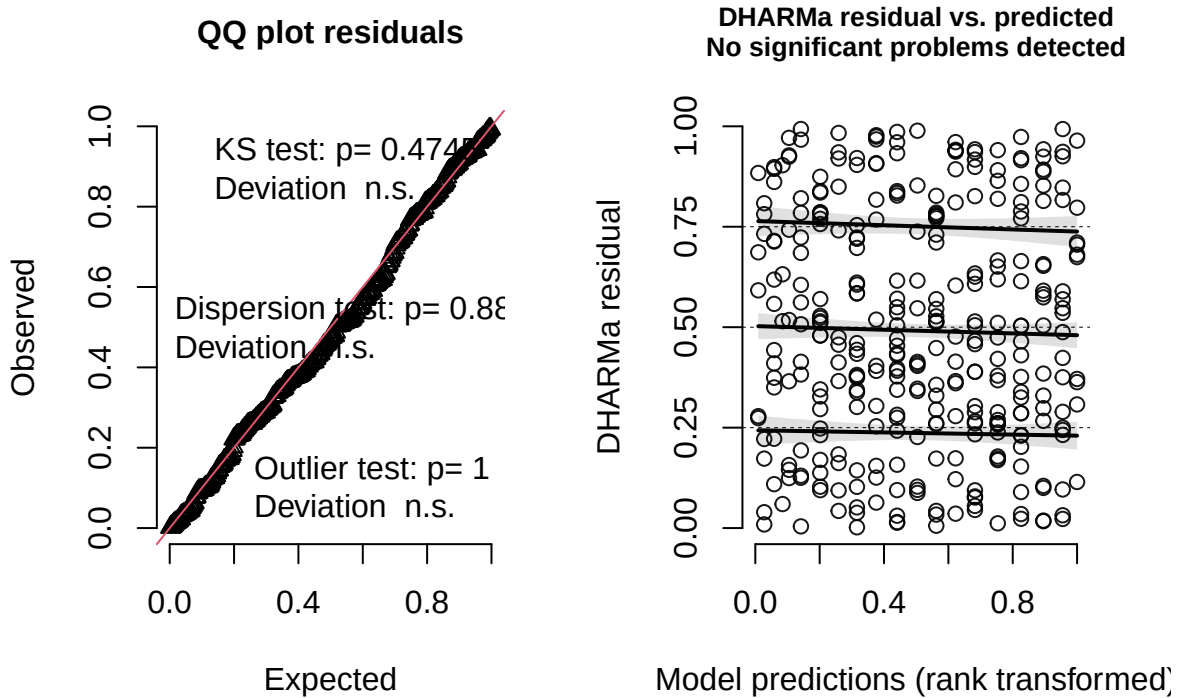
DHARMA residual



DHARMa residual



DHARMa residual



Mild concern on terr only model.

Probabilites used to generate simulation 42

Simulation 42
 Probability of mate fidelity: 0.036
 Probability of pair dispersal: 0.666
 Probability of single male dispersal: 0.69
 Probability of single female dispersal: 0.341

Simulation 43

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
359	1	-468.5546	981.1092	1066.542	291.4315	337	0.3670118	0.1198488	Poisson	BirdID+Terr
359	1	-470.2401	982.4803	1064.030	326.2985	338	0.3329991	0.1209011	Poisson	Terr
359	1	-470.6799	983.3598	1064.910	311.7595	338	0.3709657	0.1206580	Poisson	BirdID

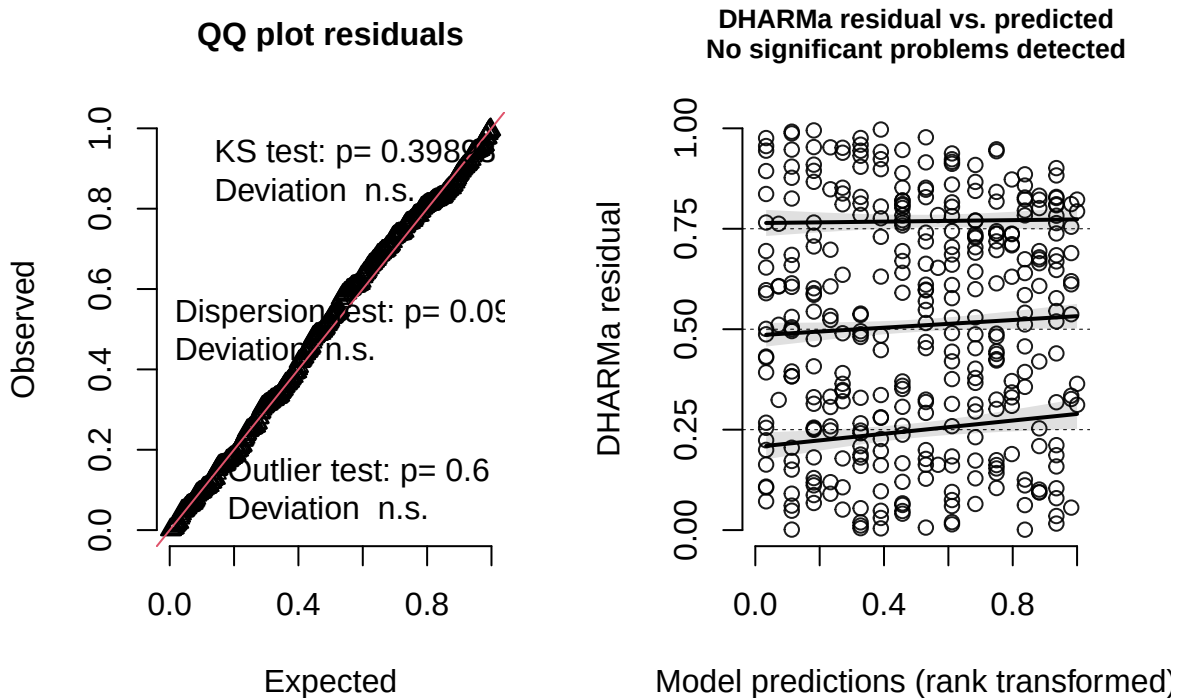
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.5093439	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.4520487	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.3710801	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.3432394	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.3226239	0.2590024	1.2456408	0.2128963	Poisson	BirdID
fixed	factor(fac.yr)2	-0.0764098	0.3239013	-0.2359046	0.8135067	Poisson	BirdID
fixed	factor(fac.yr)3	-0.4410043	0.3333588	-1.3229116	0.1858648	Poisson	BirdID
fixed	factor(fac.yr)4	-0.2579443	0.3268229	-0.7892480	0.4299671	Poisson	BirdID
fixed	factor(fac.yr)5	0.1266374	0.3069004	0.4126334	0.6798752	Poisson	BirdID
fixed	factor(fac.yr)6	-0.3114536	0.3261575	-0.9549178	0.3396193	Poisson	BirdID
fixed	factor(fac.yr)7	-0.7966286	0.3592216	-2.2176524	0.0265785	Poisson	BirdID
fixed	factor(fac.yr)8	-0.4541059	0.3280778	-1.3841409	0.1663153	Poisson	BirdID
fixed	factor(fac.yr)9	-0.3375917	0.3237790	-1.0426610	0.2971053	Poisson	BirdID
fixed	factor(fac.yr)10	-0.5302830	0.3395359	-1.5617876	0.1183380	Poisson	BirdID
fixed	factor(fac.yr)11	-1.2512094	0.4111184	-3.0434285	0.0023390	Poisson	BirdID
fixed	factor(fac.yr)12	-0.8379725	0.3684169	-2.2745228	0.0229346	Poisson	BirdID
fixed	factor(fac.yr)13	-0.5101225	0.3583617	-1.4234849	0.1545956	Poisson	BirdID
fixed	factor(fac.yr)14	-0.6407131	0.4053161	-1.5807737	0.1139298	Poisson	BirdID
fixed	factor(fac.yr)15	-0.6792528	0.4465906	-1.5209744	0.1282663	Poisson	BirdID
fixed	factor(fac.yr)16	-0.2499018	0.4129727	-0.6051292	0.5450931	Poisson	BirdID
fixed	factor(fac.yr)17	0.1939417	0.3999714	0.4848891	0.6277550	Poisson	BirdID
fixed	factor(fac.yr)18	0.2074314	0.4810700	0.4311876	0.6663320	Poisson	BirdID
fixed	factor(fac.yr)19	-0.3382789	0.6679236	-0.5064634	0.6125314	Poisson	BirdID
fixed	factor(fac.yr)20	-0.9088426	0.6667111	-1.3631731	0.1728280	Poisson	BirdID
fixed	(Intercept)	0.4179384	0.2506655	1.6673156	0.0954517	Poisson	Terr
fixed	factor(fac.yr)2	-0.1580029	0.3260140	-0.4846506	0.6279242	Poisson	Terr
fixed	factor(fac.yr)3	-0.5186392	0.3289364	-1.5767158	0.1148609	Poisson	Terr
fixed	factor(fac.yr)4	-0.2484754	0.3196393	-0.7773617	0.4369454	Poisson	Terr
fixed	factor(fac.yr)5	0.0661514	0.3006504	0.2200277	0.8258496	Poisson	Terr
fixed	factor(fac.yr)6	-0.2285048	0.3200462	-0.7139744	0.4752430	Poisson	Terr
fixed	factor(fac.yr)7	-0.8029660	0.3542316	-2.2667821	0.0234035	Poisson	Terr
fixed	factor(fac.yr)8	-0.4856370	0.3173247	-1.5304105	0.1259152	Poisson	Terr
fixed	factor(fac.yr)9	-0.2657191	0.3090250	-0.8598627	0.3898647	Poisson	Terr
fixed	factor(fac.yr)10	-0.5394957	0.3267012	-1.6513430	0.0986686	Poisson	Terr
fixed	factor(fac.yr)11	-1.1738707	0.3991844	-2.9406728	0.0032750	Poisson	Terr
fixed	factor(fac.yr)12	-0.9284243	0.3555551	-2.6111963	0.0090226	Poisson	Terr
fixed	factor(fac.yr)13	-0.5676087	0.3421599	-1.6588988	0.0971362	Poisson	Terr
fixed	factor(fac.yr)14	-0.5162473	0.3914503	-1.3188070	0.1872337	Poisson	Terr
fixed	factor(fac.yr)15	-0.7994369	0.4308642	-1.8554266	0.0635353	Poisson	Terr
fixed	factor(fac.yr)16	-0.4123129	0.3979876	-1.0359943	0.3002048	Poisson	Terr
fixed	factor(fac.yr)17	0.1511283	0.3830987	0.3944893	0.6932198	Poisson	Terr
fixed	factor(fac.yr)18	-0.0574796	0.4606360	-0.1247831	0.9006952	Poisson	Terr
fixed	factor(fac.yr)19	-0.6821139	0.6757269	-1.0094520	0.3127579	Poisson	Terr
fixed	factor(fac.yr)20	-0.9217389	0.6600772	-1.3964108	0.1625908	Poisson	Terr
fixed	(Intercept)	0.3610798	0.2606979	1.3850504	0.1660370	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.1379550	0.3323498	-0.4150897	0.6780762	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.4899441	0.3374217	-1.4520232	0.1464952	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.2513462	0.3300136	-0.7616238	0.4462846	Poisson	BirdID+Terr

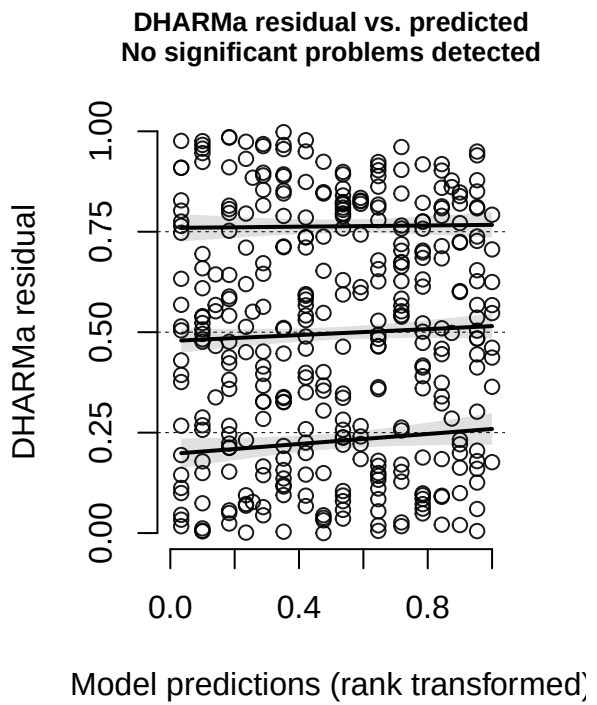
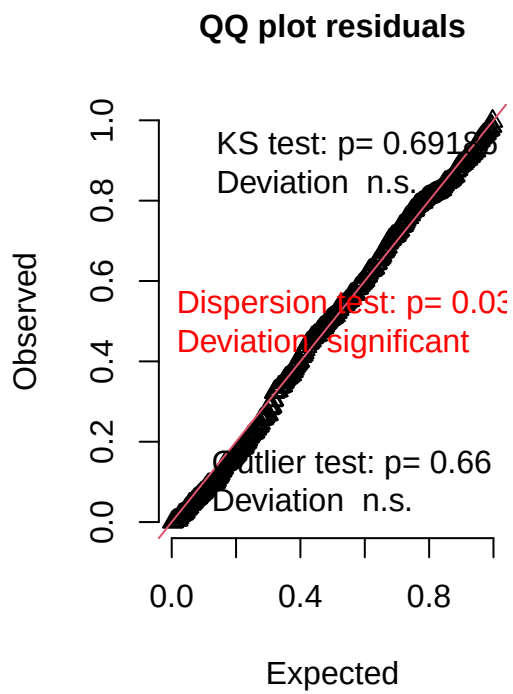
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.0744557	0.3115033	0.2390207	0.8110895	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.2778532	0.3301634	-0.8415627	0.4000328	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.8060196	0.3636324	-2.2165782	0.0266519	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.5026071	0.3297370	-1.5242664	0.1274422	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.3122341	0.3233336	-0.9656715	0.3342086	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.5647738	0.3399208	-1.6614863	0.0966158	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-1.2395619	0.4118391	-3.0098205	0.0026140	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.9252769	0.3695449	-2.5038283	0.0122858	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.5689335	0.3576824	-1.5906109	0.1116972	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.5836466	0.4060642	-1.4373260	0.1506254	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.7687137	0.4470484	-1.7195314	0.0855177	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.3500921	0.4157899	-0.8419927	0.3997920	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	0.1974452	0.4011016	0.4922574	0.6225374	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	0.0694975	0.4910988	0.1415142	0.8874637	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-0.5590308	0.6858189	-0.8151289	0.4149985	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.9146280	0.6739714	-1.3570725	0.1747582	Poisson	BirdID+Terr

Territory is performing best on BIC, while the model fit with both REs is performing best on AIC.

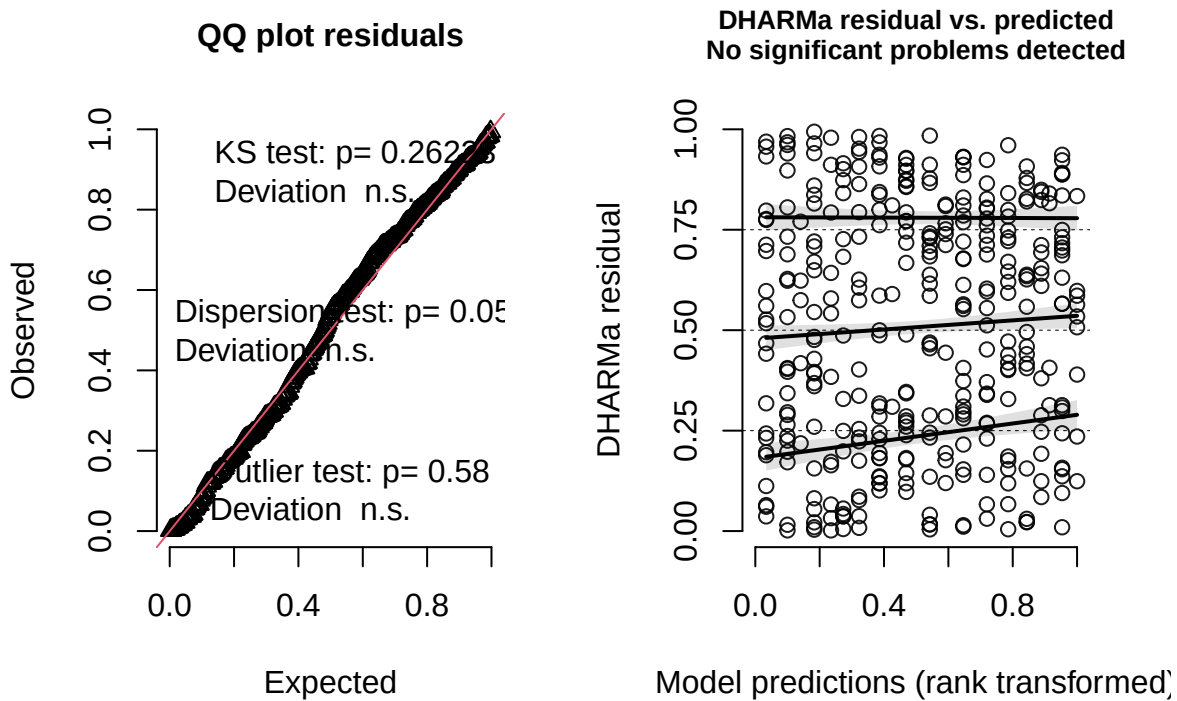
DHARMa residual



DHARMa residual



DHARMa residual



Significant dispersion test on terr only model.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
359	1	-201.9863	447.9727	533.4058	176.4391	337	0.878	Binomial	BirdID+Terr
359	1	-206.9013	455.8027	537.3524	257.9474	338	0.724	Binomial	BirdID
359	1	-211.4966	464.9932	546.5429	284.5539	338	0.720	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.826612	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.576282	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.974587	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.534292	Binomial	BirdID+Terr

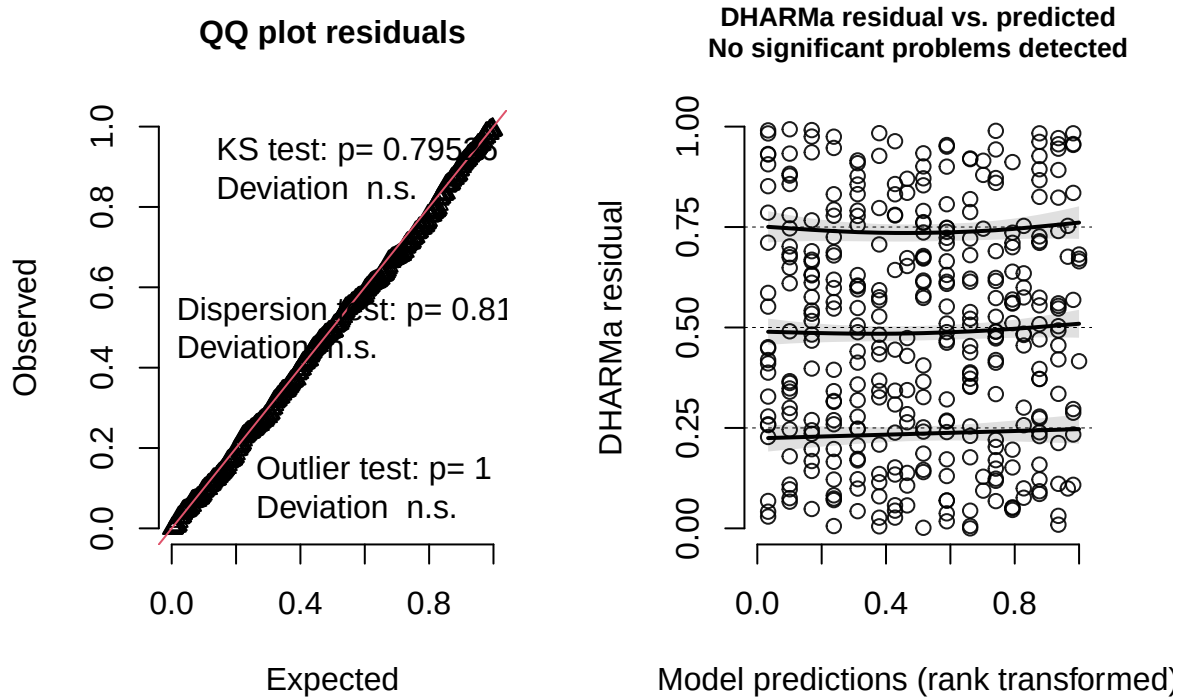
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	1.2762130	7.845314e-01	1.6267201	0.1037966	Binomial	BirdID
fixed	factor(fac.yr)2	0.5853563	1.017223e+00	0.5754455	0.5649901	Binomial	BirdID
fixed	factor(fac.yr)3	-0.4208403	9.238425e-01	-0.4555326	0.6487261	Binomial	BirdID
fixed	factor(fac.yr)4	-0.1271504	9.622464e-01	-0.1321392	0.8948742	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.3081433	9.798790e-01	0.3144708	0.7531635	Binomial	BirdID
fixed	factor(fac.yr)6	-0.3221663	9.669939e-01	-0.3331627	0.7390115	Binomial	BirdID
fixed	factor(fac.yr)7	-1.8449012	9.434225e-01	-1.9555409	0.0505193	Binomial	BirdID
fixed	factor(fac.yr)8	-1.0374142	9.493903e-01	-1.0927163	0.2745184	Binomial	BirdID
fixed	factor(fac.yr)9	-0.3856085	9.598674e-01	-0.4017310	0.6878820	Binomial	BirdID
fixed	factor(fac.yr)10	-1.5467048	9.559190e-01	-1.6180291	0.1056563	Binomial	BirdID
fixed	factor(fac.yr)11	-2.3257785	9.875774e-01	-2.3550343	0.0185210	Binomial	BirdID
fixed	factor(fac.yr)12	-1.7092834	9.716643e-01	-1.7591296	0.0785555	Binomial	BirdID
fixed	factor(fac.yr)13	-0.9952405	1.000662e+00	-0.9945820	0.3199396	Binomial	BirdID
fixed	factor(fac.yr)14	-0.8218961	1.063475e+00	-0.7728403	0.4396169	Binomial	BirdID
fixed	factor(fac.yr)15	-0.6734913	1.098014e+00	-0.6133723	0.5396302	Binomial	BirdID
fixed	factor(fac.yr)16	0.0179850	1.155238e+00	0.0155682	0.9875789	Binomial	BirdID
fixed	factor(fac.yr)17	1.6806335	1.569433e+00	1.0708538	0.2842352	Binomial	BirdID
fixed	factor(fac.yr)18	19.7873903	1.551978e+04	0.0012750	0.9989827	Binomial	BirdID
fixed	factor(fac.yr)19	0.7804554	1.733634e+00	0.4501846	0.6525773	Binomial	BirdID
fixed	factor(fac.yr)20	-0.2942772	1.452441e+00	-0.2026088	0.8394408	Binomial	BirdID
fixed	(Intercept)	1.3899082	6.968440e-01	1.9945758	0.0460892	Binomial	Terr
fixed	factor(fac.yr)2	0.4028417	9.641517e-01	0.4178199	0.6760788	Binomial	Terr
fixed	factor(fac.yr)3	-0.2057806	8.666636e-01	-0.2374400	0.8123155	Binomial	Terr
fixed	factor(fac.yr)4	0.2942558	8.696587e-01	0.3383578	0.7350936	Binomial	Terr
fixed	factor(fac.yr)5	0.3611501	8.858725e-01	0.4076773	0.6835106	Binomial	Terr
fixed	factor(fac.yr)6	0.2330373	8.664736e-01	0.2689491	0.7879689	Binomial	Terr
fixed	factor(fac.yr)7	-1.4777279	8.588813e-01	-1.7205262	0.0853368	Binomial	Terr
fixed	factor(fac.yr)8	-0.8993372	8.292580e-01	-1.0845083	0.2781395	Binomial	Terr
fixed	factor(fac.yr)9	-0.2740606	8.117183e-01	-0.3376302	0.7356419	Binomial	Terr
fixed	factor(fac.yr)10	-1.5961417	8.452506e-01	-1.8883650	0.0589770	Binomial	Terr
fixed	factor(fac.yr)11	-1.9092827	8.648904e-01	-2.2075429	0.0272762	Binomial	Terr
fixed	factor(fac.yr)12	-2.1234886	8.564514e-01	-2.4794036	0.0131602	Binomial	Terr
fixed	factor(fac.yr)13	-1.5380581	8.810023e-01	-1.7458049	0.0808449	Binomial	Terr
fixed	factor(fac.yr)14	-0.5641495	9.154307e-01	-0.6162668	0.5377185	Binomial	Terr
fixed	factor(fac.yr)15	-1.0899655	1.023924e+00	-1.0644980	0.2871031	Binomial	Terr
fixed	factor(fac.yr)16	-0.2881784	1.065031e+00	-0.2705820	0.7867125	Binomial	Terr
fixed	factor(fac.yr)17	1.6502328	1.472455e+00	1.1207359	0.2624003	Binomial	Terr
fixed	factor(fac.yr)18	10.9050902	2.228070e+02	0.0489441	0.9609638	Binomial	Terr
fixed	factor(fac.yr)19	-0.4933362	1.798767e+00	-0.2742635	0.7838821	Binomial	Terr
fixed	factor(fac.yr)20	-0.0092935	1.329951e+00	-0.0069878	0.9944246	Binomial	Terr
fixed	(Intercept)	1.7671717	9.226727e-01	1.9152748	0.0554575	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	0.4017685	1.172390e+00	0.3426919	0.7318303	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-0.4502549	1.088464e+00	-0.4136608	0.6791225	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	0.0040110	1.123387e+00	0.0035705	0.9971512	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	0.2146787	1.149616e+00	0.1867395	0.8518649	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-0.3125893	1.124662e+00	-0.2779407	0.7810579	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-2.2571023	1.114169e+00	-2.0258166	0.0427836	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-1.3445177	1.107592e+00	-1.2139108	0.2247818	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-0.4562881	1.099155e+00	-0.4151262	0.6780495	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-2.1885340	1.116710e+00	-1.9598057	0.0500185	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-2.8895296	1.160689e+00	-2.4894953	0.0127925	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-2.6335035	1.131861e+00	-2.3267025	0.0199811	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-1.7183106	1.181893e+00	-1.4538632	0.1459842	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-0.6750186	1.230754e+00	-0.5484592	0.5833766	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-1.1198155	1.328394e+00	-0.8429842	0.3992373	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	0.0267398	1.392024e+00	0.0192093	0.9846741	Binomial	BirdID+Terr

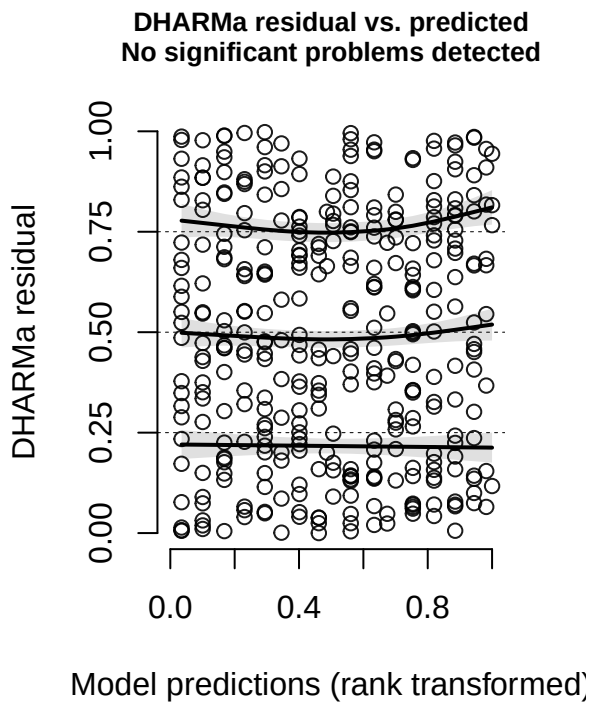
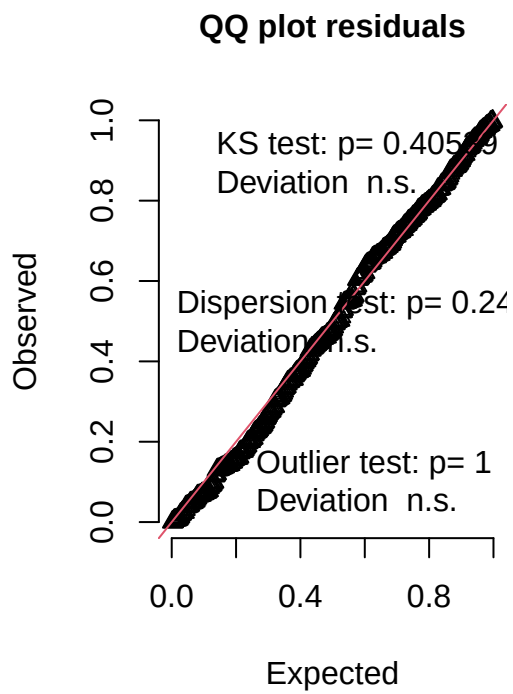
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	1.9396079	1.880355e+00	1.0315115	0.3023010	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	14.8857087	1.790671e+03	0.0083129	0.9933673	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	0.5357554	2.178661e+00	0.2459104	0.8057516	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-0.0220802	1.662295e+00	-0.0132830	0.9894020	Binomial	BirdID+Terr

Similar to the other binomial models.

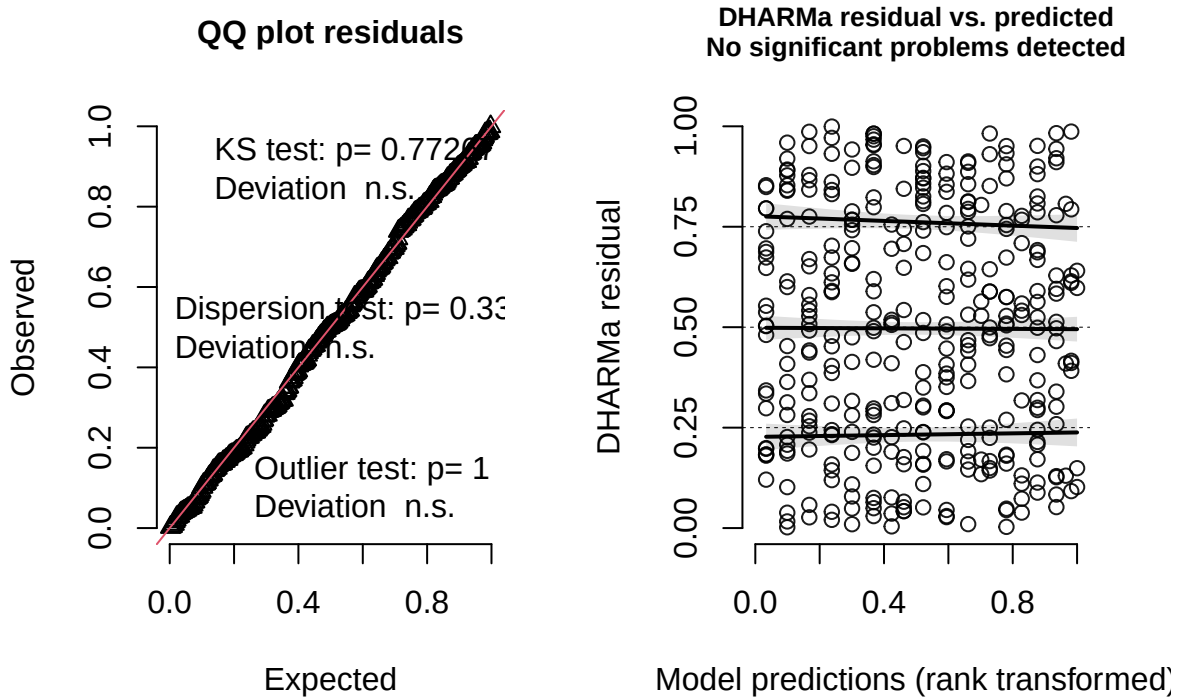
DHARMa residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Probabilites used to generate simulation 43

Simulation 43

Probability of mate fidelity: 0.881

Probability of pair dispersal: 0.896

Probability of single male dispersal: 0.455

Probability of single female dispersal: 0.82

Simulation 44

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
357	1	-478.1953	998.3906	1079.823	375.1778	336	0.2679497	0.1836492	Poisson	Terr
357	1	-478.1952	1000.3904	1085.701	375.2340	335	0.2666790	0.1828601	Poisson	BirdID+Terr
357	1	-480.0553	1002.1107	1083.543	420.7606	336	0.1993853	0.1925886	Poisson	BirdID

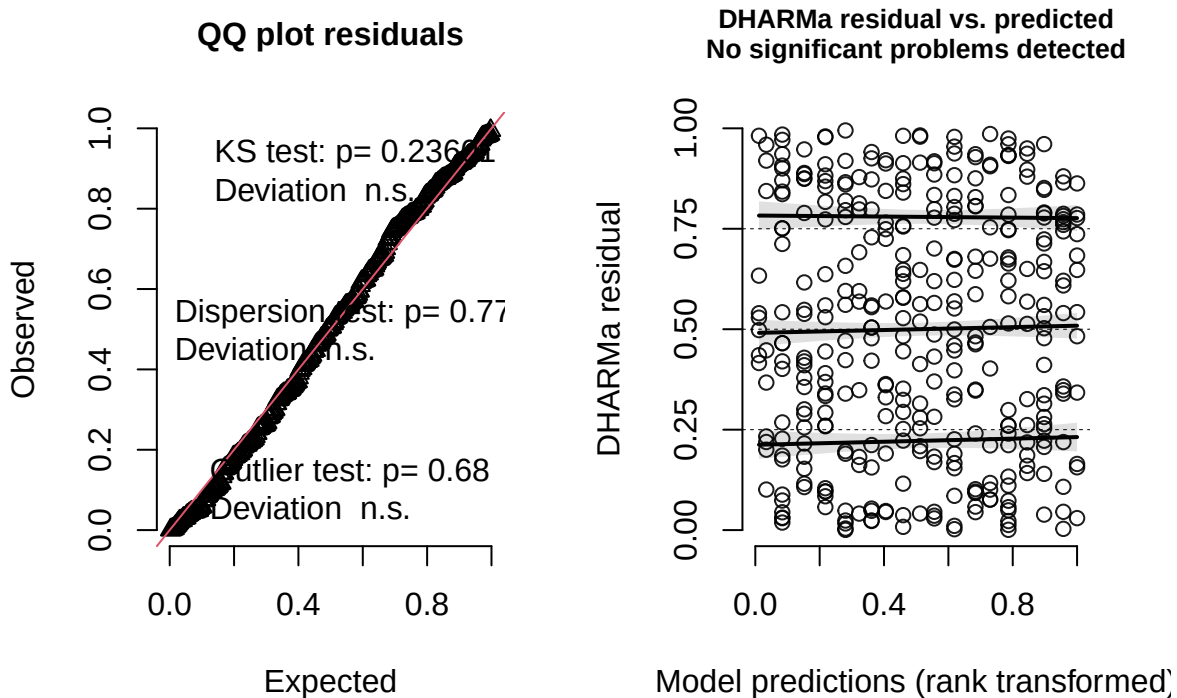
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.0737123	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.2694553	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.0008553	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.2692123	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.7702009	0.1975182	3.8993929	0.0000964	Poisson	BirdID
fixed	factor(fac.yr)2	-0.6208068	0.2942066	-2.1101048	0.0348493	Poisson	BirdID
fixed	factor(fac.yr)3	-0.9393415	0.3131021	-3.0001120	0.0026988	Poisson	BirdID
fixed	factor(fac.yr)4	-0.5997336	0.2670566	-2.2457173	0.0247221	Poisson	BirdID
fixed	factor(fac.yr)5	-0.5224855	0.2586387	-2.0201367	0.0433692	Poisson	BirdID
fixed	factor(fac.yr)6	-1.0072081	0.3031995	-3.3219325	0.0008940	Poisson	BirdID
fixed	factor(fac.yr)7	-1.2574126	0.3190823	-3.9407163	0.0000812	Poisson	BirdID
fixed	factor(fac.yr)8	-0.8334125	0.3191914	-2.6110118	0.0090275	Poisson	BirdID
fixed	factor(fac.yr)9	-0.2387140	0.2522226	-0.9464416	0.3439234	Poisson	BirdID
fixed	factor(fac.yr)10	-0.5911017	0.2846239	-2.0767815	0.0378217	Poisson	BirdID
fixed	factor(fac.yr)11	-1.0891120	0.3191811	-3.4122069	0.0006444	Poisson	BirdID
fixed	factor(fac.yr)12	-0.7290771	0.2816069	-2.5889890	0.0096258	Poisson	BirdID
fixed	factor(fac.yr)13	-0.1366321	0.2622921	-0.5209158	0.6024254	Poisson	BirdID
fixed	factor(fac.yr)14	-0.3442016	0.2992532	-1.1502018	0.2500608	Poisson	BirdID
fixed	factor(fac.yr)15	-0.7747001	0.3331767	-2.3251926	0.0200617	Poisson	BirdID
fixed	factor(fac.yr)16	-0.6333120	0.3259777	-1.9428080	0.0520394	Poisson	BirdID
fixed	factor(fac.yr)17	-0.5476400	0.3260316	-1.6797142	0.0930129	Poisson	BirdID
fixed	factor(fac.yr)18	-0.8791062	0.3885177	-2.2627187	0.0236530	Poisson	BirdID
fixed	factor(fac.yr)19	-1.5852380	0.5387587	-2.9423899	0.0032569	Poisson	BirdID
fixed	factor(fac.yr)20	-2.0271524	0.7356058	-2.7557592	0.0058556	Poisson	BirdID
fixed	(Intercept)	0.7842860	0.2106552	3.7230798	0.0001968	Poisson	Terr
fixed	factor(fac.yr)2	-0.6470020	0.3080093	-2.1005922	0.0356768	Poisson	Terr
fixed	factor(fac.yr)3	-1.0077094	0.3271217	-3.0805331	0.0020663	Poisson	Terr
fixed	factor(fac.yr)4	-0.6675666	0.2778920	-2.4022515	0.0162945	Poisson	Terr
fixed	factor(fac.yr)5	-0.5752719	0.2702173	-2.1289228	0.0332606	Poisson	Terr
fixed	factor(fac.yr)6	-1.0615840	0.3157460	-3.3621450	0.0007734	Poisson	Terr
fixed	factor(fac.yr)7	-1.3074624	0.3309902	-3.9501549	0.0000781	Poisson	Terr
fixed	factor(fac.yr)8	-0.8509315	0.3336770	-2.5501655	0.0107672	Poisson	Terr
fixed	factor(fac.yr)9	-0.2464382	0.2663943	-0.9250878	0.3549202	Poisson	Terr
fixed	factor(fac.yr)10	-0.6397848	0.2964246	-2.1583390	0.0309015	Poisson	Terr
fixed	factor(fac.yr)11	-1.1138332	0.3318719	-3.3562144	0.0007902	Poisson	Terr
fixed	factor(fac.yr)12	-0.7470381	0.2942820	-2.5385108	0.0111325	Poisson	Terr
fixed	factor(fac.yr)13	-0.1862538	0.2759724	-0.6749003	0.4997391	Poisson	Terr
fixed	factor(fac.yr)14	-0.3842780	0.3147748	-1.2208031	0.2221606	Poisson	Terr
fixed	factor(fac.yr)15	-0.8099073	0.3461764	-2.3395800	0.0193054	Poisson	Terr
fixed	factor(fac.yr)16	-0.6753601	0.3394810	-1.9893900	0.0466582	Poisson	Terr
fixed	factor(fac.yr)17	-0.5417795	0.3397663	-1.5945652	0.1108095	Poisson	Terr
fixed	factor(fac.yr)18	-0.8443123	0.4024316	-2.0980268	0.0359028	Poisson	Terr
fixed	factor(fac.yr)19	-1.6339984	0.5523318	-2.9583636	0.0030928	Poisson	Terr
fixed	factor(fac.yr)20	-2.0611770	0.7496449	-2.7495380	0.0059679	Poisson	Terr
fixed	(Intercept)	0.7853367	0.2105490	3.7299466	0.0001915	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.6483286	0.3079562	-2.1052622	0.0352685	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-1.0088711	0.3270647	-3.0846225	0.0020381	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.6689792	0.2778400	-2.4077856	0.0160496	Poisson	BirdID+Terr

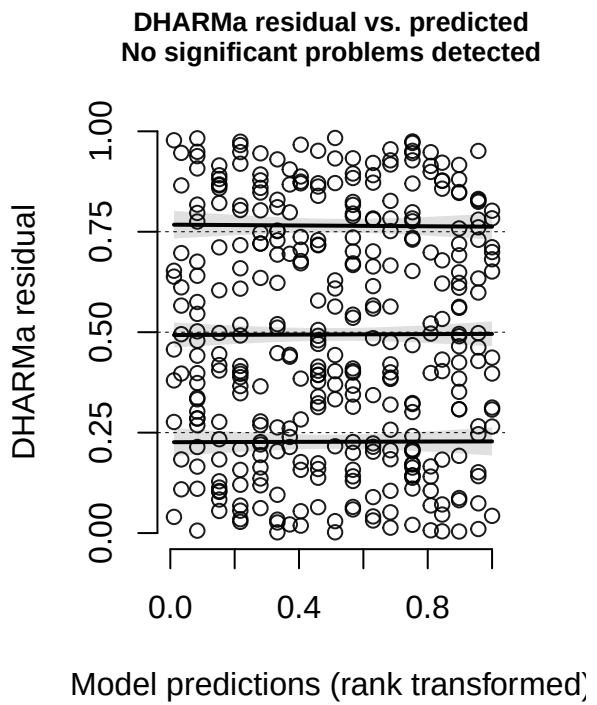
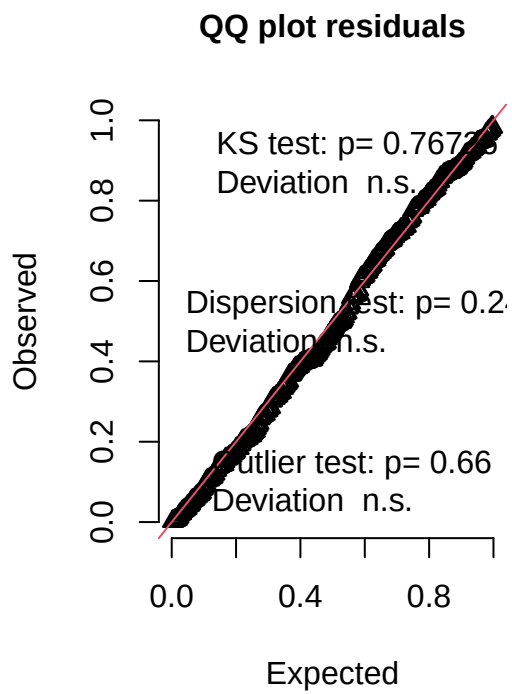
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.5758905	0.2701168	-2.1320051	0.0330064	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-1.0620601	0.3156314	-3.3648750	0.0007658	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-1.3086579	0.3309402	-3.9543639	0.0000767	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.8520358	0.3336096	-2.5539908	0.0106496	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.2470224	0.2662864	-0.9276567	0.3535857	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.6398187	0.2962846	-2.1594735	0.0308135	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-1.1142338	0.3317419	-3.3587372	0.0007830	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.7487512	0.2942483	-2.5446238	0.0109396	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.1869311	0.2758721	-0.6776008	0.4980248	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.3848223	0.3146604	-1.2229768	0.2213385	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.8118023	0.3461962	-2.3449194	0.0190312	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.6762198	0.3393973	-1.9924138	0.0463257	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.5433121	0.3397383	-1.5992076	0.1097745	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.8457023	0.4024087	-2.1016002	0.0355883	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.6351154	0.5523207	-2.9604456	0.0030719	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-2.0590036	0.7485484	-2.7506618	0.0059475	Poisson	BirdID+Terr

Territory only model is performing well on all model selection criteria.

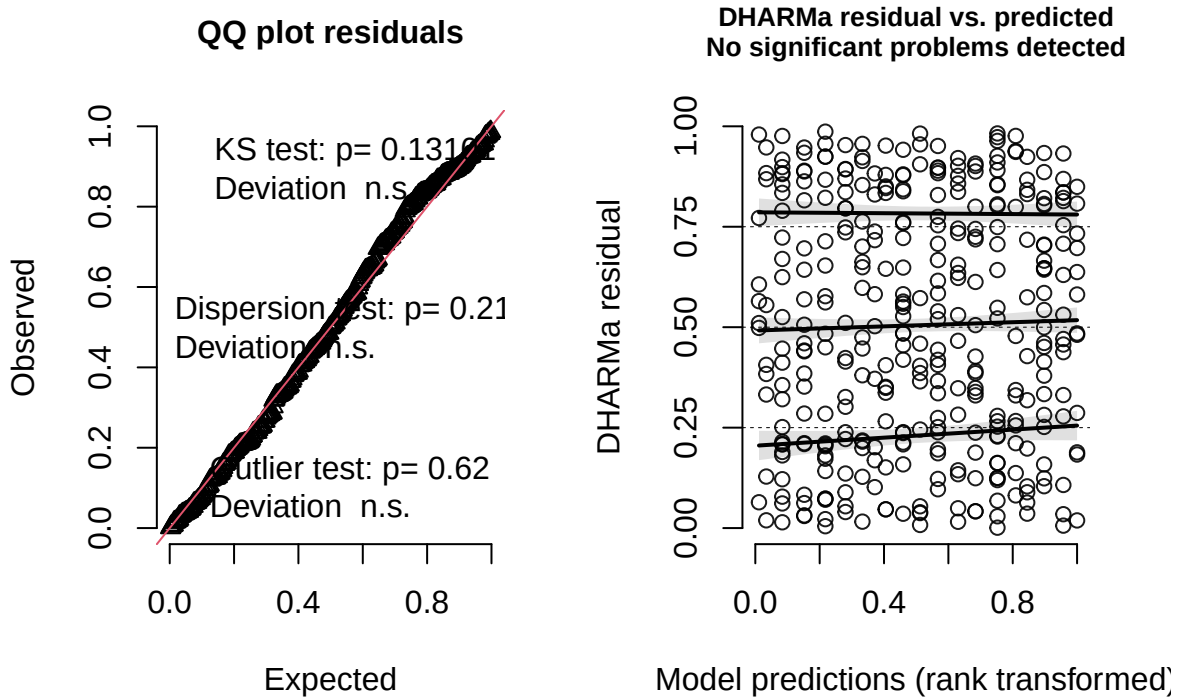
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
357	1	-209.6560	463.3120	548.6221	261.6397	335	0.731	Binomial	BirdID+Terr
357	1	-212.3686	466.7372	548.1697	348.5843	336	0.528	Binomial	Terr
357	1	-213.6189	469.2378	550.6703	353.0127	336	0.539	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.8923039	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	0.9003005	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.0892995	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.0606557	Binomial	BirdID+Terr

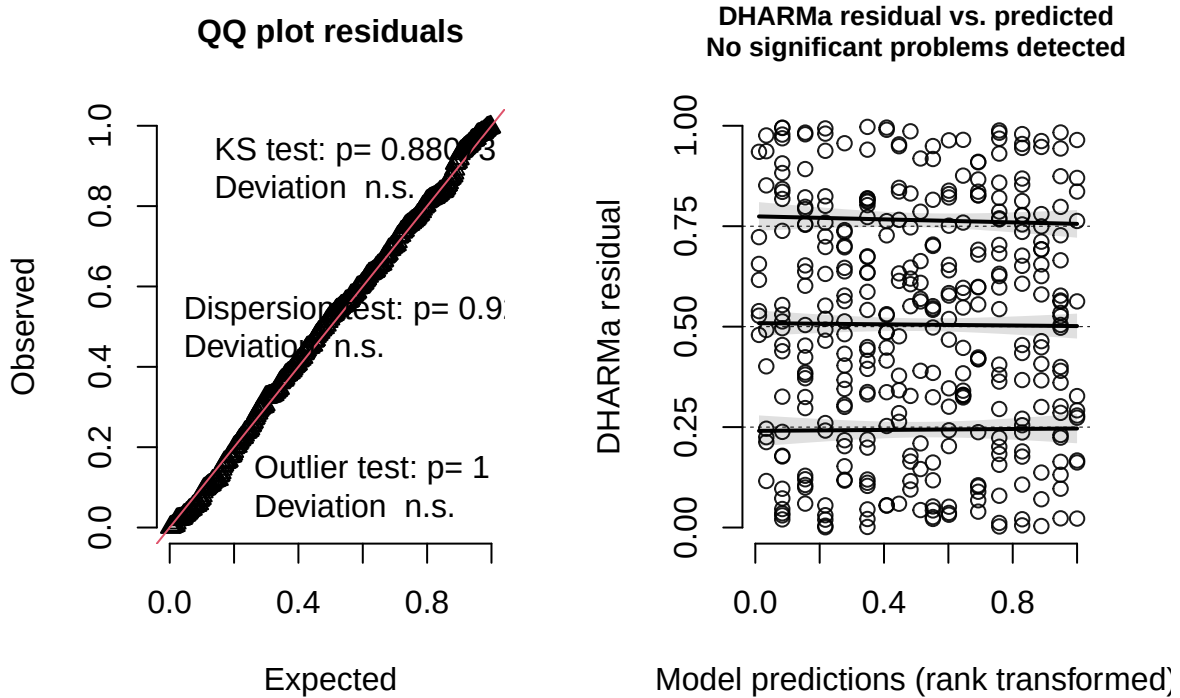
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	2.5728022	1.110468	2.3168627	0.0205112	Binomial	BirdID
fixed	factor(fac.yr)2	-1.7790988	1.217187	-1.4616482	0.1438376	Binomial	BirdID
fixed	factor(fac.yr)3	-2.6973581	1.204121	-2.2401051	0.0250841	Binomial	BirdID
fixed	factor(fac.yr)4	-1.4515796	1.203347	-1.2062849	0.2277077	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-2.4070502	1.180929	-2.0382675	0.0415232	Binomial	BirdID
fixed	factor(fac.yr)6	-2.7886775	1.196003	-2.3316652	0.0197183	Binomial	BirdID
fixed	factor(fac.yr)7	-2.8903124	1.190043	-2.4287465	0.0151511	Binomial	BirdID
fixed	factor(fac.yr)8	-1.9002282	1.238424	-1.5343919	0.1249333	Binomial	BirdID
fixed	factor(fac.yr)9	-0.4722172	1.298357	-0.3637037	0.7160793	Binomial	BirdID
fixed	factor(fac.yr)10	-1.5666025	1.230938	-1.2726895	0.2031283	Binomial	BirdID
fixed	factor(fac.yr)11	-2.6428730	1.204168	-2.1947708	0.0281800	Binomial	BirdID
fixed	factor(fac.yr)12	-1.4072162	1.212527	-1.1605647	0.2458190	Binomial	BirdID
fixed	factor(fac.yr)13	-0.7914292	1.308464	-0.6048535	0.5452763	Binomial	BirdID
fixed	factor(fac.yr)14	-1.7665023	1.289827	-1.3695653	0.1708226	Binomial	BirdID
fixed	factor(fac.yr)15	-2.4009232	1.258900	-1.9071600	0.0564999	Binomial	BirdID
fixed	factor(fac.yr)16	-2.2492198	1.275706	-1.7631177	0.0778806	Binomial	BirdID
fixed	factor(fac.yr)17	-2.1492276	1.285757	-1.6715665	0.0946098	Binomial	BirdID
fixed	factor(fac.yr)18	-2.1255590	1.325643	-1.6034179	0.1088424	Binomial	BirdID
fixed	factor(fac.yr)19	-3.4521856	1.365426	-2.5282847	0.0114621	Binomial	BirdID
fixed	factor(fac.yr)20	-4.7219720	1.655763	-2.8518404	0.0043467	Binomial	BirdID
fixed	(Intercept)	2.9317862	1.109175	2.6432135	0.0082123	Binomial	Terr
fixed	factor(fac.yr)2	-2.0963789	1.235215	-1.6971780	0.0896630	Binomial	Terr
fixed	factor(fac.yr)3	-3.0832410	1.208192	-2.5519462	0.0107123	Binomial	Terr
fixed	factor(fac.yr)4	-1.8430834	1.199528	-1.5365067	0.1244142	Binomial	Terr
fixed	factor(fac.yr)5	-2.7965509	1.176543	-2.3769212	0.0174578	Binomial	Terr
fixed	factor(fac.yr)6	-3.1529884	1.193534	-2.6417251	0.0082485	Binomial	Terr
fixed	factor(fac.yr)7	-3.3033157	1.189498	-2.7770675	0.0054852	Binomial	Terr
fixed	factor(fac.yr)8	-2.0093072	1.241020	-1.6190769	0.1054307	Binomial	Terr
fixed	factor(fac.yr)9	-0.4448247	1.296328	-0.3431421	0.7314916	Binomial	Terr
fixed	factor(fac.yr)10	-1.7810882	1.224607	-1.4544157	0.1458311	Binomial	Terr
fixed	factor(fac.yr)11	-2.9930859	1.200918	-2.4923326	0.0126907	Binomial	Terr
fixed	factor(fac.yr)12	-1.7236035	1.208336	-1.4264268	0.1537452	Binomial	Terr
fixed	factor(fac.yr)13	-1.0492720	1.299744	-0.8072914	0.4194986	Binomial	Terr
fixed	factor(fac.yr)14	-2.0525430	1.289755	-1.5914204	0.1115150	Binomial	Terr
fixed	factor(fac.yr)15	-2.5684417	1.251307	-2.0526077	0.0401106	Binomial	Terr
fixed	factor(fac.yr)16	-2.3538084	1.272614	-1.8495851	0.0643734	Binomial	Terr
fixed	factor(fac.yr)17	-2.2471607	1.275794	-1.7613818	0.0781738	Binomial	Terr
fixed	factor(fac.yr)18	-2.1077779	1.311872	-1.6066943	0.1081214	Binomial	Terr
fixed	factor(fac.yr)19	-3.6732298	1.352461	-2.7159602	0.0066084	Binomial	Terr
fixed	factor(fac.yr)20	-4.6863253	1.610845	-2.9092335	0.0036232	Binomial	Terr
fixed	(Intercept)	3.1899664	1.217131	2.6208900	0.0087701	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-2.2006960	1.338184	-1.6445390	0.1000649	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-3.6020267	1.317024	-2.7349736	0.0062385	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-1.9601813	1.309342	-1.4970733	0.1343742	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-3.0446300	1.289670	-2.3607820	0.0182364	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-3.5380527	1.317241	-2.6859572	0.0072322	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-3.6035554	1.312973	-2.7445766	0.0060589	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-2.1741524	1.371727	-1.5849746	0.1129721	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-0.4088948	1.435524	-0.2848401	0.7757667	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-1.8882364	1.354326	-1.3942257	0.1632495	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-3.1381791	1.327499	-2.3639788	0.0180798	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-1.5562014	1.331956	-1.1683575	0.2426626	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-0.9005372	1.442224	-0.6244088	0.5323592	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-2.1059286	1.433172	-1.4694177	0.1417195	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-2.9619928	1.390905	-2.1295441	0.0332093	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-2.7177928	1.416441	-1.9187471	0.0550163	Binomial	BirdID+Terr

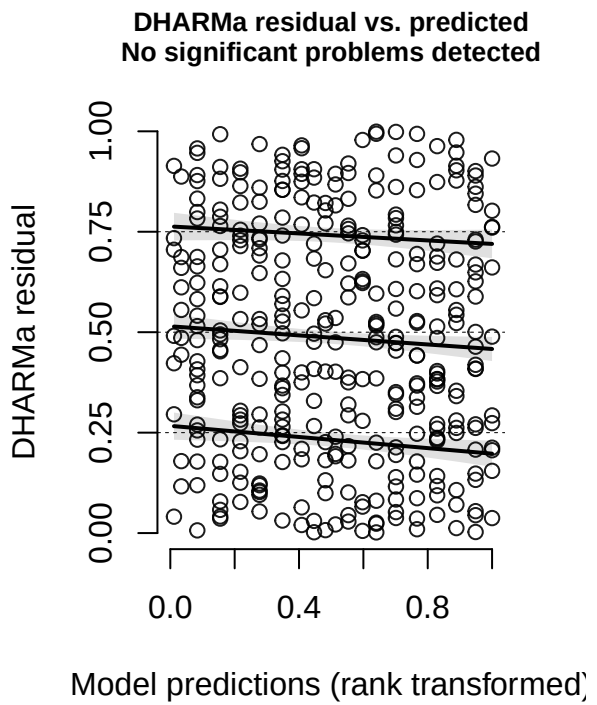
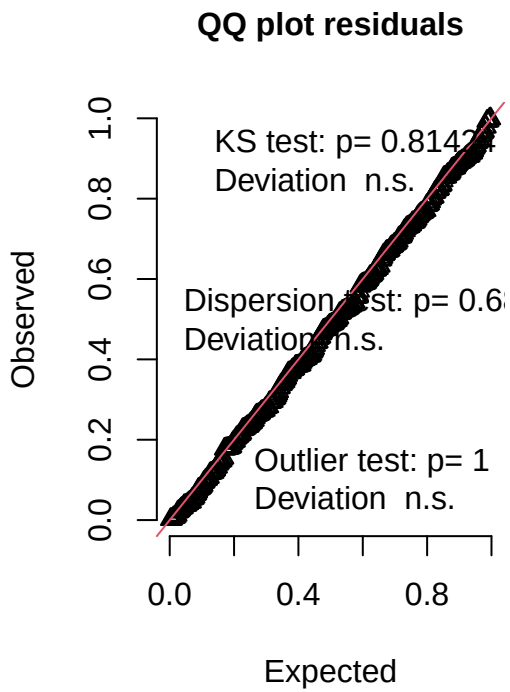
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-2.3716668	1.425405	-1.6638549	0.0961414	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-2.3630386	1.476548	-1.6003806	0.1095142	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-4.2176205	1.526890	-2.7622293	0.0057408	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-5.4966997	1.830120	-3.0034639	0.0026693	Binomial	BirdID+Terr

Territory only model is performing best on BIC, and not by a significant margin.

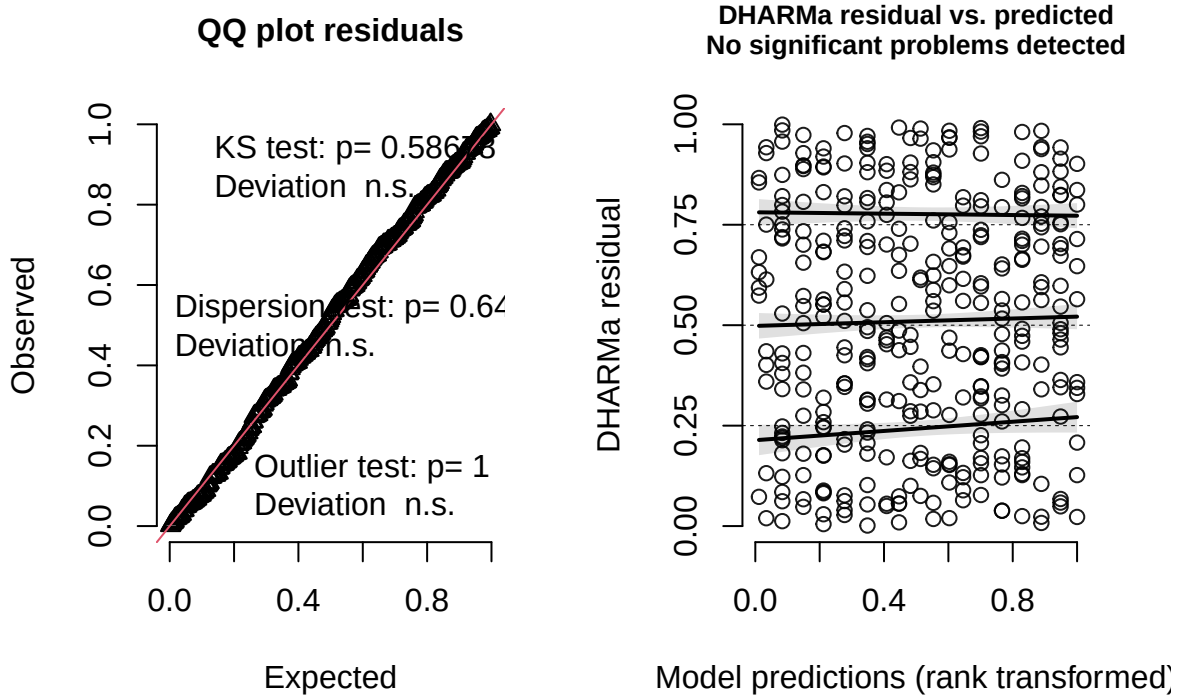
DHARMa residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Probabilites used to generate simulation 44

Simulation 44

Probability of mate fidelity: 0.4

Probability of pair dispersal: 0.794

Probability of single male dispersal: 0.951

Probability of single female dispersal: 0.64

Simulation 45

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
310	1	-434.3874	910.7749	989.2429	332.1708	289	0.1899410	0.1096374	Poisson	Terr
310	1	-435.2995	912.5990	991.0670	358.0235	289	0.1453064	0.1197103	Poisson	BirdID
310	1	-434.3818	912.7637	994.9683	330.4567	288	0.1926846	0.1094931	Poisson	BirdID+Terr

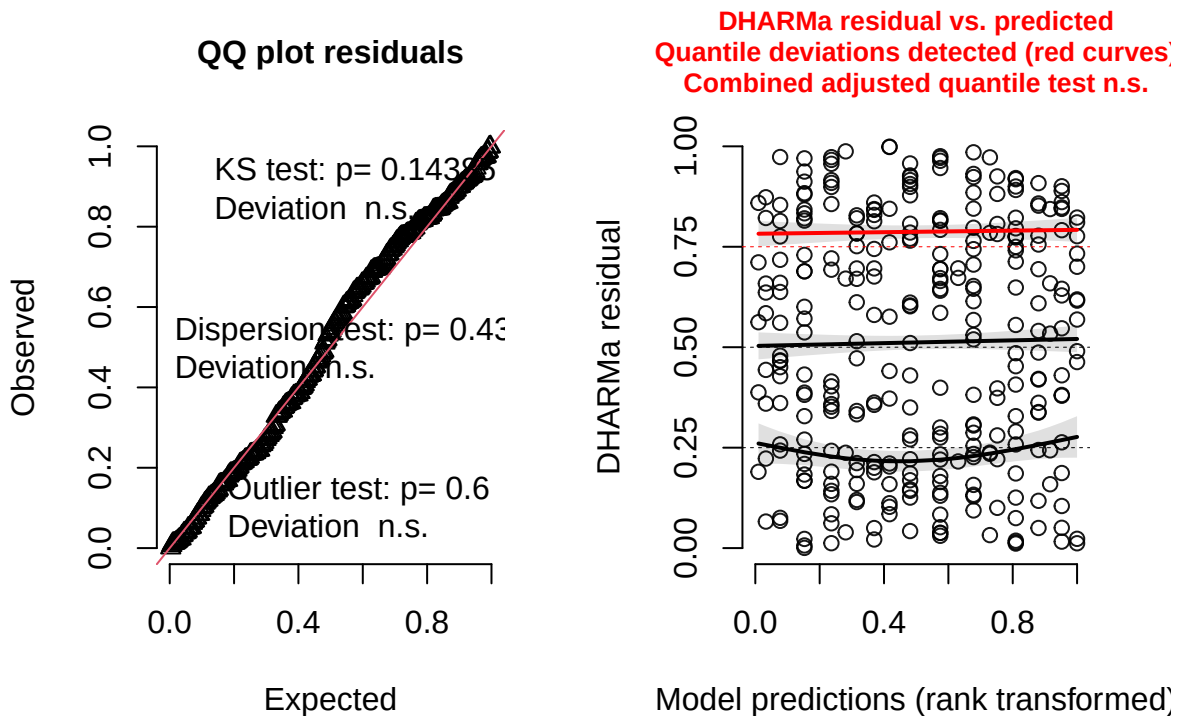
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.1335607	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.2402154	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.0608855	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.2373260	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.7618894	0.2004903	3.8001310	0.0001446	Poisson	BirdID
fixed	factor(fac.yr)2	-0.6270291	0.3276838	-1.9135188	0.0556817	Poisson	BirdID
fixed	factor(fac.yr)3	-0.9561869	0.3276871	-2.9179876	0.0035230	Poisson	BirdID
fixed	factor(fac.yr)4	-0.3534620	0.2815853	-1.2552576	0.2093853	Poisson	BirdID
fixed	factor(fac.yr)5	-0.3713711	0.2575945	-1.4416889	0.1493901	Poisson	BirdID
fixed	factor(fac.yr)6	-0.4498146	0.2617327	-1.7186027	0.0856867	Poisson	BirdID
fixed	factor(fac.yr)7	-0.9269797	0.2873800	-3.2256238	0.0012570	Poisson	BirdID
fixed	factor(fac.yr)8	-0.5701186	0.2687365	-2.1214778	0.0338816	Poisson	BirdID
fixed	factor(fac.yr)9	-0.5154918	0.2566011	-2.0089230	0.0445453	Poisson	BirdID
fixed	factor(fac.yr)10	-0.8617229	0.2979740	-2.8919398	0.0038287	Poisson	BirdID
fixed	factor(fac.yr)11	-0.7672996	0.3225512	-2.3788458	0.0173669	Poisson	BirdID
fixed	factor(fac.yr)12	-0.7676736	0.3169854	-2.4217947	0.0154441	Poisson	BirdID
fixed	factor(fac.yr)13	-0.2625862	0.2778696	-0.9449978	0.3446600	Poisson	BirdID
fixed	factor(fac.yr)14	-0.3828560	0.3457391	-1.1073553	0.2681404	Poisson	BirdID
fixed	factor(fac.yr)15	-1.6685353	0.7396720	-2.2557775	0.0240846	Poisson	BirdID
fixed	factor(fac.yr)16	-1.0155873	0.4313744	-2.3543059	0.0185573	Poisson	BirdID
fixed	factor(fac.yr)17	-0.2696568	0.4118126	-0.6548046	0.5125936	Poisson	BirdID
fixed	factor(fac.yr)18	-0.4367917	0.4331402	-1.0084303	0.3132479	Poisson	BirdID
fixed	factor(fac.yr)19	-0.7691023	0.5443973	-1.4127593	0.1577265	Poisson	BirdID
fixed	factor(fac.yr)20	-0.4768012	0.5460455	-0.8731894	0.3825598	Poisson	BirdID
fixed	(Intercept)	0.7479612	0.2089562	3.5795118	0.0003442	Poisson	Terr
fixed	factor(fac.yr)2	-0.6041366	0.3380704	-1.7870143	0.0739352	Poisson	Terr
fixed	factor(fac.yr)3	-0.9352056	0.3364469	-2.7796533	0.0054417	Poisson	Terr
fixed	factor(fac.yr)4	-0.3148651	0.2930349	-1.0744969	0.2826000	Poisson	Terr
fixed	factor(fac.yr)5	-0.3456925	0.2671569	-1.2939680	0.1956765	Poisson	Terr
fixed	factor(fac.yr)6	-0.4583653	0.2704425	-1.6948718	0.0900997	Poisson	Terr
fixed	factor(fac.yr)7	-0.9084316	0.2958402	-3.0706833	0.0021357	Poisson	Terr
fixed	factor(fac.yr)8	-0.5700723	0.2768354	-2.0592465	0.0394706	Poisson	Terr
fixed	factor(fac.yr)9	-0.4806553	0.2648302	-1.8149566	0.0695306	Poisson	Terr
fixed	factor(fac.yr)10	-0.8594186	0.3045545	-2.8218880	0.0047742	Poisson	Terr
fixed	factor(fac.yr)11	-0.7553741	0.3321688	-2.2740672	0.0229619	Poisson	Terr
fixed	factor(fac.yr)12	-0.7332901	0.3250963	-2.2556091	0.0240951	Poisson	Terr
fixed	factor(fac.yr)13	-0.2823398	0.2871976	-0.9830857	0.3255653	Poisson	Terr
fixed	factor(fac.yr)14	-0.4543858	0.3551935	-1.2792627	0.2008046	Poisson	Terr
fixed	factor(fac.yr)15	-1.6220269	0.7503955	-2.1615626	0.0306519	Poisson	Terr
fixed	factor(fac.yr)16	-0.9589764	0.4397967	-2.1804995	0.0292205	Poisson	Terr
fixed	factor(fac.yr)17	-0.3172229	0.4255301	-0.7454769	0.4559835	Poisson	Terr
fixed	factor(fac.yr)18	-0.4581391	0.4465991	-1.0258397	0.3049672	Poisson	Terr
fixed	factor(fac.yr)19	-0.8155591	0.5583182	-1.4607423	0.1440862	Poisson	Terr
fixed	factor(fac.yr)20	-0.4333581	0.5635643	-0.7689595	0.4419174	Poisson	Terr
fixed	(Intercept)	0.7505311	0.2091774	3.5880130	0.0003332	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.6080898	0.3382054	-1.7979896	0.0721787	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.9408747	0.3366535	-2.7947867	0.0051934	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.3217299	0.2932813	-1.0970008	0.2726410	Poisson	BirdID+Terr

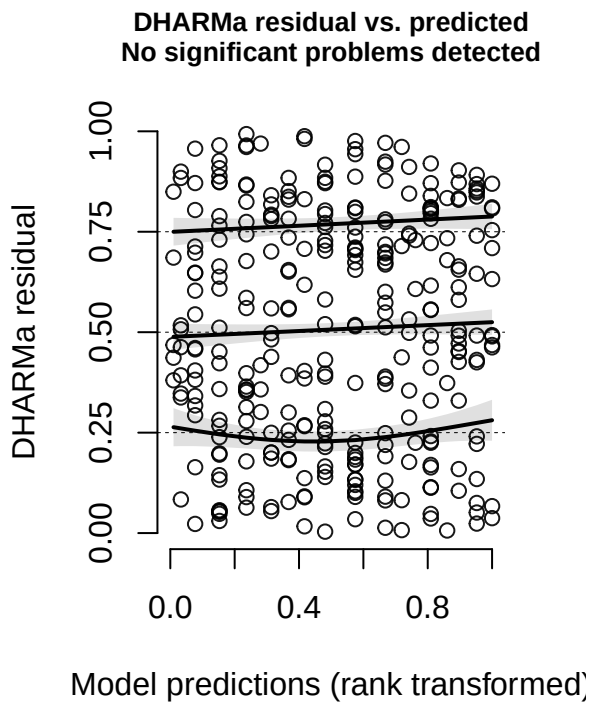
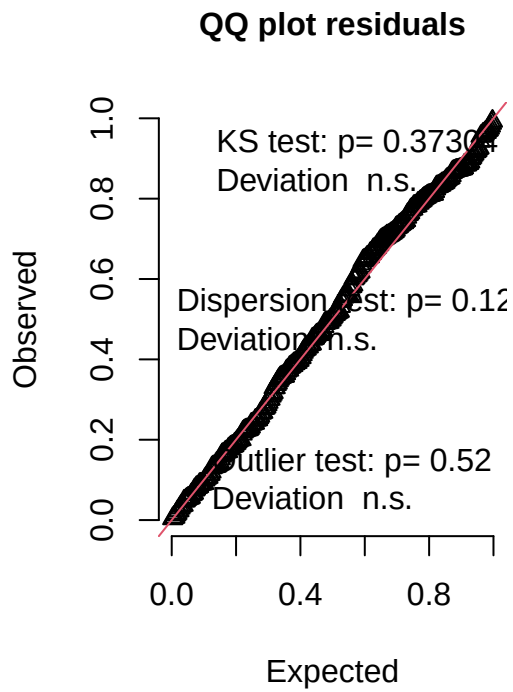
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.3517442	0.2674638	-1.3151092	0.1884732	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.4628850	0.2707513	-1.7096315	0.0873340	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.9135273	0.2961850	-3.0843137	0.0020402	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.5761621	0.2772292	-2.0782878	0.0376829	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.4857241	0.2651864	-1.8316329	0.0670061	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.8635850	0.3049469	-2.8319188	0.0046270	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.7595343	0.3326026	-2.2836087	0.0223945	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.7375424	0.3256069	-2.2651316	0.0235046	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.2848399	0.2877227	-0.9899806	0.3221836	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.4527531	0.3558854	-1.2721879	0.2033063	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-1.6247530	0.7514504	-2.1621559	0.0306062	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.9633709	0.4404951	-2.1870184	0.0287412	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.3141185	0.4263240	-0.7368070	0.4612397	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.4621166	0.4474987	-1.0326657	0.3017604	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-0.8184893	0.5592245	-1.4636149	0.1432992	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.4386544	0.5648469	-0.7765899	0.4374008	Poisson	BirdID+Terr

Terr only is performing best on AIC and BIC. R^2 s seem reasonable.

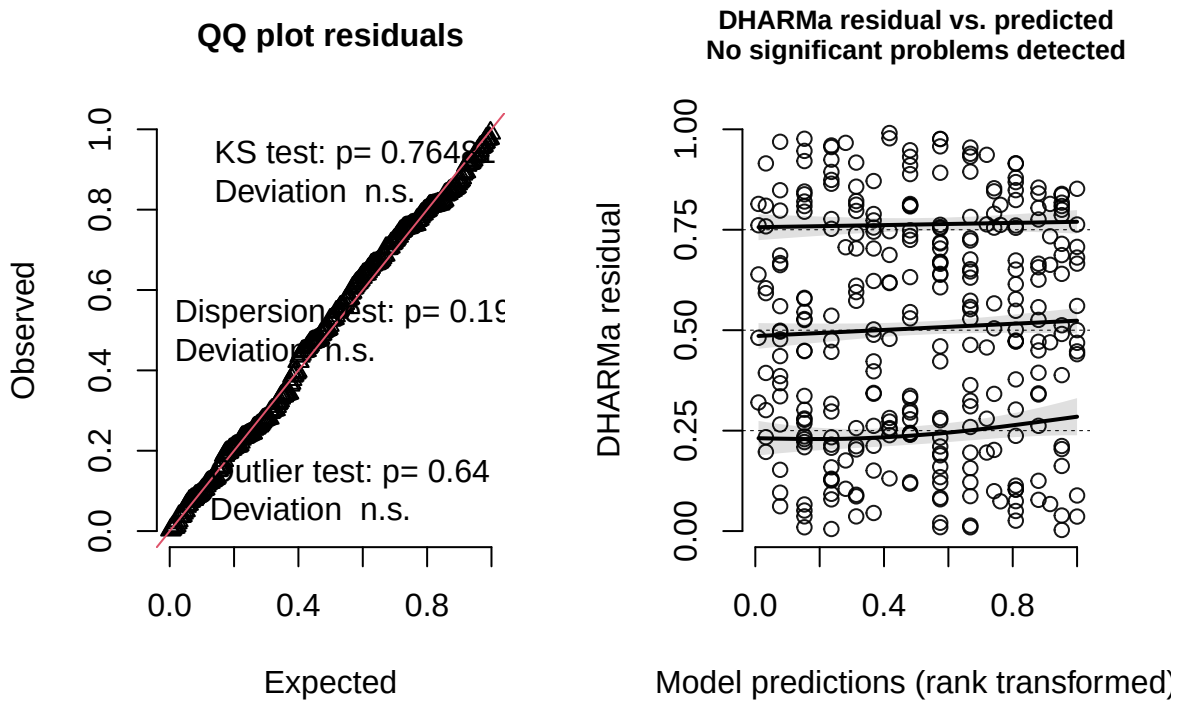
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics for territory only model look good, along with the model fit with both.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
310	1	-185.8871	415.7742	497.9788	254.1987	288	0.644	Binomial	BirdID+Terr
310	1	-187.5470	417.0940	495.5620	305.4569	289	0.435	Binomial	Terr
310	1	-188.5257	419.0514	497.5194	326.4271	289	0.351	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.7259521	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	0.9119835	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	0.8297374	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.9754262	Binomial	BirdID+Terr

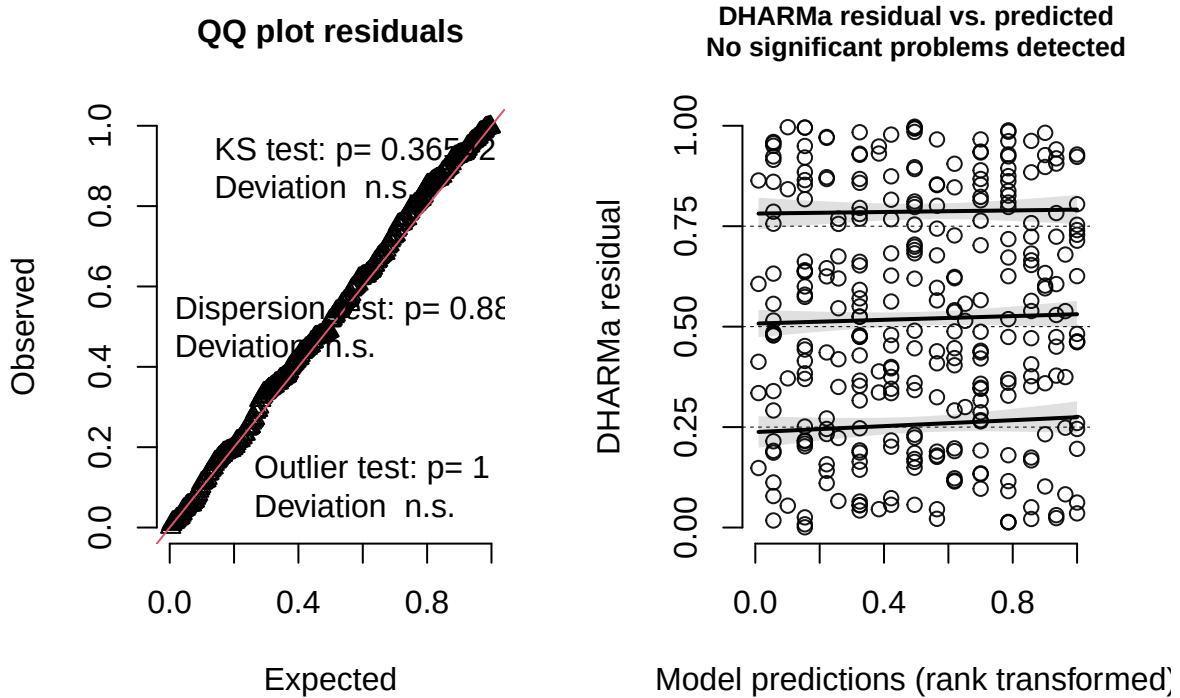
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	1.7009394	0.8206179	2.0727544	0.0381951	Binomial	BirdID
fixed	factor(fac.yr)2	-1.4302887	0.9947130	-1.4378908	0.1504650	Binomial	BirdID
fixed	factor(fac.yr)3	-1.1571700	0.9549862	-1.2117138	0.2256219	Binomial	BirdID
fixed	factor(fac.yr)4	0.3964445	1.1486403	0.3451425	0.7299873	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.4481498	0.9566432	-0.4684607	0.6394551	Binomial	BirdID
fixed	factor(fac.yr)6	-0.6486250	0.9438904	-0.6871826	0.4919677	Binomial	BirdID
fixed	factor(fac.yr)7	-1.5879849	0.9126940	-1.7398874	0.0818788	Binomial	BirdID
fixed	factor(fac.yr)8	-1.0798869	0.9309826	-1.1599431	0.2460720	Binomial	BirdID
fixed	factor(fac.yr)9	-1.3228679	0.9081045	-1.4567353	0.1451895	Binomial	BirdID
fixed	factor(fac.yr)10	-1.7821587	0.9337918	-1.9085183	0.0563243	Binomial	BirdID
fixed	factor(fac.yr)11	-1.0704339	0.9891027	-1.0822272	0.2791516	Binomial	BirdID
fixed	factor(fac.yr)12	-0.9547177	0.9858475	-0.9684233	0.3328330	Binomial	BirdID
fixed	factor(fac.yr)13	-0.2760393	1.0270999	-0.2687560	0.7881175	Binomial	BirdID
fixed	factor(fac.yr)14	-0.1317678	1.1910447	-0.1106321	0.9119081	Binomial	BirdID
fixed	factor(fac.yr)15	-1.8862965	1.2858955	-1.4669128	0.1423998	Binomial	BirdID
fixed	factor(fac.yr)16	-1.4030868	1.0963047	-1.2798329	0.2006039	Binomial	BirdID
fixed	factor(fac.yr)17	0.1551209	1.4653454	0.1058596	0.9156937	Binomial	BirdID
fixed	factor(fac.yr)18	-1.1644622	1.2693240	-0.9173877	0.3589395	Binomial	BirdID
fixed	factor(fac.yr)19	-1.6258244	1.3410691	-1.2123346	0.2253843	Binomial	BirdID
fixed	factor(fac.yr)20	-0.8861993	1.5529894	-0.5706409	0.5682431	Binomial	BirdID
fixed	(Intercept)	1.7963936	0.8565724	2.0971882	0.0359769	Binomial	Terr
fixed	factor(fac.yr)2	-1.5448760	1.0429227	-1.4812949	0.1385280	Binomial	Terr
fixed	factor(fac.yr)3	-1.1434455	1.0026539	-1.1404190	0.2541118	Binomial	Terr
fixed	factor(fac.yr)4	0.6941380	1.1985059	0.5791695	0.5624748	Binomial	Terr
fixed	factor(fac.yr)5	-0.2248902	0.9980437	-0.2253310	0.8217218	Binomial	Terr
fixed	factor(fac.yr)6	-0.6316187	0.9857186	-0.6407698	0.5216722	Binomial	Terr
fixed	factor(fac.yr)7	-1.4864972	0.9502799	-1.5642730	0.1177535	Binomial	Terr
fixed	factor(fac.yr)8	-1.0353167	0.9652539	-1.0725849	0.2834574	Binomial	Terr
fixed	factor(fac.yr)9	-1.1670760	0.9379441	-1.2442916	0.2133923	Binomial	Terr
fixed	factor(fac.yr)10	-1.8541195	0.9647296	-1.9219059	0.0546176	Binomial	Terr
fixed	factor(fac.yr)11	-1.0247741	1.0267373	-0.9980878	0.3182368	Binomial	Terr
fixed	factor(fac.yr)12	-0.9812311	1.0147568	-0.9669618	0.3335631	Binomial	Terr
fixed	factor(fac.yr)13	-0.4397314	1.0609700	-0.4144617	0.6785360	Binomial	Terr
fixed	factor(fac.yr)14	-0.3976576	1.2332324	-0.3224515	0.7471107	Binomial	Terr
fixed	factor(fac.yr)15	-1.8462714	1.3283801	-1.3898668	0.1645693	Binomial	Terr
fixed	factor(fac.yr)16	-1.2316064	1.1256827	-1.0940973	0.2739123	Binomial	Terr
fixed	factor(fac.yr)17	-0.0640010	1.4974460	-0.0427401	0.9659087	Binomial	Terr
fixed	factor(fac.yr)18	-1.3155910	1.3167074	-0.9991522	0.3177210	Binomial	Terr
fixed	factor(fac.yr)19	-1.9293030	1.3879019	-1.3900860	0.1645028	Binomial	Terr
fixed	factor(fac.yr)20	-0.5903166	1.5932310	-0.3705154	0.7109985	Binomial	Terr
fixed	(Intercept)	1.9580637	0.9251291	2.1165303	0.0342997	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-1.5644492	1.1036605	-1.4175095	0.1563340	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-1.2368207	1.0662521	-1.1599703	0.2460609	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	0.6876554	1.2905499	0.5328390	0.5941450	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-0.2956582	1.0781493	-0.2742275	0.7839098	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-0.7476444	1.0599197	-0.7053784	0.4805748	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-1.6713518	1.0241623	-1.6319208	0.1026962	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-1.2071664	1.0440242	-1.1562628	0.2475737	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-1.2662998	1.0143550	-1.2483794	0.2118921	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-2.0257023	1.0451914	-1.9381162	0.0526090	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-1.0609307	1.1083446	-0.9572209	0.3384558	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-0.9507623	1.1038759	-0.8612946	0.3890758	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-0.3589510	1.1478653	-0.3127118	0.7544996	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-0.1662354	1.3331785	-0.1246910	0.9007682	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-1.7620293	1.4535842	-1.2121962	0.2254373	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-1.2869490	1.2253282	-1.0502892	0.2935852	Binomial	BirdID+Terr

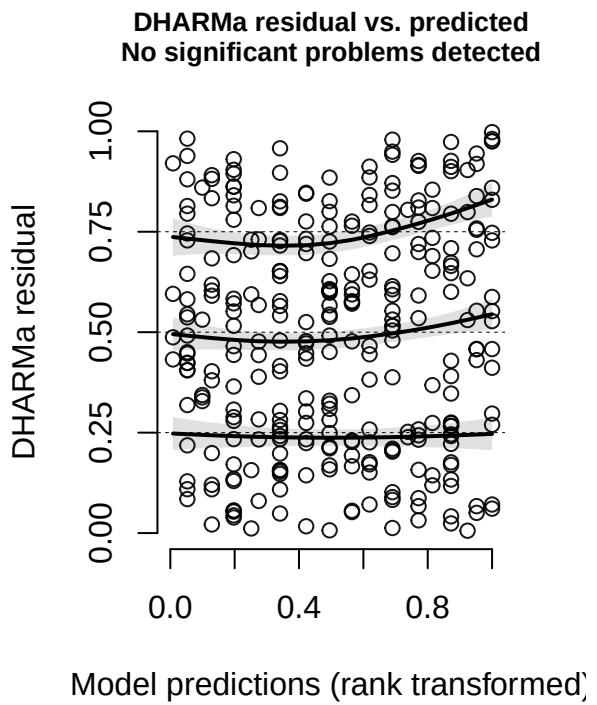
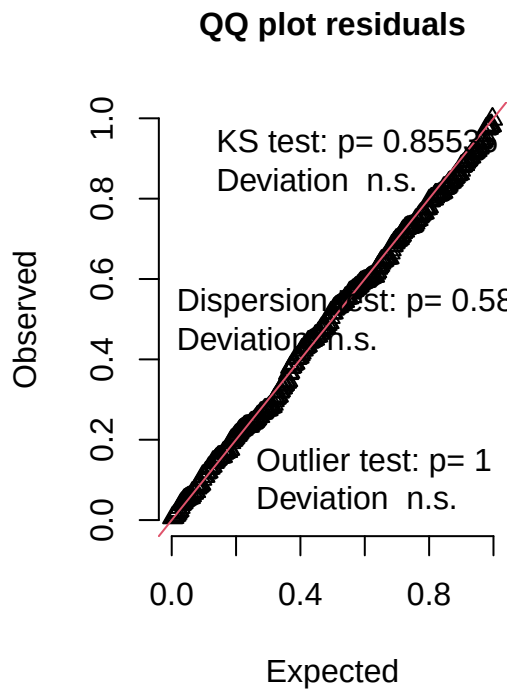
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	0.3569791	1.6582938	0.2152689	0.8295576	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-1.3750892	1.4190811	-0.9689997	0.3325453	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-2.0509713	1.4959222	-1.3710414	0.1703620	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-0.6140245	1.7427967	-0.3523214	0.7245973	Binomial	BirdID+Terr

Terr only model is performing best on BIC.

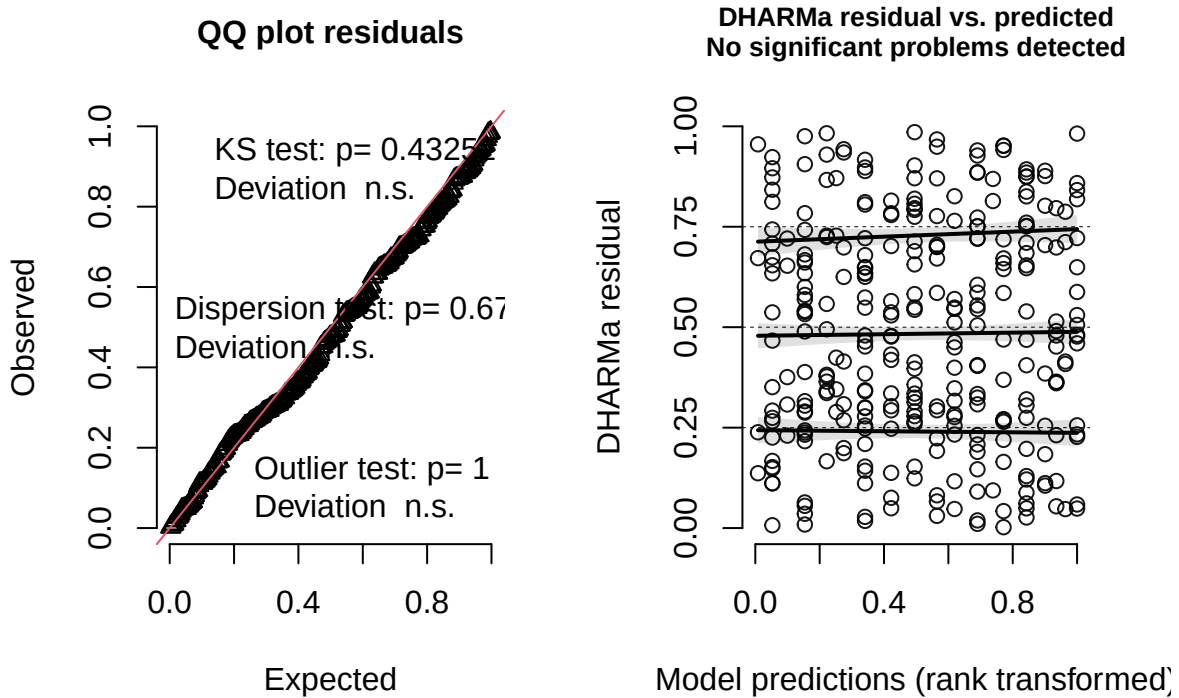
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Probabilites used to generate simulation 45

Simulation 45
 Probability of mate fidelity: 0.66
 Probability of pair dispersal: 0.56
 Probability of single male dispersal: 0.287
 Probability of single female dispersal: 0.695

Simulation 46

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
335	1	-414.9193	871.8386	951.9353	293.9723	314	0.3861621	0.1724013	Poisson	Terr
335	1	-414.9200	873.8401	957.7509	293.9575	313	0.3854569	0.1718602	Poisson	BirdID+Terr
335	1	-422.8569	887.7138	967.8105	334.5012	314	0.3222104	0.1790694	Poisson	BirdID

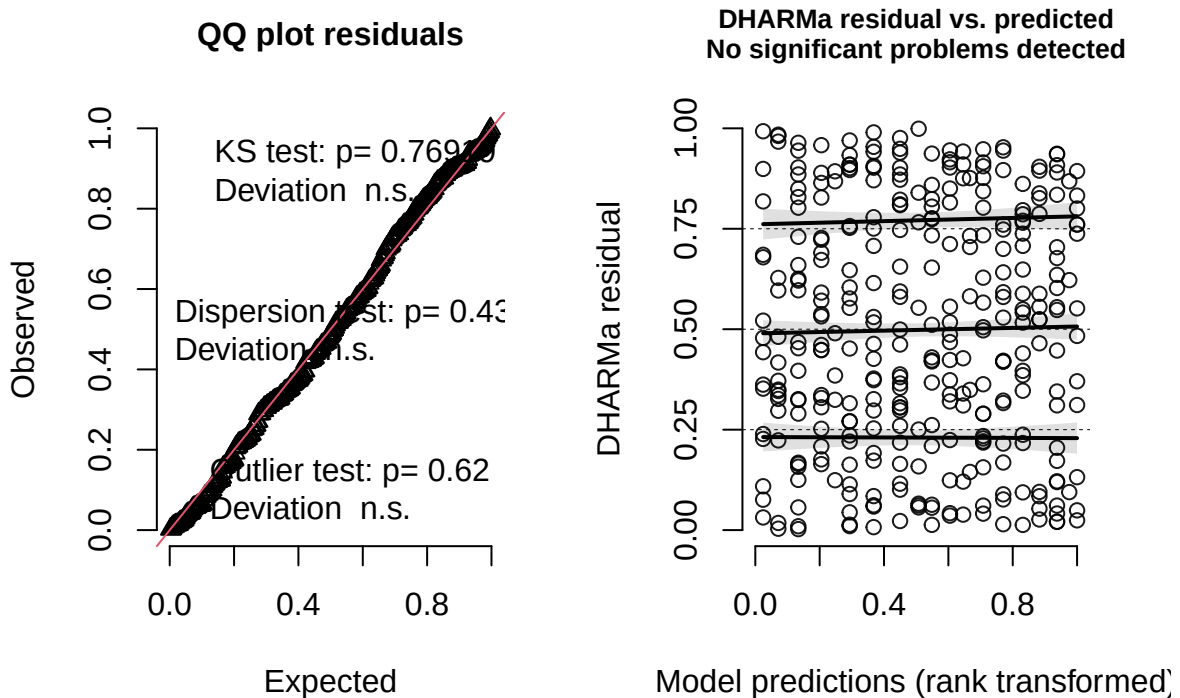
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.3845389	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.4863069	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.0009079	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.4863230	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.5965042	0.2227345	2.6780960	0.0074042	Poisson	BirdID
fixed	factor(fac.yr)2	-0.8711534	0.3170365	-2.7478012	0.0059996	Poisson	BirdID
fixed	factor(fac.yr)3	-0.4190003	0.3042127	-1.3773268	0.1684112	Poisson	BirdID
fixed	factor(fac.yr)4	-0.7385359	0.3347717	-2.2060883	0.0273778	Poisson	BirdID
fixed	factor(fac.yr)5	-0.5732785	0.3123484	-1.8353813	0.0664492	Poisson	BirdID
fixed	factor(fac.yr)6	-0.7512745	0.3018792	-2.4886591	0.0128226	Poisson	BirdID
fixed	factor(fac.yr)7	-1.1771343	0.3508688	-3.3549131	0.0007939	Poisson	BirdID
fixed	factor(fac.yr)8	-0.7845375	0.3175088	-2.4709160	0.0134767	Poisson	BirdID
fixed	factor(fac.yr)9	-0.2650102	0.2945310	-0.8997704	0.3682425	Poisson	BirdID
fixed	factor(fac.yr)10	-0.4978567	0.3170986	-1.5700375	0.1164064	Poisson	BirdID
fixed	factor(fac.yr)11	-1.1378418	0.3591032	-3.1685646	0.0015319	Poisson	BirdID
fixed	factor(fac.yr)12	-0.6704116	0.3279241	-2.0444108	0.0409130	Poisson	BirdID
fixed	factor(fac.yr)13	-0.3309170	0.3318450	-0.9972035	0.3186658	Poisson	BirdID
fixed	factor(fac.yr)14	-1.6208392	0.4795772	-3.3797250	0.0007256	Poisson	BirdID
fixed	factor(fac.yr)15	-1.9341502	0.5620888	-3.4410047	0.0005796	Poisson	BirdID
fixed	factor(fac.yr)16	-0.6611497	0.4394611	-1.5044556	0.1324641	Poisson	BirdID
fixed	factor(fac.yr)17	-0.6244753	0.4916522	-1.2701565	0.2040289	Poisson	BirdID
fixed	factor(fac.yr)18	-0.7388413	0.4403085	-1.6780081	0.0933455	Poisson	BirdID
fixed	factor(fac.yr)19	-1.0704516	0.5734030	-1.8668399	0.0619240	Poisson	BirdID
fixed	factor(fac.yr)20	-0.2615587	0.5404072	-0.4840030	0.6283837	Poisson	BirdID
fixed	(Intercept)	0.5471626	0.2299958	2.3790114	0.0173591	Poisson	Terr
fixed	factor(fac.yr)2	-0.8300404	0.3364006	-2.4674164	0.0136092	Poisson	Terr
fixed	factor(fac.yr)3	-0.4106423	0.3157021	-1.3007270	0.1933519	Poisson	Terr
fixed	factor(fac.yr)4	-0.7157023	0.3456938	-2.0703360	0.0384209	Poisson	Terr
fixed	factor(fac.yr)5	-0.4685124	0.3117600	-1.5027986	0.1328910	Poisson	Terr
fixed	factor(fac.yr)6	-0.6890298	0.3107868	-2.2170497	0.0266197	Poisson	Terr
fixed	factor(fac.yr)7	-1.0469937	0.3512936	-2.9803947	0.0028788	Poisson	Terr
fixed	factor(fac.yr)8	-0.6830266	0.3193106	-2.1390668	0.0324303	Poisson	Terr
fixed	factor(fac.yr)9	-0.1789966	0.2953516	-0.6060456	0.5444844	Poisson	Terr
fixed	factor(fac.yr)10	-0.4969641	0.3186223	-1.5597279	0.1188242	Poisson	Terr
fixed	factor(fac.yr)11	-1.0842751	0.3633745	-2.9839054	0.0028459	Poisson	Terr
fixed	factor(fac.yr)12	-0.6082703	0.3269733	-1.8603059	0.0628423	Poisson	Terr
fixed	factor(fac.yr)13	-0.2637483	0.3382798	-0.7796752	0.4355821	Poisson	Terr
fixed	factor(fac.yr)14	-1.6129842	0.4822202	-3.3449123	0.0008231	Poisson	Terr
fixed	factor(fac.yr)15	-1.9387894	0.5662332	-3.4240120	0.0006170	Poisson	Terr
fixed	factor(fac.yr)16	-0.7022432	0.4513546	-1.5558570	0.1197421	Poisson	Terr
fixed	factor(fac.yr)17	-0.6815429	0.5033151	-1.3541079	0.1757020	Poisson	Terr
fixed	factor(fac.yr)18	-0.8063879	0.4448207	-1.8128382	0.0698568	Poisson	Terr
fixed	factor(fac.yr)19	-0.8638172	0.5853630	-1.4756948	0.1400258	Poisson	Terr
fixed	factor(fac.yr)20	0.0906086	0.5510480	0.1644296	0.8693930	Poisson	Terr
fixed	(Intercept)	0.5394240	0.2305172	2.3400602	0.0192806	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.8216092	0.3367127	-2.4400895	0.0146836	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.4020867	0.3160329	-1.2722937	0.2032688	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.7068360	0.3459361	-2.0432562	0.0410271	Poisson	BirdID+Terr

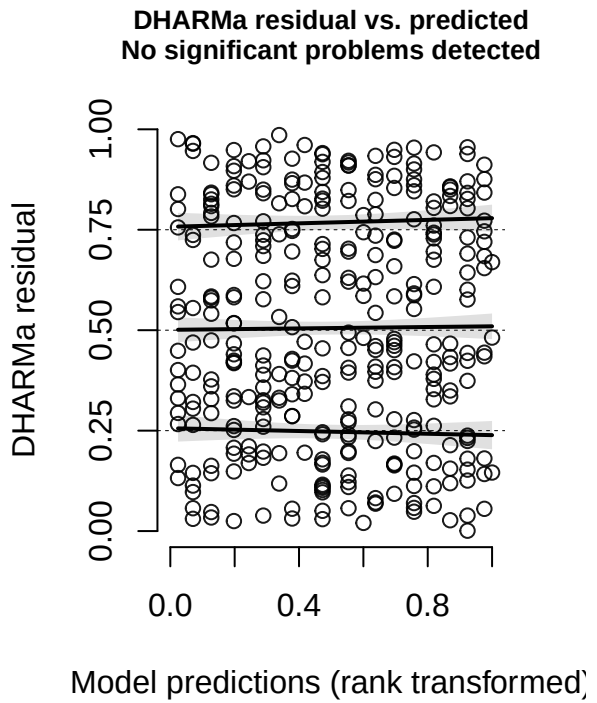
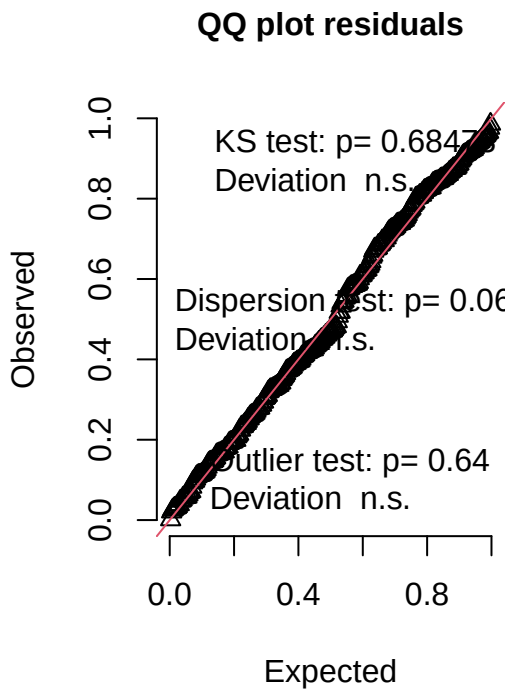
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.4604230	0.3120942	-1.4752691	0.1401402	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.6801829	0.3111223	-2.1862234	0.0287993	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-1.0395396	0.3516509	-2.9561689	0.0031149	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.6756403	0.3197054	-2.1133215	0.0345733	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.1709234	0.2957356	-0.5779599	0.5632912	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.4886604	0.3189565	-1.5320597	0.1255077	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-1.0758290	0.3636231	-2.9586379	0.0030900	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.6004356	0.3273197	-1.8344008	0.0665945	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.2555238	0.3386189	-0.7546058	0.4504855	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-1.6045422	0.4823686	-3.3263817	0.0008798	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-1.9308556	0.5664363	-3.4087780	0.0006525	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.6915683	0.4512669	-1.5325039	0.1253981	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.6705176	0.5030668	-1.3328600	0.1825778	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.7968031	0.4448739	-1.7910762	0.0732811	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-0.8532363	0.5850335	-1.4584401	0.1447193	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	0.0990352	0.5511280	0.1796954	0.8573917	Poisson	BirdID+Terr

Terr only model is performing very well on all criteria, is the best on 3 of them.

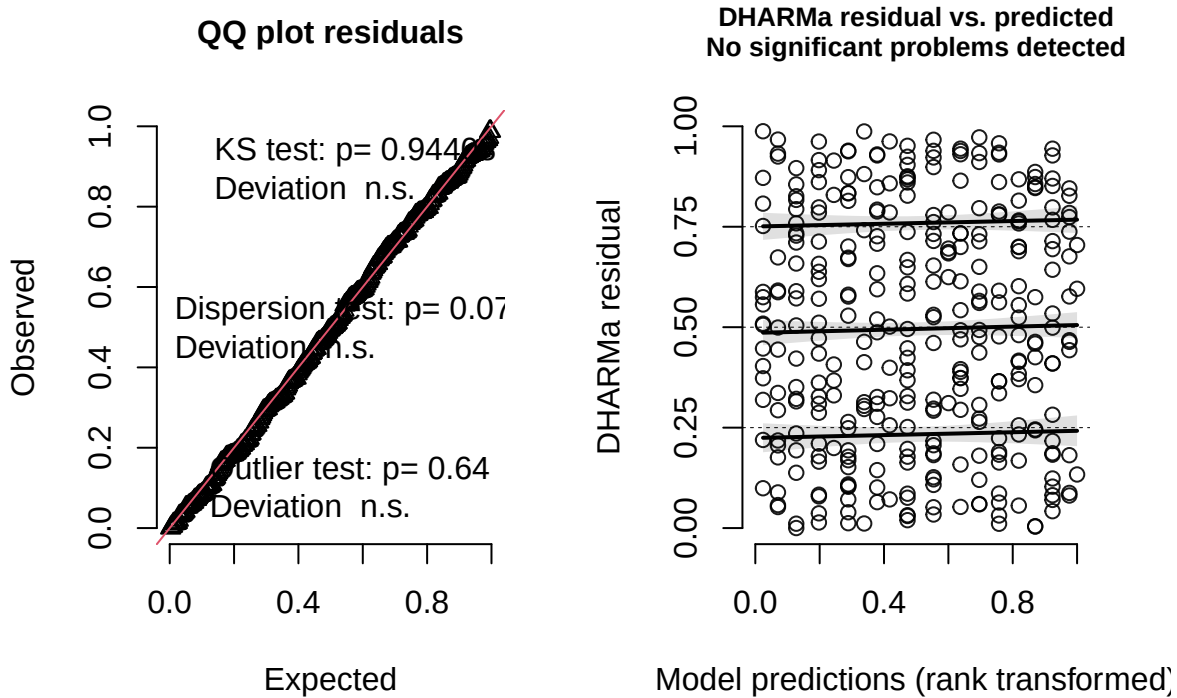
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
335	1	-194.8802	431.7603	511.8571	262.4797	314	0.702	Binomial	Terr
335	1	-193.9756	431.9511	515.8620	224.6070	313	0.788	Binomial	BirdID+Terr
335	1	-203.8967	449.7933	529.8901	292.1888	314	0.684	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.3087037	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.5490393	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	0.8577746	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.5637044	Binomial	BirdID+Terr

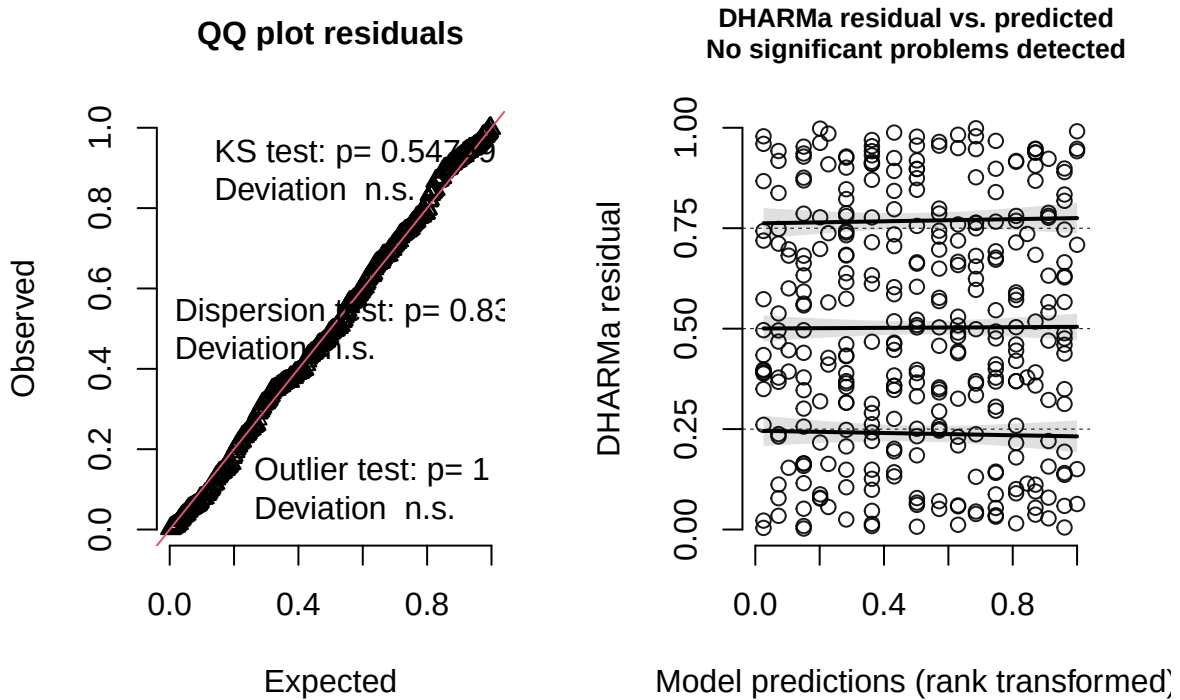
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	1.5627805	0.8110370	1.9268917	0.0539931	Binomial	BirdID
fixed	factor(fac.yr)2	-1.4656628	0.9048043	-1.6198672	0.1052608	Binomial	BirdID
fixed	factor(fac.yr)3	0.3678694	1.0270631	0.3581761	0.7202116	Binomial	BirdID
fixed	factor(fac.yr)4	-1.0575261	0.9791946	-1.0799958	0.2801441	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-1.1699667	0.9595029	-1.2193467	0.2227126	Binomial	BirdID
fixed	factor(fac.yr)6	-1.5299179	0.9236549	-1.6563739	0.0976461	Binomial	BirdID
fixed	factor(fac.yr)7	-1.8001730	0.9404037	-1.9142555	0.0555875	Binomial	BirdID
fixed	factor(fac.yr)8	-1.3356804	0.9374484	-1.4248041	0.1542138	Binomial	BirdID
fixed	factor(fac.yr)9	-0.7691604	0.9633092	-0.7984564	0.4246057	Binomial	BirdID
fixed	factor(fac.yr)10	-0.7469506	0.9865108	-0.7571641	0.4489515	Binomial	BirdID
fixed	factor(fac.yr)11	-1.3807576	0.9667113	-1.4283041	0.1532044	Binomial	BirdID
fixed	factor(fac.yr)12	-1.0297416	0.9962936	-1.0335724	0.3013361	Binomial	BirdID
fixed	factor(fac.yr)13	-0.4362114	1.0875120	-0.4011095	0.6883395	Binomial	BirdID
fixed	factor(fac.yr)14	-3.4448632	1.1340053	-3.0377840	0.0023832	Binomial	BirdID
fixed	factor(fac.yr)15	-3.1998990	1.1312663	-2.8285992	0.0046752	Binomial	BirdID
fixed	factor(fac.yr)16	-1.5696005	1.1876076	-1.3216490	0.1862851	Binomial	BirdID
fixed	factor(fac.yr)17	0.5497269	1.5682901	0.3505263	0.7259438	Binomial	BirdID
fixed	factor(fac.yr)18	-1.6411110	1.1563743	-1.4191867	0.1558446	Binomial	BirdID
fixed	factor(fac.yr)19	-2.3099967	1.3196043	-1.7505223	0.0800282	Binomial	BirdID
fixed	factor(fac.yr)20	-0.5914483	1.6943783	-0.3490651	0.7270404	Binomial	BirdID
fixed	(Intercept)	1.5350153	0.7923202	1.9373673	0.0527005	Binomial	Terr
fixed	factor(fac.yr)2	-1.1934853	0.9221567	-1.2942326	0.1955851	Binomial	Terr
fixed	factor(fac.yr)3	0.4033540	1.0501711	0.3840841	0.7009161	Binomial	Terr
fixed	factor(fac.yr)4	-0.6430379	0.9801067	-0.6560897	0.5117664	Binomial	Terr
fixed	factor(fac.yr)5	-0.5128561	0.9402899	-0.5454234	0.5854624	Binomial	Terr
fixed	factor(fac.yr)6	-1.2259040	0.9193316	-1.3334732	0.1823766	Binomial	Terr
fixed	factor(fac.yr)7	-1.5051822	0.9154461	-1.6442063	0.1001336	Binomial	Terr
fixed	factor(fac.yr)8	-0.9191012	0.9182781	-1.0008964	0.3168769	Binomial	Terr
fixed	factor(fac.yr)9	-0.2311712	0.9568339	-0.2416001	0.8090900	Binomial	Terr
fixed	factor(fac.yr)10	-0.2436165	0.9931780	-0.2452899	0.8062320	Binomial	Terr
fixed	factor(fac.yr)11	-0.8563932	0.9595020	-0.8925393	0.3721040	Binomial	Terr
fixed	factor(fac.yr)12	-0.4218117	0.9801989	-0.4303328	0.6669536	Binomial	Terr
fixed	factor(fac.yr)13	0.2678434	1.0634004	0.2518745	0.8011381	Binomial	Terr
fixed	factor(fac.yr)14	-3.4099109	1.1371379	-2.9986785	0.0027115	Binomial	Terr
fixed	factor(fac.yr)15	-3.2025503	1.1186407	-2.8628945	0.0041979	Binomial	Terr
fixed	factor(fac.yr)16	-1.6071849	1.1695217	-1.3742240	0.1693721	Binomial	Terr
fixed	factor(fac.yr)17	0.6547551	1.5644499	0.4185210	0.6755663	Binomial	Terr
fixed	factor(fac.yr)18	-1.8624641	1.1481697	-1.6221157	0.1047786	Binomial	Terr
fixed	factor(fac.yr)19	-1.8560038	1.3561129	-1.3686204	0.1711180	Binomial	Terr
fixed	factor(fac.yr)20	0.1626463	1.7356473	0.0937093	0.9253401	Binomial	Terr
fixed	(Intercept)	1.5363584	0.8527034	1.8017500	0.0715848	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-1.1688155	0.9718621	-1.2026557	0.2291096	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	0.6399717	1.1284857	0.5671066	0.5706417	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-0.6649759	1.0519902	-0.6321123	0.5273135	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-0.5323301	1.0065342	-0.5288743	0.5968927	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-1.2784371	0.9887625	-1.2929668	0.1960226	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-1.4791642	0.9861938	-1.4998718	0.1336476	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-0.9067323	0.9891403	-0.9166872	0.3593066	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-0.1146415	1.0297333	-0.1113312	0.9113537	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-0.2026344	1.0617105	-0.1908565	0.8486380	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-0.8376462	1.0334793	-0.8105108	0.4176467	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-0.4330375	1.0586657	-0.4090408	0.6825097	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	0.3504198	1.1529452	0.3039345	0.7611778	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-3.6573779	1.2399597	-2.9495942	0.0031819	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-3.4463882	1.2218923	-2.8205336	0.0047944	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-1.6505247	1.2721970	-1.2973814	0.1945000	Binomial	BirdID+Terr

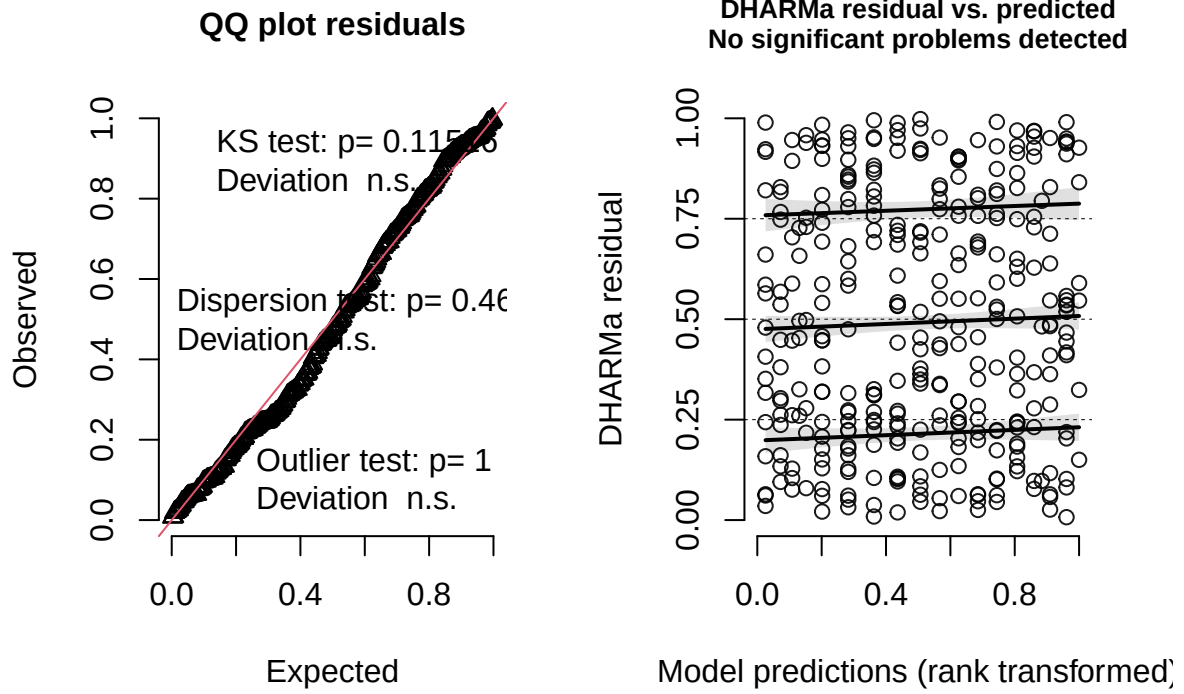
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	0.8605090	1.6881853	0.5097243	0.6102446	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-1.8870669	1.2567781	-1.5015116	0.1332233	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-2.0780745	1.4557116	-1.4275317	0.1534267	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	0.1827527	1.9341725	0.0944862	0.9247229	Binomial	BirdID+Terr

Terr only model is performing best on AIC and BIC, and well on Kappa.

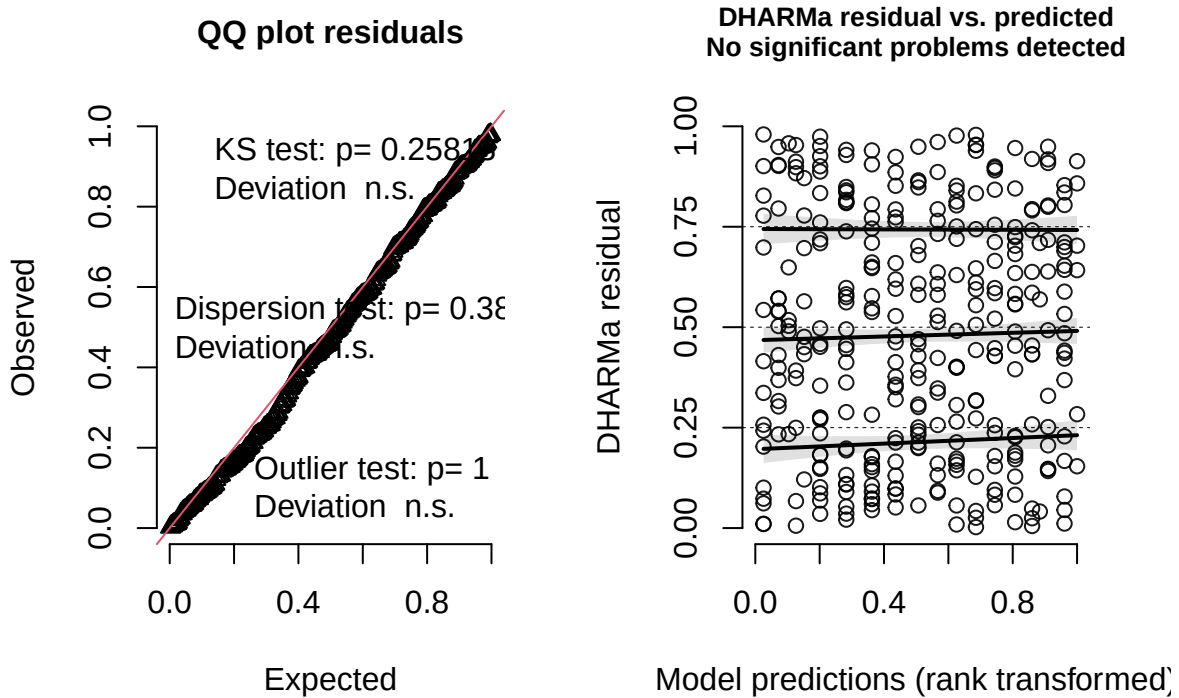
DHARMA residual



DHARMa residual



DHARMA residual



Diagnostics look good.

Probabilites used to generate simulation 46

Simulation 46

Probability of mate fidelity: 0.432

Probability of pair dispersal: 0.878

Probability of single male dispersal: 0.66

Probability of single female dispersal: 0.96

Simulation 47

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
370	1	- 446.8962	935.7924	1017.976	346.8840	349	0.8835832	0.8461327	Poisson	Terr
370	1	- 445.9625	935.9250	1022.022	322.5573	348	0.8844627	0.8445779	Poisson	BirdID+Terr
370	1	- 446.9775	935.9550	1018.139	336.2637	349	0.8919723	0.8535327	Poisson	BirdID

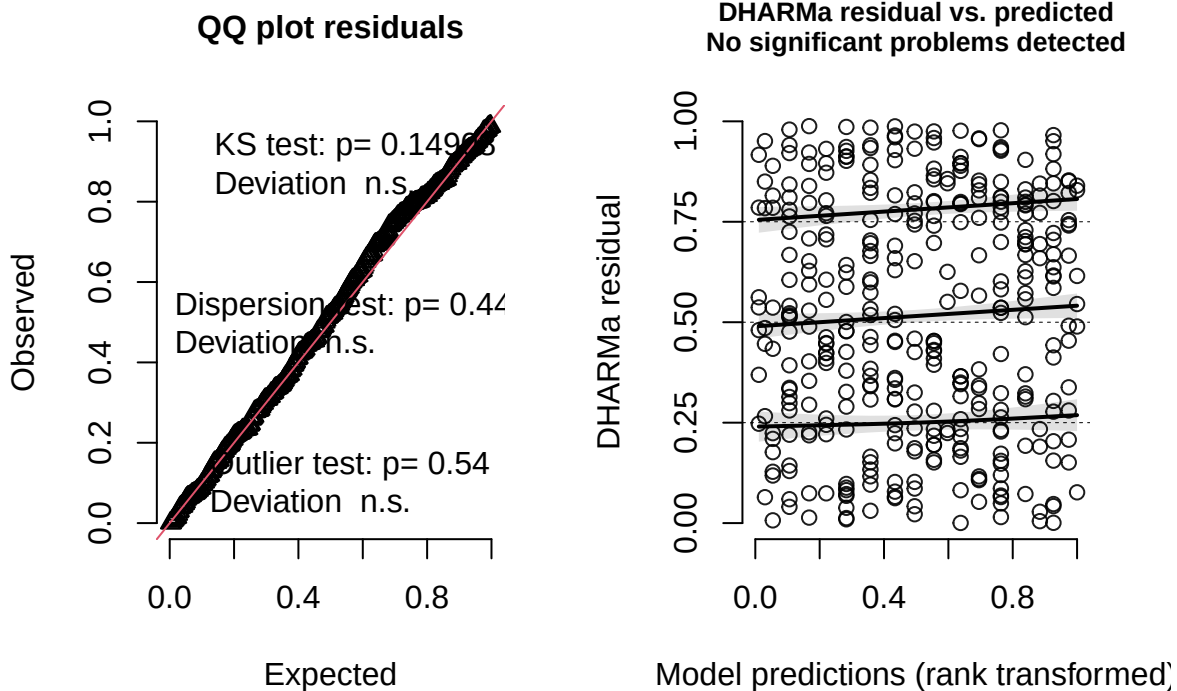
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.5140692	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.4747237	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.3552882	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.3558832	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.4689335	0.2426829	1.9322887	0.0533239	Poisson	BirdID
fixed	factor(fac.yr)2	-0.6845377	0.3480434	-1.9668168	0.0492043	Poisson	BirdID
fixed	factor(fac.yr)3	-1.1587358	0.3935730	-2.9441446	0.0032385	Poisson	BirdID
fixed	factor(fac.yr)4	-0.6189475	0.3290961	-1.8807501	0.0600059	Poisson	BirdID
fixed	factor(fac.yr)5	-0.0996290	0.2995845	-0.3325572	0.7394685	Poisson	BirdID
fixed	factor(fac.yr)6	-0.3906927	0.3042787	-1.2839965	0.1991432	Poisson	BirdID
fixed	factor(fac.yr)7	-1.0031942	0.3296145	-3.0435375	0.0023381	Poisson	BirdID
fixed	factor(fac.yr)8	-0.5422059	0.3052338	-1.7763628	0.0756731	Poisson	BirdID
fixed	factor(fac.yr)9	-0.6475446	0.3250480	-1.9921505	0.0463545	Poisson	BirdID
fixed	factor(fac.yr)10	-0.7383986	0.3337504	-2.2124273	0.0269372	Poisson	BirdID
fixed	factor(fac.yr)11	-1.8210850	0.4437053	-4.1042668	0.0000406	Poisson	BirdID
fixed	factor(fac.yr)12	-1.0133476	0.3591786	-2.8212913	0.0047831	Poisson	BirdID
fixed	factor(fac.yr)13	-0.6072267	0.3568688	-1.7015404	0.0888416	Poisson	BirdID
fixed	factor(fac.yr)14	-1.1684087	0.4199933	-2.7819700	0.0054030	Poisson	BirdID
fixed	factor(fac.yr)15	-2.2684461	0.7760828	-2.9229435	0.0034674	Poisson	BirdID
fixed	factor(fac.yr)16	-0.2227375	0.4096042	-0.5437871	0.5865880	Poisson	BirdID
fixed	factor(fac.yr)17	-0.6205490	0.5155563	-1.2036494	0.2287251	Poisson	BirdID
fixed	factor(fac.yr)18	0.0356353	0.4260394	0.0836432	0.9333401	Poisson	BirdID
fixed	factor(fac.yr)19	-2.5883448	1.0662146	-2.4276020	0.0151990	Poisson	BirdID
fixed	factor(fac.yr)20	-18.1032072	2392.5364317	-0.0075665	0.9939628	Poisson	BirdID
fixed	(Intercept)	0.5832351	0.2335000	2.4977948	0.0124969	Poisson	Terr
fixed	factor(fac.yr)2	-0.7331258	0.3513674	-2.0864935	0.0369339	Poisson	Terr
fixed	factor(fac.yr)3	-1.1843789	0.3926316	-3.0165143	0.0025570	Poisson	Terr
fixed	factor(fac.yr)4	-0.6765876	0.3134313	-2.1586473	0.0308775	Poisson	Terr
fixed	factor(fac.yr)5	-0.0026522	0.2953743	-0.0089791	0.9928358	Poisson	Terr
fixed	factor(fac.yr)6	-0.2781797	0.2949081	-0.9432756	0.3455399	Poisson	Terr
fixed	factor(fac.yr)7	-0.9441437	0.3165510	-2.9825962	0.0028581	Poisson	Terr
fixed	factor(fac.yr)8	-0.5946756	0.2927362	-2.0314384	0.0422105	Poisson	Terr
fixed	factor(fac.yr)9	-0.6191982	0.3149056	-1.9662979	0.0492642	Poisson	Terr
fixed	factor(fac.yr)10	-0.7473171	0.3202047	-2.3338728	0.0196024	Poisson	Terr
fixed	factor(fac.yr)11	-1.9003364	0.4347500	-4.3711011	0.0000124	Poisson	Terr
fixed	factor(fac.yr)12	-1.0920521	0.3466216	-3.1505595	0.0016296	Poisson	Terr
fixed	factor(fac.yr)13	-0.6445462	0.3454598	-1.8657630	0.0620745	Poisson	Terr
fixed	factor(fac.yr)14	-1.3095511	0.4058692	-3.2265349	0.0012530	Poisson	Terr
fixed	factor(fac.yr)15	-2.3549310	0.7624735	-3.0885414	0.0020114	Poisson	Terr
fixed	factor(fac.yr)16	-0.3581811	0.3921480	-0.9133826	0.3610414	Poisson	Terr
fixed	factor(fac.yr)17	-0.7914097	0.4892654	-1.6175468	0.1057603	Poisson	Terr
fixed	factor(fac.yr)18	-0.1395239	0.3983047	-0.3502944	0.7261178	Poisson	Terr
fixed	factor(fac.yr)19	-2.5855599	1.0479403	-2.4672778	0.0136145	Poisson	Terr
fixed	factor(fac.yr)20	-16.8256309	1365.6581513	-0.0123205	0.9901699	Poisson	Terr
fixed	(Intercept)	0.5136419	0.2430750	2.1131000	0.0345922	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.6943999	0.3555363	-1.9531056	0.0508071	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-1.1659508	0.3988364	-2.9233810	0.0034625	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.6474486	0.3264706	-1.9831760	0.0473478	Poisson	BirdID+Terr

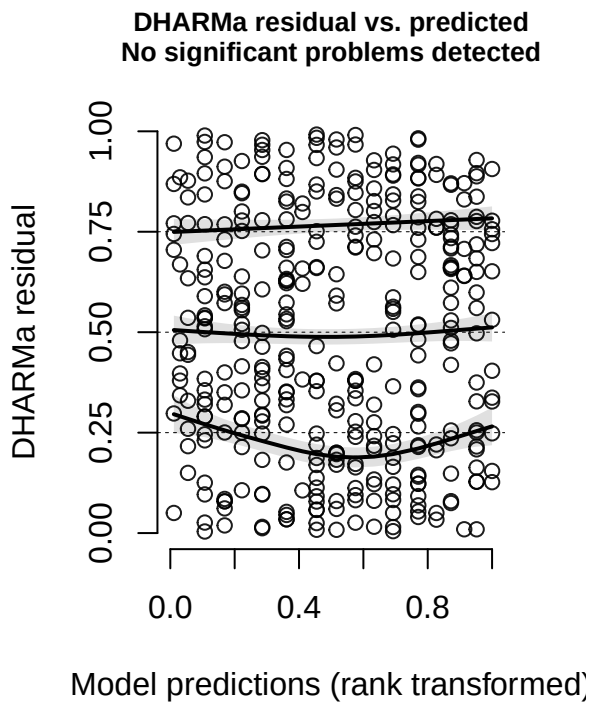
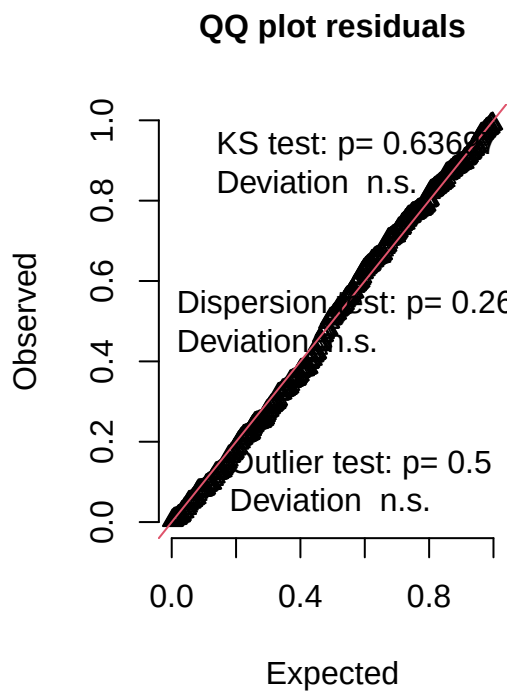
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.0491924	0.3050784	-0.1612451	0.8719004	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.3503496	0.3059769	-1.1450198	0.2522010	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.9864053	0.3291076	-2.9972117	0.0027246	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.5798852	0.3050841	-1.9007387	0.0573362	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.6155164	0.3269856	-1.8823959	0.0597823	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.7328259	0.3330659	-2.2002432	0.0277896	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-1.8512094	0.4448083	-4.1618144	0.0000316	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-1.0577708	0.3589310	-2.9470035	0.0032087	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.6321879	0.3581505	-1.7651461	0.0775392	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-1.2431174	0.4197332	-2.9616847	0.0030596	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-2.2998483	0.7768569	-2.9604529	0.0030719	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.2703049	0.4121027	-0.6559165	0.5118779	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.7164685	0.5130845	-1.3963947	0.1625957	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.0559400	0.4219050	-0.1325891	0.8945183	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-2.5828252	1.0641733	-2.4270720	0.0152212	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-17.2799281	1684.4228495	-0.0102587	0.9918149	Poisson	BirdID+Terr

Terr seems to be marginally better on the criteria than the other models, except for marginal R^2 .

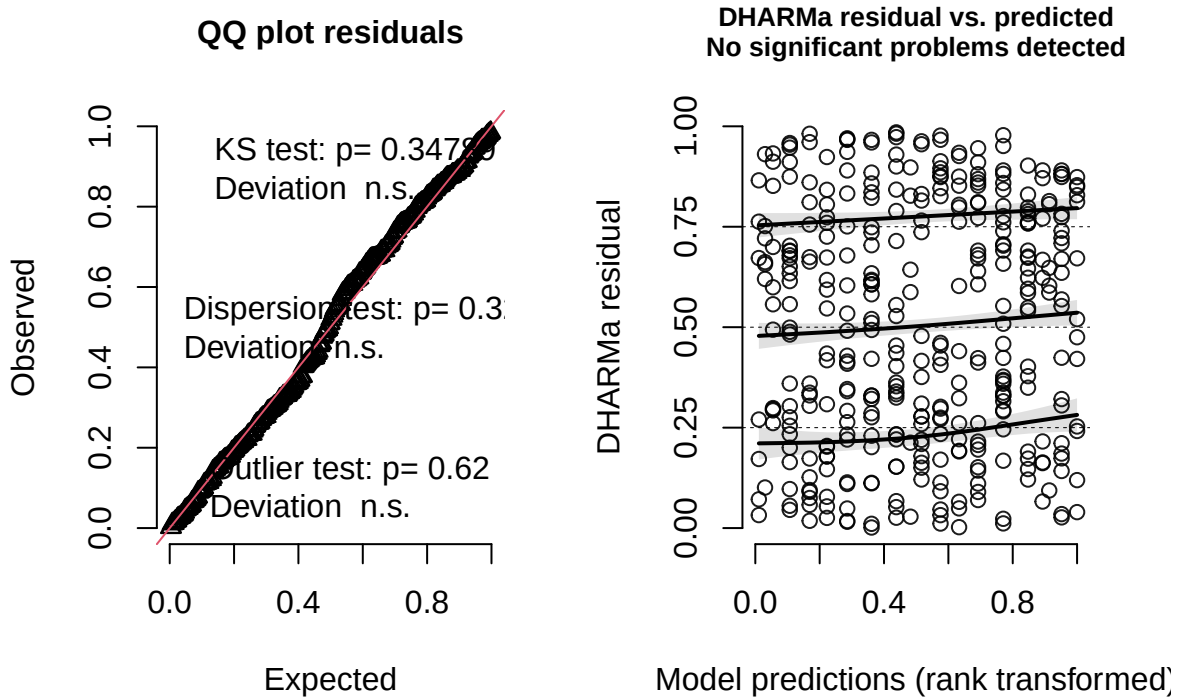
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
370	1	-209.6486	463.2971	549.3942	223.6142	348	0.827	Binomial	BirdID+Terr
370	1	-213.1926	468.3852	550.5688	291.0970	349	0.687	Binomial	BirdID
370	1	-214.7716	471.5433	553.7268	306.8585	349	0.692	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.481567	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.369684	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.431326	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.296912	Binomial	BirdID+Terr

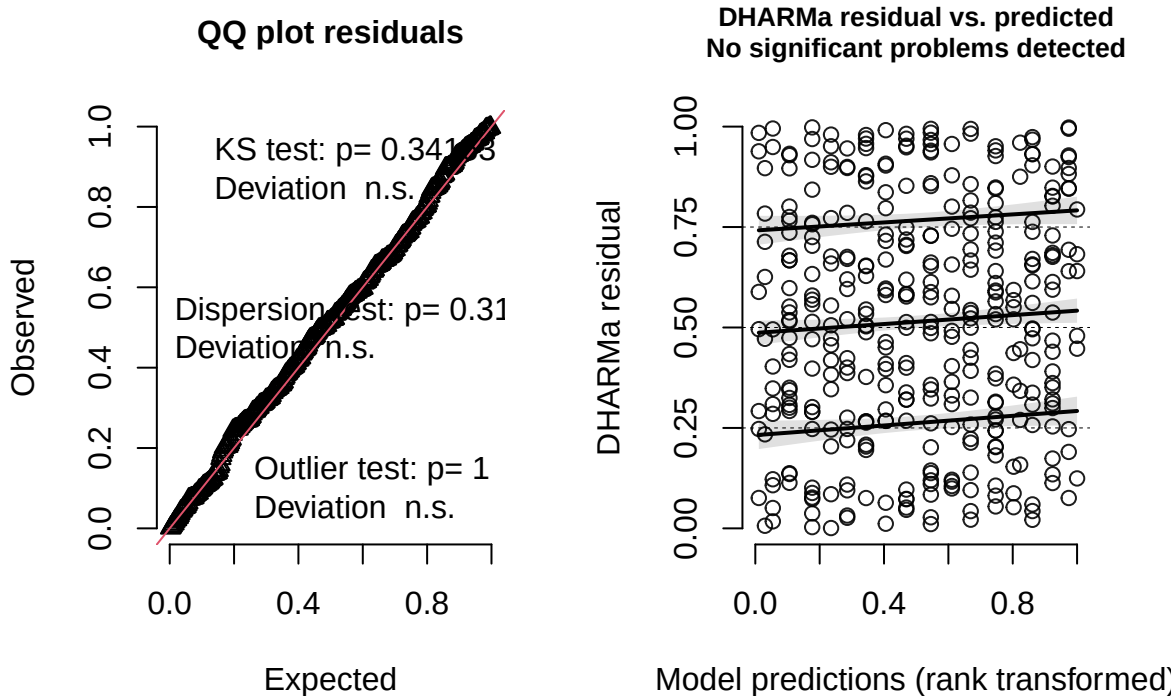
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	1.9125622	0.9666685	1.9785087	0.0478713	Binomial	BirdID
fixed	factor(fac.yr)2	-2.3758341	1.0983230	-2.1631470	0.0305299	Binomial	BirdID
fixed	factor(fac.yr)3	-2.3363245	1.0968343	-2.1300615	0.0331665	Binomial	BirdID
fixed	factor(fac.yr)4	-2.3641133	1.0858890	-2.1771224	0.0294714	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.7524164	1.1500055	-0.6542720	0.5129366	Binomial	BirdID
fixed	factor(fac.yr)6	-0.5548281	1.1615824	-0.4776485	0.6329004	Binomial	BirdID
fixed	factor(fac.yr)7	-1.9809381	1.0649844	-1.8600629	0.0628766	Binomial	BirdID
fixed	factor(fac.yr)8	-1.6736998	1.0644352	-1.5723830	0.1158618	Binomial	BirdID
fixed	factor(fac.yr)9	-1.7424844	1.0937487	-1.5931305	0.1111309	Binomial	BirdID
fixed	factor(fac.yr)10	-2.3036624	1.0951449	-2.1035230	0.0354201	Binomial	BirdID
fixed	factor(fac.yr)11	-3.5162511	1.1336974	-3.1015782	0.0019249	Binomial	BirdID
fixed	factor(fac.yr)12	-2.6490911	1.1066053	-2.3938899	0.0166708	Binomial	BirdID
fixed	factor(fac.yr)13	-1.7709972	1.1480050	-1.5426739	0.1229099	Binomial	BirdID
fixed	factor(fac.yr)14	-2.4247461	1.1472322	-2.1135618	0.0345527	Binomial	BirdID
fixed	factor(fac.yr)15	-3.6974208	1.3626312	-2.7134421	0.0066588	Binomial	BirdID
fixed	factor(fac.yr)16	-1.4079177	1.2636673	-1.1141522	0.2652139	Binomial	BirdID
fixed	factor(fac.yr)17	-0.9686752	1.4592418	-0.6638209	0.5068050	Binomial	BirdID
fixed	factor(fac.yr)18	0.0039496	1.7084095	0.0023119	0.9981554	Binomial	BirdID
fixed	factor(fac.yr)19	-4.5709843	1.7204872	-2.6567964	0.0078887	Binomial	BirdID
fixed	factor(fac.yr)20	-10.2026923	17.3377197	-0.5884679	0.5562182	Binomial	BirdID
fixed	(Intercept)	2.2102105	0.8952294	2.4688762	0.0135538	Binomial	Terr
fixed	factor(fac.yr)2	-2.3330632	1.0294786	-2.2662571	0.0234356	Binomial	Terr
fixed	factor(fac.yr)3	-2.1826886	1.0318173	-2.1153827	0.0343973	Binomial	Terr
fixed	factor(fac.yr)4	-2.3152256	1.0114421	-2.2890342	0.0220774	Binomial	Terr
fixed	factor(fac.yr)5	-0.2980646	1.0763887	-0.2769116	0.7818480	Binomial	Terr
fixed	factor(fac.yr)6	-0.0141956	1.1080326	-0.0128115	0.9897782	Binomial	Terr
fixed	factor(fac.yr)7	-1.7613124	0.9755526	-1.8054509	0.0710042	Binomial	Terr
fixed	factor(fac.yr)8	-1.8477073	0.9796687	-1.8860533	0.0592878	Binomial	Terr
fixed	factor(fac.yr)9	-1.7710590	1.0079549	-1.7570816	0.0789039	Binomial	Terr
fixed	factor(fac.yr)10	-2.3907706	1.0049429	-2.3790114	0.0173591	Binomial	Terr
fixed	factor(fac.yr)11	-3.9010070	1.0503952	-3.7138471	0.0002041	Binomial	Terr
fixed	factor(fac.yr)12	-3.0446370	1.0201074	-2.9846241	0.0028393	Binomial	Terr
fixed	factor(fac.yr)13	-2.0723445	1.0635934	-1.9484367	0.0513627	Binomial	Terr
fixed	factor(fac.yr)14	-2.8492486	1.0704219	-2.6617996	0.0077724	Binomial	Terr
fixed	factor(fac.yr)15	-4.0671494	1.3097572	-3.1052697	0.0019011	Binomial	Terr
fixed	factor(fac.yr)16	-1.9184509	1.1889022	-1.6136323	0.1066072	Binomial	Terr
fixed	factor(fac.yr)17	-1.6505645	1.3699019	-1.2048779	0.2282504	Binomial	Terr
fixed	factor(fac.yr)18	-0.1387052	1.5980388	-0.0867971	0.9308328	Binomial	Terr
fixed	factor(fac.yr)19	-4.4452411	1.5426129	-2.8816309	0.0039562	Binomial	Terr
fixed	factor(fac.yr)20	-11.2588758	34.3331815	-0.3279299	0.7429647	Binomial	Terr
fixed	(Intercept)	2.2389099	1.0718108	2.0889040	0.0367164	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-2.5666920	1.2136175	-2.1149102	0.0344376	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-2.4836936	1.2171827	-2.0405266	0.0412979	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-2.6525945	1.1977061	-2.2147291	0.0267787	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-0.4918749	1.2798033	-0.3843363	0.7007292	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-0.3848579	1.2992164	-0.2962231	0.7670597	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-2.0970451	1.1663136	-1.7980113	0.0721752	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-1.8355342	1.1771906	-1.5592497	0.1189373	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-1.8758136	1.2076519	-1.5532734	0.1203578	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-2.5412600	1.2129736	-2.0950662	0.0361651	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-4.1936173	1.2633434	-3.3194597	0.0009019	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-3.2319256	1.2333751	-2.6203914	0.0087829	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-2.1050445	1.2848596	-1.6383460	0.1013495	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-2.7948114	1.2828861	-2.1785343	0.0293663	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-4.0960067	1.5535952	-2.6364697	0.0083774	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-1.5143678	1.4107030	-1.0734845	0.2830538	Binomial	BirdID+Terr

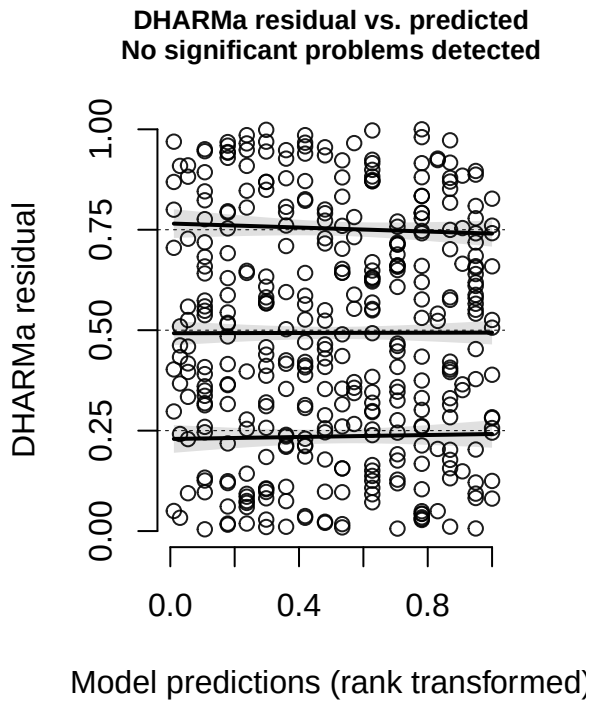
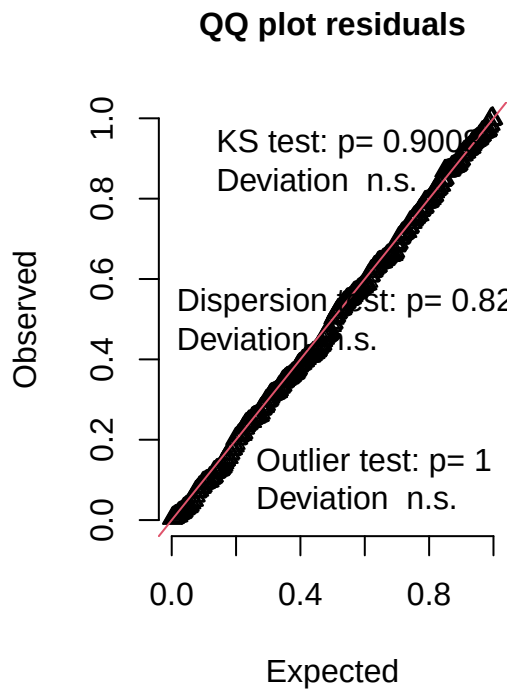
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-1.6805249	1.6163895	-1.0396782	0.2984894	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	0.3868918	2.0163564	0.1918767	0.8478388	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-5.2646854	1.9191199	-2.7432812	0.0060829	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-17.5655644	632.2038432	-0.0277847	0.9778339	Binomial	BirdID+Terr

Similar to the other binomial models.

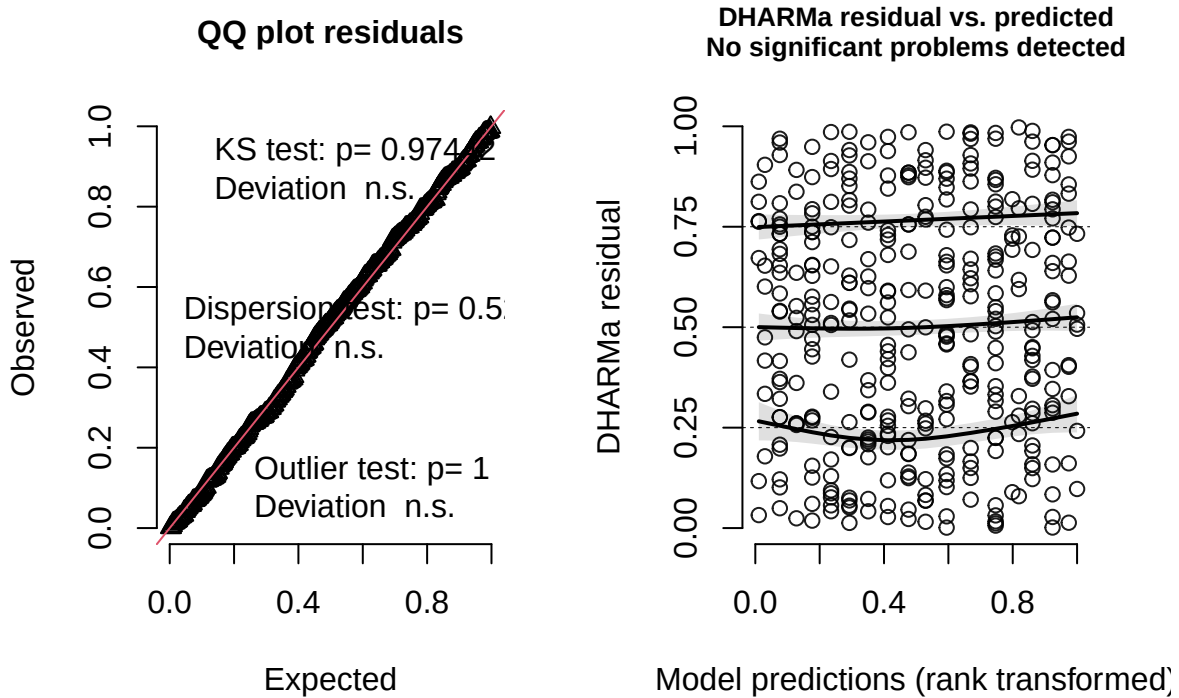
DHARMa residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Probabilites used to generate simulation 47

Simulation 47

Probability of mate fidelity: 0.562

Probability of pair dispersal: 0.05

Probability of single male dispersal: 0.901

Probability of single female dispersal: 0.184

Simulation 48

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
366	1	-457.7814	957.5628	1039.518	324.0064	345	0.5133778	0.3823967	Poisson	Terr
366	1	-457.7809	959.5618	1045.420	323.9553	344	0.5112226	0.3809358	Poisson	BirdID+Terr
366	1	-464.7492	971.4985	1053.454	364.9825	345	0.4969610	0.4161062	Poisson	BirdID

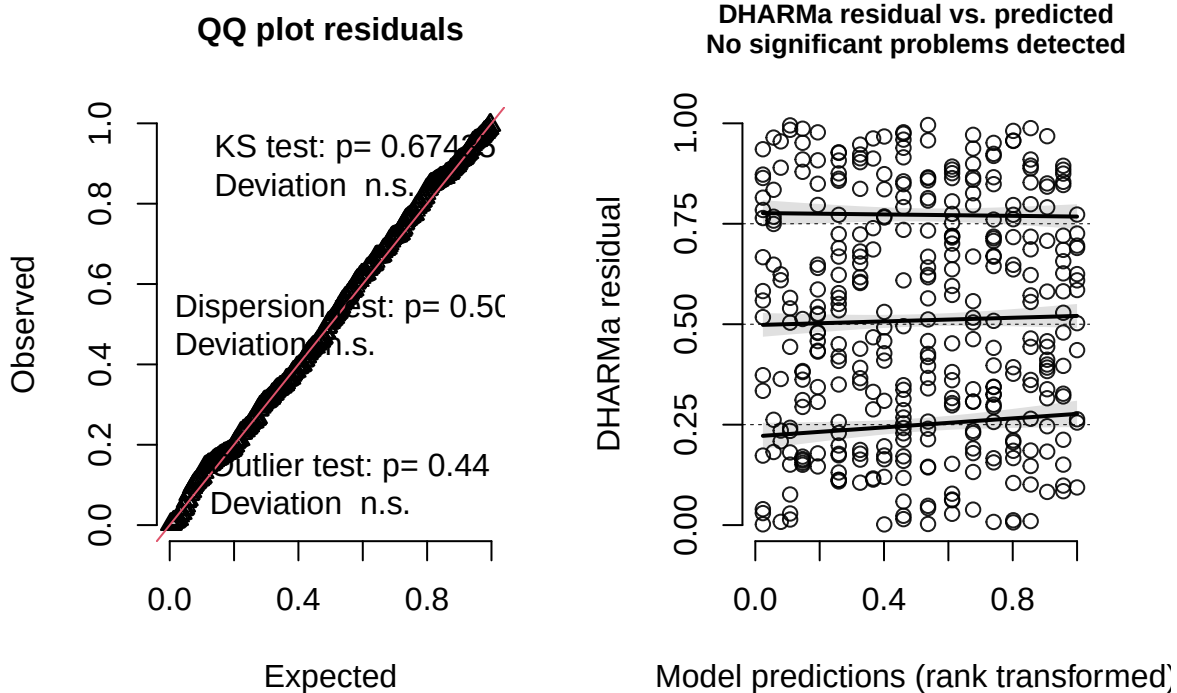
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.3278891	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.4222598	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.0011408	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.4225635	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.9610856	0.1975814	4.864251	0.0000011	Poisson	BirdID
fixed	factor(fac.yr)2	-0.6576200	0.2840089	-2.315491	0.0205861	Poisson	BirdID
fixed	factor(fac.yr)3	-1.4158527	0.3464255	-4.087033	0.0000437	Poisson	BirdID
fixed	factor(fac.yr)4	-1.1618450	0.3129389	-3.712689	0.0002051	Poisson	BirdID
fixed	factor(fac.yr)5	-0.5673326	0.2710964	-2.092734	0.0363729	Poisson	BirdID
fixed	factor(fac.yr)6	-0.7568221	0.2816532	-2.687071	0.0072082	Poisson	BirdID
fixed	factor(fac.yr)7	-0.9027972	0.2750396	-3.282426	0.0010292	Poisson	BirdID
fixed	factor(fac.yr)8	-0.7904112	0.2713049	-2.913369	0.0035755	Poisson	BirdID
fixed	factor(fac.yr)9	-0.9908425	0.2781045	-3.562842	0.0003669	Poisson	BirdID
fixed	factor(fac.yr)10	-0.9652179	0.2758462	-3.499116	0.0004668	Poisson	BirdID
fixed	factor(fac.yr)11	-1.3858454	0.3056897	-4.533503	0.0000058	Poisson	BirdID
fixed	factor(fac.yr)12	-0.8050809	0.2851699	-2.823162	0.0047553	Poisson	BirdID
fixed	factor(fac.yr)13	-0.7610479	0.2909942	-2.615337	0.0089140	Poisson	BirdID
fixed	factor(fac.yr)14	-1.0743418	0.3354992	-3.202218	0.0013637	Poisson	BirdID
fixed	factor(fac.yr)15	-3.7984813	1.0344234	-3.672076	0.0002406	Poisson	BirdID
fixed	factor(fac.yr)16	-1.9475752	0.5010281	-3.887157	0.0001014	Poisson	BirdID
fixed	factor(fac.yr)17	-1.4776463	0.4002821	-3.691513	0.0002229	Poisson	BirdID
fixed	factor(fac.yr)18	-1.1325543	0.4210214	-2.690016	0.0071449	Poisson	BirdID
fixed	factor(fac.yr)19	-2.0201584	0.6257624	-3.228316	0.0012452	Poisson	BirdID
fixed	factor(fac.yr)20	-3.0763263	1.0387275	-2.961630	0.0030602	Poisson	BirdID
fixed	(Intercept)	0.8753512	0.2035030	4.301417	0.0000170	Poisson	Terr
fixed	factor(fac.yr)2	-0.5142358	0.2957329	-1.738852	0.0820607	Poisson	Terr
fixed	factor(fac.yr)3	-1.3827888	0.3535340	-3.911332	0.0000918	Poisson	Terr
fixed	factor(fac.yr)4	-1.0717626	0.3193005	-3.356596	0.0007891	Poisson	Terr
fixed	factor(fac.yr)5	-0.4117538	0.2799396	-1.470866	0.1413272	Poisson	Terr
fixed	factor(fac.yr)6	-0.7346086	0.2849611	-2.577926	0.0099395	Poisson	Terr
fixed	factor(fac.yr)7	-0.8087523	0.2740757	-2.950836	0.0031692	Poisson	Terr
fixed	factor(fac.yr)8	-0.7239885	0.2751645	-2.631112	0.0085106	Poisson	Terr
fixed	factor(fac.yr)9	-0.8642851	0.2773187	-3.116576	0.0018296	Poisson	Terr
fixed	factor(fac.yr)10	-0.9340926	0.2787784	-3.350664	0.0008062	Poisson	Terr
fixed	factor(fac.yr)11	-1.3409526	0.3007234	-4.459090	0.0000082	Poisson	Terr
fixed	factor(fac.yr)12	-0.8175808	0.2849061	-2.869650	0.0041093	Poisson	Terr
fixed	factor(fac.yr)13	-0.5985971	0.2971202	-2.014663	0.0439400	Poisson	Terr
fixed	factor(fac.yr)14	-0.8907409	0.3419169	-2.605138	0.0091837	Poisson	Terr
fixed	factor(fac.yr)15	-3.5996219	1.0433750	-3.449979	0.0005606	Poisson	Terr
fixed	factor(fac.yr)16	-1.8696341	0.5069804	-3.687784	0.0002262	Poisson	Terr
fixed	factor(fac.yr)17	-1.3856203	0.3993900	-3.469341	0.0005217	Poisson	Terr
fixed	factor(fac.yr)18	-1.0538859	0.4165431	-2.530076	0.0114038	Poisson	Terr
fixed	factor(fac.yr)19	-1.7994673	0.6334583	-2.840704	0.0045014	Poisson	Terr
fixed	factor(fac.yr)20	-2.9332462	1.0464057	-2.803163	0.0050604	Poisson	Terr
fixed	(Intercept)	0.8788120	0.2033113	4.322495	0.0000154	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.5183553	0.2956555	-1.753241	0.0795607	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-1.3852252	0.3533083	-3.920726	0.0000883	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-1.0737590	0.3190530	-3.365457	0.0007642	Poisson	BirdID+Terr

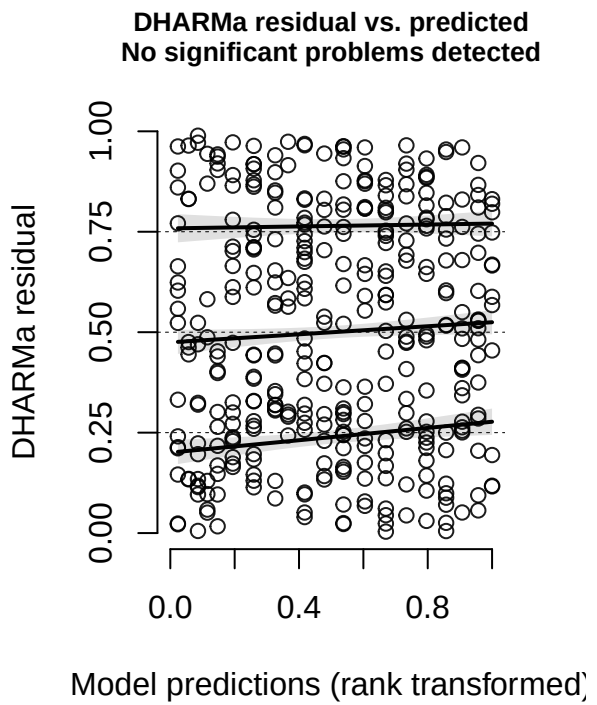
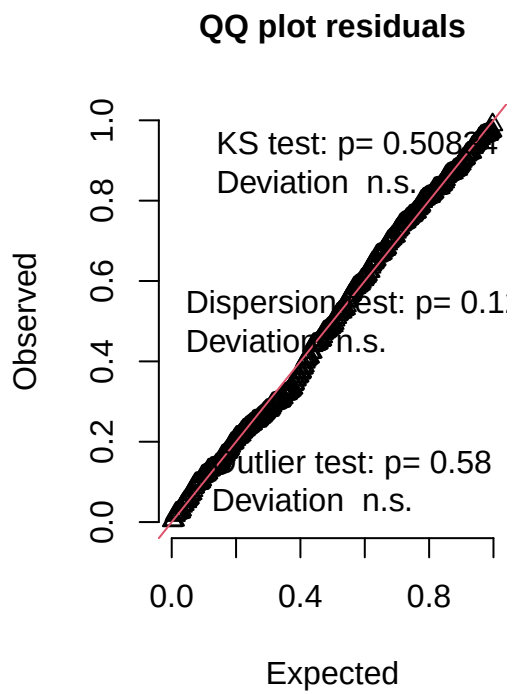
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.4155032	0.2798223	-1.484883	0.1375750	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.7386054	0.2848578	-2.592891	0.0095173	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.8128371	0.2739728	-2.966853	0.0030086	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.7274758	0.2750231	-2.645145	0.0081656	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.8677456	0.2771751	-3.130677	0.0017440	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.9385587	0.2786944	-3.367698	0.0007580	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-1.3450658	0.3006503	-4.473854	0.0000077	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.8208520	0.2847529	-2.882682	0.0039431	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.6018452	0.2969828	-2.026532	0.0427103	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.8942809	0.3418214	-2.616222	0.0088909	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-3.6074187	1.0454983	-3.450430	0.0005597	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-1.8723228	0.5067207	-3.694980	0.0002199	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-1.3913912	0.3996146	-3.481833	0.0004980	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-1.0616444	0.4170925	-2.545345	0.0109170	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.8018008	0.6330980	-2.846006	0.0044271	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-2.9424697	1.0491759	-2.804553	0.0050386	Poisson	BirdID+Terr

Terr only model is performing the best on AIC, BIC, and conditional R^2 .

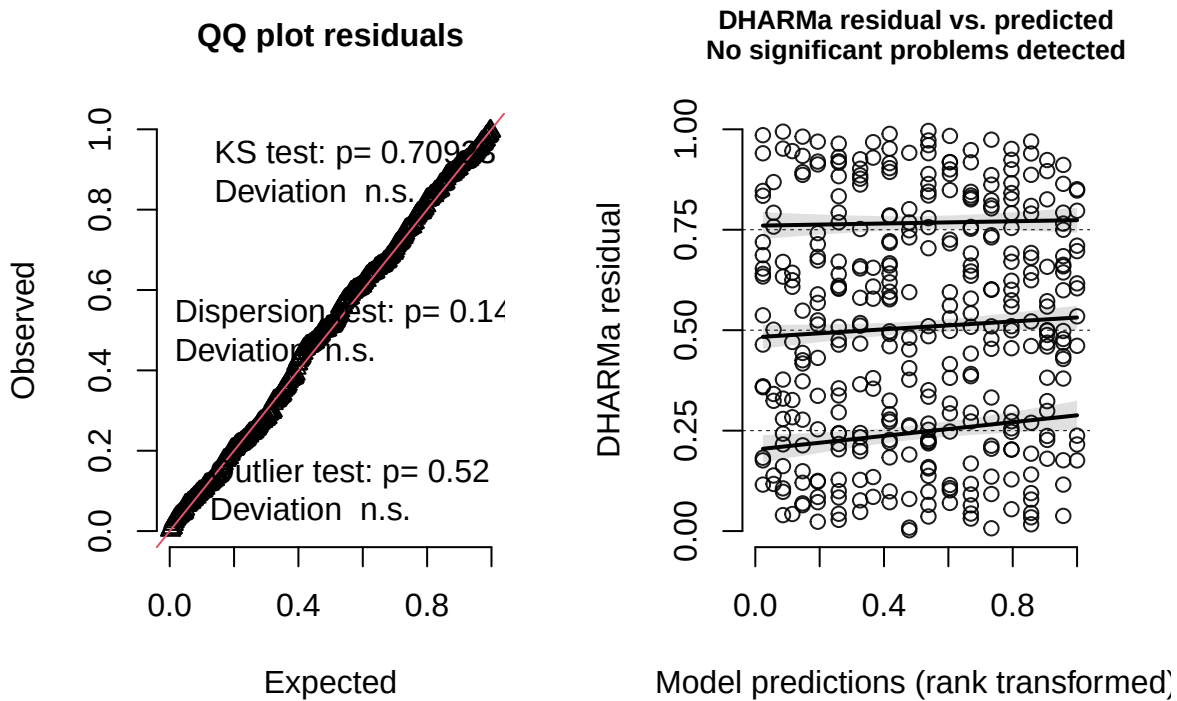
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
366	1	-204.1135	450.2270	532.1823	326.1442	345	0.610	Binomial	BirdID
366	1	-204.1131	452.2262	538.0841	325.9966	344	0.610	Binomial	BirdID+Terr
366	1	-207.7860	457.5720	539.5273	356.9196	345	0.539	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.0236943	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	0.7635592	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.0253415	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.0016764	Binomial	BirdID+Terr

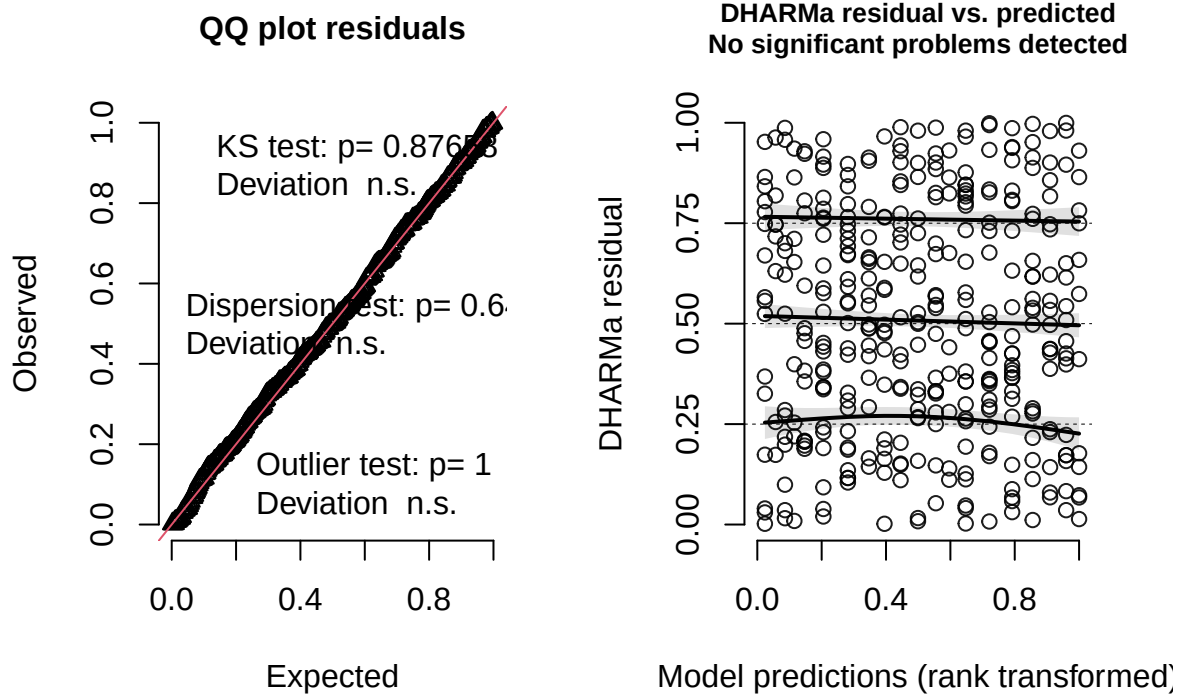
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	16.50268	1141.986	0.0144509	0.9884703	Binomial	BirdID
fixed	factor(fac.yr)2	-14.05721	1141.986	-0.0123094	0.9901787	Binomial	BirdID
fixed	factor(fac.yr)3	-16.61030	1141.986	-0.0145451	0.9883951	Binomial	BirdID
fixed	factor(fac.yr)4	-16.19033	1141.986	-0.0141773	0.9886885	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-14.77309	1141.986	-0.0129363	0.9896786	Binomial	BirdID
fixed	factor(fac.yr)6	-15.54227	1141.986	-0.0136099	0.9891412	Binomial	BirdID
fixed	factor(fac.yr)7	-15.68700	1141.986	-0.0137366	0.9890401	Binomial	BirdID
fixed	factor(fac.yr)8	-15.69514	1141.986	-0.0137437	0.9890344	Binomial	BirdID
fixed	factor(fac.yr)9	-16.77673	1141.986	-0.0146908	0.9882788	Binomial	BirdID
fixed	factor(fac.yr)10	-15.71826	1141.986	-0.0137640	0.9890183	Binomial	BirdID
fixed	factor(fac.yr)11	-16.71286	1141.986	-0.0146349	0.9883235	Binomial	BirdID
fixed	factor(fac.yr)12	-16.13090	1141.986	-0.0141253	0.9887300	Binomial	BirdID
fixed	factor(fac.yr)13	-16.17639	1141.986	-0.0141651	0.9886982	Binomial	BirdID
fixed	factor(fac.yr)14	-16.21635	1141.986	-0.0142001	0.9886703	Binomial	BirdID
fixed	factor(fac.yr)15	-19.61163	1141.986	-0.0171733	0.9862984	Binomial	BirdID
fixed	factor(fac.yr)16	-18.36857	1141.986	-0.0160848	0.9871668	Binomial	BirdID
fixed	factor(fac.yr)17	-17.47772	1141.986	-0.0153047	0.9877891	Binomial	BirdID
fixed	factor(fac.yr)18	-16.05587	1141.986	-0.0140596	0.9887824	Binomial	BirdID
fixed	factor(fac.yr)19	-17.78122	1141.986	-0.0155704	0.9875771	Binomial	BirdID
fixed	factor(fac.yr)20	-18.61561	1141.986	-0.0163011	0.9869942	Binomial	BirdID
fixed	(Intercept)	17.50291	1744.771	0.0100316	0.9919960	Binomial	Terr
fixed	factor(fac.yr)2	-15.15487	1744.771	-0.0086859	0.9930698	Binomial	Terr
fixed	factor(fac.yr)3	-17.66736	1744.771	-0.0101259	0.9919208	Binomial	Terr
fixed	factor(fac.yr)4	-17.04923	1744.771	-0.0097716	0.9922035	Binomial	Terr
fixed	factor(fac.yr)5	-15.59830	1744.771	-0.0089400	0.9928670	Binomial	Terr
fixed	factor(fac.yr)6	-16.62532	1744.771	-0.0095287	0.9923974	Binomial	Terr
fixed	factor(fac.yr)7	-16.63675	1744.771	-0.0095352	0.9923921	Binomial	Terr
fixed	factor(fac.yr)8	-16.68190	1744.771	-0.0095611	0.9923715	Binomial	Terr
fixed	factor(fac.yr)9	-17.51502	1744.771	-0.0100386	0.9919905	Binomial	Terr
fixed	factor(fac.yr)10	-16.65342	1744.771	-0.0095448	0.9923845	Binomial	Terr
fixed	factor(fac.yr)11	-17.49315	1744.771	-0.0100260	0.9920005	Binomial	Terr
fixed	factor(fac.yr)12	-17.04689	1744.771	-0.0097703	0.9922046	Binomial	Terr
fixed	factor(fac.yr)13	-16.94281	1744.771	-0.0097106	0.9922522	Binomial	Terr
fixed	factor(fac.yr)14	-16.97505	1744.771	-0.0097291	0.9922374	Binomial	Terr
fixed	factor(fac.yr)15	-20.27038	1744.772	-0.0116178	0.9907306	Binomial	Terr
fixed	factor(fac.yr)16	-19.29781	1744.771	-0.0110604	0.9911753	Binomial	Terr
fixed	factor(fac.yr)17	-18.49471	1744.771	-0.0106001	0.9915425	Binomial	Terr
fixed	factor(fac.yr)18	-17.28205	1744.771	-0.0099050	0.9920970	Binomial	Terr
fixed	factor(fac.yr)19	-18.46801	1744.771	-0.0105848	0.9915547	Binomial	Terr
fixed	factor(fac.yr)20	-19.43534	1744.772	-0.0111392	0.9911124	Binomial	Terr
fixed	(Intercept)	17.24378	1654.149	0.0104246	0.9916826	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-14.79610	1654.150	-0.0089448	0.9928631	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-17.34845	1654.149	-0.0104878	0.9916321	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-16.93451	1654.149	-0.0102376	0.9918317	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-15.50553	1654.149	-0.0093737	0.9925210	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-16.28299	1654.149	-0.0098437	0.9921460	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-16.43023	1654.149	-0.0099327	0.9920750	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-16.43869	1654.149	-0.0099379	0.9920709	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-17.51690	1654.149	-0.0105897	0.9915508	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-16.46210	1654.149	-0.0099520	0.9920596	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-17.45299	1654.149	-0.0105510	0.9915816	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-16.87079	1654.149	-0.0101991	0.9918625	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-16.91912	1654.149	-0.0102283	0.9918391	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-16.94826	1654.149	-0.0102459	0.9918251	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-20.35671	1654.150	-0.0123064	0.9901811	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-19.10137	1654.150	-0.0115475	0.9907866	Binomial	BirdID+Terr

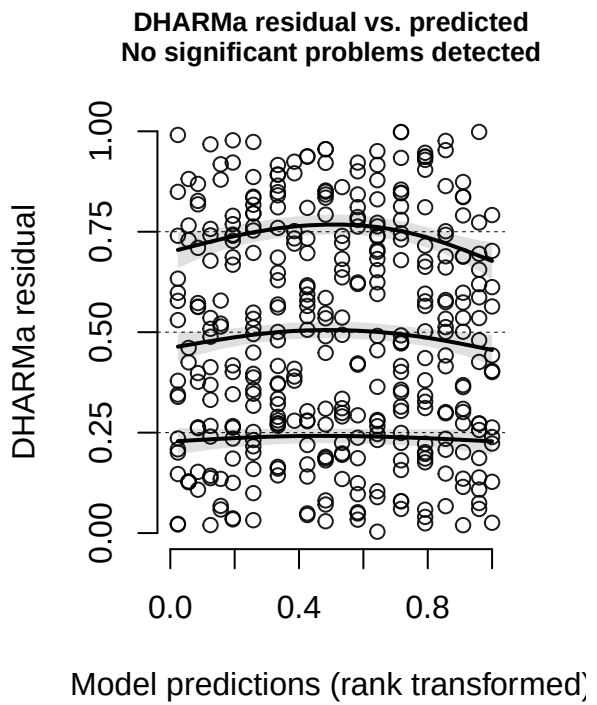
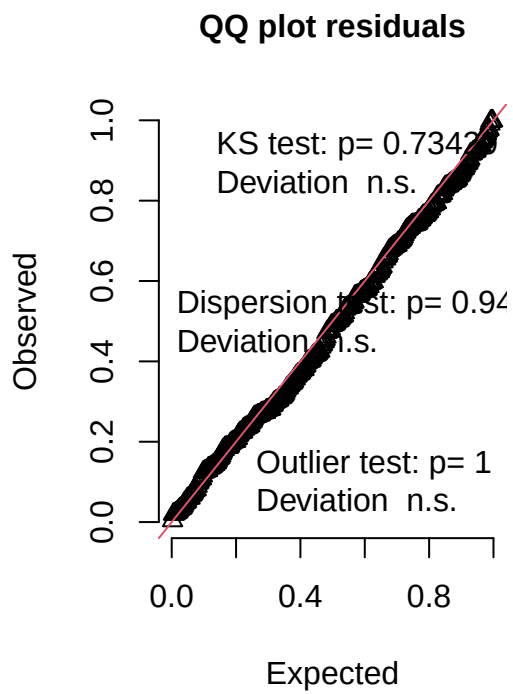
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-18.21388	1654.149	-0.0110110	0.9912146	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-16.78703	1654.150	-0.0101484	0.9919029	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-18.51691	1654.150	-0.0111942	0.9910685	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-19.35667	1654.150	-0.0117019	0.9906635	Binomial	BirdID+Terr

Terr only did not perform as well.

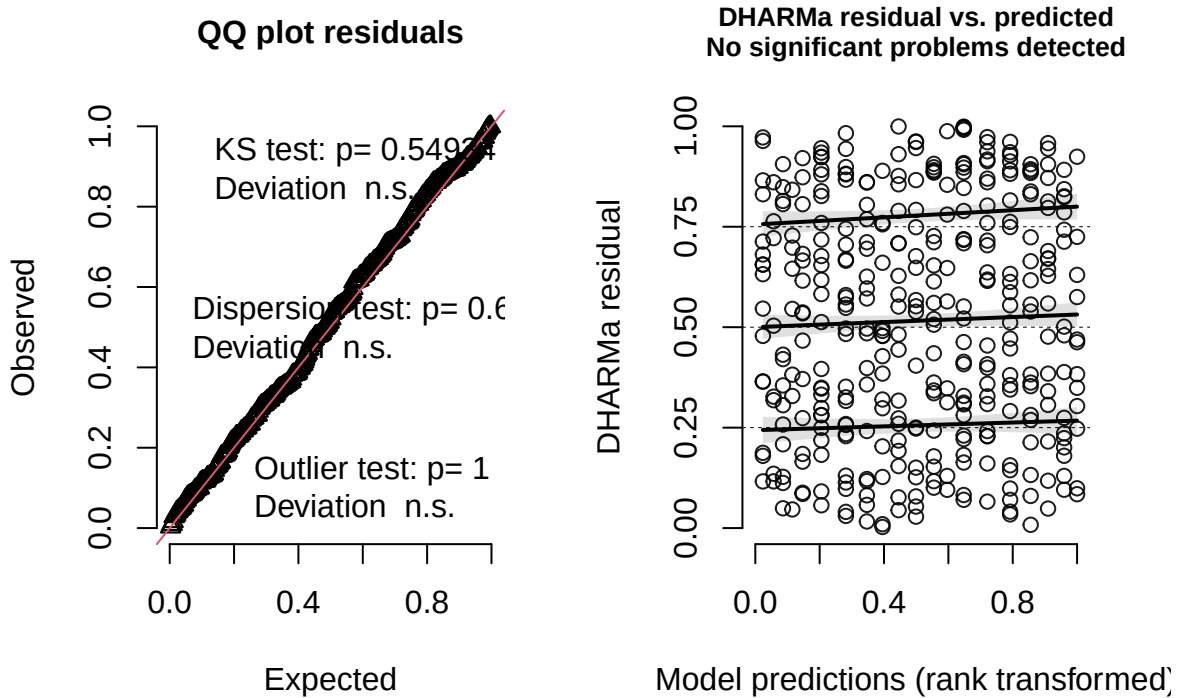
DHARMa residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Probabilites used to generate simulation 48

Simulation 48
 Probability of mate fidelity: 0.356
 Probability of pair dispersal: 0.446
 Probability of single male dispersal: 0.847
 Probability of single female dispersal: 0.859

Simulation 49

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
358	1	-450.4488	944.8977	1030.269	319.0530	336	0.321834	0.0760573	Poisson	BirdID+Terr
358	1	-452.0525	946.1050	1027.596	352.2955	337	0.284736	0.0811222	Poisson	Terr
358	1	-453.8362	949.6724	1031.164	353.1363	337	0.302677	0.0852626	Poisson	BirdID

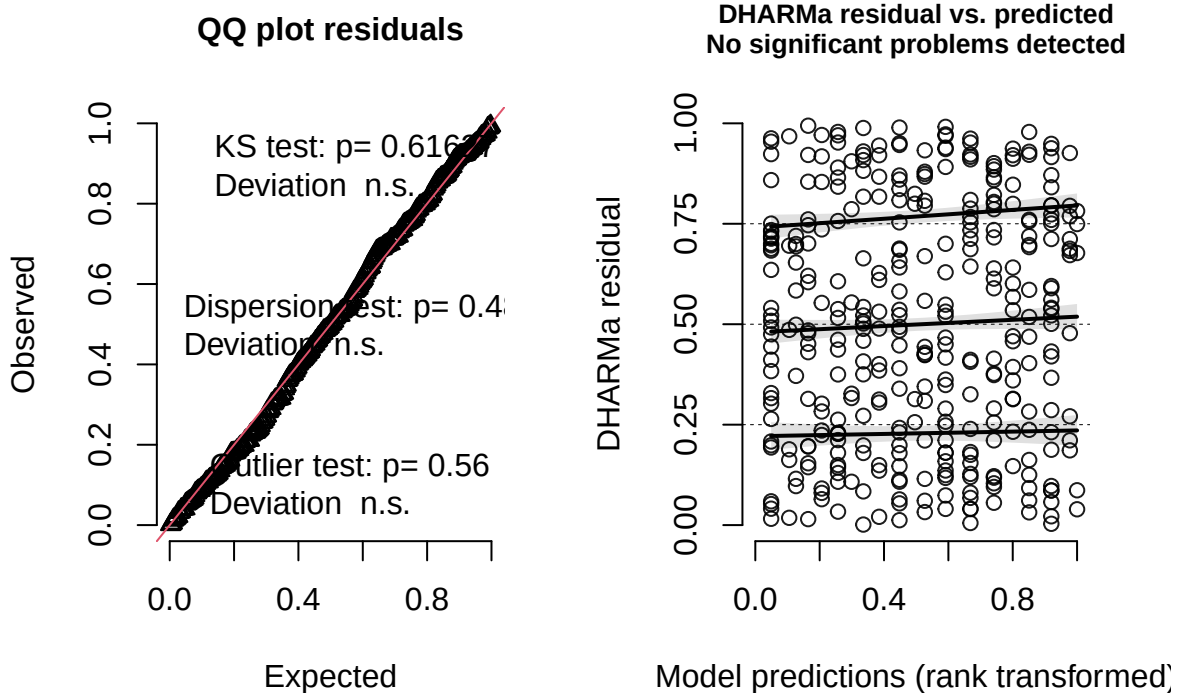
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd____(Intercept)	0.4746382	Poisson	BirdID
ran_pars	Terrs	sd____(Intercept)	0.4521700	Poisson	Terr
ran_pars	birdID	sd____(Intercept)	0.3399435	Poisson	BirdID+Terr
ran_pars	Terrs	sd____(Intercept)	0.3844644	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.2598349	0.2827331	0.9190113	0.3580897	Poisson	BirdID
fixed	factor(fac.yr)2	-0.2615402	0.3583622	-0.7298207	0.4654998	Poisson	BirdID
fixed	factor(fac.yr)3	-0.6638079	0.3843703	-1.7270011	0.0841675	Poisson	BirdID
fixed	factor(fac.yr)4	-0.2803768	0.3457153	-0.8110049	0.4173629	Poisson	BirdID
fixed	factor(fac.yr)5	-0.3791752	0.3520621	-1.0770122	0.2814748	Poisson	BirdID
fixed	factor(fac.yr)6	-0.0625270	0.3324319	-0.1880897	0.8508063	Poisson	BirdID
fixed	factor(fac.yr)7	-1.0113677	0.3776885	-2.6777823	0.0074111	Poisson	BirdID
fixed	factor(fac.yr)8	-0.5877569	0.3560134	-1.6509403	0.0987508	Poisson	BirdID
fixed	factor(fac.yr)9	-0.3252303	0.3526949	-0.9221294	0.3564611	Poisson	BirdID
fixed	factor(fac.yr)10	-0.6928711	0.3845241	-1.8018924	0.0715623	Poisson	BirdID
fixed	factor(fac.yr)11	-0.6604585	0.4202715	-1.5715042	0.1160656	Poisson	BirdID
fixed	factor(fac.yr)12	-0.3847991	0.3829034	-1.0049509	0.3149205	Poisson	BirdID
fixed	factor(fac.yr)13	-0.1464952	0.3634562	-0.4030616	0.6869029	Poisson	BirdID
fixed	factor(fac.yr)14	-0.7572666	0.4189700	-1.8074482	0.0706924	Poisson	BirdID
fixed	factor(fac.yr)15	-0.7022362	0.4554843	-1.5417352	0.1231380	Poisson	BirdID
fixed	factor(fac.yr)16	-0.7599683	0.5190258	-1.4642207	0.1431336	Poisson	BirdID
fixed	factor(fac.yr)17	0.2328577	0.4833247	0.4817831	0.6299600	Poisson	BirdID
fixed	factor(fac.yr)18	-0.9053439	0.6725897	-1.3460568	0.1782842	Poisson	BirdID
fixed	factor(fac.yr)19	-0.6783308	0.6054262	-1.1204186	0.2625354	Poisson	BirdID
fixed	factor(fac.yr)20	-0.4054291	0.6140856	-0.6602160	0.5091152	Poisson	BirdID
fixed	(Intercept)	0.2506881	0.2724800	0.9200236	0.3575604	Poisson	Terr
fixed	factor(fac.yr)2	-0.1795846	0.3641250	-0.4931949	0.6218749	Poisson	Terr
fixed	factor(fac.yr)3	-0.5600711	0.3786020	-1.4793137	0.1390565	Poisson	Terr
fixed	factor(fac.yr)4	-0.2626563	0.3355075	-0.7828628	0.4337077	Poisson	Terr
fixed	factor(fac.yr)5	-0.4092131	0.3338006	-1.2259209	0.2202284	Poisson	Terr
fixed	factor(fac.yr)6	0.0202856	0.3182953	0.0637320	0.9491836	Poisson	Terr
fixed	factor(fac.yr)7	-0.9530101	0.3652783	-2.6089971	0.0090808	Poisson	Terr
fixed	factor(fac.yr)8	-0.5146367	0.3429759	-1.5005043	0.1334838	Poisson	Terr
fixed	factor(fac.yr)9	-0.3635006	0.3387791	-1.0729724	0.2832835	Poisson	Terr
fixed	factor(fac.yr)10	-0.6267769	0.3737818	-1.6768525	0.0935713	Poisson	Terr
fixed	factor(fac.yr)11	-0.5259149	0.4132859	-1.2725210	0.2031881	Poisson	Terr
fixed	factor(fac.yr)12	-0.4694138	0.3671805	-1.2784279	0.2010986	Poisson	Terr
fixed	factor(fac.yr)13	-0.1387398	0.3478198	-0.3988842	0.6899785	Poisson	Terr
fixed	factor(fac.yr)14	-0.7480413	0.4047999	-1.8479287	0.0646127	Poisson	Terr
fixed	factor(fac.yr)15	-0.5315511	0.4499256	-1.1814201	0.2374359	Poisson	Terr
fixed	factor(fac.yr)16	-0.6672944	0.5112655	-1.3051818	0.1918310	Poisson	Terr
fixed	factor(fac.yr)17	0.1780064	0.4797951	0.3710050	0.7106338	Poisson	Terr
fixed	factor(fac.yr)18	-0.8331358	0.6613716	-1.2597092	0.2077743	Poisson	Terr
fixed	factor(fac.yr)19	-0.8869950	0.5930934	-1.4955402	0.1347735	Poisson	Terr
fixed	factor(fac.yr)20	-0.2941137	0.6119603	-0.4806091	0.6307944	Poisson	Terr
fixed	(Intercept)	0.2056429	0.2874818	0.7153252	0.4744081	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.2153178	0.3733946	-0.5766495	0.5641763	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.5996190	0.3913652	-1.5321215	0.1254925	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.2699344	0.3510564	-0.7689202	0.4419407	Poisson	BirdID+Terr

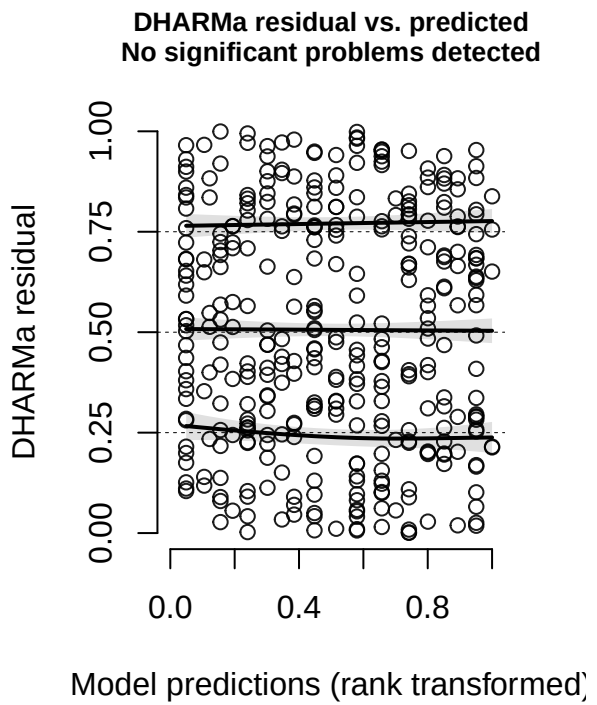
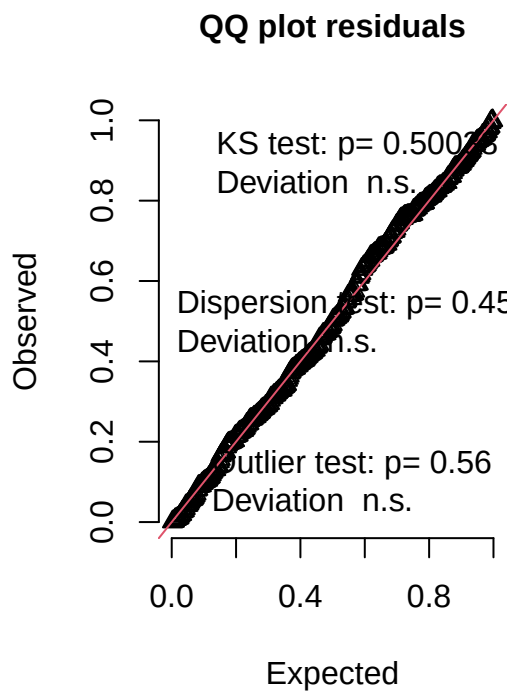
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.3953884	0.3525736	-1.1214352	0.2621027	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	0.0045579	0.3368190	0.0135322	0.9892032	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.9576479	0.3815405	-2.5099510	0.0120748	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.5288860	0.3600746	-1.4688236	0.1418806	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.3433477	0.3563789	-0.9634345	0.3353295	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.6311451	0.3898872	-1.6187889	0.1054927	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.5481672	0.4299664	-1.2749069	0.2023422	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.4277611	0.3851513	-1.1106313	0.2667271	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.1155373	0.3671734	-0.3146668	0.7530146	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.7410093	0.4228071	-1.7525942	0.0796717	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.5725772	0.4659762	-1.2287691	0.2191584	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.6980112	0.5273067	-1.3237290	0.1855931	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	0.2306827	0.5006120	0.4608014	0.6449411	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.7911501	0.6805280	-1.1625534	0.2450108	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-0.8031811	0.6145171	-1.3070120	0.1912087	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.3390050	0.6319858	-0.5364124	0.5916736	Poisson	BirdID+Terr

Terr only model is performing best only on AIC

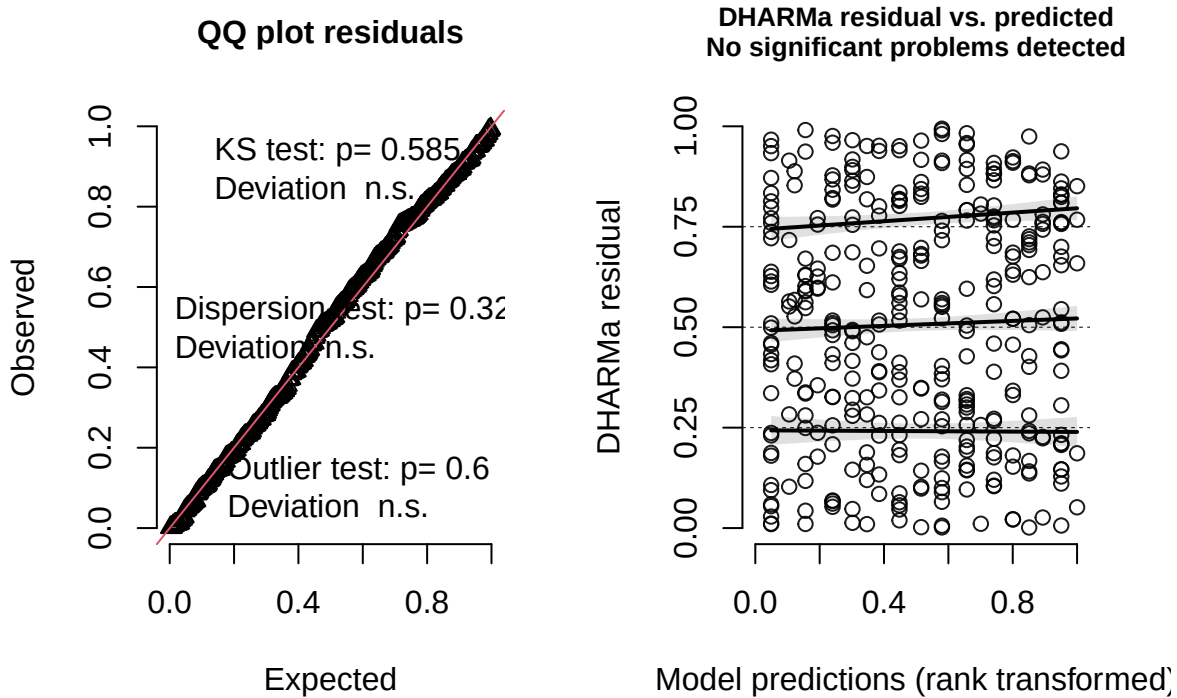
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
358	1	-226.4816	496.9632	582.3349	302.2250	336	0.689	Binomial	BirdID+Terr
358	1	-228.0657	498.1314	579.6226	345.6806	337	0.559	Binomial	BirdID
358	1	-232.3518	506.7036	588.1948	385.7574	337	0.494	Binomial	Terr

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.1896685	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	0.8646813	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.1796240	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.7700712	Binomial	BirdID+Terr

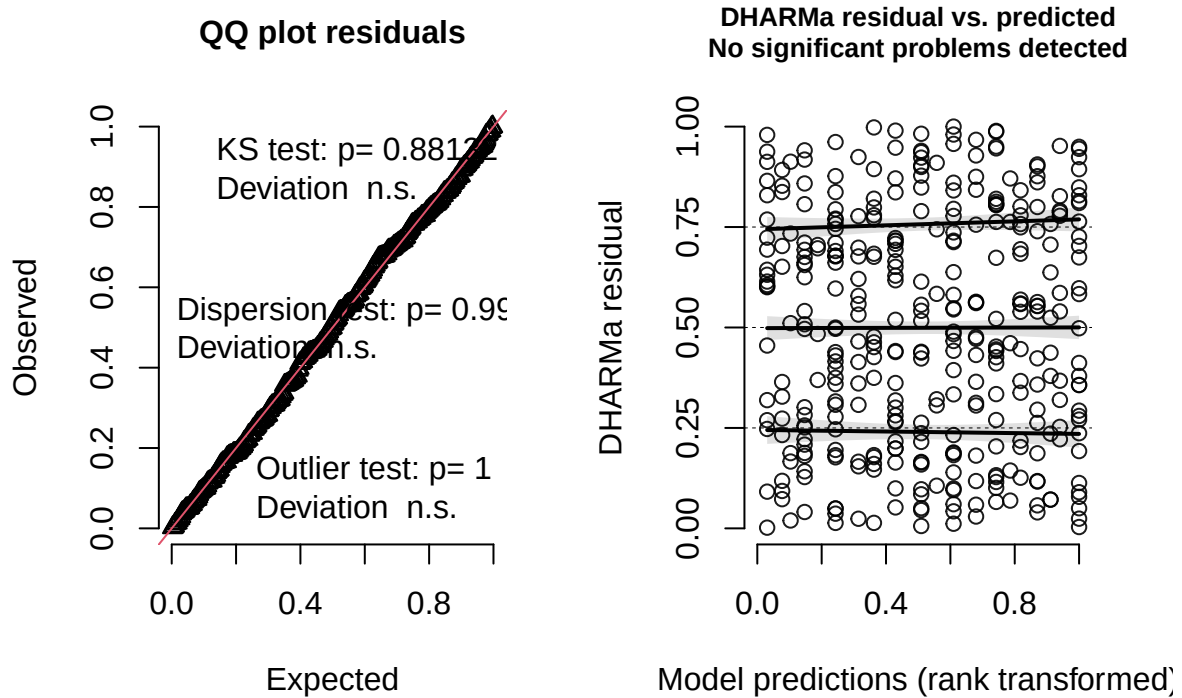
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.9585610	0.7943537	1.2067180	0.2275408	Binomial	BirdID
fixed	factor(fac.yr)2	-0.2079156	0.9803434	-0.2120845	0.8320411	Binomial	BirdID
fixed	factor(fac.yr)3	-1.6930413	0.9384065	-1.8041662	0.0712053	Binomial	BirdID
fixed	factor(fac.yr)4	-0.3407050	0.9363266	-0.3638741	0.7159520	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-1.1304931	0.9105632	-1.2415317	0.2144094	Binomial	BirdID
fixed	factor(fac.yr)6	0.1017812	0.9212282	0.1104842	0.9120254	Binomial	BirdID
fixed	factor(fac.yr)7	-1.2736983	0.8932965	-1.4258404	0.1539144	Binomial	BirdID
fixed	factor(fac.yr)8	-0.6158624	0.9055990	-0.6800608	0.4964659	Binomial	BirdID
fixed	factor(fac.yr)9	-0.8526117	0.9149862	-0.9318301	0.3514243	Binomial	BirdID
fixed	factor(fac.yr)10	-1.4688321	0.9390456	-1.5641754	0.1177764	Binomial	BirdID
fixed	factor(fac.yr)11	-1.2725669	1.0097733	-1.2602501	0.2075792	Binomial	BirdID
fixed	factor(fac.yr)12	-0.4069831	0.9749893	-0.4174231	0.6763689	Binomial	BirdID
fixed	factor(fac.yr)13	-0.1671030	0.9727935	-0.1717764	0.8636133	Binomial	BirdID
fixed	factor(fac.yr)14	-1.1564320	0.9747356	-1.1864058	0.2354621	Binomial	BirdID
fixed	factor(fac.yr)15	-1.4989067	1.0556580	-1.4198790	0.1556429	Binomial	BirdID
fixed	factor(fac.yr)16	-0.0920819	1.1531556	-0.0798521	0.9363549	Binomial	BirdID
fixed	factor(fac.yr)17	-0.2357655	1.3569187	-0.1737506	0.8620614	Binomial	BirdID
fixed	factor(fac.yr)18	-1.4689249	1.2854896	-1.1426968	0.2531645	Binomial	BirdID
fixed	factor(fac.yr)19	-0.8058614	1.2571472	-0.6410239	0.5215072	Binomial	BirdID
fixed	factor(fac.yr)20	-1.3146611	1.3953771	-0.9421547	0.3461134	Binomial	BirdID
fixed	(Intercept)	0.8790959	0.7369087	1.1929508	0.2328886	Binomial	Terr
fixed	factor(fac.yr)2	0.0983049	0.9321412	0.1054614	0.9160097	Binomial	Terr
fixed	factor(fac.yr)3	-1.1476988	0.8781112	-1.3070085	0.1912098	Binomial	Terr
fixed	factor(fac.yr)4	-0.0359392	0.8686291	-0.0413746	0.9669973	Binomial	Terr
fixed	factor(fac.yr)5	-1.0385659	0.8351720	-1.2435354	0.2136706	Binomial	Terr
fixed	factor(fac.yr)6	0.2174980	0.8521019	0.2552488	0.7985309	Binomial	Terr
fixed	factor(fac.yr)7	-1.0921949	0.8244492	-1.3247571	0.1852517	Binomial	Terr
fixed	factor(fac.yr)8	-0.4737230	0.8335745	-0.5683032	0.5698291	Binomial	Terr
fixed	factor(fac.yr)9	-0.9596823	0.8448939	-1.1358614	0.2560146	Binomial	Terr
fixed	factor(fac.yr)10	-1.3853263	0.8699852	-1.5923562	0.1113047	Binomial	Terr
fixed	factor(fac.yr)11	-0.9855047	0.9278095	-1.0621843	0.2881520	Binomial	Terr
fixed	factor(fac.yr)12	-0.4961578	0.8978062	-0.5526335	0.5805144	Binomial	Terr
fixed	factor(fac.yr)13	-0.2099048	0.8983786	-0.2336485	0.8152579	Binomial	Terr
fixed	factor(fac.yr)14	-1.2377029	0.8937336	-1.3848677	0.1660929	Binomial	Terr
fixed	factor(fac.yr)15	-1.1023979	0.9793157	-1.1256819	0.2603002	Binomial	Terr
fixed	factor(fac.yr)16	0.0654272	1.0630332	0.0615476	0.9509231	Binomial	Terr
fixed	factor(fac.yr)17	-0.3043340	1.2340159	-0.2466208	0.8052017	Binomial	Terr
fixed	factor(fac.yr)18	-1.3237233	1.1770925	-1.1245704	0.2607711	Binomial	Terr
fixed	factor(fac.yr)19	-1.0344139	1.1584880	-0.8928999	0.3719108	Binomial	Terr
fixed	factor(fac.yr)20	-0.9797721	1.3146185	-0.7452901	0.4560964	Binomial	Terr
fixed	(Intercept)	0.8867230	0.8405986	1.0548708	0.2914844	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-0.0455134	1.0371566	-0.0438828	0.9649978	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-1.6862733	0.9905470	-1.7023657	0.0886868	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-0.2160039	0.9854920	-0.2191838	0.8265069	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-1.1207128	0.9558543	-1.1724724	0.2410075	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	0.3124198	0.9713835	0.3216235	0.7477379	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-1.2142943	0.9446547	-1.2854372	0.1986395	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-0.4936214	0.9546568	-0.5170668	0.6051095	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-0.8462491	0.9645178	-0.8773805	0.3802800	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-1.4293527	0.9965848	-1.4342510	0.1515006	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-1.1083204	1.0652098	-1.0404714	0.2981209	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-0.3555945	1.0285847	-0.3457124	0.7295589	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	0.0025509	1.0252932	0.0024880	0.9980149	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-1.2139603	1.0232877	-1.1863334	0.2354907	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-1.2826795	1.1198567	-1.1453961	0.2520451	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	0.1016999	1.2123684	0.0838853	0.9331476	Binomial	BirdID+Terr

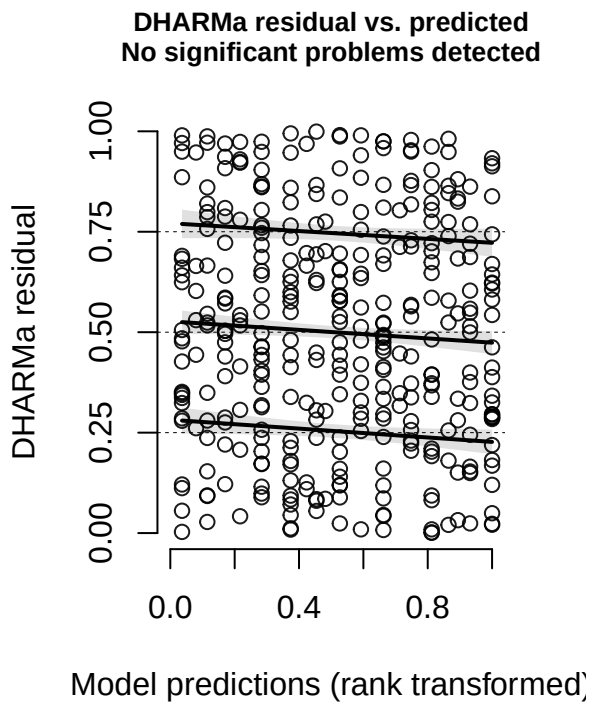
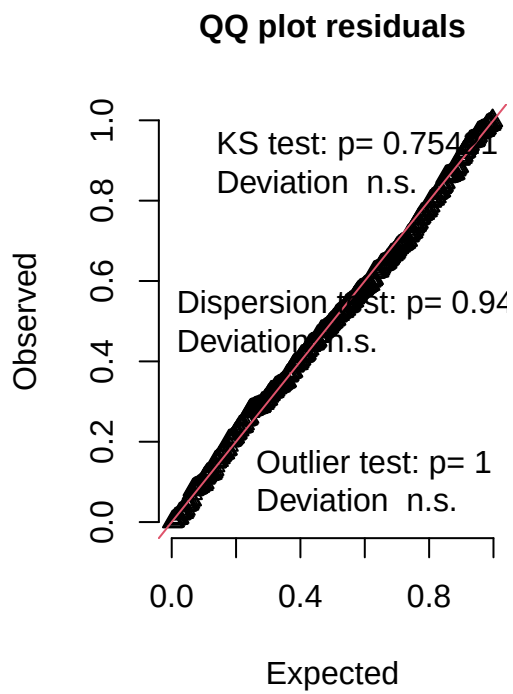
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-0.0784487	1.4213778	-0.0551920	0.9559855	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-1.2102104	1.3354759	-0.9062016	0.3648291	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-0.8208185	1.3405081	-0.6123189	0.5403268	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-1.2090836	1.4909773	-0.8109336	0.4174038	Binomial	BirdID+Terr

Similar to the other binomial models.

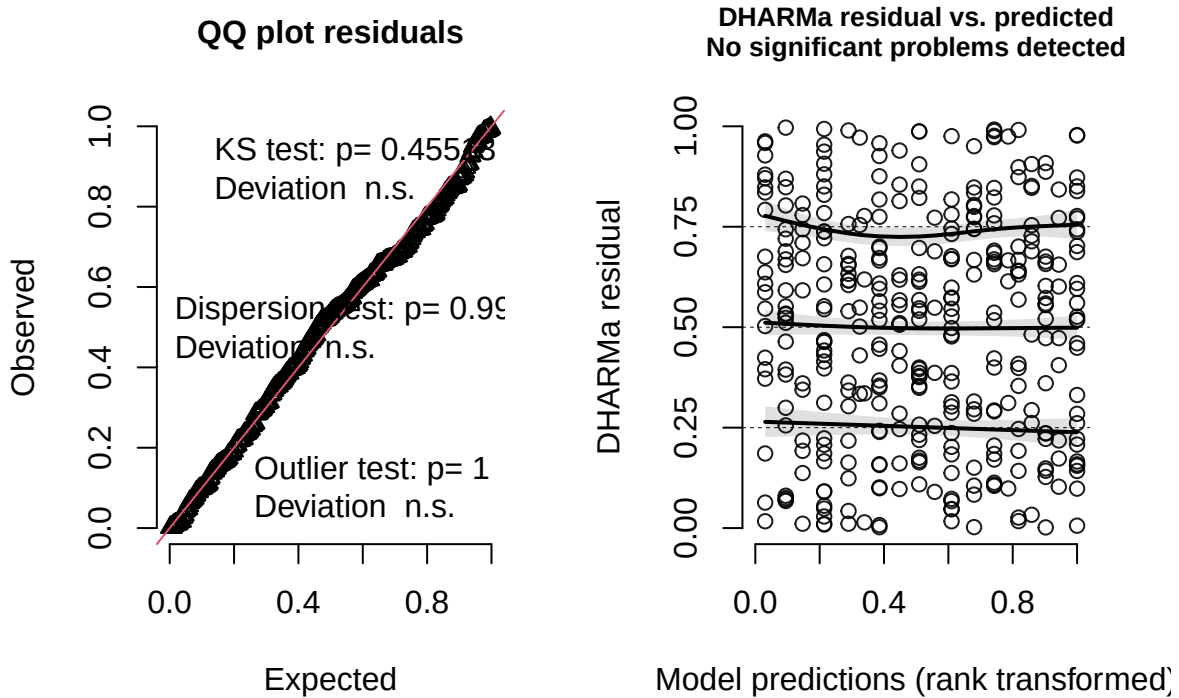
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Probabilities used to generate simulation 49

Simulation 49
 Probability of mate fidelity: 0.317
 Probability of pair dispersal: 0.632
 Probability of single male dispersal: 0.885
 Probability of single female dispersal: 0.294

Simulation 50

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
340	1	-439.6553	921.3106	1001.718	284.7572	319	0.4488573	0.1576052	Poisson	BirdID
340	1	-439.5776	923.1552	1007.392	277.6613	318	0.4504477	0.1565491	Poisson	BirdID+Terr
340	1	-446.1381	934.2761	1014.684	319.0294	319	0.3990395	0.1624344	Poisson	Terr

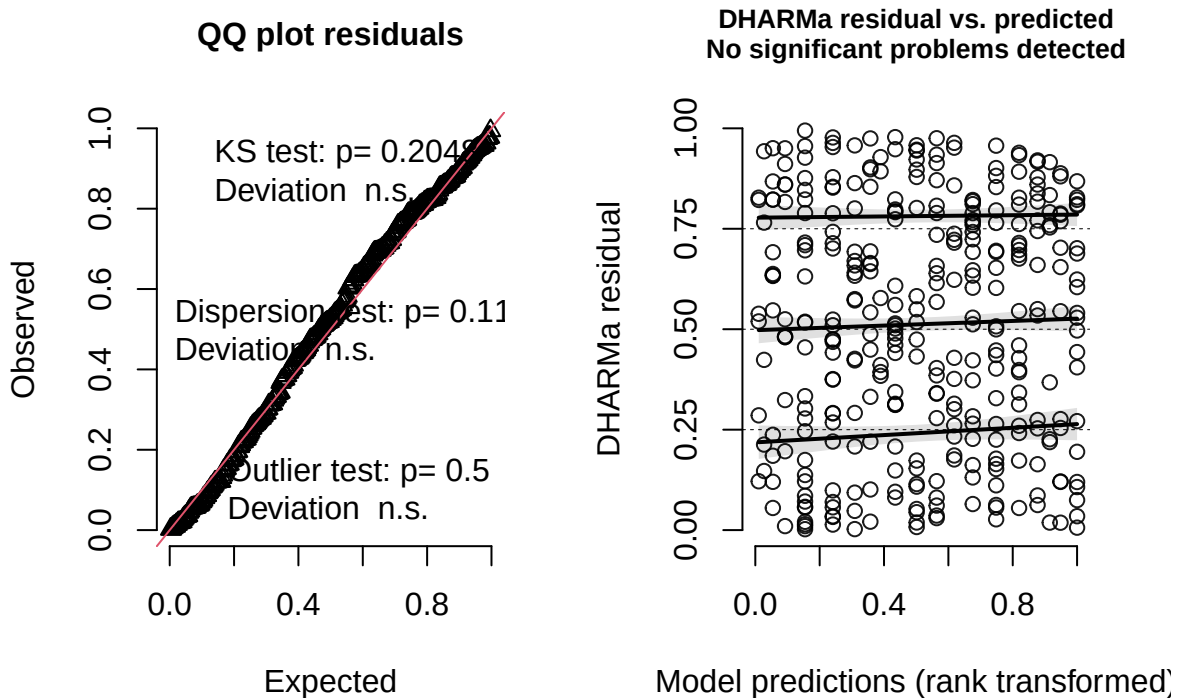
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.5887812	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.4953425	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.5629065	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.1681396	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.4903424	0.2439575	2.0099499	0.0444365	Poisson	BirdID
fixed	factor(fac.yr)2	-0.9545858	0.3307111	-2.8864647	0.0038960	Poisson	BirdID
fixed	factor(fac.yr)3	-0.9615238	0.3196696	-3.0078678	0.0026309	Poisson	BirdID
fixed	factor(fac.yr)4	-0.4352262	0.3045310	-1.4291686	0.1529558	Poisson	BirdID
fixed	factor(fac.yr)5	0.0701850	0.3002990	0.2337171	0.8152046	Poisson	BirdID
fixed	factor(fac.yr)6	-0.4513421	0.3229317	-1.3976396	0.1622213	Poisson	BirdID
fixed	factor(fac.yr)7	-0.7590580	0.3402195	-2.2310830	0.0256756	Poisson	BirdID
fixed	factor(fac.yr)8	-0.7308918	0.3490781	-2.0937773	0.0362798	Poisson	BirdID
fixed	factor(fac.yr)9	-0.3791995	0.3209522	-1.1814830	0.2374109	Poisson	BirdID
fixed	factor(fac.yr)10	-0.5050341	0.3565036	-1.4166309	0.1565909	Poisson	BirdID
fixed	factor(fac.yr)11	-0.9228771	0.3906871	-2.3621898	0.0181673	Poisson	BirdID
fixed	factor(fac.yr)12	-0.6588487	0.3526708	-1.8681692	0.0617385	Poisson	BirdID
fixed	factor(fac.yr)13	-0.2468714	0.3357388	-0.7353077	0.4621521	Poisson	BirdID
fixed	factor(fac.yr)14	-0.7950592	0.3877160	-2.0506226	0.0403037	Poisson	BirdID
fixed	factor(fac.yr)15	-1.5142965	0.5002381	-3.0271517	0.0024687	Poisson	BirdID
fixed	factor(fac.yr)16	-1.3651193	0.5026787	-2.7156895	0.0066138	Poisson	BirdID
fixed	factor(fac.yr)17	-0.1219905	0.3891769	-0.3134577	0.7539330	Poisson	BirdID
fixed	factor(fac.yr)18	-0.7614575	0.5135713	-1.4826717	0.1381617	Poisson	BirdID
fixed	factor(fac.yr)19	-1.7371955	0.7821952	-2.2209233	0.0263562	Poisson	BirdID
fixed	factor(fac.yr)20	-1.8827885	0.7855124	-2.3968922	0.0165348	Poisson	BirdID
fixed	(Intercept)	0.5965978	0.2325024	2.5659859	0.0102883	Poisson	Terr
fixed	factor(fac.yr)2	-1.0604868	0.3306135	-3.2076334	0.0013383	Poisson	Terr
fixed	factor(fac.yr)3	-1.0200558	0.3292983	-3.0976647	0.0019505	Poisson	Terr
fixed	factor(fac.yr)4	-0.6213876	0.3095532	-2.0073690	0.0447104	Poisson	Terr
fixed	factor(fac.yr)5	-0.1182674	0.2975771	-0.3974344	0.6910471	Poisson	Terr
fixed	factor(fac.yr)6	-0.4790437	0.3095199	-1.5476989	0.1216948	Poisson	Terr
fixed	factor(fac.yr)7	-0.6788937	0.3262150	-2.0811237	0.0374226	Poisson	Terr
fixed	factor(fac.yr)8	-0.6873402	0.3370618	-2.0392115	0.0414289	Poisson	Terr
fixed	factor(fac.yr)9	-0.2170802	0.3079090	-0.7050139	0.4808016	Poisson	Terr
fixed	factor(fac.yr)10	-0.2723721	0.3447352	-0.7900907	0.4294748	Poisson	Terr
fixed	factor(fac.yr)11	-0.8567393	0.3741800	-2.2896445	0.0220419	Poisson	Terr
fixed	factor(fac.yr)12	-0.6420449	0.3310216	-1.9395859	0.0524300	Poisson	Terr
fixed	factor(fac.yr)13	-0.2358302	0.3037901	-0.7762931	0.4375759	Poisson	Terr
fixed	factor(fac.yr)14	-0.7552302	0.3630835	-2.0800454	0.0375214	Poisson	Terr
fixed	factor(fac.yr)15	-1.4222805	0.4831752	-2.9436122	0.0032441	Poisson	Terr
fixed	factor(fac.yr)16	-1.3285622	0.4882416	-2.7211166	0.0065062	Poisson	Terr
fixed	factor(fac.yr)17	0.0100159	0.3625501	0.0276264	0.9779602	Poisson	Terr
fixed	factor(fac.yr)18	-0.5865766	0.4962632	-1.1819869	0.2372109	Poisson	Terr
fixed	factor(fac.yr)19	-1.4654575	0.7608322	-1.9261245	0.0540888	Poisson	Terr
fixed	factor(fac.yr)20	-1.6375012	0.7763286	-2.1092887	0.0349197	Poisson	Terr
fixed	(Intercept)	0.4996109	0.2462305	2.0290369	0.0424545	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.9613093	0.3322623	-2.8932239	0.0038131	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.9669591	0.3242120	-2.9824902	0.0028591	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.4599906	0.3093447	-1.4869837	0.1370191	Poisson	BirdID+Terr

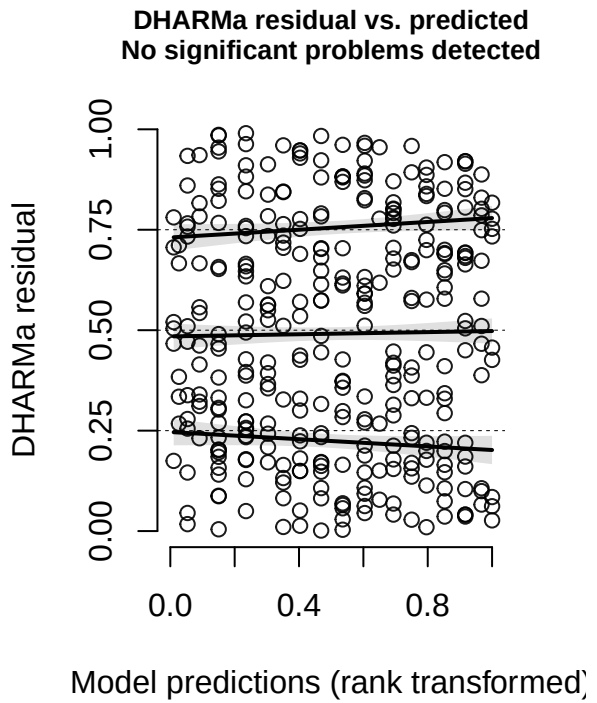
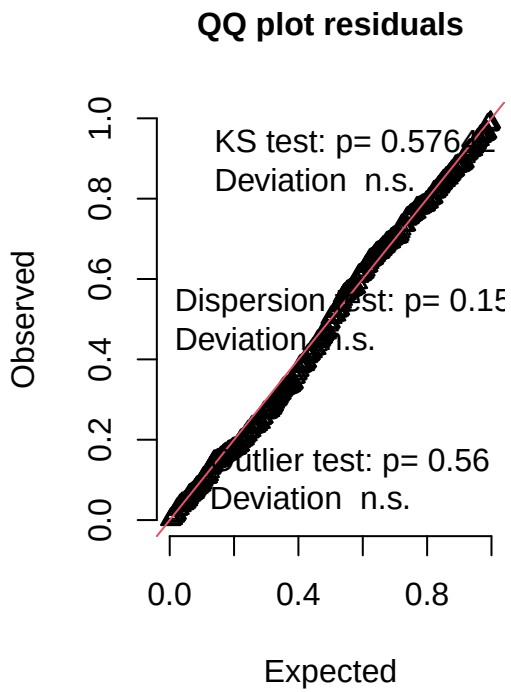
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.0530036	0.3049253	0.1738248	0.8620032	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	-0.4592220	0.3255501	-1.4106030	0.1583617	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.7568416	0.3429585	-2.2068023	0.0273279	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.7395488	0.3529581	-2.0952877	0.0361454	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	-0.3666776	0.3243679	-1.1304376	0.2582919	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.4967420	0.3598010	-1.3806022	0.1674013	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.9200156	0.3928621	-2.3418282	0.0191895	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.6553829	0.3547908	-1.8472373	0.0647128	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	-0.2612280	0.3367615	-0.7757061	0.4379225	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.7972632	0.3895102	-2.0468351	0.0406743	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-1.5096403	0.5029119	-3.0017988	0.0026839	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-1.3576090	0.5051709	-2.6874254	0.0072005	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.1043642	0.3931999	-0.2654228	0.7906838	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.7580446	0.5197141	-1.4585800	0.1446807	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-1.7240027	0.7838866	-2.1993013	0.0278565	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-1.8551626	0.7889333	-2.3514823	0.0186988	Poisson	BirdID+Terr

Not a great day for the territory only model.

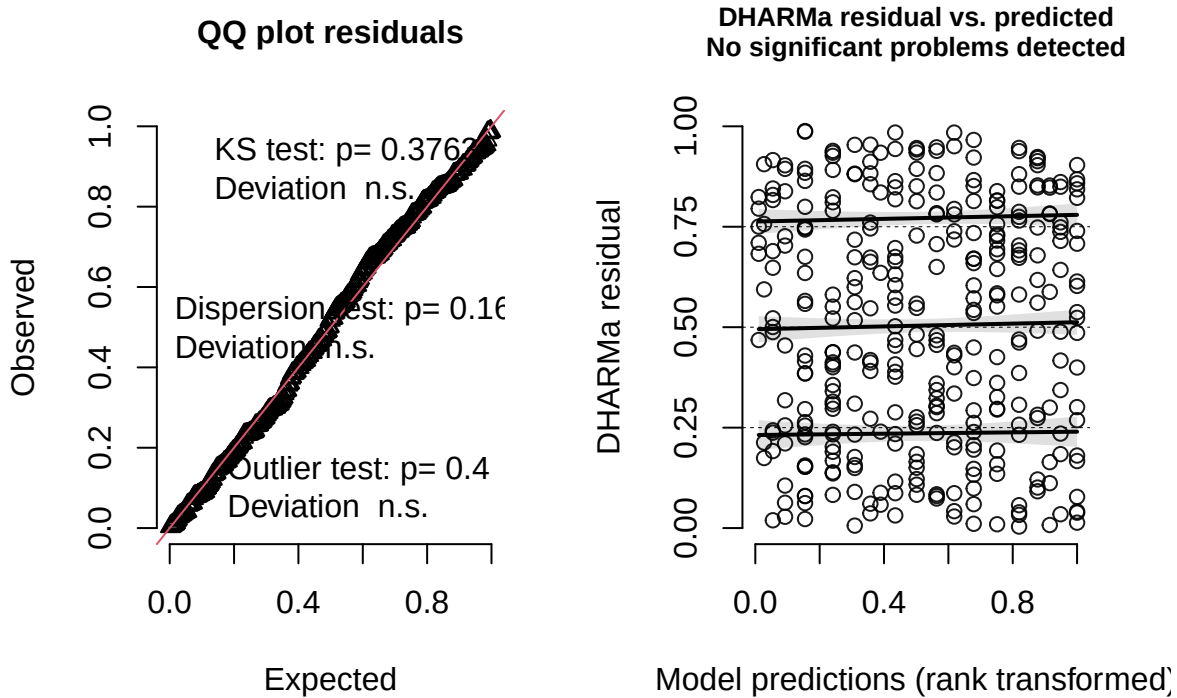
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
340	1	-193.8402	431.6804	515.9172	169.4439	318	0.886	Binomial	BirdID+Terr
340	1	-200.1624	442.3247	522.7326	255.0568	319	0.717	Binomial	Terr
340	1	-200.3884	442.7769	523.1847	251.1475	319	0.710	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	1.781454	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.791110	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	1.765294	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.763852	Binomial	BirdID+Terr

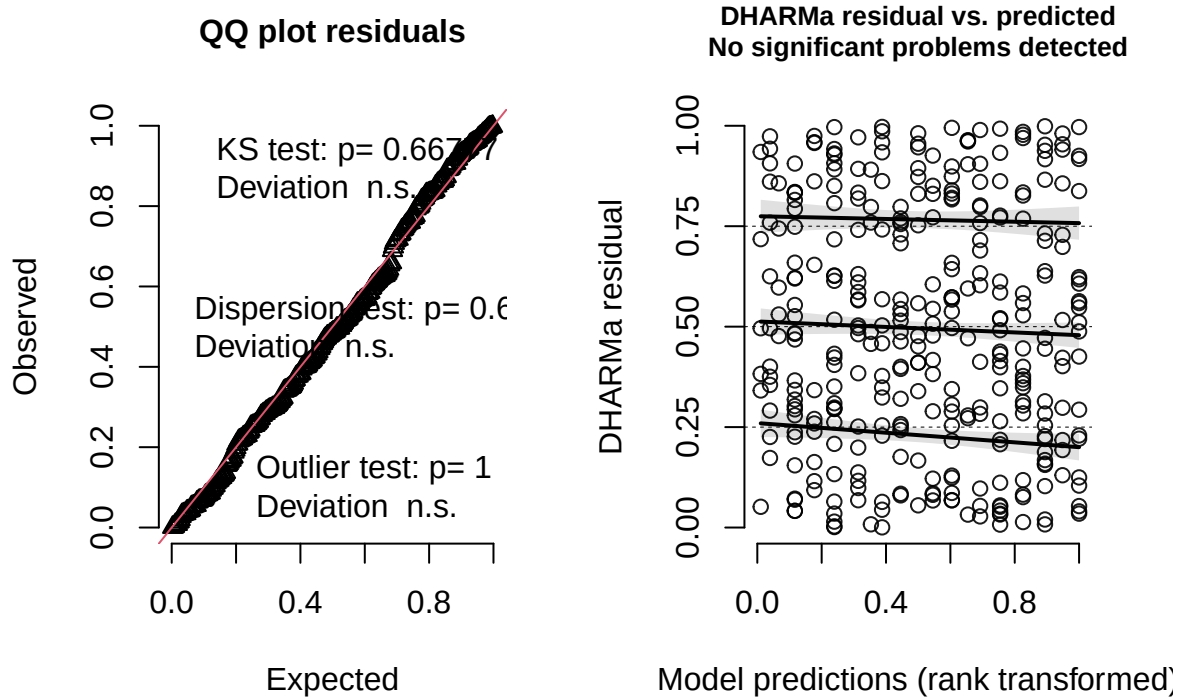
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	1.6691036	0.9054972	1.8433007	0.0652851	Binomial	BirdID
fixed	factor(fac.yr)2	-2.1529513	0.9991182	-2.1548514	0.0311735	Binomial	BirdID
fixed	factor(fac.yr)3	-1.8563871	0.9869827	-1.8808710	0.0599895	Binomial	BirdID
fixed	factor(fac.yr)4	-0.7083070	1.0303314	-0.6874555	0.4917958	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.1427789	1.0914984	0.1308100	0.8959256	Binomial	BirdID
fixed	factor(fac.yr)6	-0.0872109	1.0689996	-0.0815818	0.9349793	Binomial	BirdID
fixed	factor(fac.yr)7	-0.9921502	1.0655913	-0.9310795	0.3518125	Binomial	BirdID
fixed	factor(fac.yr)8	-1.8490156	1.0905333	-1.6955151	0.0899778	Binomial	BirdID
fixed	factor(fac.yr)9	-0.5687952	1.0937083	-0.5200612	0.6030209	Binomial	BirdID
fixed	factor(fac.yr)10	-1.0912026	1.1545340	-0.9451454	0.3445846	Binomial	BirdID
fixed	factor(fac.yr)11	-1.4804278	1.1132557	-1.3298183	0.1835781	Binomial	BirdID
fixed	factor(fac.yr)12	-1.3563744	1.0845677	-1.2506130	0.2110757	Binomial	BirdID
fixed	factor(fac.yr)13	-0.7271220	1.1885693	-0.6117624	0.5406950	Binomial	BirdID
fixed	factor(fac.yr)14	-1.2982749	1.1511559	-1.1278011	0.2594039	Binomial	BirdID
fixed	factor(fac.yr)15	-2.7617967	1.1839190	-2.3327581	0.0196608	Binomial	BirdID
fixed	factor(fac.yr)16	-2.1171671	1.1900045	-1.7791253	0.0752192	Binomial	BirdID
fixed	factor(fac.yr)17	-0.8130871	1.3002234	-0.6253442	0.5317452	Binomial	BirdID
fixed	factor(fac.yr)18	-1.7452344	1.4273709	-1.2226915	0.2214462	Binomial	BirdID
fixed	factor(fac.yr)19	-2.7345296	1.5118035	-1.8087864	0.0704842	Binomial	BirdID
fixed	factor(fac.yr)20	-3.0271478	1.4764743	-2.0502543	0.0403396	Binomial	BirdID
fixed	(Intercept)	1.6598441	0.9051531	1.8337716	0.0666879	Binomial	Terr
fixed	factor(fac.yr)2	-2.1940102	1.0129328	-2.1659978	0.0303114	Binomial	Terr
fixed	factor(fac.yr)3	-1.8445904	1.0240555	-1.8012602	0.0716619	Binomial	Terr
fixed	factor(fac.yr)4	-0.7553750	1.0429439	-0.7242720	0.4688988	Binomial	Terr
fixed	factor(fac.yr)5	0.2415668	1.0901171	0.2215971	0.8246275	Binomial	Terr
fixed	factor(fac.yr)6	-0.0036655	1.0842776	-0.0033806	0.9973027	Binomial	Terr
fixed	factor(fac.yr)7	-0.5026770	1.0617805	-0.4734284	0.6359076	Binomial	Terr
fixed	factor(fac.yr)8	-1.4913539	1.0855936	-1.3737681	0.1695137	Binomial	Terr
fixed	factor(fac.yr)9	0.0641535	1.0749511	0.0596804	0.9524102	Binomial	Terr
fixed	factor(fac.yr)10	-0.0465992	1.1268378	-0.0413539	0.9670137	Binomial	Terr
fixed	factor(fac.yr)11	-0.9931347	1.0961196	-0.9060459	0.3649116	Binomial	Terr
fixed	factor(fac.yr)12	-1.0375928	1.0589278	-0.9798523	0.3271590	Binomial	Terr
fixed	factor(fac.yr)13	-0.6438318	1.1275725	-0.5709893	0.5680069	Binomial	Terr
fixed	factor(fac.yr)14	-0.6866092	1.0837592	-0.6335441	0.5263784	Binomial	Terr
fixed	factor(fac.yr)15	-1.8021109	1.1593257	-1.5544475	0.1200777	Binomial	Terr
fixed	factor(fac.yr)16	-1.5691940	1.1609152	-1.3516870	0.1764755	Binomial	Terr
fixed	factor(fac.yr)17	0.1280603	1.2176287	0.1051719	0.9162394	Binomial	Terr
fixed	factor(fac.yr)18	-1.3076848	1.3760275	-0.9503333	0.3419429	Binomial	Terr
fixed	factor(fac.yr)19	-1.1148716	1.4410691	-0.7736420	0.4391425	Binomial	Terr
fixed	factor(fac.yr)20	-2.2536984	1.5110039	-1.4915239	0.1358240	Binomial	Terr
fixed	(Intercept)	2.0842928	1.1779428	1.7694347	0.0768214	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-2.6345524	1.2744775	-2.0671627	0.0387188	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-2.2006477	1.2804953	-1.7185910	0.0856889	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-0.7010526	1.3197073	-0.5312183	0.5952675	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	0.5000902	1.3852552	0.3610094	0.7180924	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	0.0815752	1.3700446	0.0595420	0.9525204	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-0.7444406	1.3661072	-0.5449357	0.5857977	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-2.0365013	1.4019932	-1.4525757	0.1463416	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-0.3641715	1.3787715	-0.2641275	0.7916816	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-0.6828010	1.4486996	-0.4713200	0.6374123	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-1.5405457	1.4160374	-1.0879273	0.2766272	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-1.4256129	1.3706323	-1.0401133	0.2982873	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-1.2458853	1.4681389	-0.8486154	0.3960953	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-1.3819797	1.4134334	-0.9777466	0.3281997	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-2.7065245	1.4947955	-1.8106320	0.0701978	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-2.1504186	1.4722030	-1.4606808	0.1441031	Binomial	BirdID+Terr

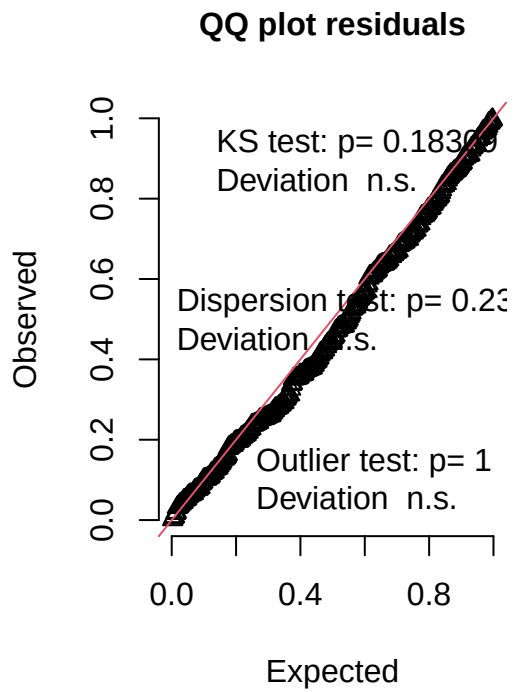
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-0.2677239	1.5824679	-0.1691812	0.8656541	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-1.9689920	1.7892829	-1.1004364	0.2711420	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-2.4582908	1.9215429	-1.2793317	0.2007803	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-3.5192035	1.9790851	-1.7781971	0.0753715	Binomial	BirdID+Terr

The model fit with both REs is best optimizing selection criteria.

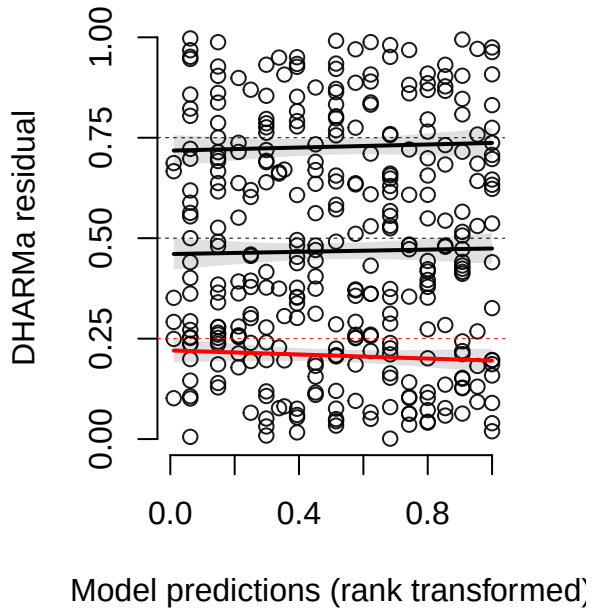
DHARMA residual



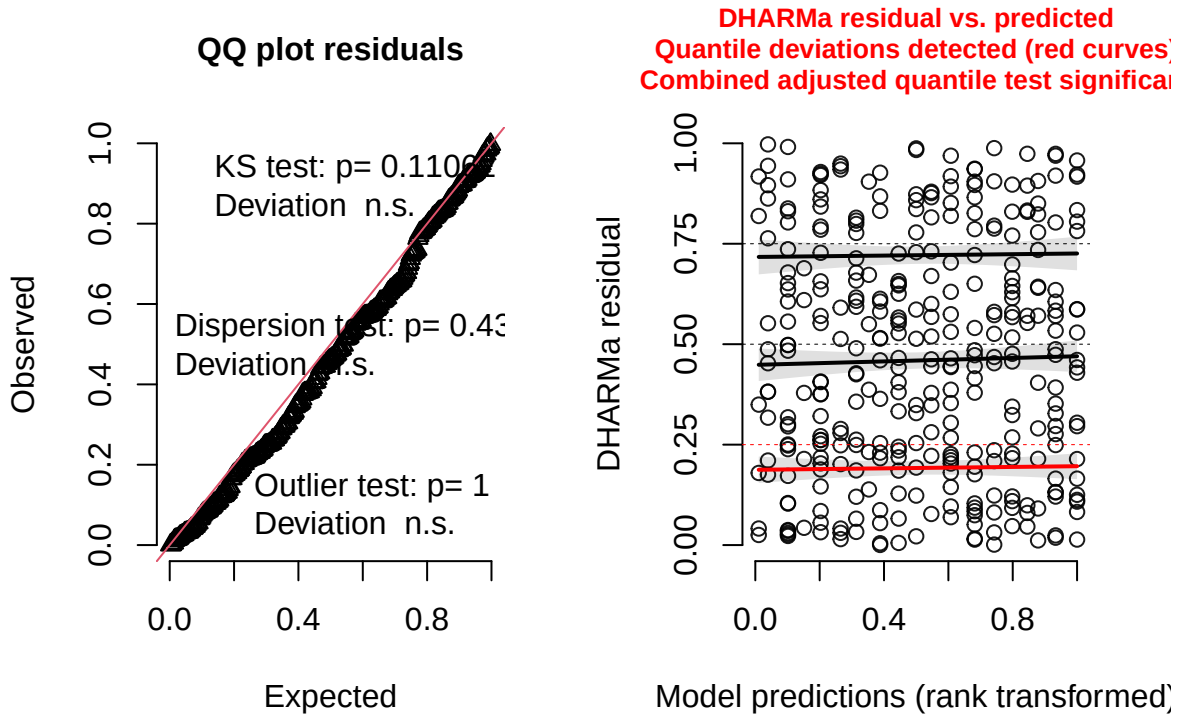
DHARMa residual



DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



DHARMa residual



Diagnostics are a bit concerning.

Probabilities used to generate simulation 50

Simulation 50
 Probability of mate fidelity: 0.486
 Probability of pair dispersal: 0.385
 Probability of single male dispersal: 0.989
 Probability of single female dispersal: 0.554

Simulation 51

Number of fledgling as the response (Poisson)

boundary (singular) fit: see help('isSingular')

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	conditional_R2	marginal_R2	family	RE
308	1	-435.0472	912.0944	990.4265	291.2566	287	0.2166428	0.0798514	Poisson	Terr
308	1	-435.0472	914.0944	996.1566	291.2569	286	0.2166275	0.0798349	Poisson	BirdID+Terr

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	condtional_R2	marginal_R2	Family	RE
308	1	-437.1891	916.3782	994.7103	321.9805	287	0.1576396	0.0861474	Poisson	BirdID

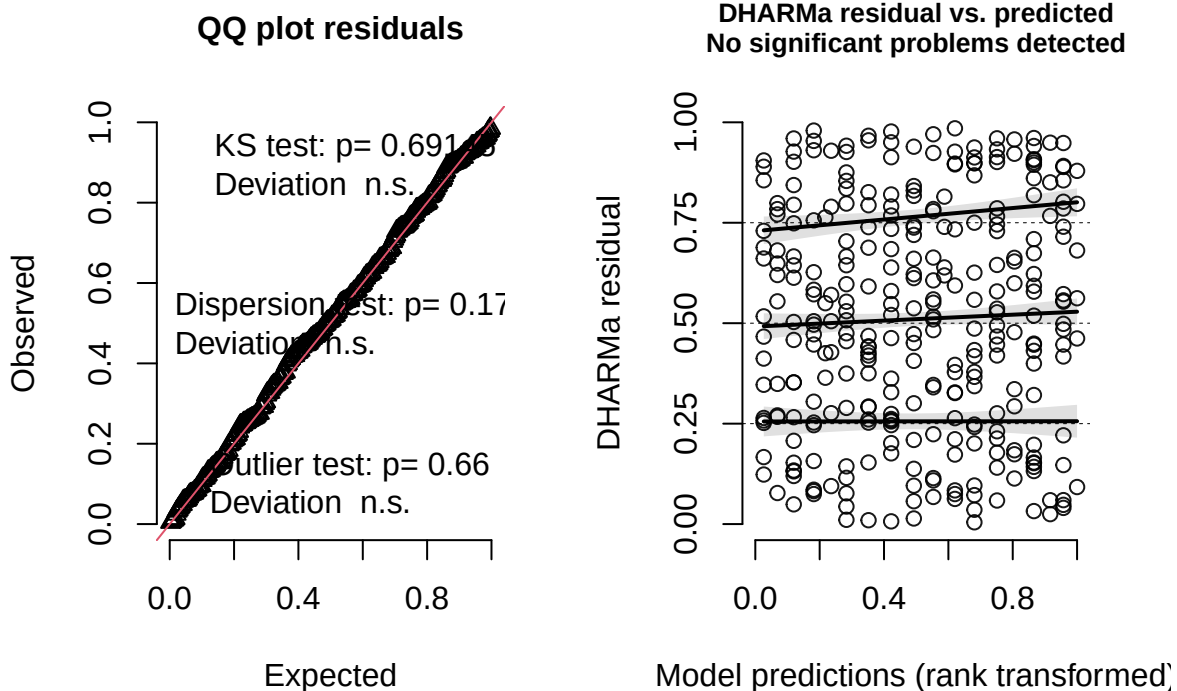
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.2206429	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.3154314	Poisson	Terr
ran_pars	birdID	sd__(Intercept)	0.0001117	Poisson	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	0.3154300	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.4022986	0.2527534	1.5916642	0.1114602	Poisson	BirdID
fixed	factor(fac.yr)2	-0.1006444	0.3232550	-0.3113468	0.7555370	Poisson	BirdID
fixed	factor(fac.yr)3	-0.4959841	0.3510752	-1.4127572	0.1577271	Poisson	BirdID
fixed	factor(fac.yr)4	-0.2002936	0.3234049	-0.6193277	0.5357005	Poisson	BirdID
fixed	factor(fac.yr)5	-0.0713549	0.3115101	-0.2290613	0.8188213	Poisson	BirdID
fixed	factor(fac.yr)6	-0.0726610	0.3164017	-0.2296480	0.8183653	Poisson	BirdID
fixed	factor(fac.yr)7	-0.5050723	0.3494050	-1.4455211	0.1483116	Poisson	BirdID
fixed	factor(fac.yr)8	-0.2476327	0.3215577	-0.7701036	0.4412385	Poisson	BirdID
fixed	factor(fac.yr)9	0.0507328	0.3005803	0.1687830	0.8659674	Poisson	BirdID
fixed	factor(fac.yr)10	-0.2691546	0.3262246	-0.8250591	0.4093380	Poisson	BirdID
fixed	factor(fac.yr)11	-0.8021354	0.4099758	-1.9565434	0.0504012	Poisson	BirdID
fixed	factor(fac.yr)12	-0.2659779	0.3319616	-0.8012310	0.4229980	Poisson	BirdID
fixed	factor(fac.yr)13	0.1320314	0.3098039	0.4261774	0.6699786	Poisson	BirdID
fixed	factor(fac.yr)14	-0.0776376	0.3357517	-0.2312350	0.8171322	Poisson	BirdID
fixed	factor(fac.yr)15	-0.5122716	0.4111312	-1.2460050	0.2127626	Poisson	BirdID
fixed	factor(fac.yr)16	-0.4348701	0.4635387	-0.9381528	0.3481659	Poisson	BirdID
fixed	factor(fac.yr)17	0.1284272	0.4298937	0.2987417	0.7651371	Poisson	BirdID
fixed	factor(fac.yr)18	0.2391880	0.3950042	0.6055329	0.5448250	Poisson	BirdID
fixed	factor(fac.yr)19	-0.4721524	0.5731997	-0.8237136	0.4101023	Poisson	BirdID
fixed	factor(fac.yr)20	-0.0872040	0.4942564	-0.1764348	0.8599524	Poisson	BirdID
fixed	(Intercept)	0.3393536	0.2594960	1.3077415	0.1909610	Poisson	Terr
fixed	factor(fac.yr)2	-0.0543585	0.3335916	-0.1629493	0.8705583	Poisson	Terr
fixed	factor(fac.yr)3	-0.4328949	0.3625672	-1.1939712	0.2324892	Poisson	Terr
fixed	factor(fac.yr)4	-0.1544985	0.3331248	-0.4637855	0.6428014	Poisson	Terr
fixed	factor(fac.yr)5	0.0022213	0.3212538	0.0069145	0.9944831	Poisson	Terr
fixed	factor(fac.yr)6	0.0236173	0.3229066	0.0731397	0.9416949	Poisson	Terr
fixed	factor(fac.yr)7	-0.4306244	0.3547401	-1.2139151	0.2247801	Poisson	Terr
fixed	factor(fac.yr)8	-0.1980352	0.3284173	-0.6029987	0.5465095	Poisson	Terr
fixed	factor(fac.yr)9	0.1002319	0.3070590	0.3264256	0.7441023	Poisson	Terr
fixed	factor(fac.yr)10	-0.2099137	0.3339654	-0.6285493	0.5296442	Poisson	Terr
fixed	factor(fac.yr)11	-0.7910500	0.4133802	-1.9136139	0.0556695	Poisson	Terr
fixed	factor(fac.yr)12	-0.2773692	0.3356730	-0.8263077	0.4086295	Poisson	Terr
fixed	factor(fac.yr)13	0.1491820	0.3154035	0.4729879	0.6362218	Poisson	Terr
fixed	factor(fac.yr)14	-0.0943865	0.3408333	-0.2769285	0.7818350	Poisson	Terr
fixed	factor(fac.yr)15	-0.4010259	0.4208012	-0.9530057	0.3405872	Poisson	Terr
fixed	factor(fac.yr)16	-0.3669106	0.4706727	-0.7795451	0.4356587	Poisson	Terr
fixed	factor(fac.yr)17	0.2227645	0.4437216	0.5020367	0.6156417	Poisson	Terr
fixed	factor(fac.yr)18	0.2984778	0.4027557	0.7410889	0.4586396	Poisson	Terr

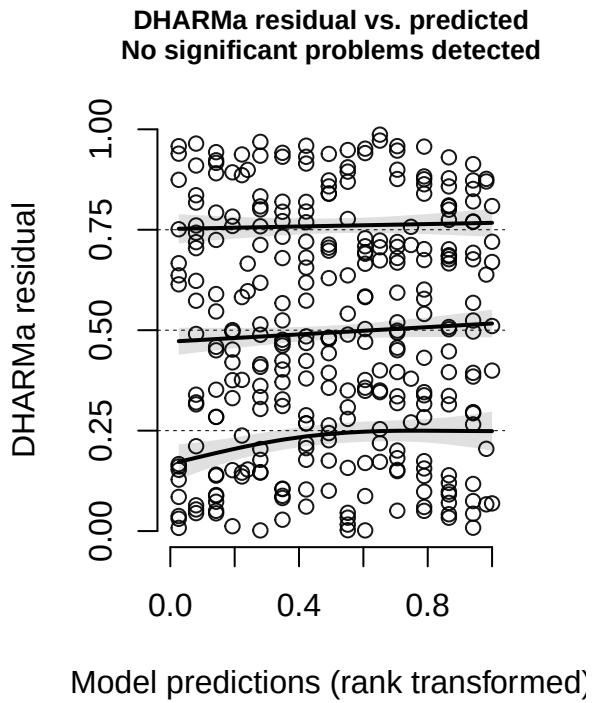
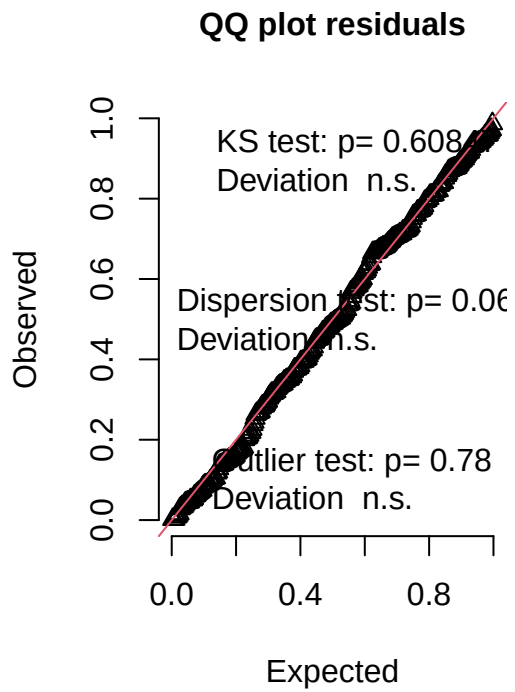
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)19	-0.3562688	0.5796381	-0.6146401	0.5387924	Poisson	Terr
fixed	factor(fac.yr)20	0.0044970	0.5039223	0.0089240	0.9928798	Poisson	Terr
fixed	(Intercept)	0.3391596	0.2595165	1.3068904	0.1912499	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.0541527	0.3336066	-0.1623252	0.8710498	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.4327117	0.3625826	-1.1934153	0.2327068	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.1543254	0.3331419	-0.4632422	0.6431908	Poisson	BirdID+Terr
fixed	factor(fac.yr)5	0.0023581	0.3212730	0.0073398	0.9941437	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	0.0238365	0.3229216	0.0738151	0.9411575	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.4303987	0.3547525	-1.2132366	0.2250394	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	-0.1978689	0.3284348	-0.6024604	0.5468677	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	0.1003471	0.3070791	0.3267792	0.7438349	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.2097055	0.3339804	-0.6278974	0.5300711	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.7907488	0.4133796	-1.9128879	0.0557624	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.2771429	0.3356867	-0.8256001	0.4090310	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	0.1493923	0.3154195	0.4736303	0.6357635	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.0941405	0.3408455	-0.2761970	0.7823968	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-0.4008029	0.4208102	-0.9524554	0.3408661	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.3668128	0.4706973	-0.7792967	0.4358050	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	0.2228783	0.4437417	0.5022704	0.6154773	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	0.2987258	0.4027630	0.7416911	0.4582745	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-0.3560117	0.5796329	-0.6142020	0.5390818	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	0.0046818	0.5039329	0.0092905	0.9925874	Poisson	BirdID+Terr

Territory only model seems to be performing the best on all criteria

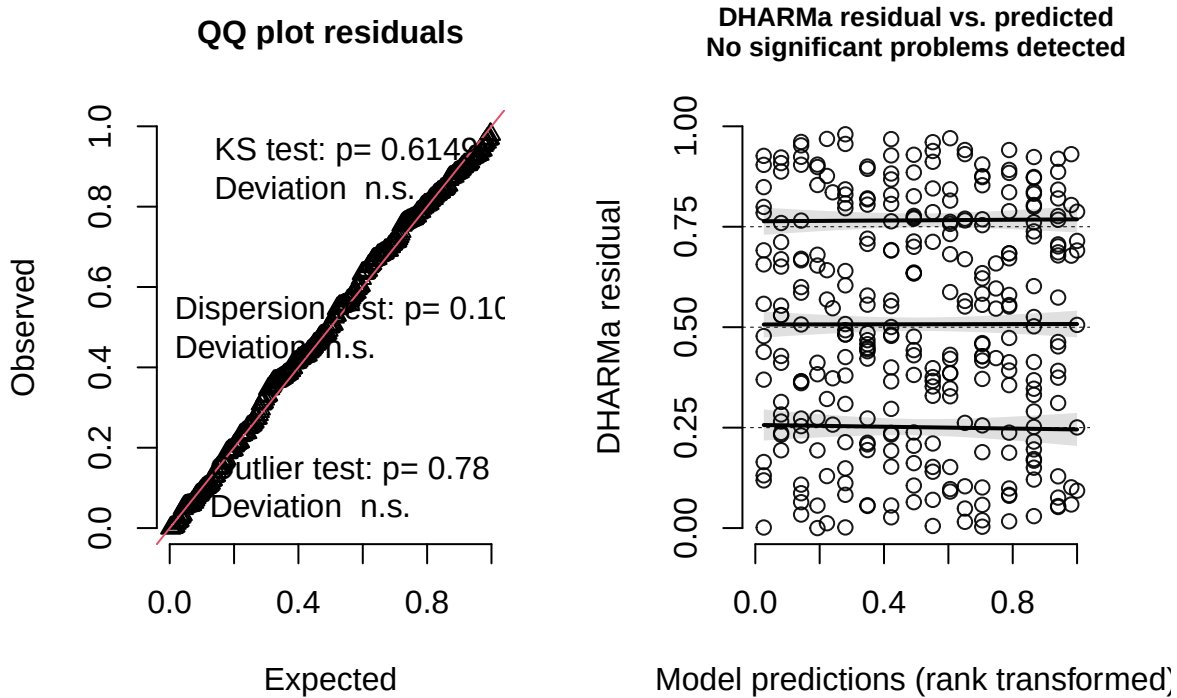
DHARMa residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
308	1	-173.5017	389.0034	467.3355	255.7888	287	0.471	Binomial	Terr
308	1	-173.1120	390.2241	472.2863	238.9954	286	0.552	Binomial	BirdID+Terr
308	1	-176.9103	395.8206	474.1527	290.2922	287	0.416	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.9182797	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.2174709	Binomial	Terr
ran_pars	birdID	sd__(Intercept)	0.5861577	Binomial	BirdID+Terr
ran_pars	Terrs	sd__(Intercept)	1.1743214	Binomial	BirdID+Terr

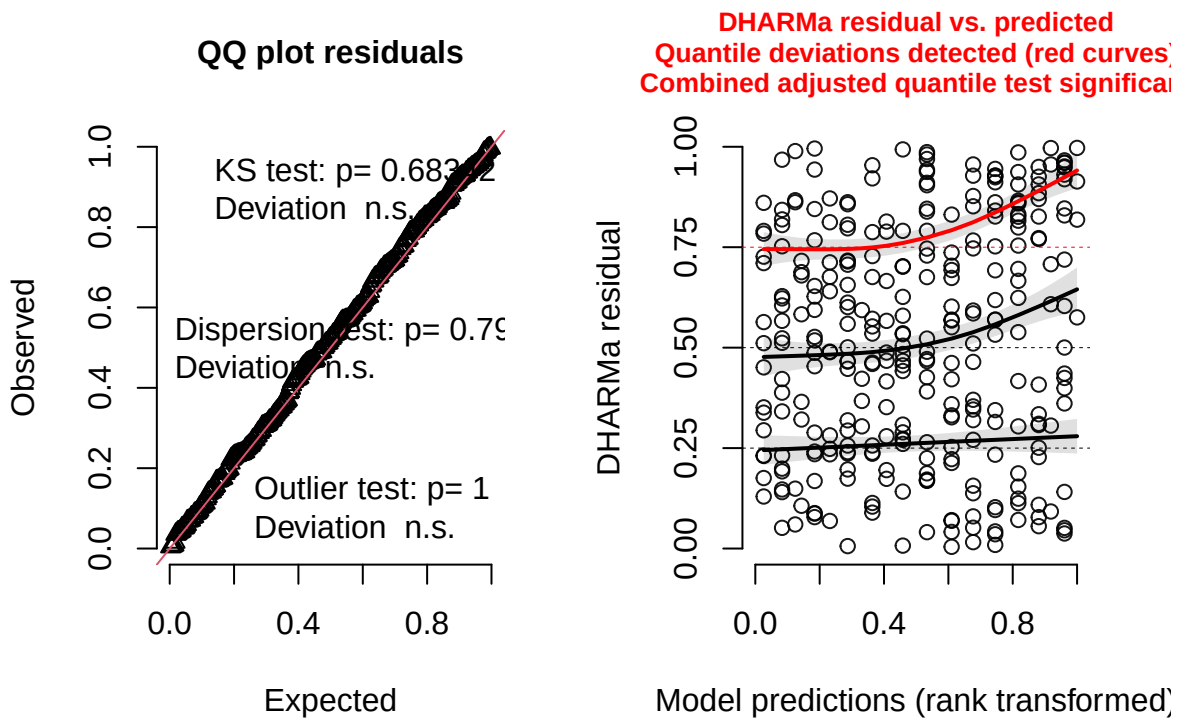
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	1.0096613	0.7497083	1.3467388	0.1780644	Binomial	BirdID
fixed	factor(fac.yr)2	0.5829821	1.0006312	0.5826144	0.5601529	Binomial	BirdID
fixed	factor(fac.yr)3	-0.6097349	0.8981885	-0.6788495	0.4972332	Binomial	BirdID
fixed	factor(fac.yr)4	0.2917909	0.9360919	0.3117118	0.7552596	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.4873281	0.9359483	0.5206785	0.6025908	Binomial	BirdID
fixed	factor(fac.yr)6	0.2728992	0.9226049	0.2957921	0.7673888	Binomial	BirdID
fixed	factor(fac.yr)7	-0.9157249	0.8981719	-1.0195430	0.3079452	Binomial	BirdID
fixed	factor(fac.yr)8	-0.2019302	0.8968429	-0.2251567	0.8218574	Binomial	BirdID
fixed	factor(fac.yr)9	0.2713013	0.8995028	0.3016125	0.7629475	Binomial	BirdID
fixed	factor(fac.yr)10	0.3129198	0.9223682	0.3392569	0.7344162	Binomial	BirdID
fixed	factor(fac.yr)11	-1.0336388	0.9435840	-1.0954390	0.2733243	Binomial	BirdID
fixed	factor(fac.yr)12	0.1026673	0.9233674	0.1111879	0.9114673	Binomial	BirdID
fixed	factor(fac.yr)13	0.6768909	0.9737205	0.6951593	0.4869555	Binomial	BirdID
fixed	factor(fac.yr)14	-0.0345665	0.9605445	-0.0359864	0.9712932	Binomial	BirdID
fixed	factor(fac.yr)15	-0.4514859	1.0185386	-0.4432683	0.6575717	Binomial	BirdID
fixed	factor(fac.yr)16	-0.7826361	1.1209753	-0.6981743	0.4850682	Binomial	BirdID
fixed	factor(fac.yr)17	-0.8197885	1.2482446	-0.6567531	0.5113397	Binomial	BirdID
fixed	factor(fac.yr)18	0.6367139	1.4345371	0.4438463	0.6571537	Binomial	BirdID
fixed	factor(fac.yr)19	-0.0988239	1.4894066	-0.0663512	0.9470983	Binomial	BirdID
fixed	factor(fac.yr)20	14.8515353	1604.2523869	0.0092576	0.9926136	Binomial	BirdID
fixed	(Intercept)	1.0541670	0.7932788	1.3288733	0.1838898	Binomial	Terr
fixed	factor(fac.yr)2	0.8670917	1.0594001	0.8184742	0.4130865	Binomial	Terr
fixed	factor(fac.yr)3	-0.3173858	0.9658200	-0.3286180	0.7424444	Binomial	Terr
fixed	factor(fac.yr)4	0.5349977	0.9929348	0.5388044	0.5900218	Binomial	Terr
fixed	factor(fac.yr)5	0.8347534	0.9894067	0.8436908	0.3988422	Binomial	Terr
fixed	factor(fac.yr)6	0.5405759	0.9578943	0.5643377	0.5725243	Binomial	Terr
fixed	factor(fac.yr)7	-0.8748532	0.9426517	-0.9280769	0.3533677	Binomial	Terr
fixed	factor(fac.yr)8	-0.1601450	0.9378292	-0.1707613	0.8644114	Binomial	Terr
fixed	factor(fac.yr)9	0.2288117	0.9418663	0.2429344	0.8080562	Binomial	Terr
fixed	factor(fac.yr)10	0.4367515	0.9752053	0.4478559	0.6542572	Binomial	Terr
fixed	factor(fac.yr)11	-1.2016955	0.9800953	-1.2261006	0.2201608	Binomial	Terr
fixed	factor(fac.yr)12	0.0478815	0.9634760	0.0496966	0.9603642	Binomial	Terr
fixed	factor(fac.yr)13	0.8865278	1.0272243	0.8630323	0.3881197	Binomial	Terr
fixed	factor(fac.yr)14	0.0433405	1.0084049	0.0429793	0.9657181	Binomial	Terr
fixed	factor(fac.yr)15	0.0013476	1.0649825	0.0012653	0.9989904	Binomial	Terr
fixed	factor(fac.yr)16	-0.6683473	1.1666507	-0.5728770	0.5667280	Binomial	Terr
fixed	factor(fac.yr)17	-0.4736198	1.3536290	-0.3498889	0.7264221	Binomial	Terr
fixed	factor(fac.yr)18	0.6973739	1.4654779	0.4758679	0.6341685	Binomial	Terr
fixed	factor(fac.yr)19	0.0277587	1.4688875	0.0188978	0.9849227	Binomial	Terr
fixed	factor(fac.yr)20	15.0290001	1521.8954102	0.0098752	0.9921209	Binomial	Terr
fixed	(Intercept)	1.1094614	0.8224842	1.3489152	0.1773642	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	0.8157239	1.0838214	0.7526368	0.4516682	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-0.4357956	0.9894556	-0.4404397	0.6596187	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	0.4882657	1.0243097	0.4766778	0.6335915	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	0.7886690	1.0193834	0.7736725	0.4391245	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	0.5456328	0.9914810	0.5503210	0.5820993	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-0.9288803	0.9756131	-0.9520990	0.3410468	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	-0.1726660	0.9728503	-0.1774847	0.8591277	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	0.2462384	0.9745564	0.2526672	0.8005254	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	0.4487721	1.0091530	0.4447017	0.6565353	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-1.2407150	1.0178776	-1.2189236	0.2228732	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	0.0667806	0.9990002	0.0668475	0.9467031	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	0.9019878	1.0622815	0.8491043	0.3958232	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	0.0275360	1.0473909	0.0262901	0.9790259	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-0.1562493	1.1013073	-0.1418762	0.8871778	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-0.7858380	1.2103892	-0.6492440	0.5161806	Binomial	BirdID+Terr

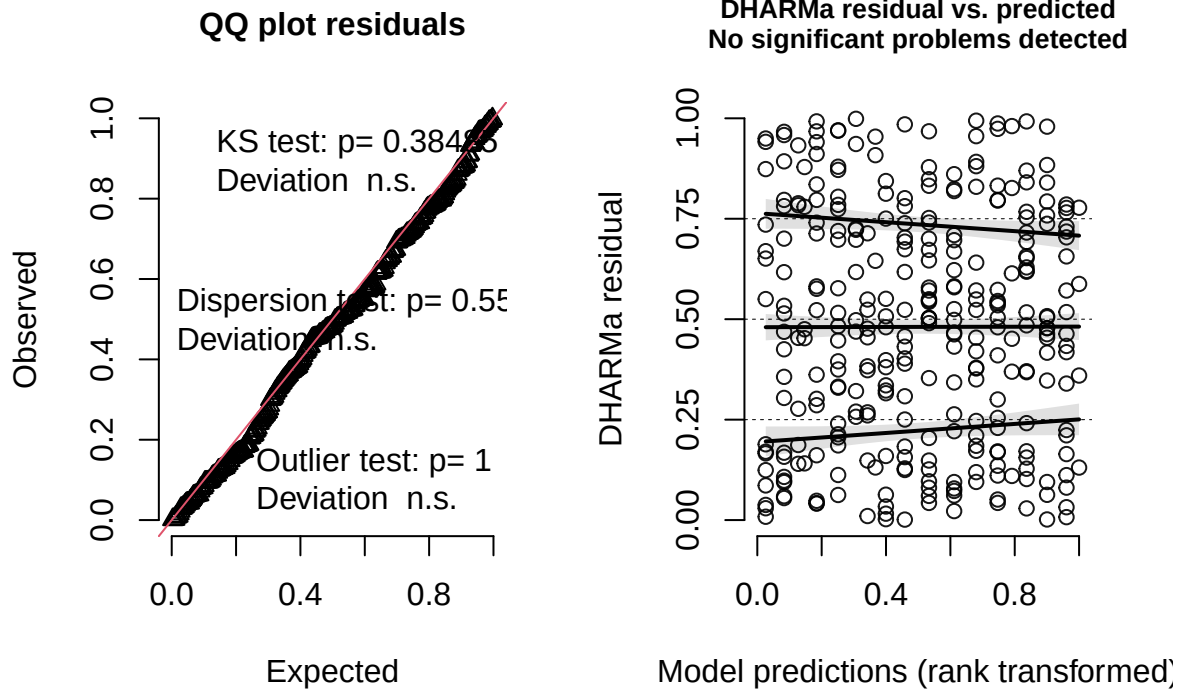
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-0.6001151	1.4025195	-0.4278836	0.6687359	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	0.6667570	1.5192211	0.4388808	0.6607479	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	-0.1292014	1.5208102	-0.0849556	0.9322967	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	11.8936299	342.3997370	0.0347361	0.9722902	Binomial	BirdID+Terr

Terr only does well on all but Kappa.

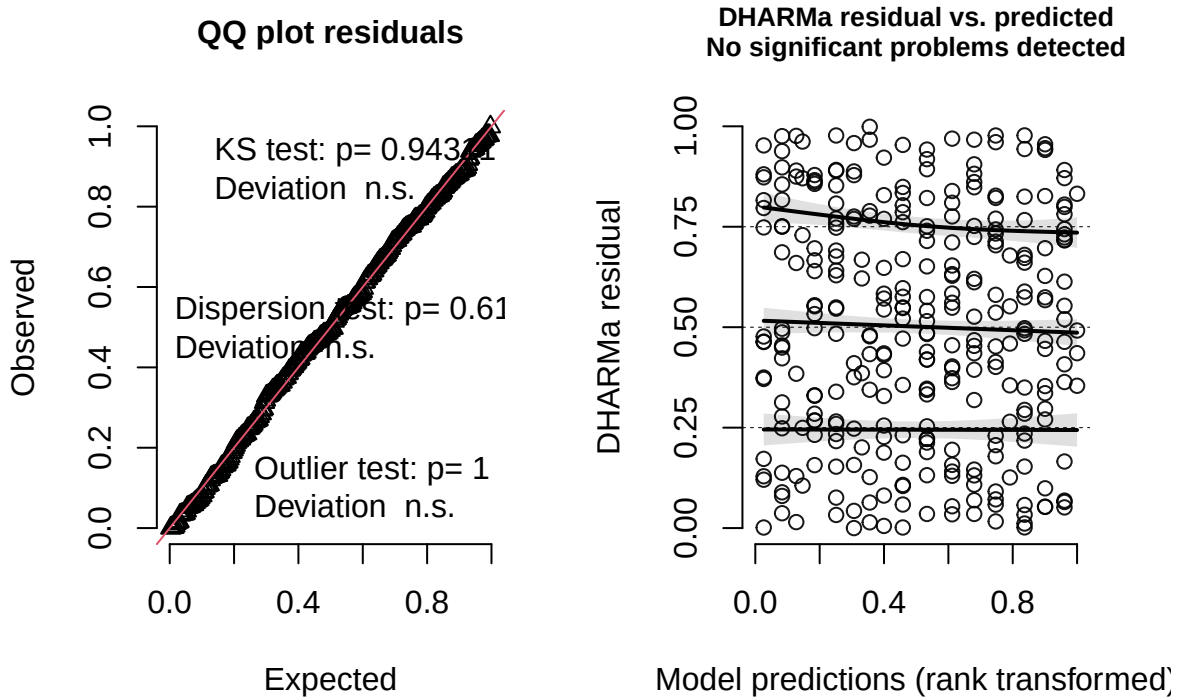
DHARMa residual



DHARMa residual



DHARMa residual



Terr only diagnostics look good.

Probabilites used to generate simulation 51

Simulation 51
 Probability of mate fidelity: 0.1
 Probability of pair dispersal: 0.9
 Probability of single male dispersal: 0.95
 Probability of single female dispersal: 0.95

I would say the territory effect is present here, as it was intended to be.

Simulation 52

Number of fledgling as the response (Poisson)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Family	RE
276	1	-395.4679	832.9358	908.9642	270.0346	255	Poisson	Terr
276	1	-395.4679	834.9359	914.5847	270.0255	254	Poisson	BirdID+Terr
276	1	-396.5615	835.1231	911.1515	305.7231	255	Poisson	BirdID

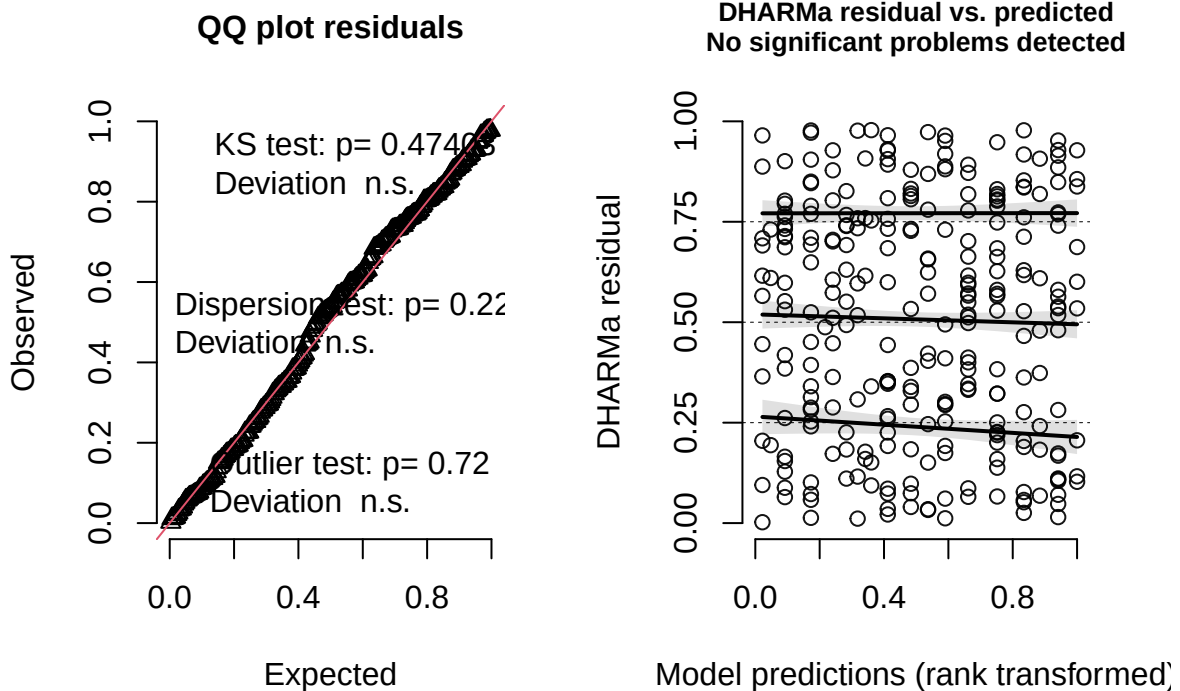
effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.0000002	Poisson	BirdID
ran_pars	Terrs	sd__(Intercept)	0.2316630	Poisson	Terr
ran_pars	Terrs	sd__(Intercept)	0.2317023	Poisson	BirdID+Terr
ran_pars	birdID	sd__(Intercept)	0.0001622	Poisson	BirdID+Terr

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	0.4855078	0.2773501	1.7505233	0.0800281	Poisson	BirdID
fixed	factor(fac.yr)2	-0.0800427	0.3443686	-0.2324332	0.8162015	Poisson	BirdID
fixed	factor(fac.yr)3	-0.5855913	0.3599370	-1.6269271	0.1037526	Poisson	BirdID
fixed	factor(fac.yr)4	-0.1244945	0.3274540	-0.3801892	0.7038050	Poisson	BirdID
fixed	factor(fac.yr)5	-0.0287494	0.3320488	-0.0865819	0.9310039	Poisson	BirdID
fixed	factor(fac.yr)6	-0.0300323	0.3182975	-0.0943529	0.9248289	Poisson	BirdID
fixed	factor(fac.yr)7	-0.5764796	0.3529053	-1.6335251	0.1023586	Poisson	BirdID
fixed	factor(fac.yr)8	-0.0374831	0.3235751	-0.1158405	0.9077790	Poisson	BirdID
fixed	factor(fac.yr)9	0.0925700	0.3182975	0.2908286	0.7711824	Poisson	BirdID
fixed	factor(fac.yr)10	-0.1226023	0.3469890	-0.3533321	0.7238395	Poisson	BirdID
fixed	factor(fac.yr)11	-0.3984964	0.4003204	-0.9954438	0.3195205	Poisson	BirdID
fixed	factor(fac.yr)12	-0.4855078	0.4003204	-1.2127981	0.2252070	Poisson	BirdID
fixed	factor(fac.yr)13	0.1504810	0.3684381	0.4084294	0.6829584	Poisson	BirdID
fixed	factor(fac.yr)14	-0.1060182	0.3599370	-0.2945465	0.7683403	Poisson	BirdID
fixed	factor(fac.yr)15	-1.0916436	0.4935481	-2.2118282	0.0269785	Poisson	BirdID
fixed	factor(fac.yr)16	-0.3677248	0.4336291	-0.8480169	0.3964286	Poisson	BirdID
fixed	factor(fac.yr)17	-0.3031863	0.4935481	-0.6142993	0.5390176	Poisson	BirdID
fixed	factor(fac.yr)18	-0.1490356	0.4688072	-0.3179038	0.7505579	Poisson	BirdID
fixed	factor(fac.yr)19	-0.4855078	1.0377490	-0.4678470	0.6398940	Poisson	BirdID
fixed	factor(fac.yr)20	-0.8909729	0.7595545	-1.1730204	0.2407876	Poisson	BirdID
fixed	(Intercept)	0.4159867	0.2898252	1.4353018	0.1512011	Poisson	Terr
fixed	factor(fac.yr)2	-0.0513055	0.3574691	-0.1435243	0.8858761	Poisson	Terr
fixed	factor(fac.yr)3	-0.5636478	0.3723687	-1.5136818	0.1301066	Poisson	Terr
fixed	factor(fac.yr)4	-0.0840458	0.3399778	-0.2472097	0.8047460	Poisson	Terr
fixed	factor(fac.yr)5	0.0103920	0.3459766	0.0300366	0.9760379	Poisson	Terr
fixed	factor(fac.yr)6	0.0220861	0.3303553	0.0668556	0.9466967	Poisson	Terr
fixed	factor(fac.yr)7	-0.5286935	0.3659920	-1.4445493	0.1485845	Poisson	Terr
fixed	factor(fac.yr)8	0.0129436	0.3353938	0.0385924	0.9692154	Poisson	Terr
fixed	factor(fac.yr)9	0.1569882	0.3320442	0.4727931	0.6363608	Poisson	Terr
fixed	factor(fac.yr)10	-0.0820481	0.3596170	-0.2281541	0.8195264	Poisson	Terr
fixed	factor(fac.yr)11	-0.3268557	0.4159217	-0.7858587	0.4319502	Poisson	Terr
fixed	factor(fac.yr)12	-0.4735288	0.4146745	-1.1419290	0.2534835	Poisson	Terr
fixed	factor(fac.yr)13	0.1605705	0.3861977	0.4157728	0.6775763	Poisson	Terr
fixed	factor(fac.yr)14	-0.0412885	0.3756116	-0.1099233	0.9124702	Poisson	Terr
fixed	factor(fac.yr)15	-1.0104087	0.5077149	-1.9901104	0.0465788	Poisson	Terr
fixed	factor(fac.yr)16	-0.3251382	0.4502844	-0.7220729	0.4702496	Poisson	Terr
fixed	factor(fac.yr)17	-0.2221851	0.5097324	-0.4358858	0.6629196	Poisson	Terr
fixed	factor(fac.yr)18	-0.0465451	0.4847434	-0.0960201	0.9235046	Poisson	Terr
fixed	factor(fac.yr)19	-0.4476664	1.0737343	-0.4169248	0.6767334	Poisson	Terr
fixed	factor(fac.yr)20	-0.8680070	0.7742934	-1.1210311	0.2622746	Poisson	Terr
fixed	(Intercept)	0.4152847	0.2899150	1.4324361	0.1520191	Poisson	BirdID+Terr
fixed	factor(fac.yr)2	-0.0505724	0.3575401	-0.1414455	0.8875180	Poisson	BirdID+Terr
fixed	factor(fac.yr)3	-0.5629275	0.3724370	-1.5114705	0.1306686	Poisson	BirdID+Terr
fixed	factor(fac.yr)4	-0.0832923	0.3400520	-0.2449400	0.8065029	Poisson	BirdID+Terr

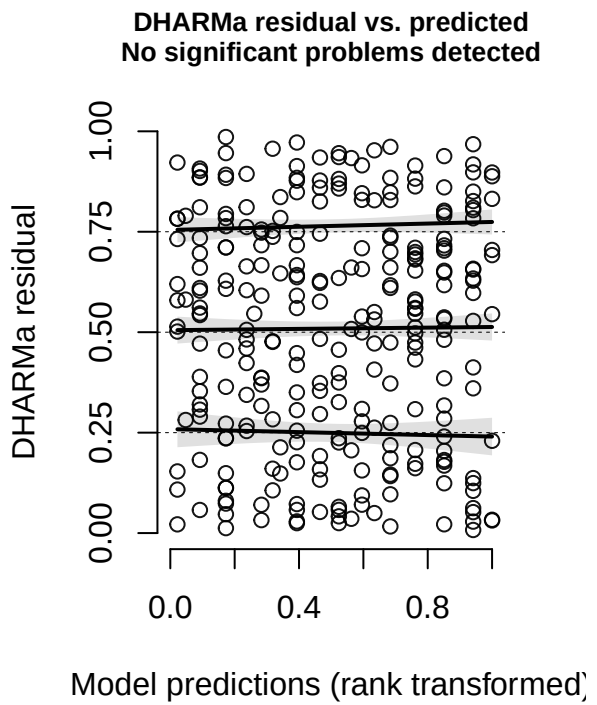
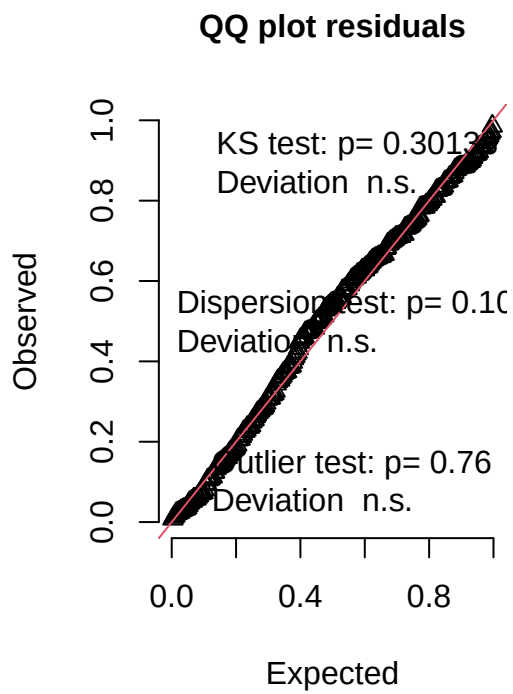
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	0.0111336	0.3460504	0.0321733	0.9743338	Poisson	BirdID+Terr
fixed	factor(fac.yr)6	0.0228276	0.3304325	0.0690840	0.9449228	Poisson	BirdID+Terr
fixed	factor(fac.yr)7	-0.5279100	0.3660583	-1.4421475	0.1492608	Poisson	BirdID+Terr
fixed	factor(fac.yr)8	0.0136180	0.3354722	0.0405934	0.9676200	Poisson	BirdID+Terr
fixed	factor(fac.yr)9	0.1576797	0.3321236	0.4747620	0.6349566	Poisson	BirdID+Terr
fixed	factor(fac.yr)10	-0.0813739	0.3596907	-0.2262331	0.8210201	Poisson	BirdID+Terr
fixed	factor(fac.yr)11	-0.3261708	0.4159880	-0.7840871	0.4329890	Poisson	BirdID+Terr
fixed	factor(fac.yr)12	-0.4728251	0.4147368	-1.1400608	0.2542610	Poisson	BirdID+Terr
fixed	factor(fac.yr)13	0.1612442	0.3862683	0.4174409	0.6763560	Poisson	BirdID+Terr
fixed	factor(fac.yr)14	-0.0405694	0.3756813	-0.1079888	0.9140046	Poisson	BirdID+Terr
fixed	factor(fac.yr)15	-1.0096927	0.5077663	-1.9884991	0.0467565	Poisson	BirdID+Terr
fixed	factor(fac.yr)16	-0.3244204	0.4503419	-0.7203869	0.4712868	Poisson	BirdID+Terr
fixed	factor(fac.yr)17	-0.2215615	0.5097970	-0.4346072	0.6638475	Poisson	BirdID+Terr
fixed	factor(fac.yr)18	-0.0458542	0.4848008	-0.0945836	0.9246455	Poisson	BirdID+Terr
fixed	factor(fac.yr)19	-0.4470688	1.0738108	-0.4163385	0.6771623	Poisson	BirdID+Terr
fixed	factor(fac.yr)20	-0.8671391	0.7742611	-1.1199570	0.2627321	Poisson	BirdID+Terr

Still, the terr effect pulls through.

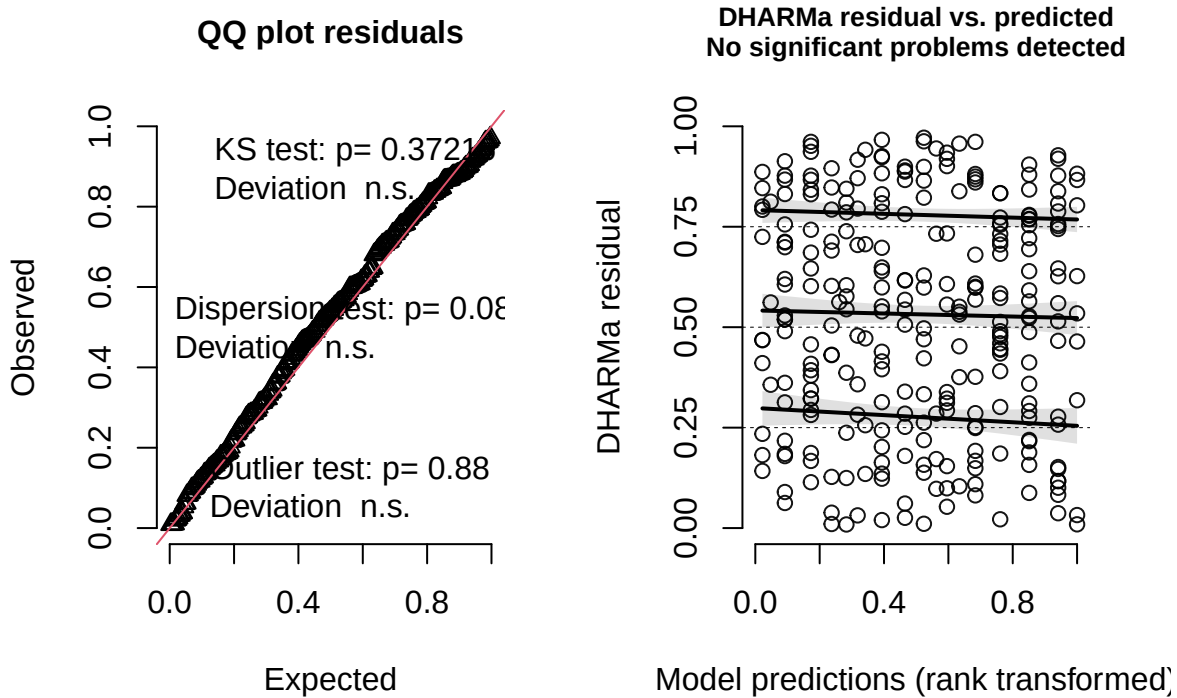
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Eggs (Binomial)

nobs	sigma	logLik	AIC	BIC	deviance	df.residual	Kappa	Family	RE
276	1	-151.0517	344.1034	420.1318	228.9504	255	0.349	Binomial	Terr
276	1	-150.6330	345.2660	424.9148	208.0686	254	0.469	Binomial	BirdID+Terr
276	1	-153.6651	349.3303	425.3587	259.1262	255	0.292	Binomial	BirdID

effect	group	term	estimate	Family	RE
ran_pars	birdID	sd__(Intercept)	0.8078711	Binomial	BirdID
ran_pars	Terrs	sd__(Intercept)	1.0991312	Binomial	Terr
ran_pars	Terrs	sd__(Intercept)	1.1220961	Binomial	BirdID+Terr
ran_pars	birdID	sd__(Intercept)	0.5911468	Binomial	BirdID+Terr

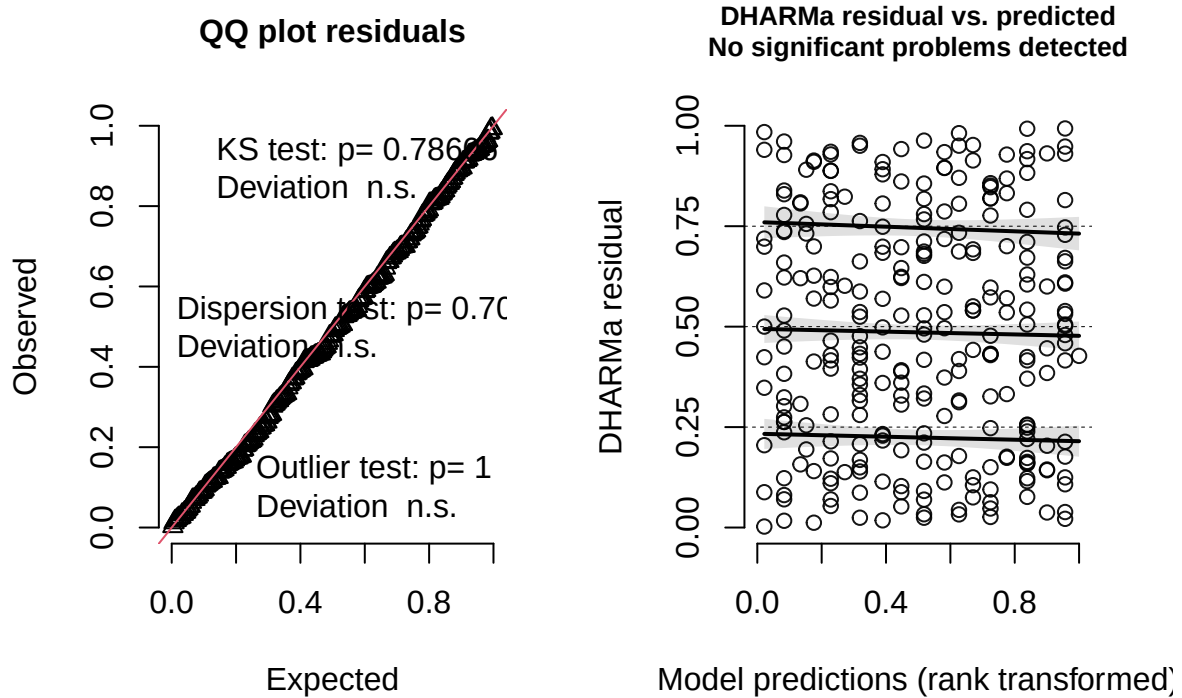
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	(Intercept)	2.2271267	1.149574	1.9373494	0.0527027	Binomial	BirdID
fixed	factor(fac.yr)2	-1.2714008	1.259844	-1.0091734	0.3128915	Binomial	BirdID
fixed	factor(fac.yr)3	-1.4836872	1.240261	-1.1962704	0.2315911	Binomial	BirdID
fixed	factor(fac.yr)4	-0.9396813	1.253371	-0.7497235	0.4534213	Binomial	BirdID

effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)5	-0.8714800	1.274443	-0.6838124	0.4940936	Binomial	BirdID
fixed	factor(fac.yr)6	-0.4585292	1.268089	-0.3615908	0.7176578	Binomial	BirdID
fixed	factor(fac.yr)7	-1.9554872	1.233149	-1.5857675	0.1127921	Binomial	BirdID
fixed	factor(fac.yr)8	0.4620134	1.404655	0.3289159	0.7422192	Binomial	BirdID
fixed	factor(fac.yr)9	-1.2856197	1.251352	-1.0273843	0.3042395	Binomial	BirdID
fixed	factor(fac.yr)10	-0.9764547	1.310898	-0.7448745	0.4563476	Binomial	BirdID
fixed	factor(fac.yr)11	-0.8723632	1.367164	-0.6380824	0.5234201	Binomial	BirdID
fixed	factor(fac.yr)12	-0.9148199	1.360378	-0.6724749	0.5012814	Binomial	BirdID
fixed	factor(fac.yr)13	-0.8568880	1.445672	-0.5927264	0.5533642	Binomial	BirdID
fixed	factor(fac.yr)14	-0.8904531	1.356527	-0.6564212	0.5115531	Binomial	BirdID
fixed	factor(fac.yr)15	-2.4733543	1.327375	-1.8633426	0.0624141	Binomial	BirdID
fixed	factor(fac.yr)16	-1.6778678	1.404005	-1.1950582	0.2320643	Binomial	BirdID
fixed	factor(fac.yr)17	-1.8947905	1.517141	-1.2489222	0.2116935	Binomial	BirdID
fixed	factor(fac.yr)18	-1.8977420	1.516981	-1.2509988	0.2109349	Binomial	BirdID
fixed	factor(fac.yr)19	15.9864641	8266.569068	0.0019339	0.9984570	Binomial	BirdID
fixed	factor(fac.yr)20	-1.3046573	1.751794	-0.7447551	0.4564198	Binomial	BirdID
fixed	(Intercept)	2.0049440	1.208263	1.6593609	0.0970431	Binomial	Terr
fixed	factor(fac.yr)2	-1.0564104	1.335228	-0.7911832	0.4288371	Binomial	Terr
fixed	factor(fac.yr)3	-1.2641166	1.303986	-0.9694249	0.3323332	Binomial	Terr
fixed	factor(fac.yr)4	-0.6222089	1.319204	-0.4716548	0.6371732	Binomial	Terr
fixed	factor(fac.yr)5	-0.5892730	1.341094	-0.4393974	0.6603736	Binomial	Terr
fixed	factor(fac.yr)6	-0.0132900	1.332329	-0.0099751	0.9920412	Binomial	Terr
fixed	factor(fac.yr)7	-1.8557092	1.299703	-1.4277953	0.1533508	Binomial	Terr
fixed	factor(fac.yr)8	0.8927707	1.466016	0.6089773	0.5425395	Binomial	Terr
fixed	factor(fac.yr)9	-0.9061288	1.315318	-0.6889047	0.4908832	Binomial	Terr
fixed	factor(fac.yr)10	-0.6133802	1.382866	-0.4435571	0.6573628	Binomial	Terr
fixed	factor(fac.yr)11	-0.5548420	1.439130	-0.3855398	0.6998375	Binomial	Terr
fixed	factor(fac.yr)12	-0.8300220	1.423039	-0.5832744	0.5597086	Binomial	Terr
fixed	factor(fac.yr)13	-0.4865465	1.531232	-0.3177483	0.7506759	Binomial	Terr
fixed	factor(fac.yr)14	-0.2438976	1.428542	-0.1707318	0.8644347	Binomial	Terr
fixed	factor(fac.yr)15	-2.0604444	1.391877	-1.4803347	0.1387840	Binomial	Terr
fixed	factor(fac.yr)16	-1.3383372	1.478778	-0.9050295	0.3654498	Binomial	Terr
fixed	factor(fac.yr)17	-1.1868232	1.565763	-0.7579841	0.4484605	Binomial	Terr
fixed	factor(fac.yr)18	-1.2763907	1.577845	-0.8089454	0.4185466	Binomial	Terr
fixed	factor(fac.yr)19	12.9738571	2026.286851	0.0064028	0.9948914	Binomial	Terr
fixed	factor(fac.yr)20	-1.5612134	1.810035	-0.8625322	0.3883947	Binomial	Terr
fixed	(Intercept)	2.2299956	1.286643	1.7331891	0.0830621	Binomial	BirdID+Terr
fixed	factor(fac.yr)2	-1.1924639	1.400303	-0.8515759	0.3944495	Binomial	BirdID+Terr
fixed	factor(fac.yr)3	-1.4584231	1.377479	-1.0587629	0.2897078	Binomial	BirdID+Terr
fixed	factor(fac.yr)4	-0.7239421	1.395982	-0.5185899	0.6040468	Binomial	BirdID+Terr
fixed	factor(fac.yr)5	-0.6700892	1.420512	-0.4717236	0.6371241	Binomial	BirdID+Terr
fixed	factor(fac.yr)6	-0.1099157	1.409981	-0.0779554	0.9378635	Binomial	BirdID+Terr
fixed	factor(fac.yr)7	-2.0023727	1.380337	-1.4506404	0.1468800	Binomial	BirdID+Terr
fixed	factor(fac.yr)8	0.8658630	1.553742	0.5572758	0.5773390	Binomial	BirdID+Terr
fixed	factor(fac.yr)9	-1.0641441	1.396557	-0.7619769	0.4460738	Binomial	BirdID+Terr
fixed	factor(fac.yr)10	-0.7545732	1.468472	-0.5138491	0.6073575	Binomial	BirdID+Terr
fixed	factor(fac.yr)11	-0.6304576	1.524340	-0.4135938	0.6791716	Binomial	BirdID+Terr
fixed	factor(fac.yr)12	-0.9408290	1.508966	-0.6234923	0.5329610	Binomial	BirdID+Terr
fixed	factor(fac.yr)13	-0.6423607	1.626164	-0.3950160	0.6928311	Binomial	BirdID+Terr
fixed	factor(fac.yr)14	-0.4203200	1.513954	-0.2776306	0.7812960	Binomial	BirdID+Terr
fixed	factor(fac.yr)15	-2.3259177	1.478476	-1.5731861	0.1156757	Binomial	BirdID+Terr
fixed	factor(fac.yr)16	-1.5547512	1.570558	-0.9899358	0.3222055	Binomial	BirdID+Terr

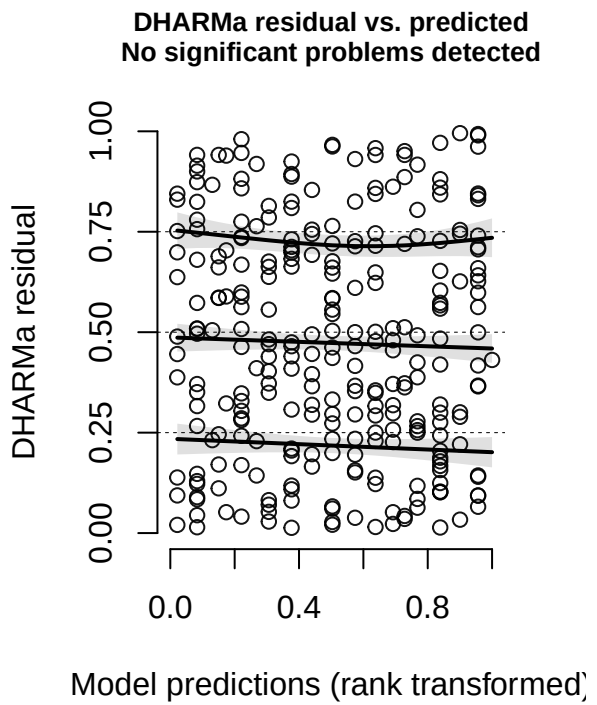
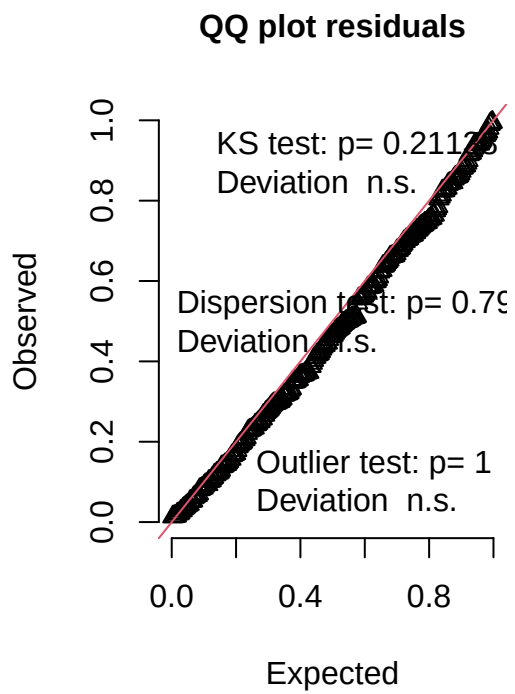
effect	term	estimate	std.error	statistic	p.value	Family	RE
fixed	factor(fac.yr)17	-1.4645290	1.647590	-0.8888914	0.3740615	Binomial	BirdID+Terr
fixed	factor(fac.yr)18	-1.4824504	1.672265	-0.8864924	0.3753523	Binomial	BirdID+Terr
fixed	factor(fac.yr)19	12.5458954	1687.926226	0.0074327	0.9940696	Binomial	BirdID+Terr
fixed	factor(fac.yr)20	-1.6200497	1.908045	-0.8490625	0.3958465	Binomial	BirdID+Terr

Terr pulls through on all but Kappa.

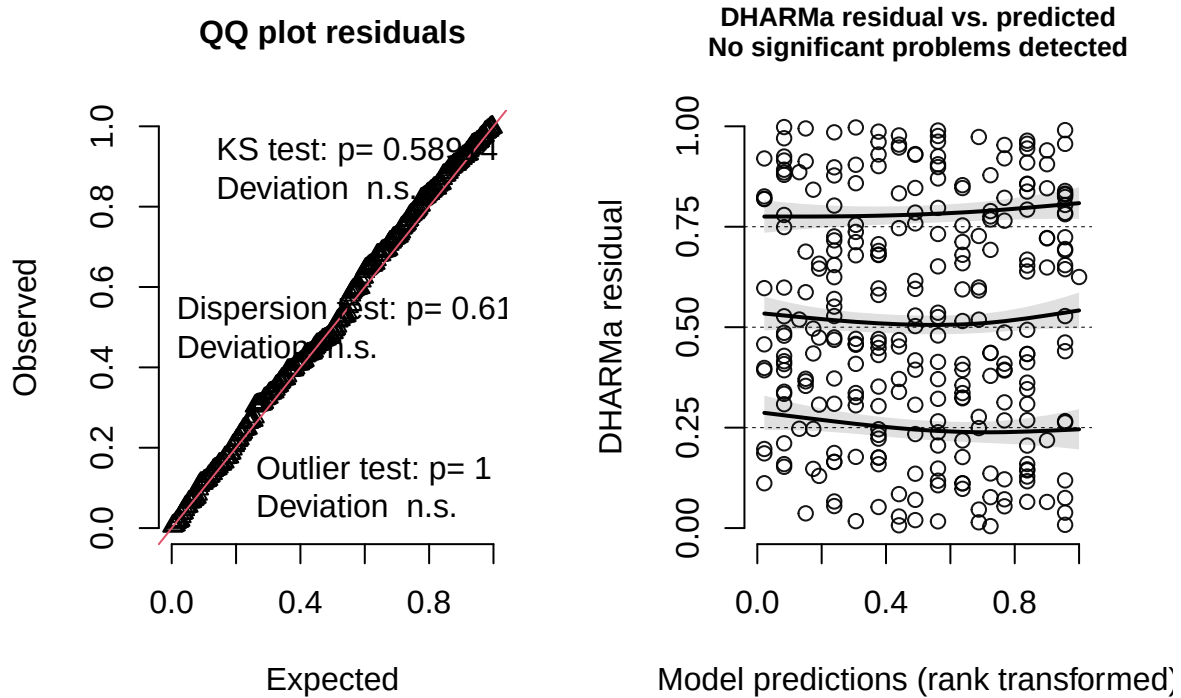
DHARMA residual



DHARMa residual



DHARMa residual



Diagnostics look good.

Probabilites used to generate simulation 52

Simulation 52

Probability of mate fidelity: 0.97

Probability of pair dispersal: 0.005

Probability of single male dispersal: 0.05

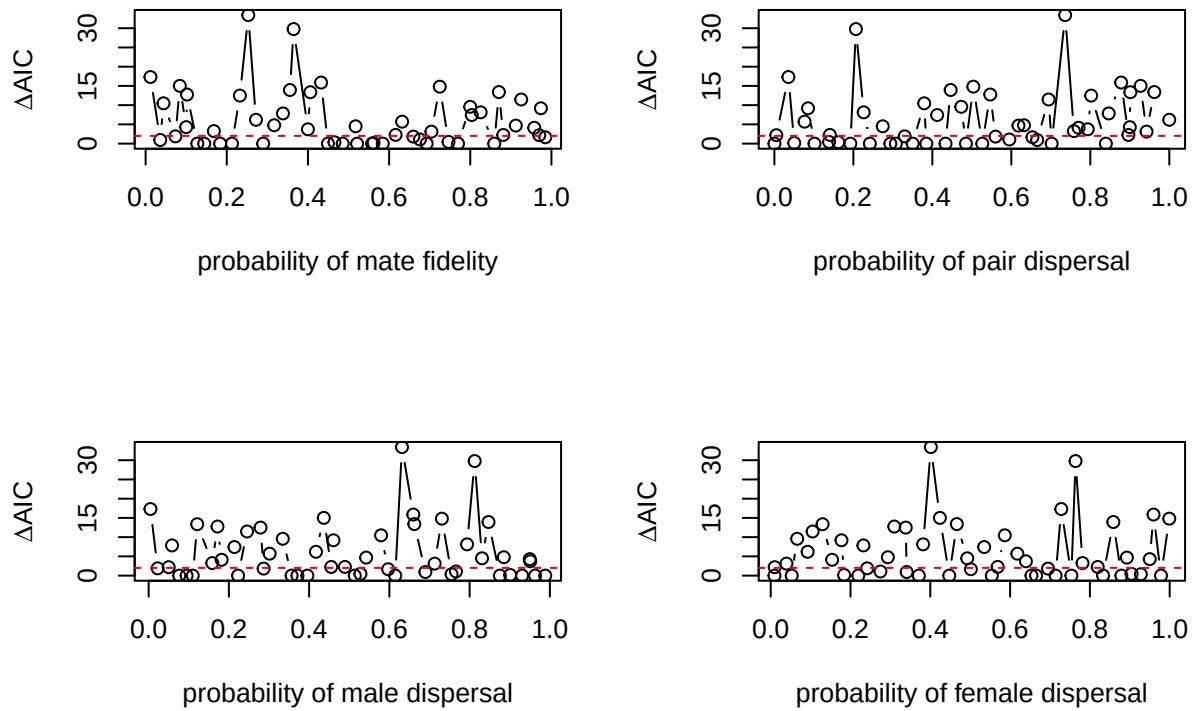
Probability of single female dispersal: 0.01

Detla AIC Plots

Here, we explore the relationships between probabilities of dispersal, and mate fidelity, and delta AIC for each response variable and random effect structure.

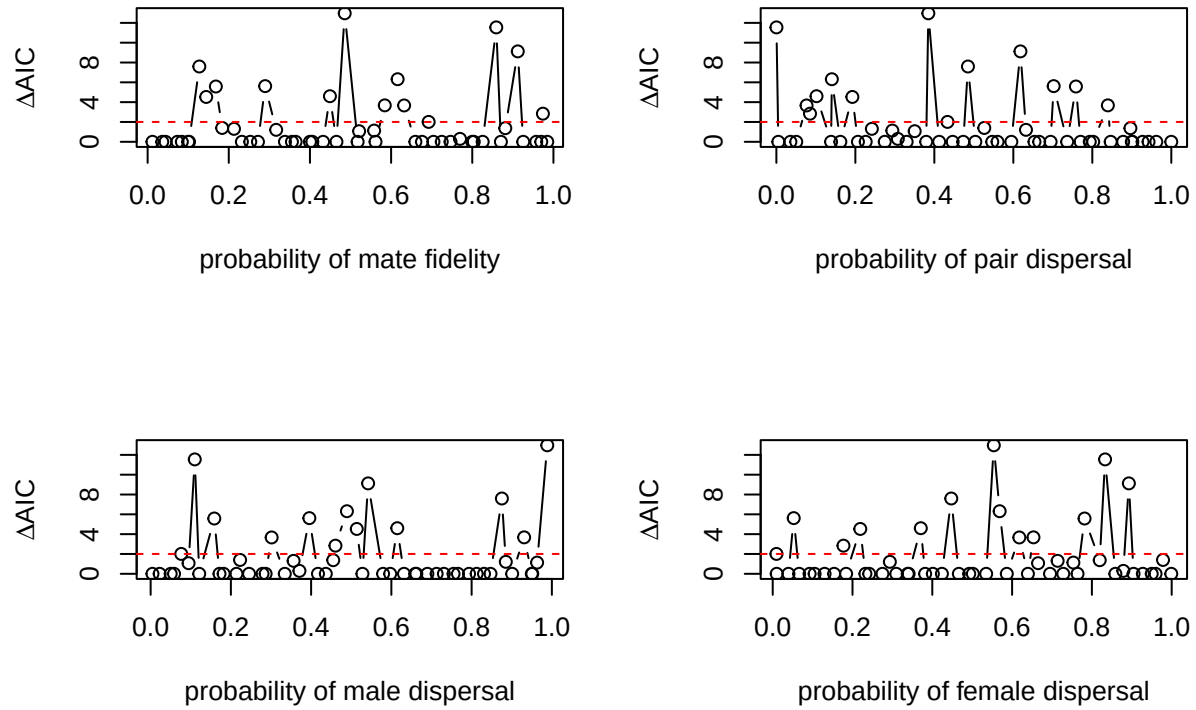
Fledge Response

Band ID



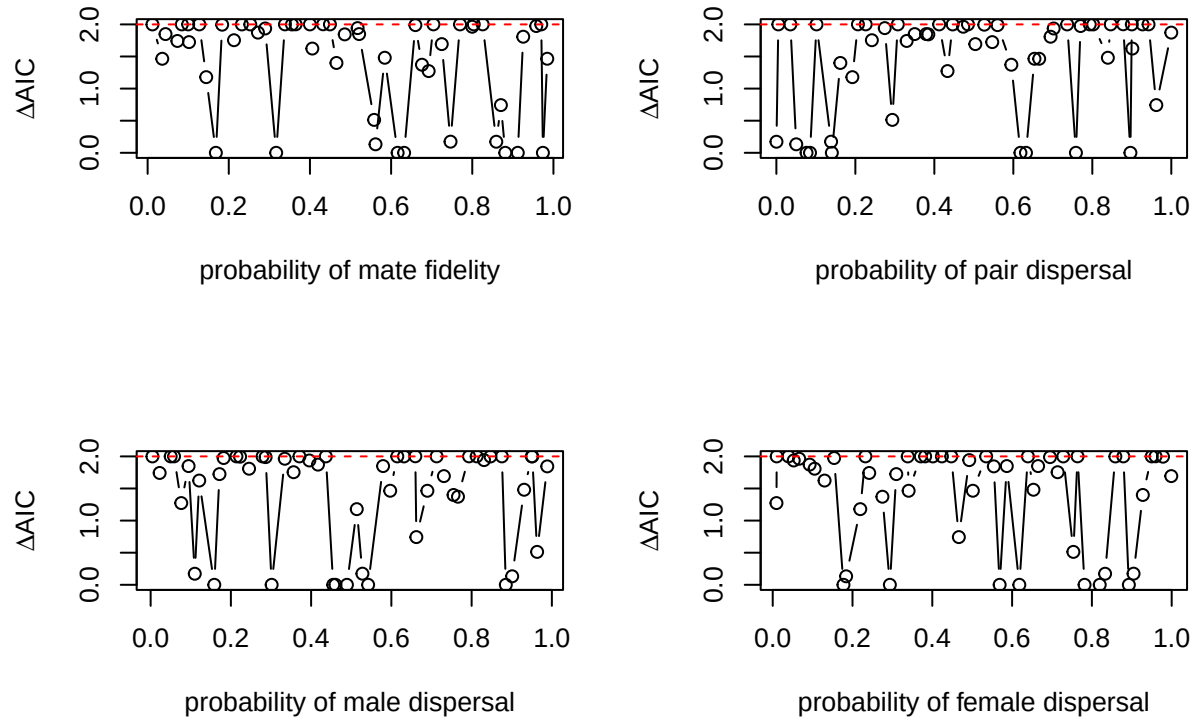
There does seem to be some low ΔAIC s when male dispersal is high, other than that, I don't see a terribly clear pattern in any of these plots.

Territory



Not a clear pattern, but it does seem that most territory only models are performing well.

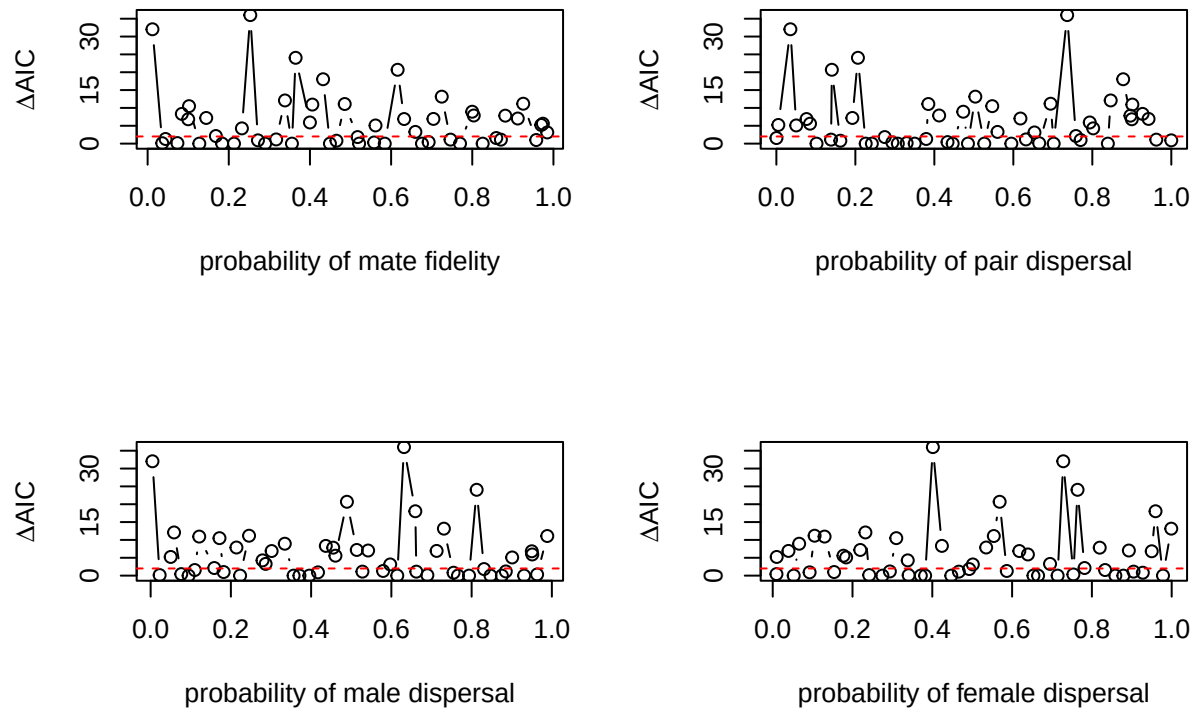
Band ID + Territory



It does seem that these models all perform well on AIC, but very few points are at 0, meaning that most of these models aren't the best on AIC for each respective data set.

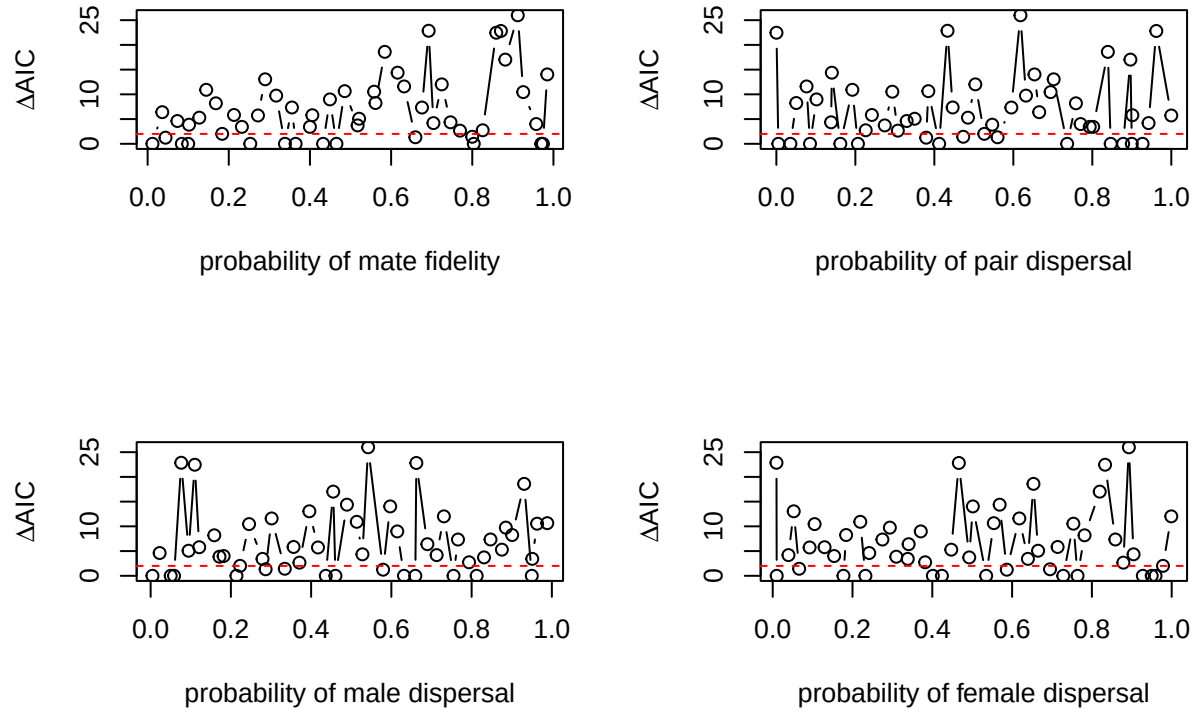
Eggs Response

Band ID



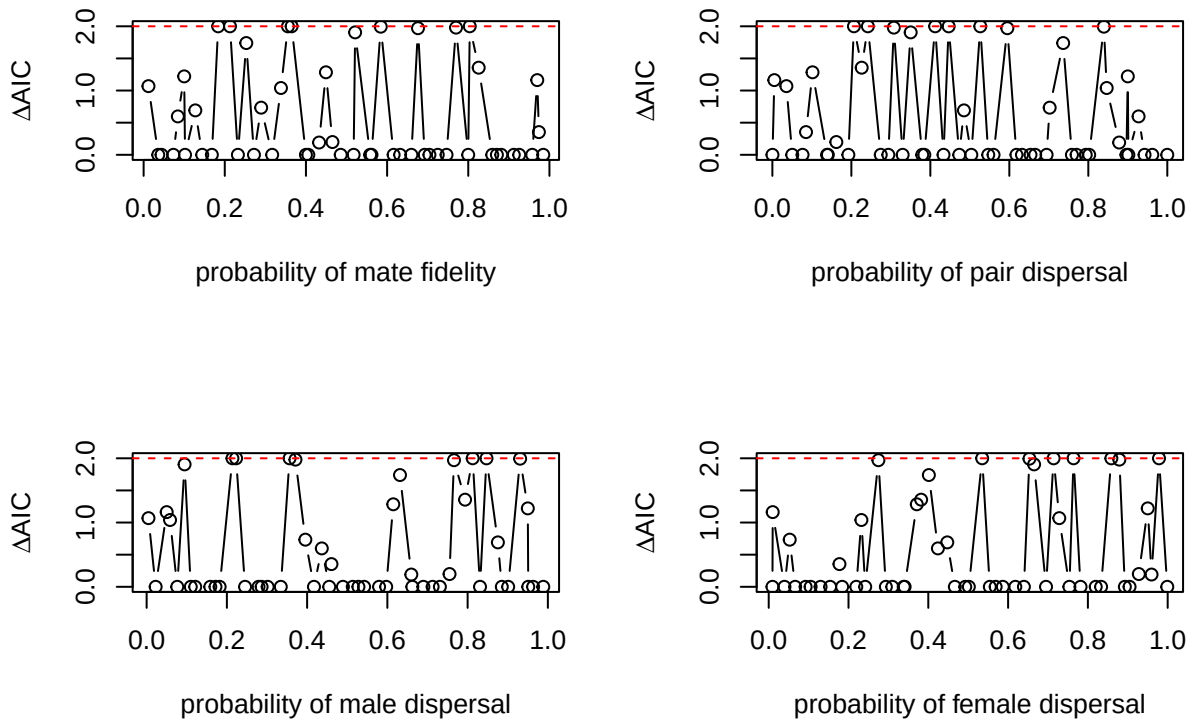
There doesn't seem to be a clear pattern, which would make sense as band ID plays no role in variation in the response.

Territory



Looking at these plots, it seems the only 2 probabilities that potentially play a role in the territory only model performing well are probabilities of female and pair dispersals, and they seem to need to be higher.

Band ID + Territory



It seems regardless of any probability setting, in the binomial response models, the model fitting both REs performed very well on AIC.

Proportion of band ID only models that best optimize AIC and BIC based on response type

	Criteria	
	AIC	BIC
Dist		
Poisson	0.25	0.29
Binomial	0.21	0.46

Proportion of territory only models that best optimize AIC and BIC based on response type

	Criteria
--	----------

	AIC	BIC
Dist		
Poisson	0.62	0.69
Binomial	0.21	0.33

Proportion of band ID + territory models that best optimize AIC and BIC based on response type

	Criteria	
	AIC	BIC
Dist		
Poisson	0.13	0.58
Binomial	0.02	0.21

Heat maps

Fledge response models

Band ID only models

```
terra 1.9.34
```

```
Attaching package: 'terra'
```

```
The following object is masked from 'package:gratia':
```

```
draw
```

```
The following objects are masked from 'package:vcd':
```

```
mosaic, sieve
```

```
The following object is masked from 'package:grid':
```

```
depth
```

```
The following object is masked from 'package:tidyr':
```

```
extract
```

```
Loading required package: spam
```

```
Spam version 2.11-4 (2026-05-28) is loaded.
```

```
Type 'help( Spam)' or 'demo( spam)' for a short introduction  
and overview of this package.
```

```
Help for individual functions is also obtained by adding the  
suffix '.spam' to the function name, e.g. 'help( chol.spam)'.
```

Attaching package: 'spam'

The following object is masked from 'package:Matrix':

det

The following object is masked from 'package:performance':

display

The following objects are masked from 'package:base':

backsolve, forwardsolve

Loading required package: viridisLite

Loading required package: RColorBrewer

Try help(fields) to get started.

Attaching package: 'fields'

The following object is masked from 'package:terra':

describe

The following object is masked from 'package:ggfortify':

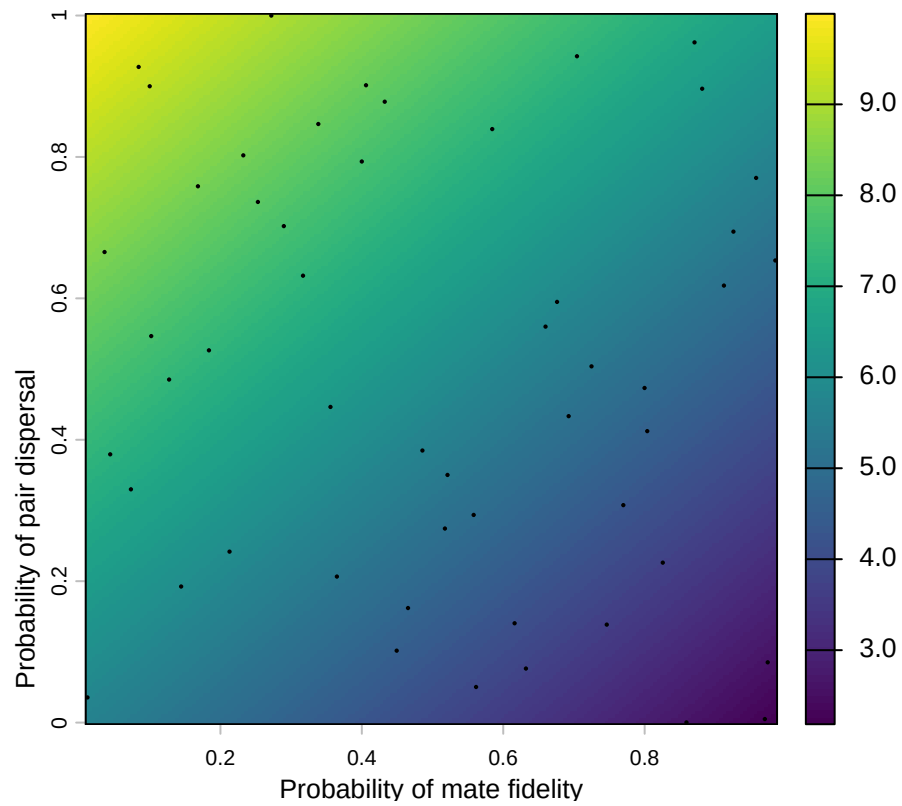
unscale

Warning:

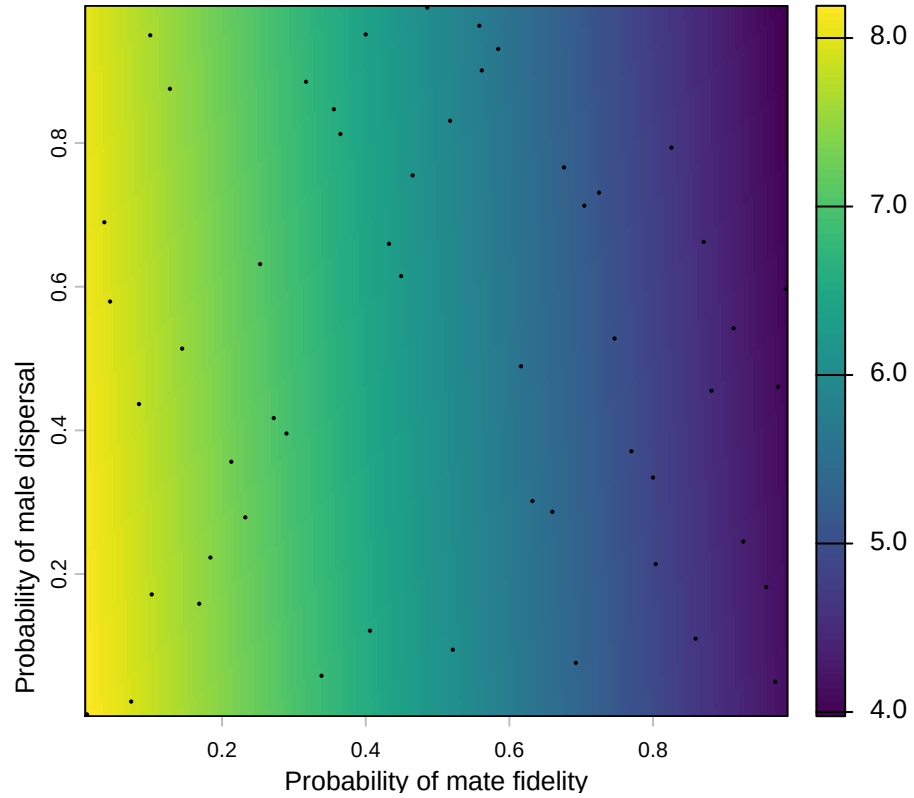
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:

(GCV) Generalized Cross-Validation

minimum at right endpoint lambda = 346.6894 (eff. df= 3.00081)



Warning:
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:
(GCV) Generalized Cross-Validation
minimum at right endpoint lambda = 260.8983 (eff. df= 3.000995)

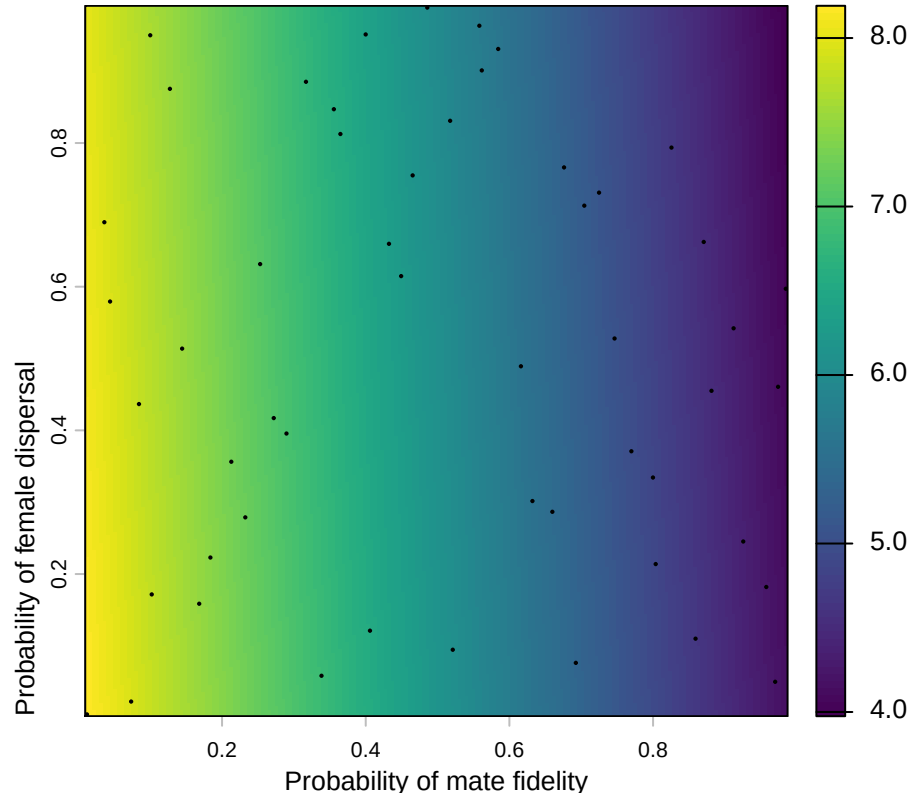


Warning:

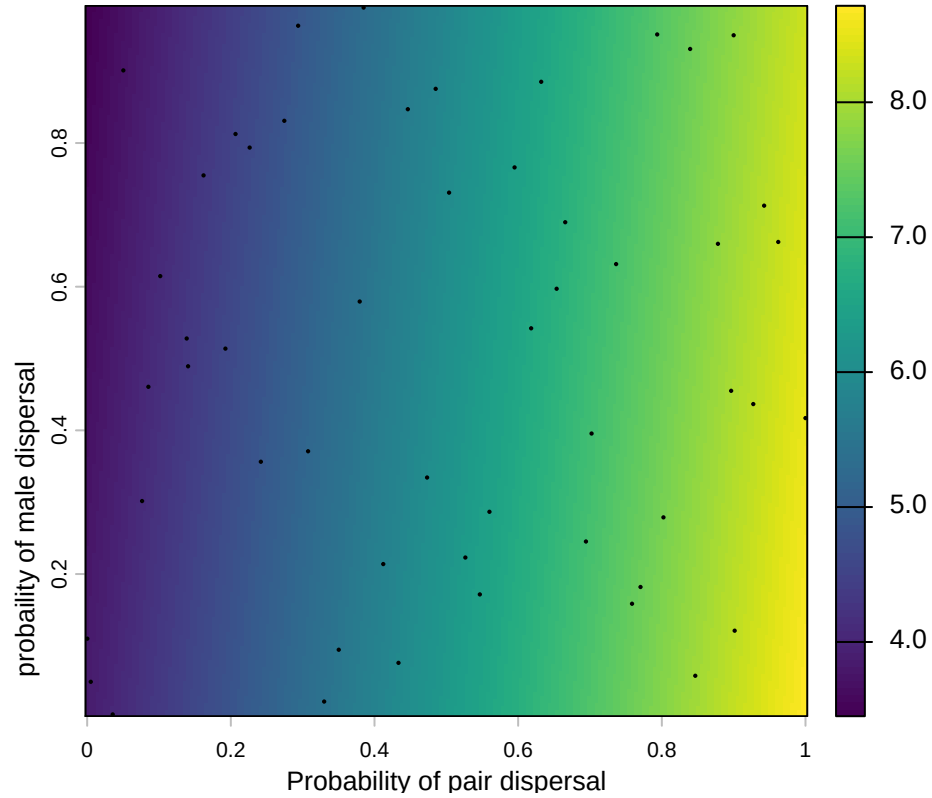
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:

(GCV) Generalized Cross-Validation

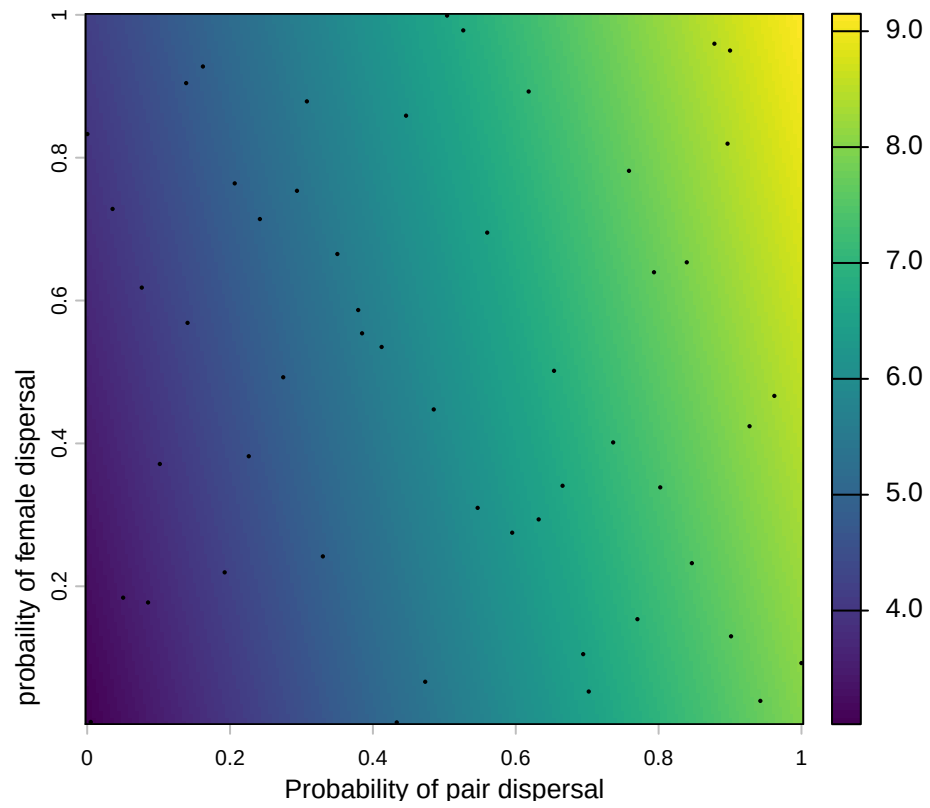
minimum at right endpoint lambda = 260.8983 (eff. df= 3.000995)

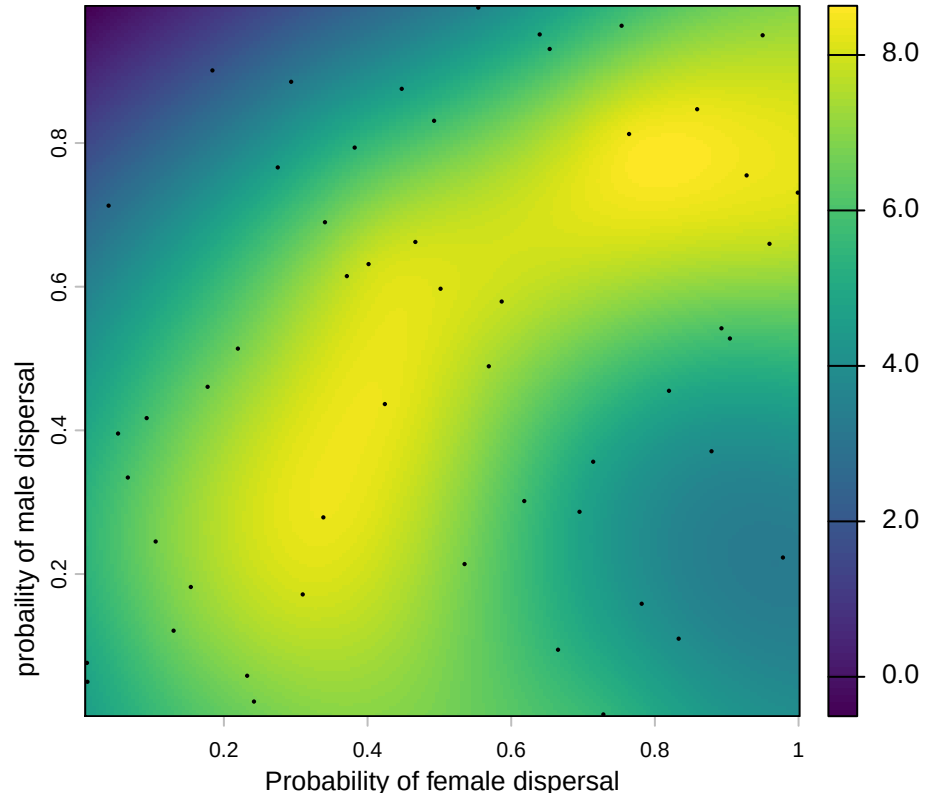


Warning:
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:
(GCV) Generalized Cross-Validation
minimum at right endpoint lambda = 275.416 (eff. df= 3.000995)



Warning:
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:
(GCV) Generalized Cross-Validation
minimum at right endpoint lambda = 282.9751 (eff. df= 3.001007)

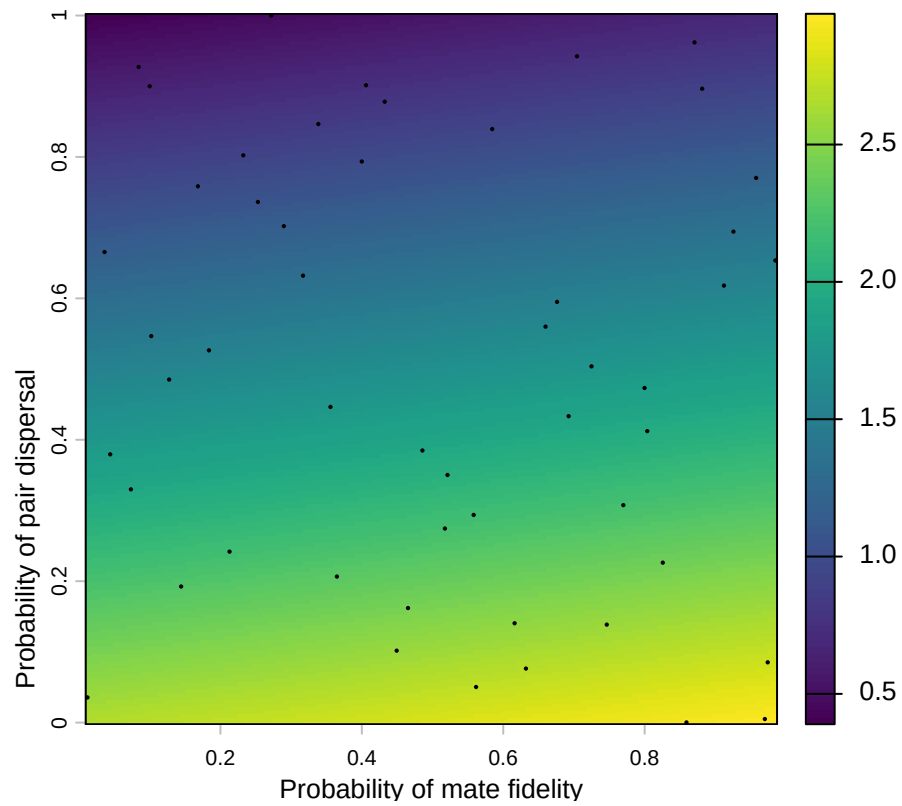




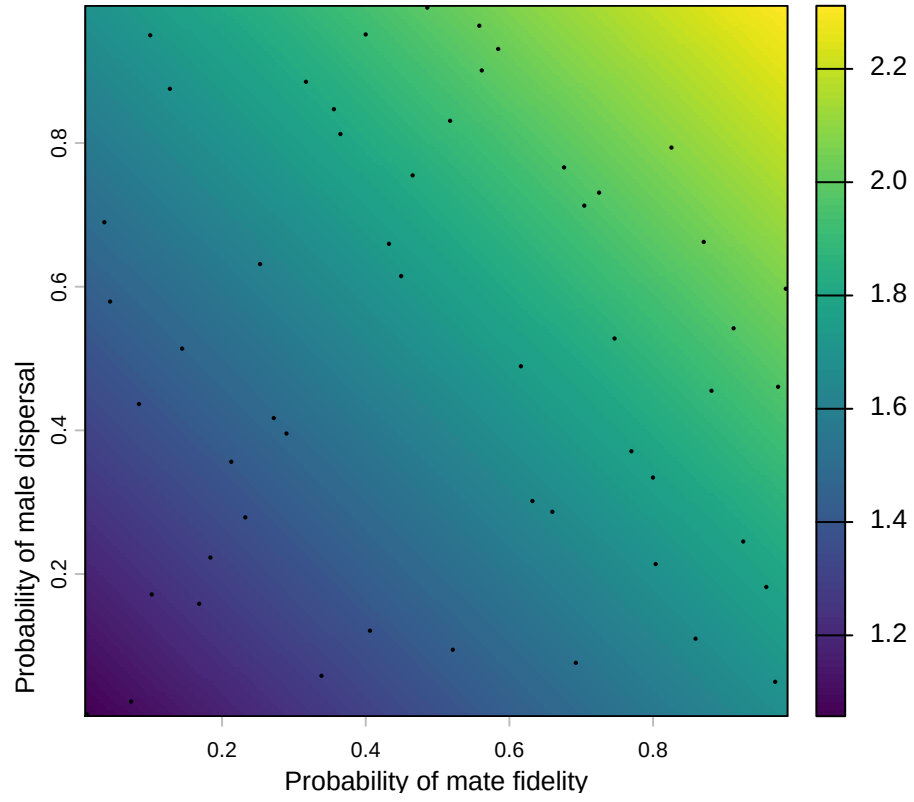
Territory only models

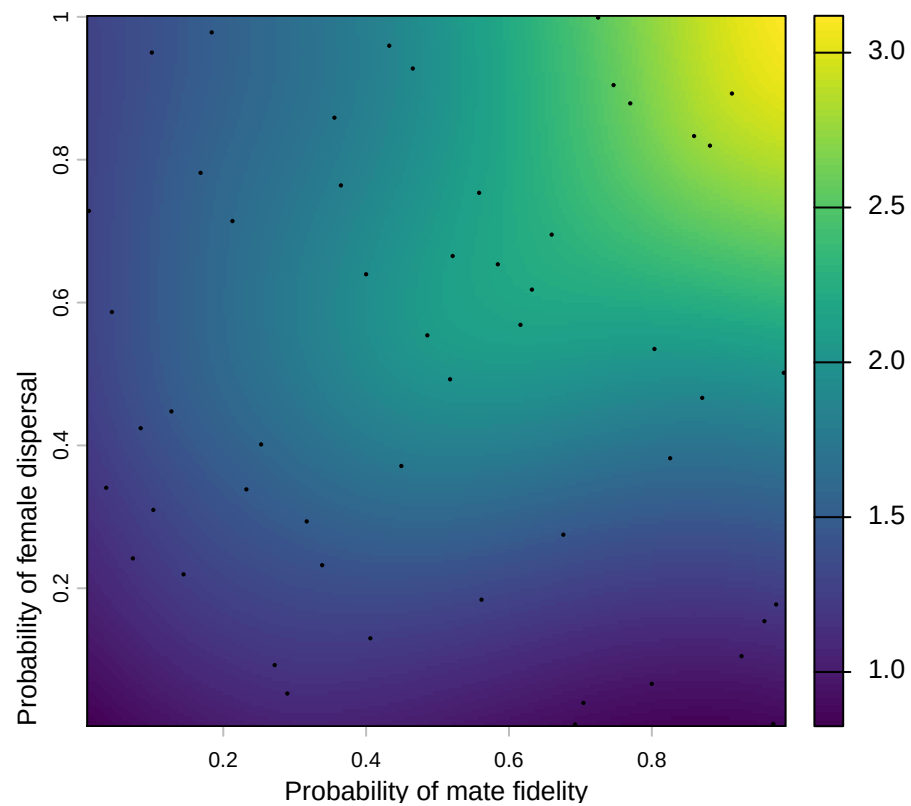
Warning:

Grid searches over lambda (nugget and sill variances) with minima at the endpoints:
 (GCV) Generalized Cross-Validation
 minimum at right endpoint lambda = 346.6894 (eff. df= 3.00081)

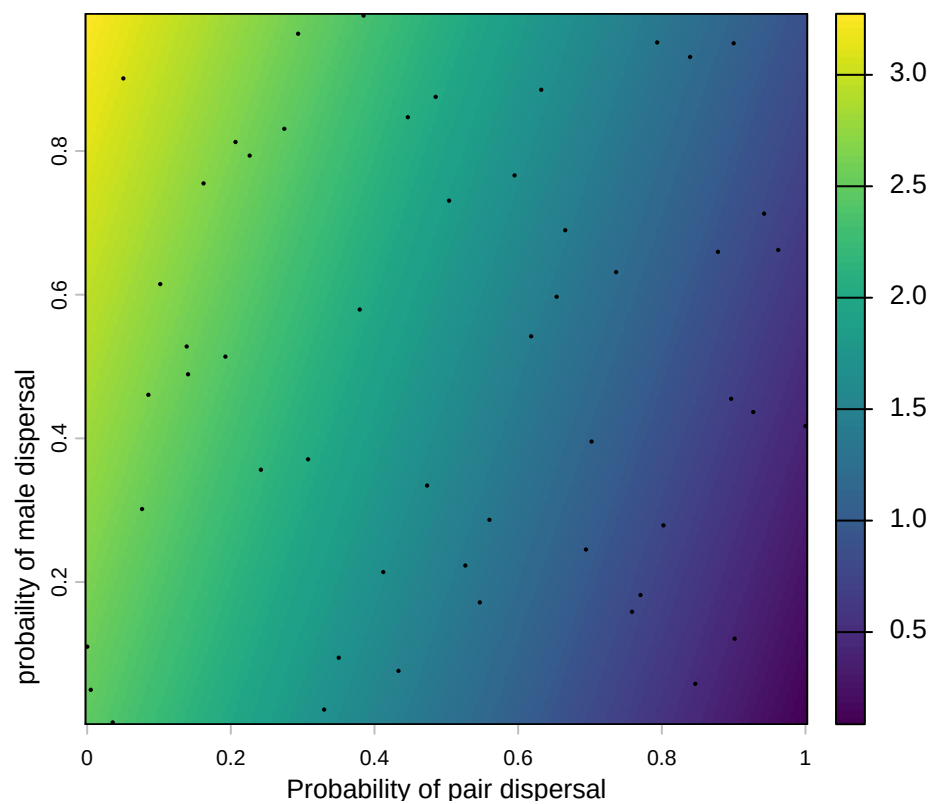


Warning:
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:
(GCV) Generalized Cross-Validation
minimum at right endpoint lambda = 260.8983 (eff. df= 3.000995)





Warning:
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:
(GCV) Generalized Cross-Validation
minimum at right endpoint lambda = 275.416 (eff. df= 3.000995)



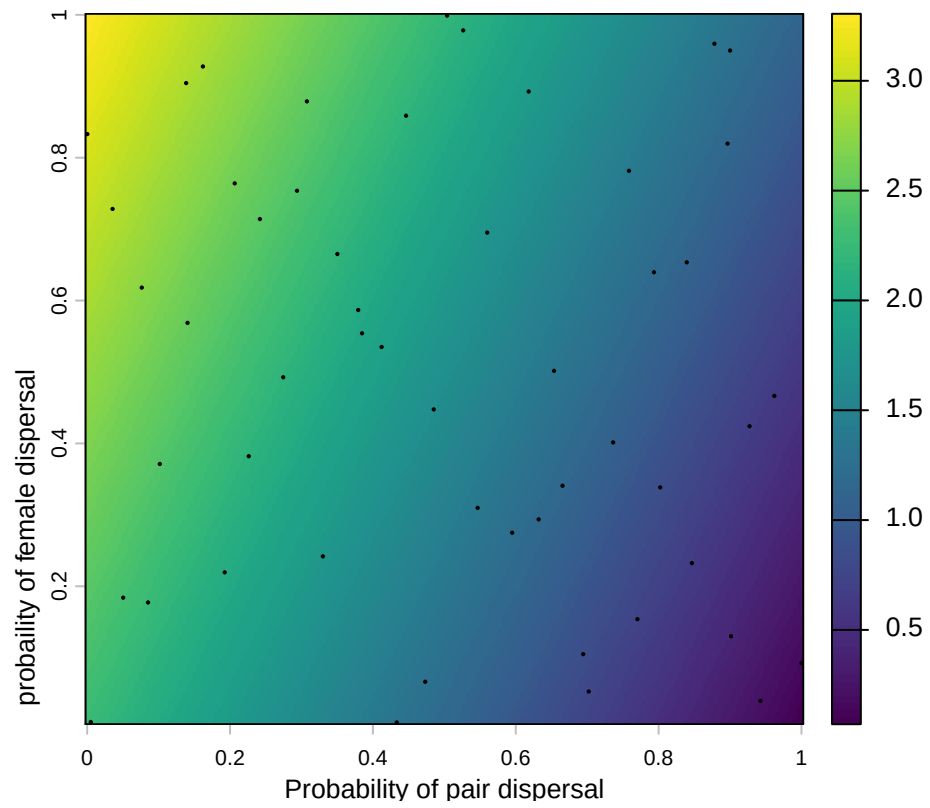
Since we fit only females in the model, it makes sense that male dispersal doesn't seem to make a huge difference in ΔAIC .

Warning:

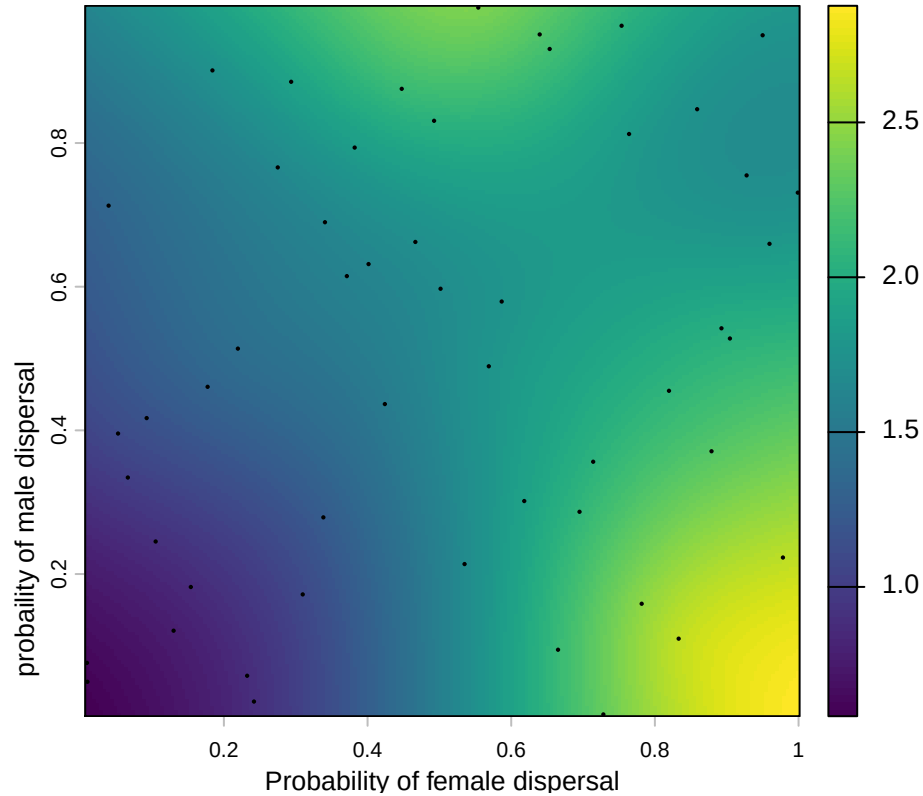
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:

(GCV) Generalized Cross-Validation

minimum at right endpoint lambda = 282.9751 (eff. df= 3.001007)



I'm surprised to see pair dispersal seems to be driving the bivariate relationship.



As before, I'm not surprised that there isn't a clear bivariate relationship here as male dispersal doesn't seem like it should play a huge role since we only fit the models with female observations.

Band ID + territory only models

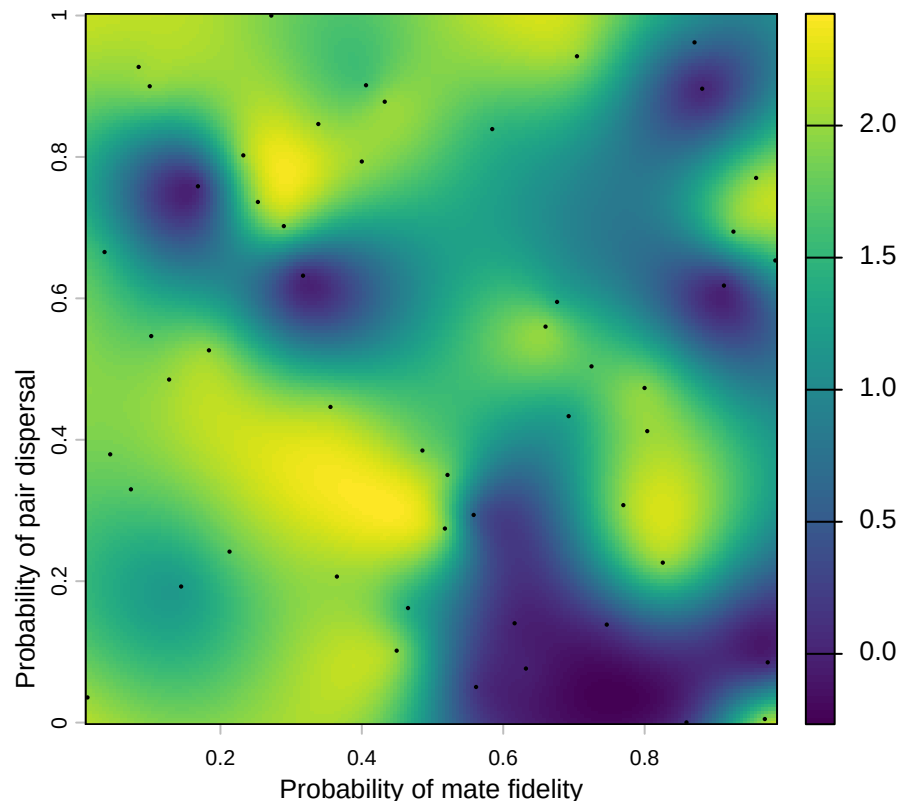
Heat maps for band ID only models

Warning:

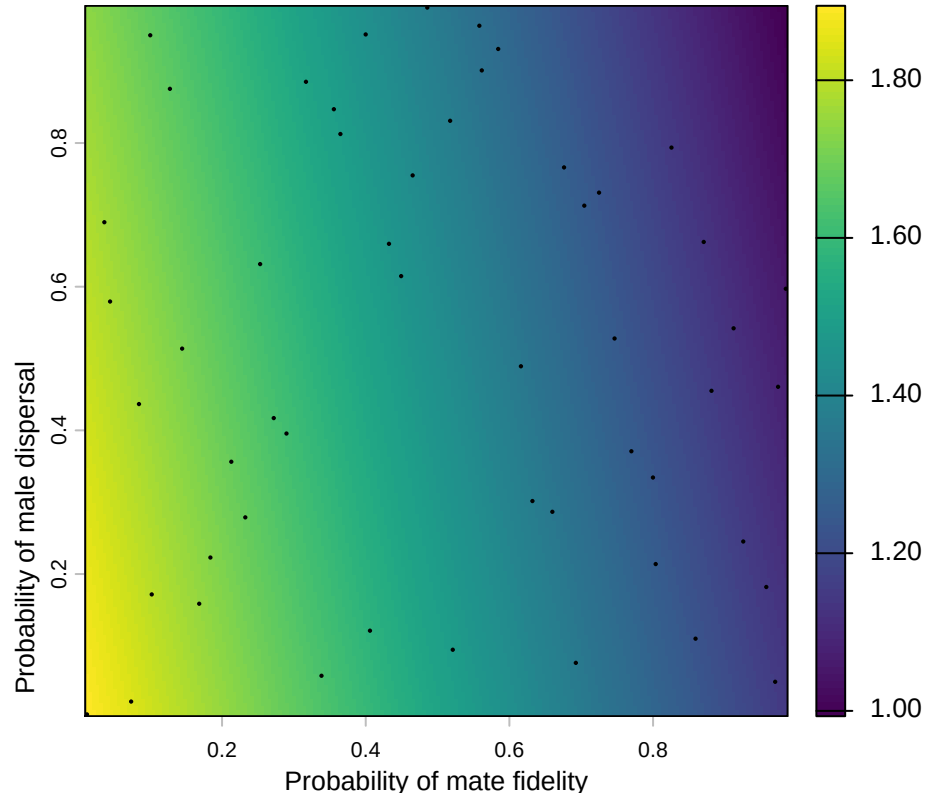
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:

(GCV) Generalized Cross-Validation

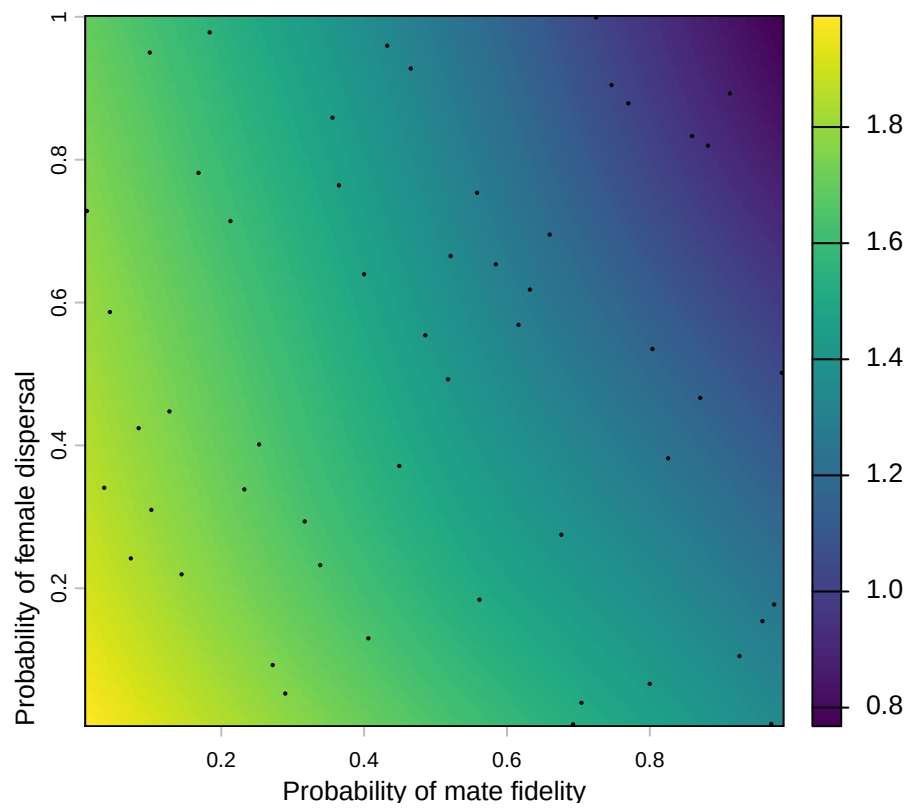
minimum at right endpoint lambda = 2.028851e-05 (eff. df= 49.39999)

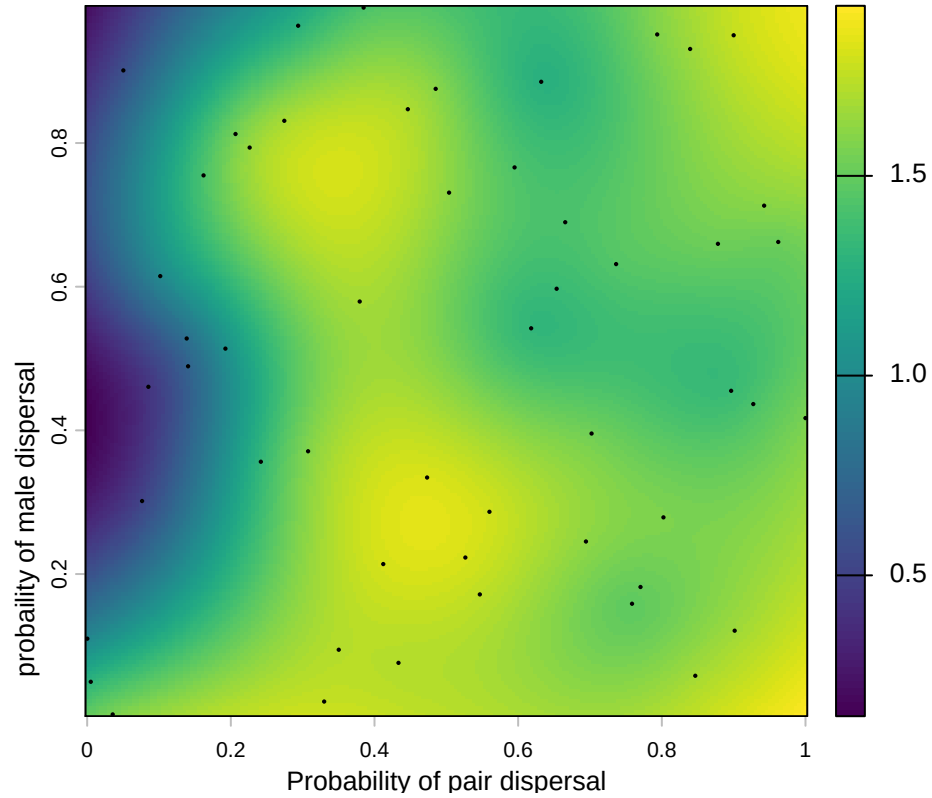


Warning:
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:
(GCV) Generalized Cross-Validation
minimum at right endpoint lambda = 260.8983 (eff. df= 3.000995)

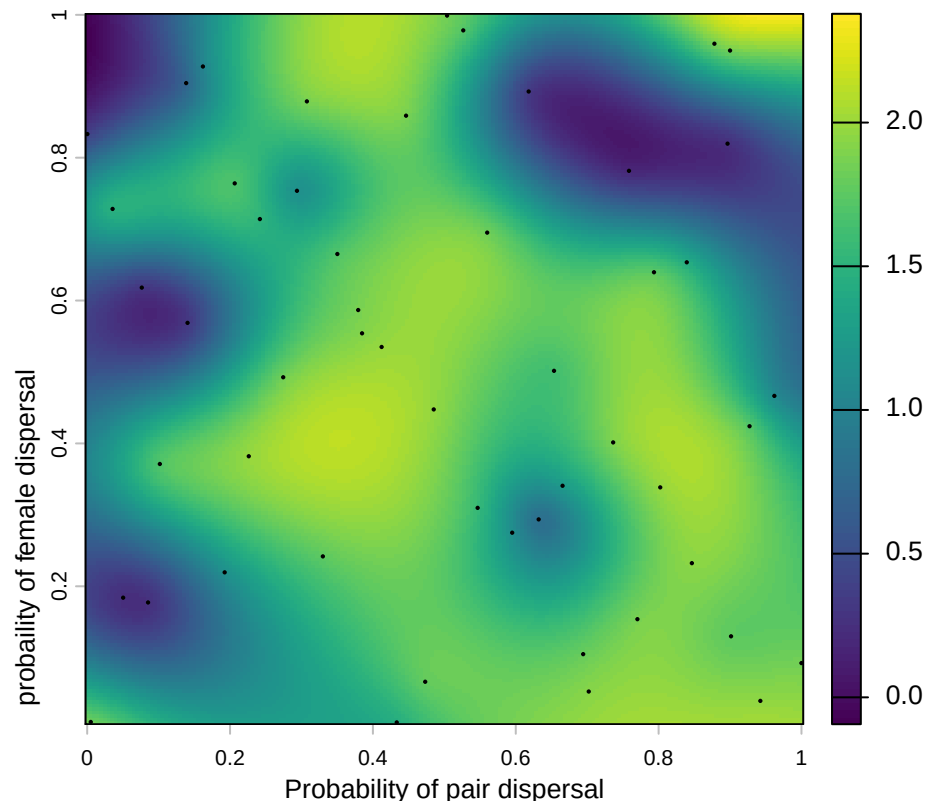


Primarily driven by mate fidelity it seems.

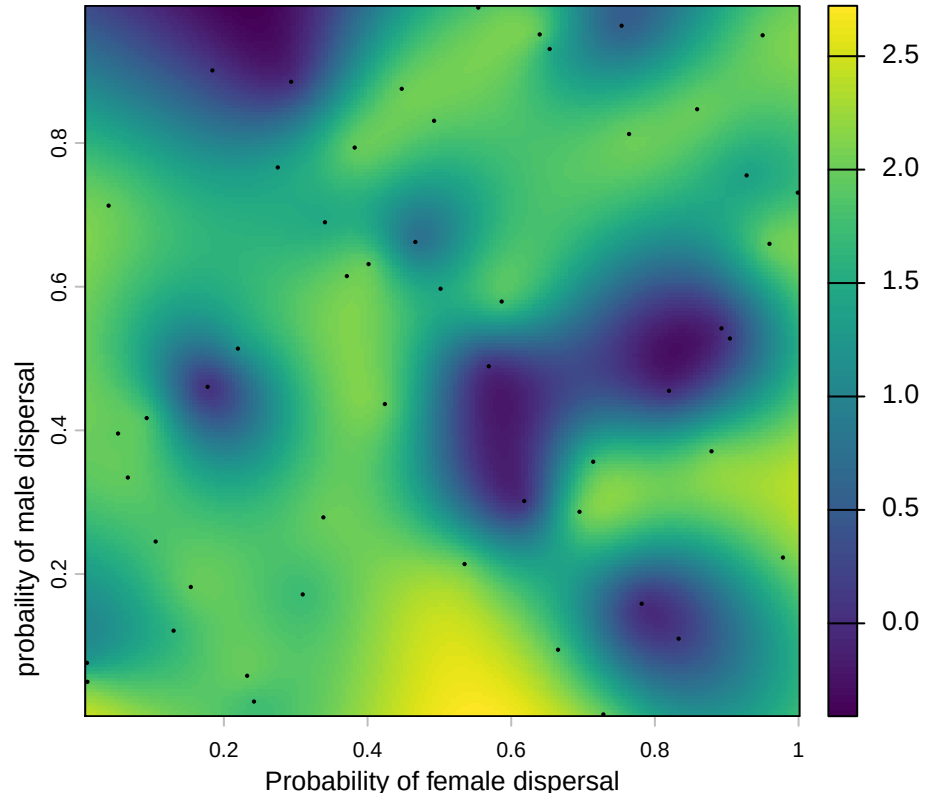




It seems as pair dispersal increases, so does Δ AIC. This makes sense to me as we would expect a model that has an effect that wasn't present in the data generation, the model won't perform as well and as pairs disperse, it should be clearer that the quality of the bird plays no role in variation of fledgling success.



Warning:
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:
(GCV) Generalized Cross-Validation
minimum at right endpoint lambda = 1.797176e-05 (eff. df= 49.4)



Eggs response

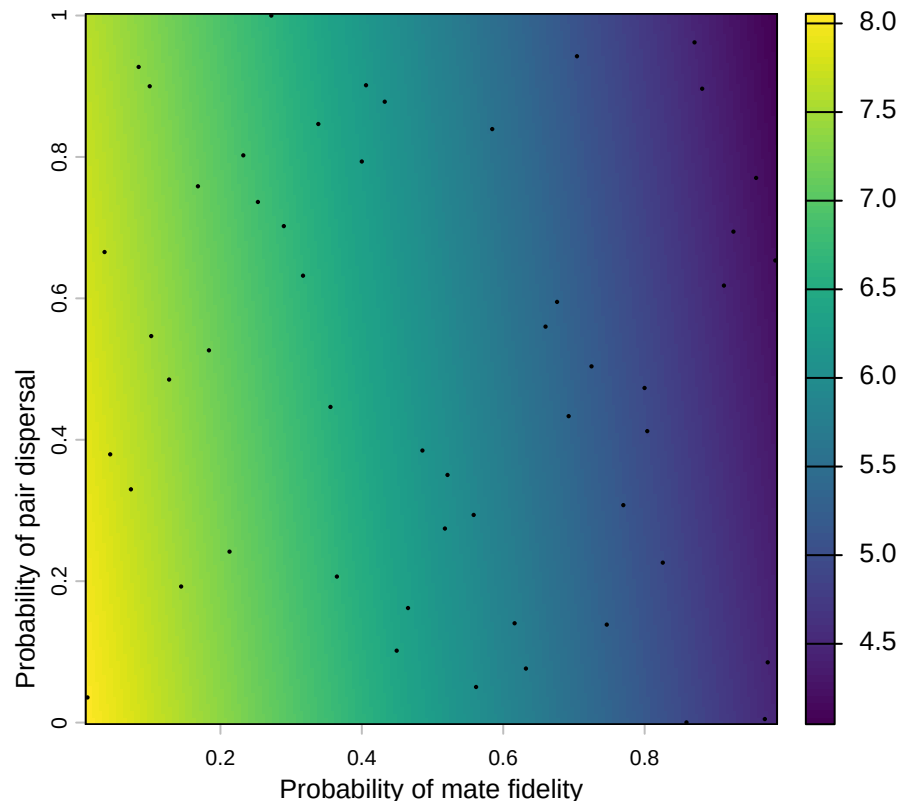
Band ID only models

Warning:

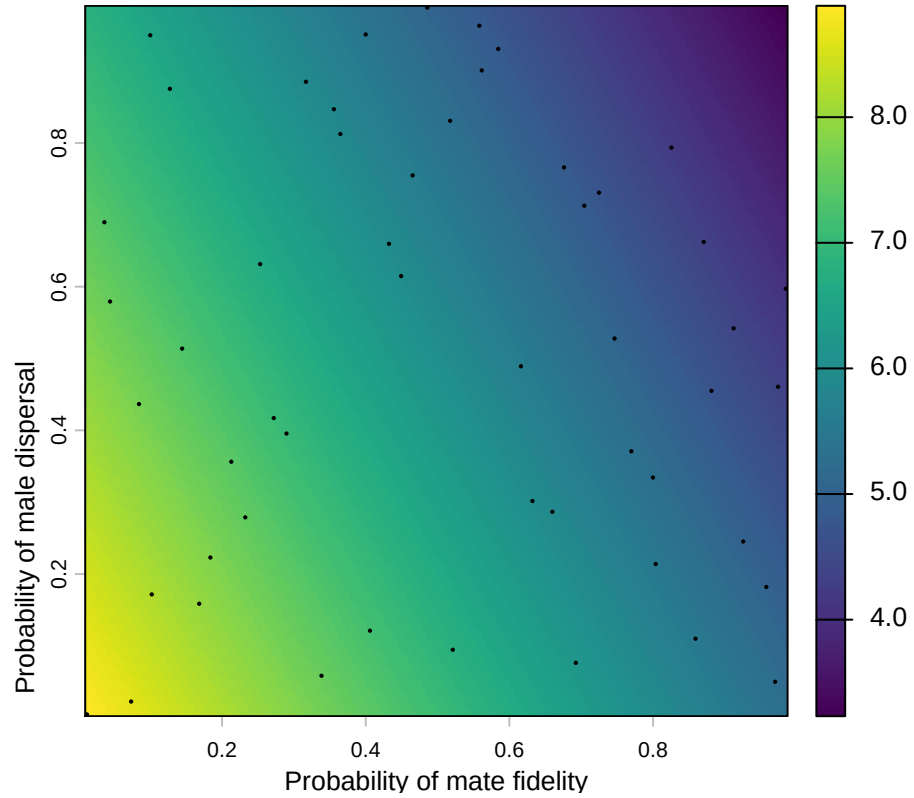
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:

(GCV) Generalized Cross-Validation

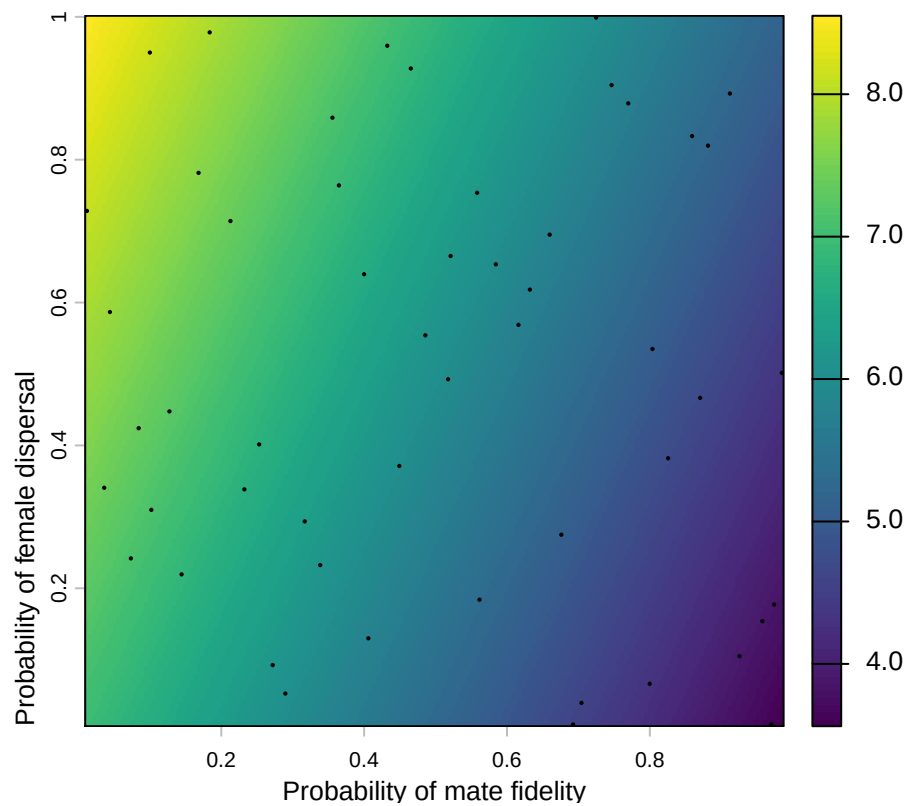
minimum at right endpoint lambda = 346.6894 (eff. df= 3.00081)



Warning:
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:
(GCV) Generalized Cross-Validation
minimum at right endpoint lambda = 260.8983 (eff. df= 3.000995)



Warning:
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:
(GCV) Generalized Cross-Validation
minimum at right endpoint lambda = 224.8003 (eff. df= 3.00124)

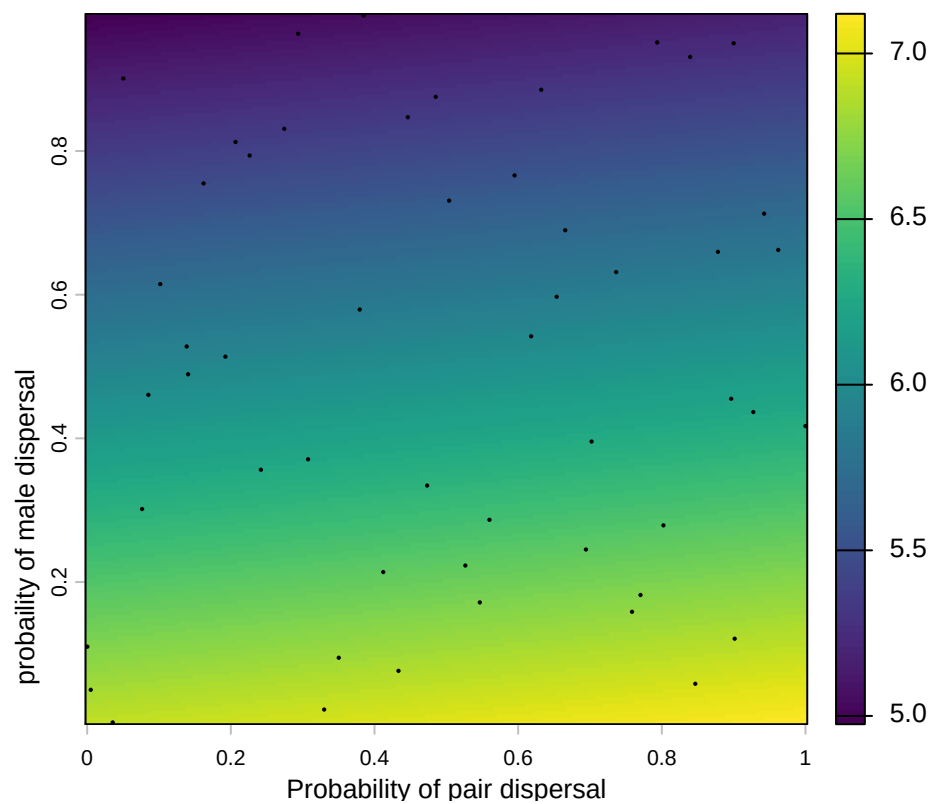


Warning:

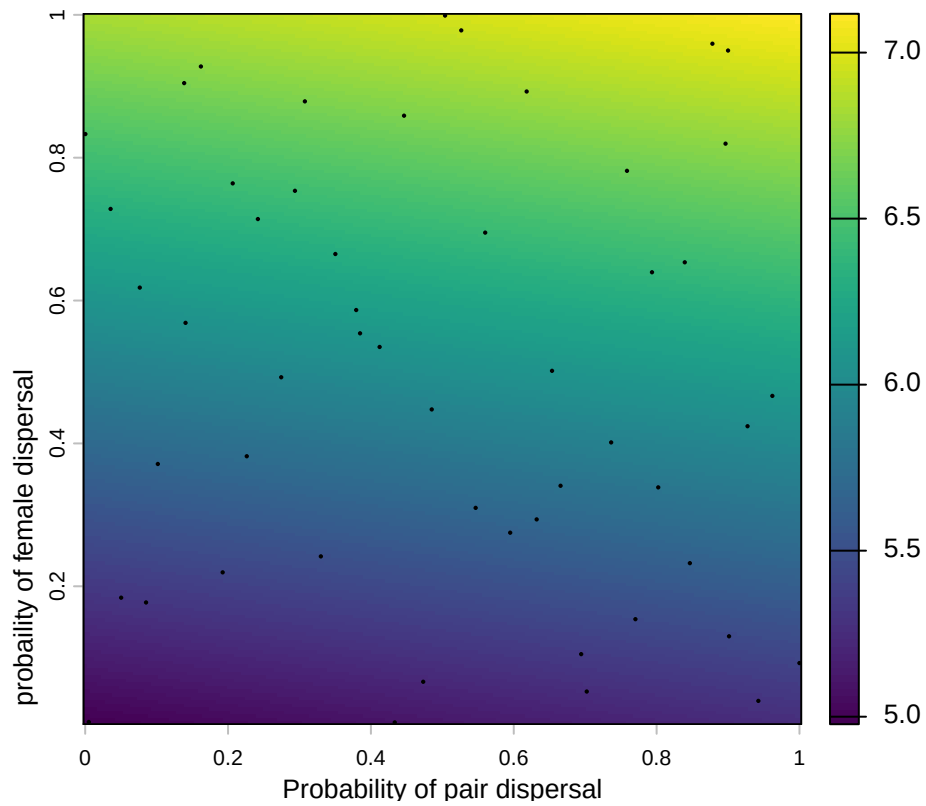
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:

(GCV) Generalized Cross-Validation

minimum at right endpoint lambda = 275.416 (eff. df= 3.000995)



Warning:
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:
(GCV) Generalized Cross-Validation
minimum at right endpoint lambda = 282.9751 (eff. df= 3.001007)



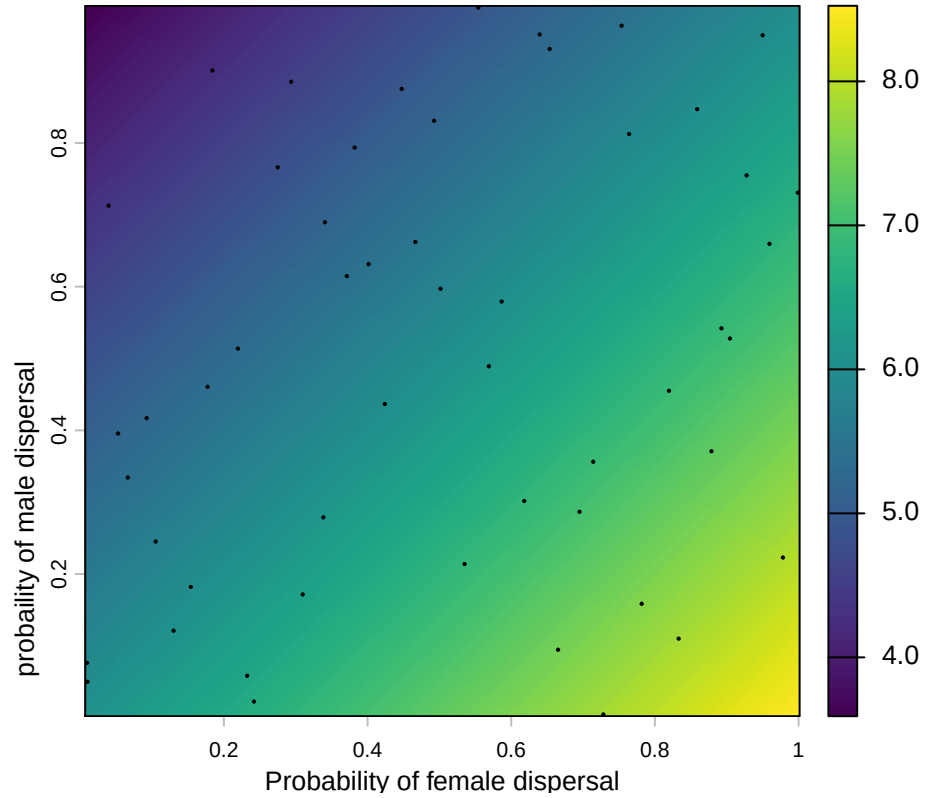
This seems to be primarily driven by female dispersal.

Warning:

Grid searches over lambda (nugget and sill variances) with minima at the endpoints:

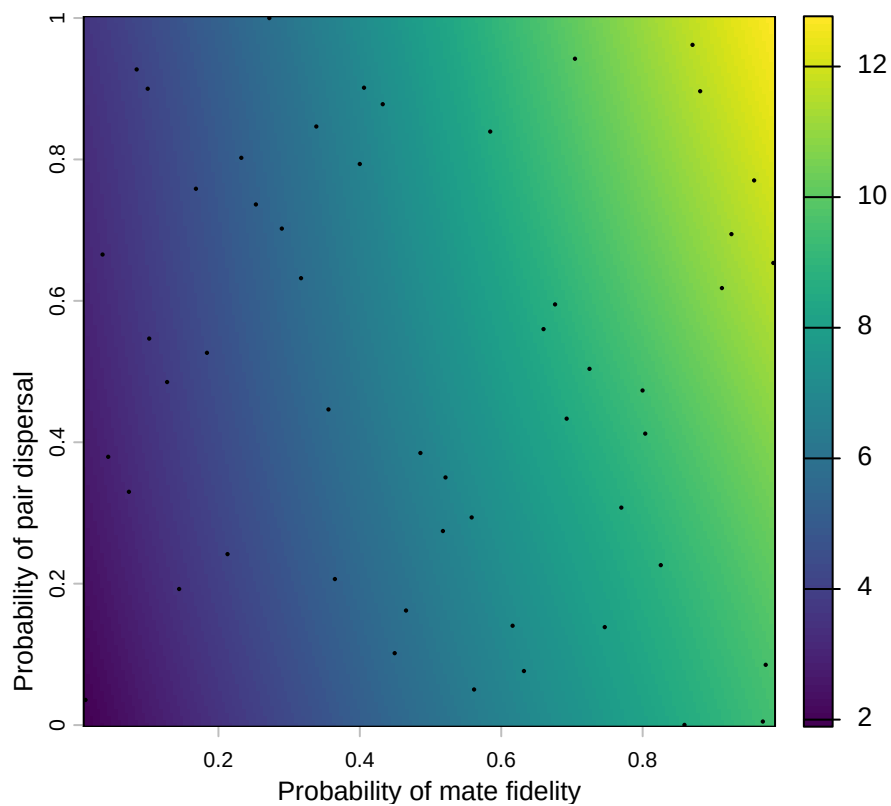
(GCV) Generalized Cross-Validation

minimum at right endpoint lambda = 268.0589 (eff. df= 3.001013)



It makes sense that as female dispersal increases, so does ΔAIC . As females disperse, we should hopefully see the territory effect a bit more clearly.

Territory only models



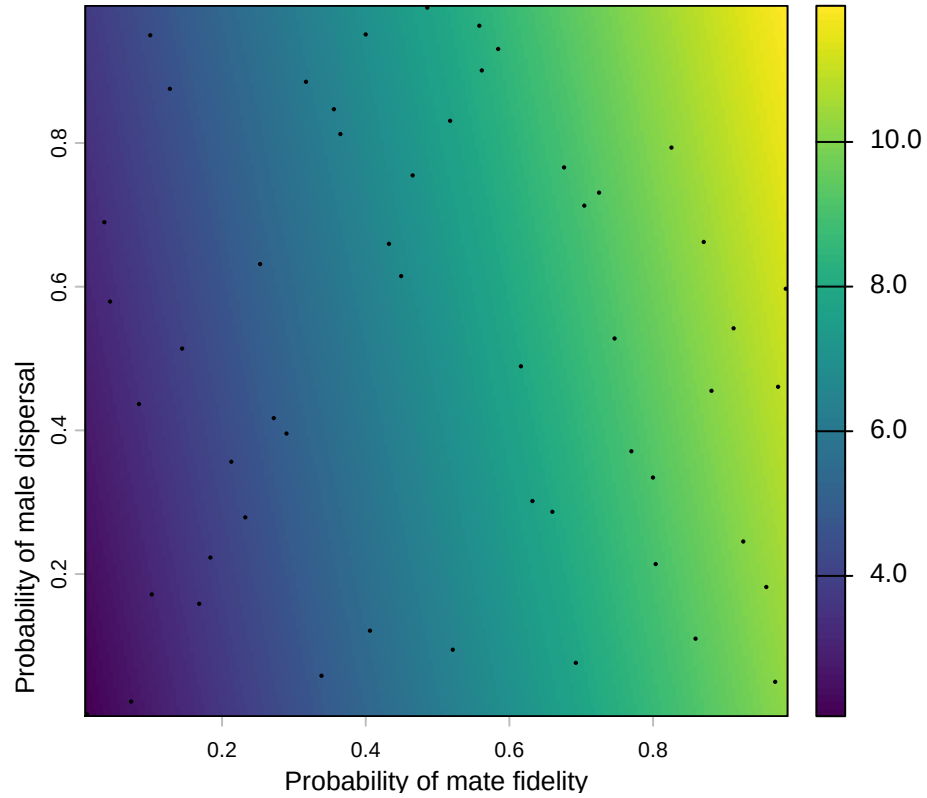
Seems to be more driven by mate fidelity and the story makes sense.

Warning:

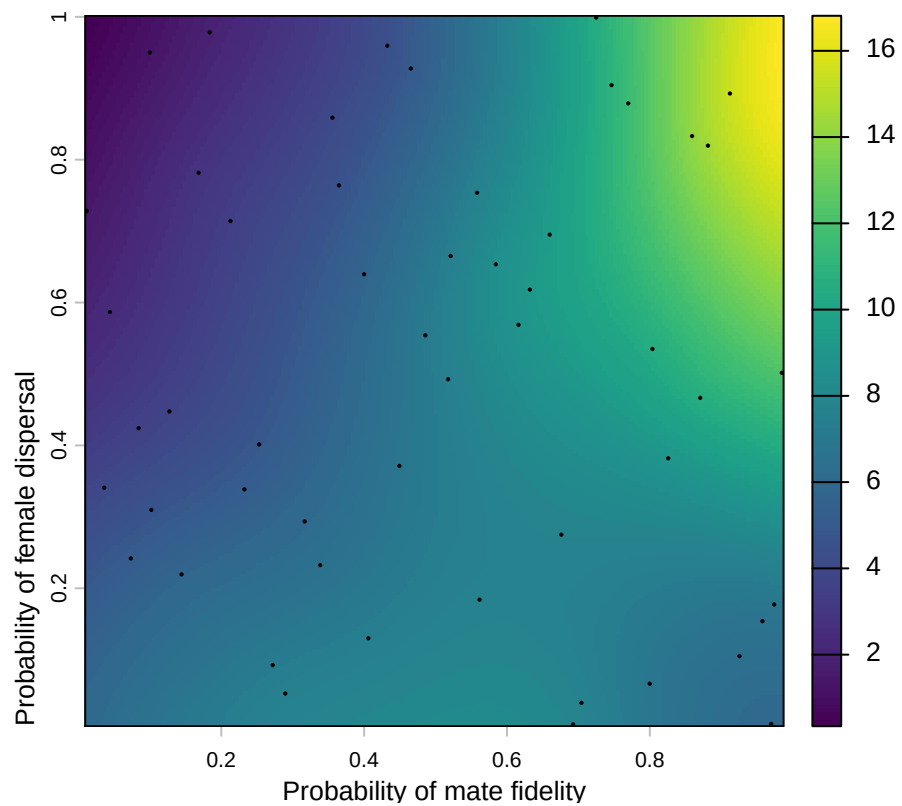
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:

(GCV) Generalized Cross-Validation

minimum at right endpoint lambda = 260.8983 (eff. df= 3.000995)



Seeing that male dispersal shouldn't play a large role, it makes sense to see mate fidelity drive the bivariate relationship.

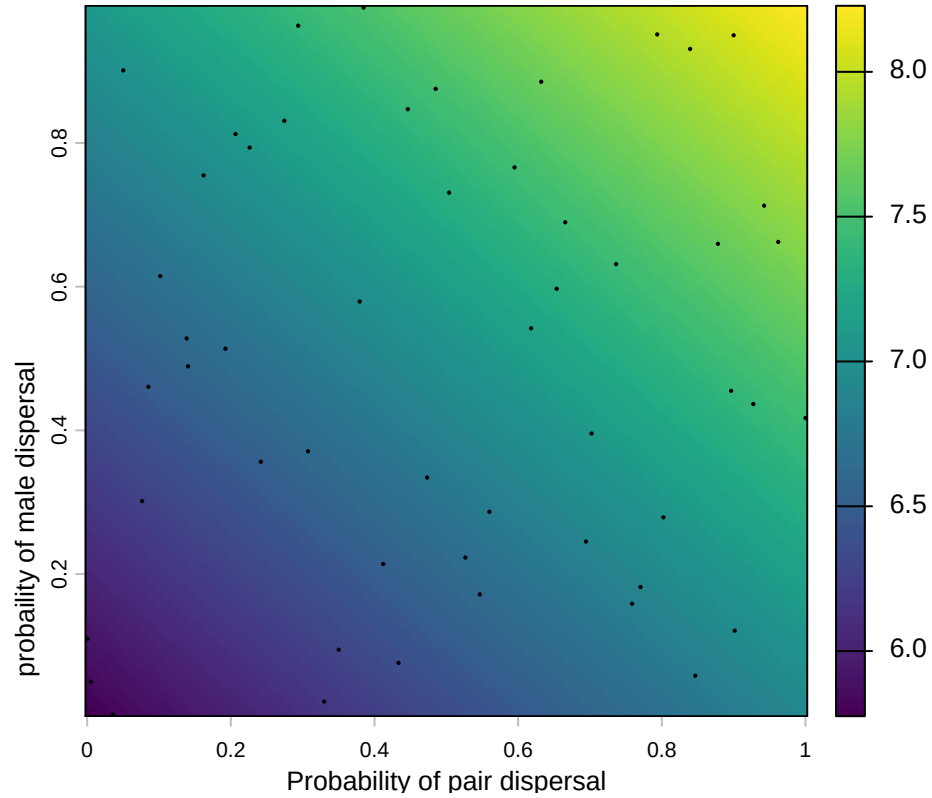


Warning:

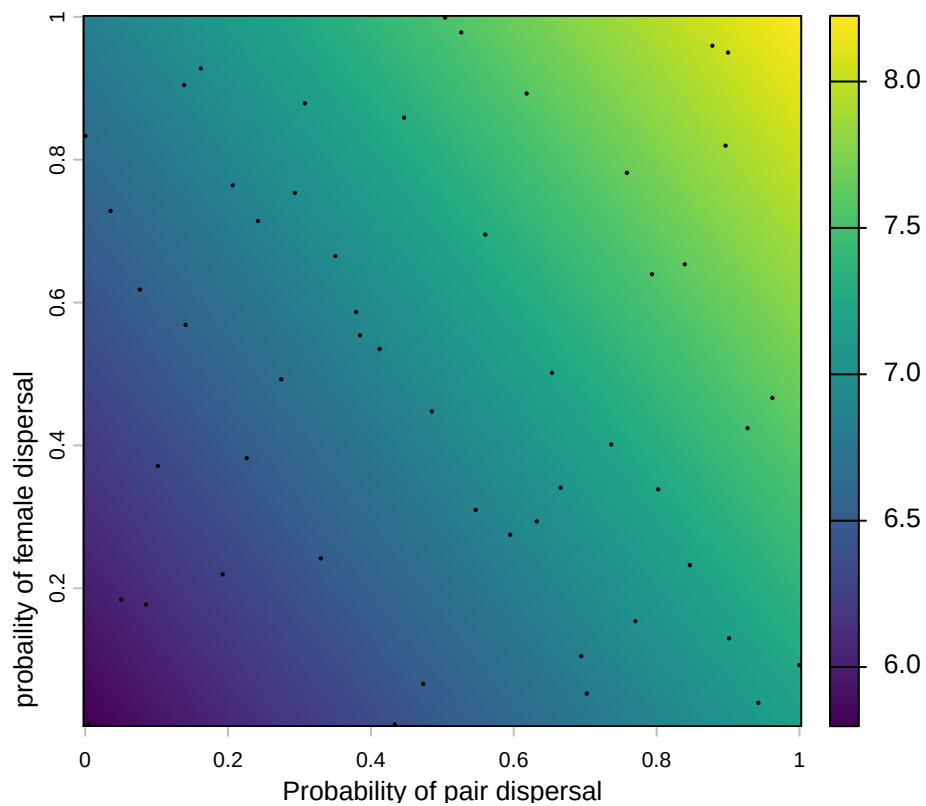
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:

(GCV) Generalized Cross-Validation

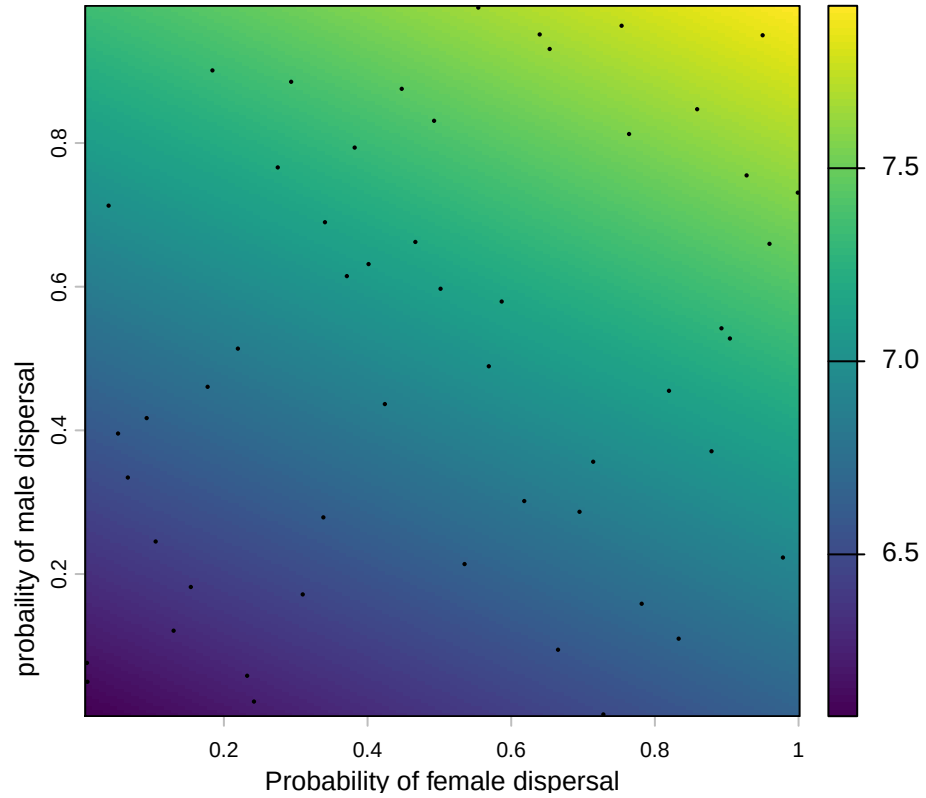
minimum at right endpoint lambda = 275.416 (eff. df= 3.000995)



Warning:
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:
(GCV) Generalized Cross-Validation
minimum at right endpoint lambda = 282.9751 (eff. df= 3.001007)



Warning:
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:
(GCV) Generalized Cross-Validation
minimum at right endpoint lambda = 268.0589 (eff. df= 3.001013)



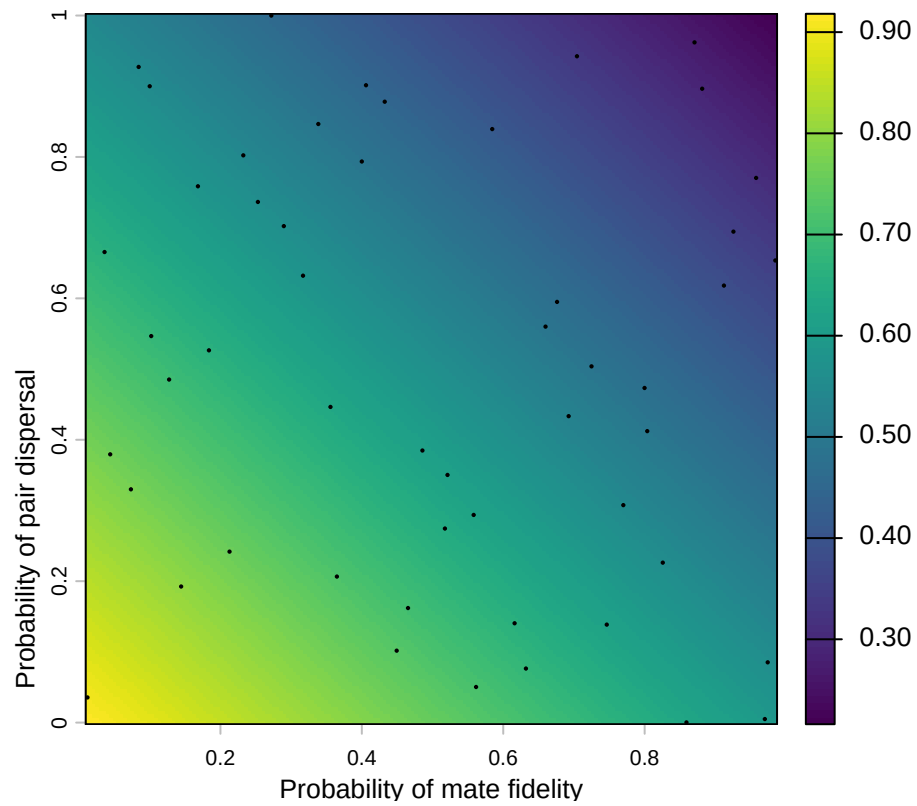
Band ID + territory only models

Warning:

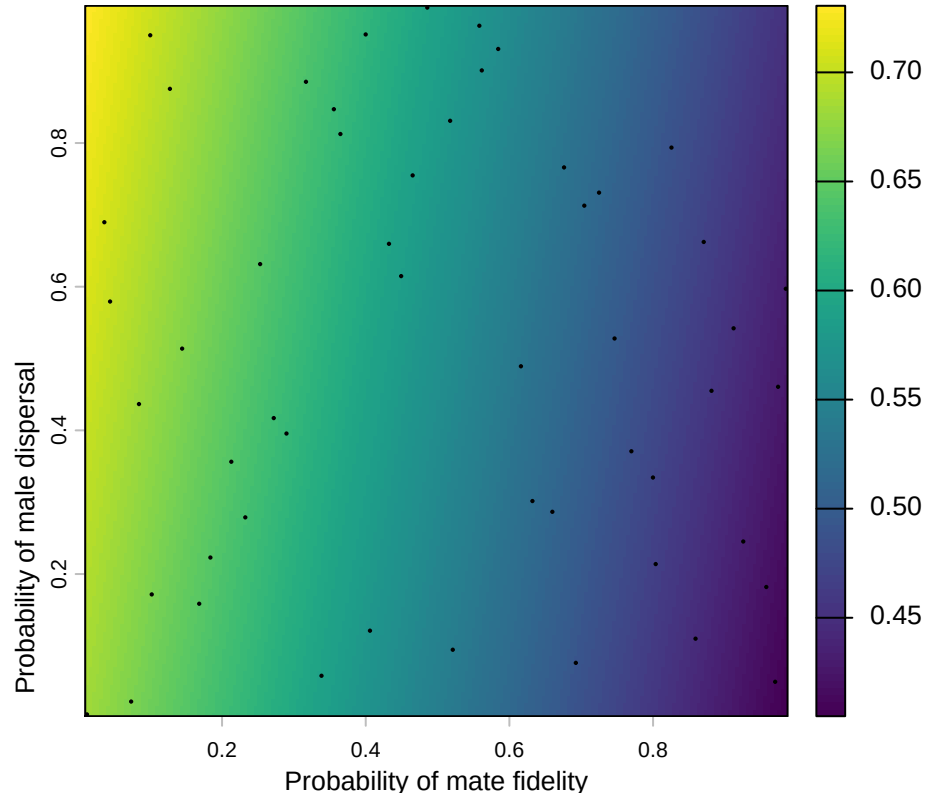
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:

(GCV) Generalized Cross-Validation

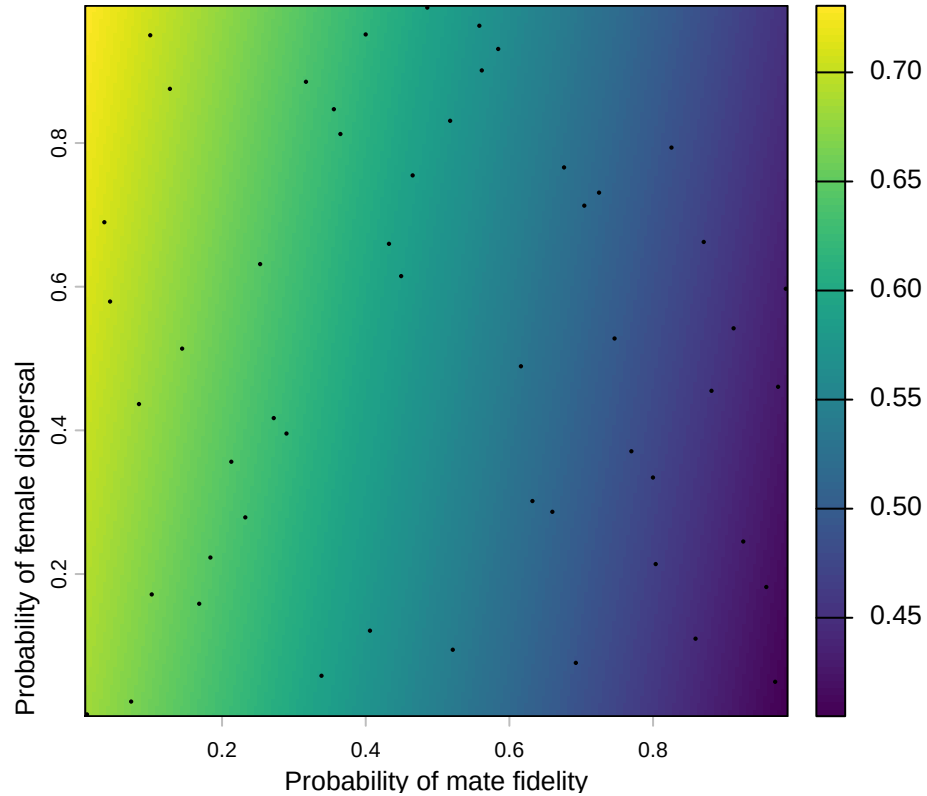
minimum at right endpoint lambda = 346.6894 (eff. df= 3.00081)



Warning:
 Grid searches over lambda (nugget and sill variances) with minima at the endpoints:
 (GCV) Generalized Cross-Validation
 minimum at right endpoint lambda = 260.8983 (eff. df= 3.000995)



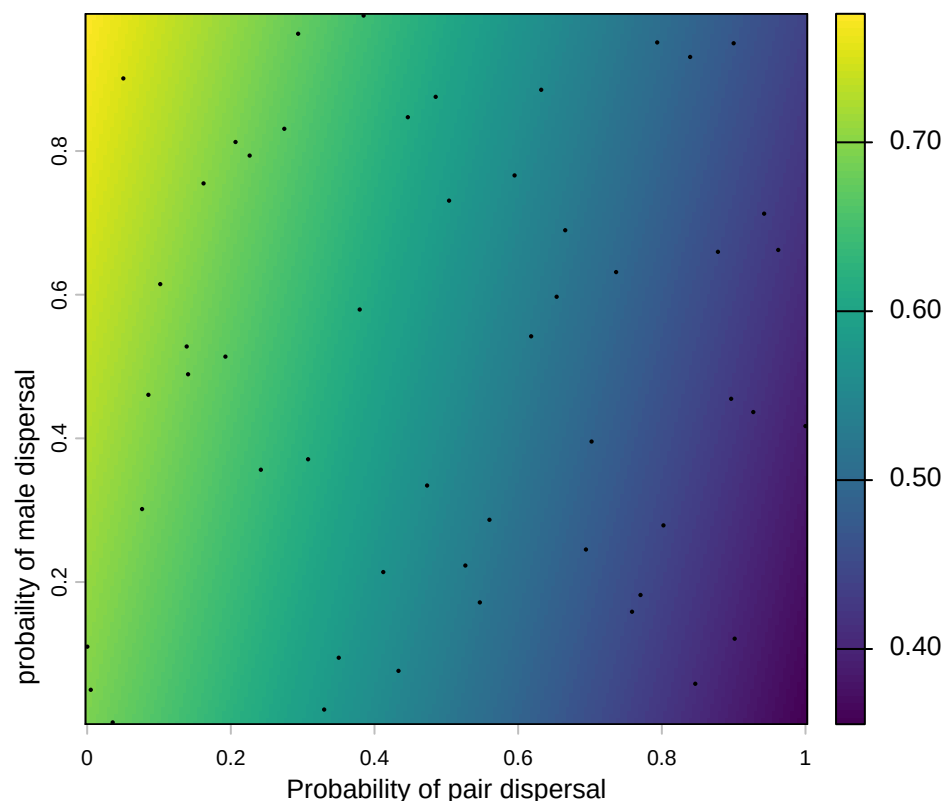
Warning:
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:
(GCV) Generalized Cross-Validation
minimum at right endpoint lambda = 260.8983 (eff. df= 3.000995)



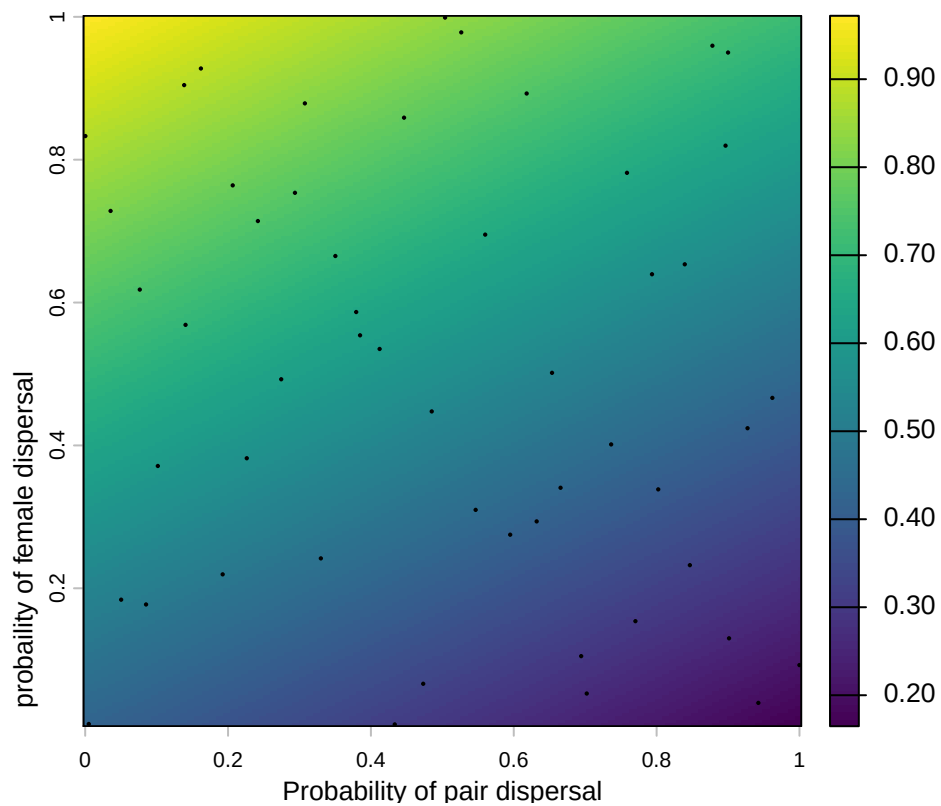
It makes sense that high mate fidelity results in lower Δ AIC. However, the model doesn't predict anything all that low.

Warning:

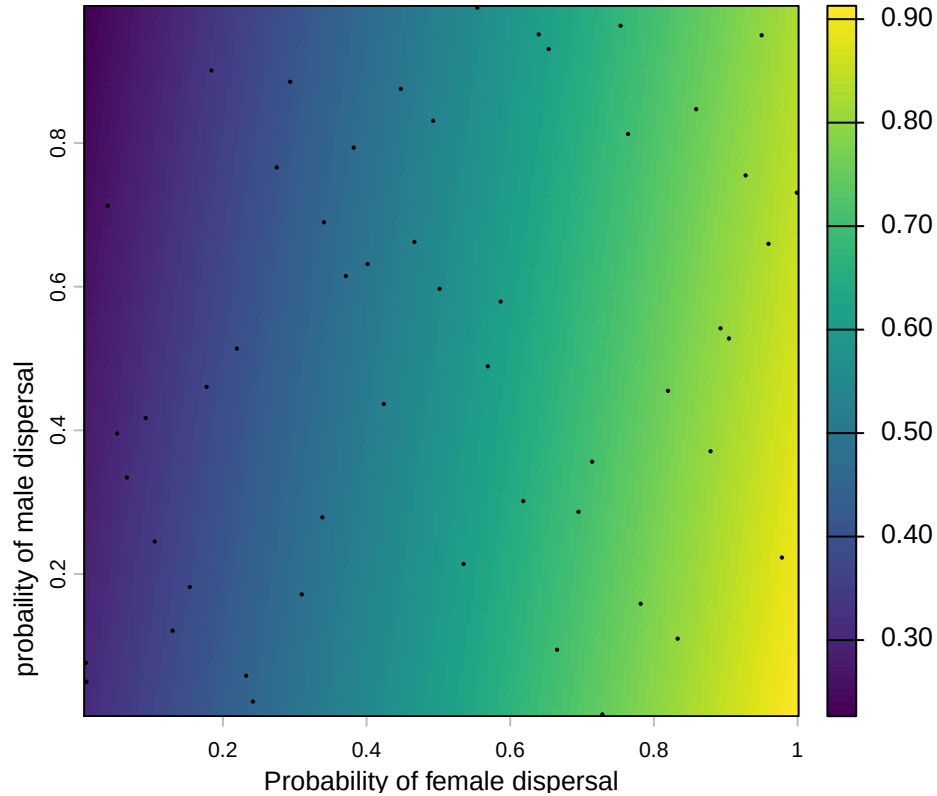
```
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:
(GCV) Generalized Cross-Validation
  minimum at right endpoint lambda = 275.416 (eff. df= 3.000995 )
```



Warning:
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:
(GCV) Generalized Cross-Validation
minimum at right endpoint lambda = 282.9751 (eff. df= 3.001007)



Warning:
Grid searches over lambda (nugget and sill variances) with minima at the endpoints:
(GCV) Generalized Cross-Validation
minimum at right endpoint lambda = 268.0589 (eff. df= 3.001013)



It seems that female dispersal is leading this relation and I feel that it makes sense that as females disperse more, that we can more easily detect the territory effect, i.e., ΔAIC gets higher. Note that ΔAIC is still quite low.

Some Summaries

0. From the summary plots and table, it seems that the Poisson models do a better job detecting the territory effect that was coded in the simulation, and the models that fit both random effects seem to still do well on model selection criteria. There seemed to be some slight patterns in ΔAIC based on certain simulation probabilistic settings, but surprisingly nothing super clear.